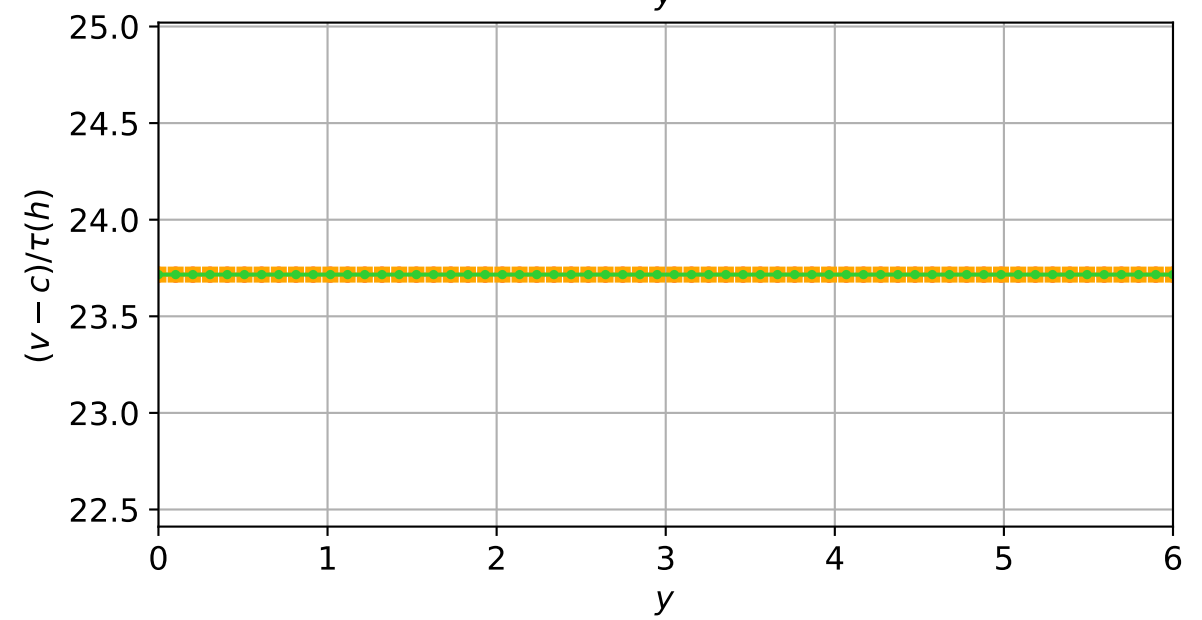
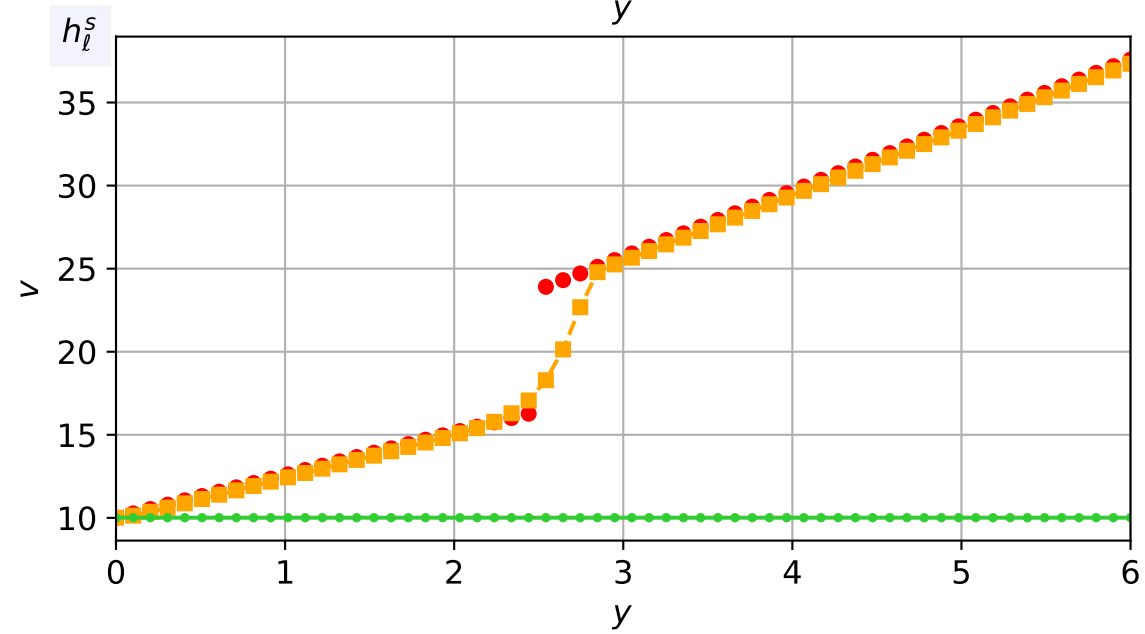
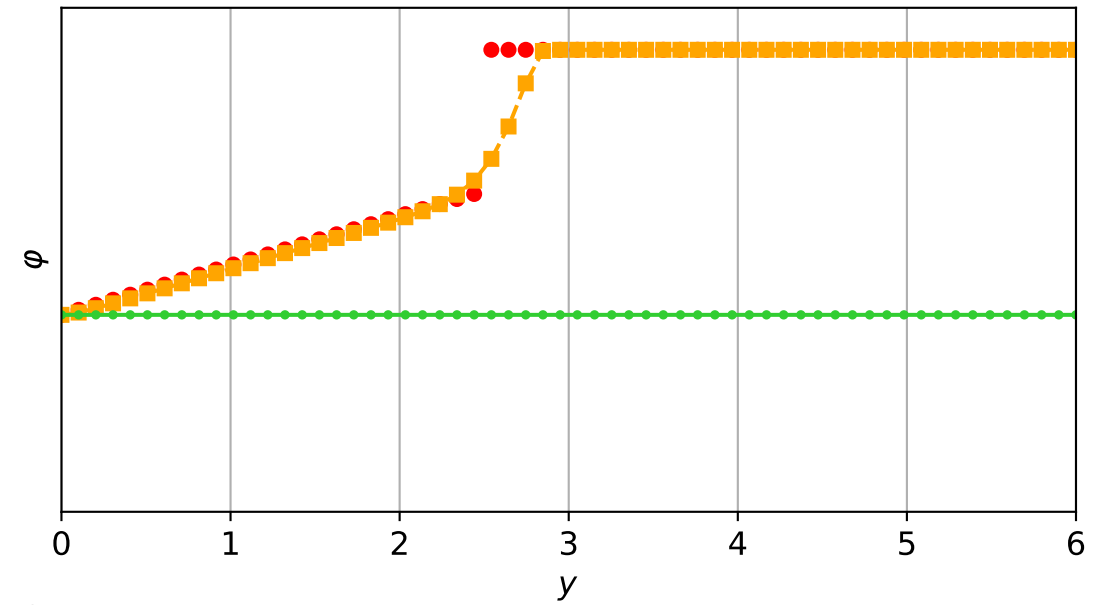
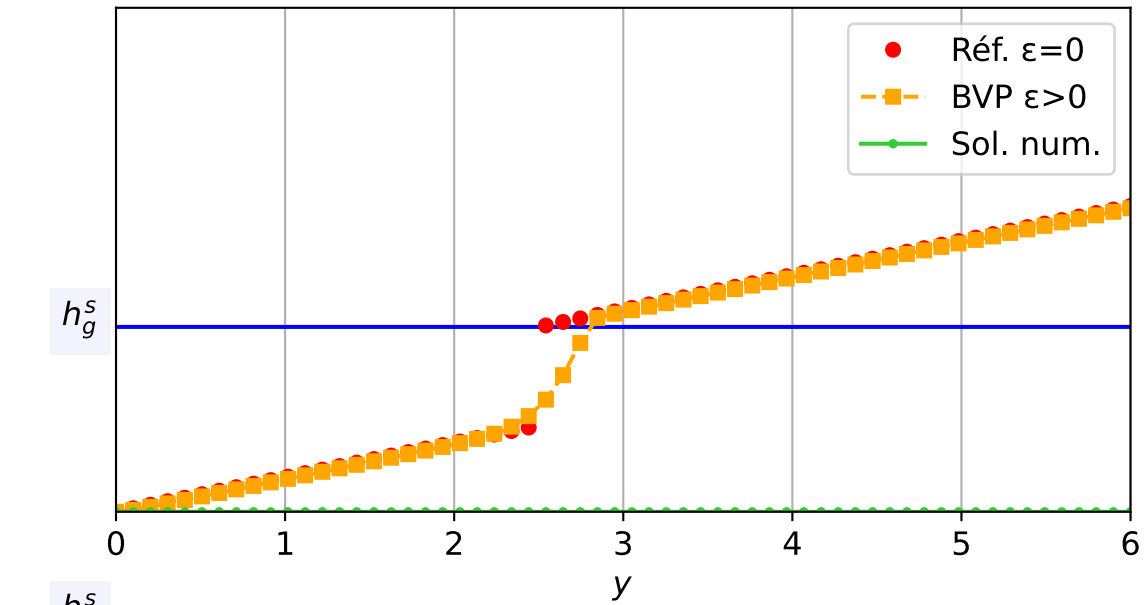
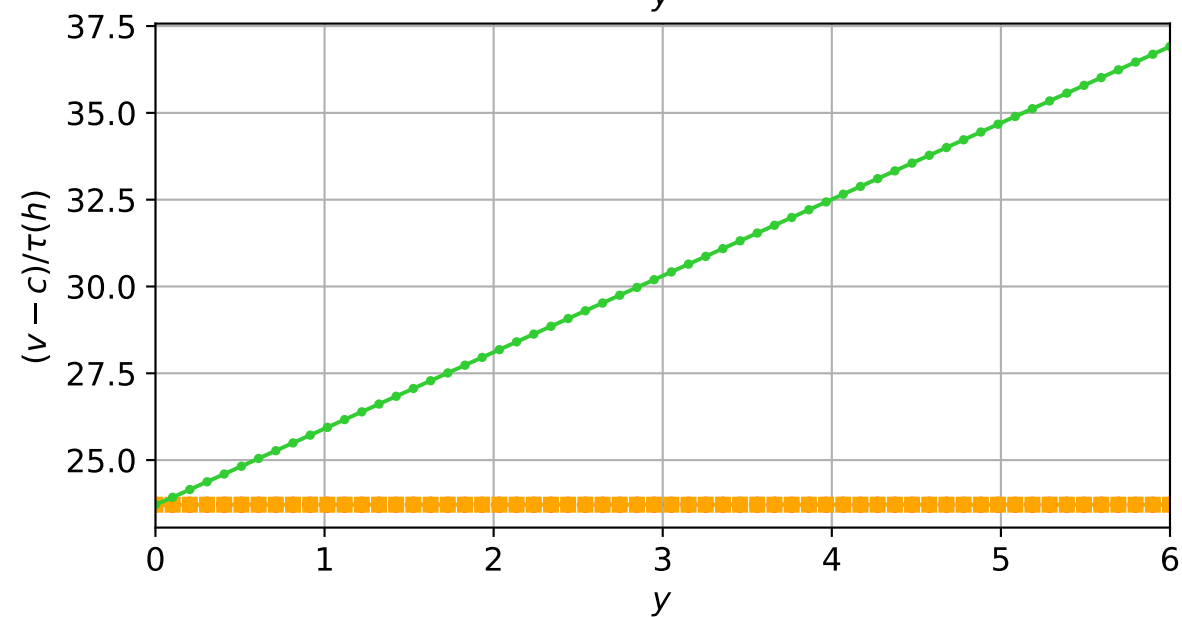
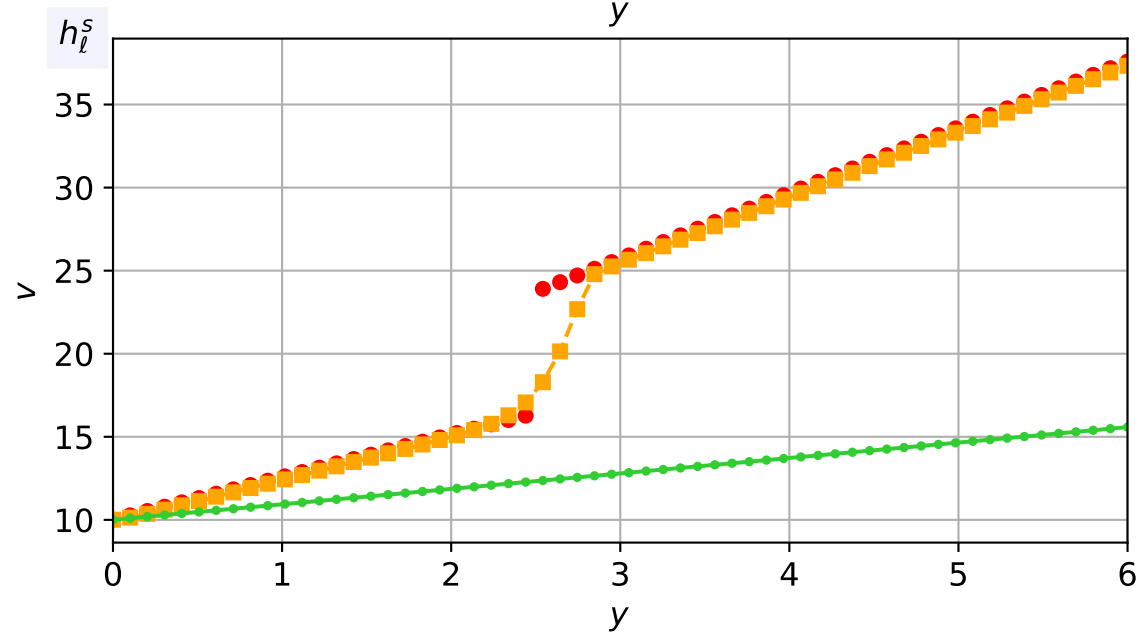
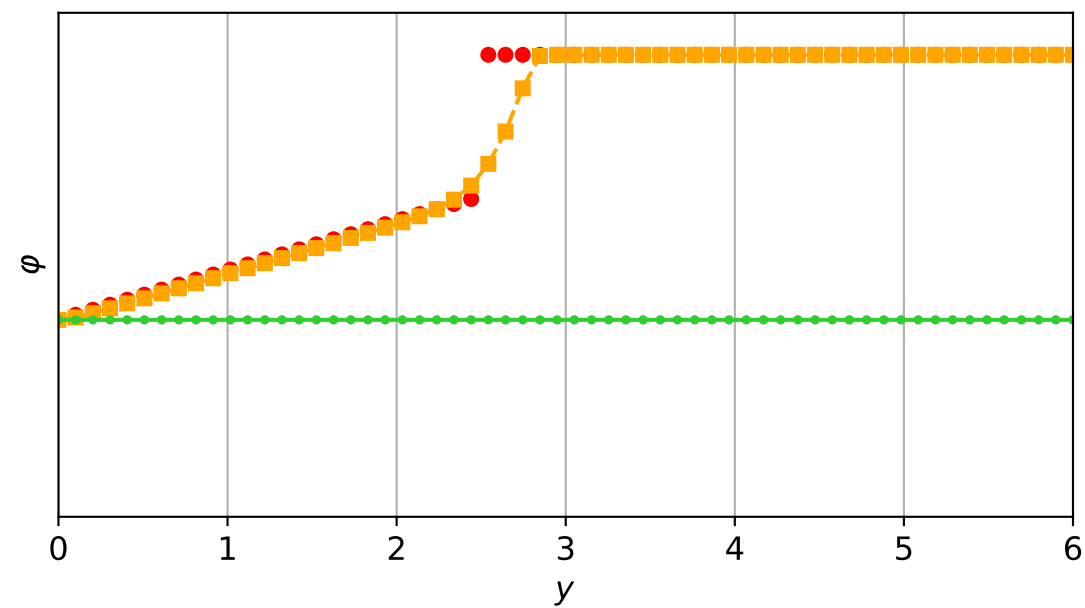
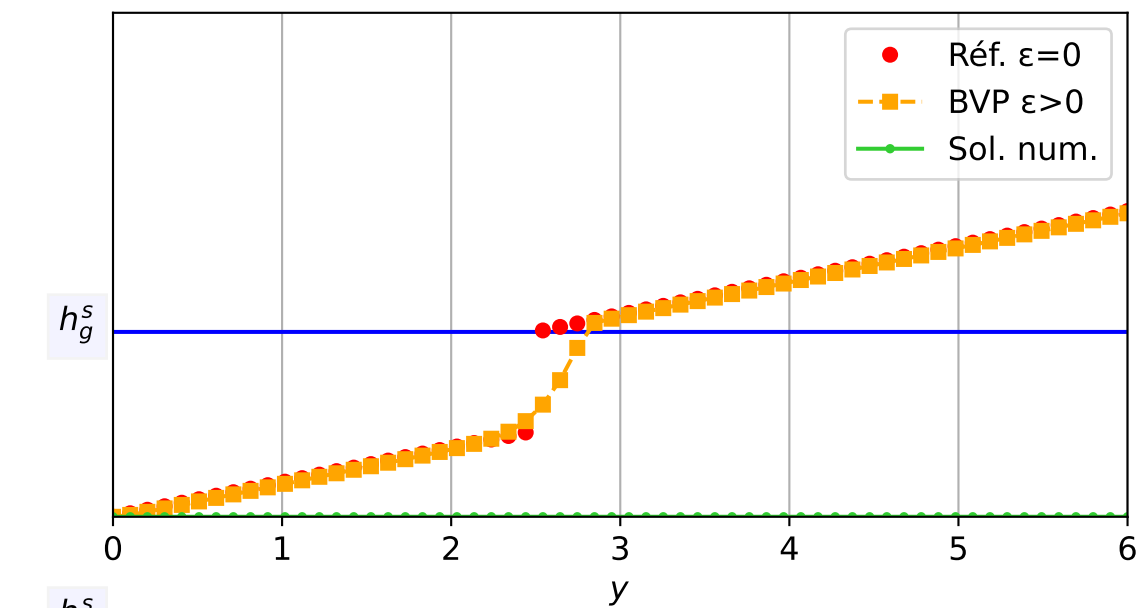


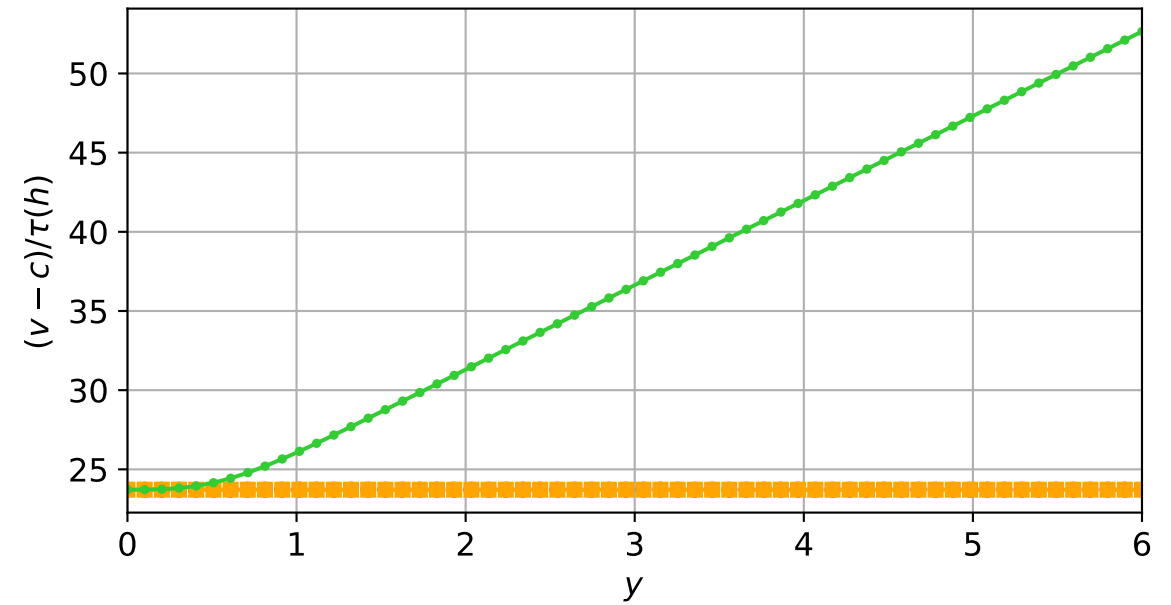
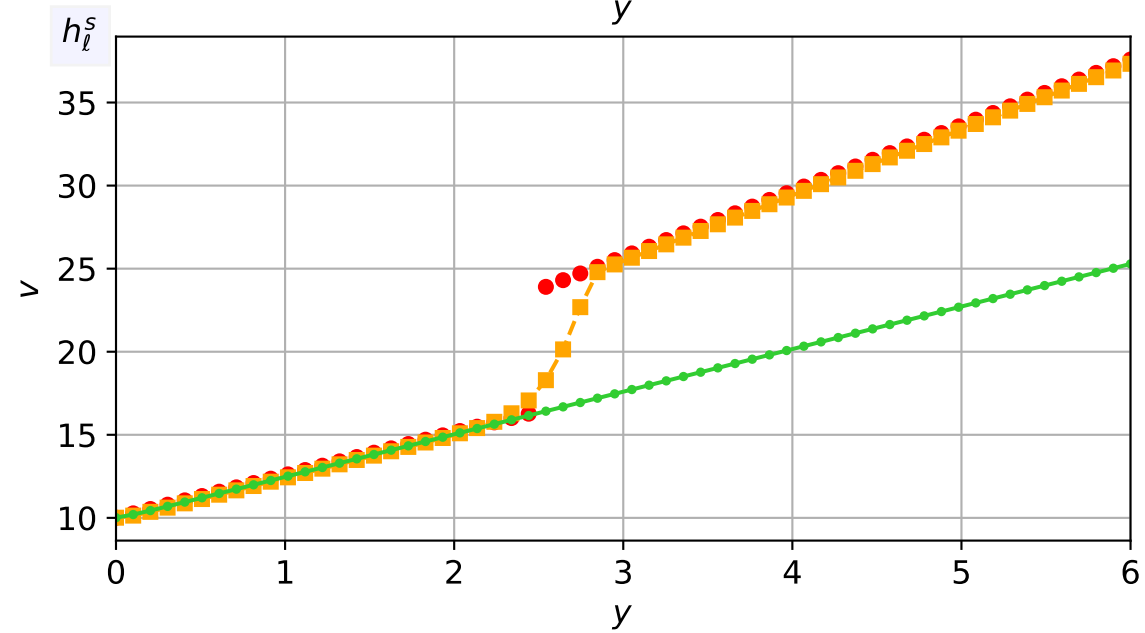
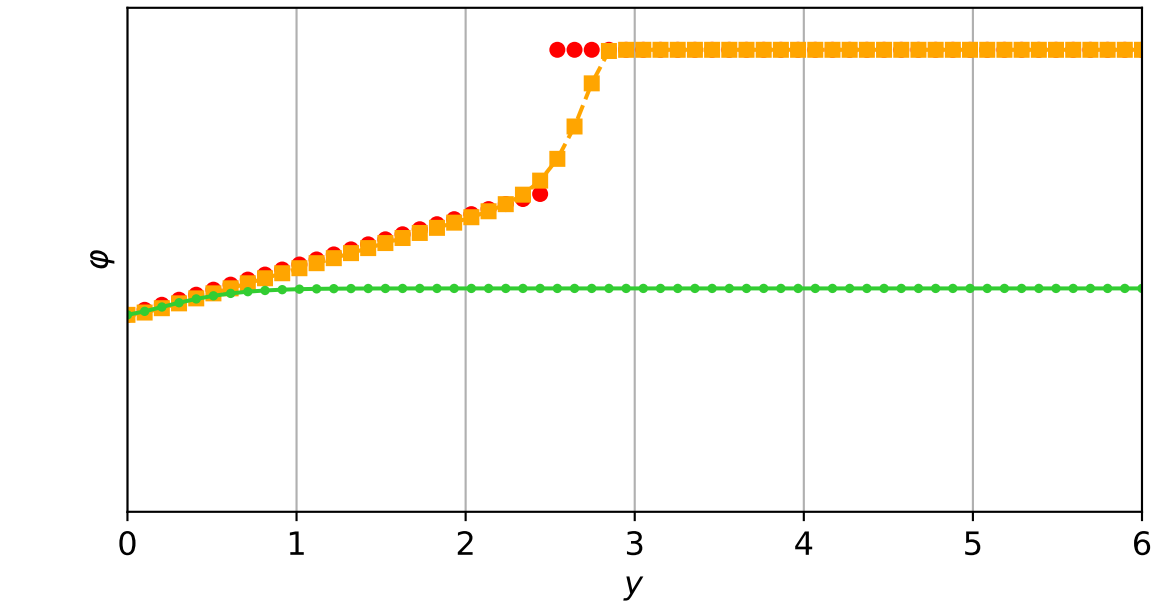
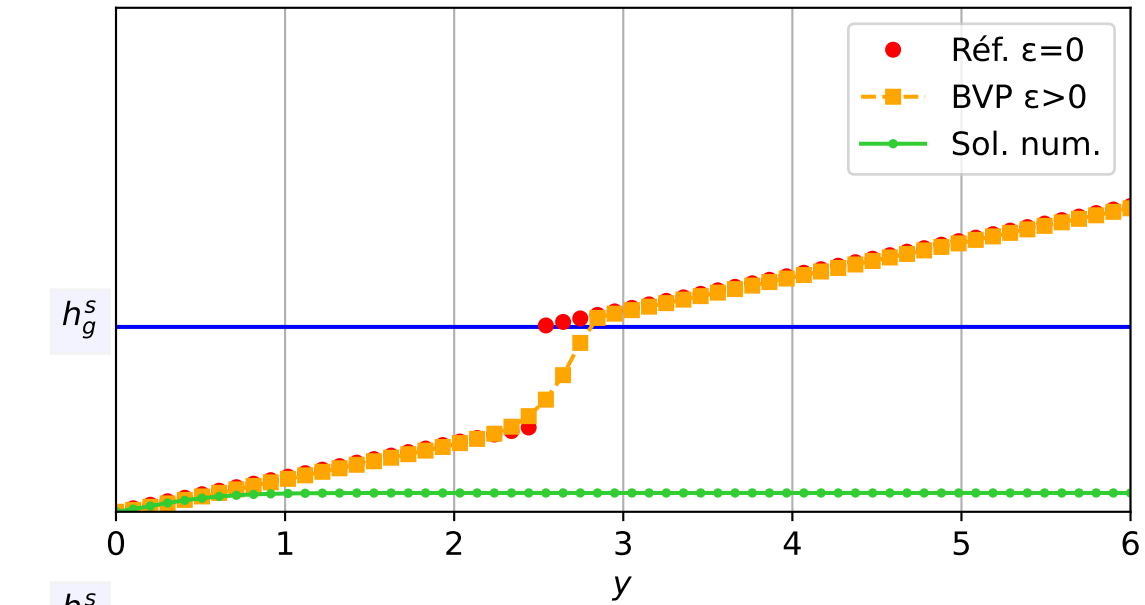
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 9.58e-04$  —  $t = 0/12$  (0 iter) — résidu eq. =  $0.00e+00$



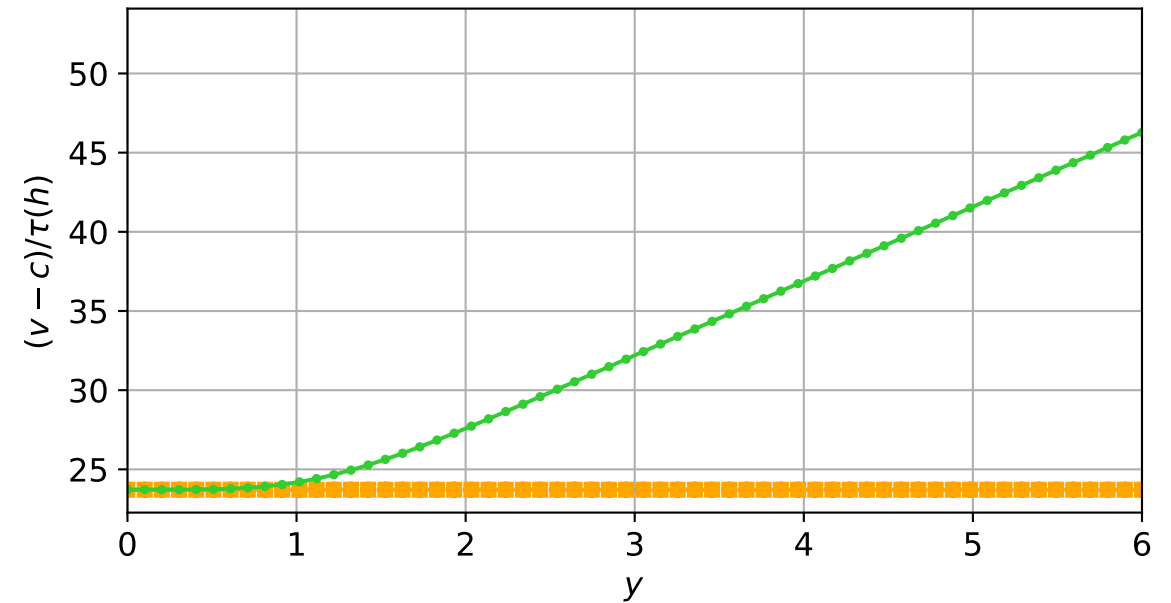
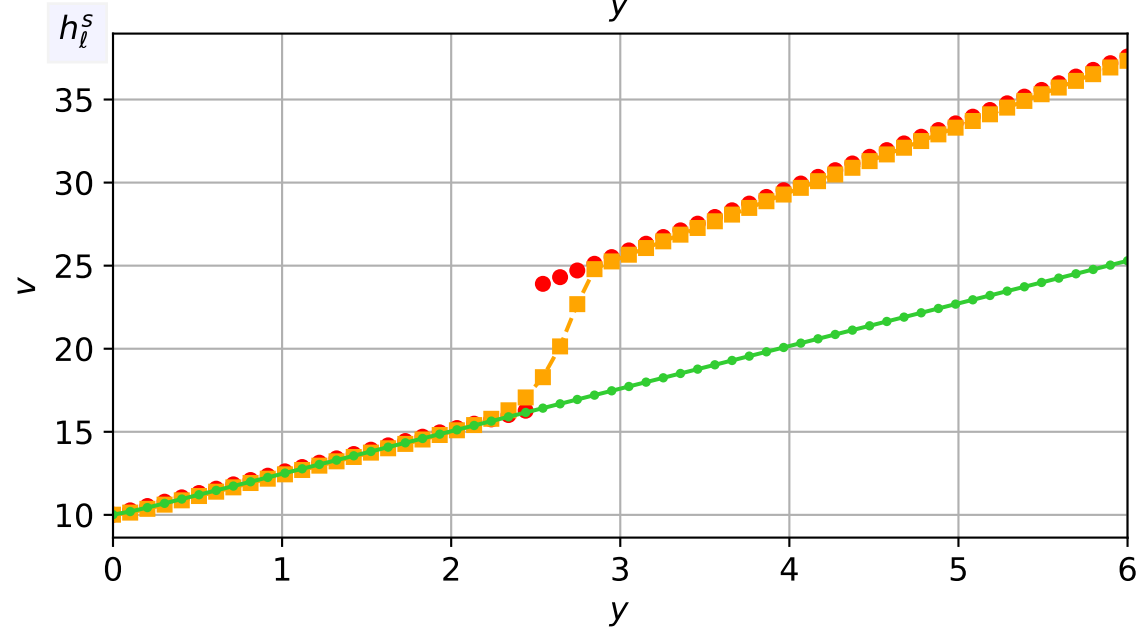
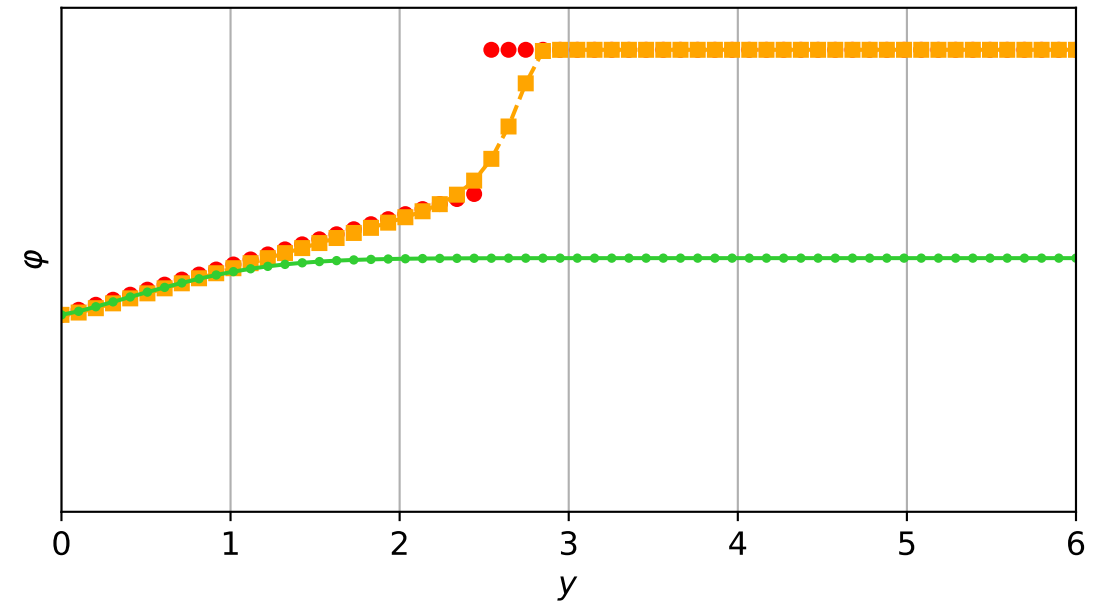
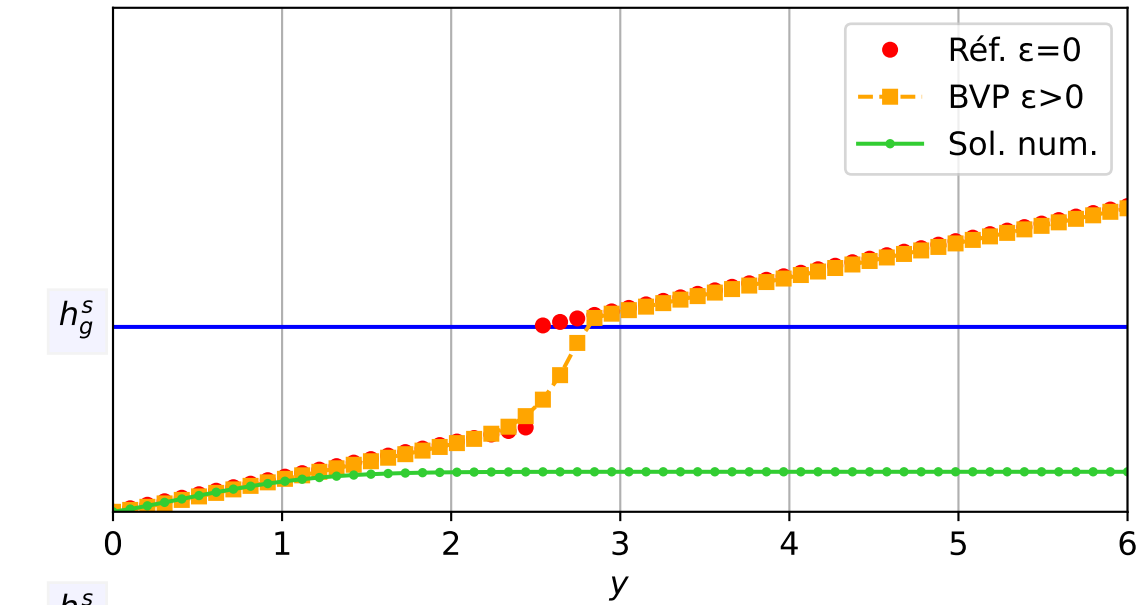
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.20e-04$  —  $t = 0.0001/12$  (1 iter) — résidu eq. =  $1.21e-10$



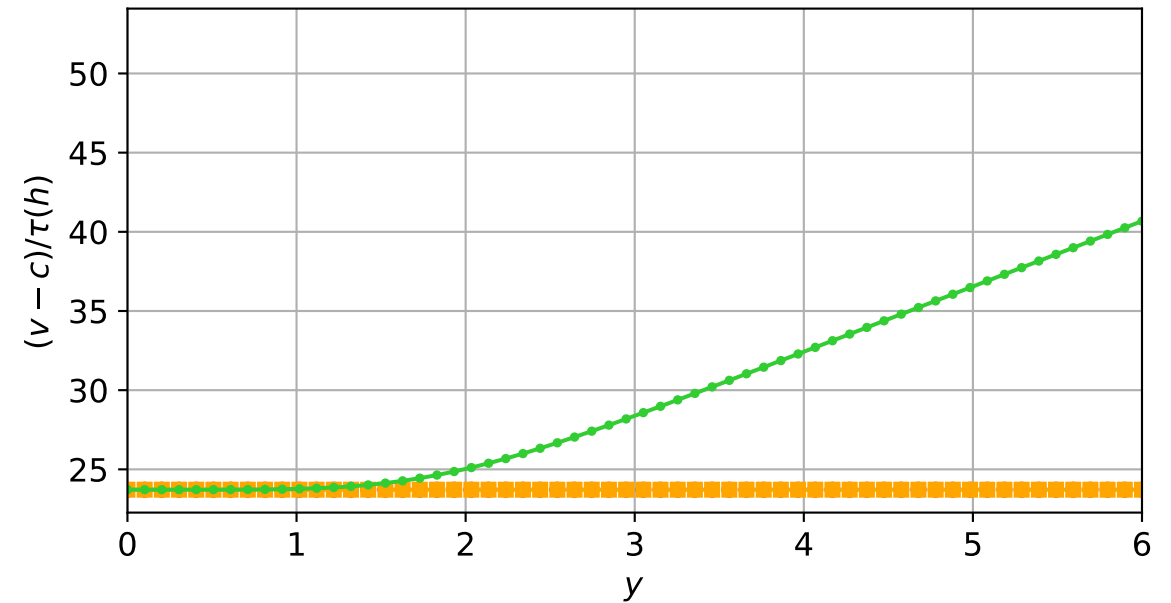
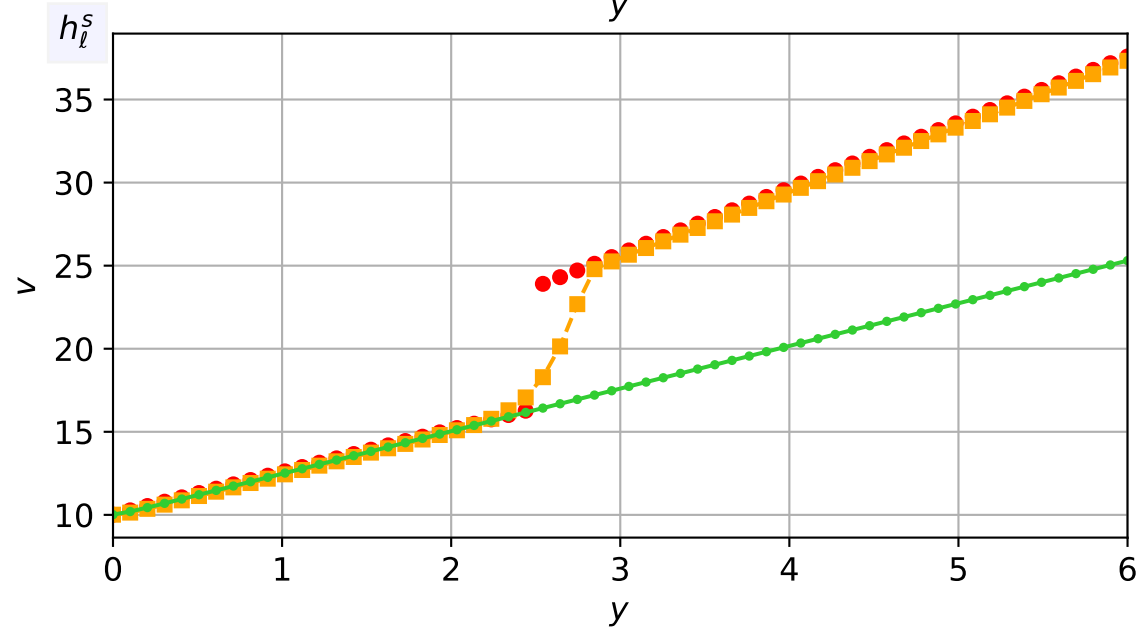
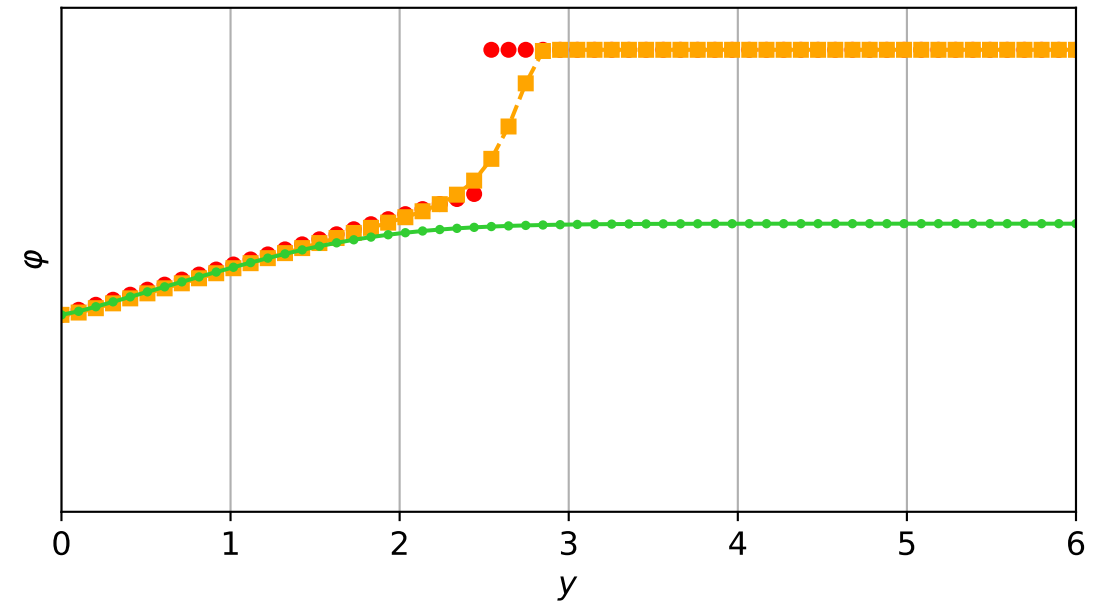
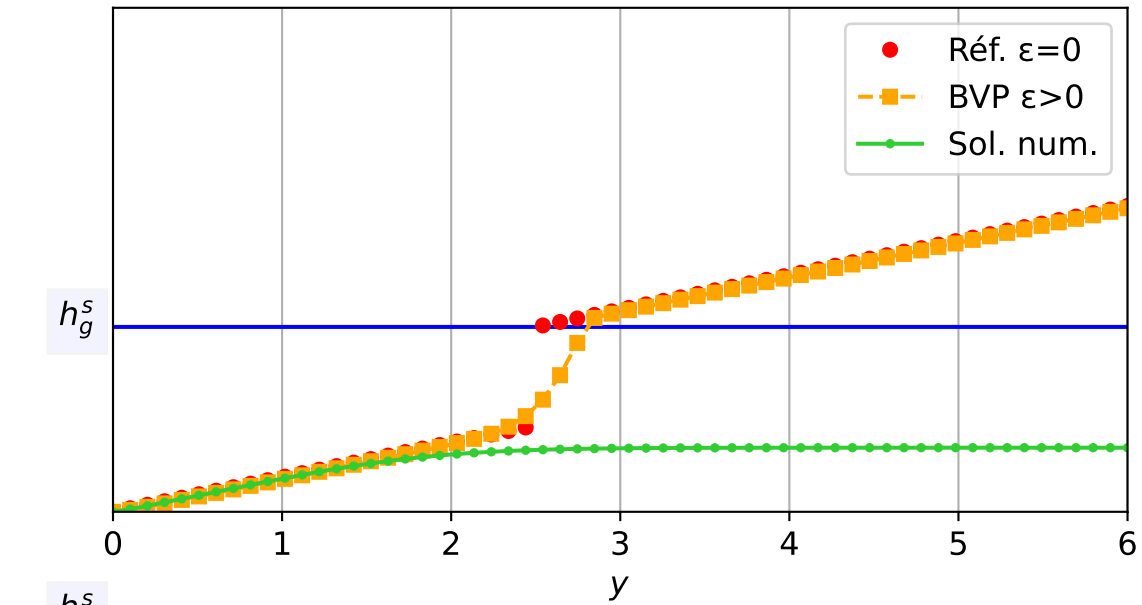
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.62e-03$  —  $t = 0.0512482/12$  (29 iter) — résidu eq. =  $1.31e-07$



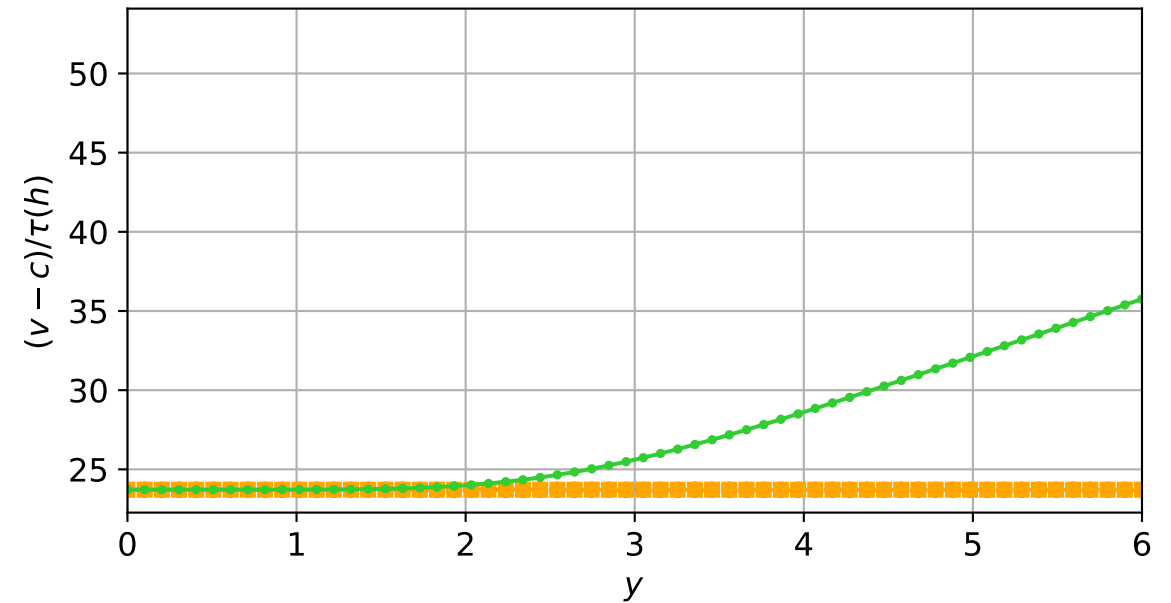
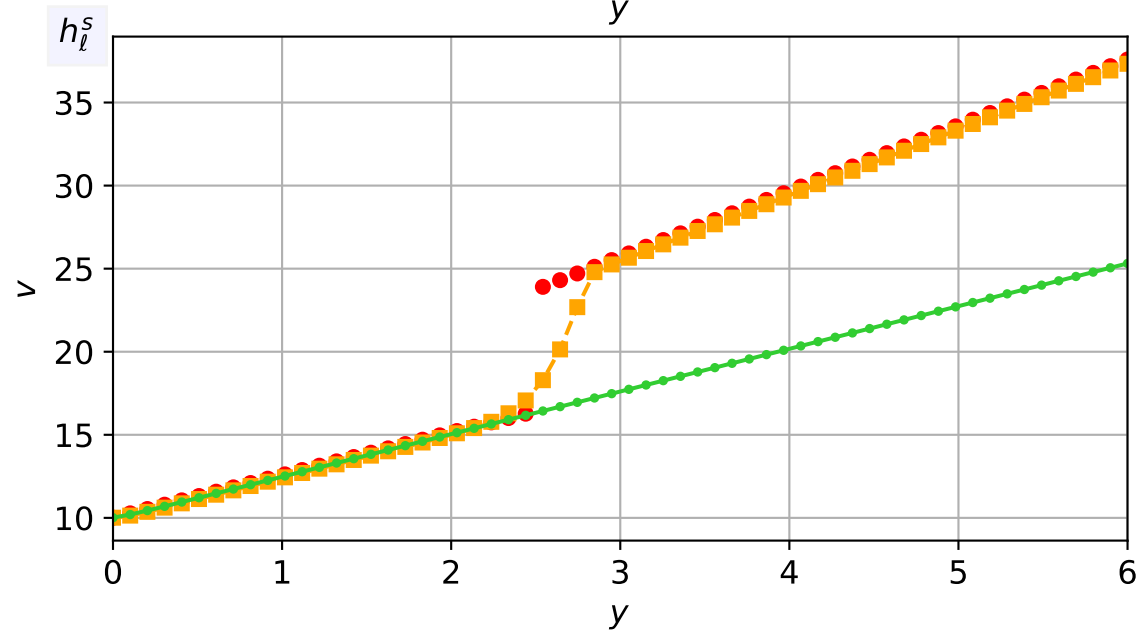
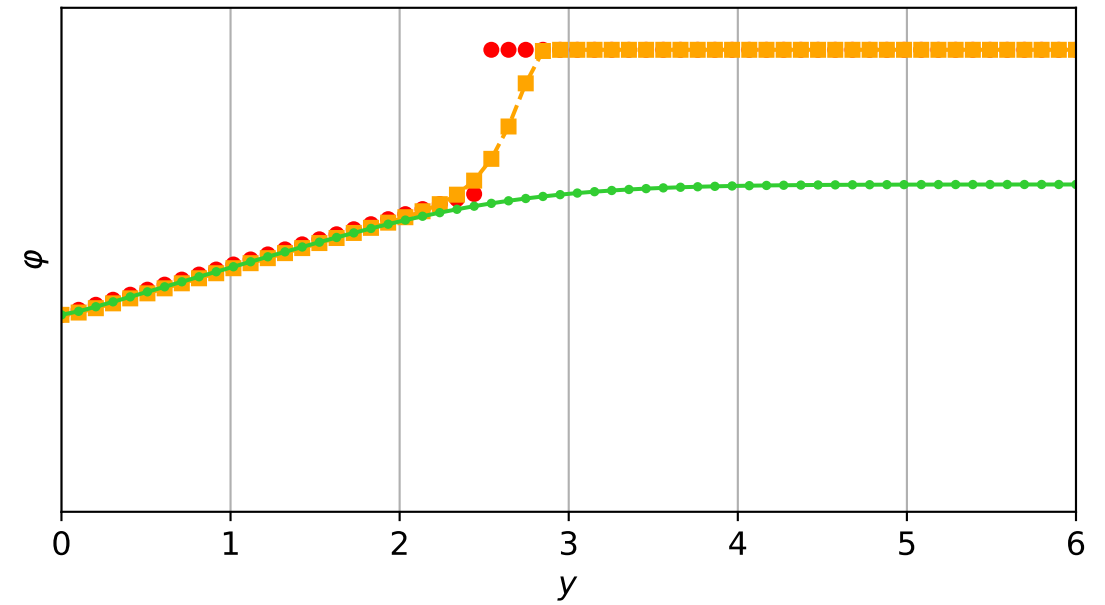
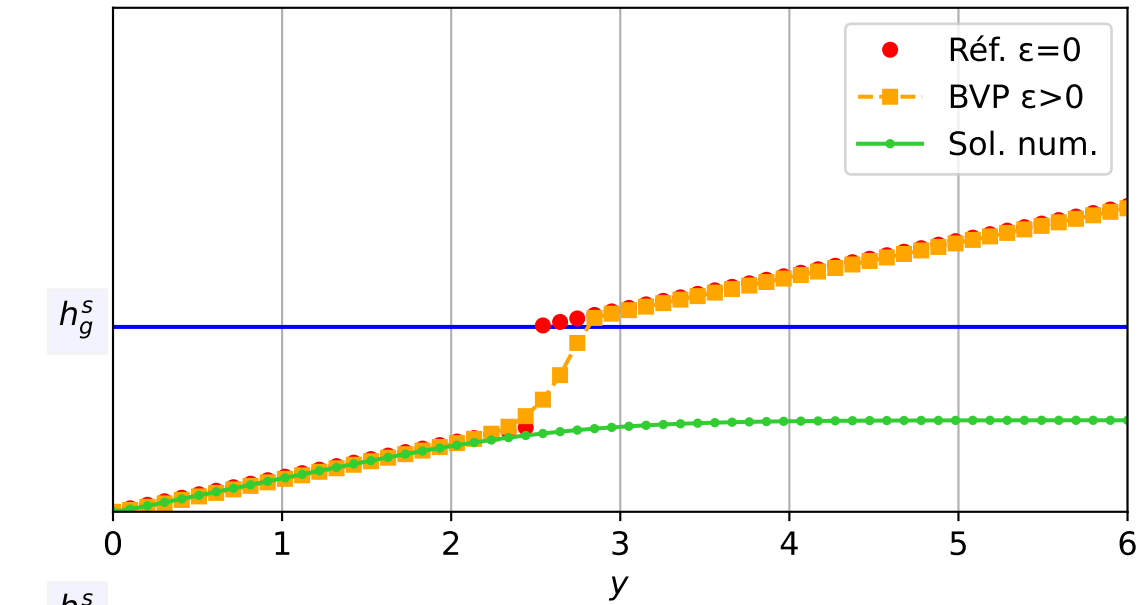
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.62e-03$  —  $t = 0.101916/12$  (43 iter) — résidu eq. =  $9.53e-08$



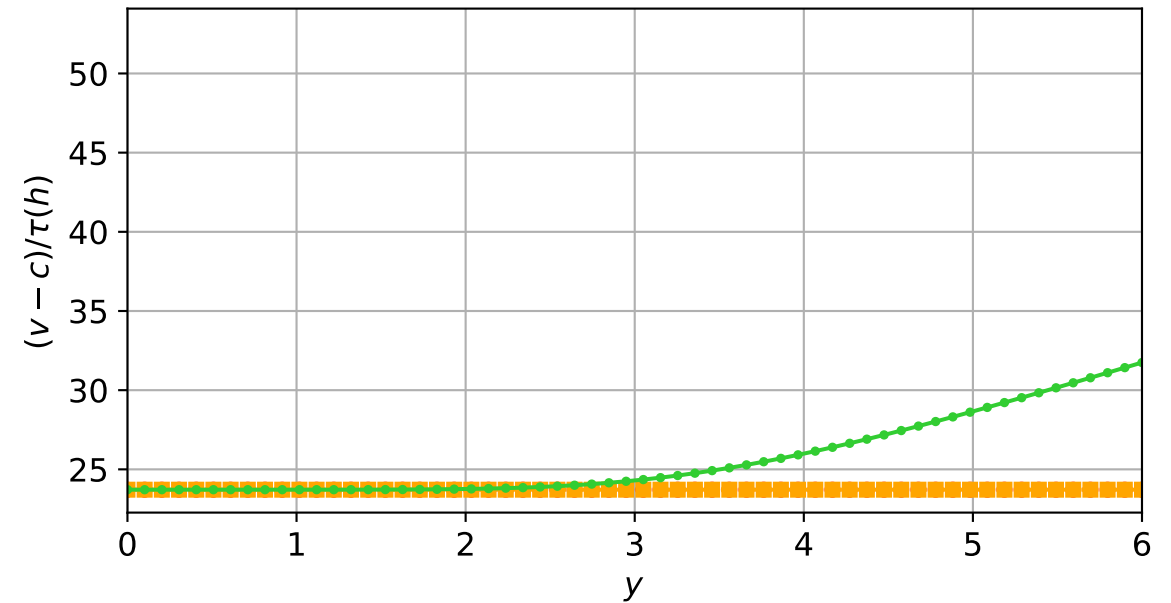
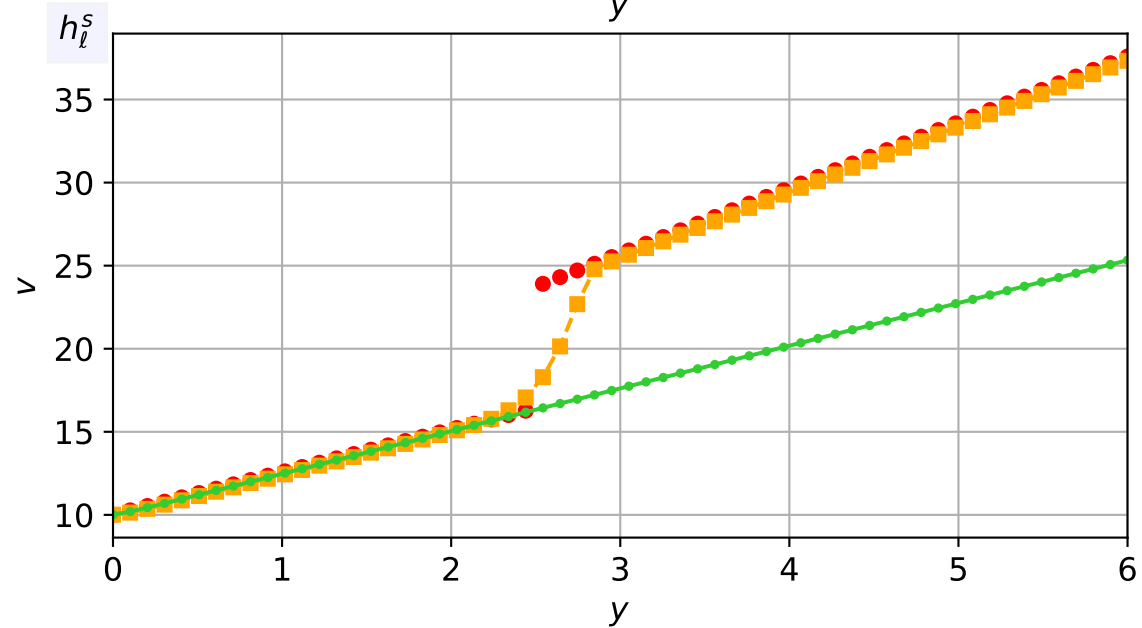
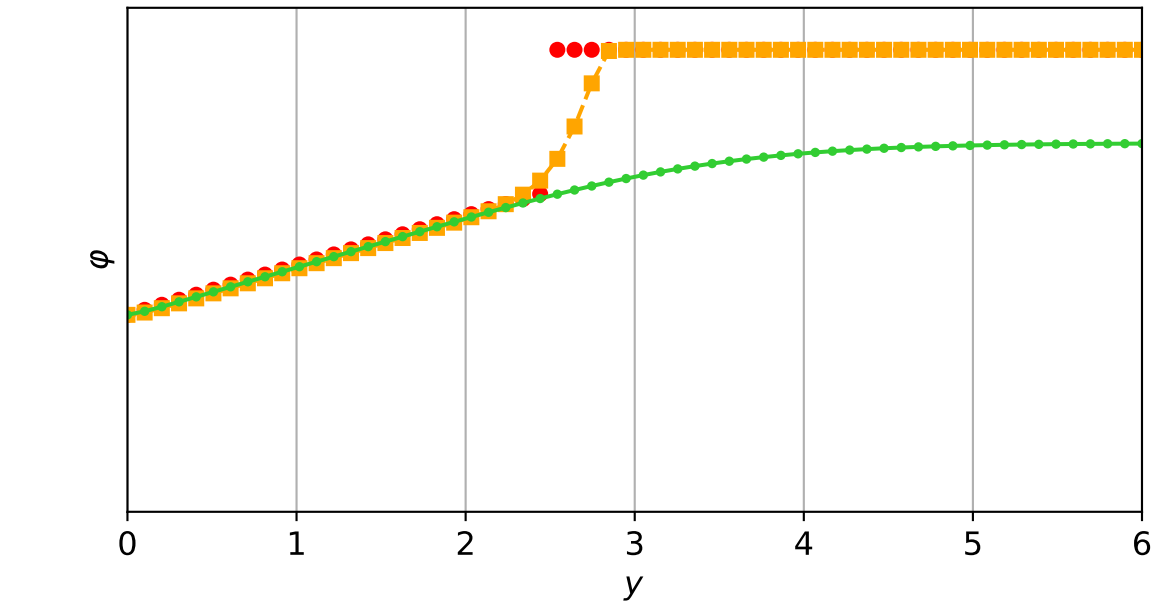
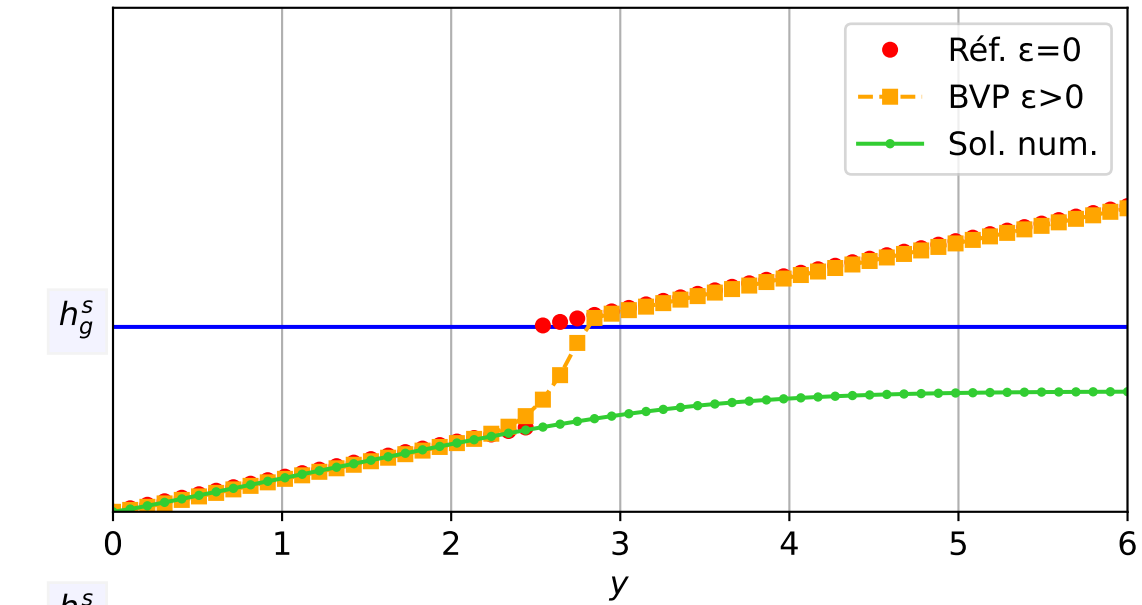
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.62e-03$  —  $t = 0.152568/12$  (57 iter) — résidu eq. =  $1.61e-07$



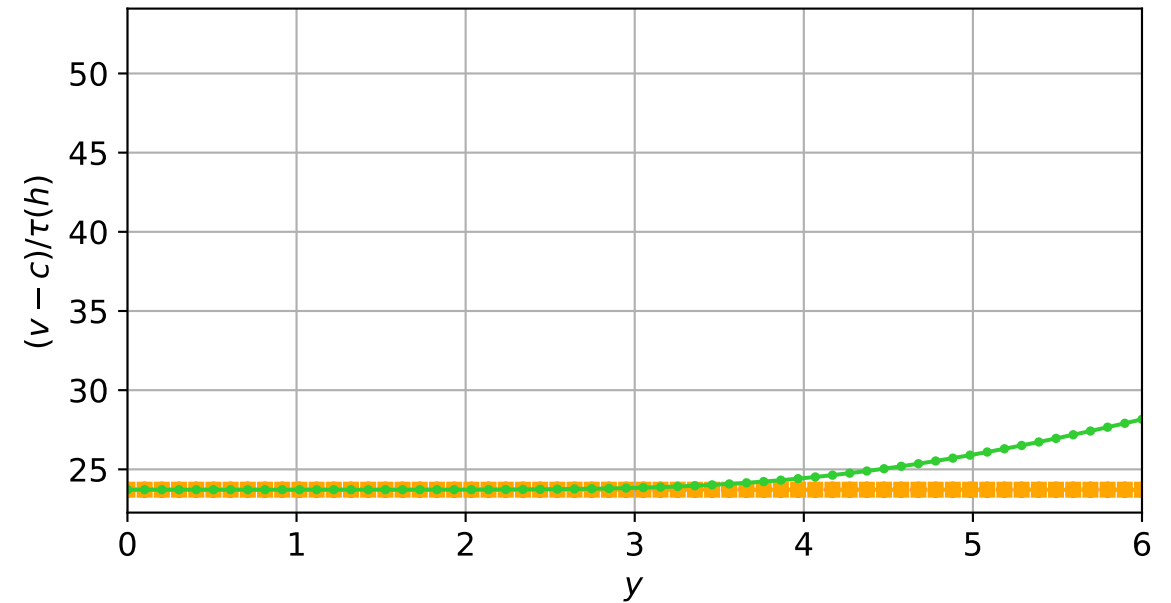
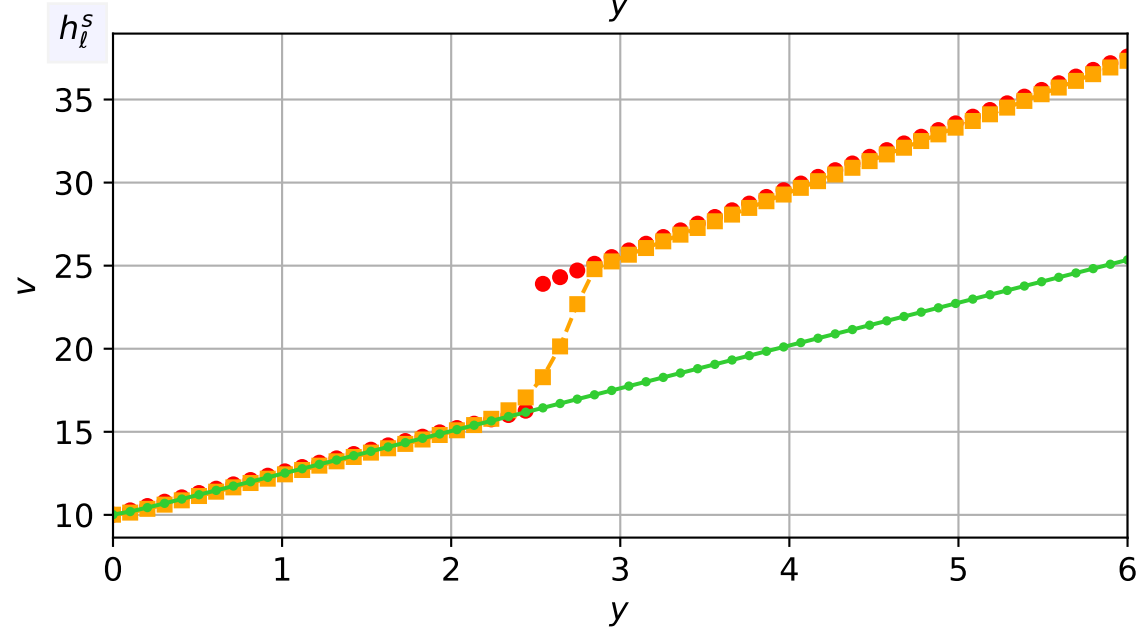
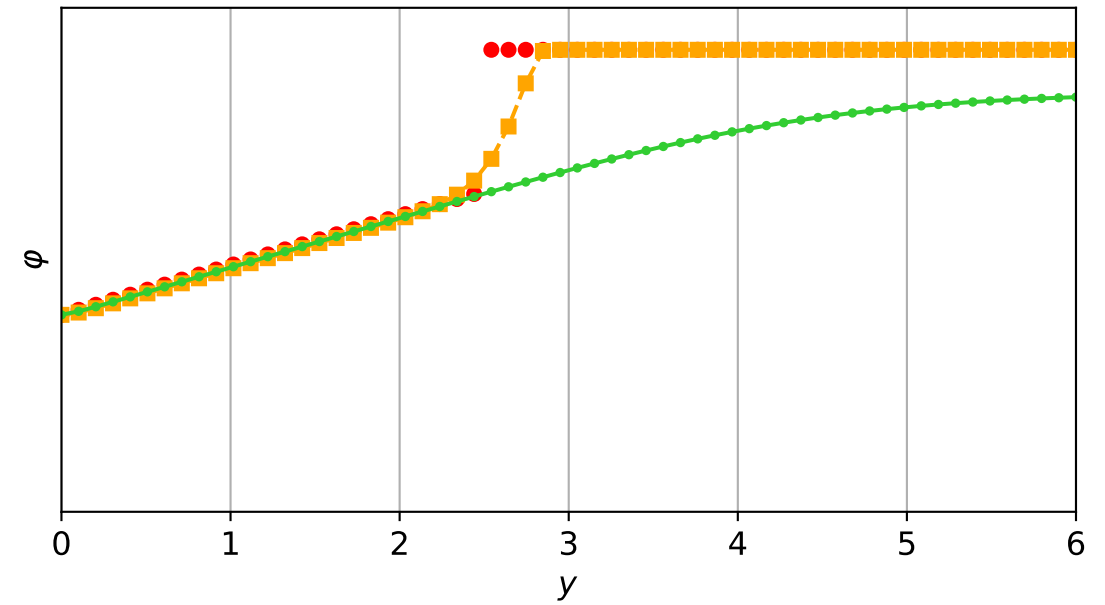
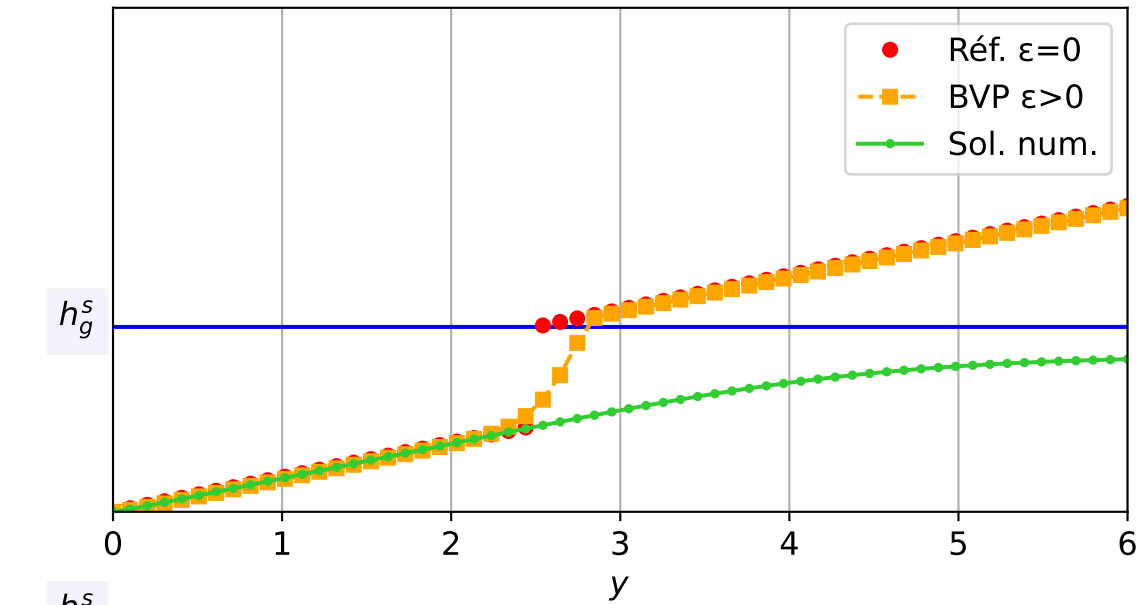
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.62e-03$  —  $t = 0.203198/12$  (71 iter) — résidu eq. =  $5.10e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.61e-03$  —  $t = 0.250191/12$  (84 iter) — résidu eq. =  $6.36e-06$

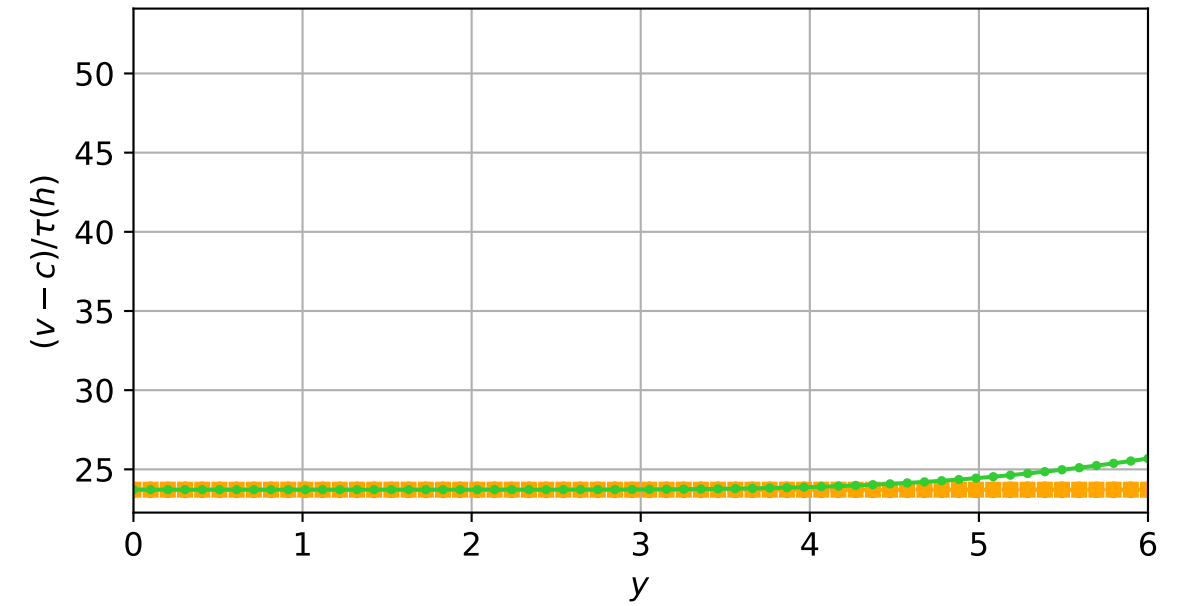
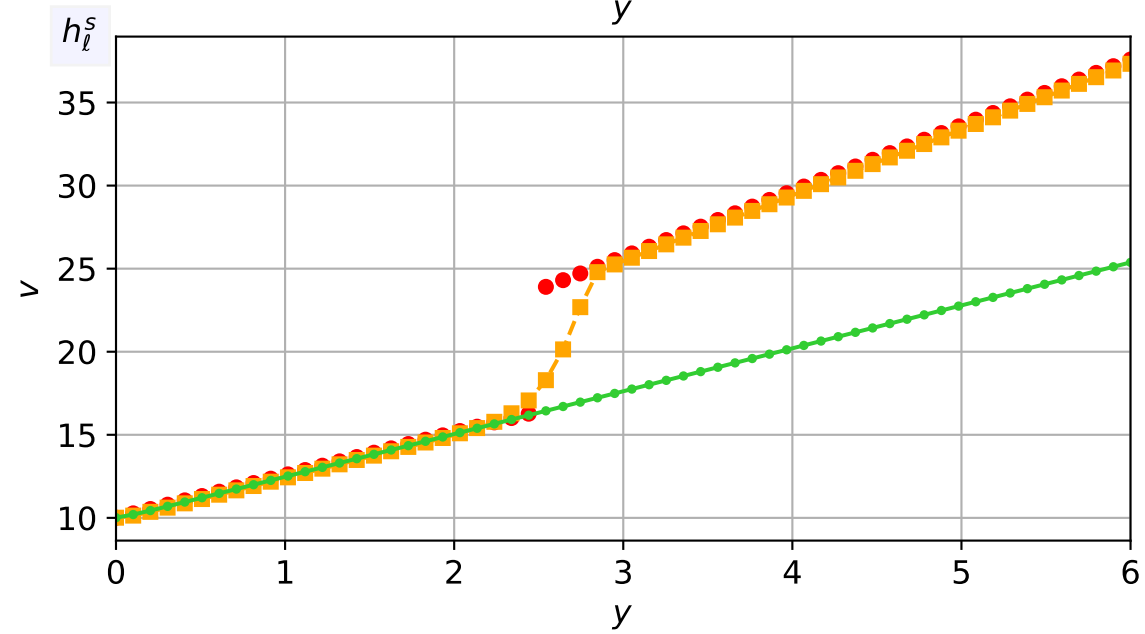
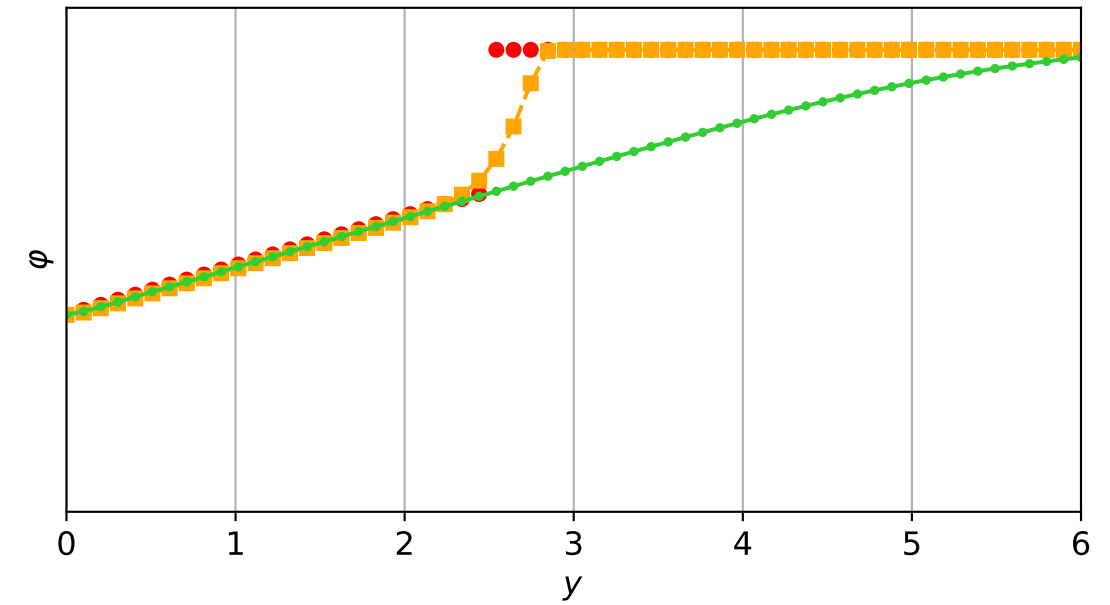
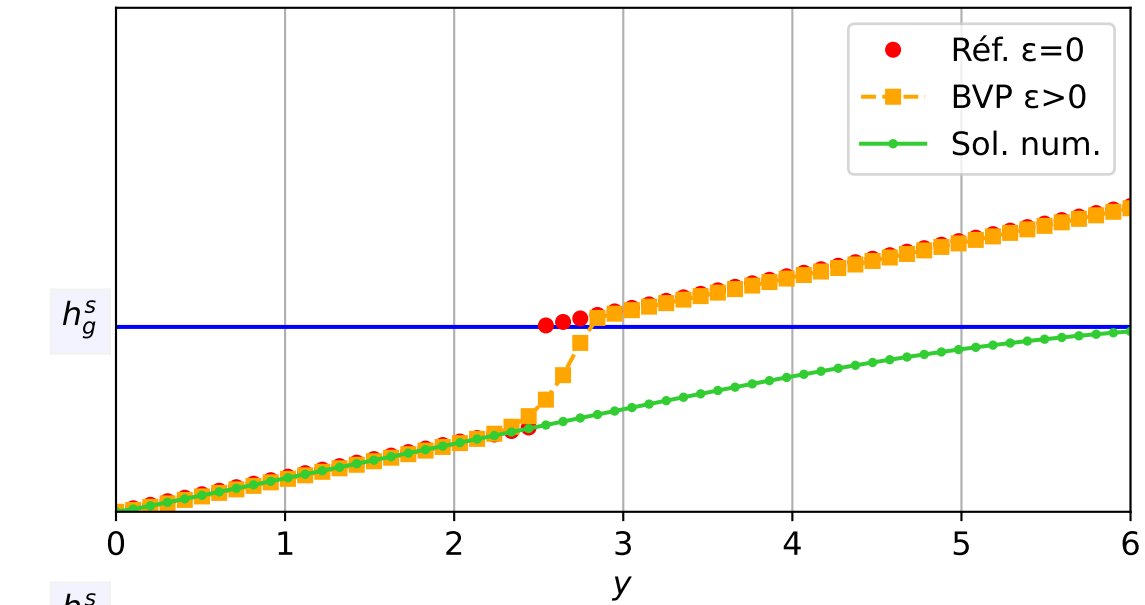


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.61e-03$  —  $t = 0.300793/12$  (98 iter) — résidu eq. =  $9.00e-06$

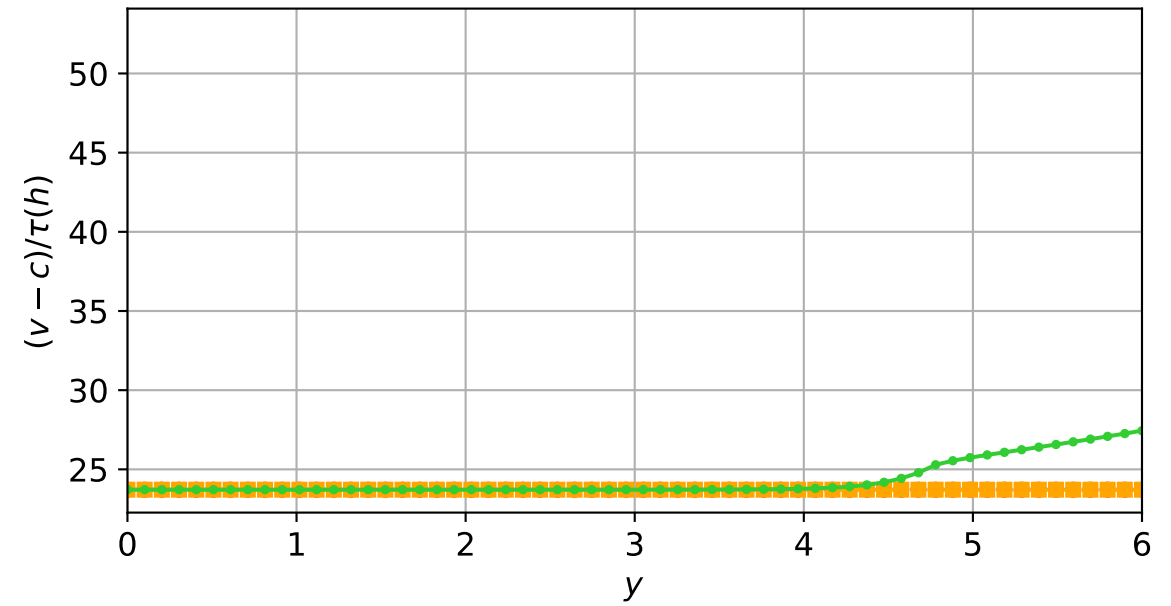
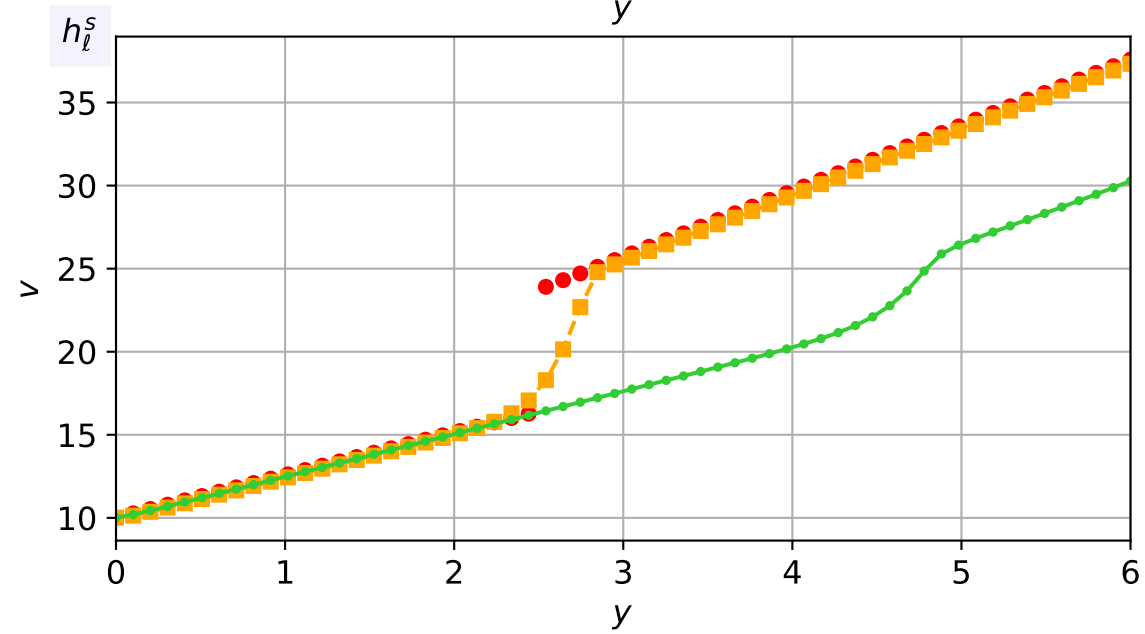
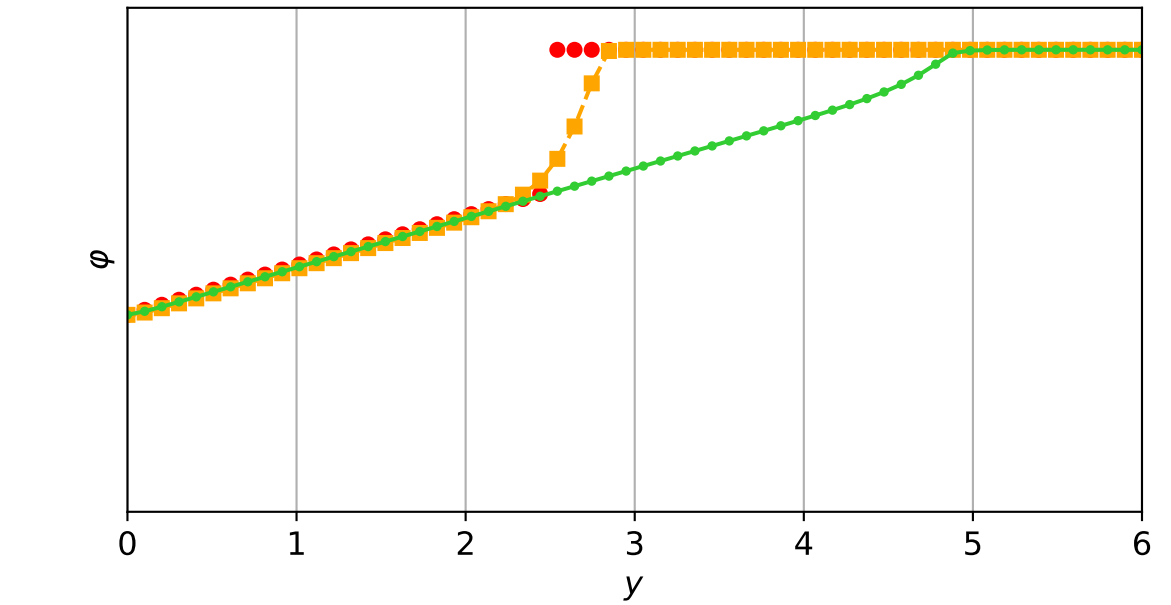
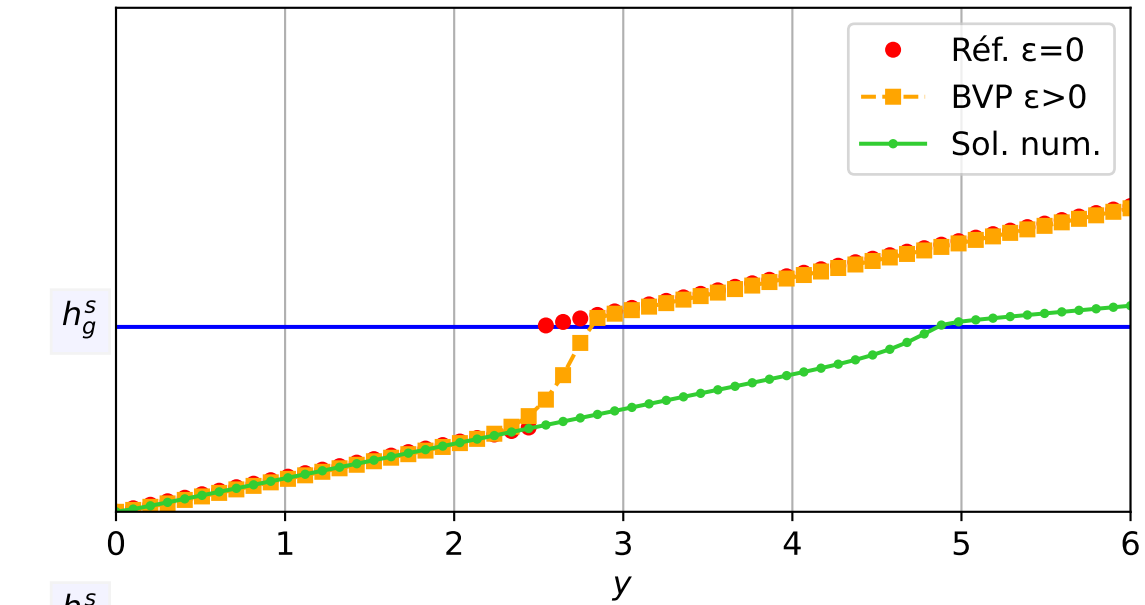




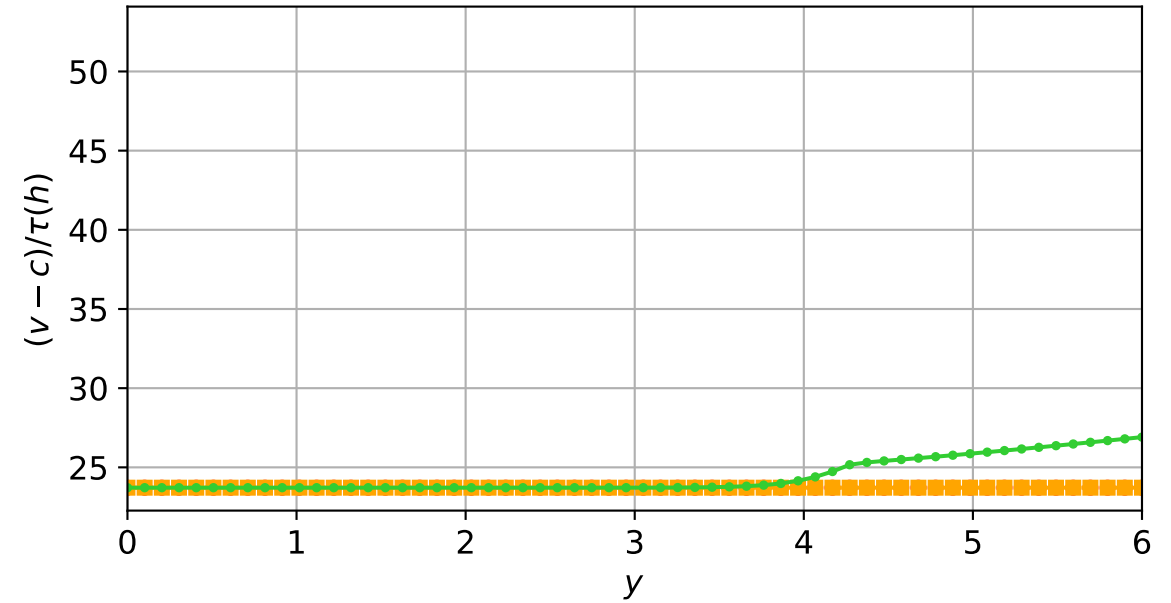
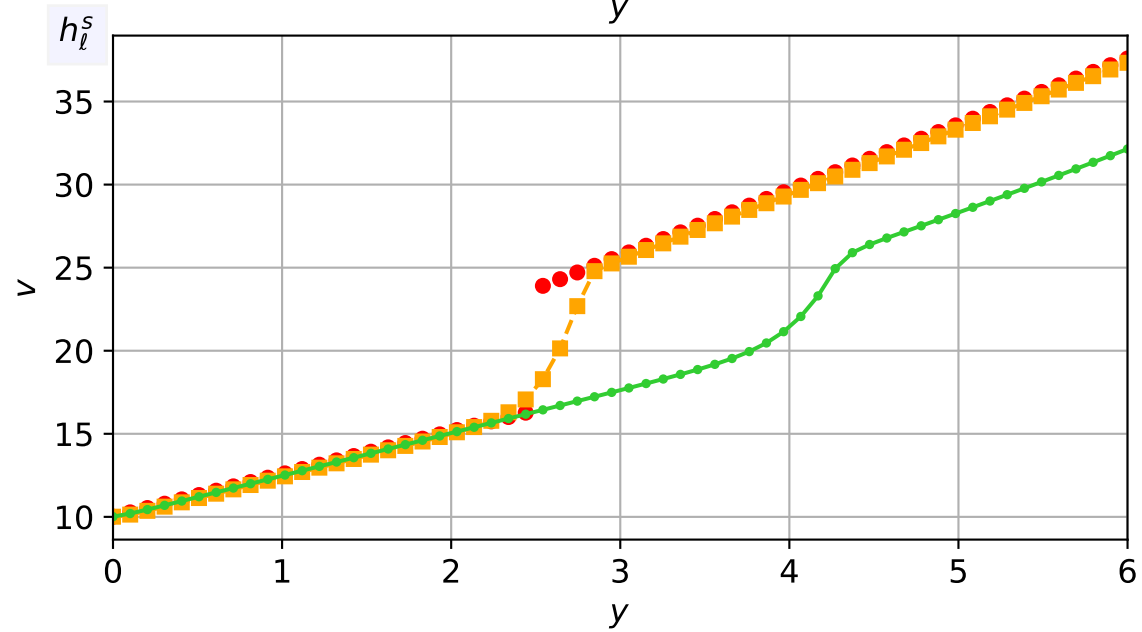
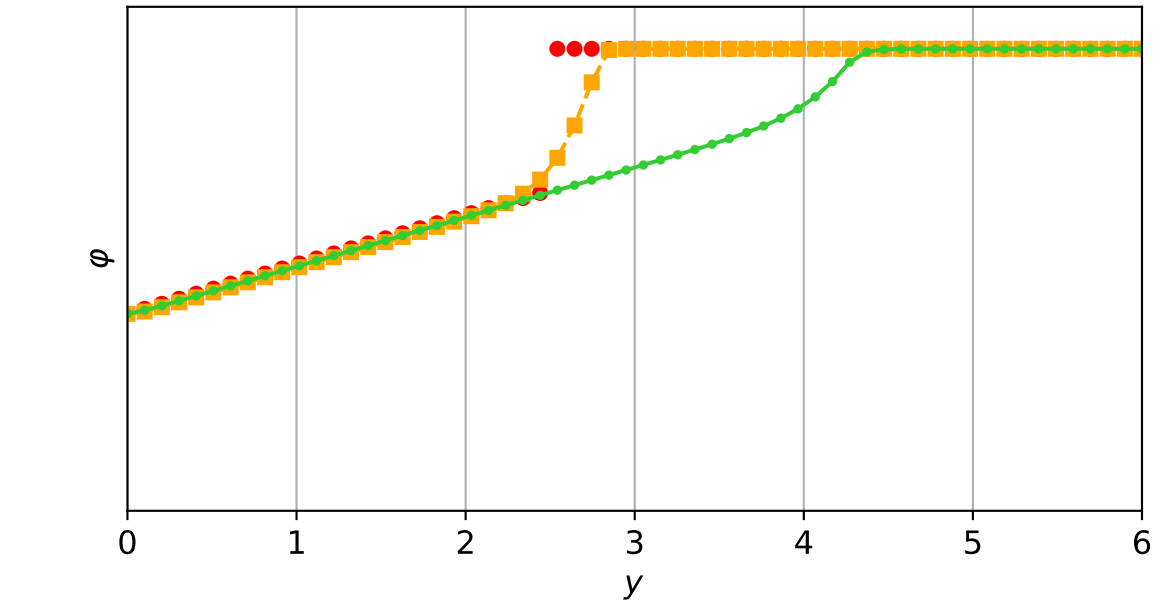
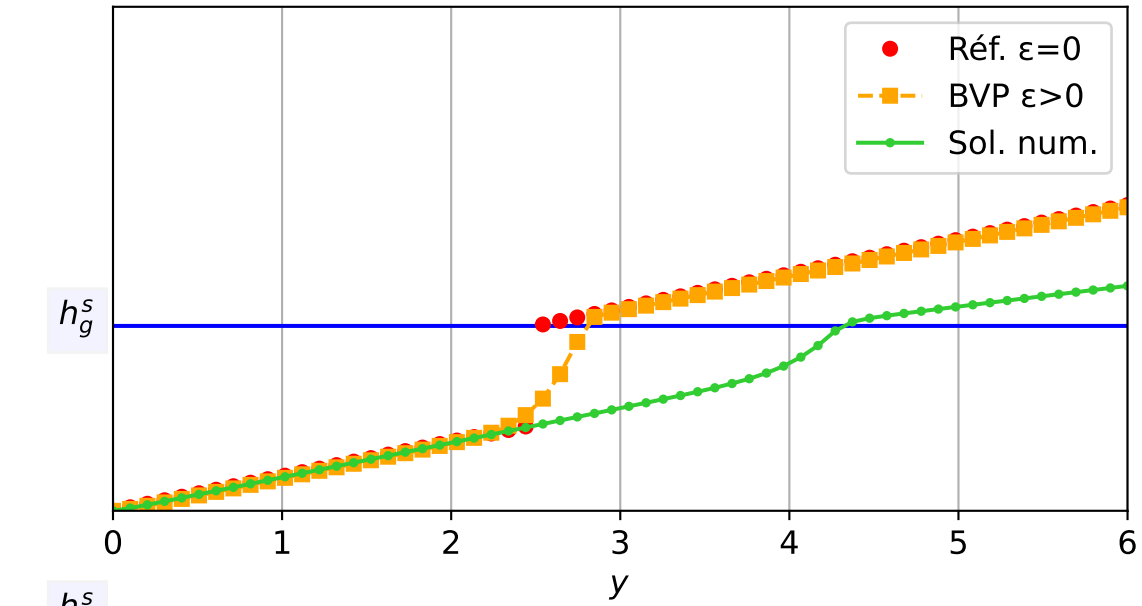
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 9.04e-04$  —  $t = 0.350492/12$  (139 iter) — résidu eq. =  $6.51e-06$



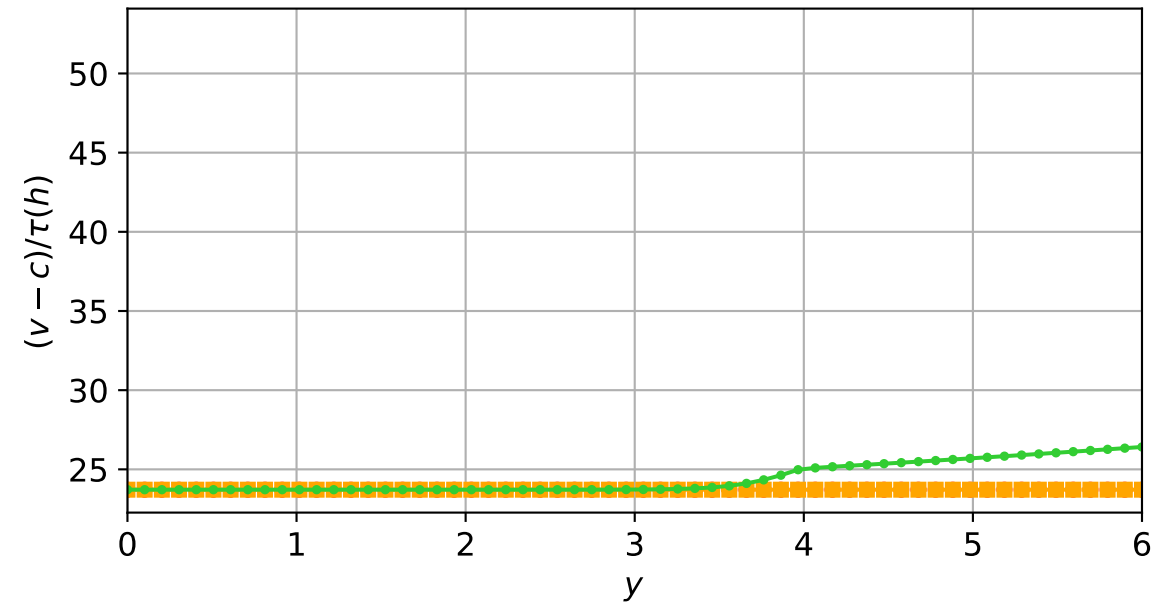
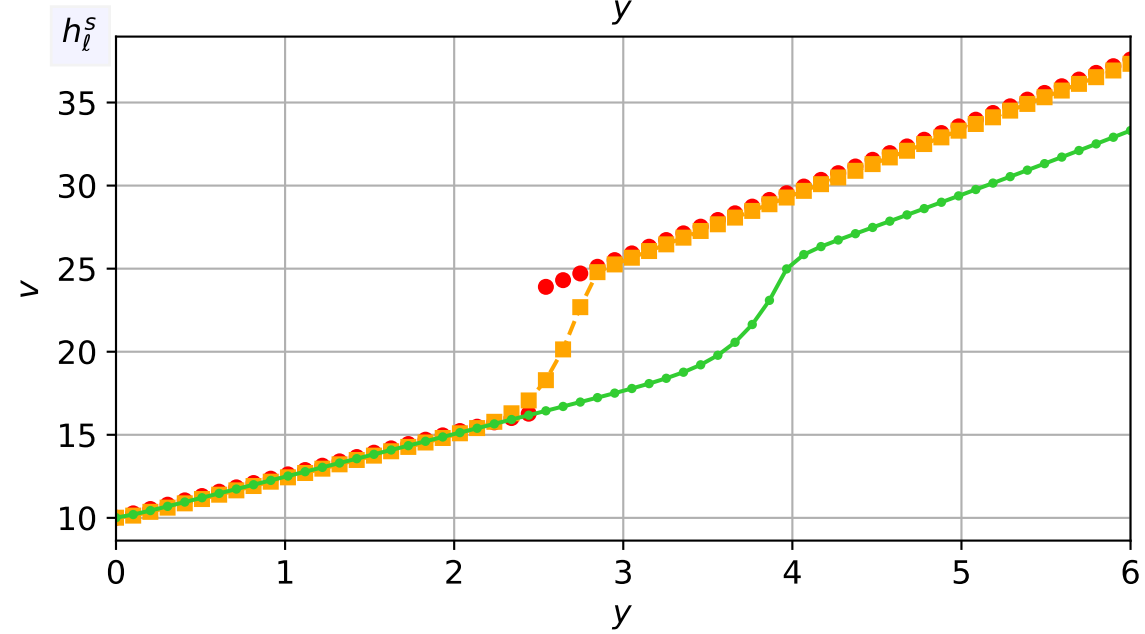
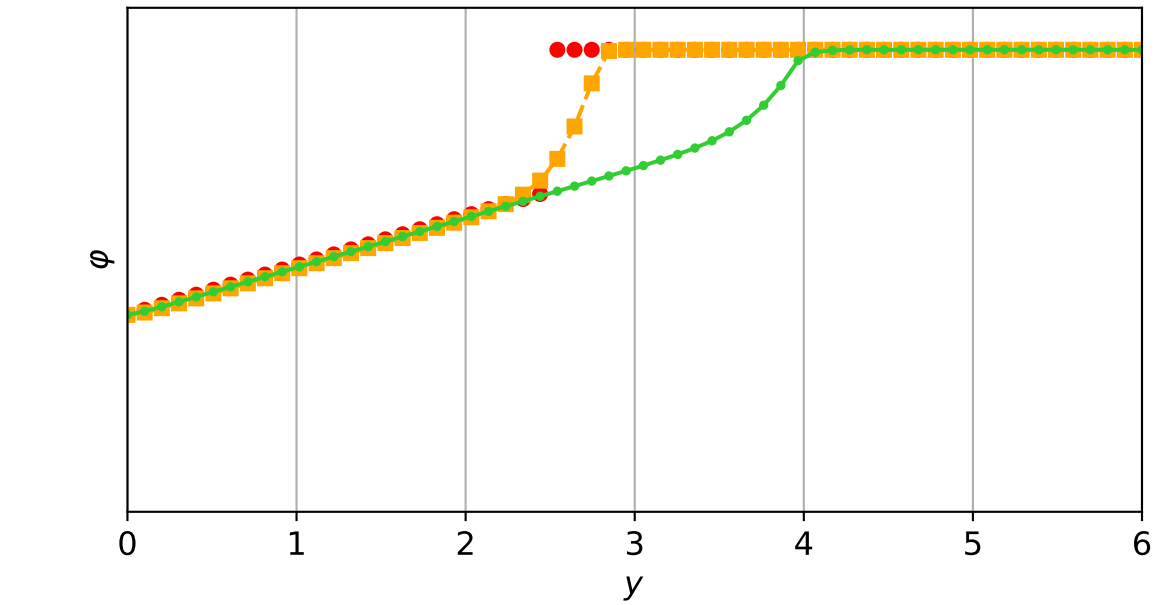
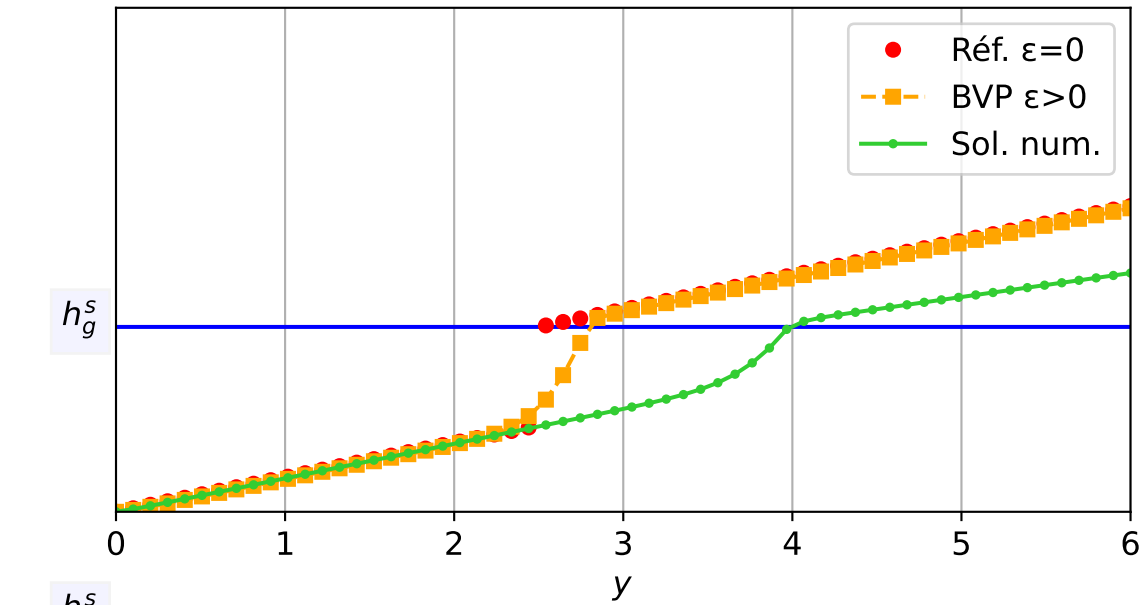
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 0.40019/12$  (322 iter) — résidu eq. =  $7.22e-06$



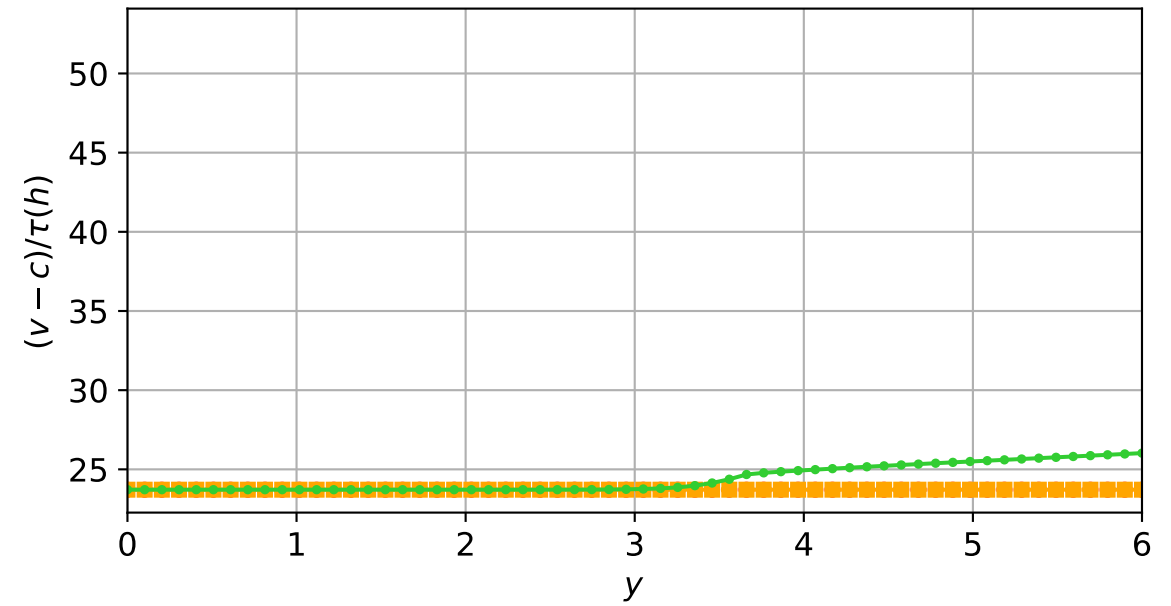
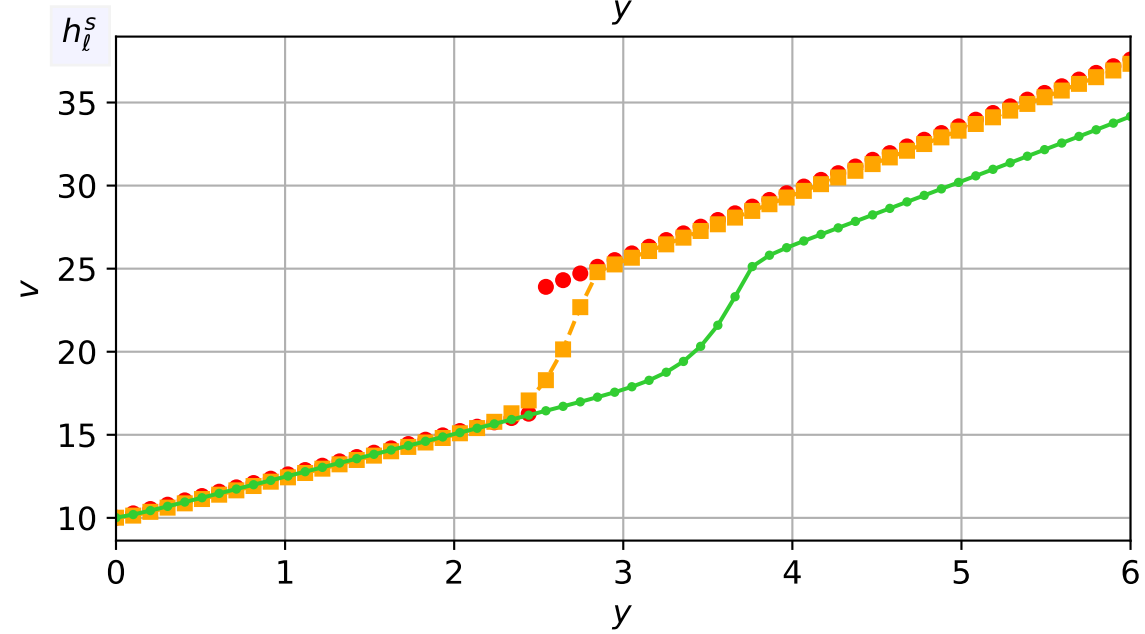
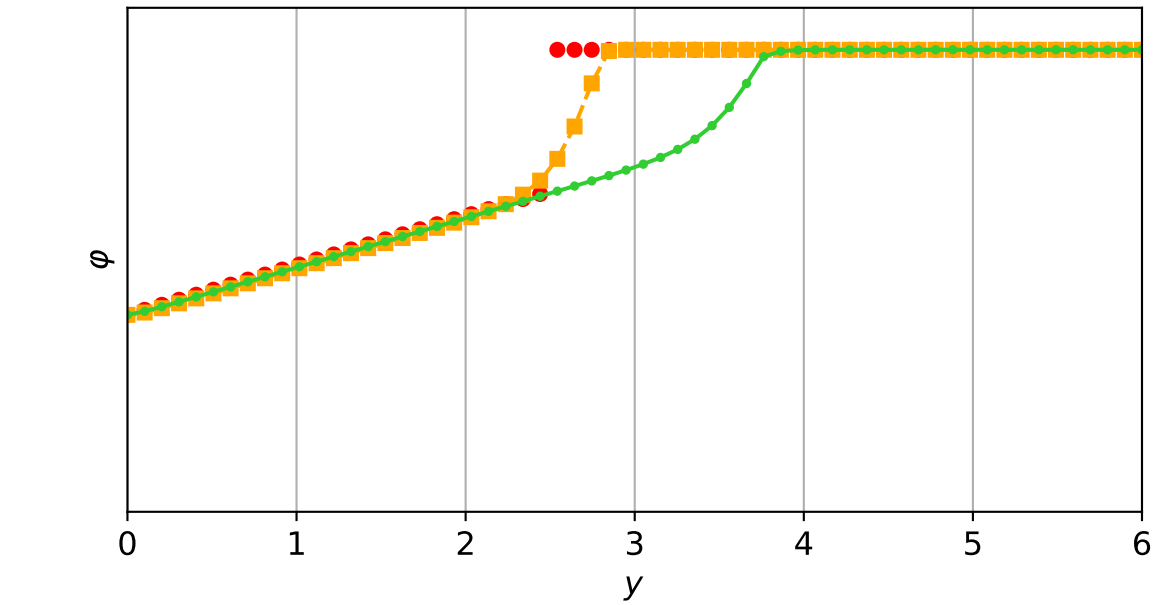
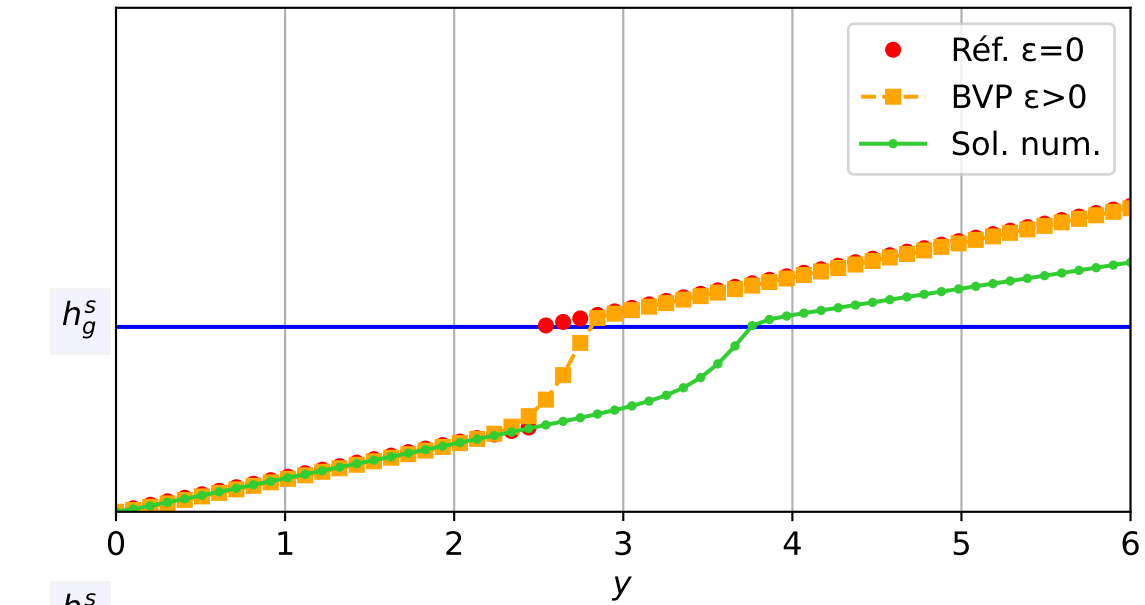
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 0.450115/12$  (543 iter) — résidu eq. =  $5.21e-06$



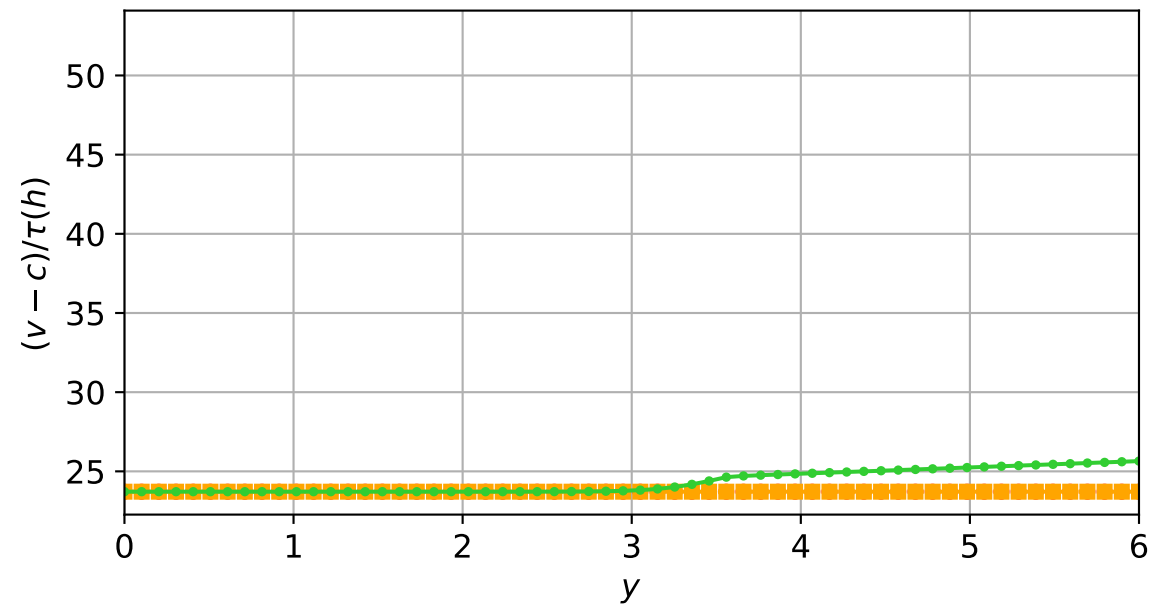
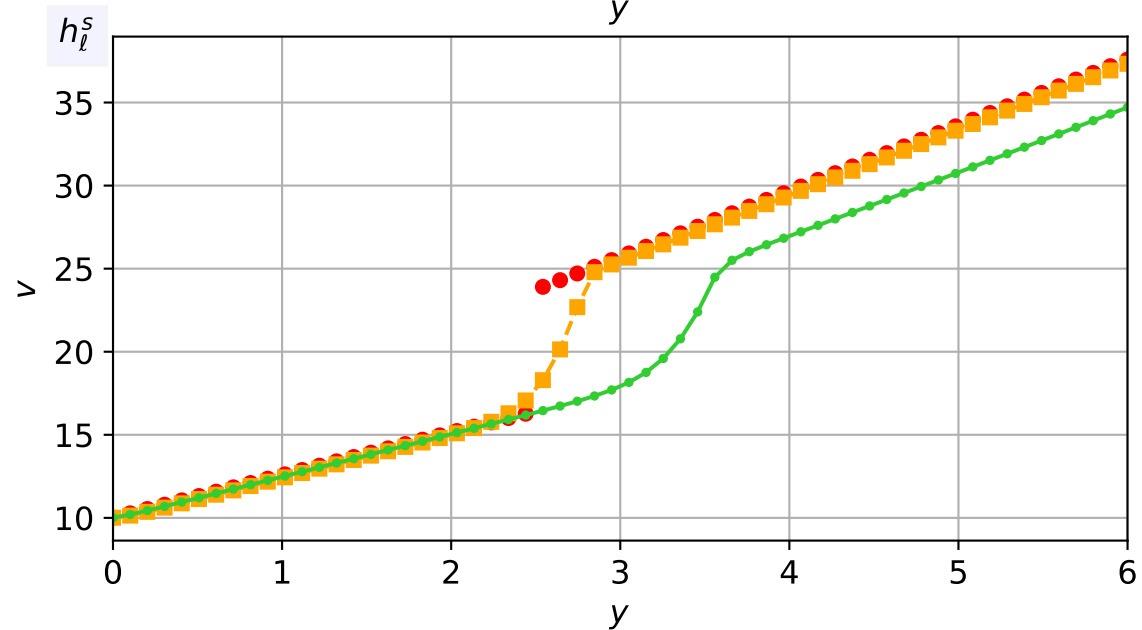
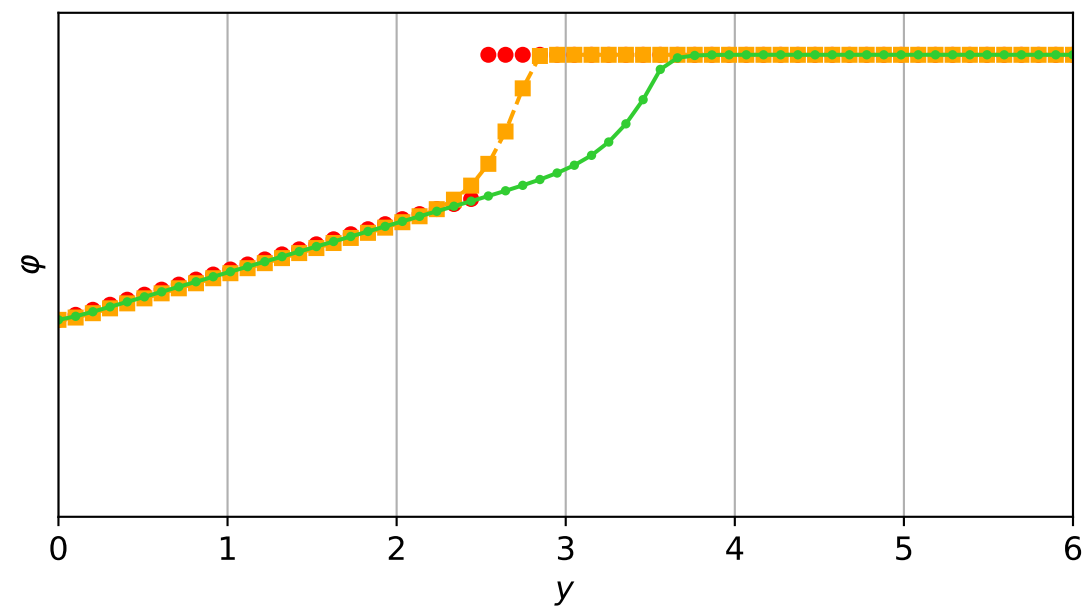
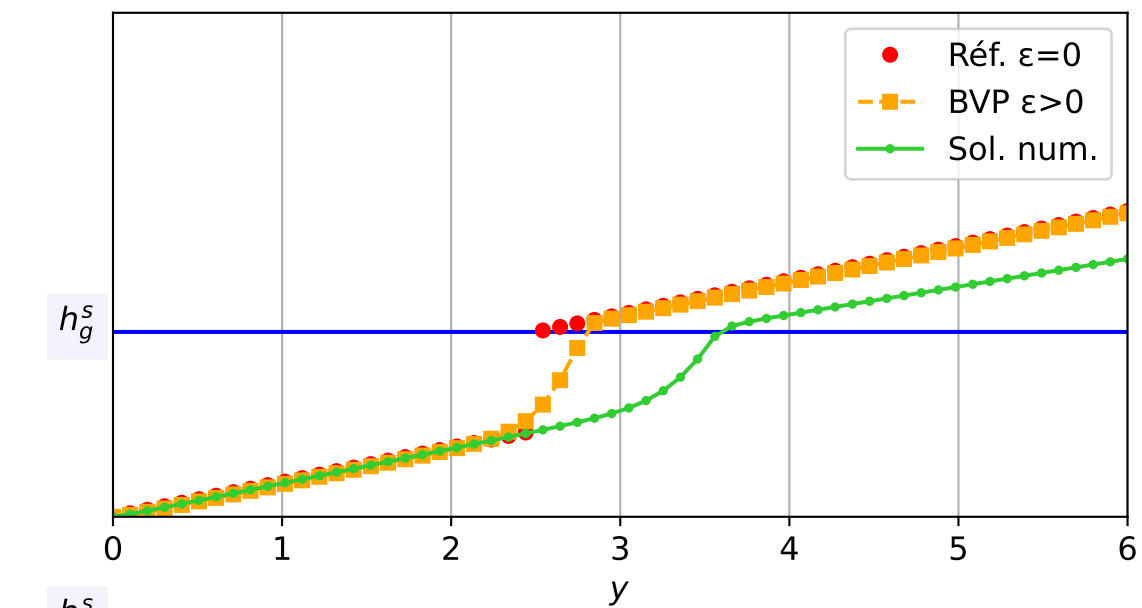
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 0.500039/12$  (764 iter) — résidu eq. =  $9.36e-06$



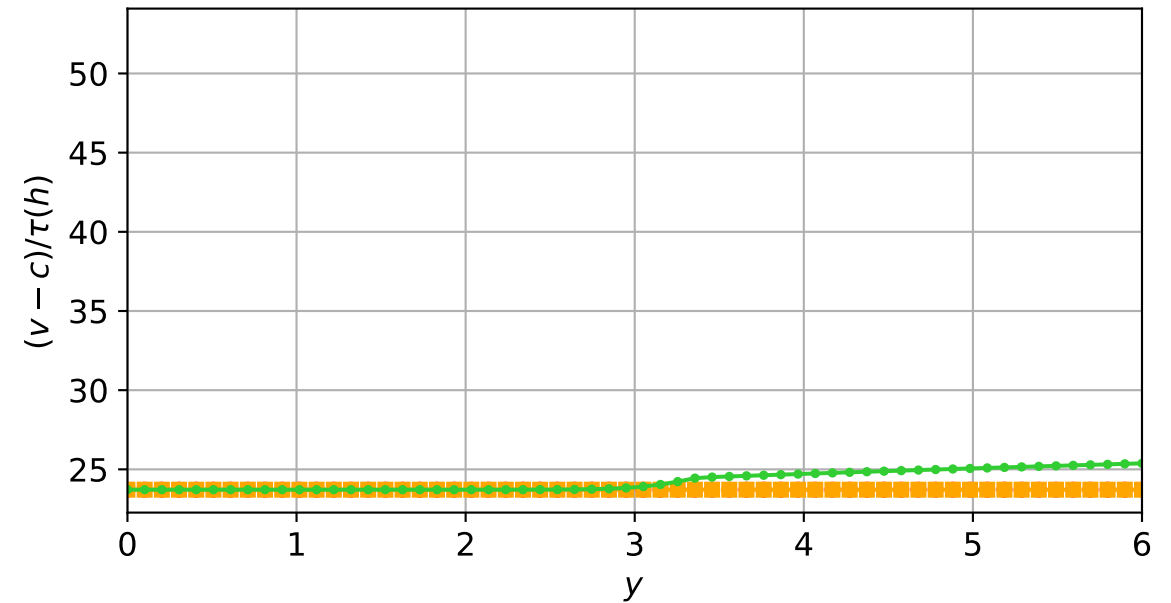
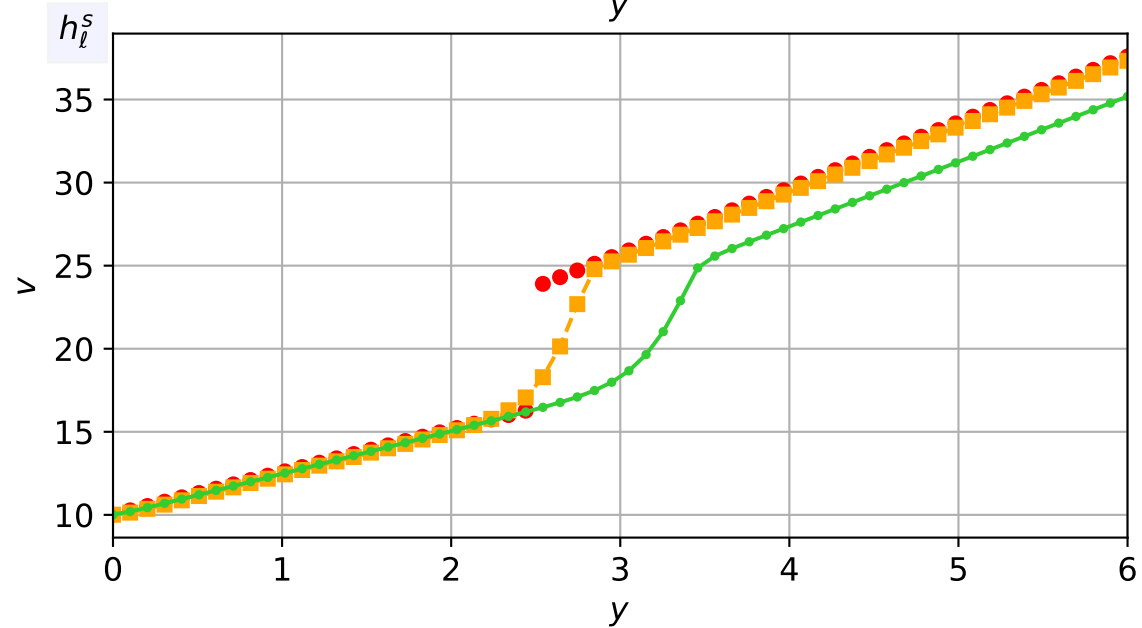
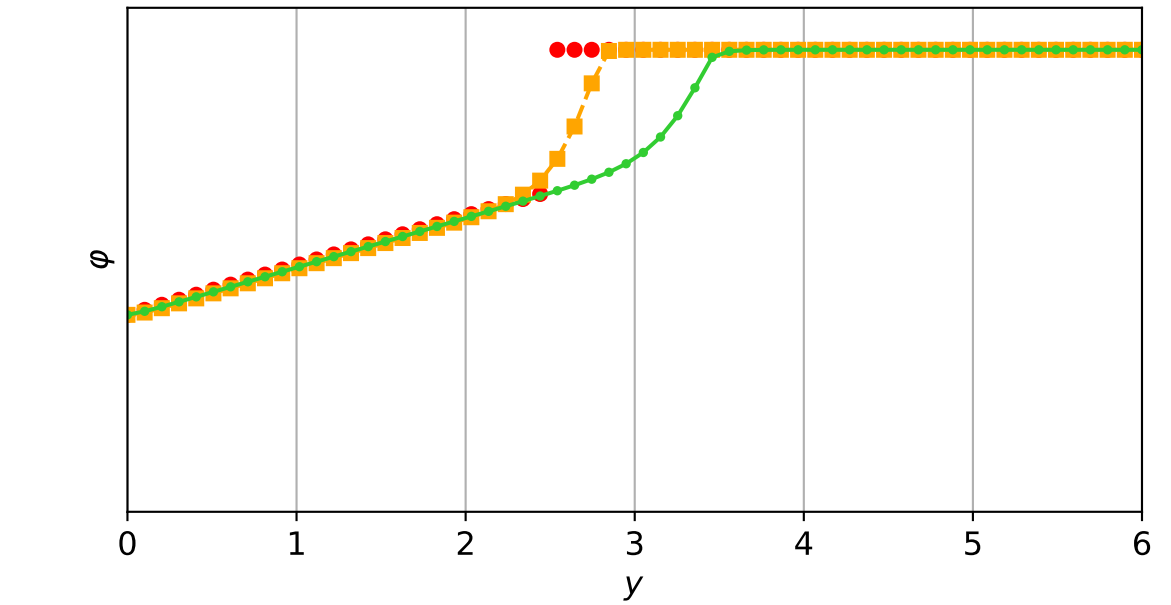
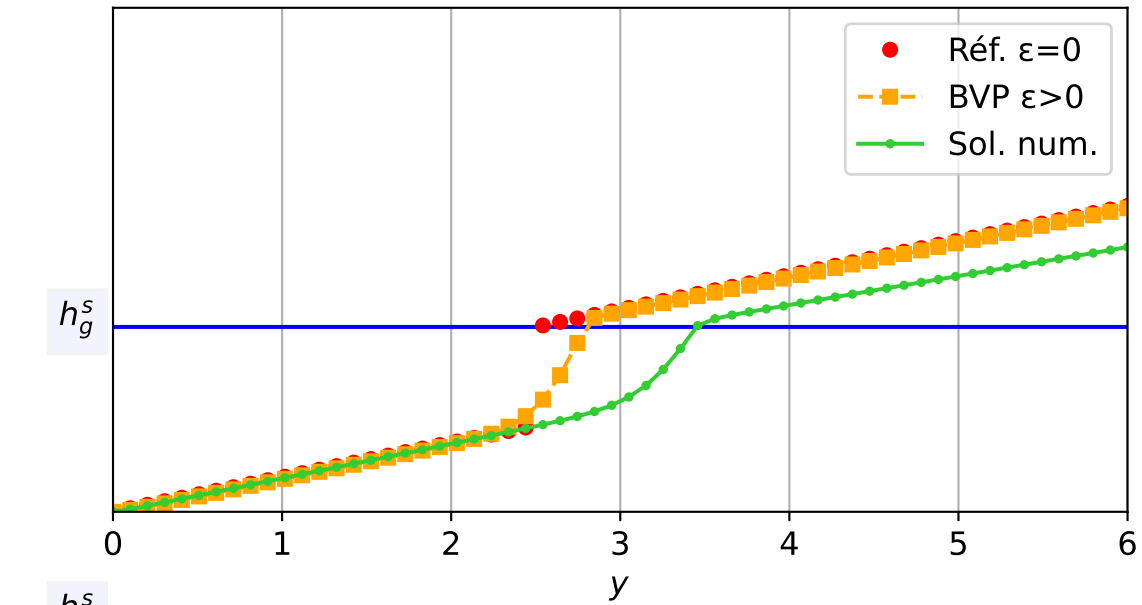
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 0.550189/12$  (986 iter) — résidu eq. =  $7.07e-06$



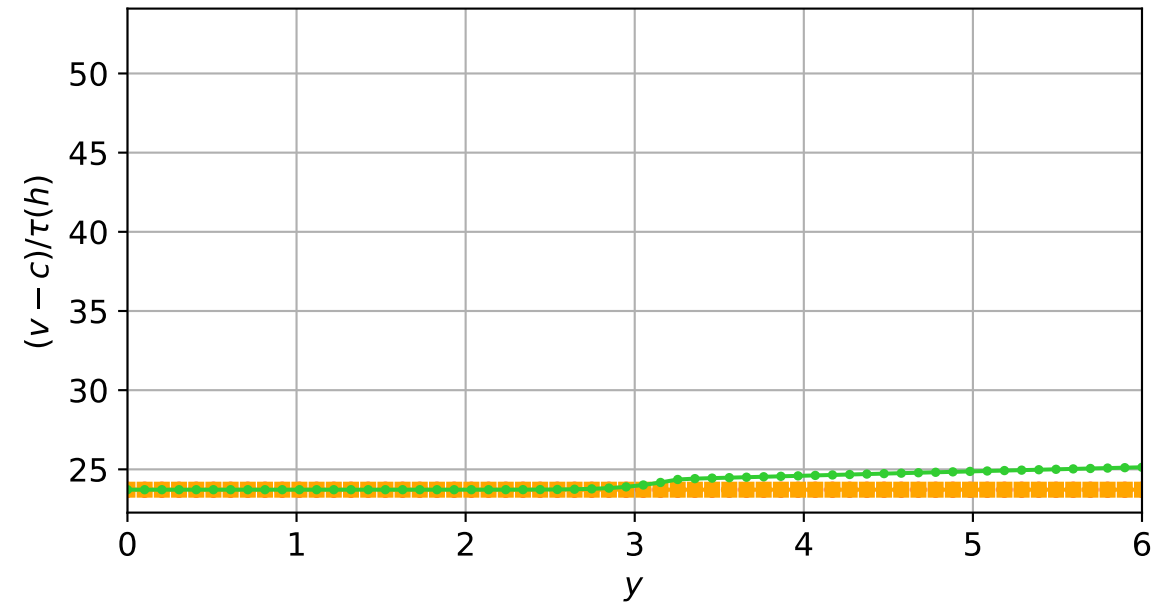
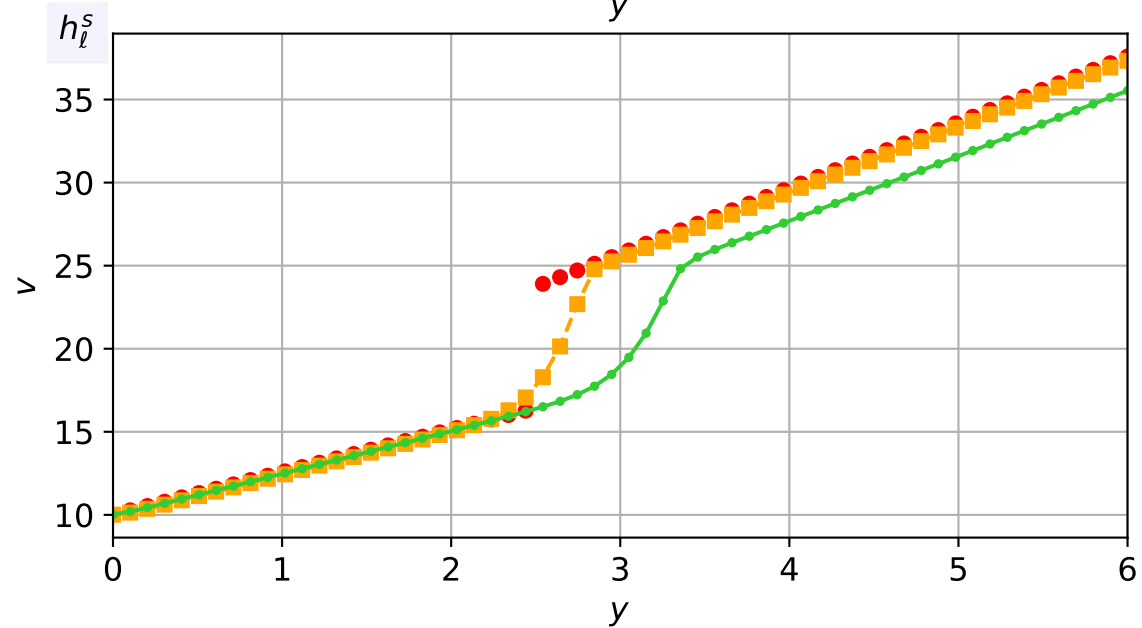
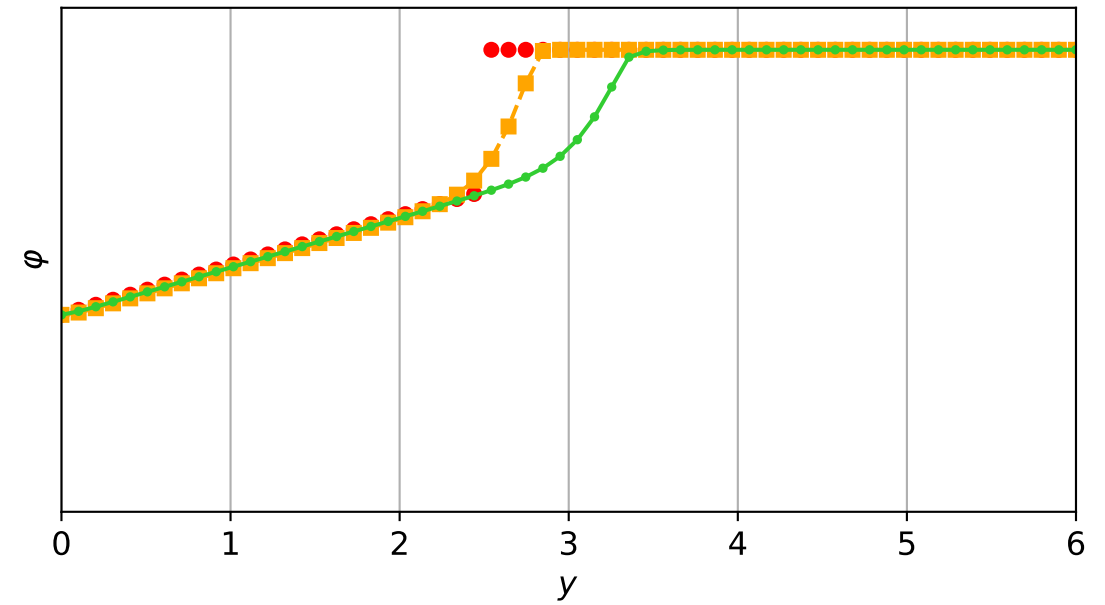
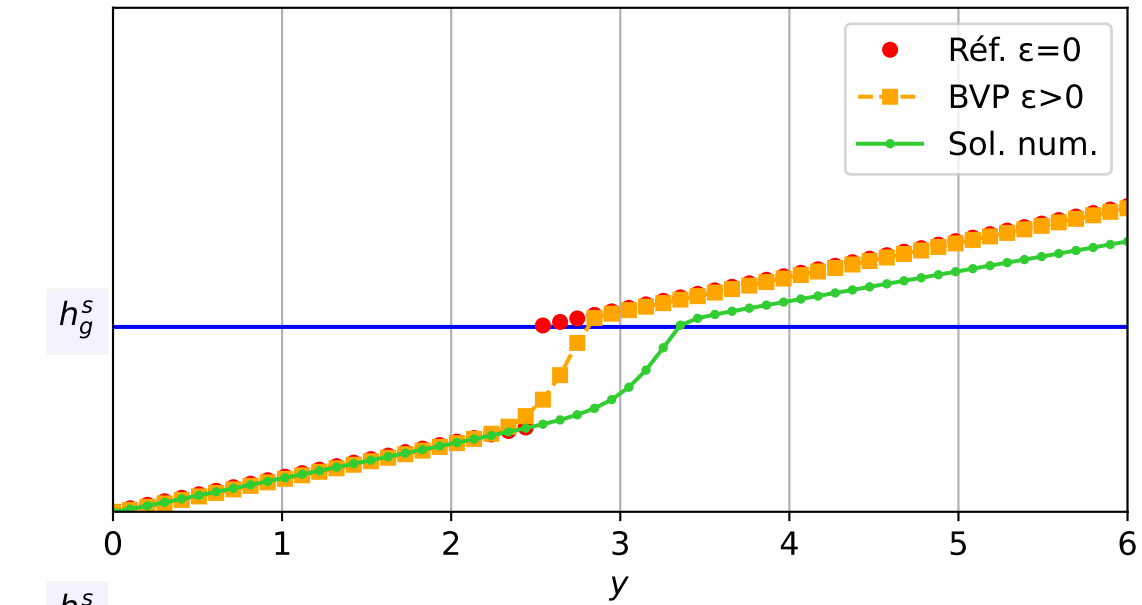
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 0.600114/12$  (1207 iter) — résidu eq. =  $5.96e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 0.650038/12$  (1428 iter) — résidu eq. =  $8.43e-06$

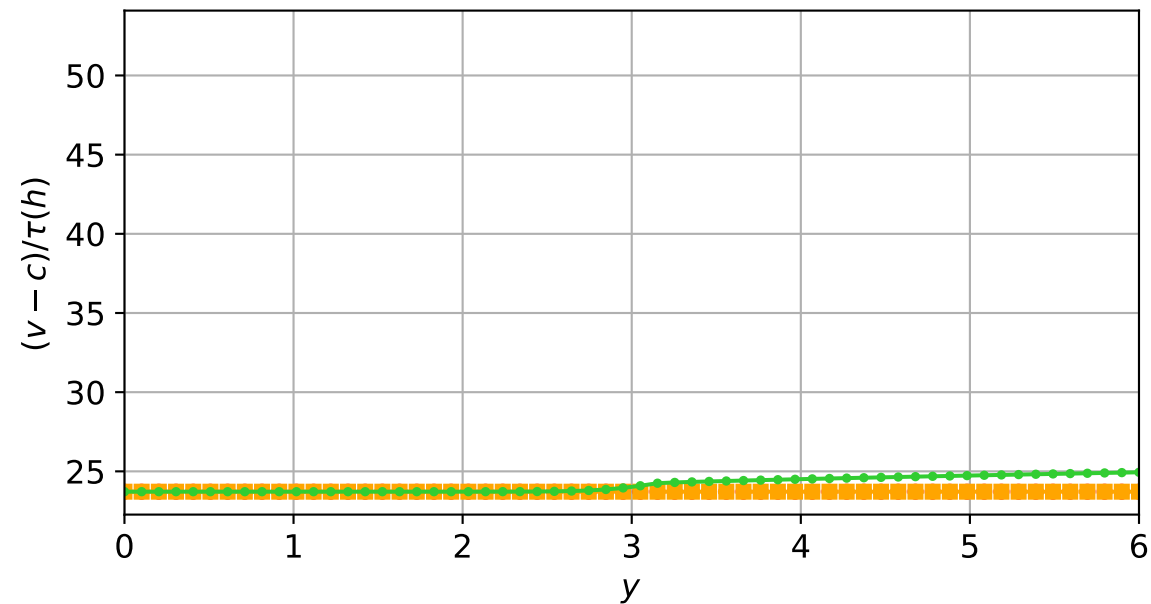
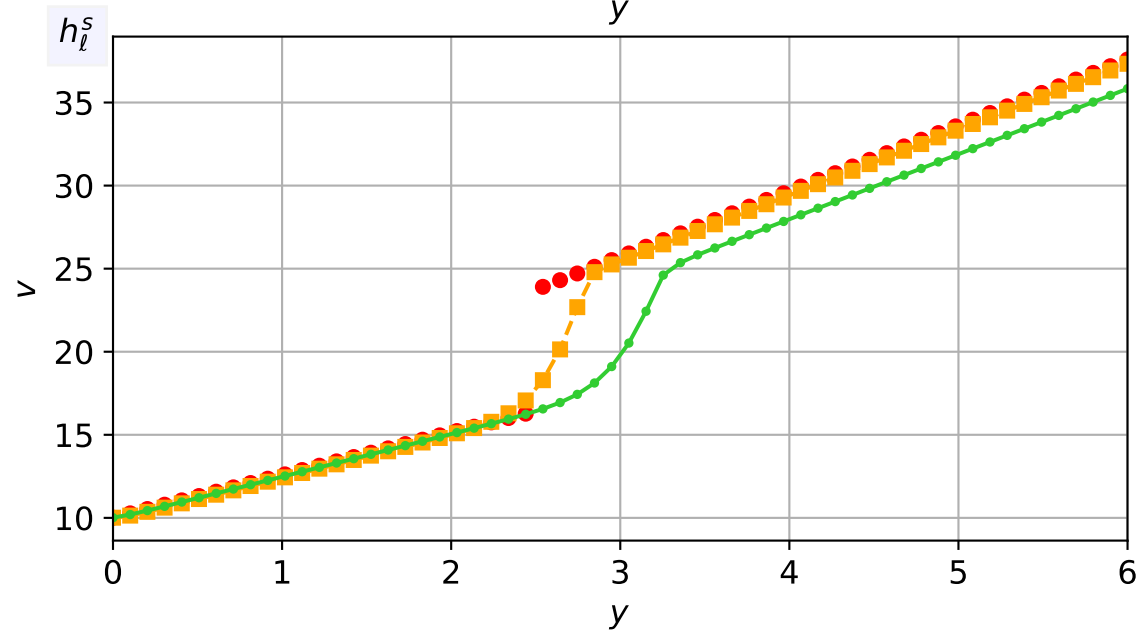
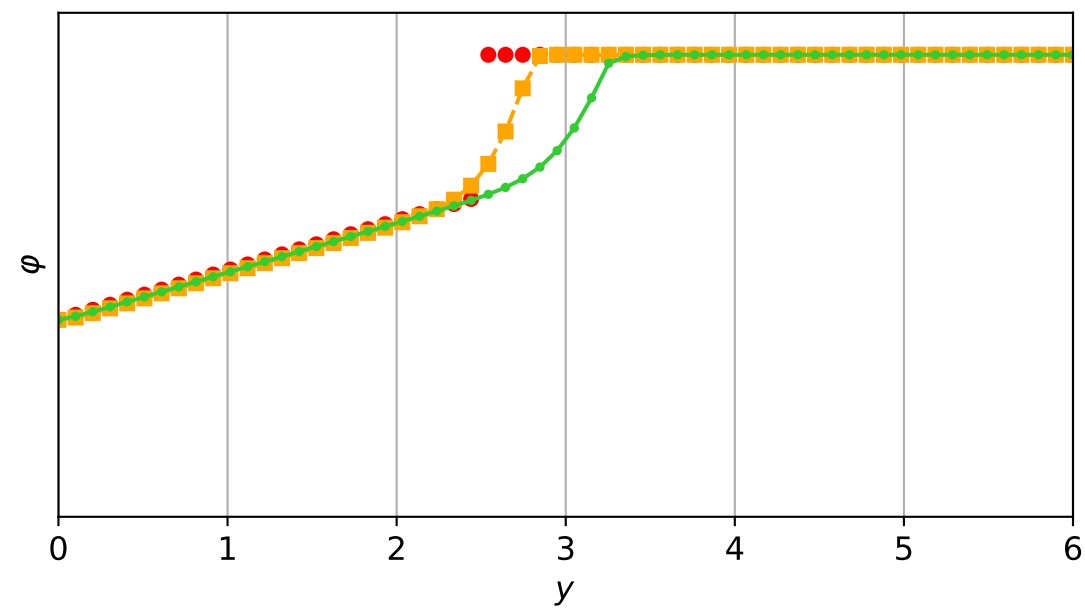
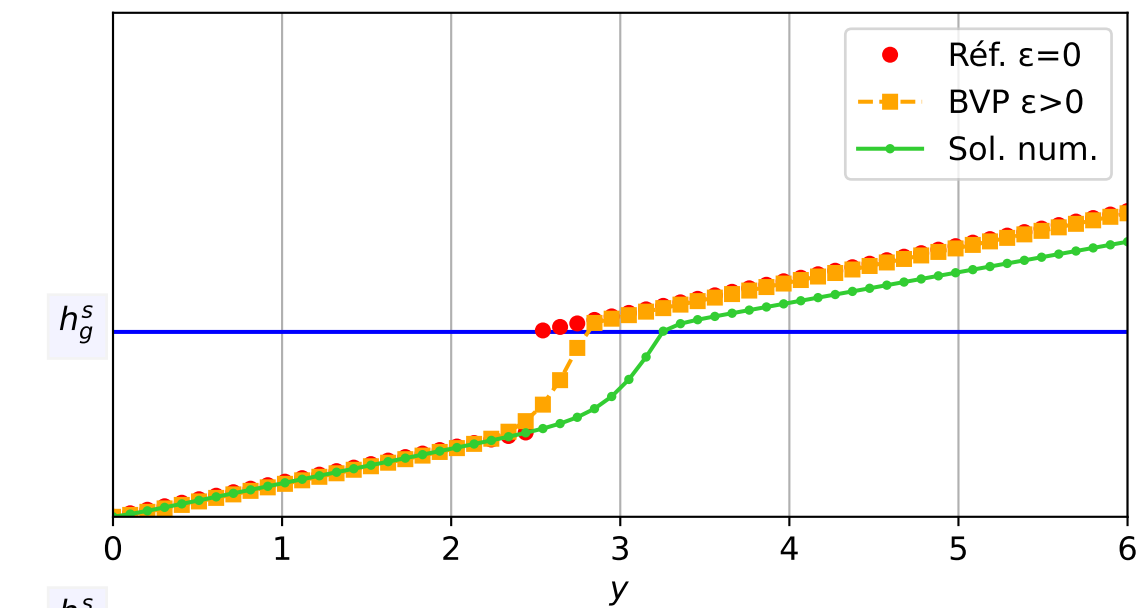


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 0.700189/12$  (1650 iter) — résidu eq. =  $6.05e-06$

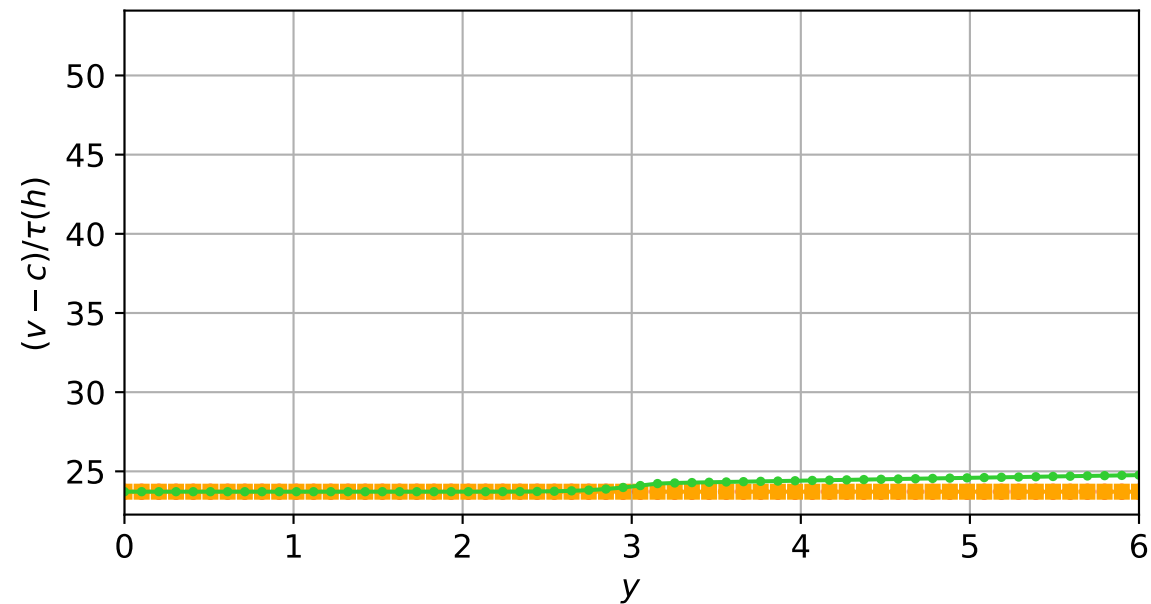
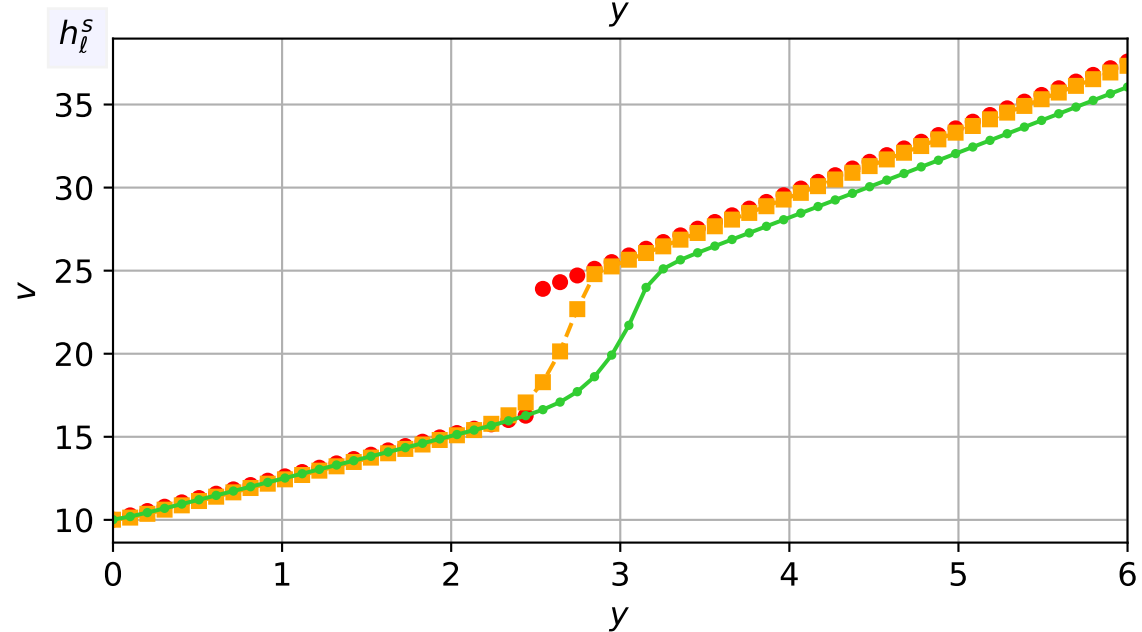
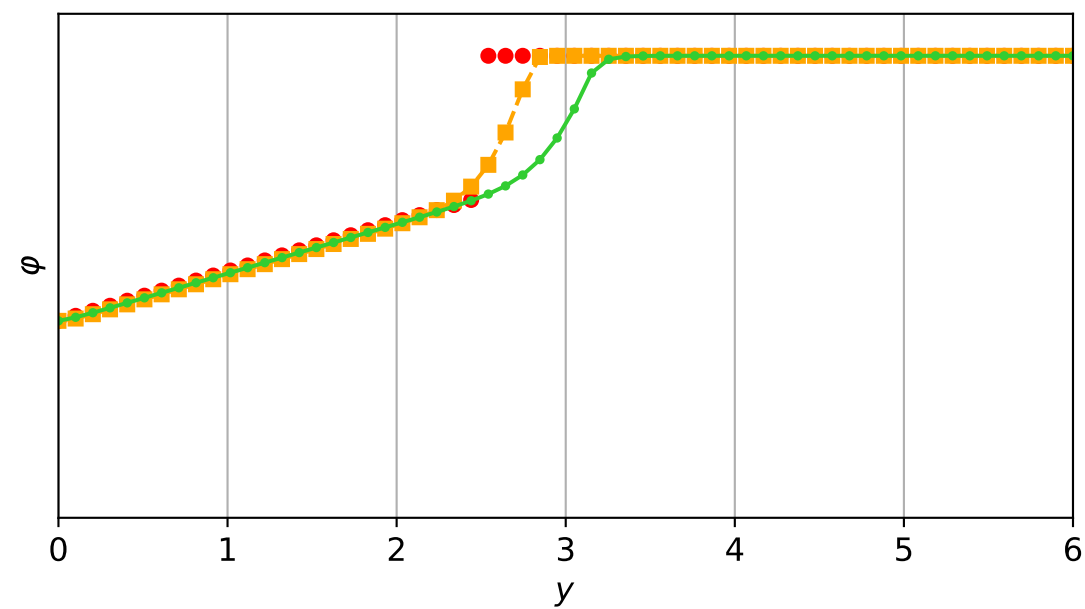
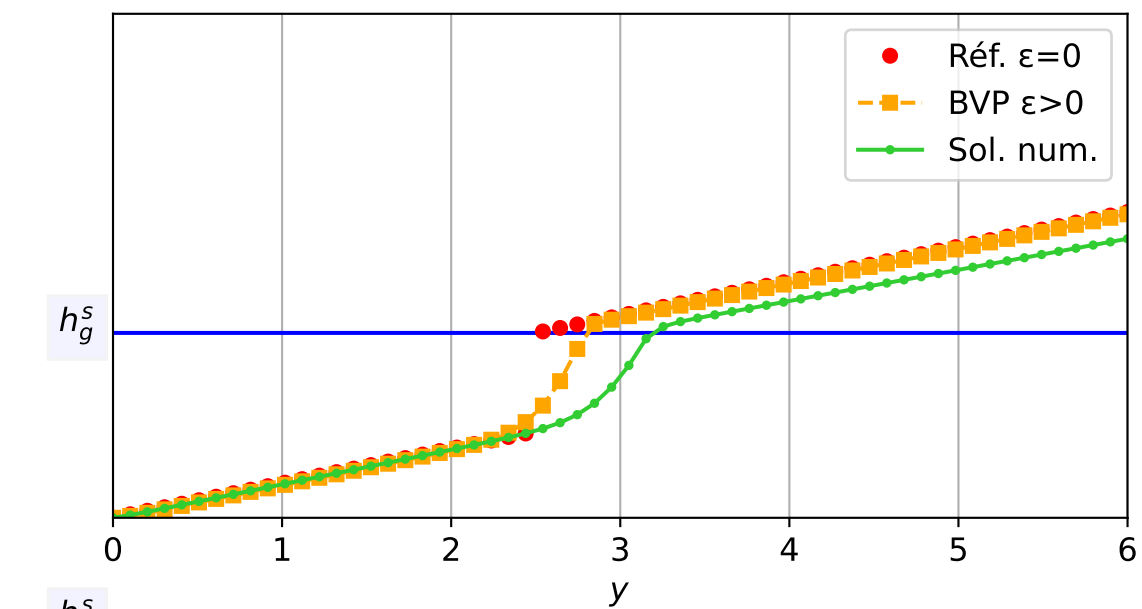




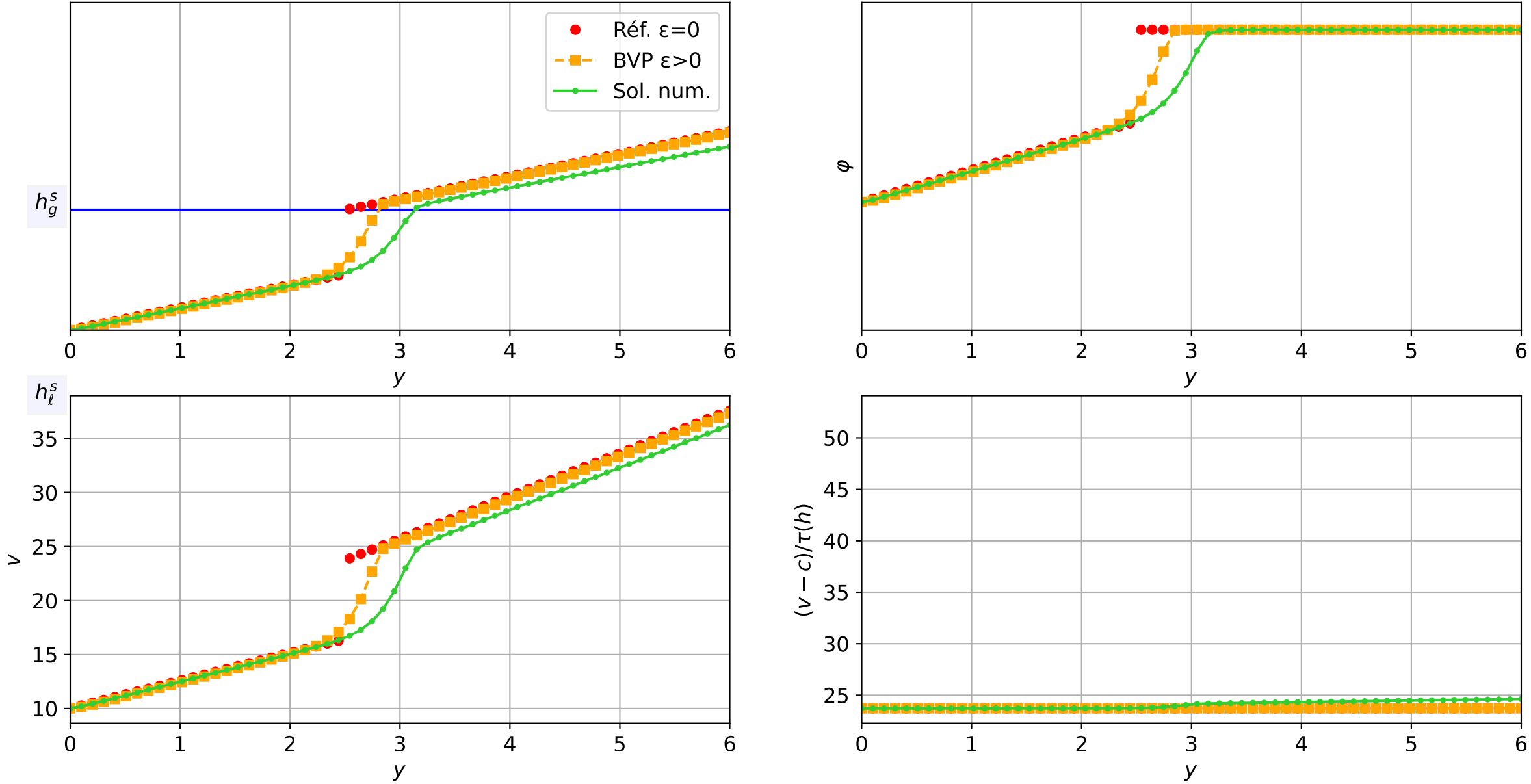
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 0.750113/12$  (1871 iter) — résidu eq. =  $6.38e-06$



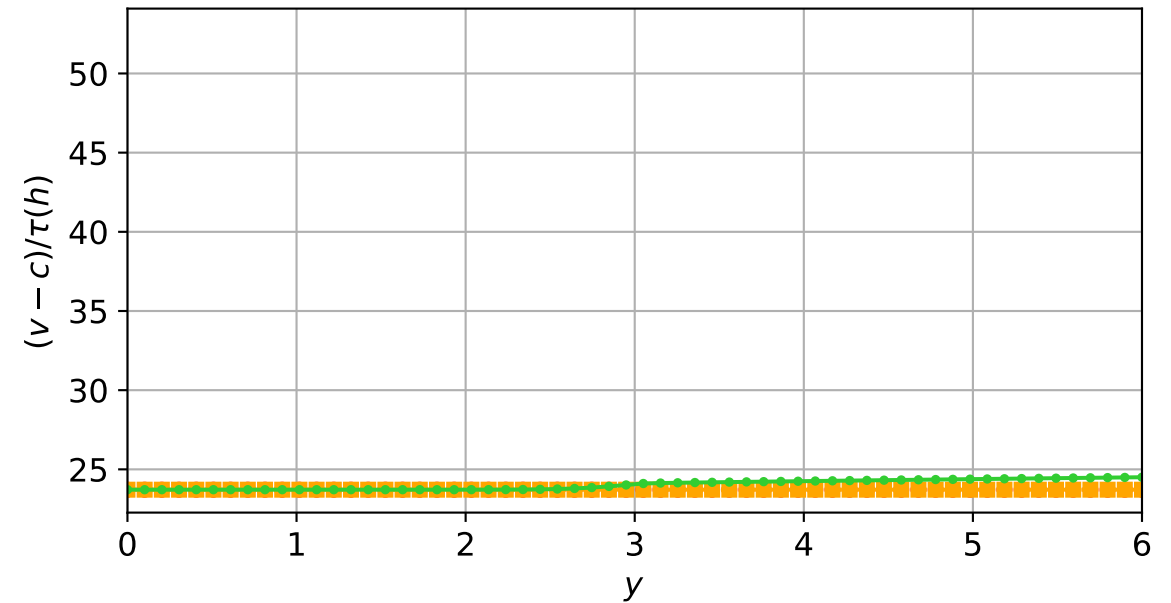
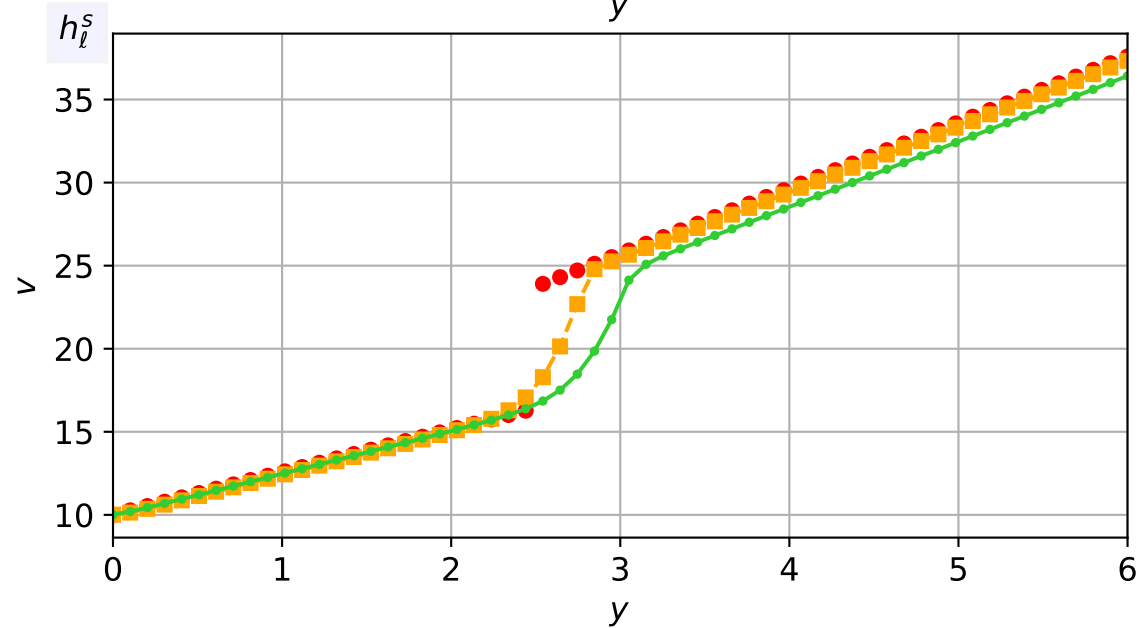
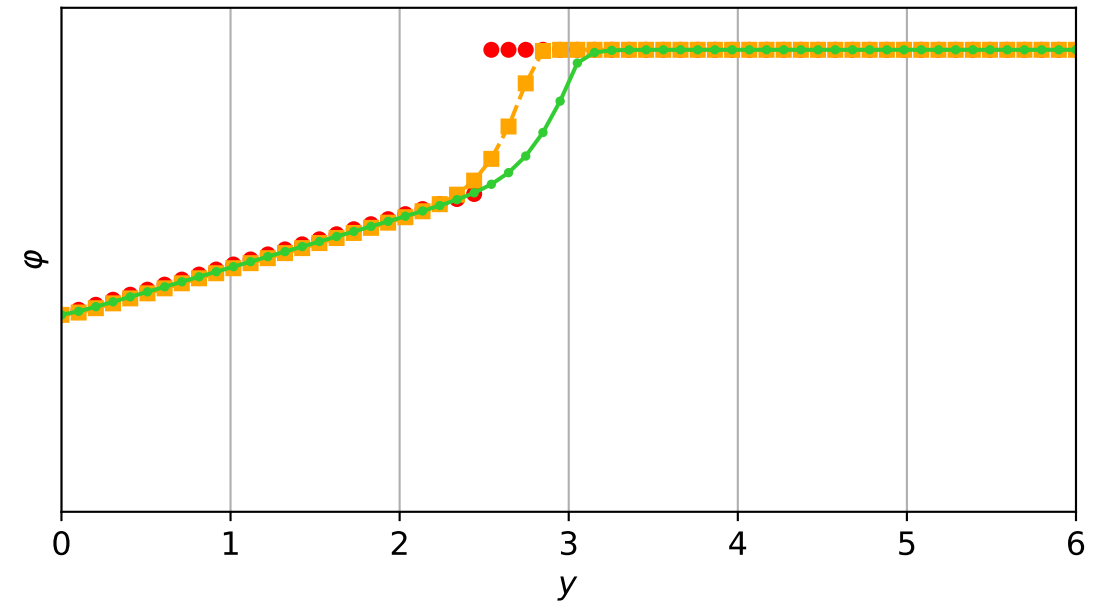
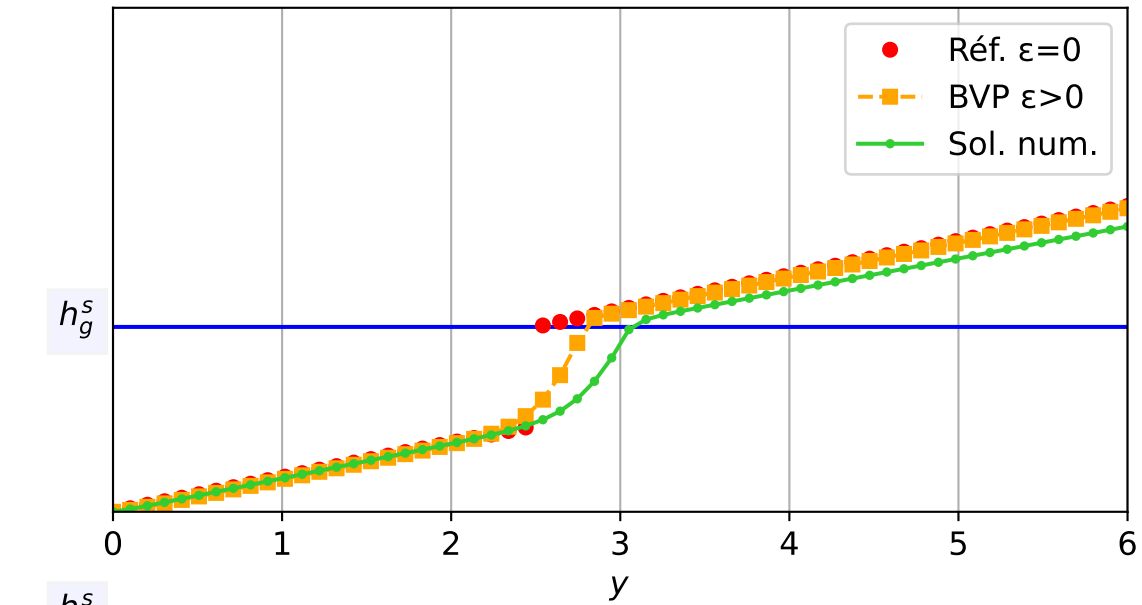
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 0.800038/12$  (2092 iter) — résidu eq. =  $5.70e-06$



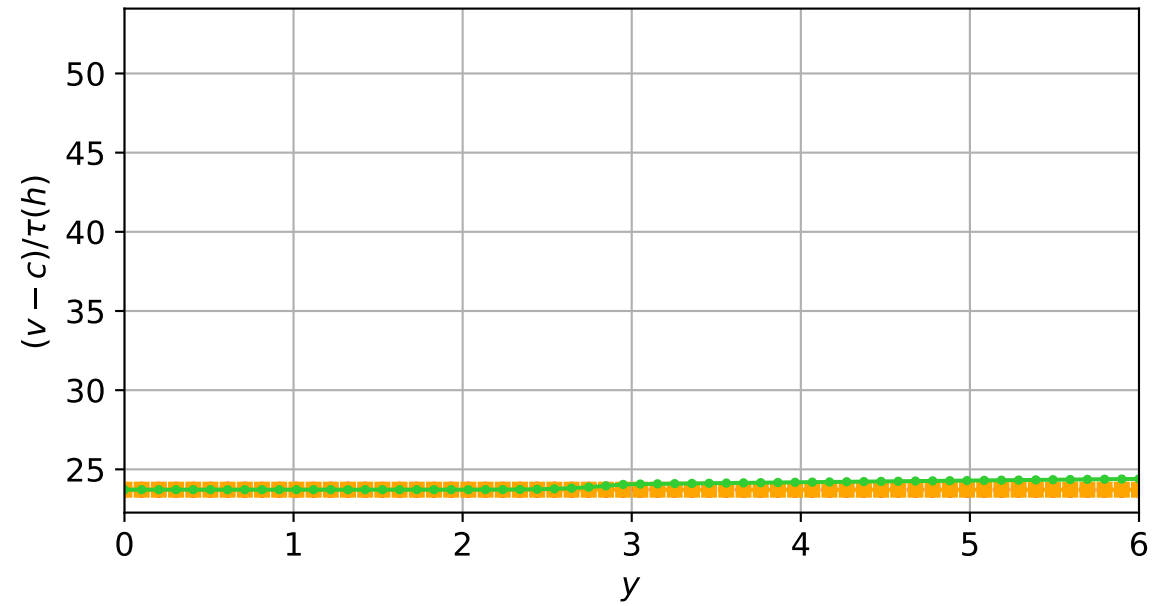
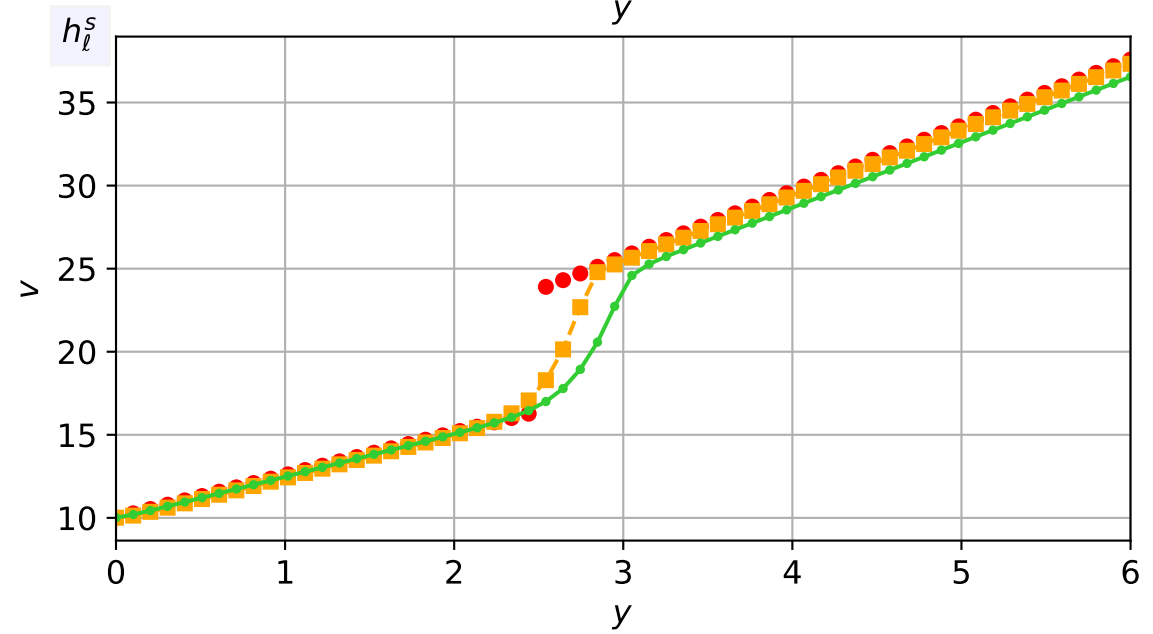
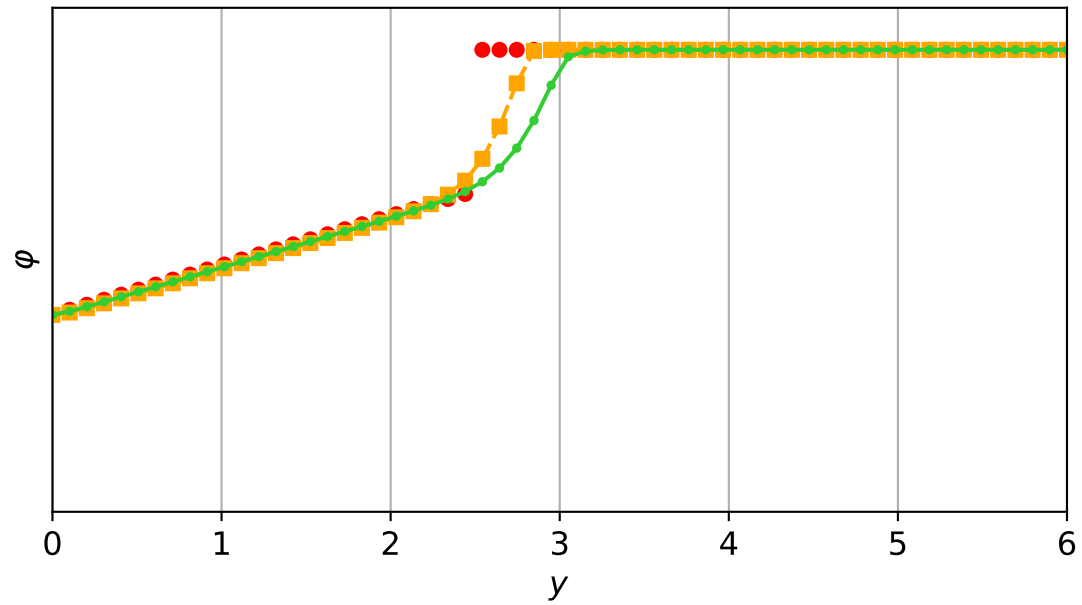
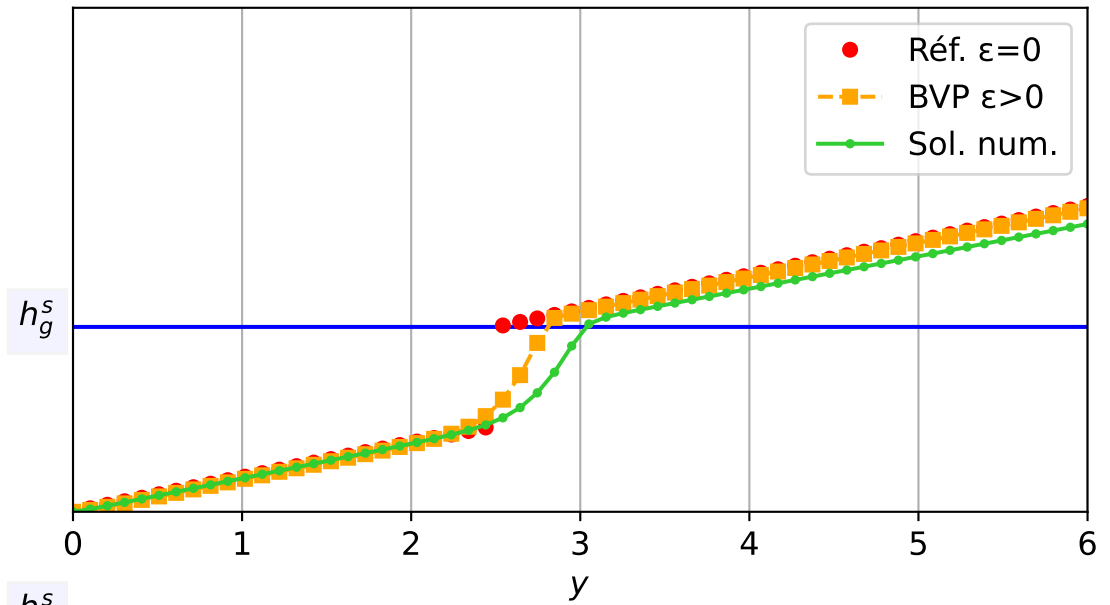
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 0.850188/12$  (2314 iter) — résidu eq. =  $9.67e-06$



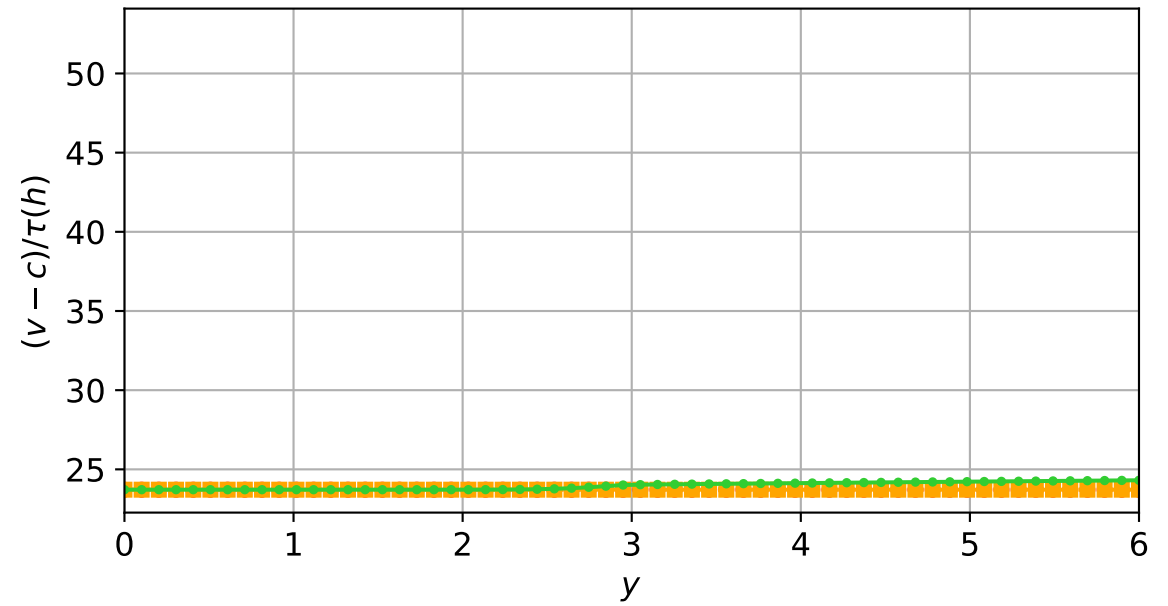
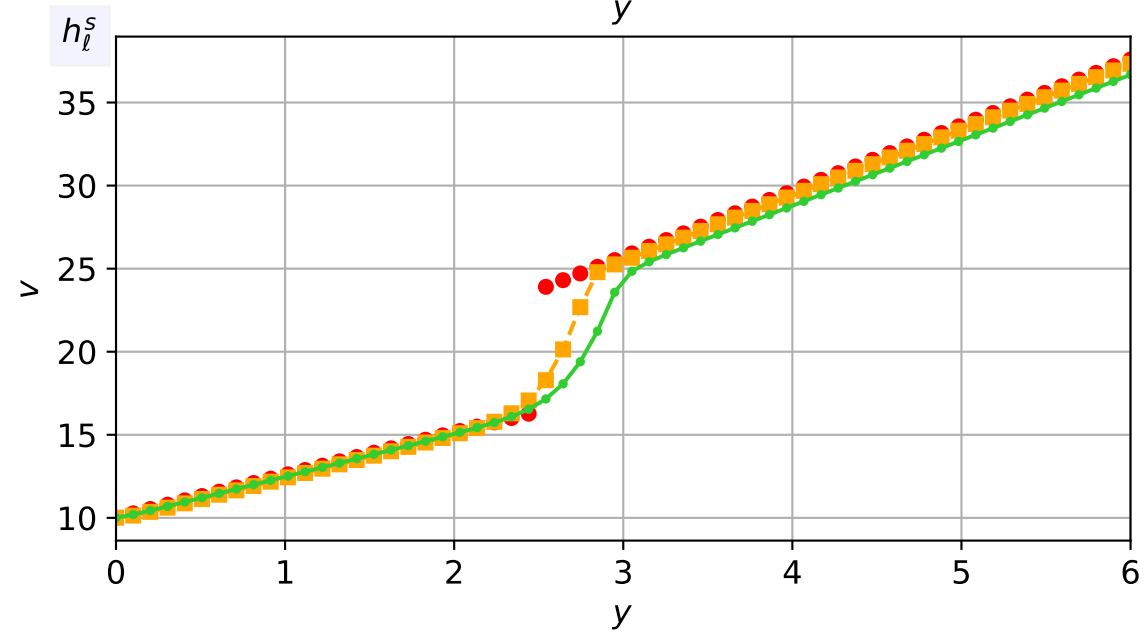
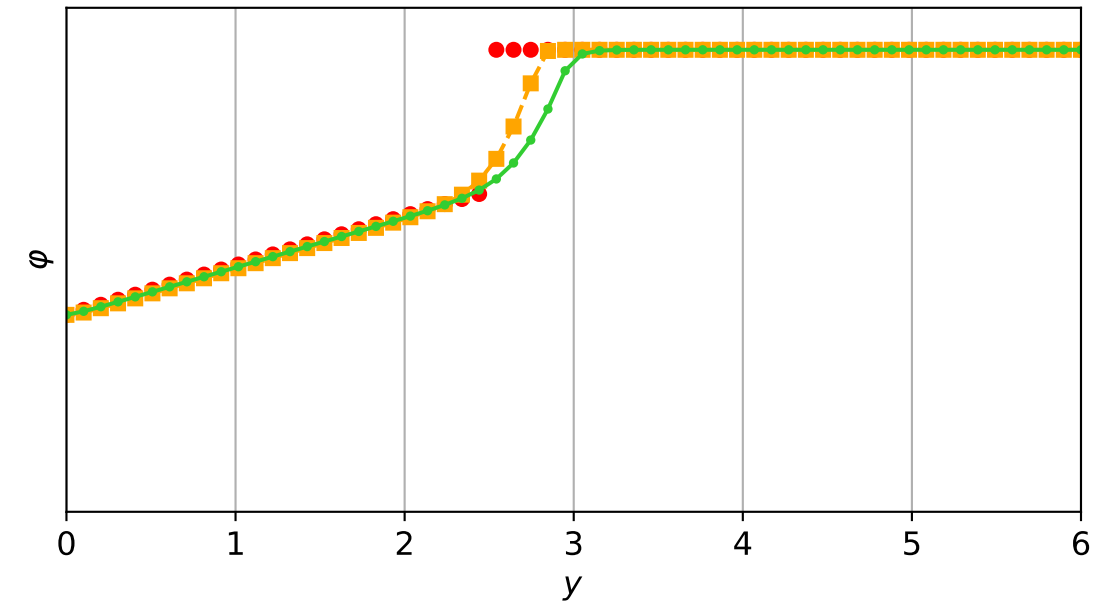
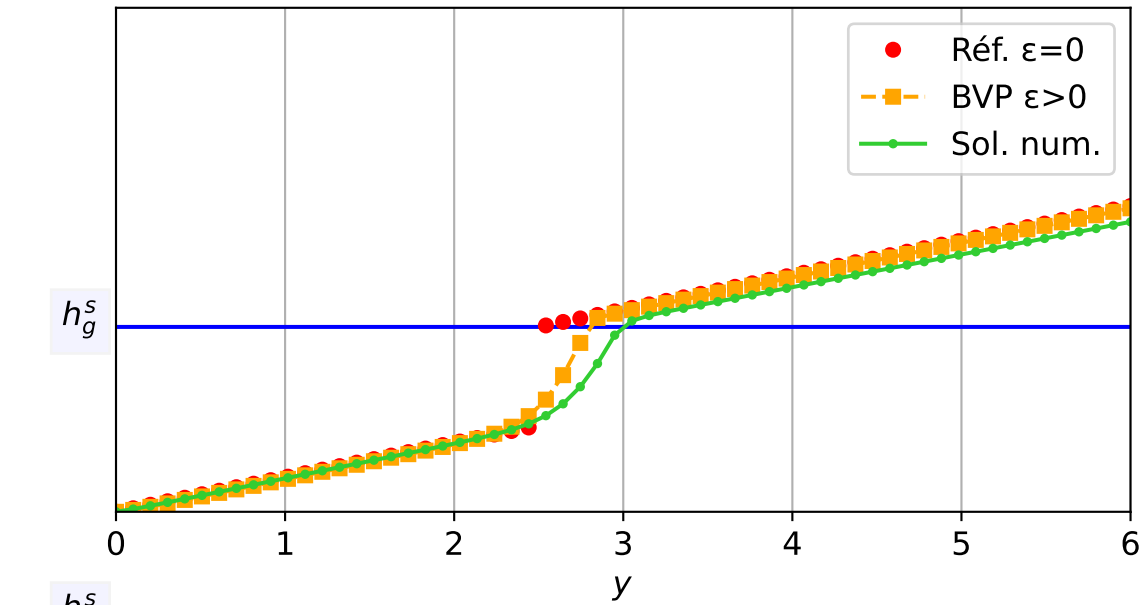
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 0.900112/12$  (2535 iter) — résidu eq. =  $7.97e-06$



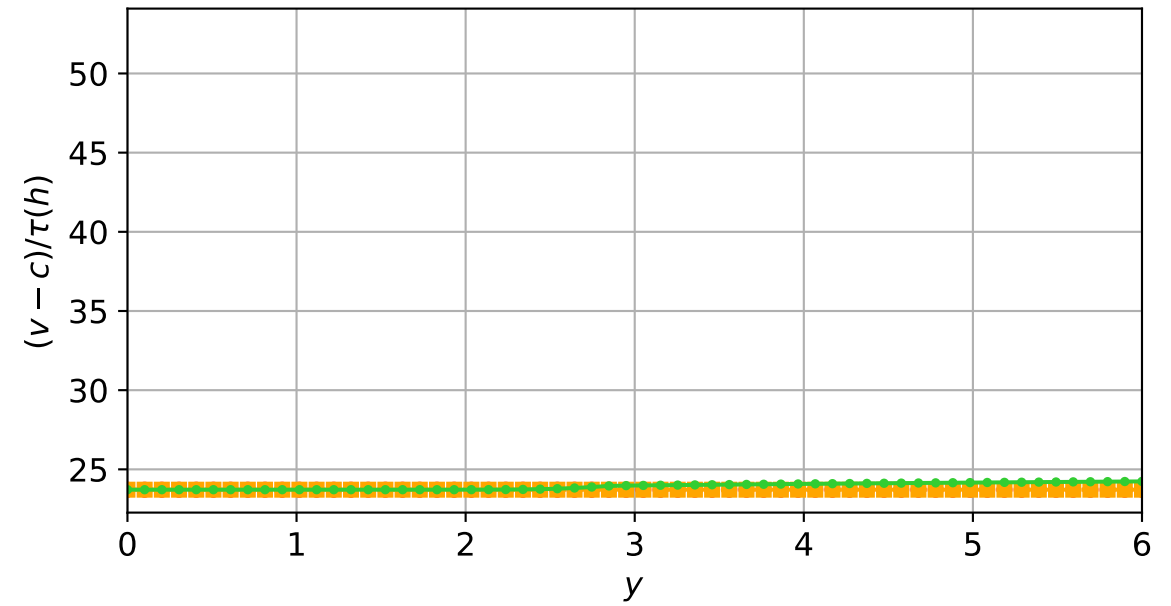
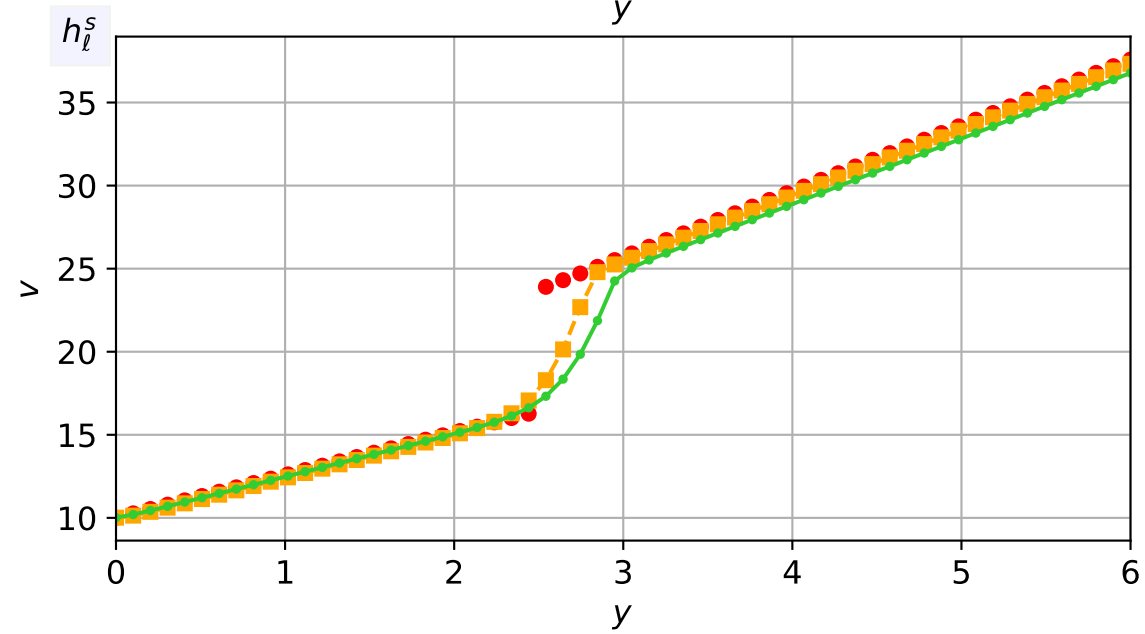
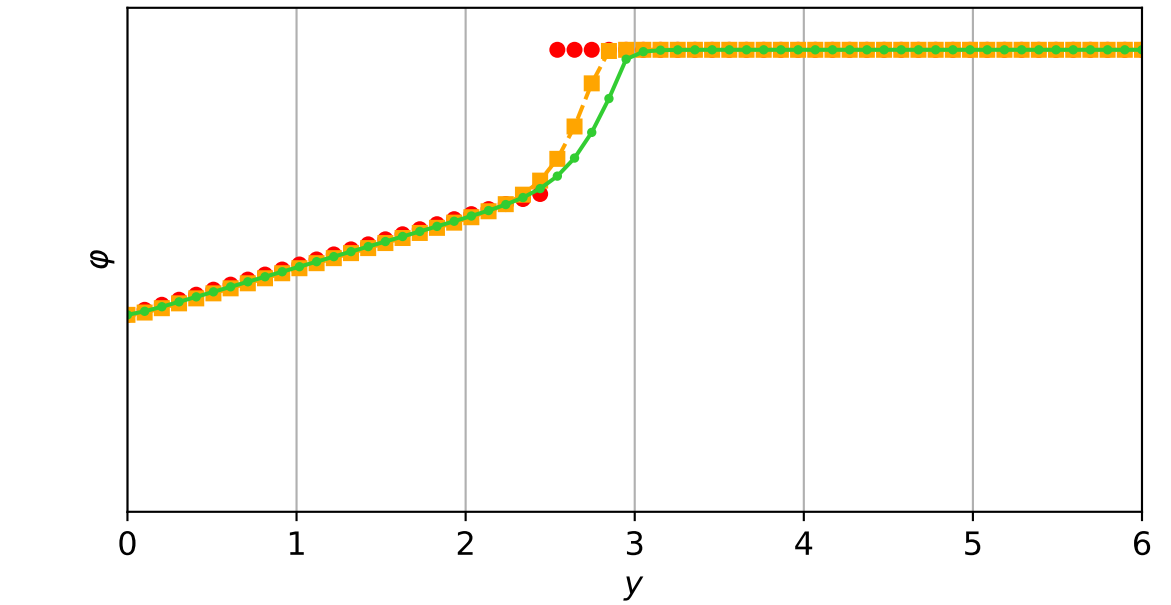
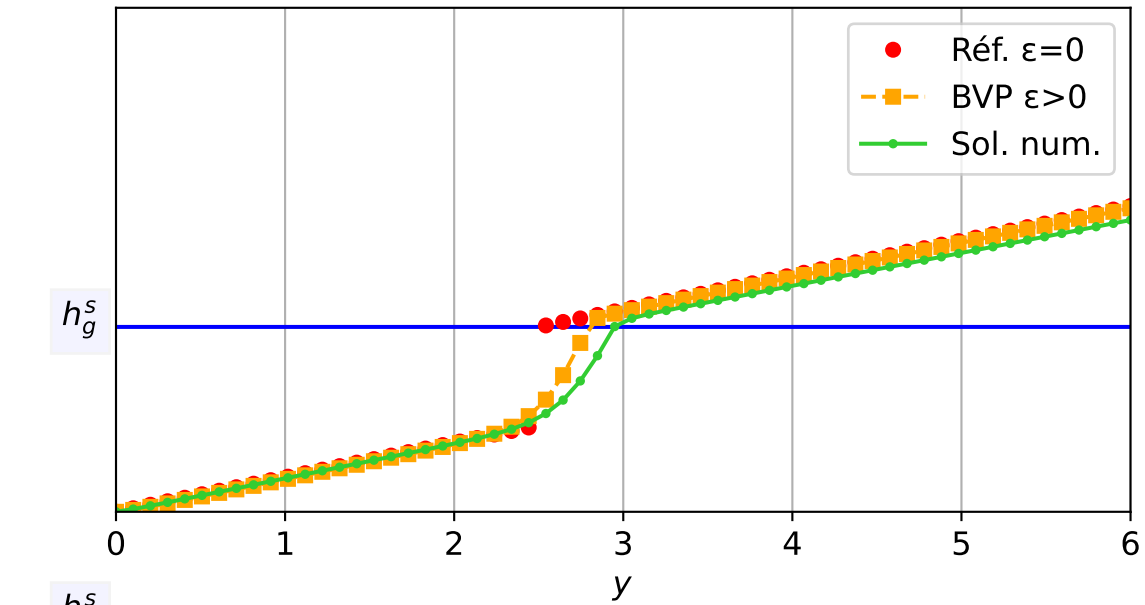
TEST-3\_MG\_AS : Validation AS | Asymptotique | eps=1e-03  
 $\varepsilon = 1\text{e-}03$ , schéma implicite, N = 60, dx = 0.1  
dt = 2.26e-04 — t = 0.950037/12 (2756 iter) — résidu eq. = 7.14e-06



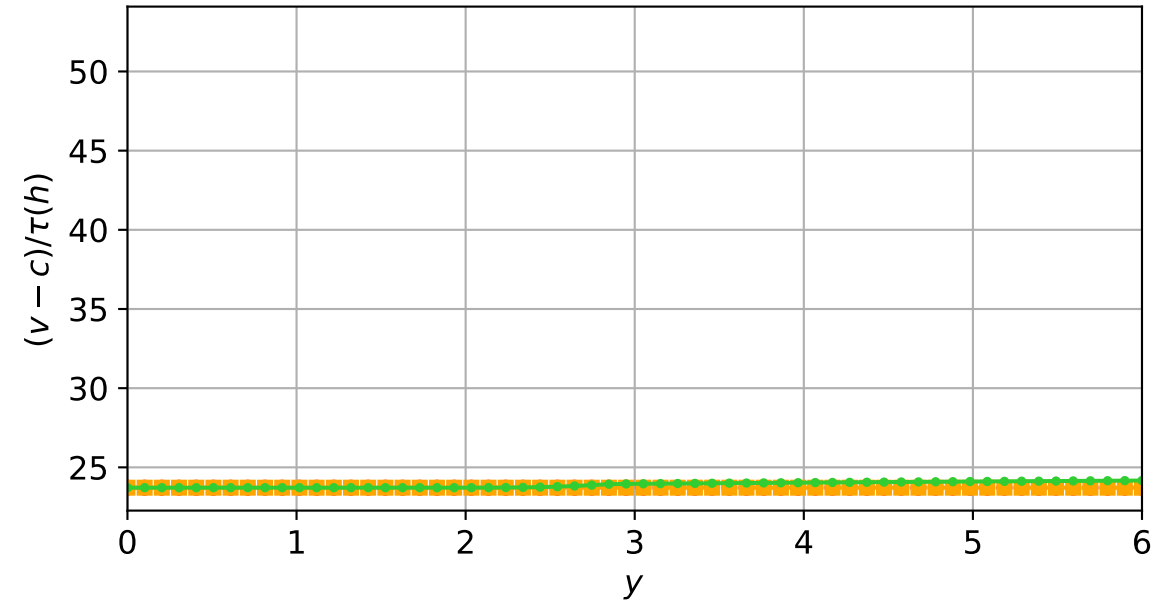
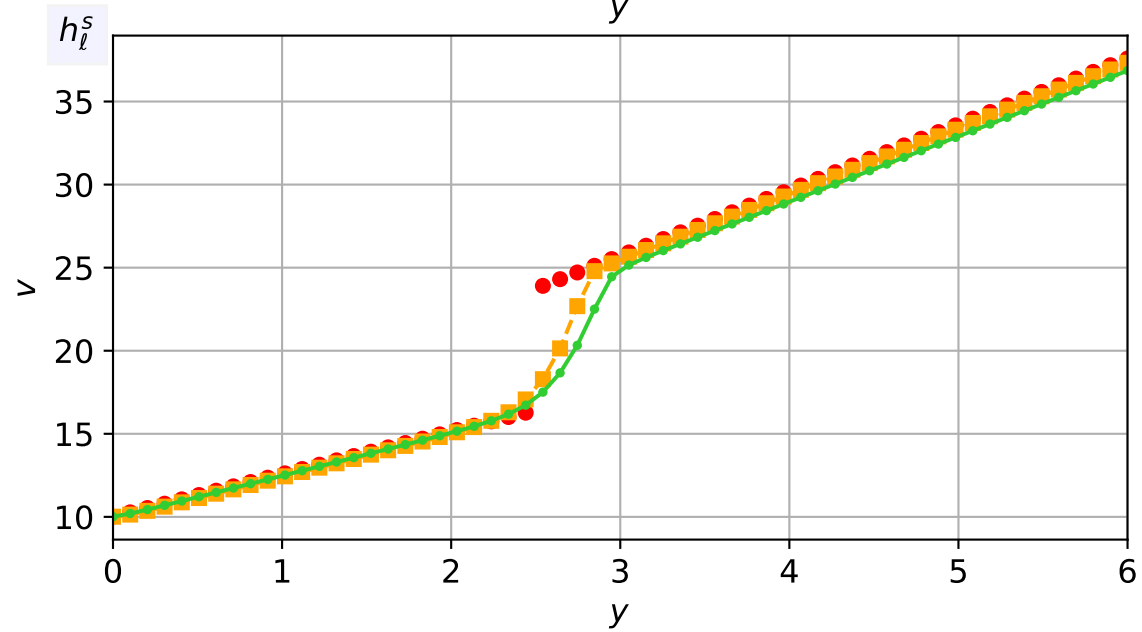
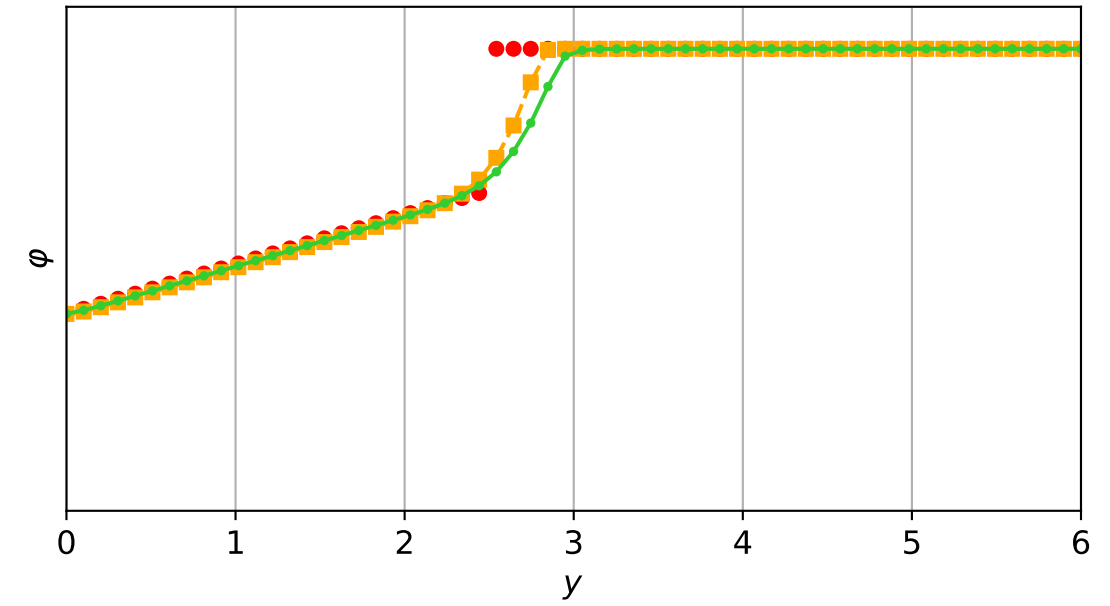
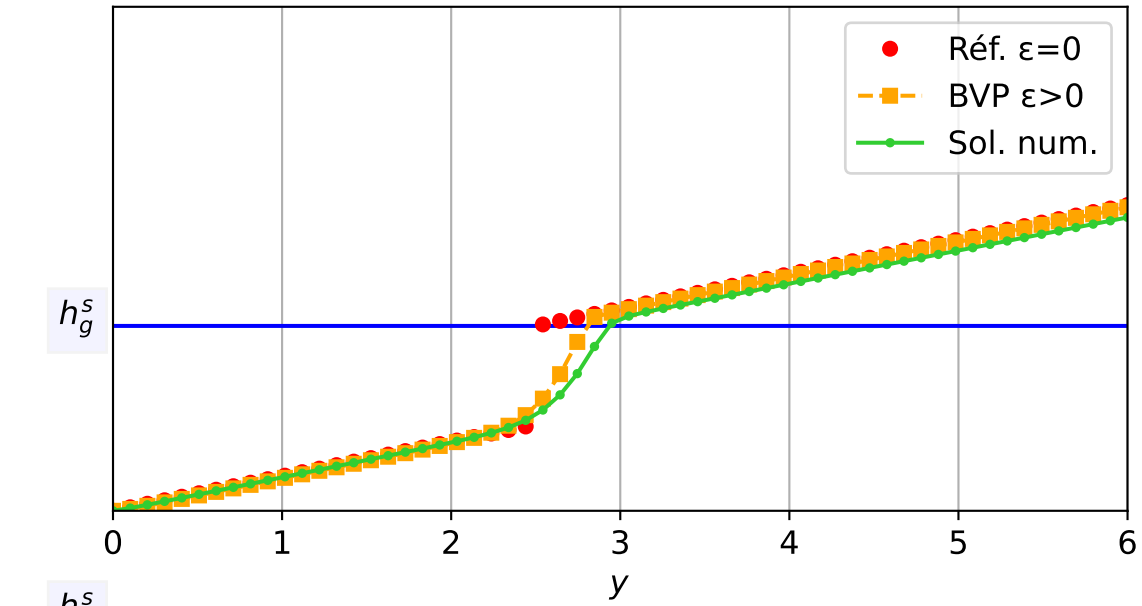
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 1.00019/12$  (2978 iter) — résidu eq. =  $5.92e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 1.05011/12$  (3199 iter) — résidu eq. =  $9.19e-06$

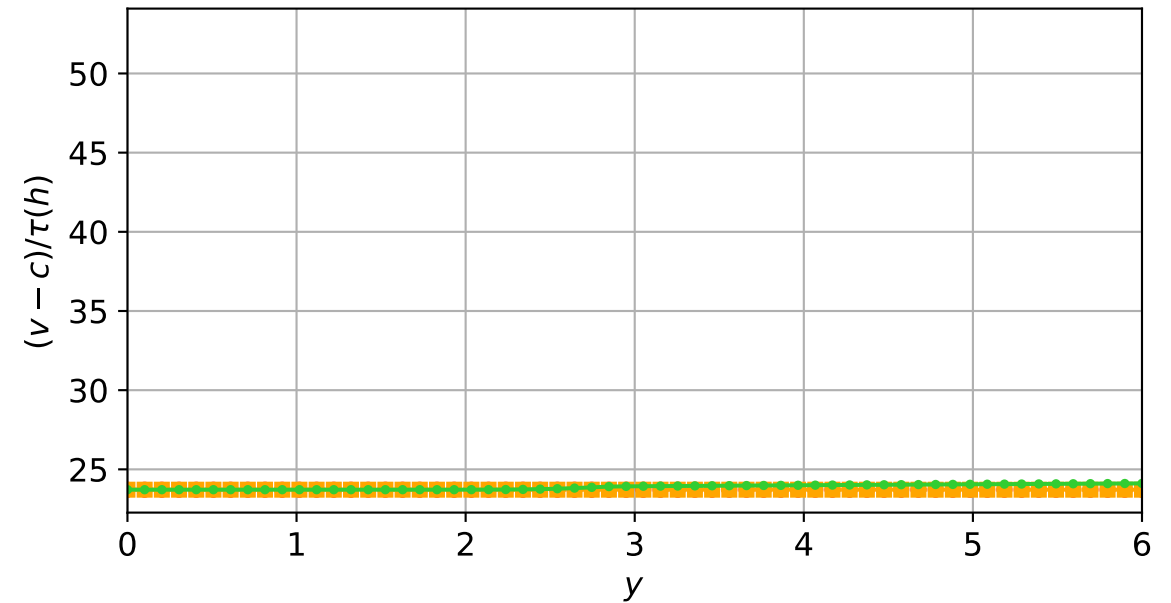
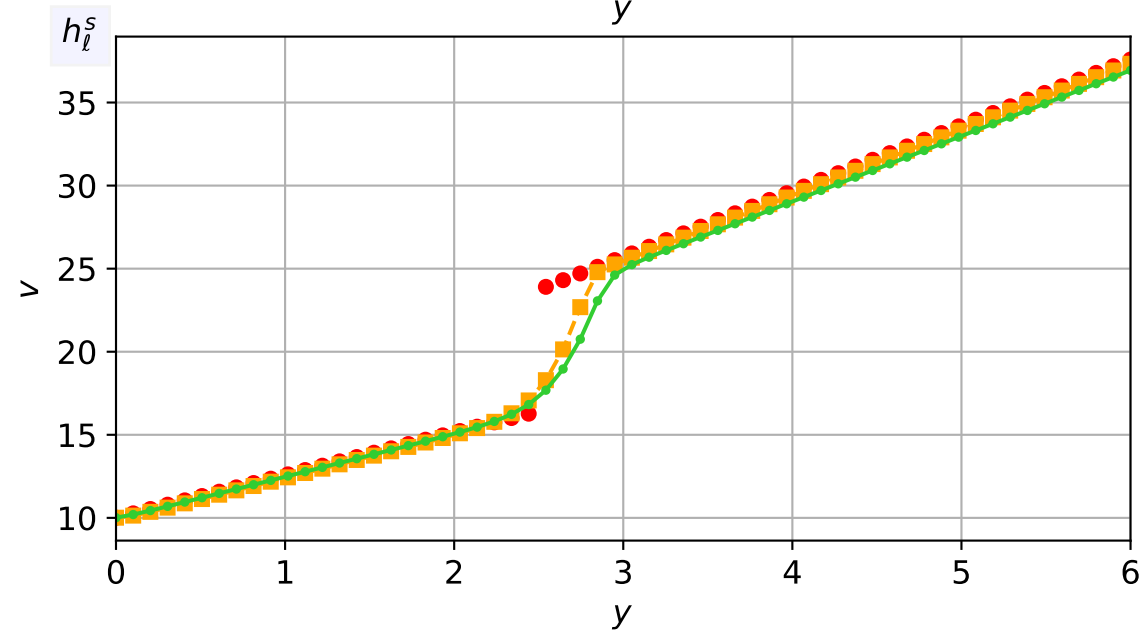
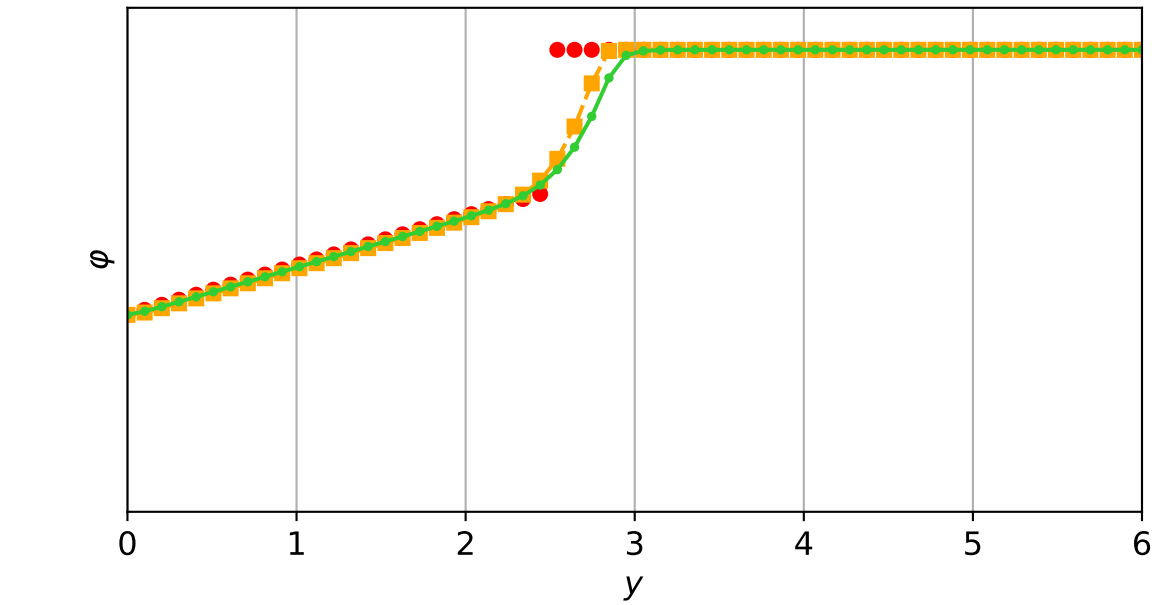
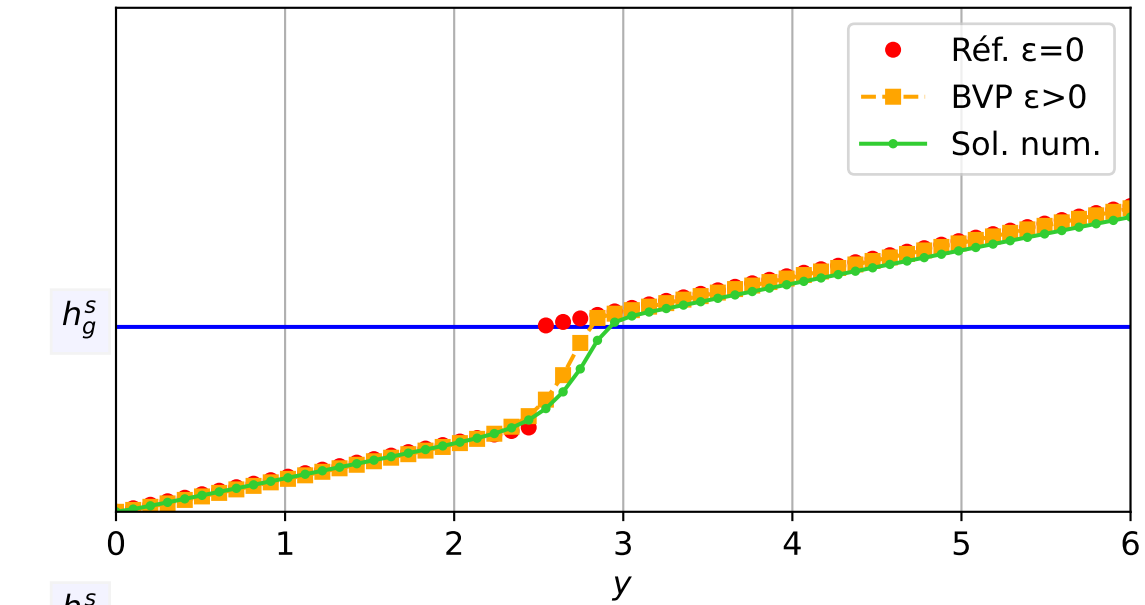


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\epsilon = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 1.10004/12$  (3420 iter) — résidu eq. =  $9.03e-06$

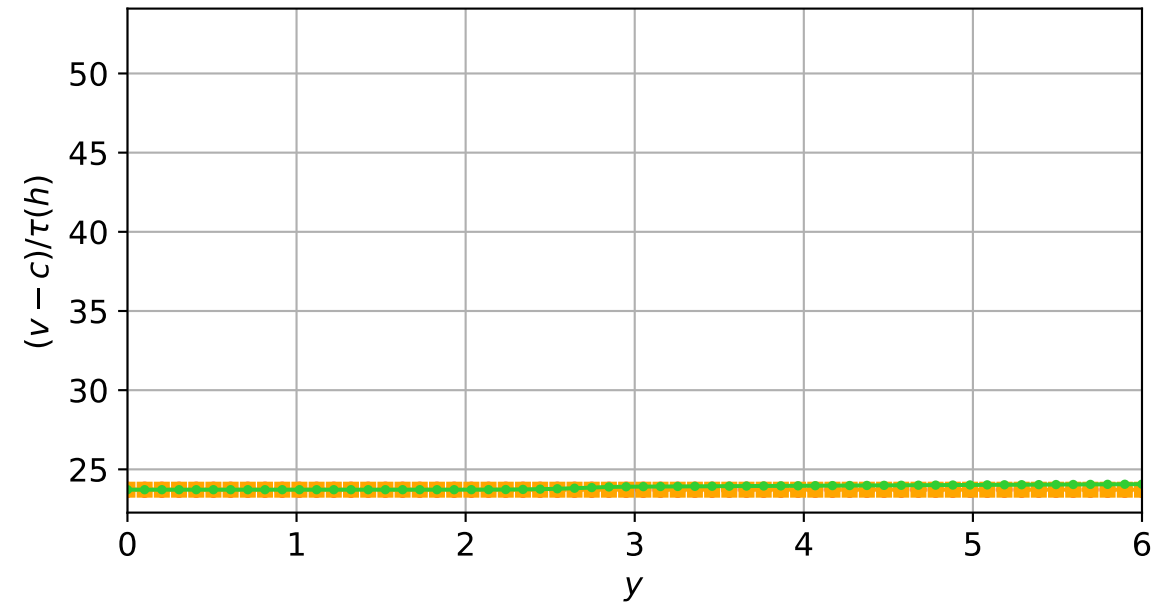
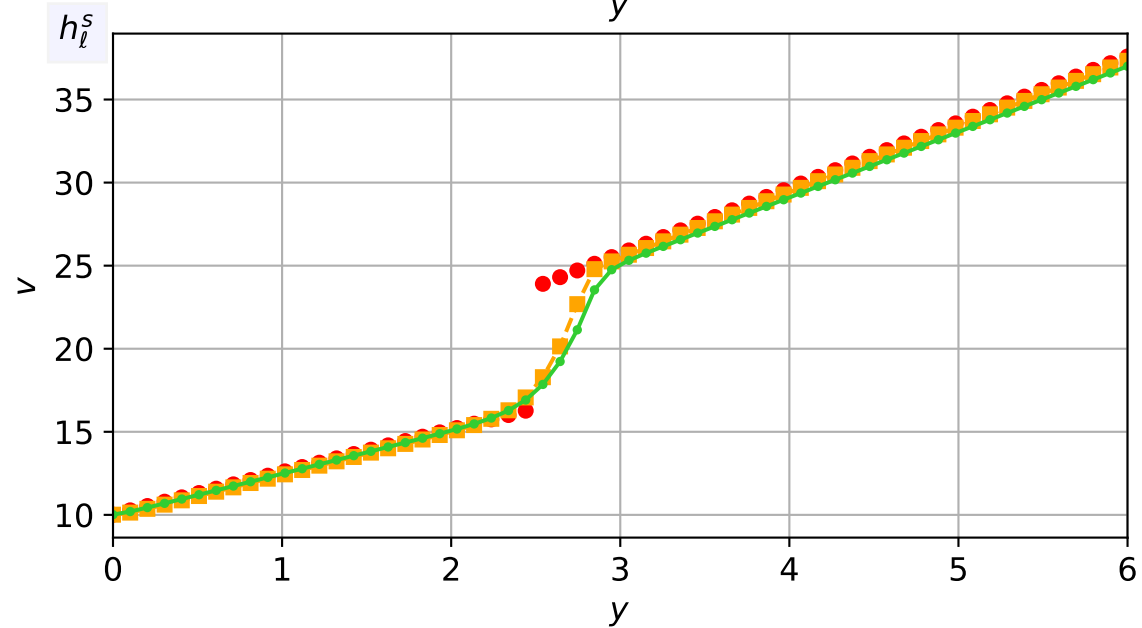
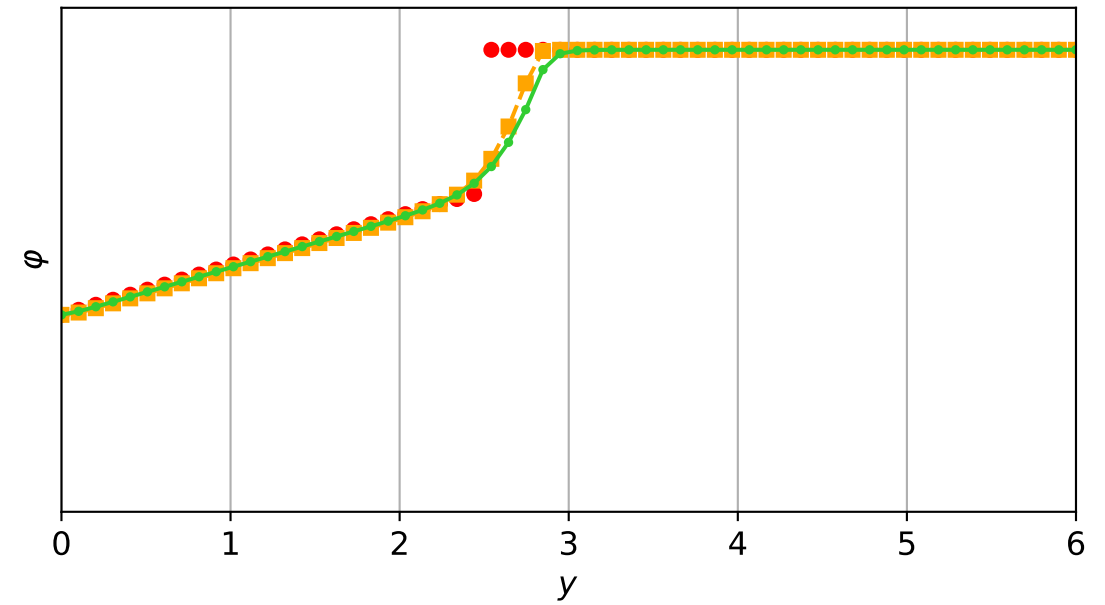
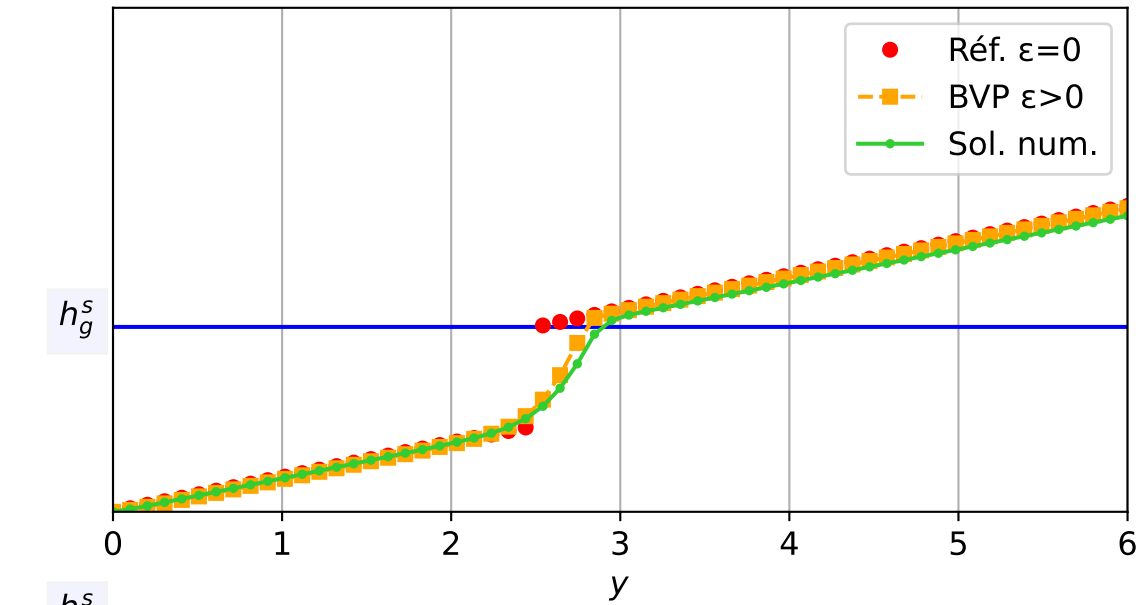




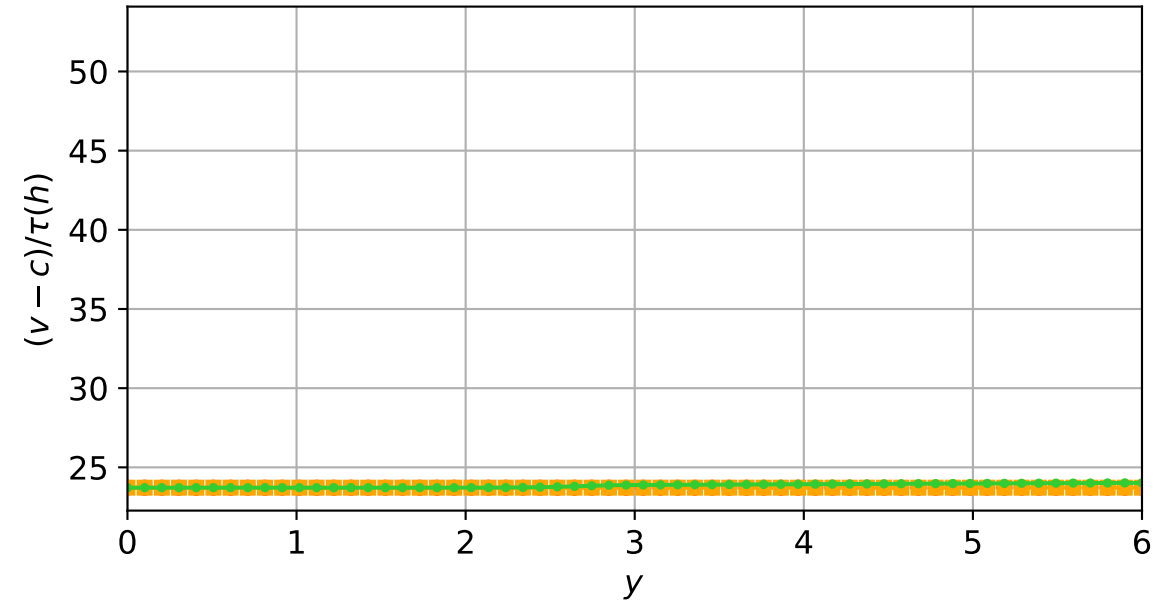
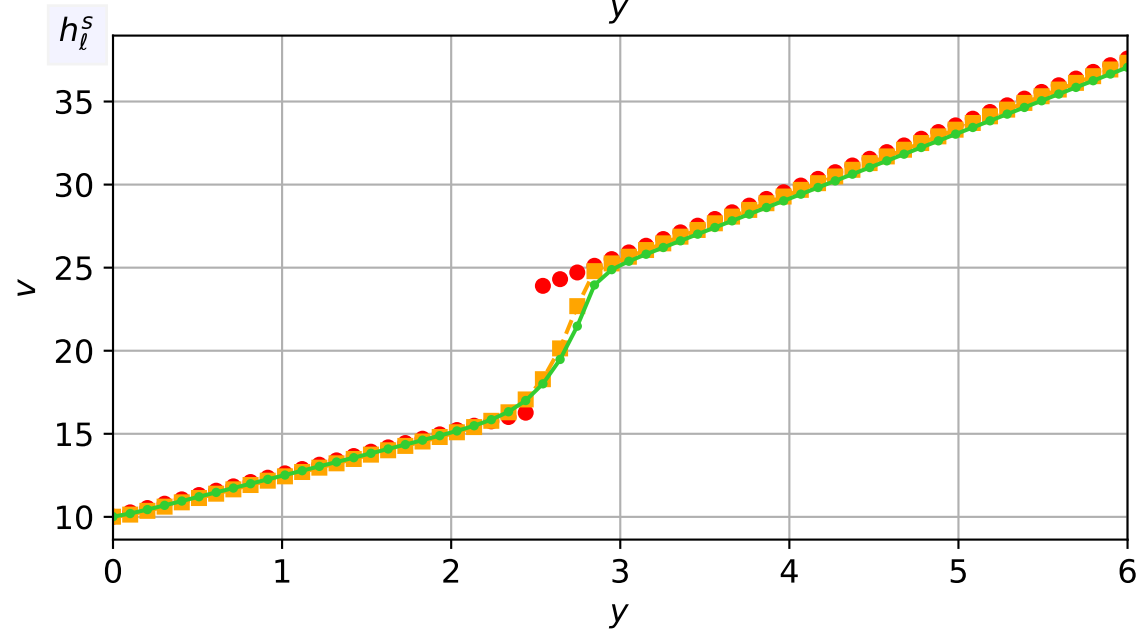
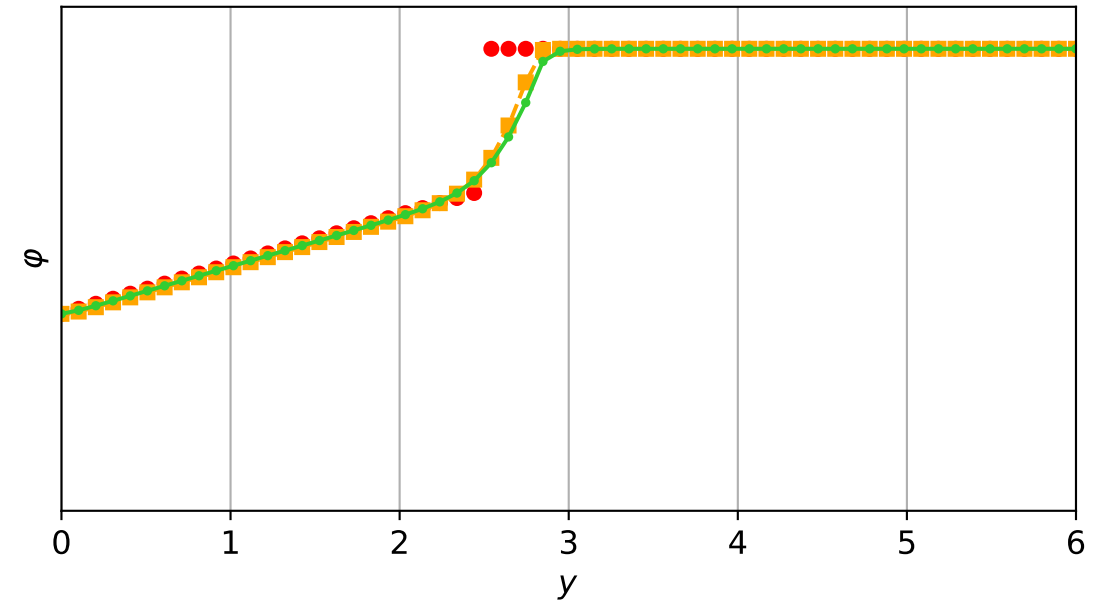
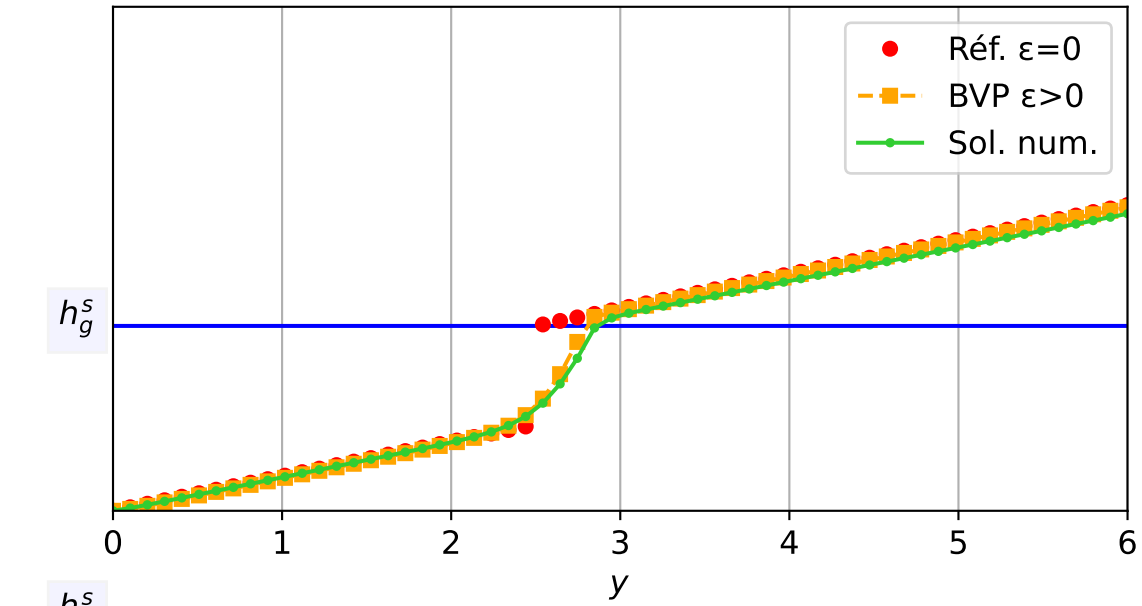
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 1.15019/12$  (3642 iter) — résidu eq. =  $7.53e-06$



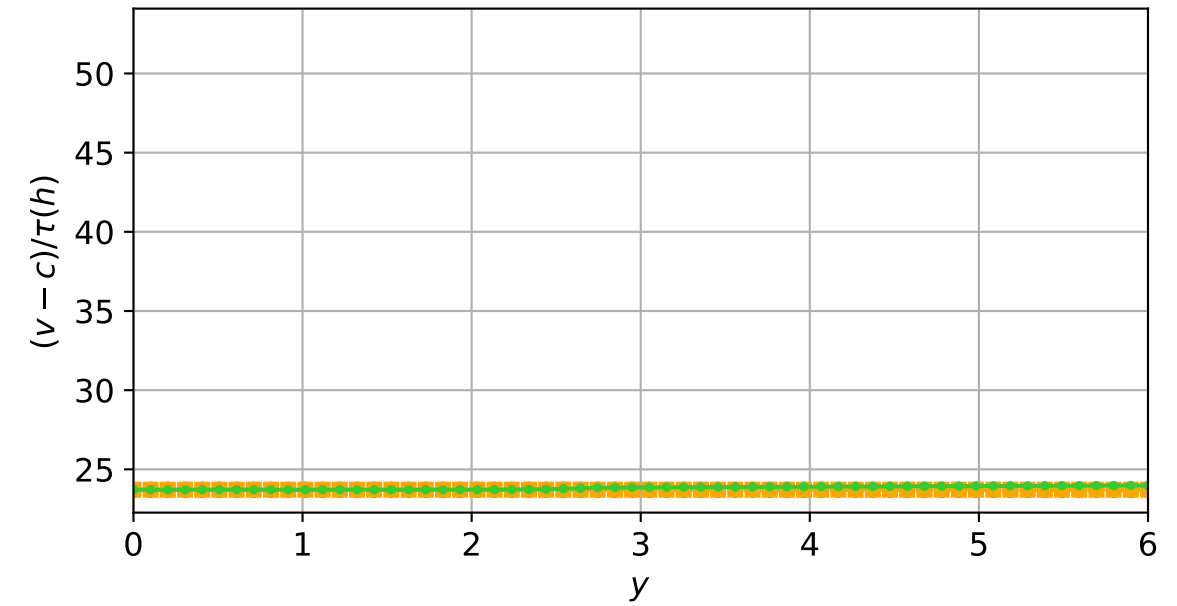
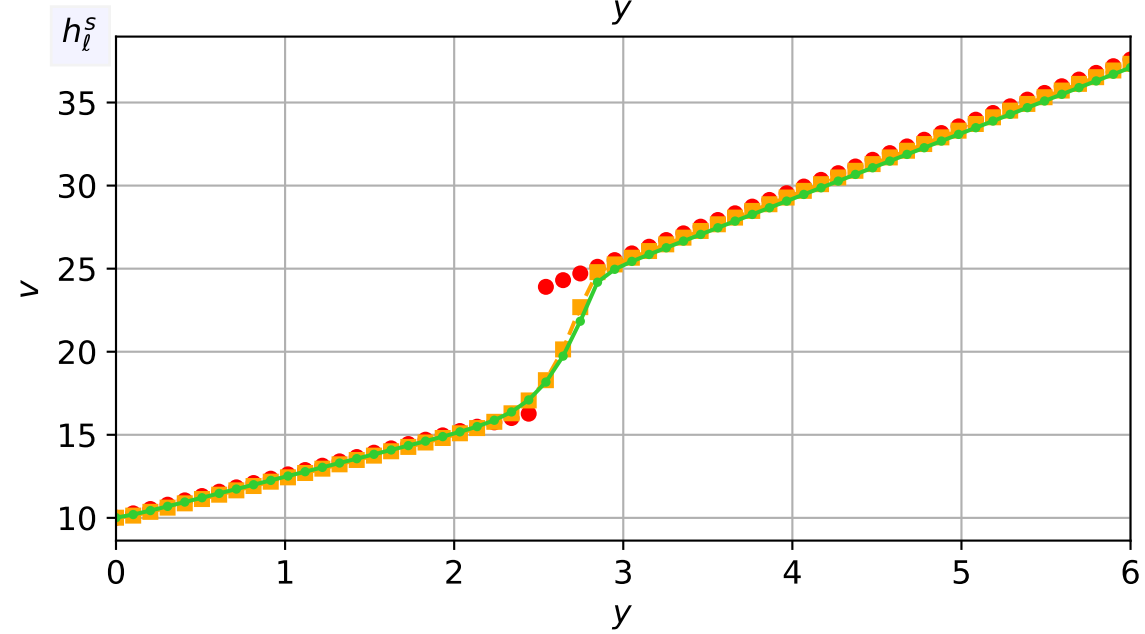
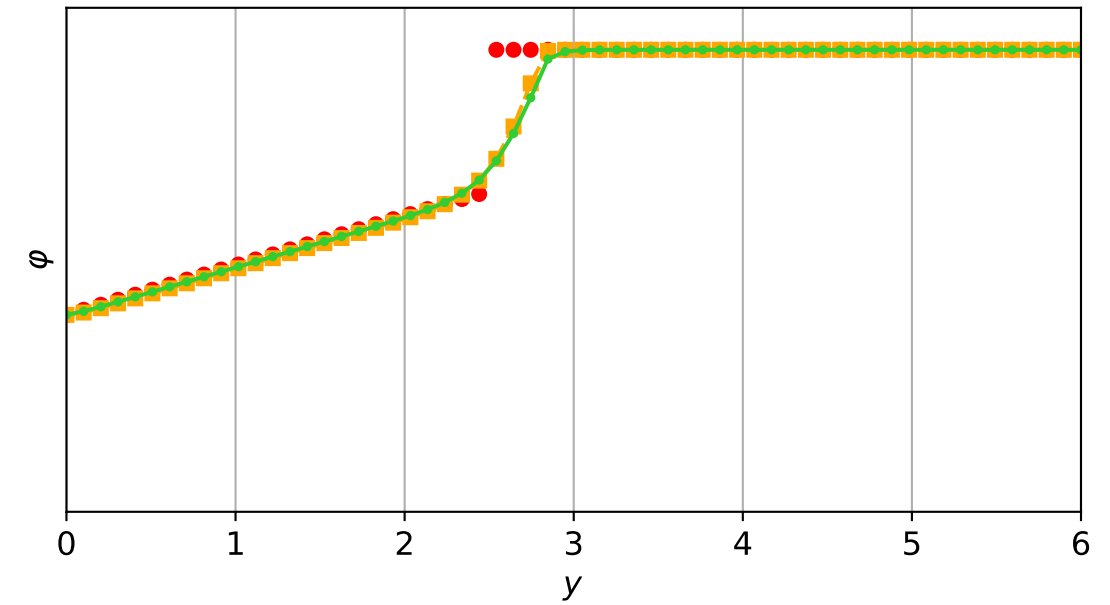
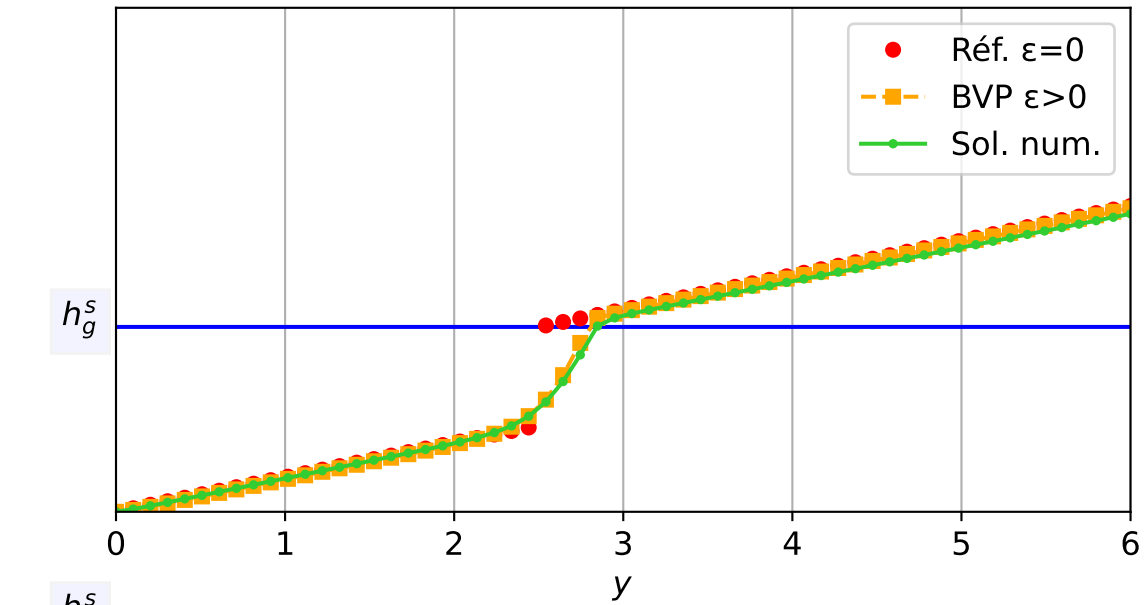
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 1.20011/12$  (3863 iter) — résidu eq. =  $6.44e-06$



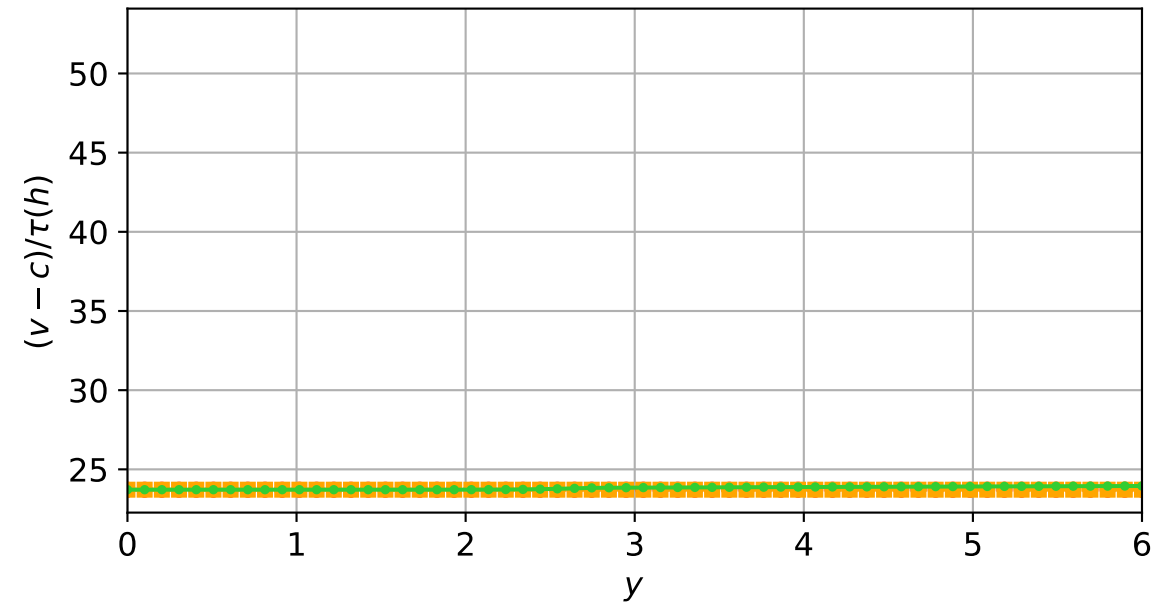
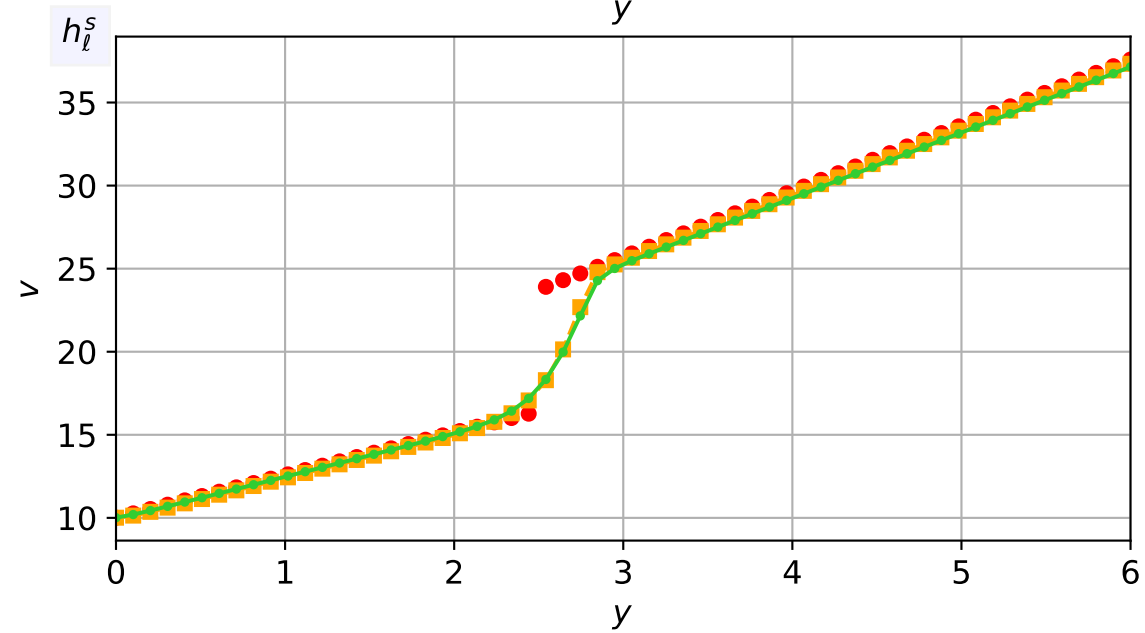
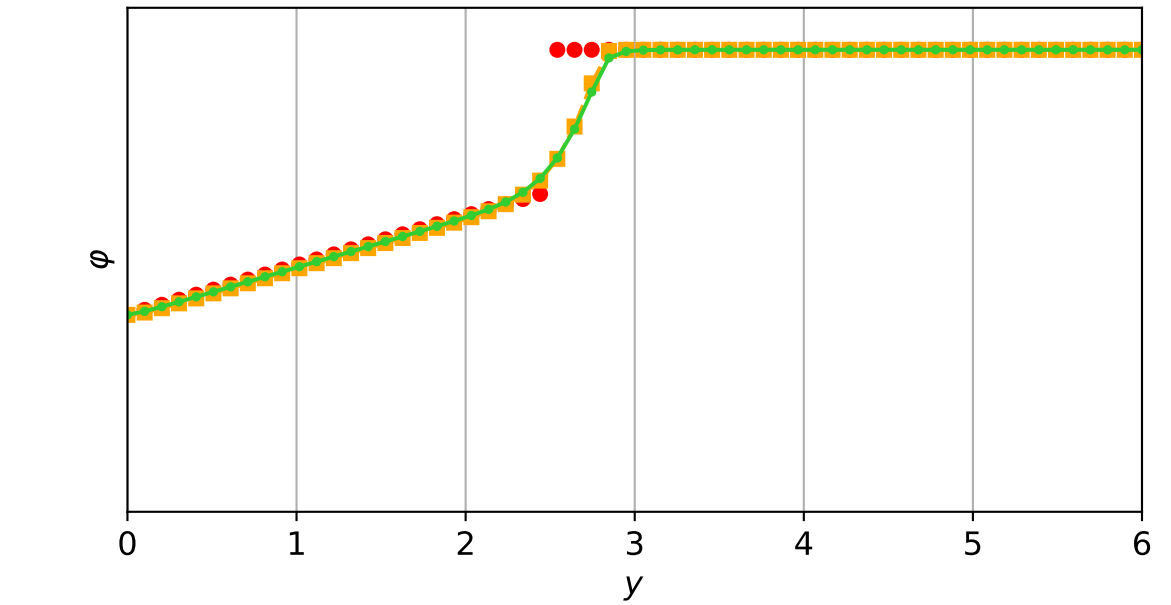
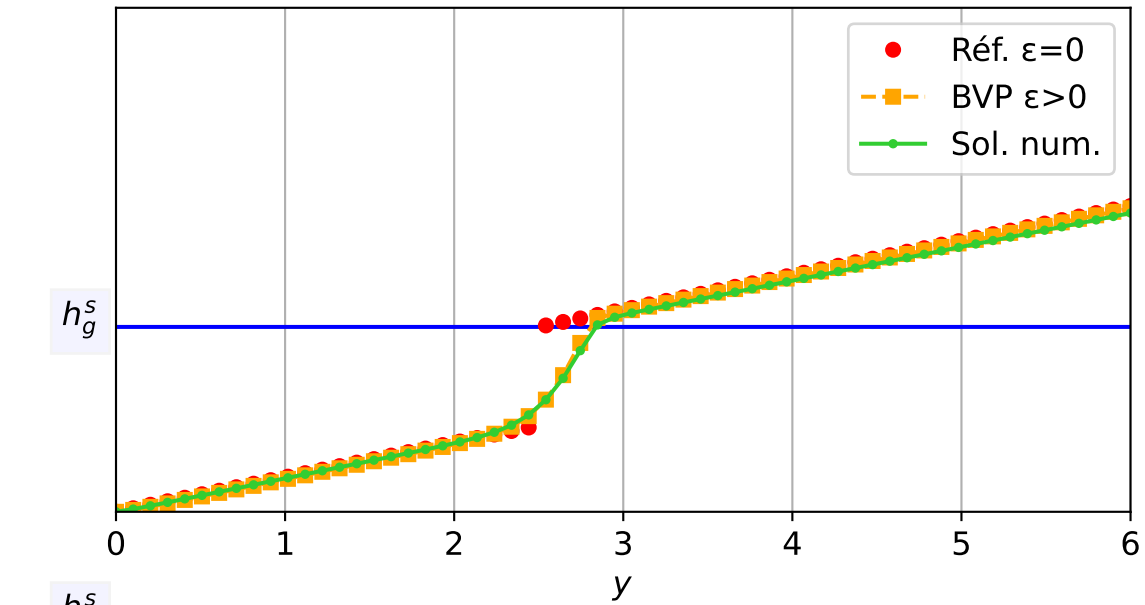
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 1.25004/12$  (4084 iter) — résidu eq. =  $5.58e-06$



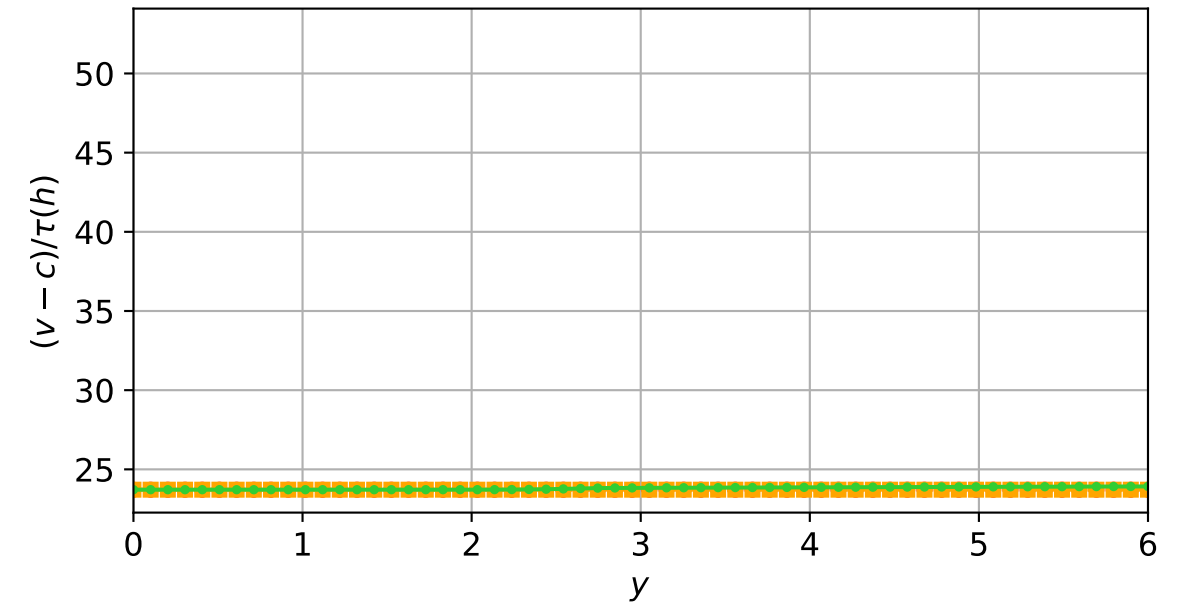
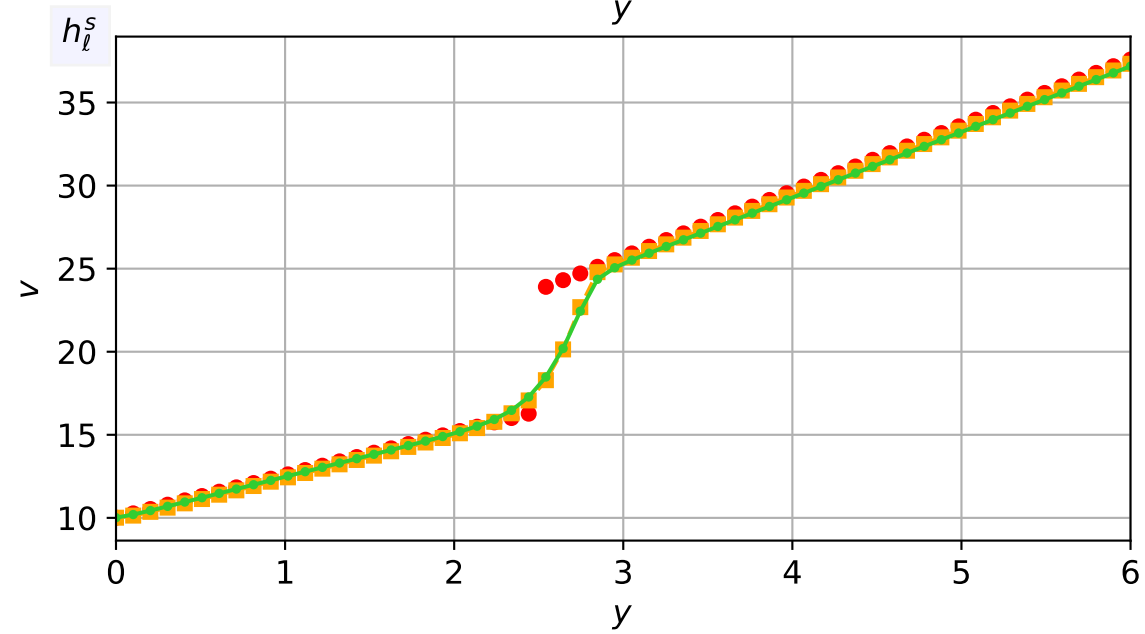
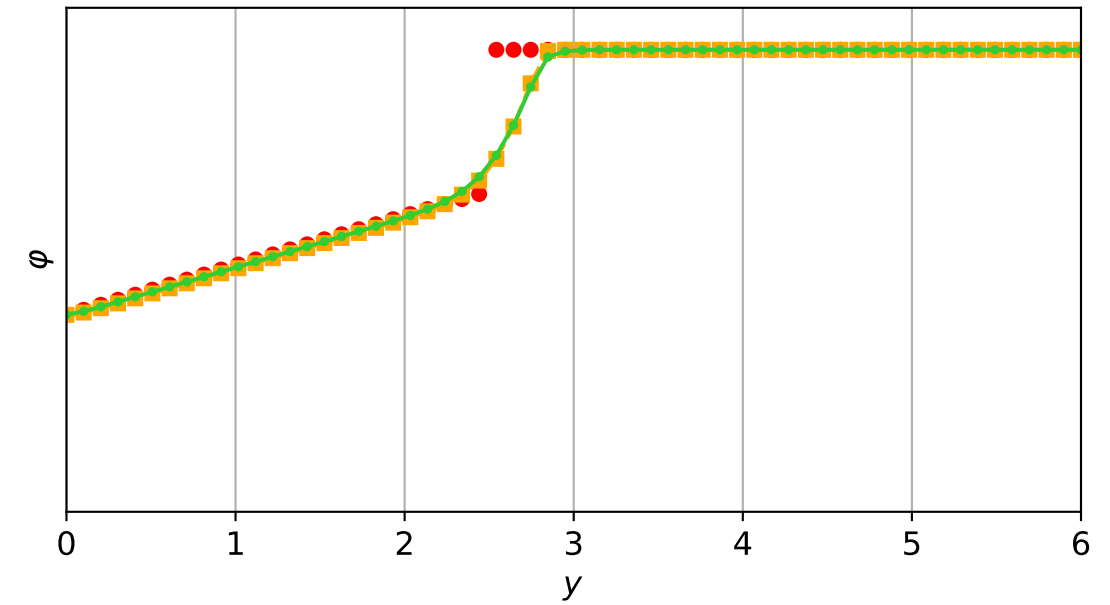
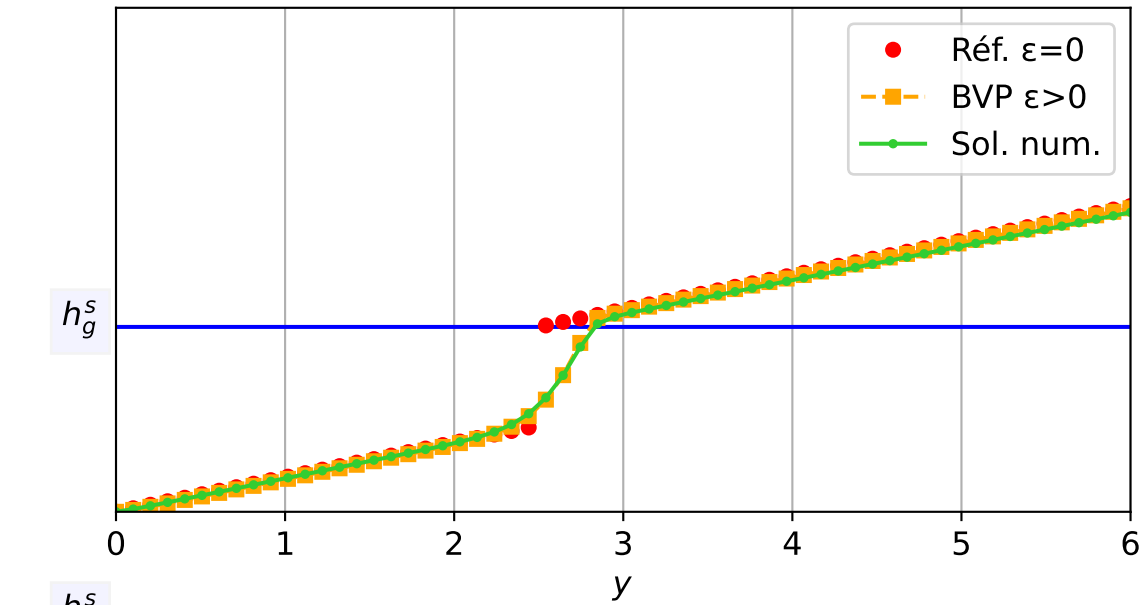
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 1.30019/12$  (4306 iter) — résidu eq. =  $9.73e-06$



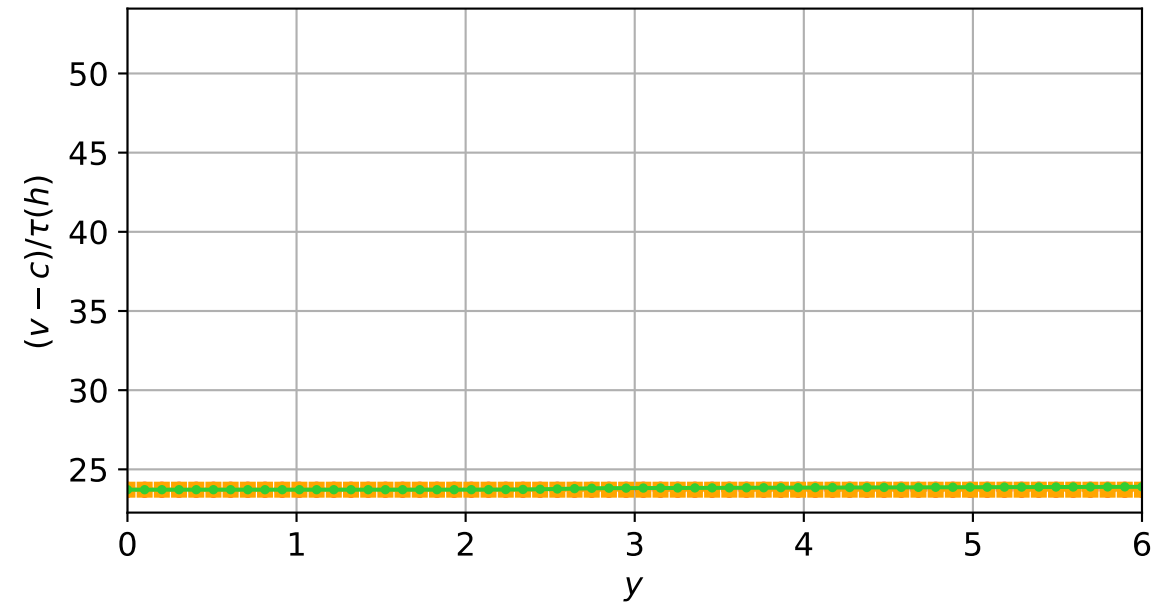
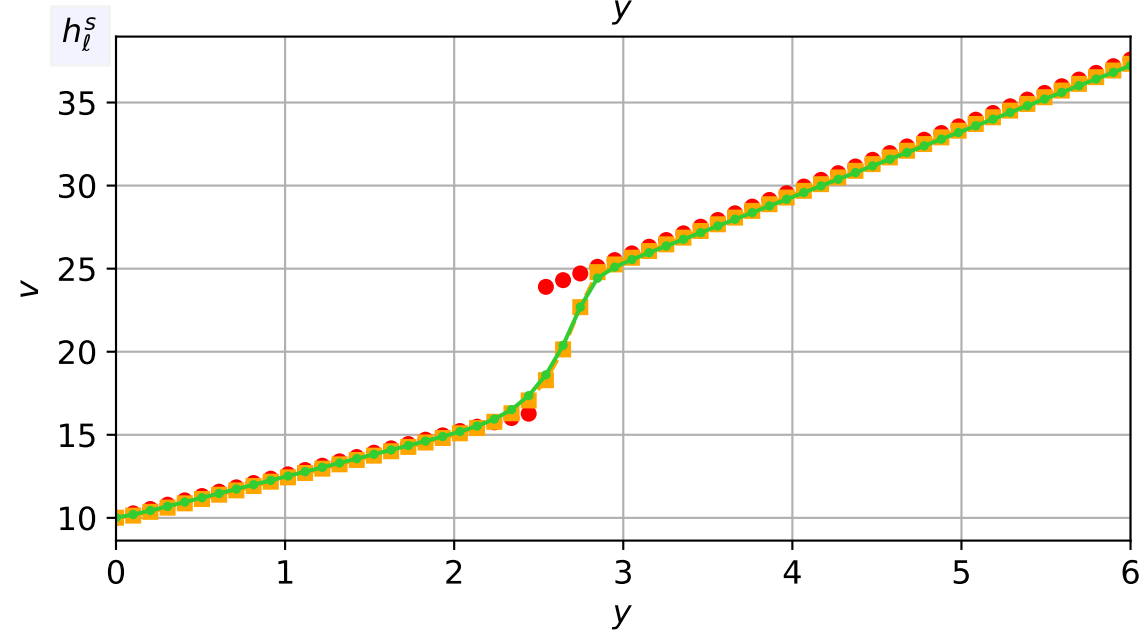
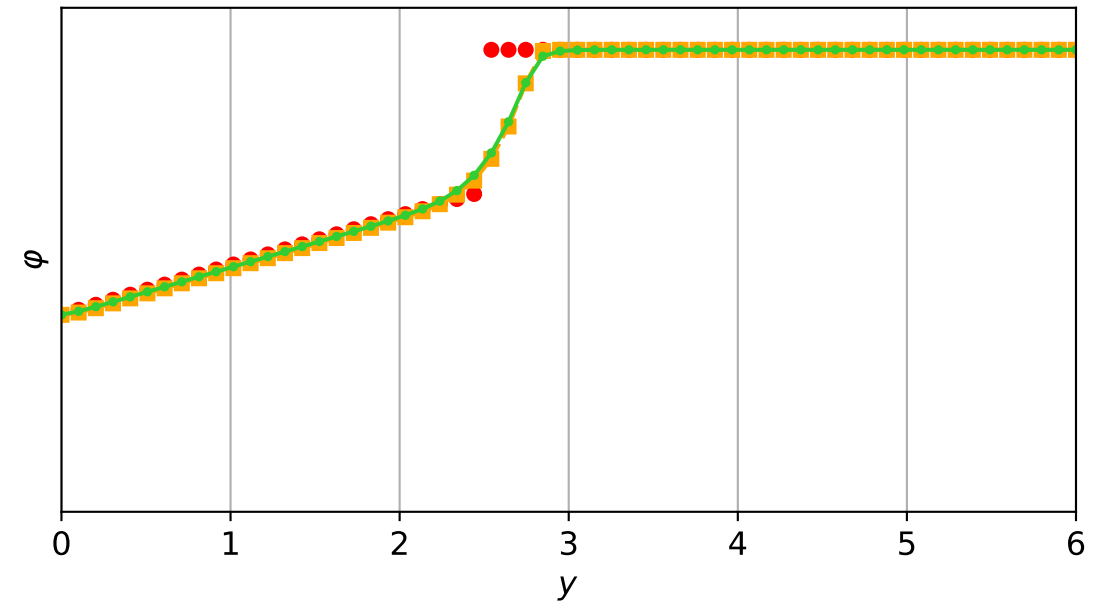
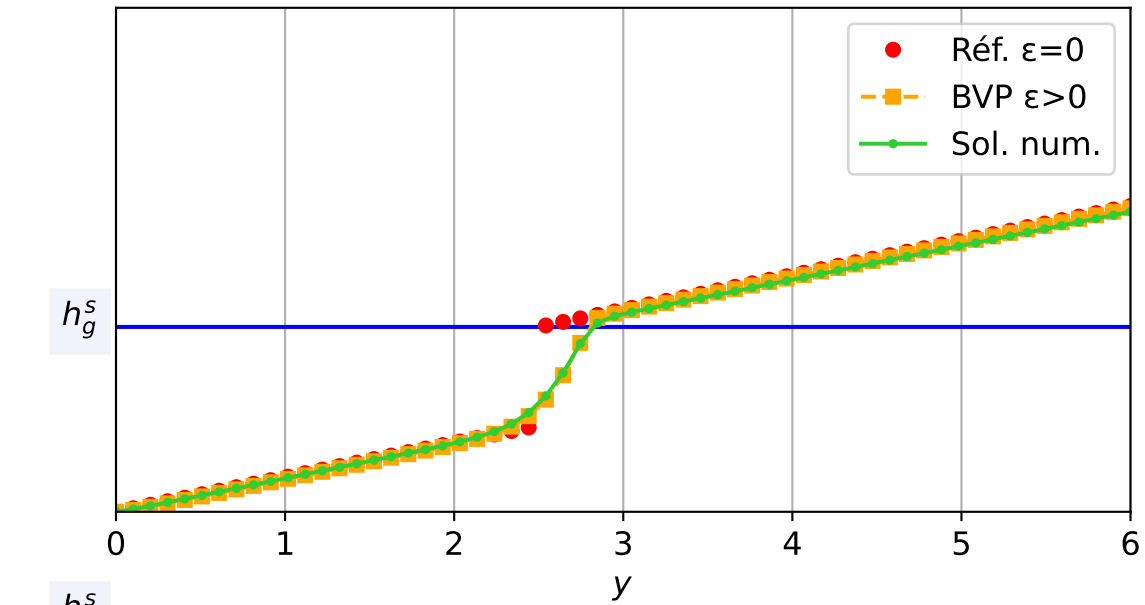
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 1.35011/12$  (4527 iter) — résidu eq. =  $8.74e-06$



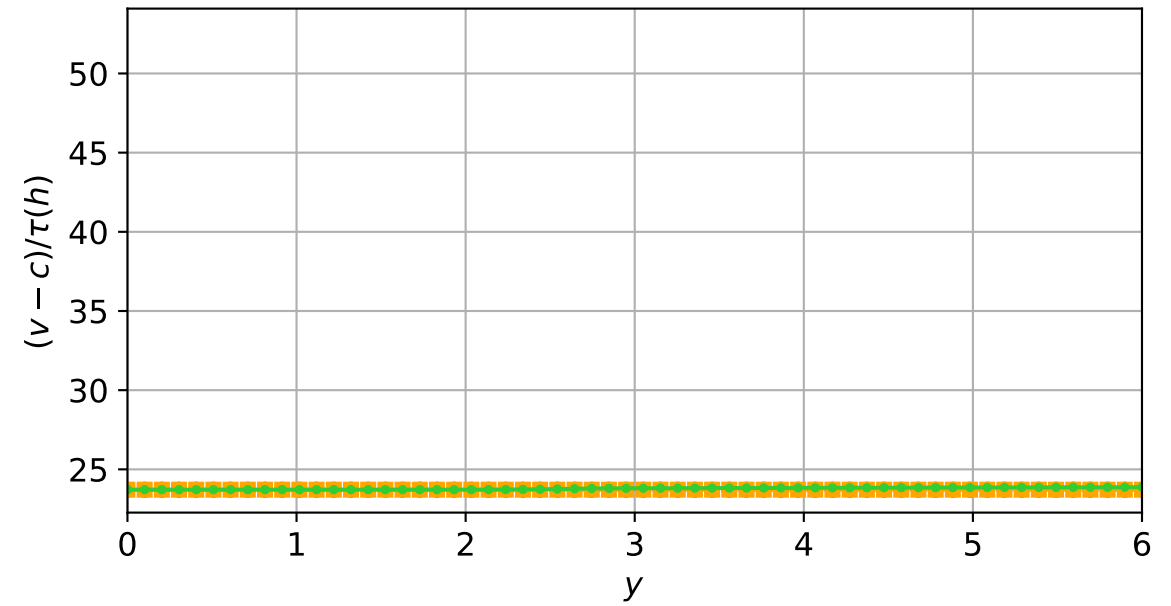
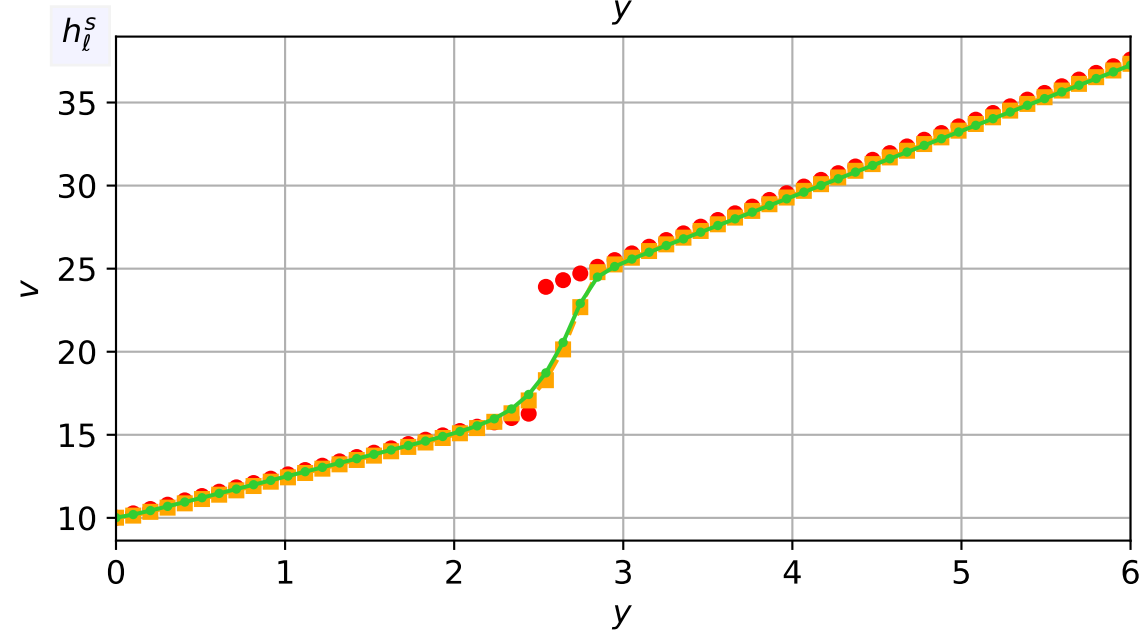
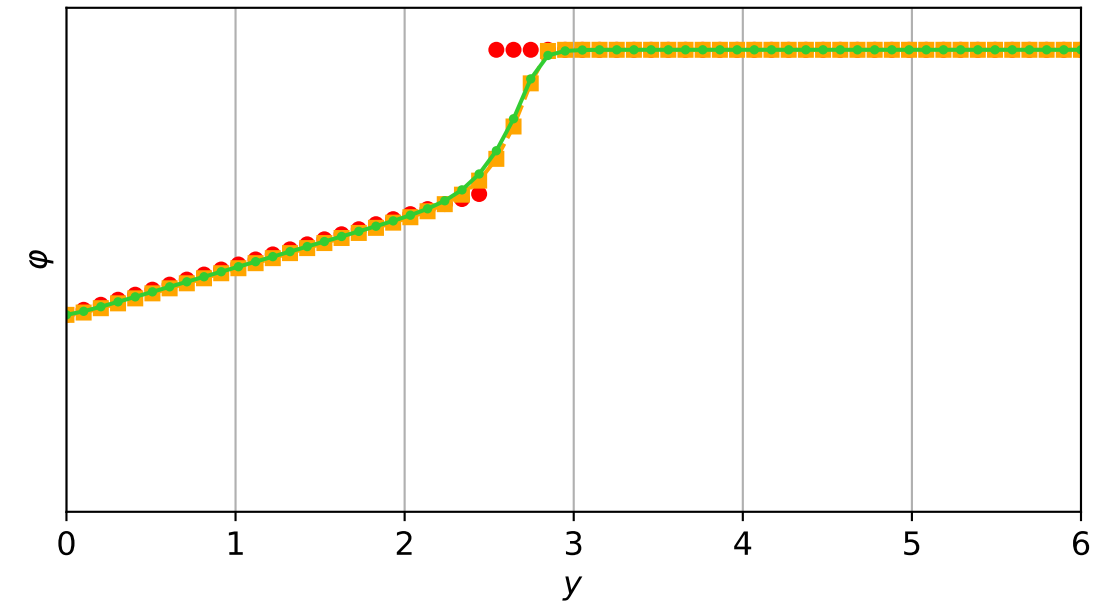
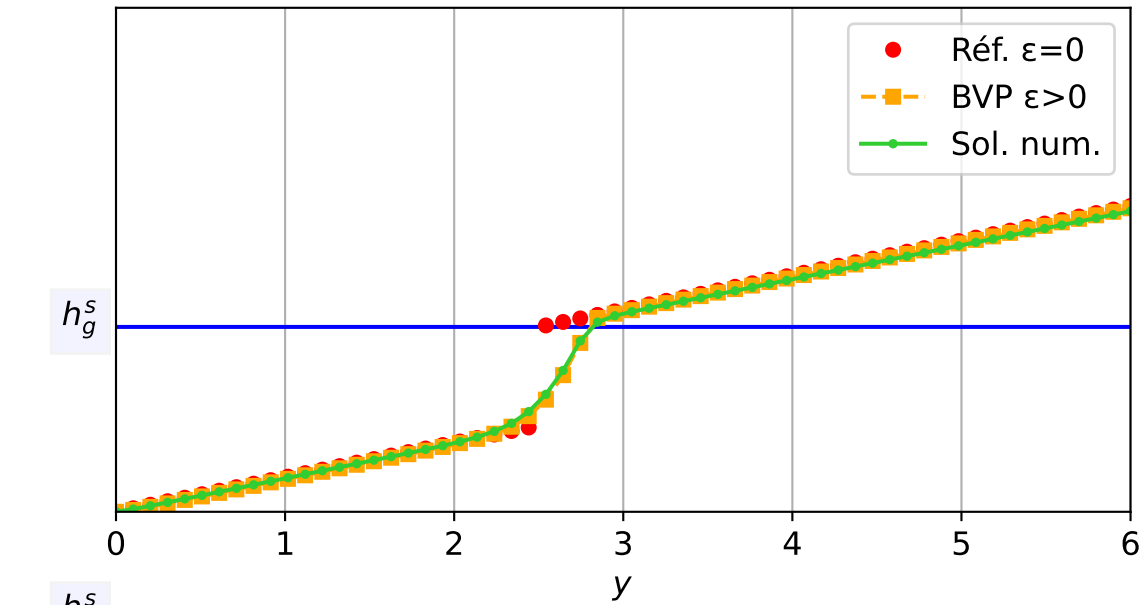
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 1.40003/12$  (4748 iter) — résidu eq. =  $7.42e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 1.45018/12$  (4970 iter) — résidu eq. =  $6.41e-06$

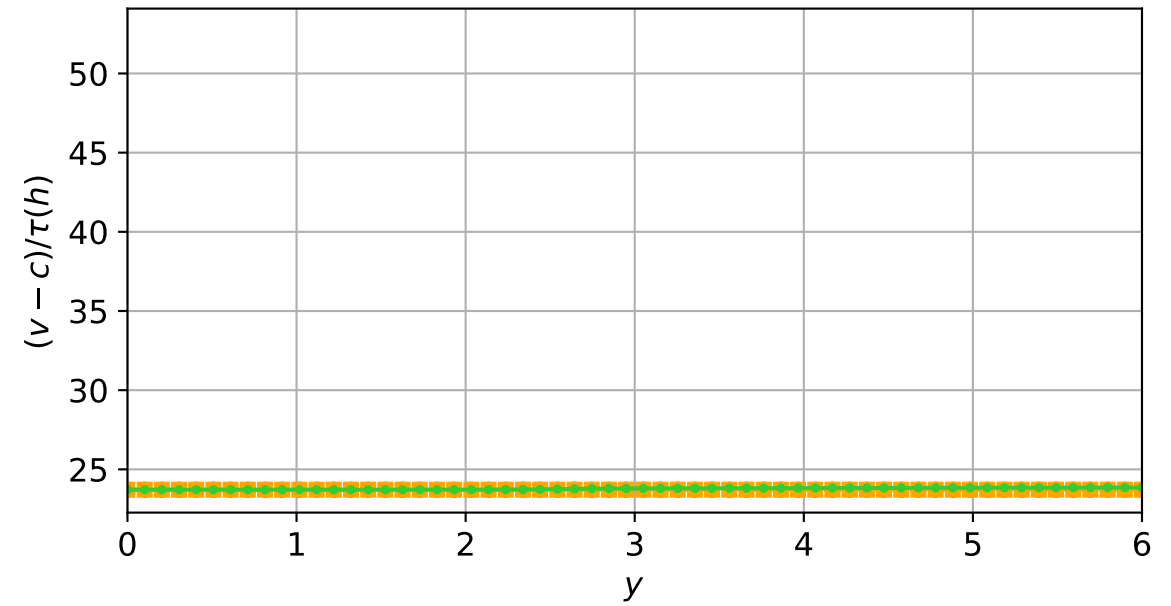
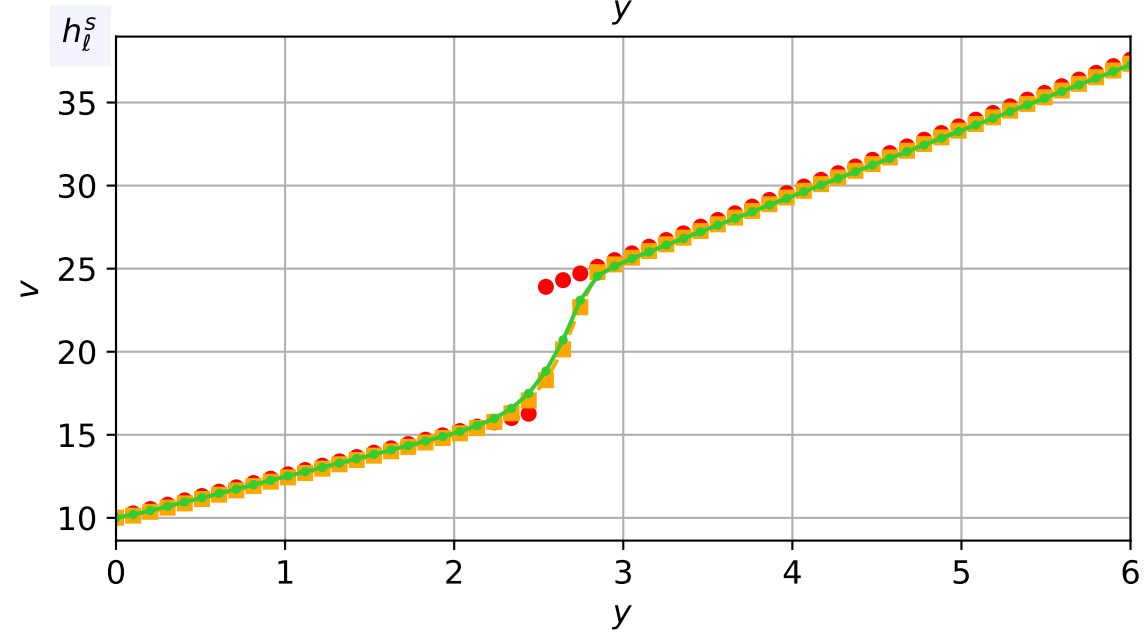
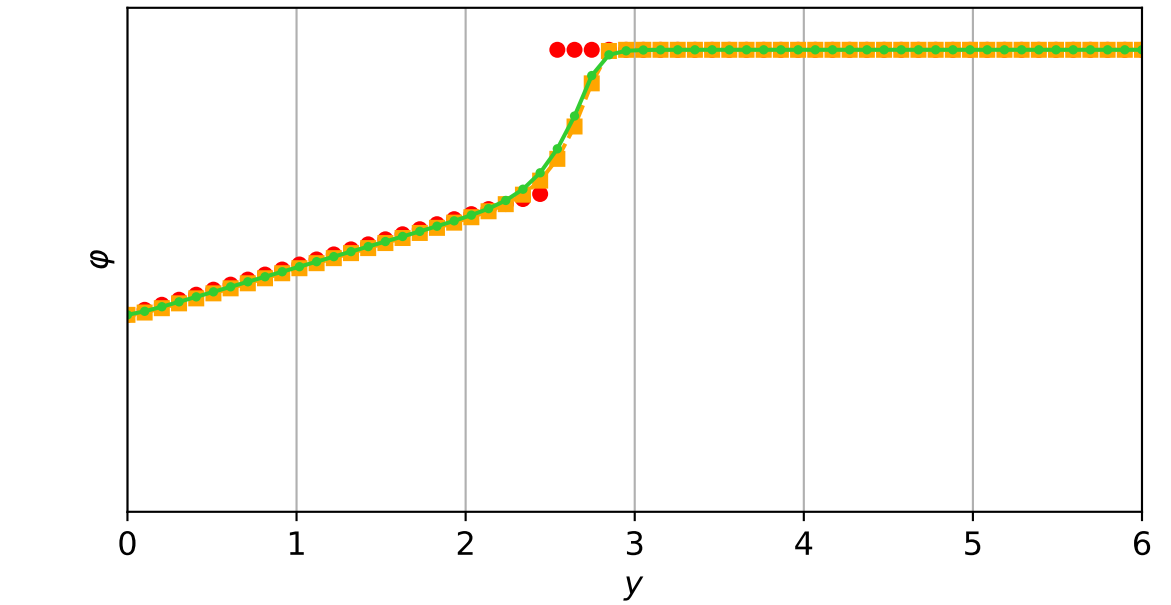
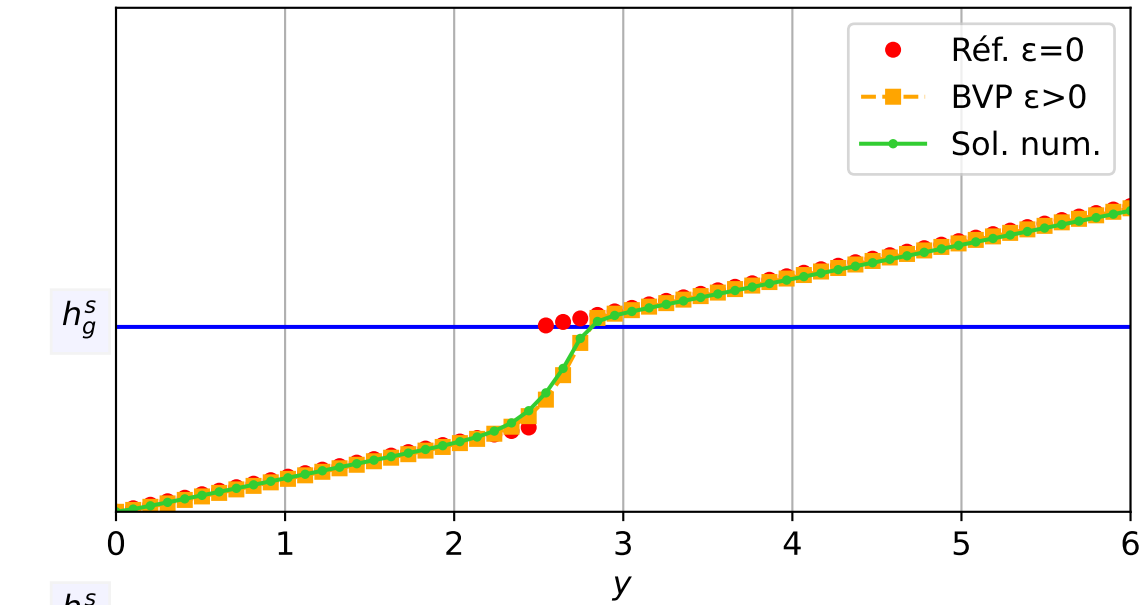


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.26e-04$  —  $t = 1.50011/12$  (5191 iter) — résidu eq. =  $5.60e-06$

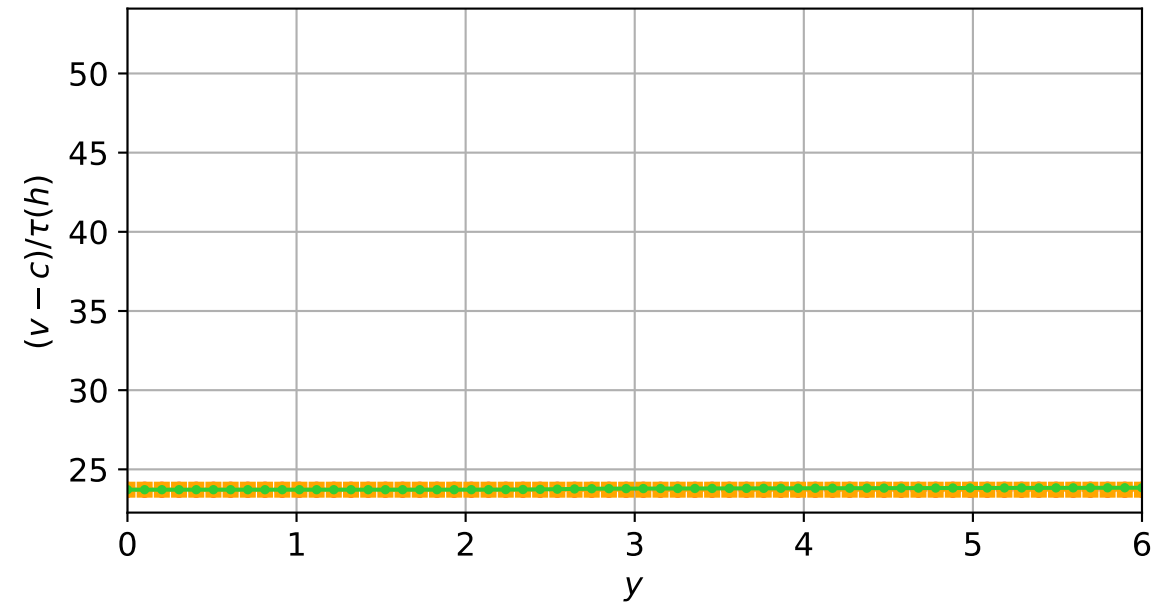
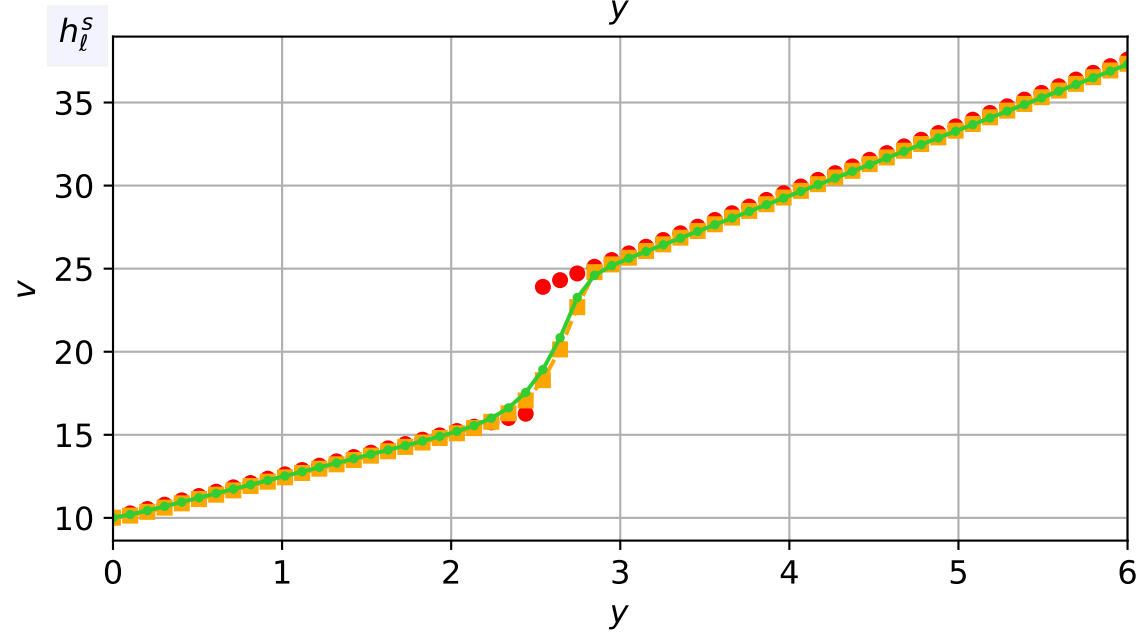
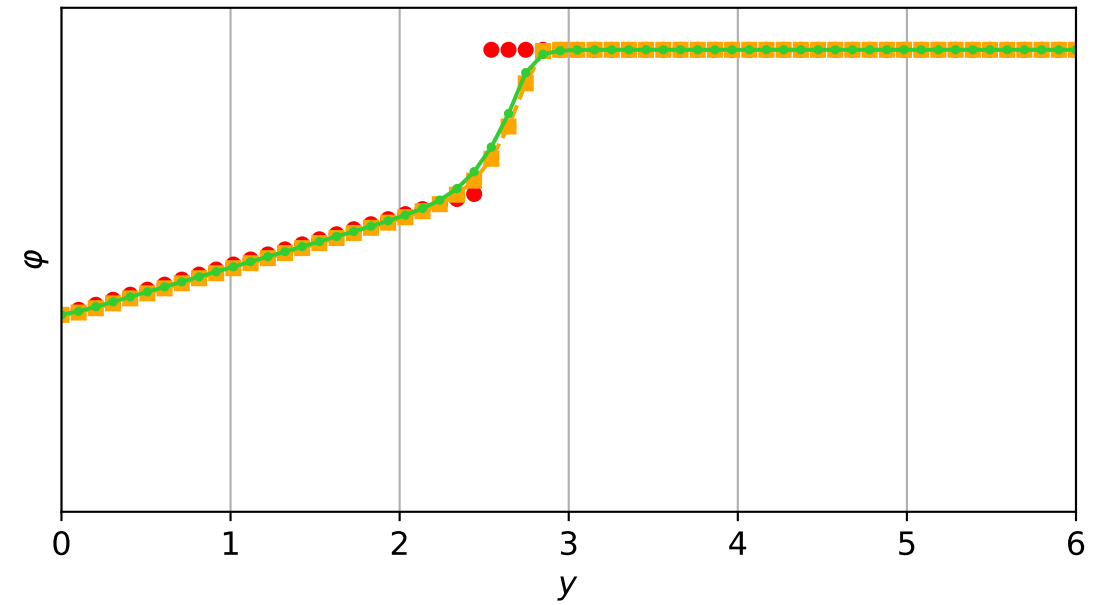
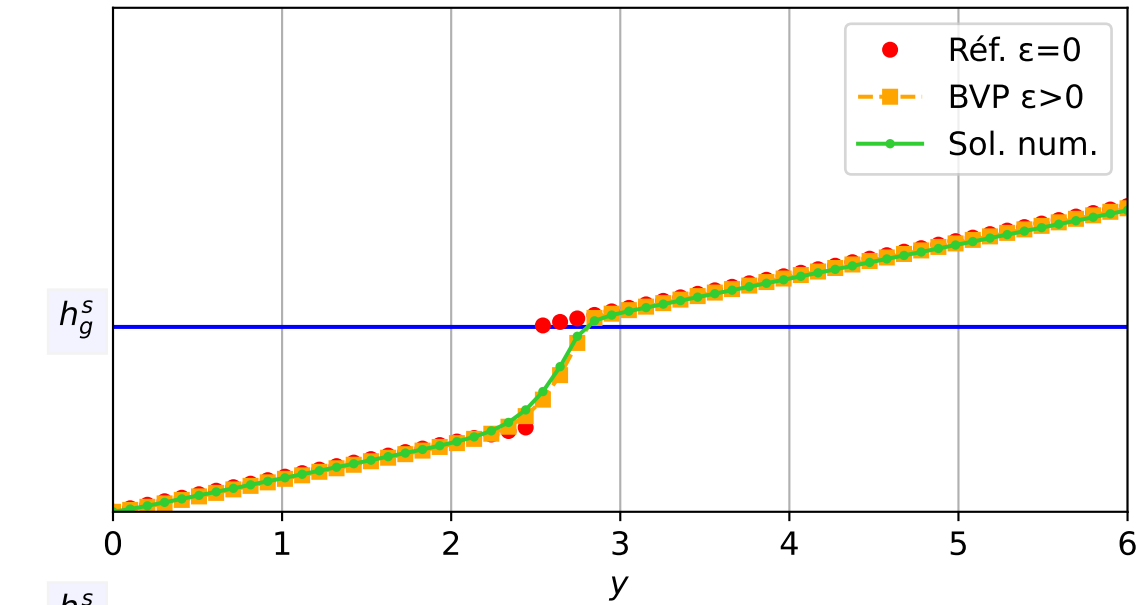




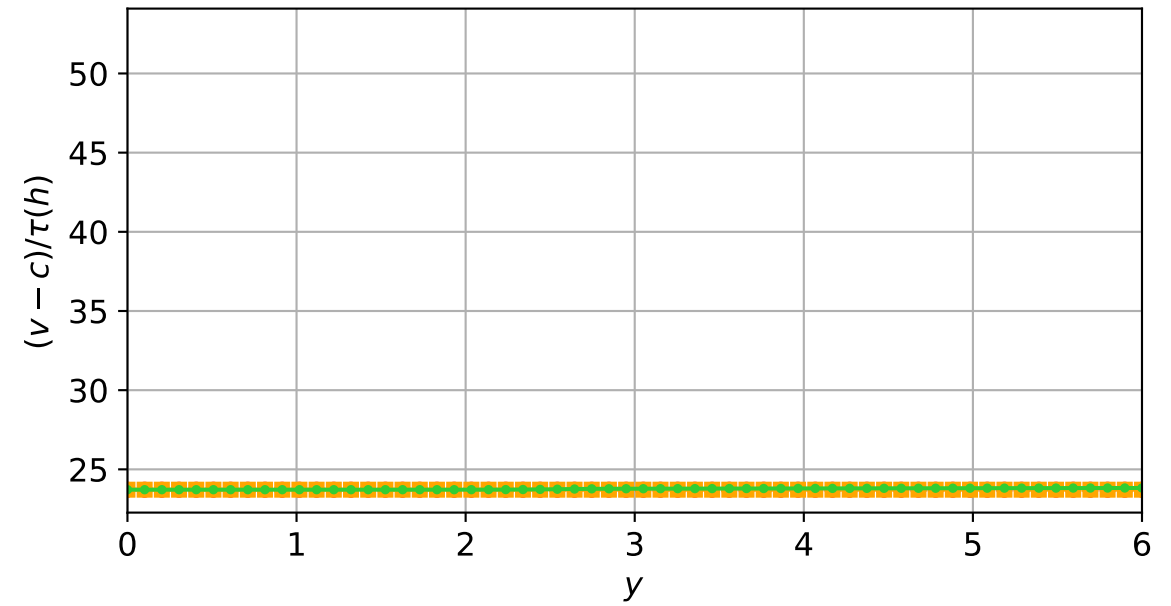
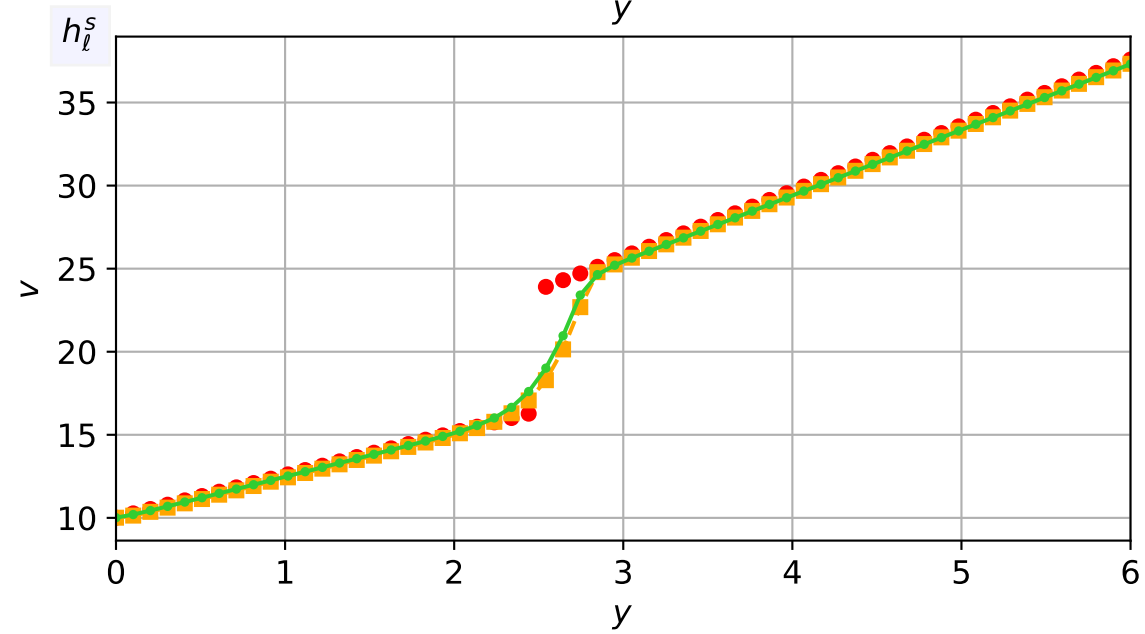
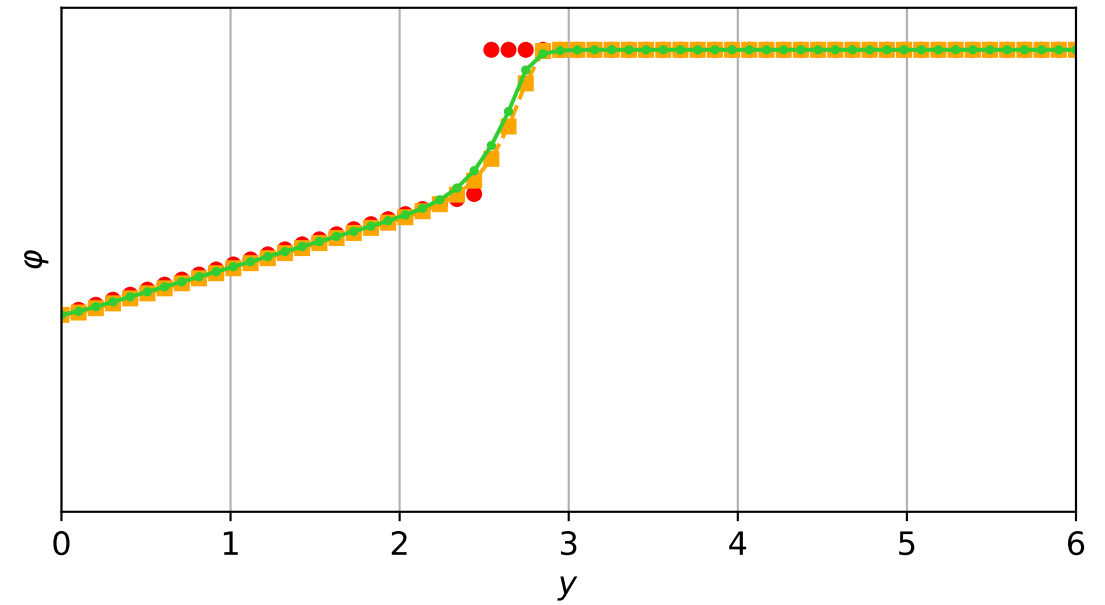
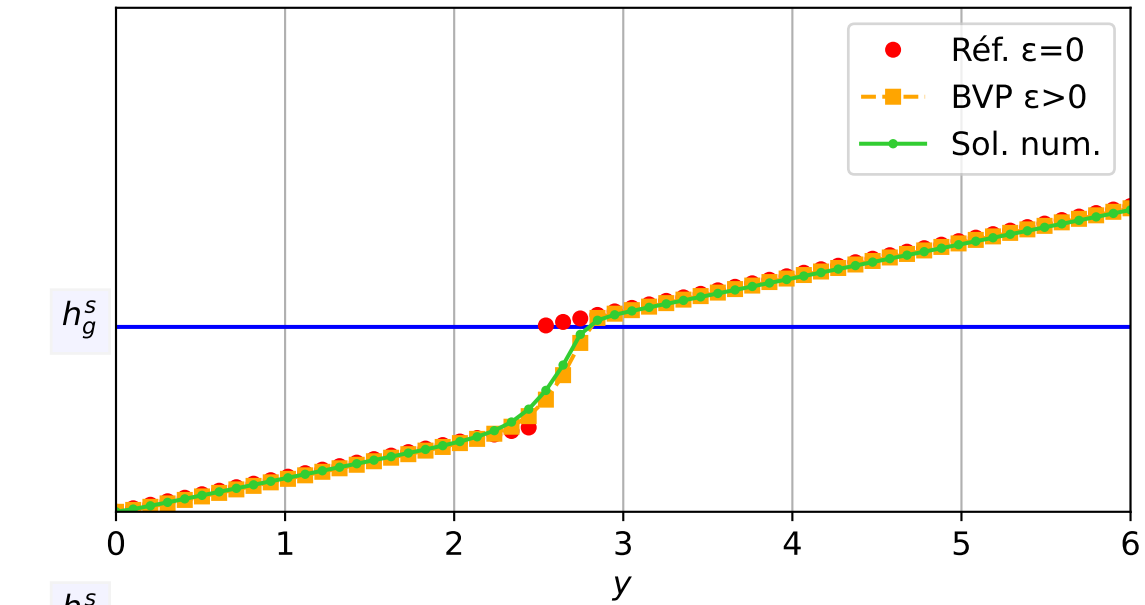
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.71e-04$  —  $t = 1.55012/12$  (5398 iter) — résidu eq. =  $7.70e-06$



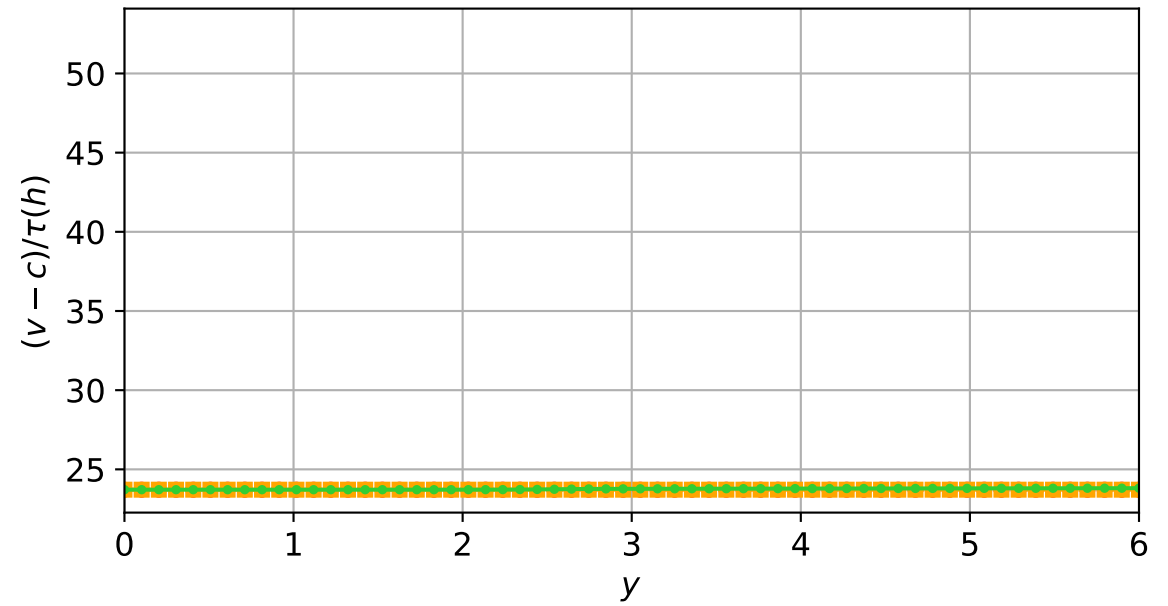
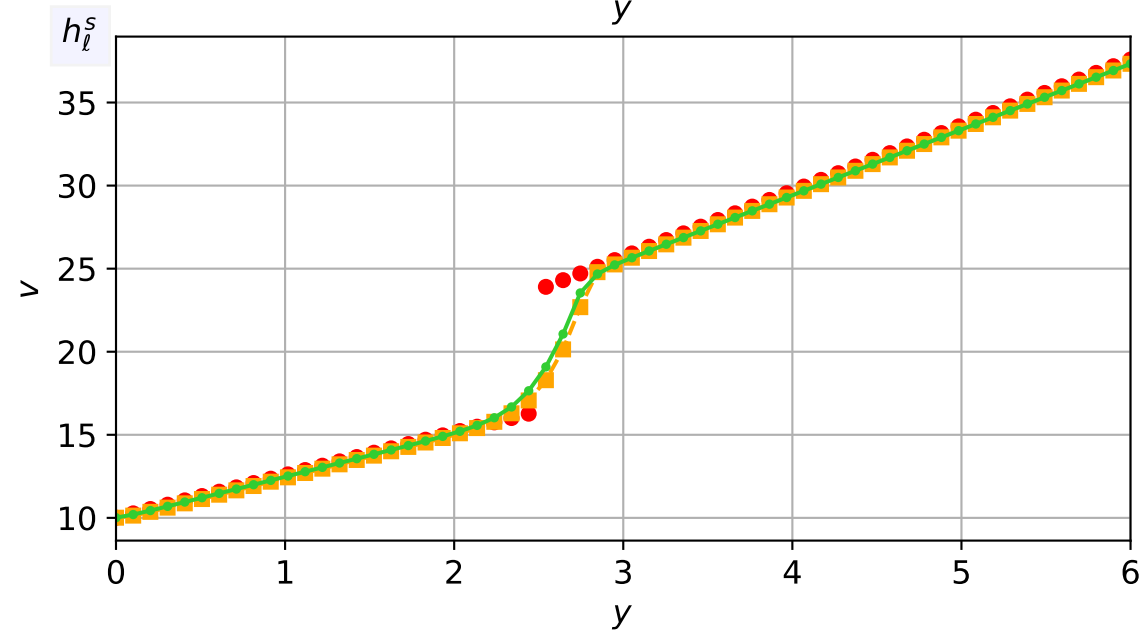
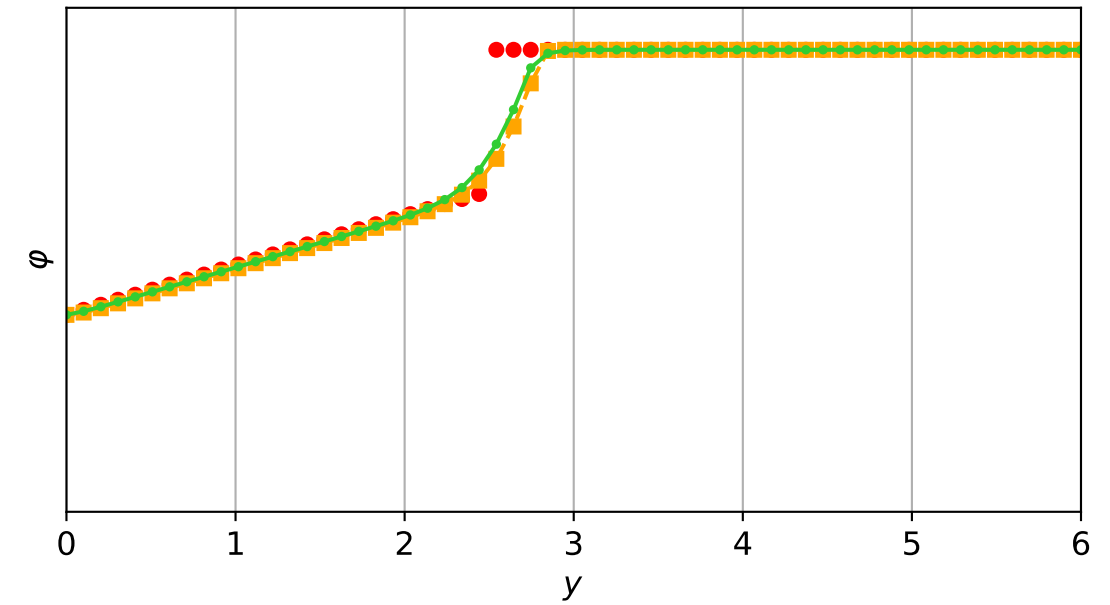
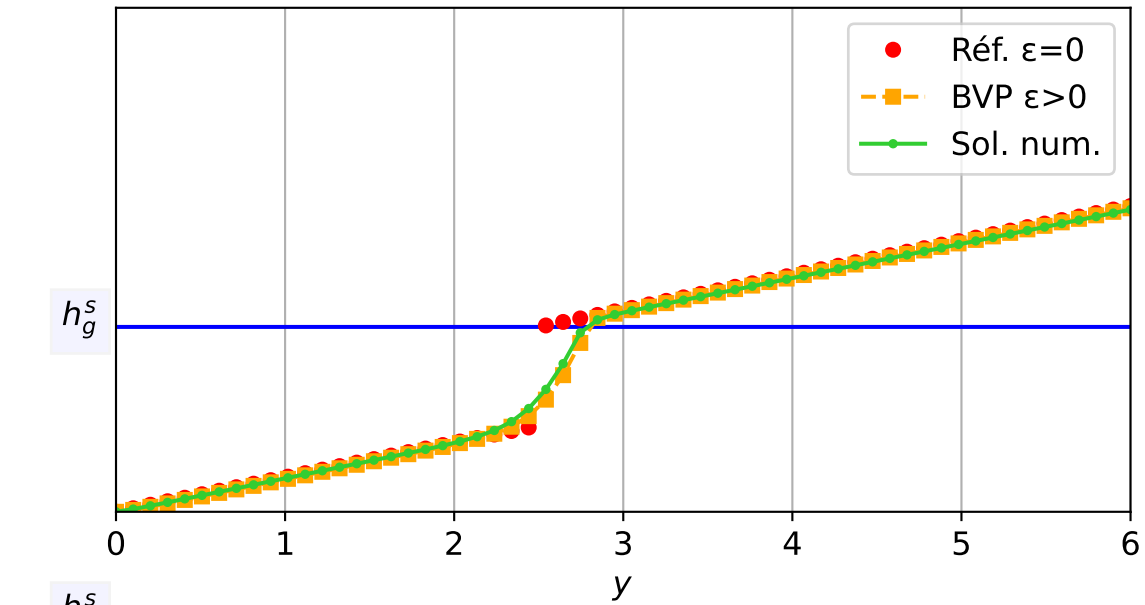
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.71e-04$  —  $t = 1.6/12$  (5582 iter) — résidu eq. =  $6.77e-06$



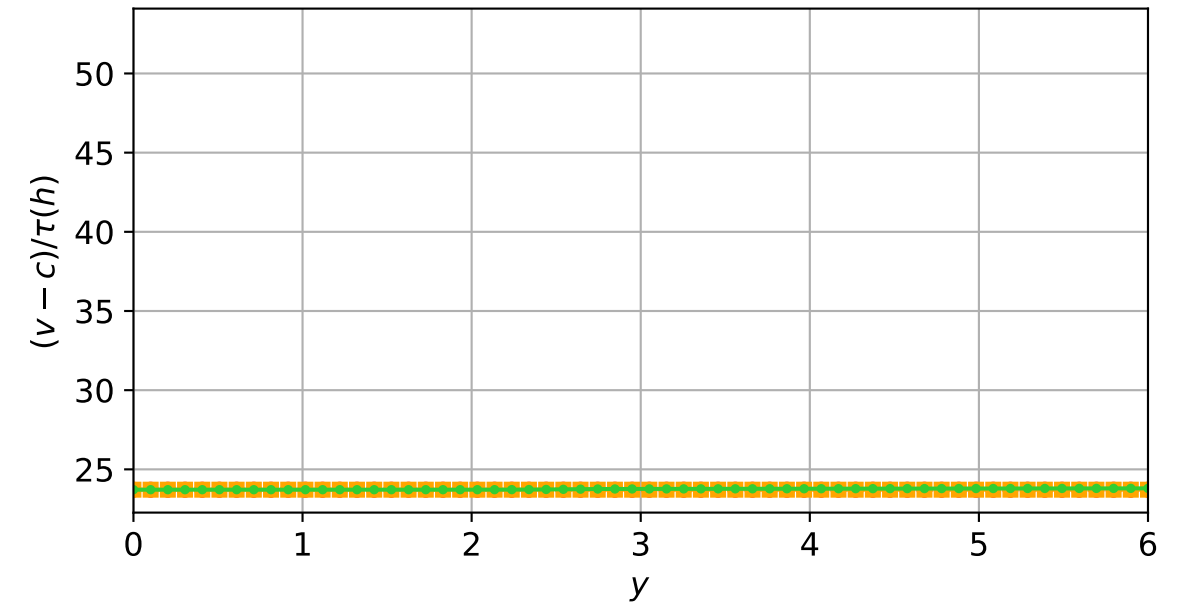
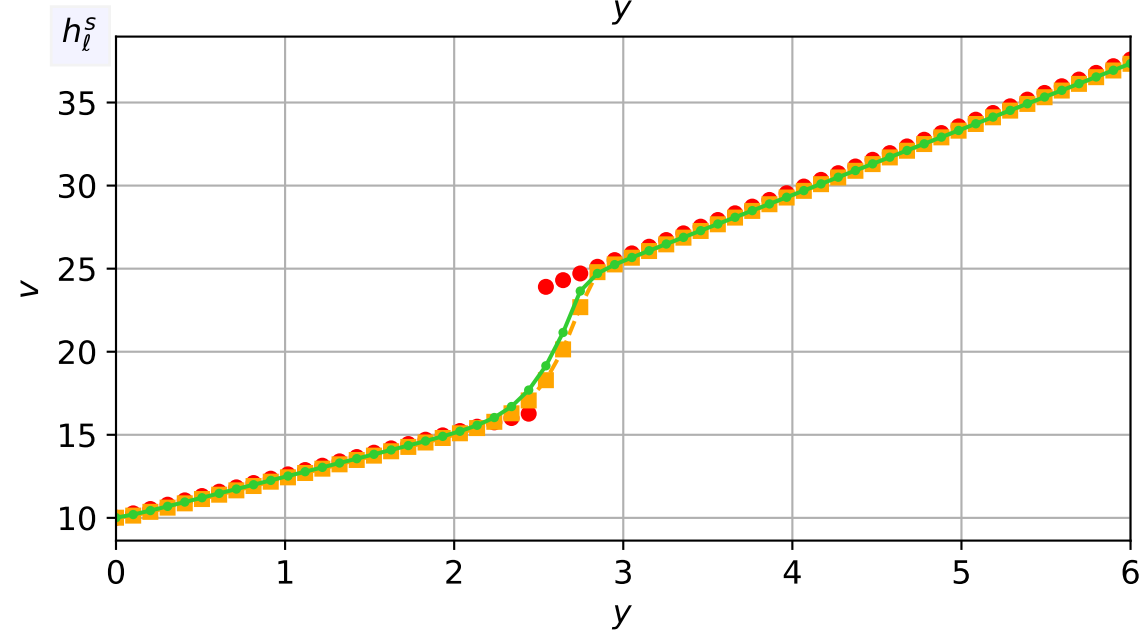
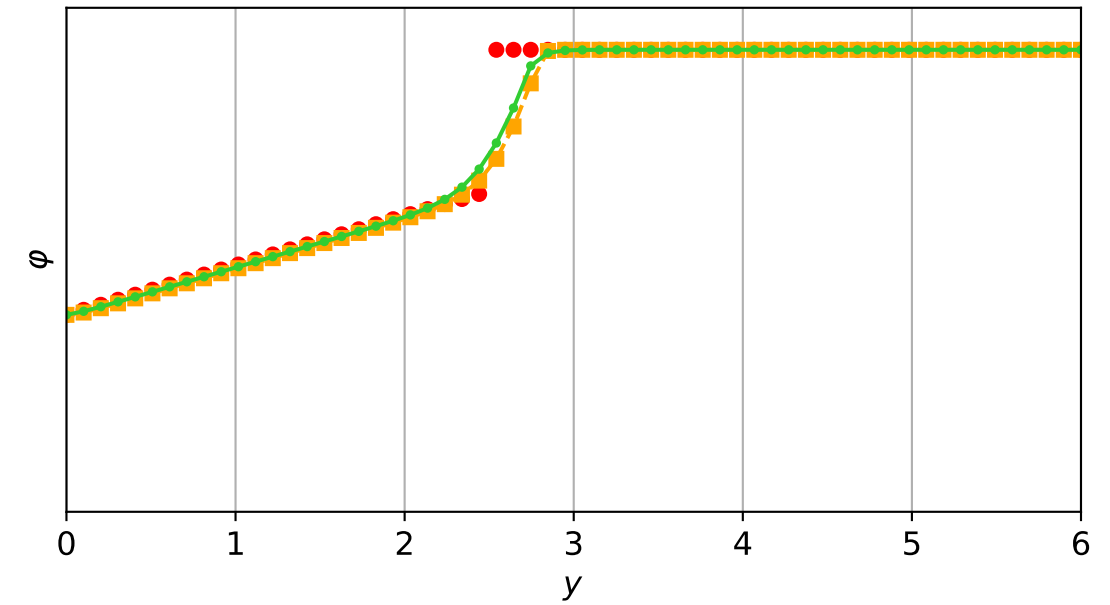
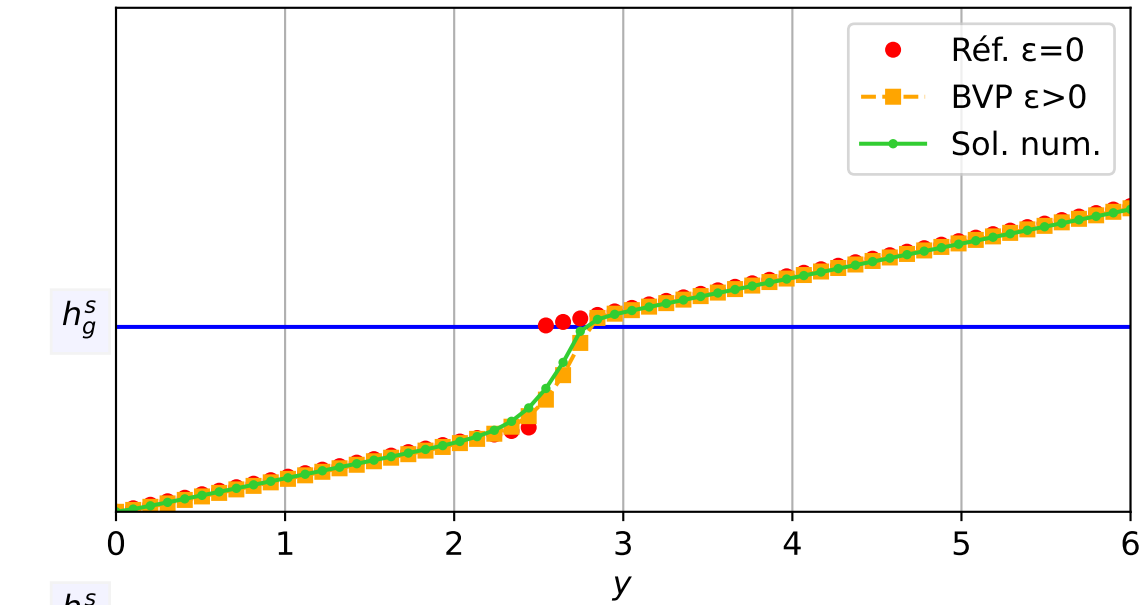
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.71e-04$  —  $t = 1.65015/12$  (5767 iter) — résidu eq. =  $5.95e-06$



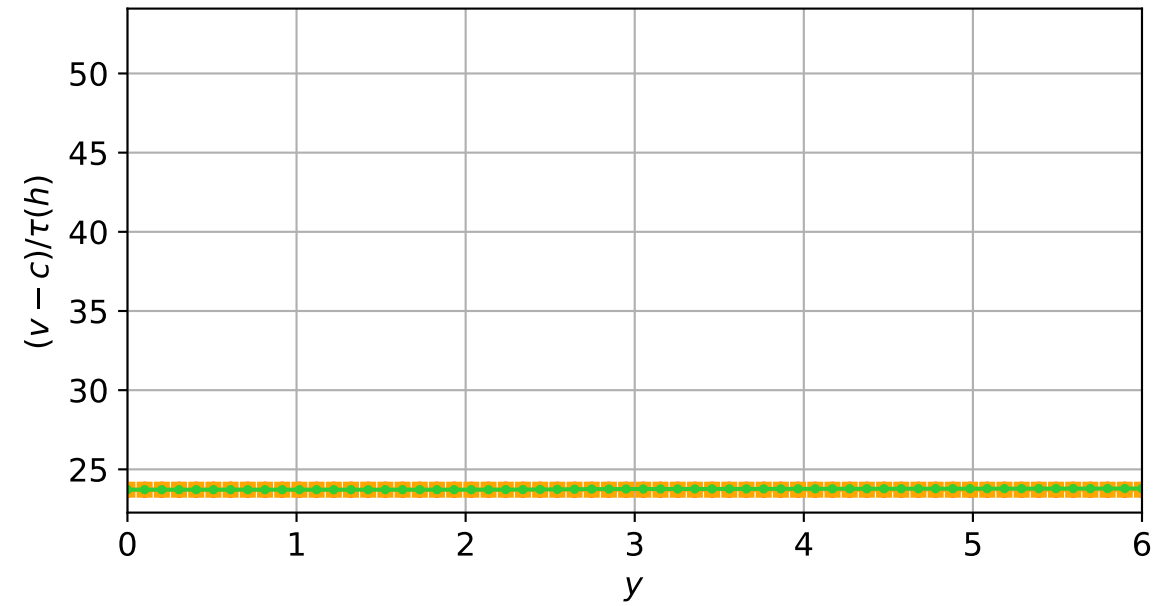
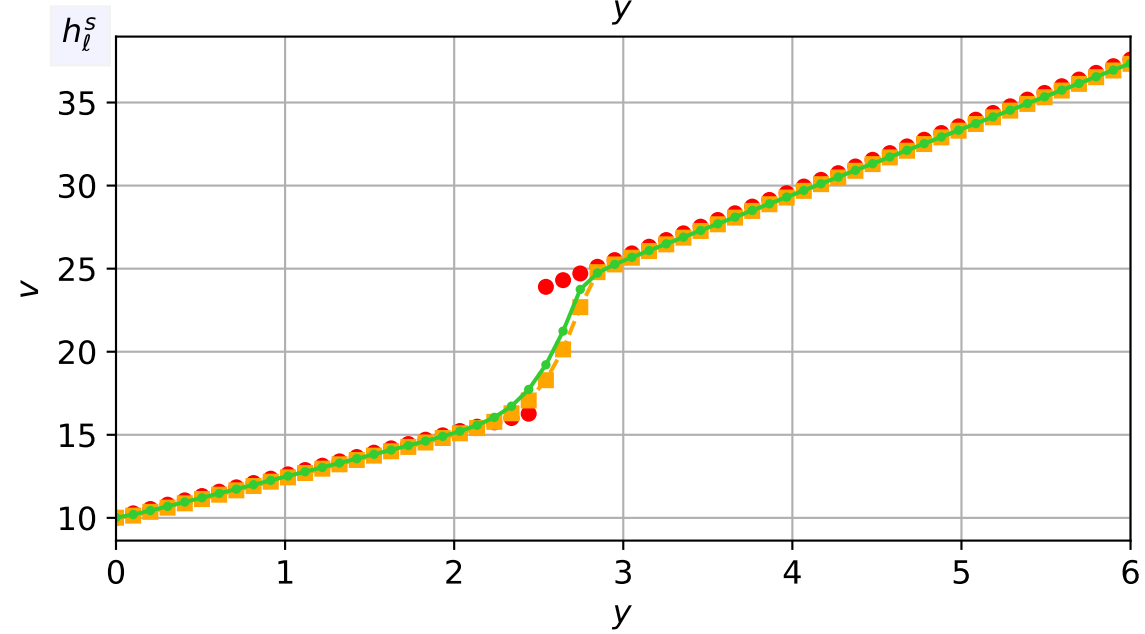
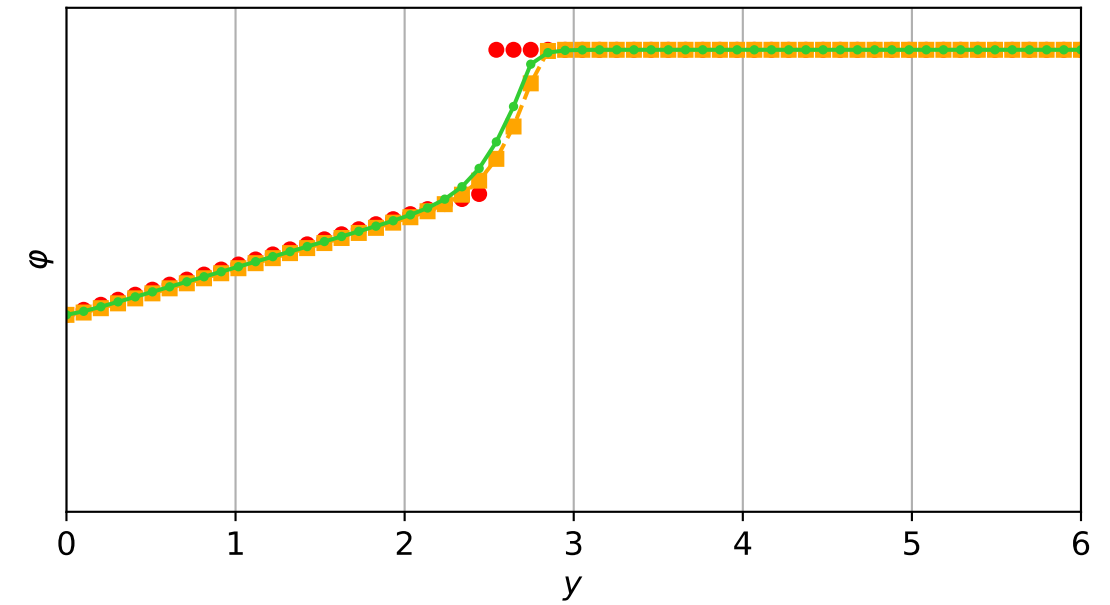
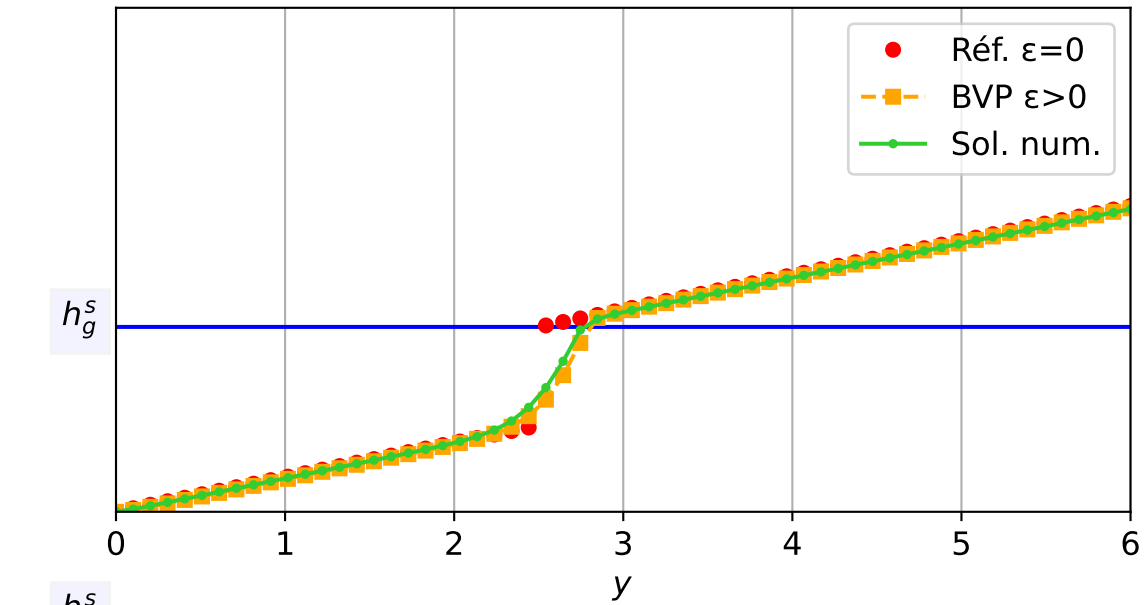
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.25e-04$  —  $t = 1.70003/12$  (5941 iter) — résidu eq. =  $7.99e-06$



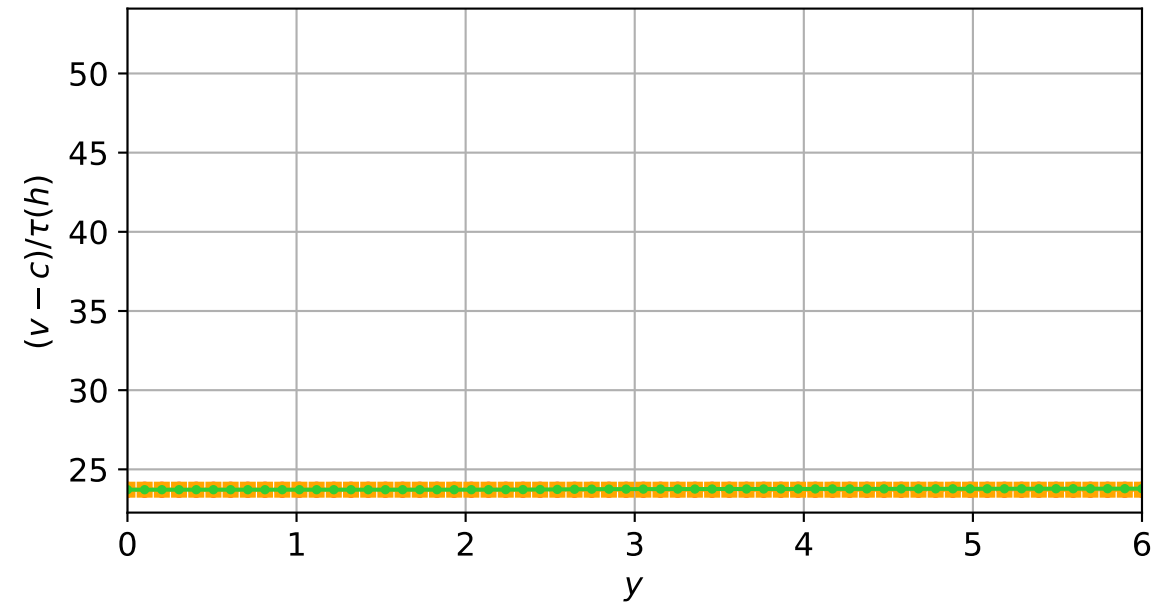
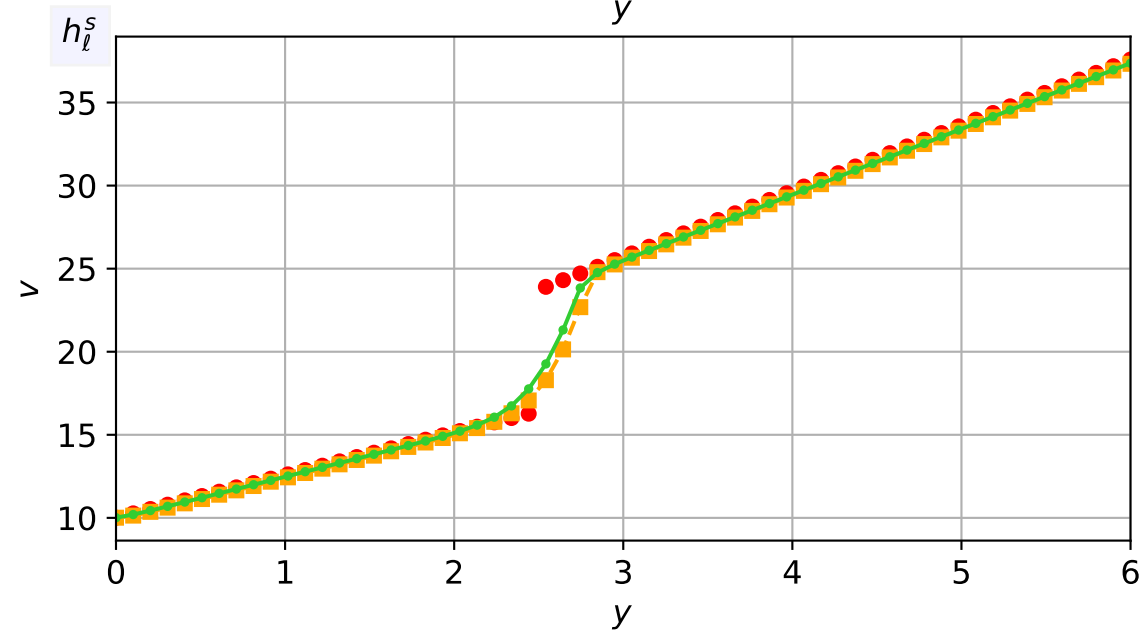
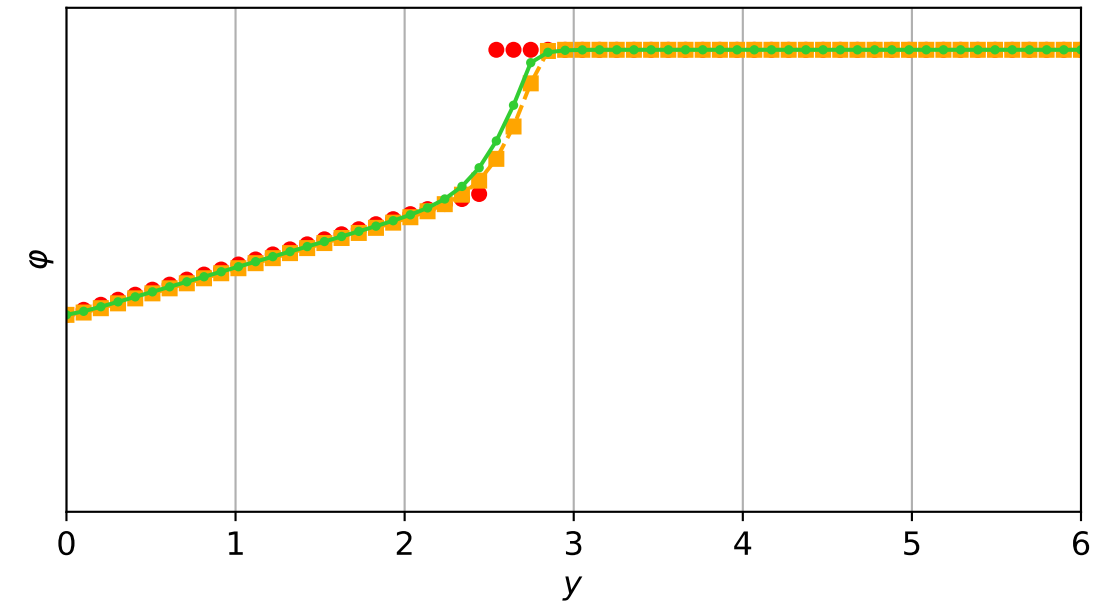
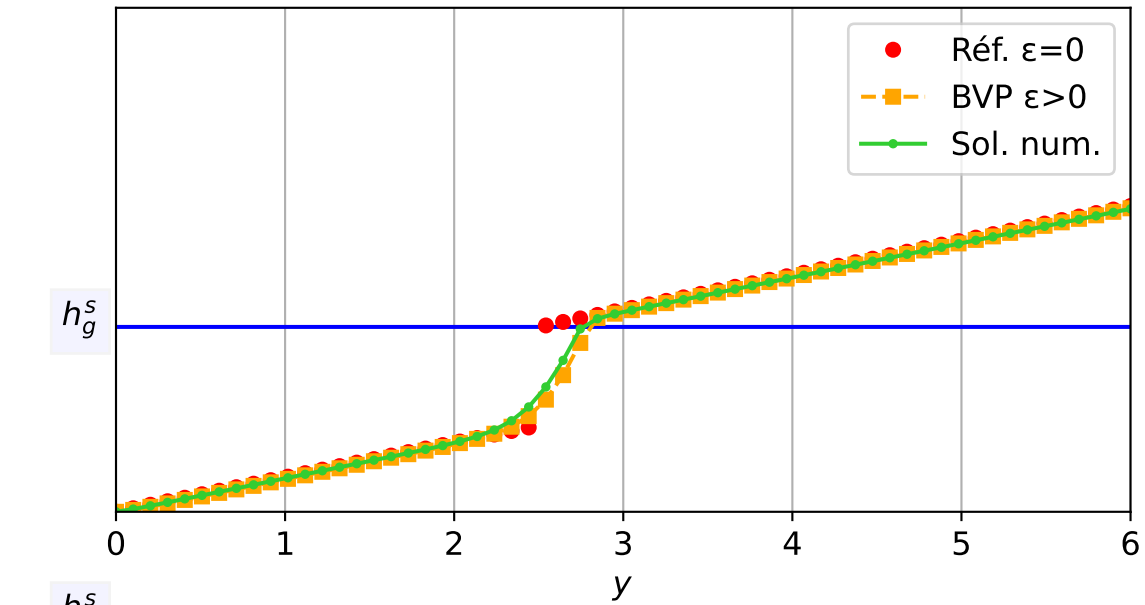
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.25e-04$  —  $t = 1.75013/12$  (6095 iter) — résidu eq. =  $7.03e-06$



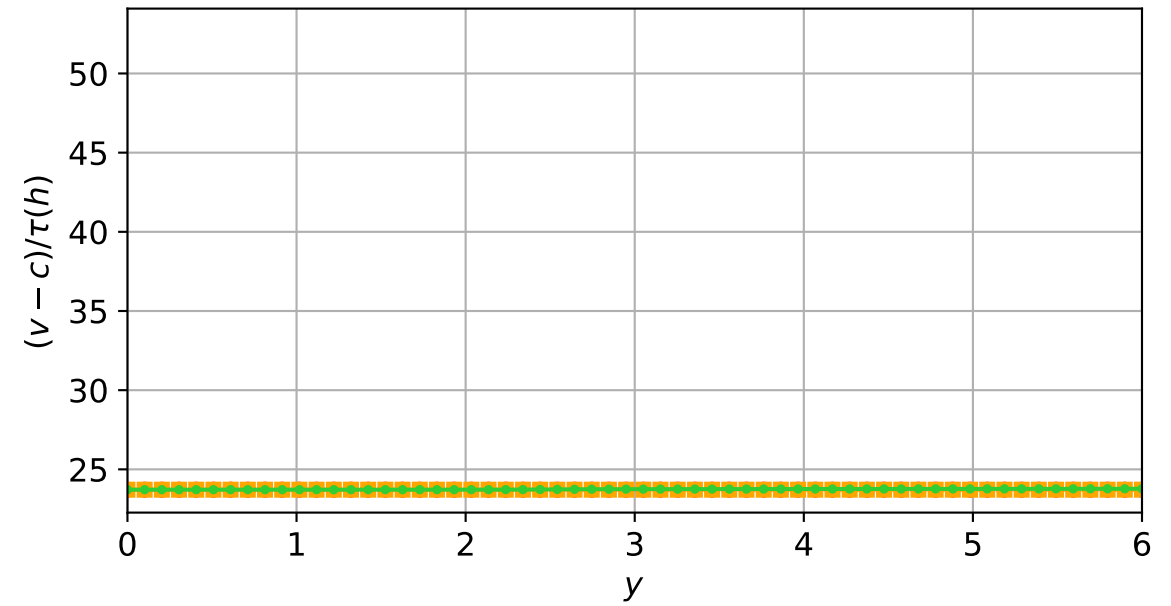
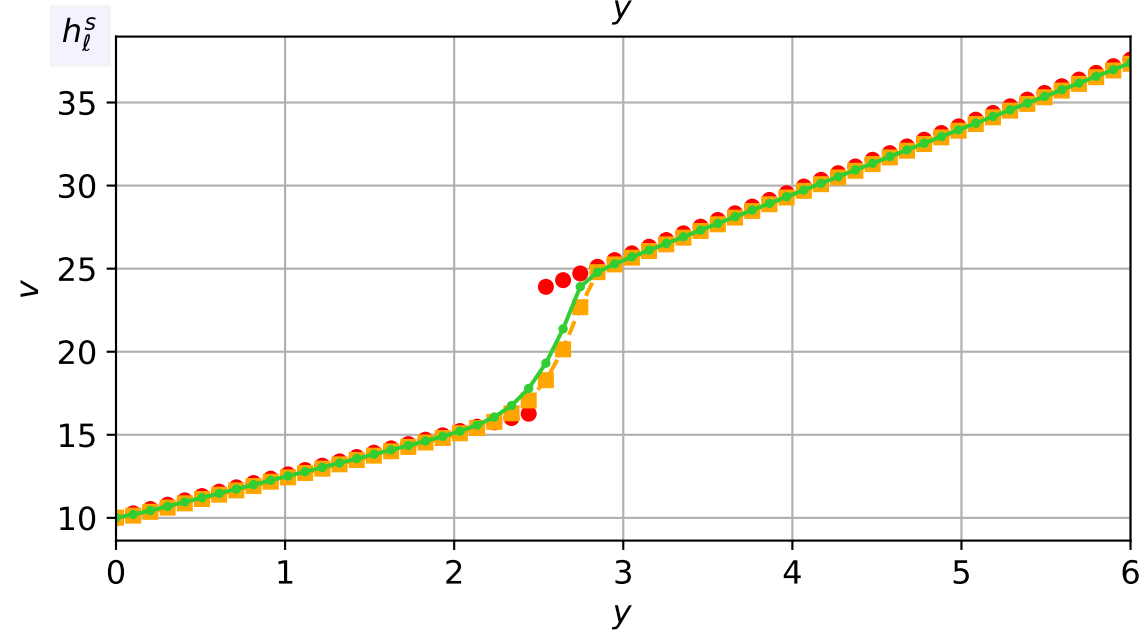
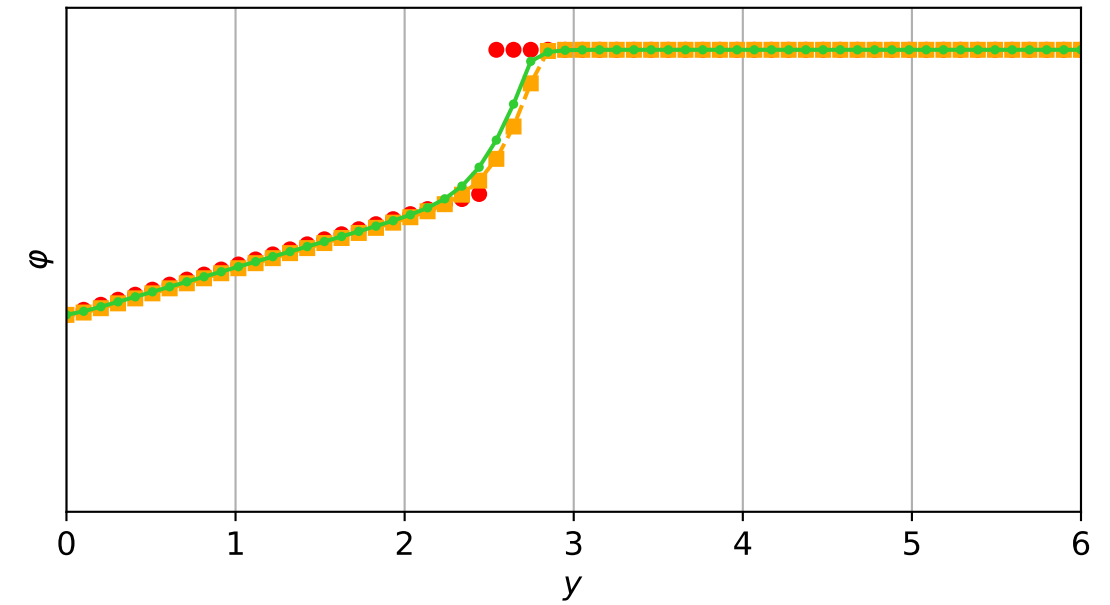
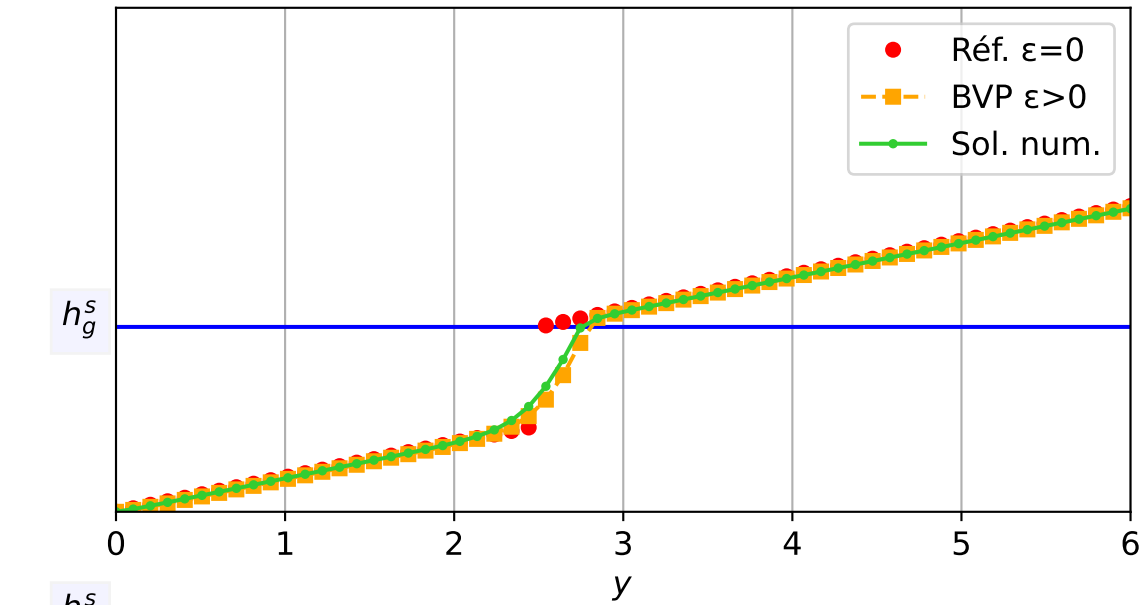
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.25e-04$  —  $t = 1.80023/12$  (6249 iter) — résidu eq. =  $6.19e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.90e-04$  —  $t = 1.85006/12$  (6392 iter) — résidu eq. =  $8.12e-06$

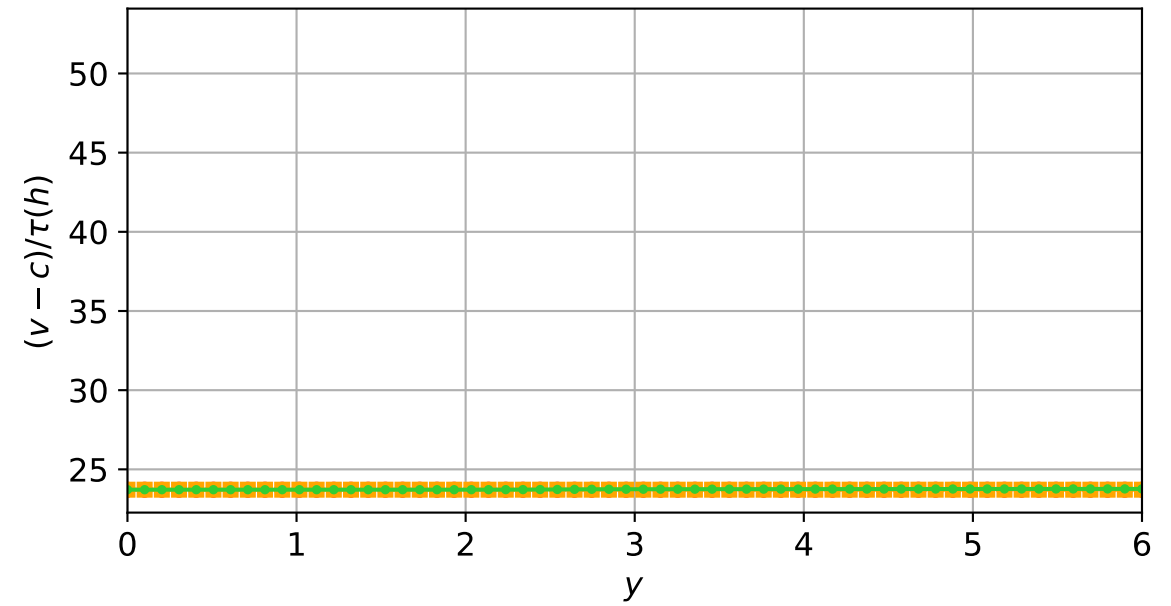
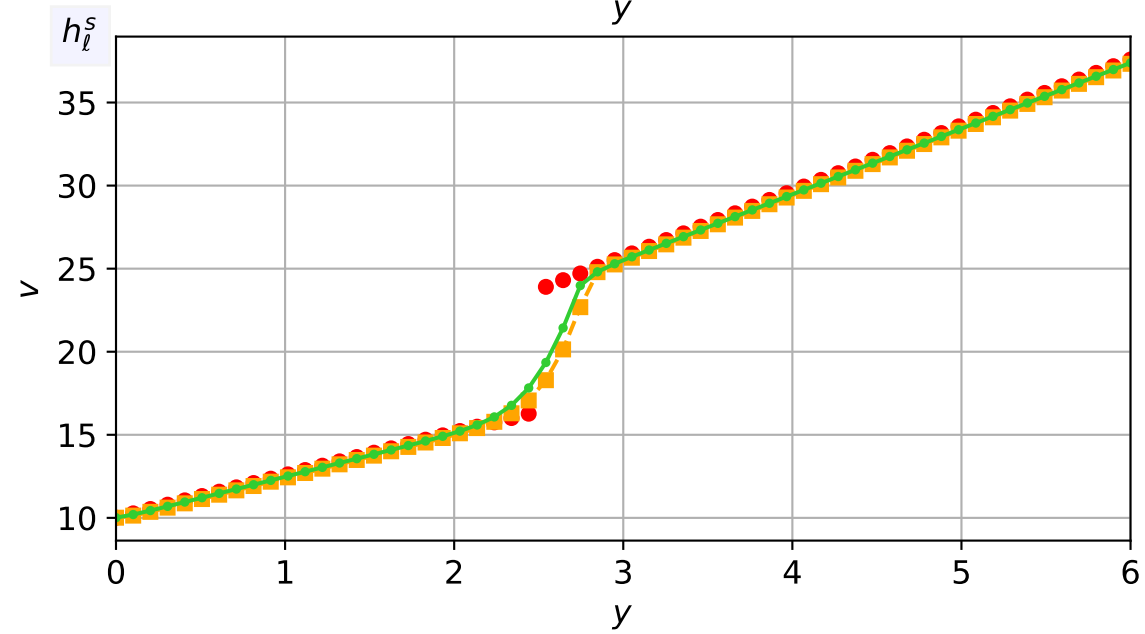
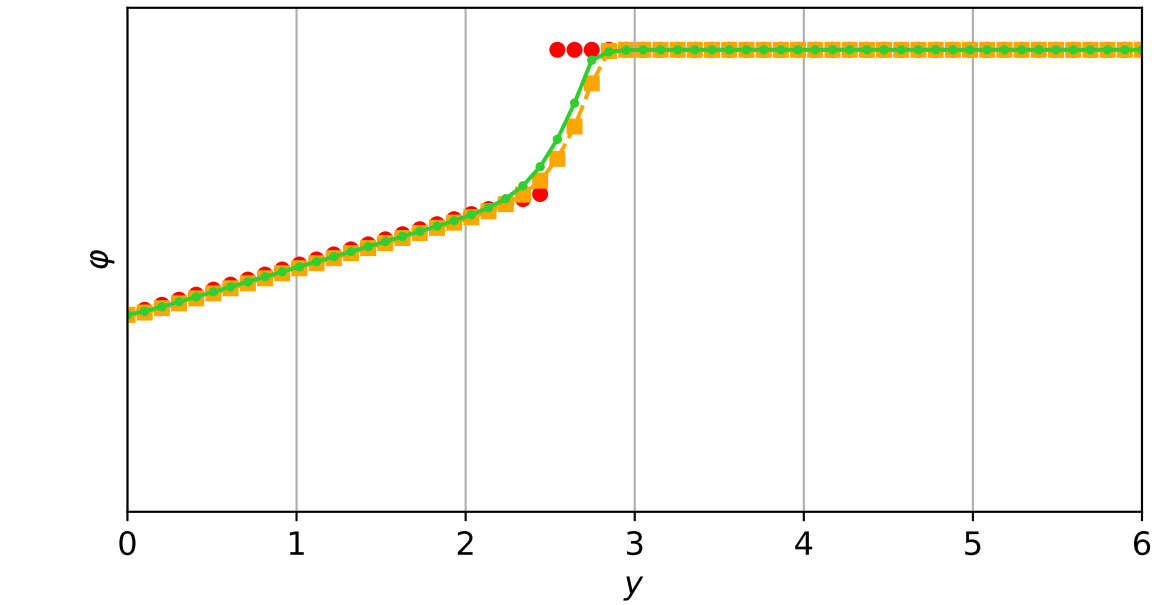
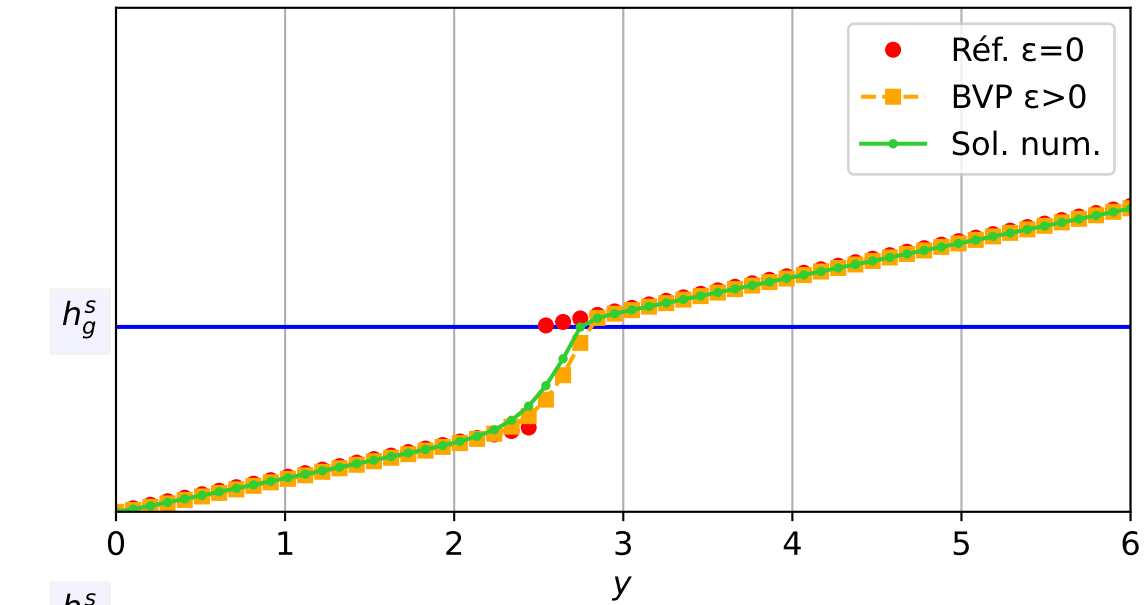


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.90e-04$  —  $t = 1.90003/12$  (6520 iter) — résidu eq. =  $7.16e-06$

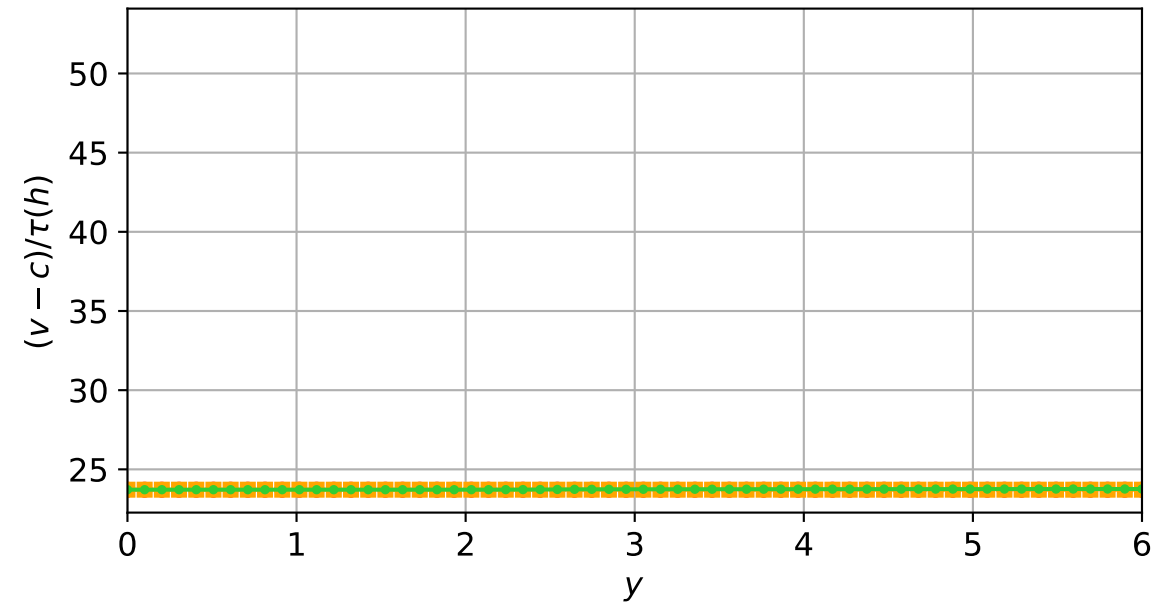
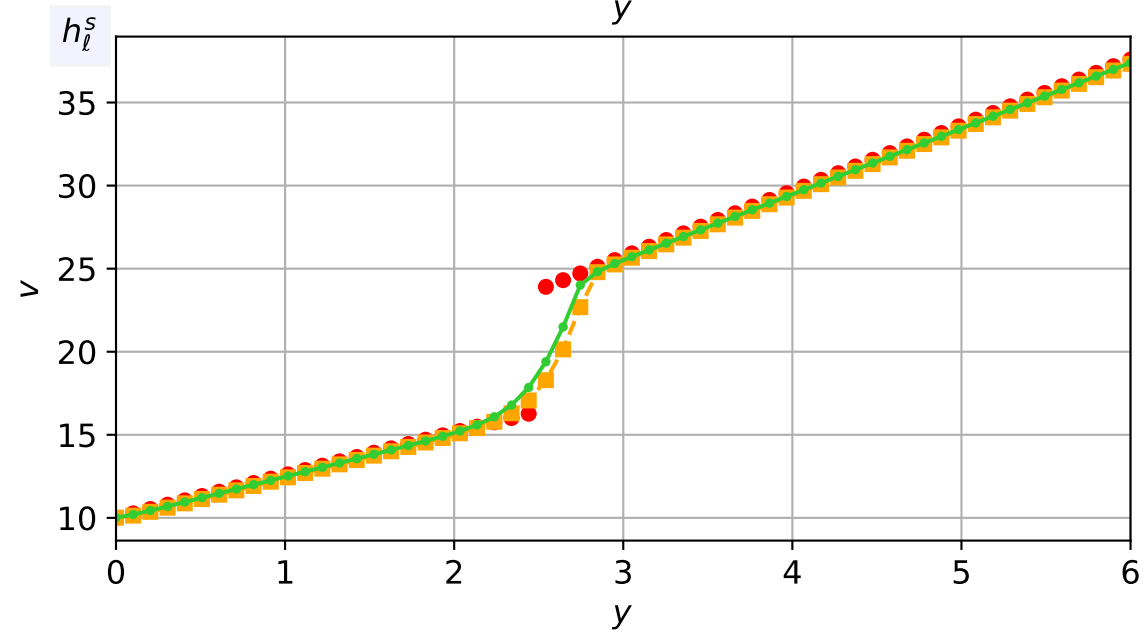
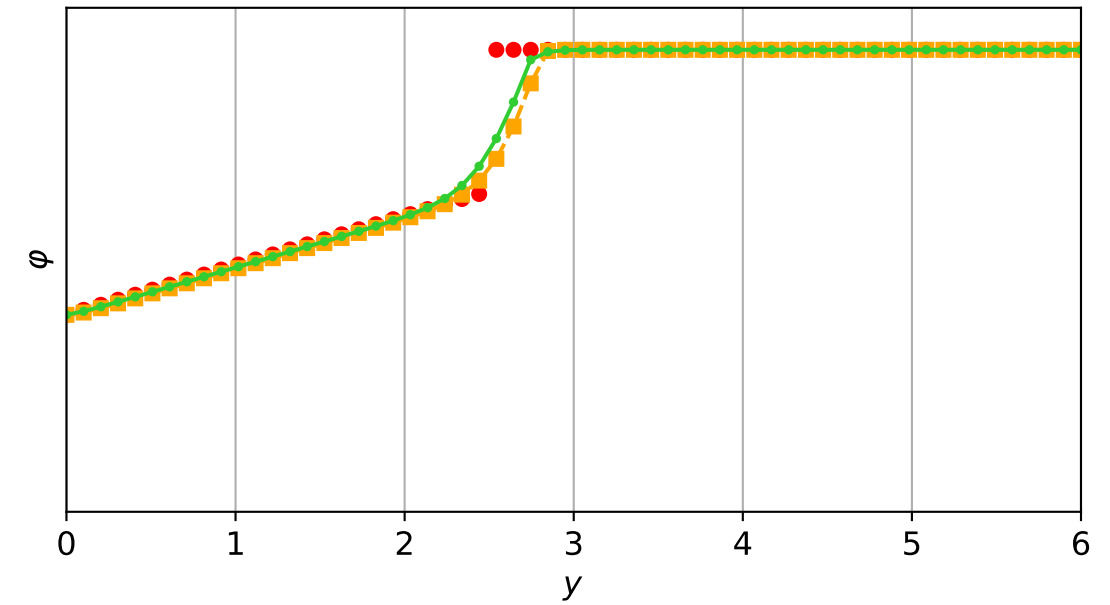
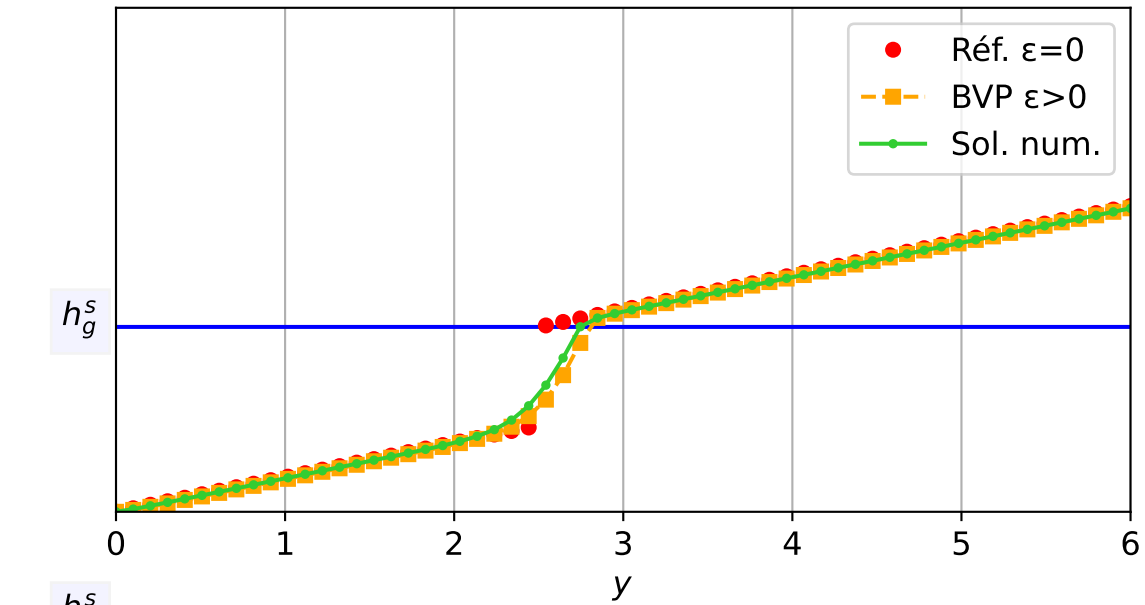




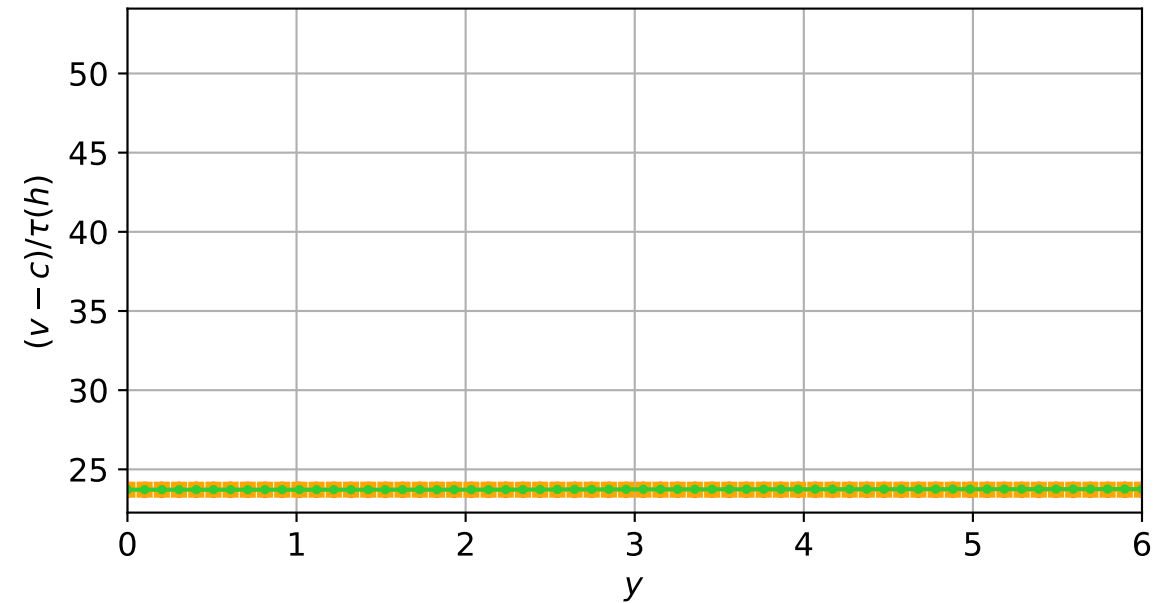
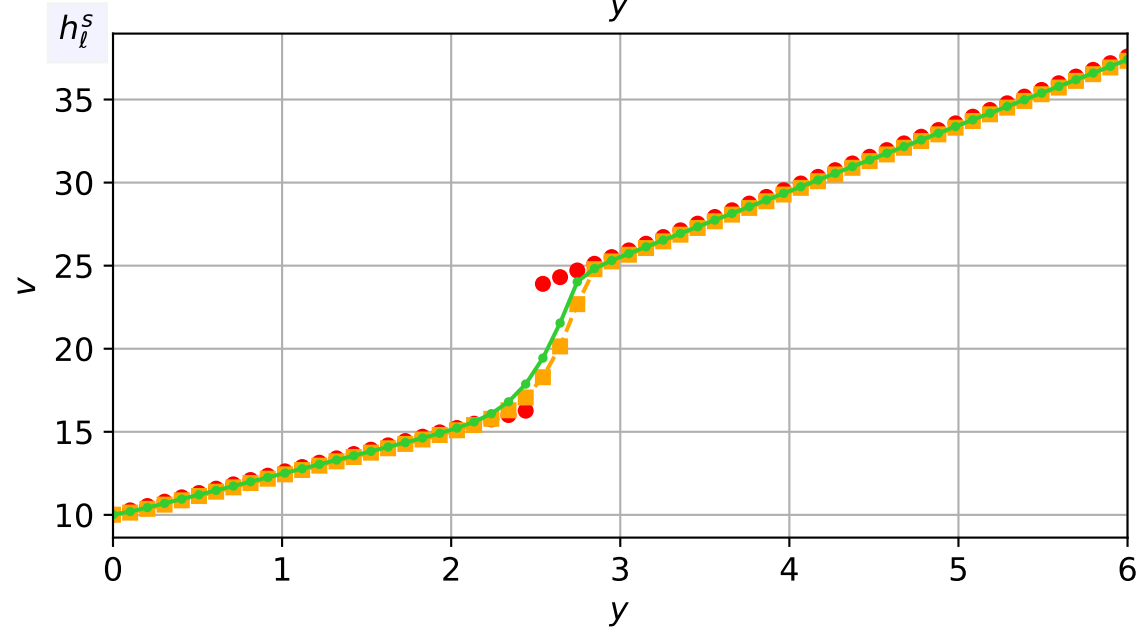
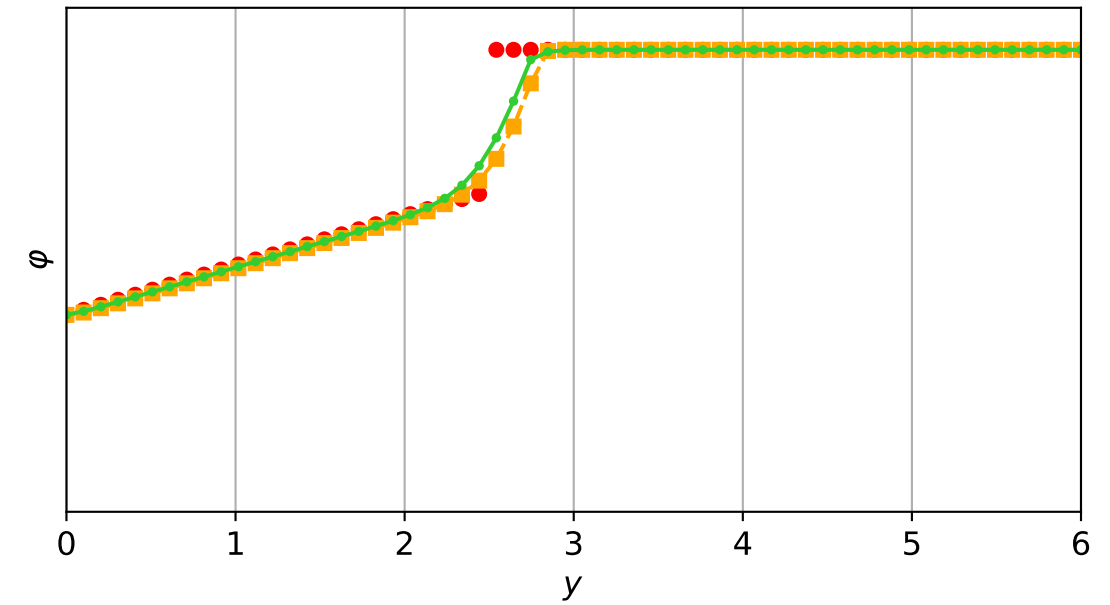
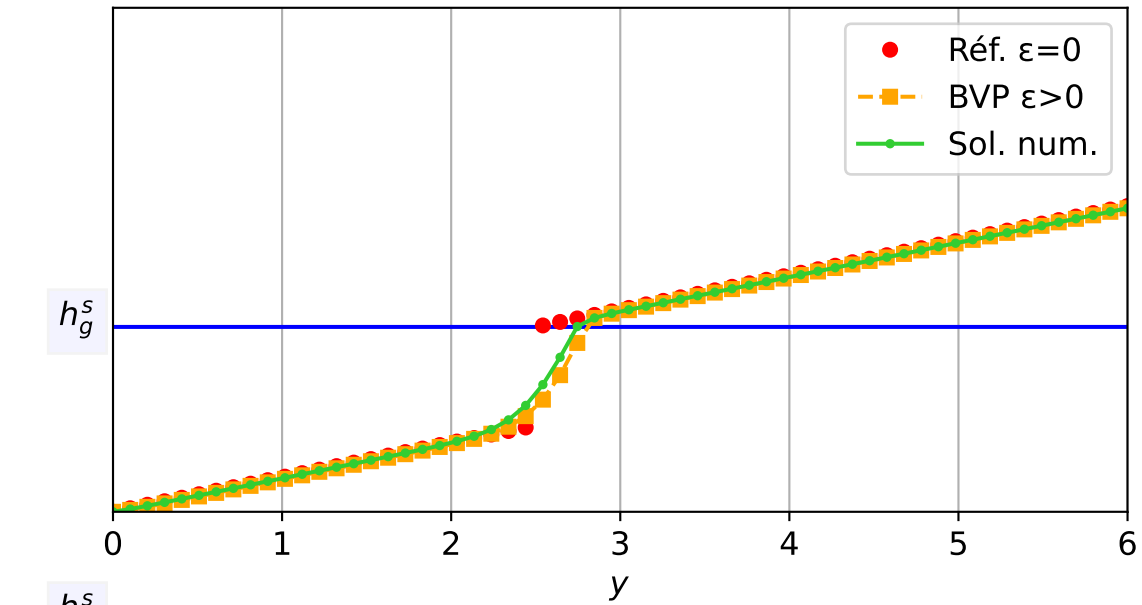
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.90e-04$  —  $t = 1.95038/12$  (6649 iter) — résidu eq. =  $6.30e-06$



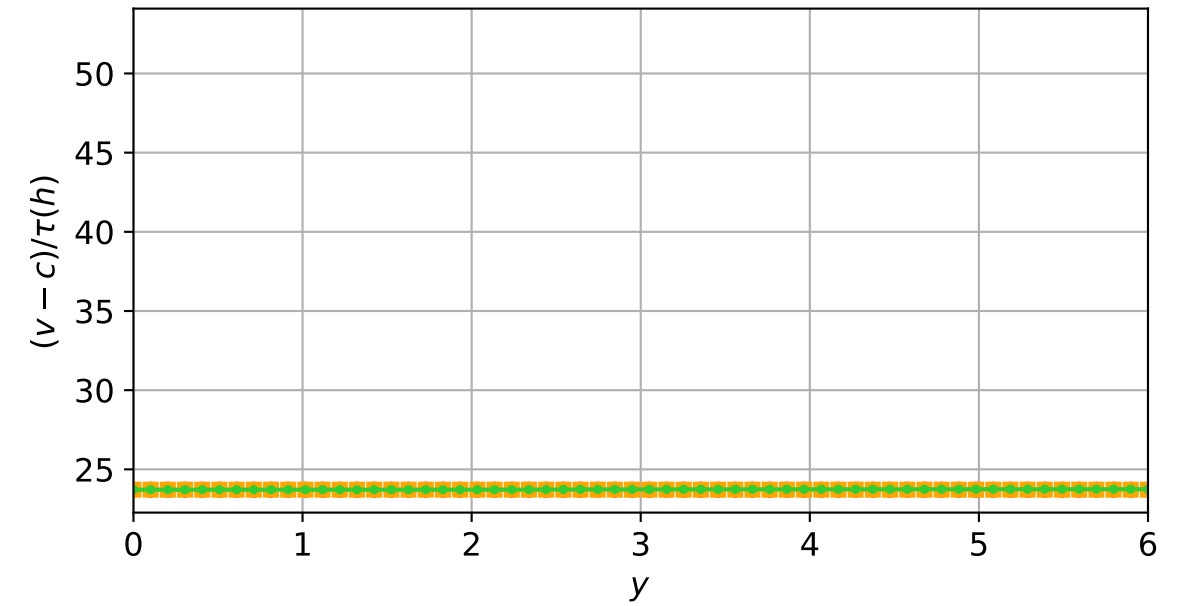
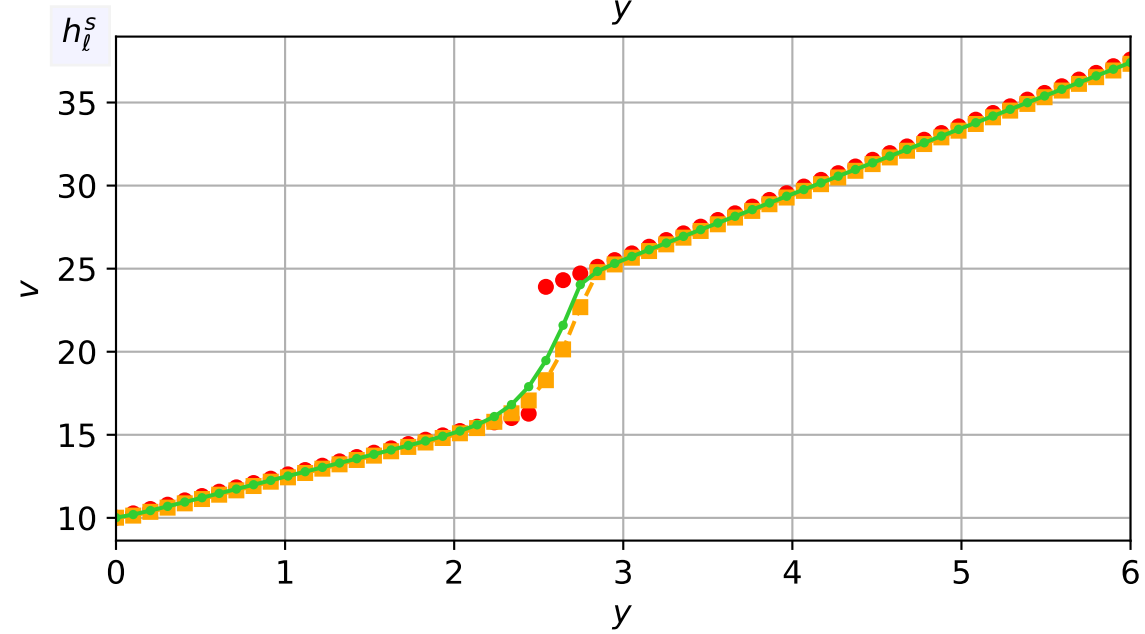
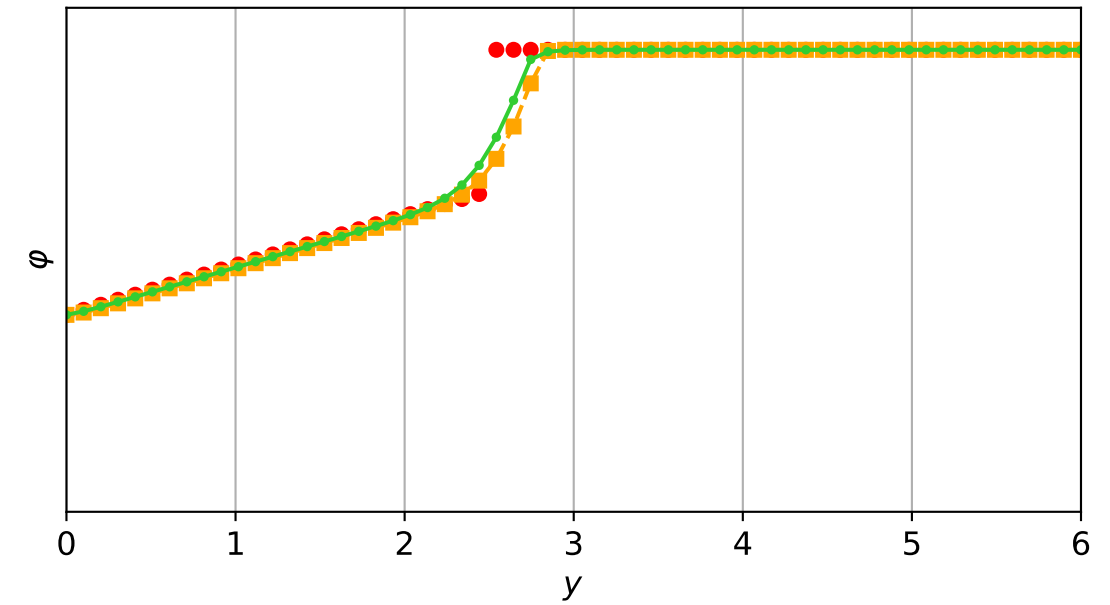
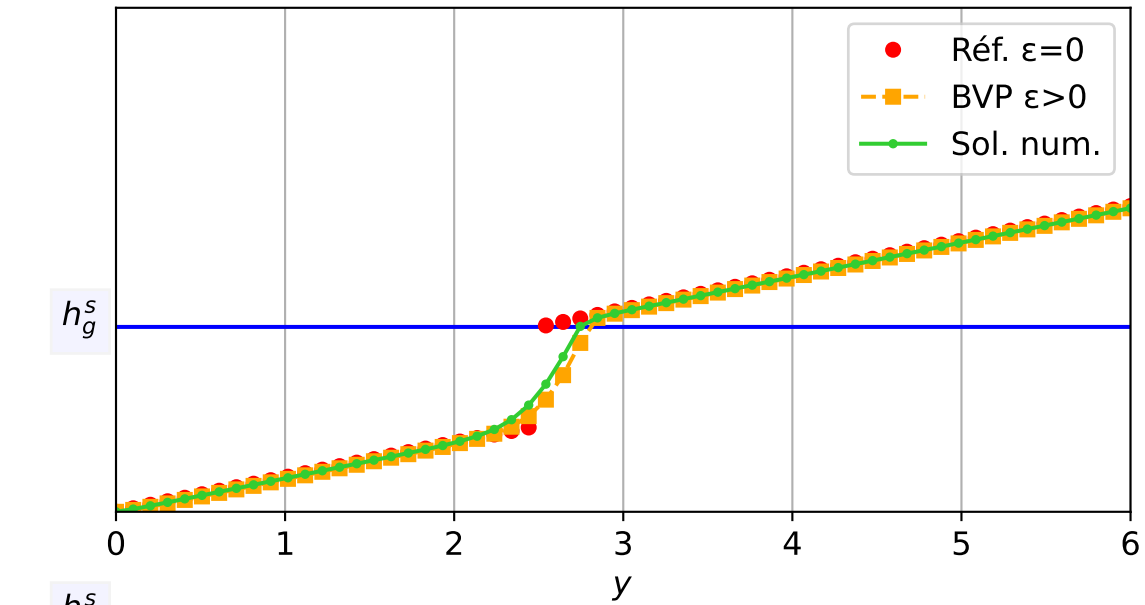
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 4.68e-04$  —  $t = 2.00035/12$  (6762 iter) — résidu eq. =  $8.55e-06$



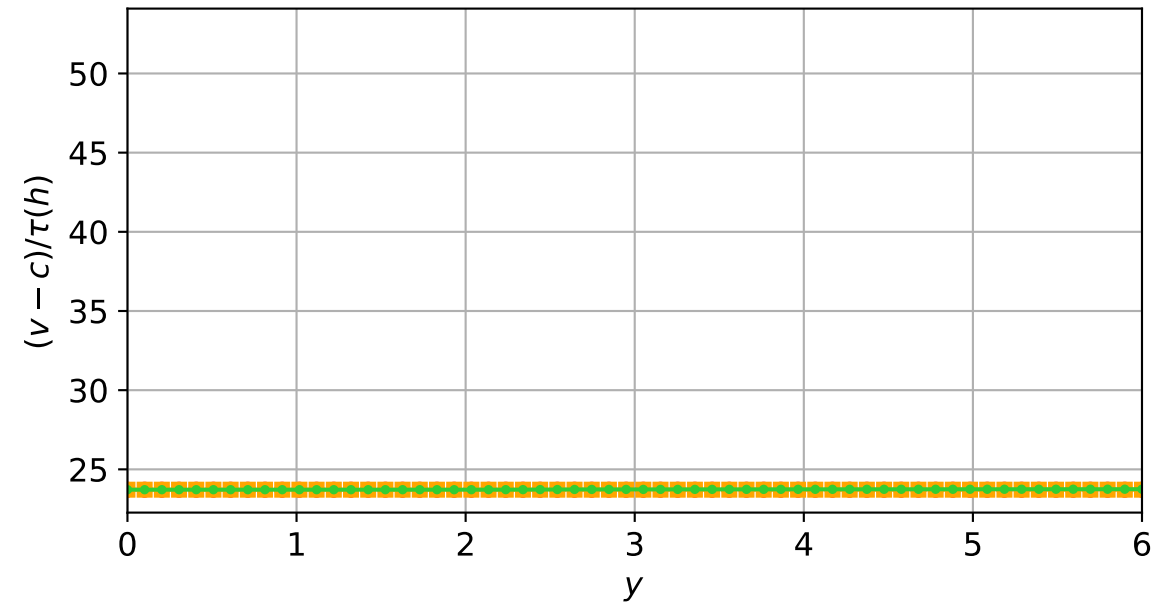
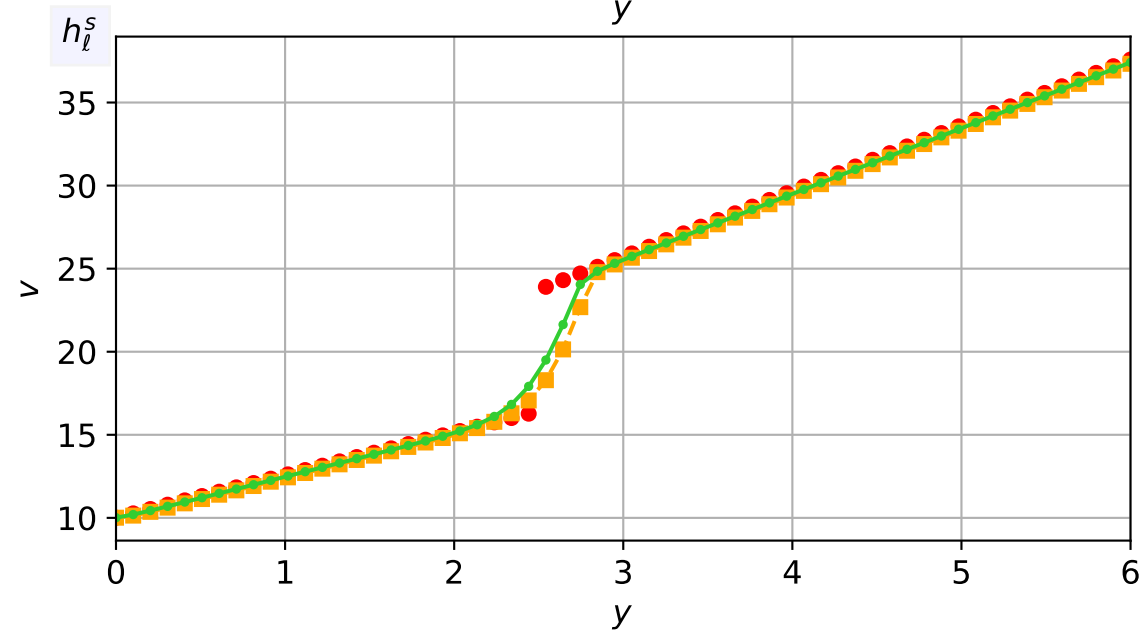
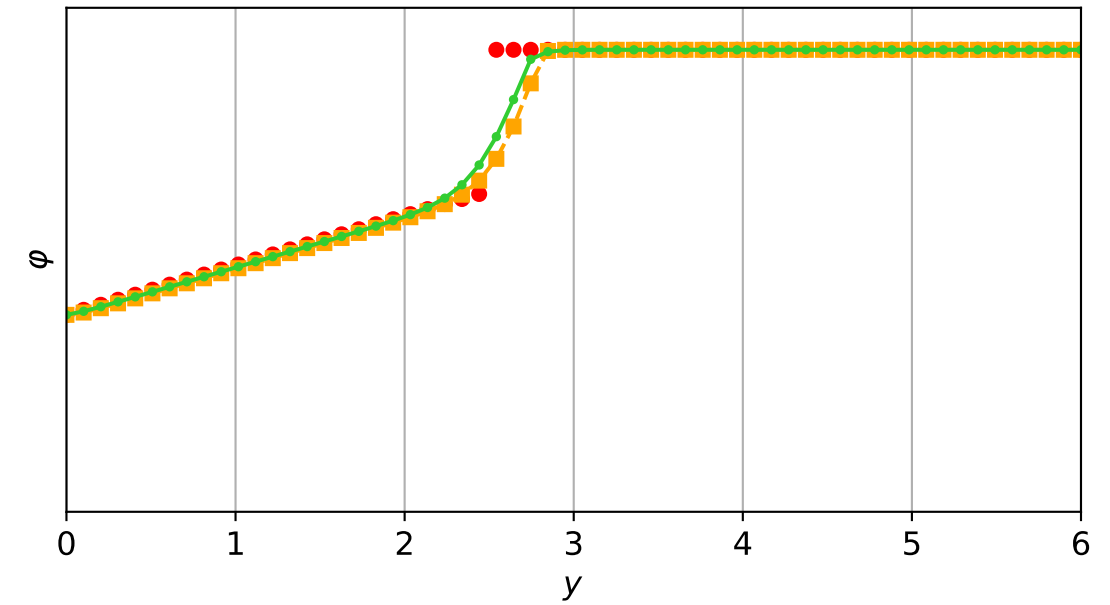
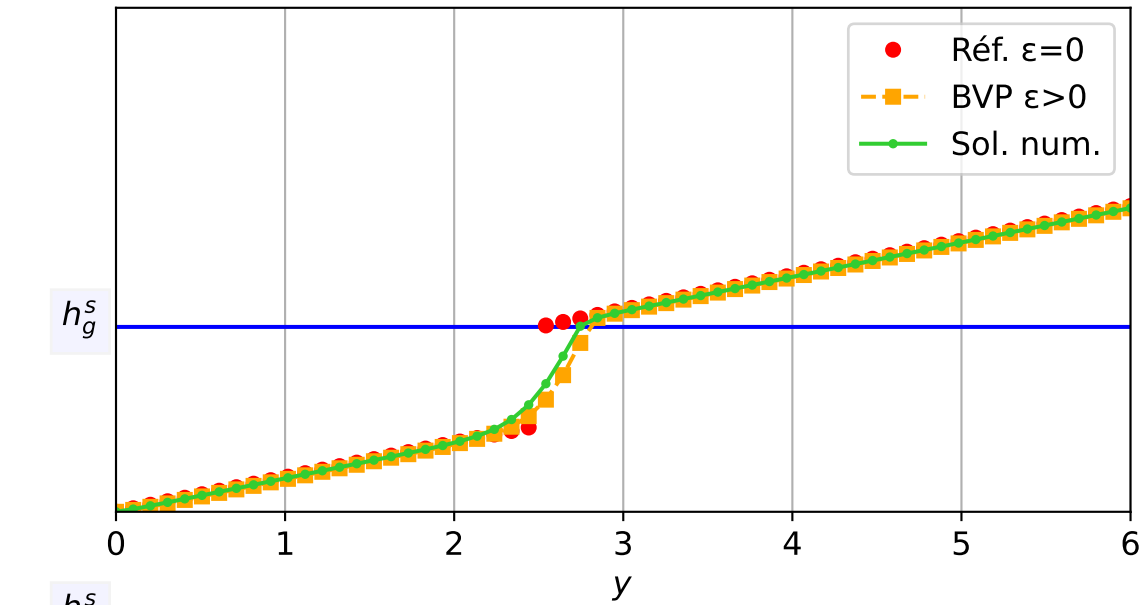
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 4.68e-04$  —  $t = 2.05/12$  (6868 iter) — résidu eq. =  $7.44e-06$



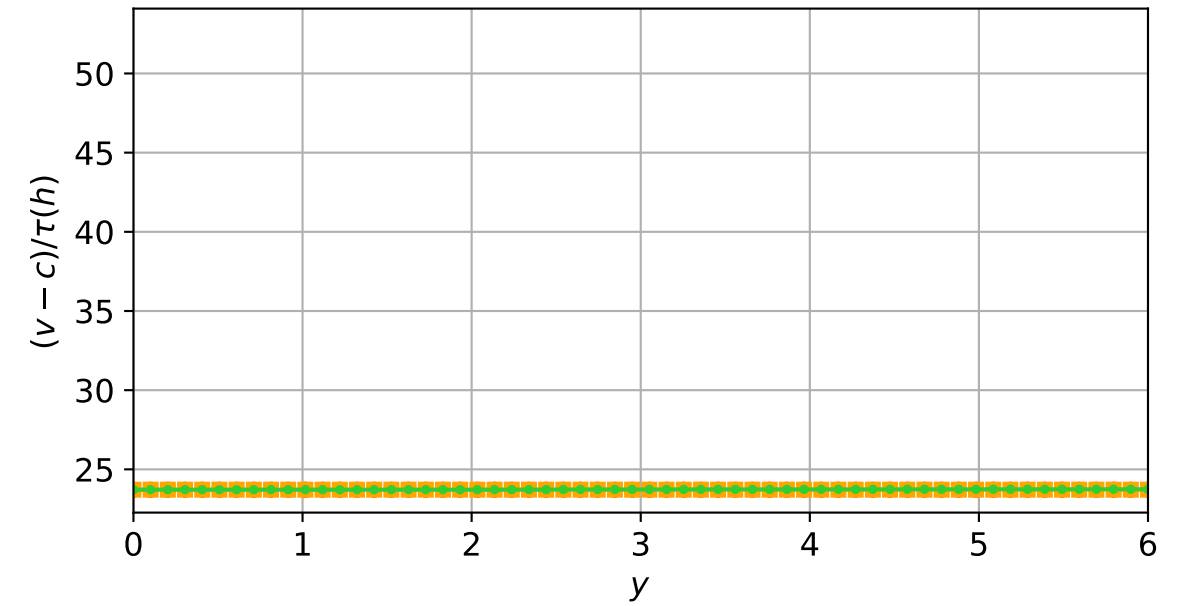
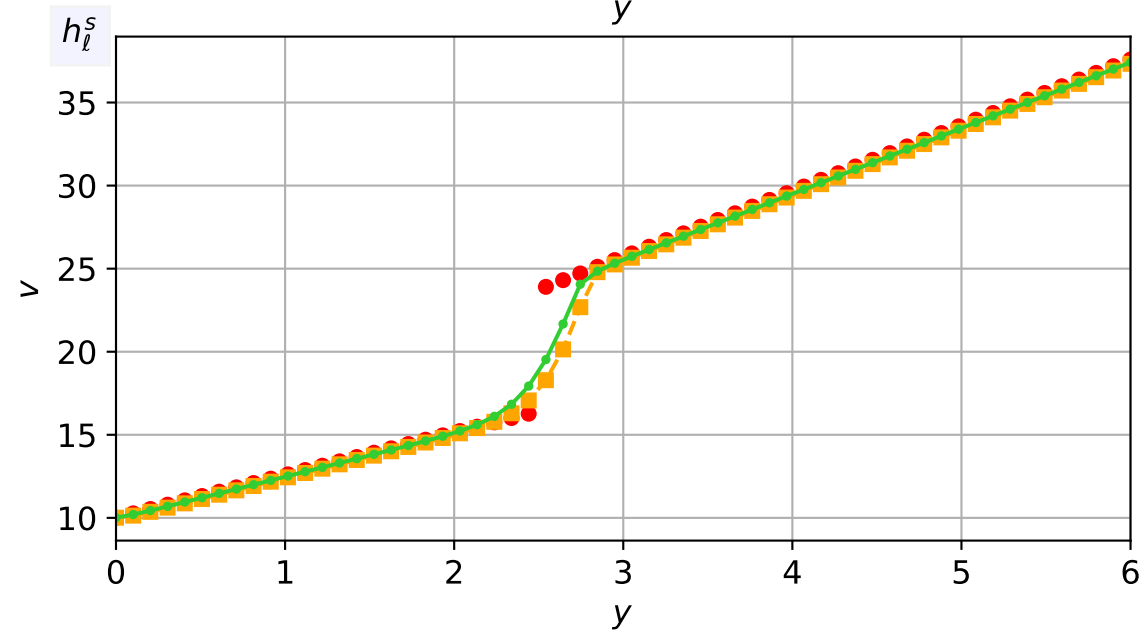
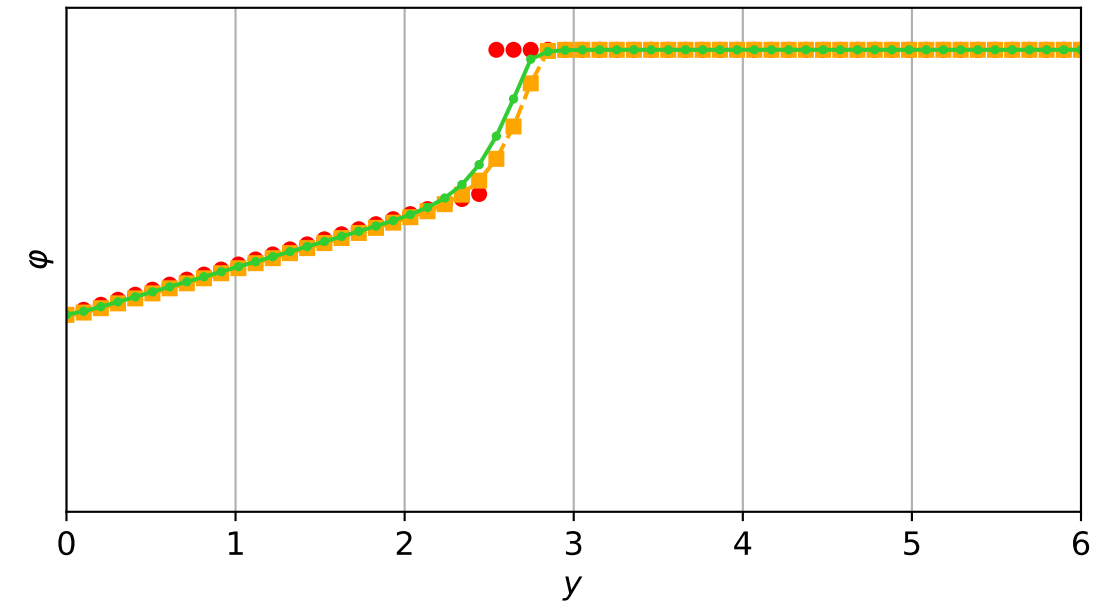
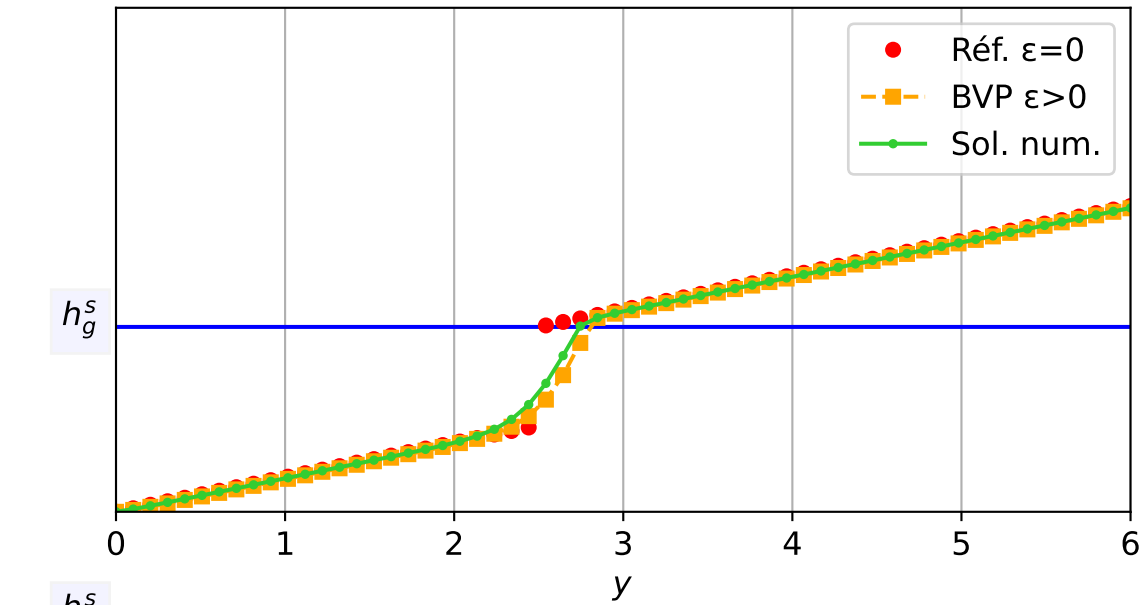
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 4.68e-04$  —  $t = 2.10013/12$  (6975 iter) — résidu eq. =  $6.36e-06$



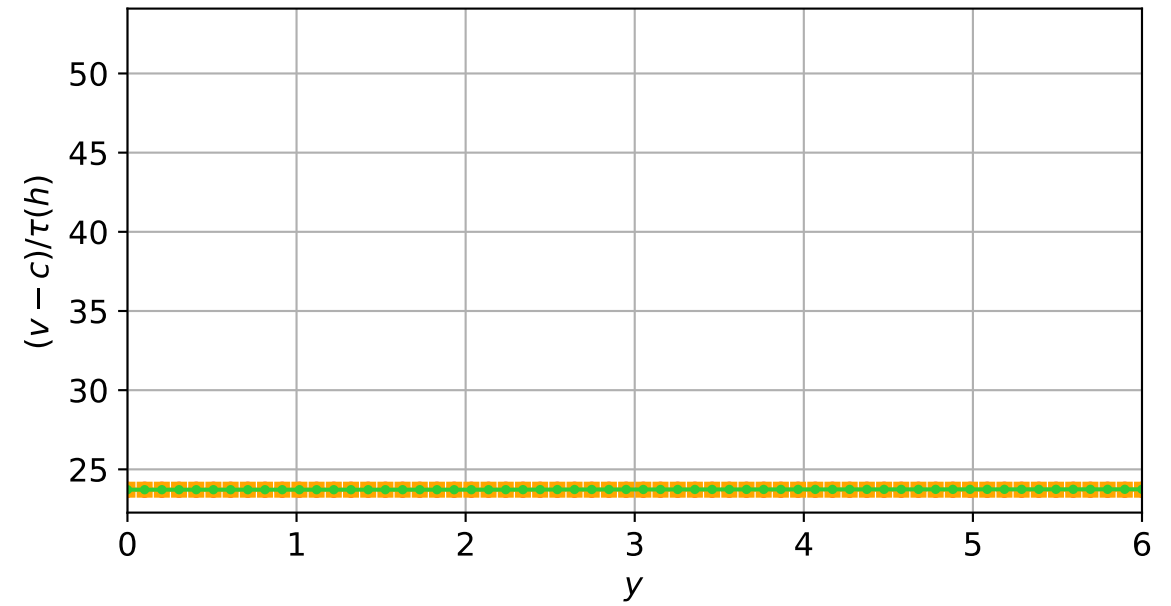
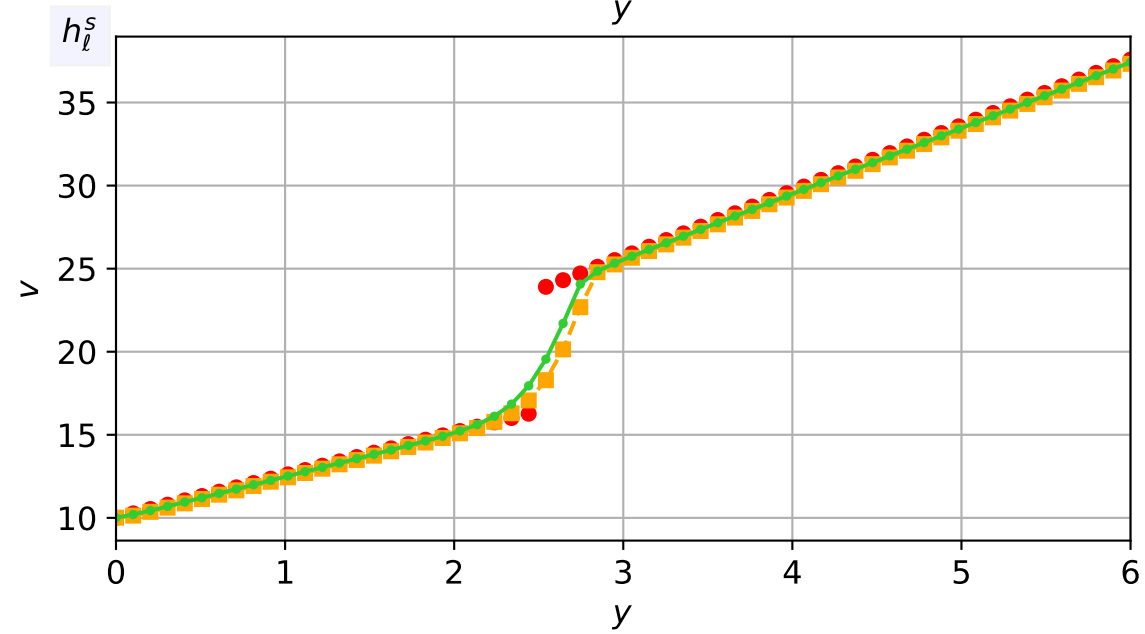
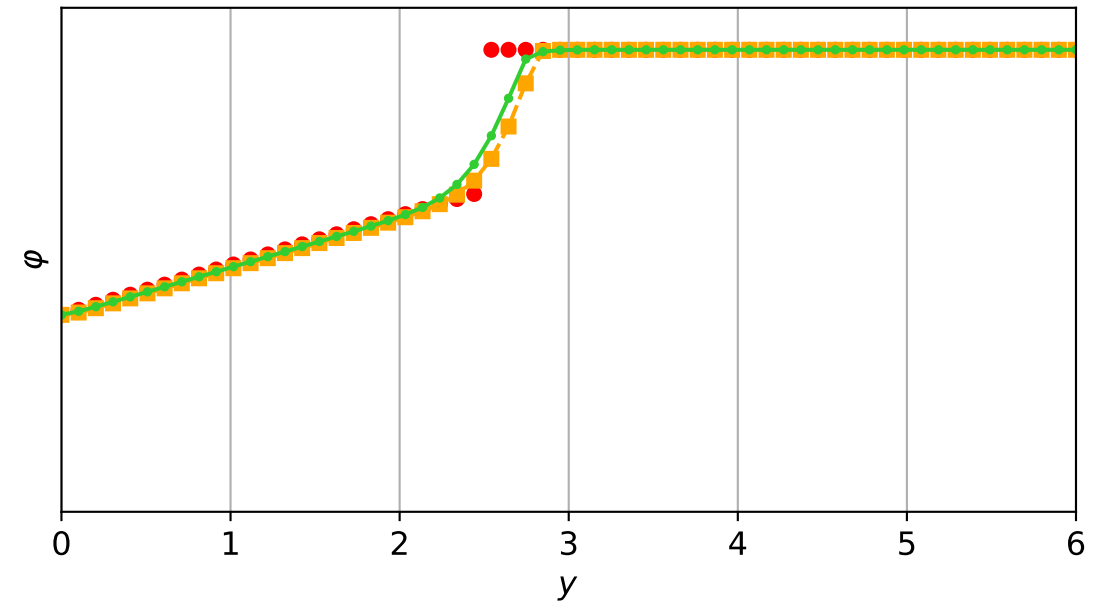
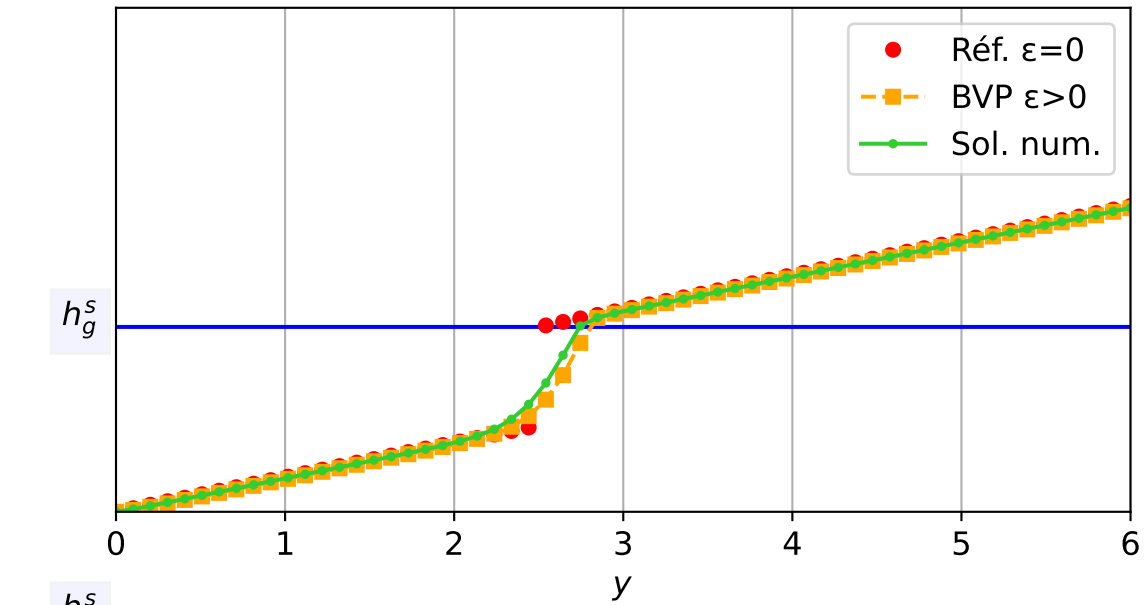
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 5.62e-04$  —  $t = 2.15025/12$  (7066 iter) — résidu eq. =  $7.89e-06$



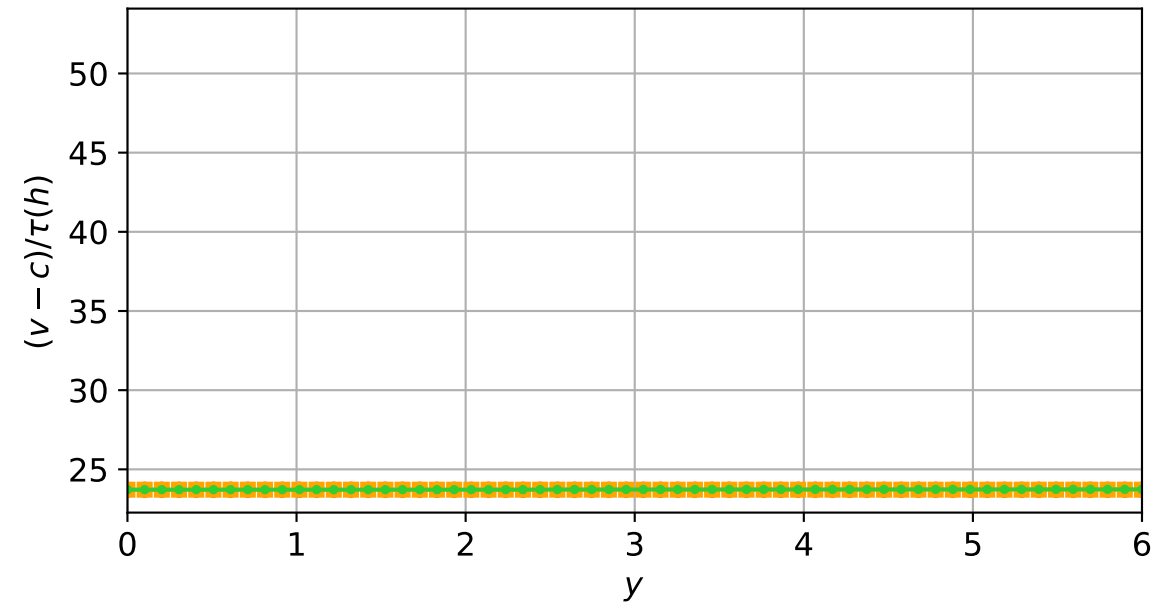
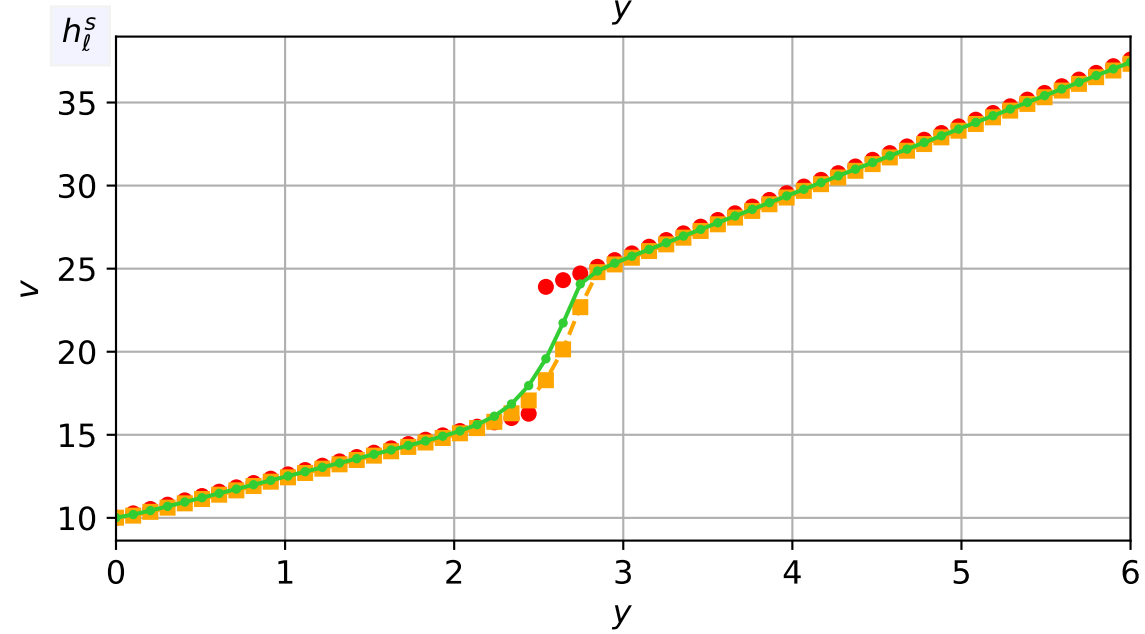
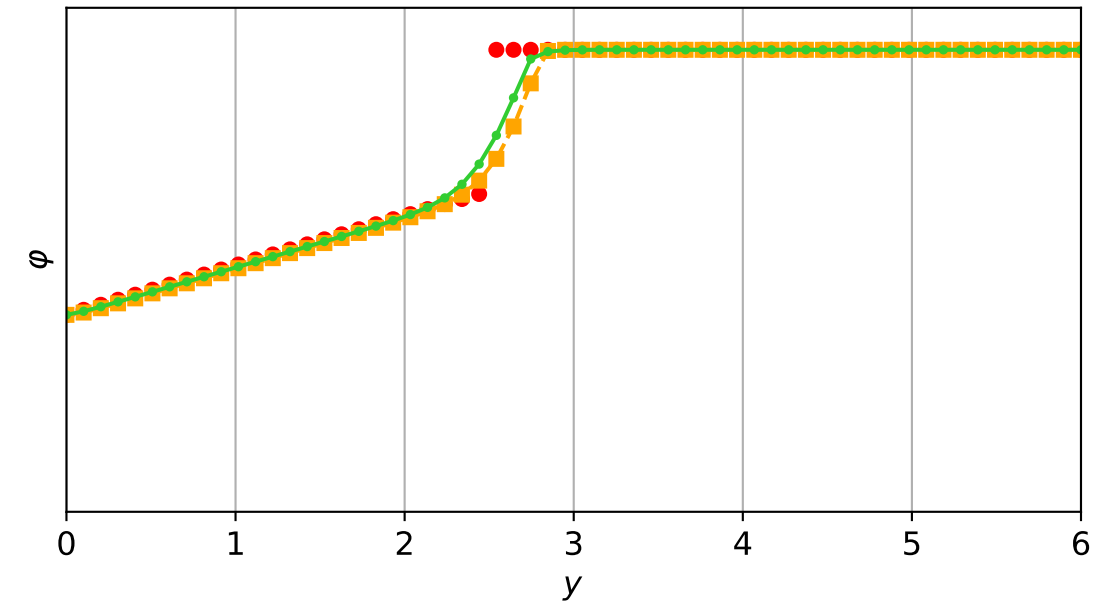
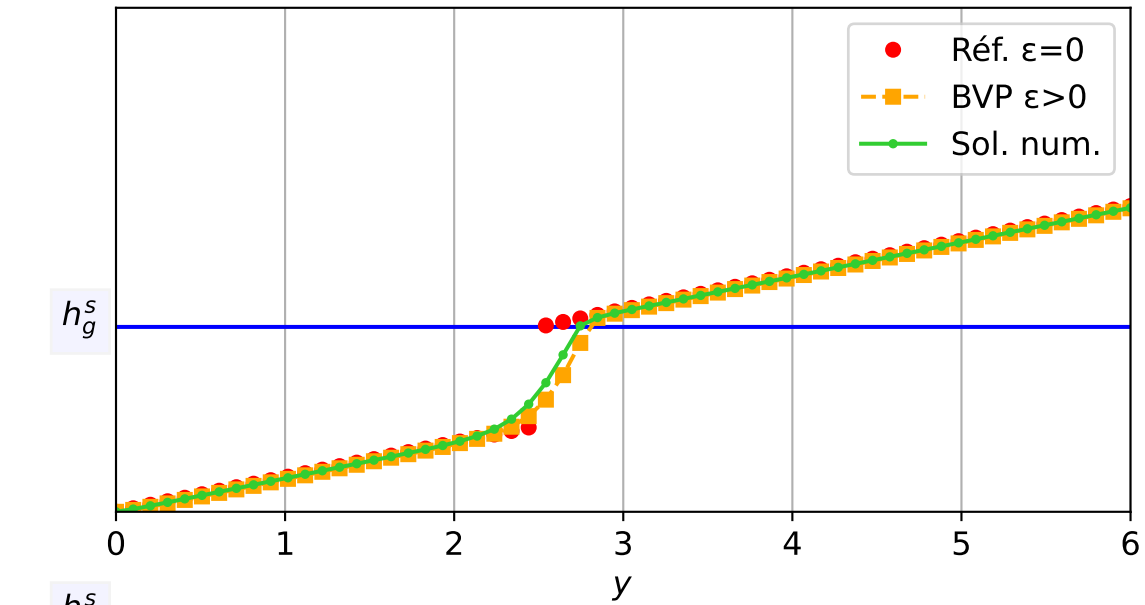
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 5.62e-04$  —  $t = 2.20028/12$  (7155 iter) — résidu eq. =  $6.91e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 6.75e-04$  —  $t = 2.2503/12$  (7236 iter) — résidu eq. =  $8.52e-06$

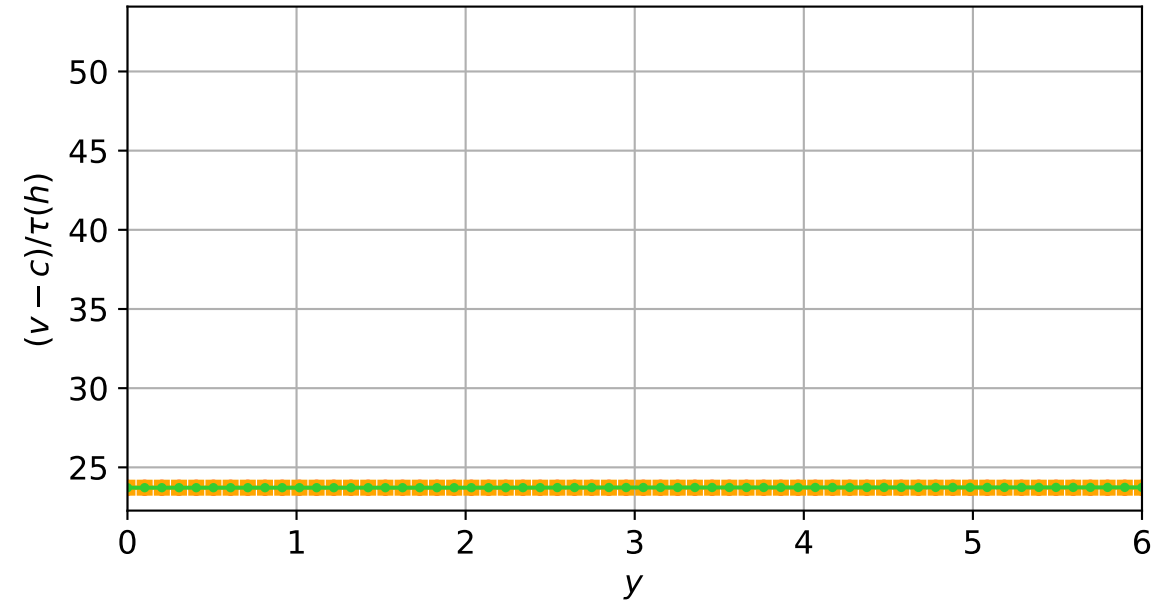
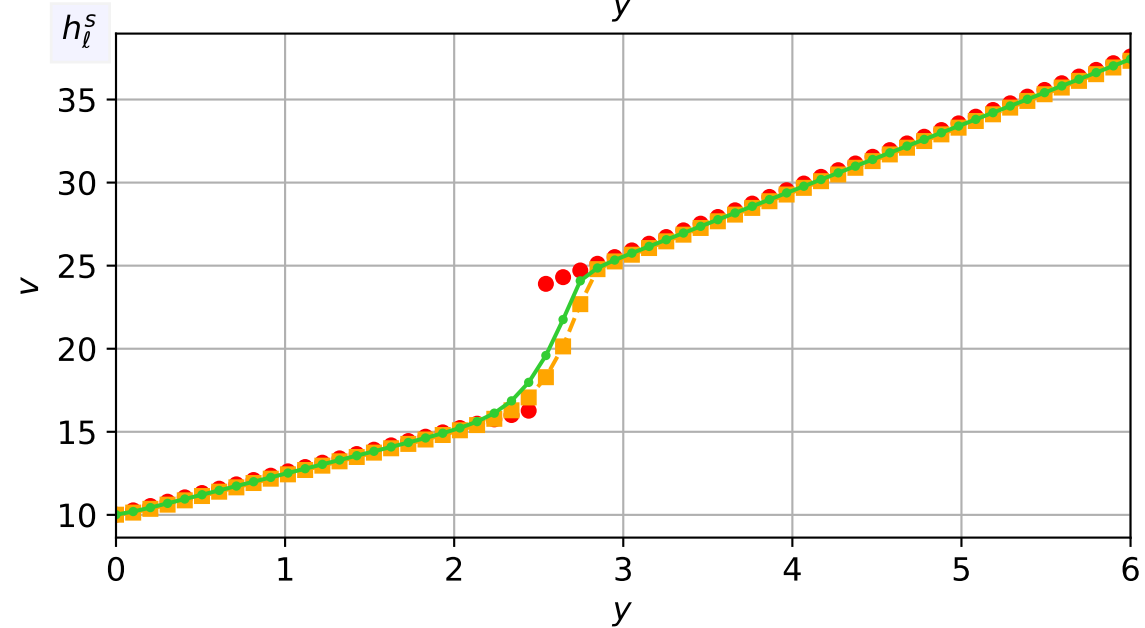
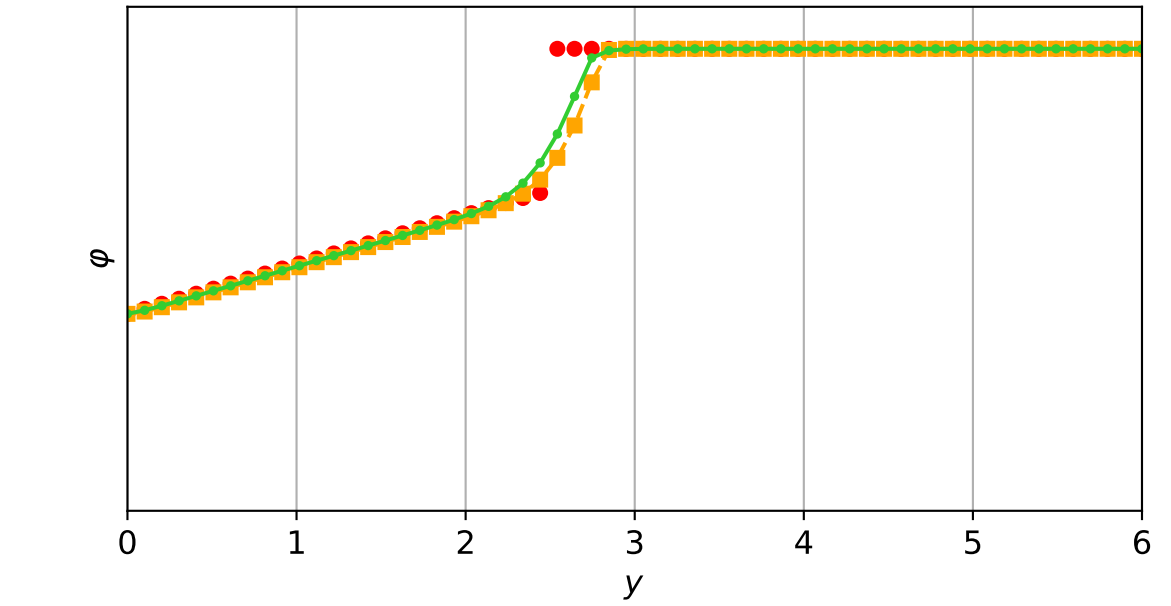
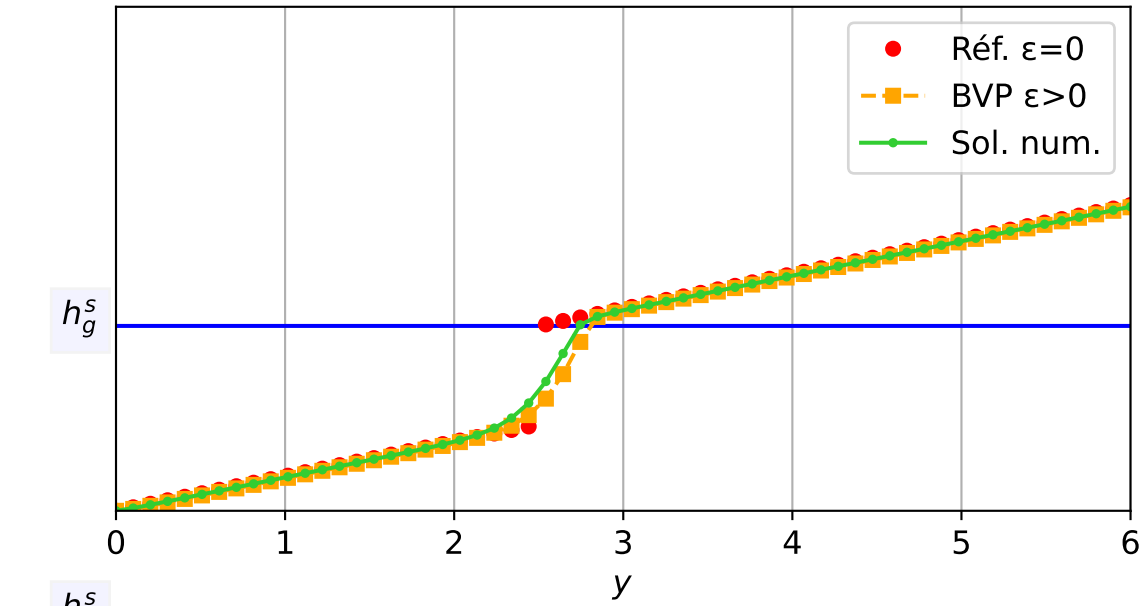


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 6.75e-04$  —  $t = 2.30022/12$  (7310 iter) — résidu eq. =  $7.51e-06$

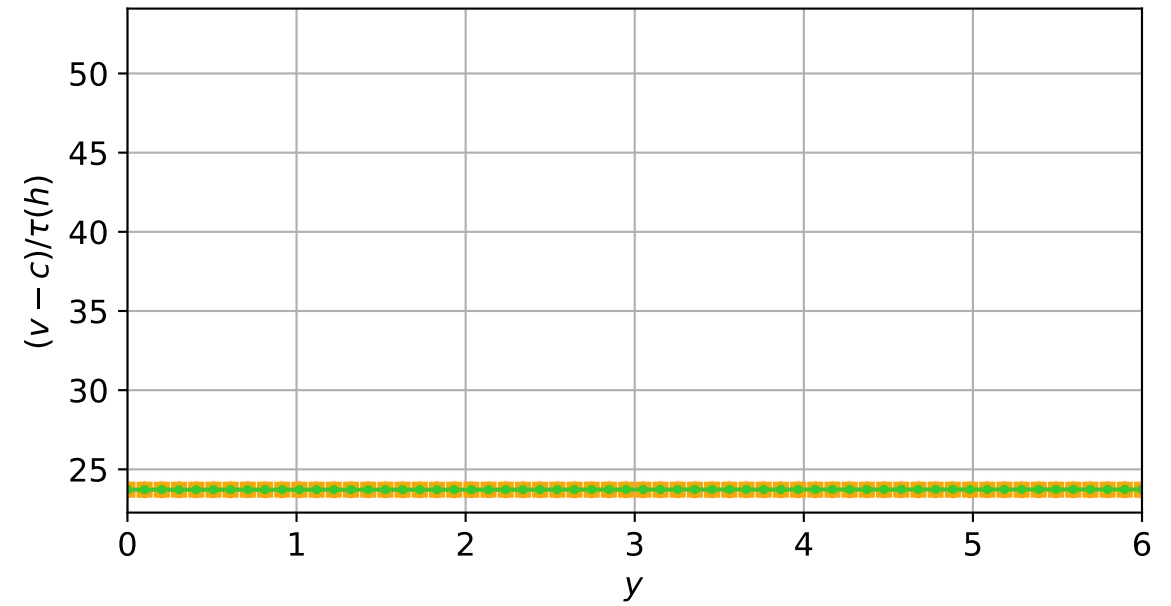
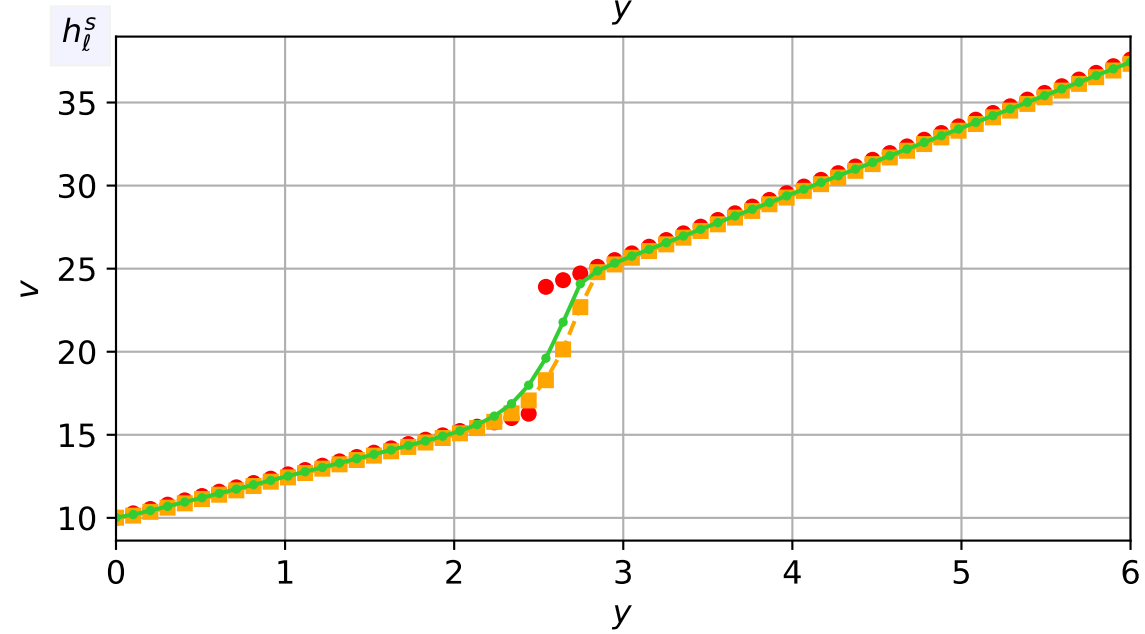
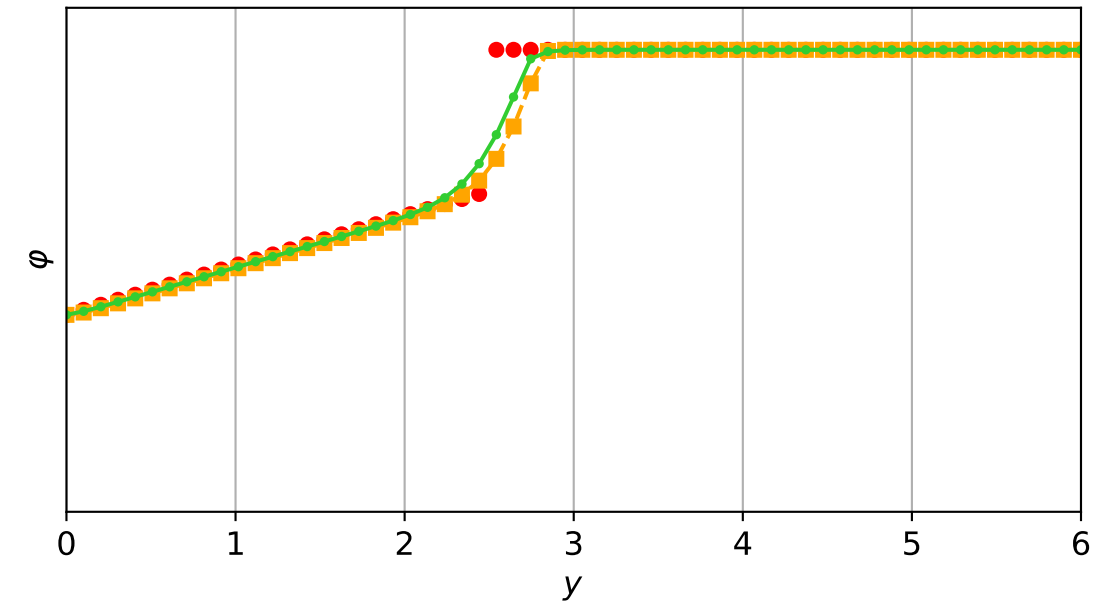
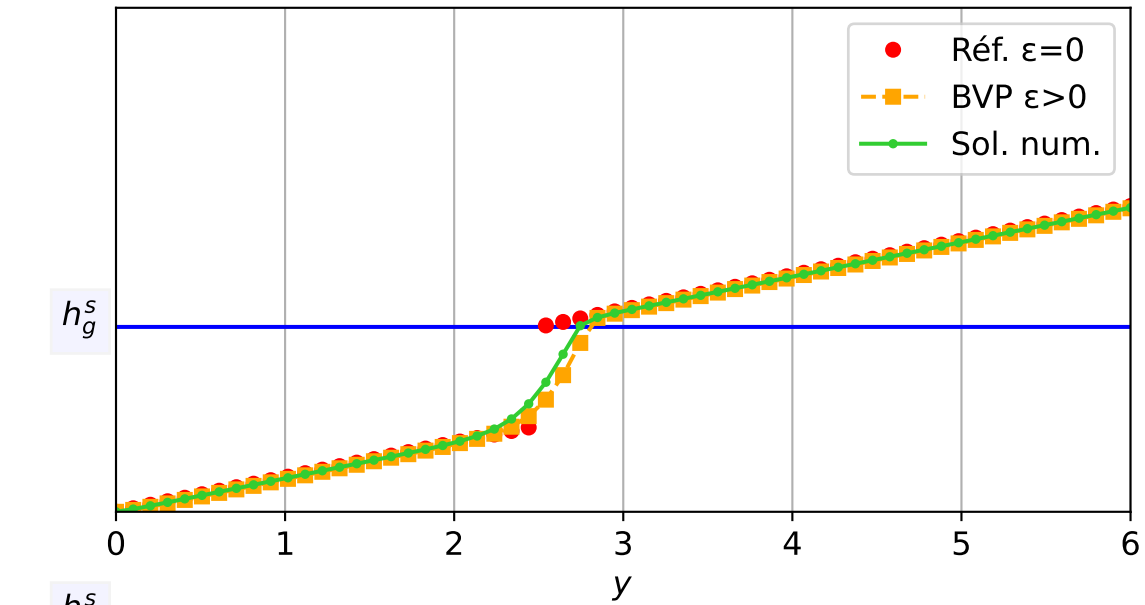




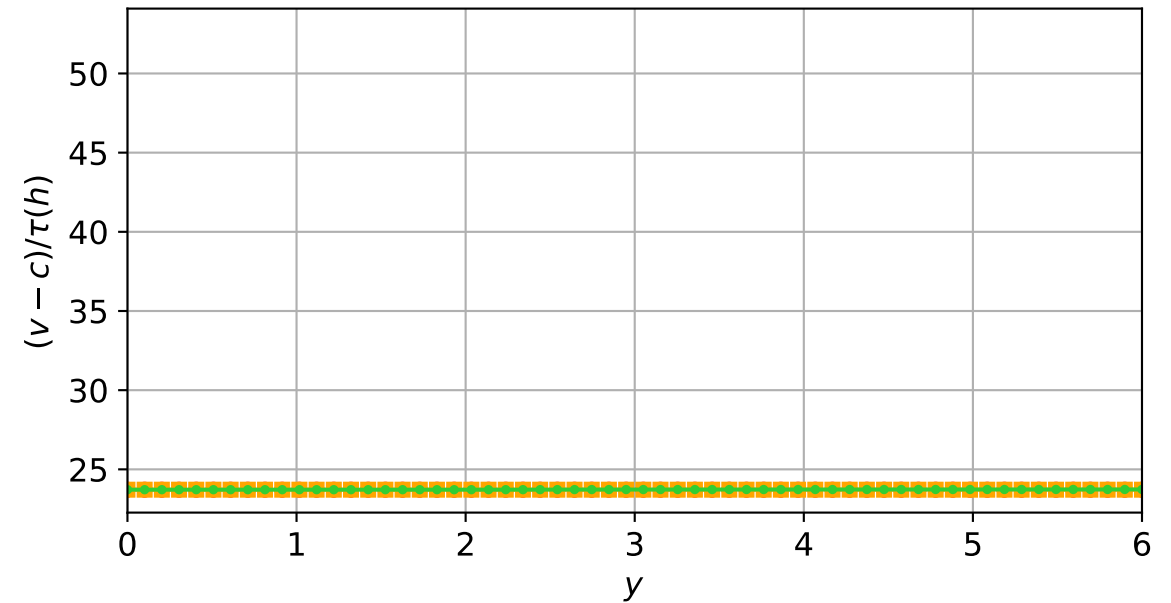
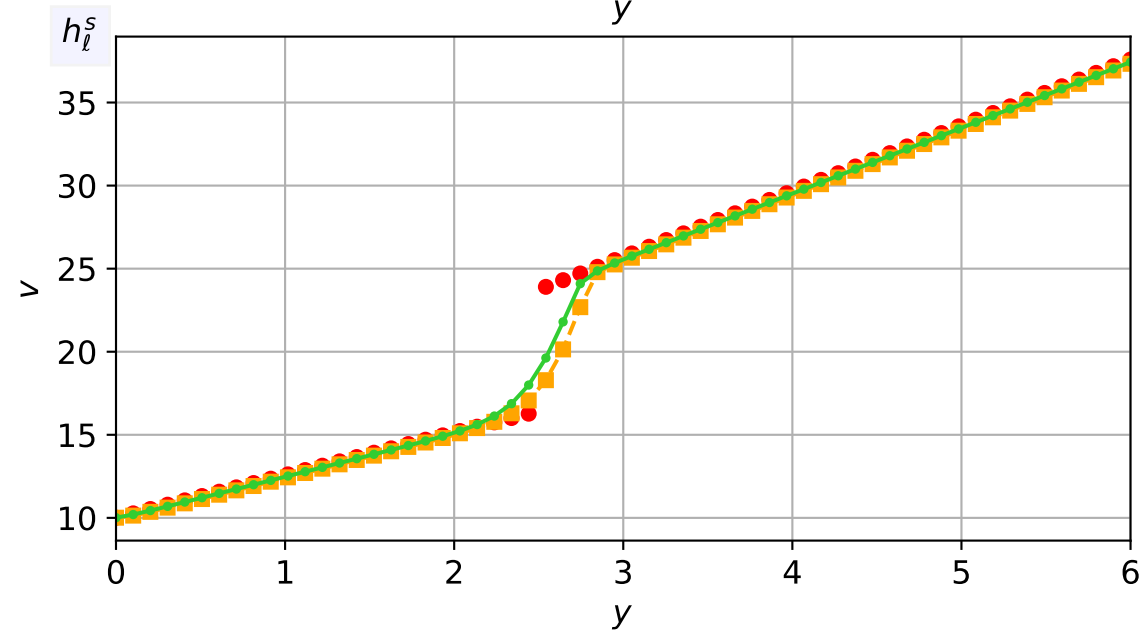
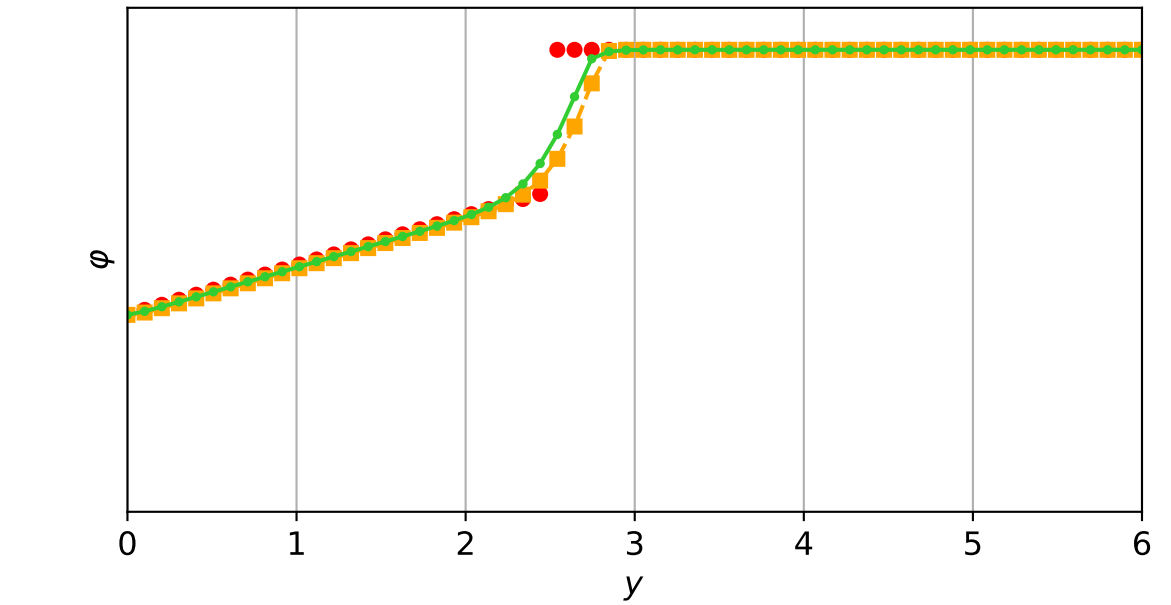
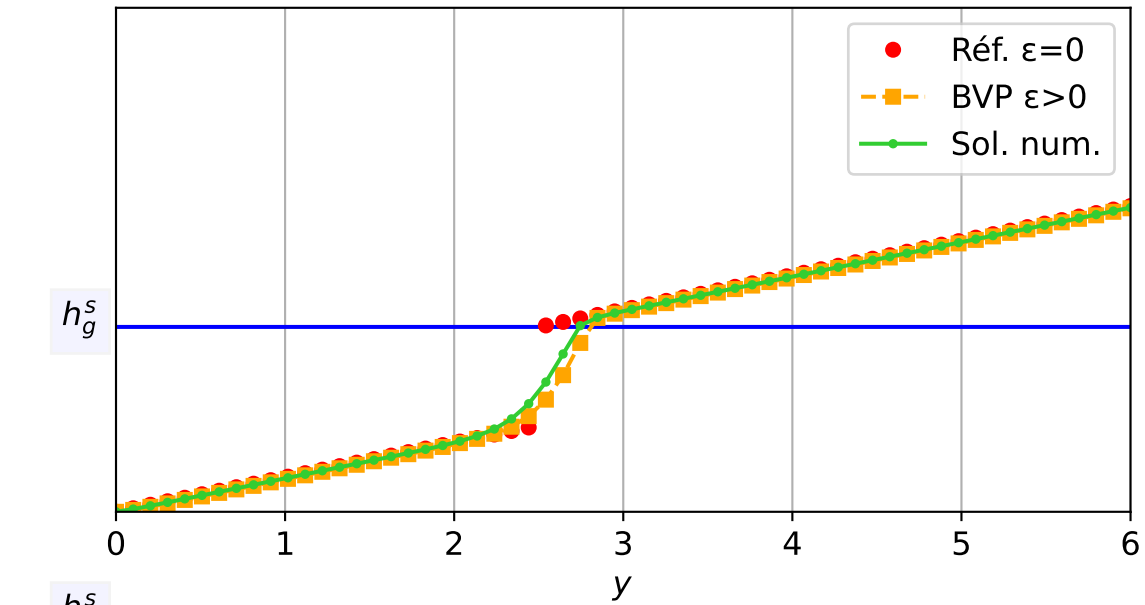
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 8.09e-04$  —  $t = 2.35/12$  (7382 iter) — résidu eq. =  $9.11e-06$



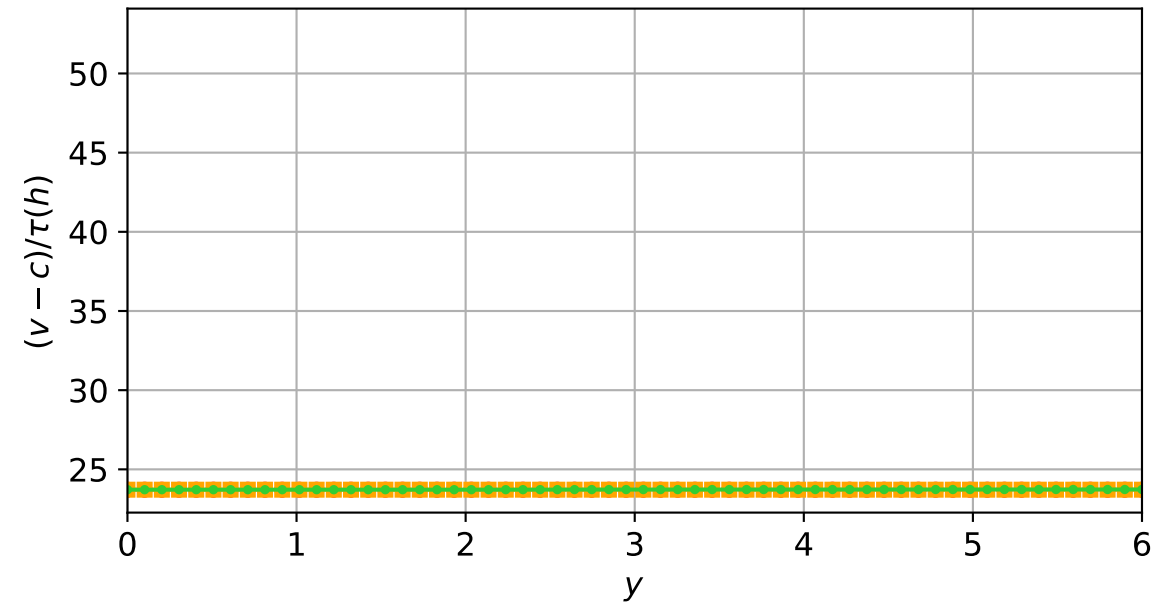
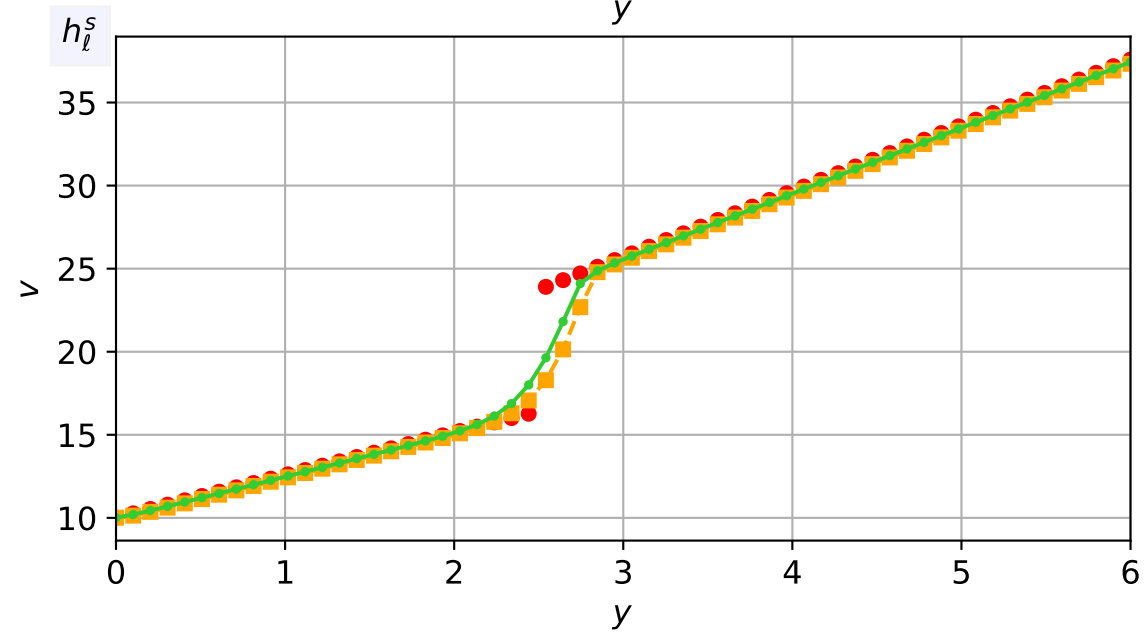
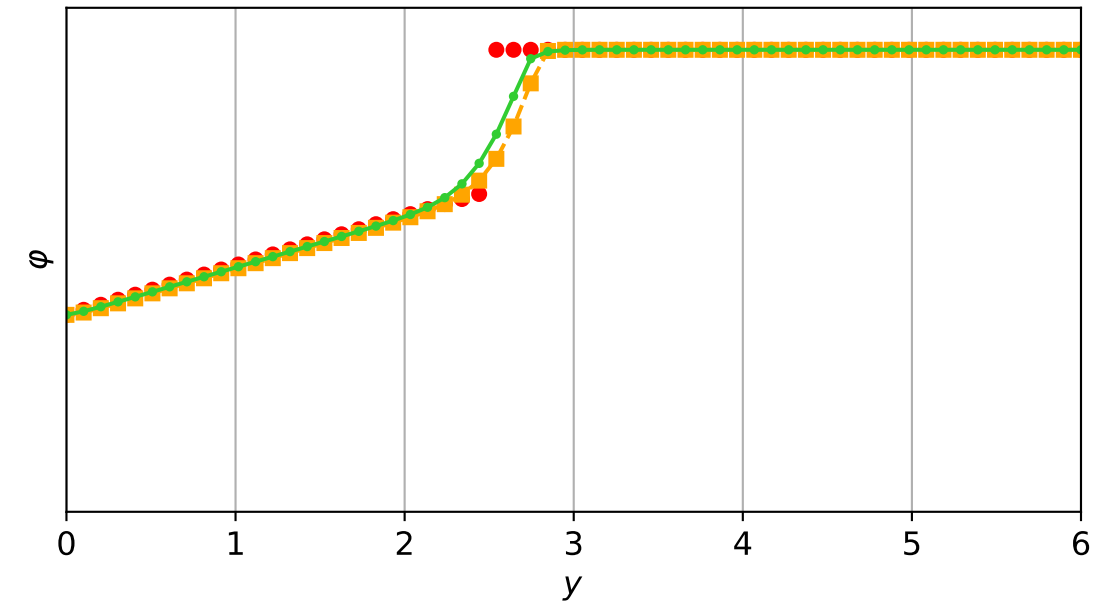
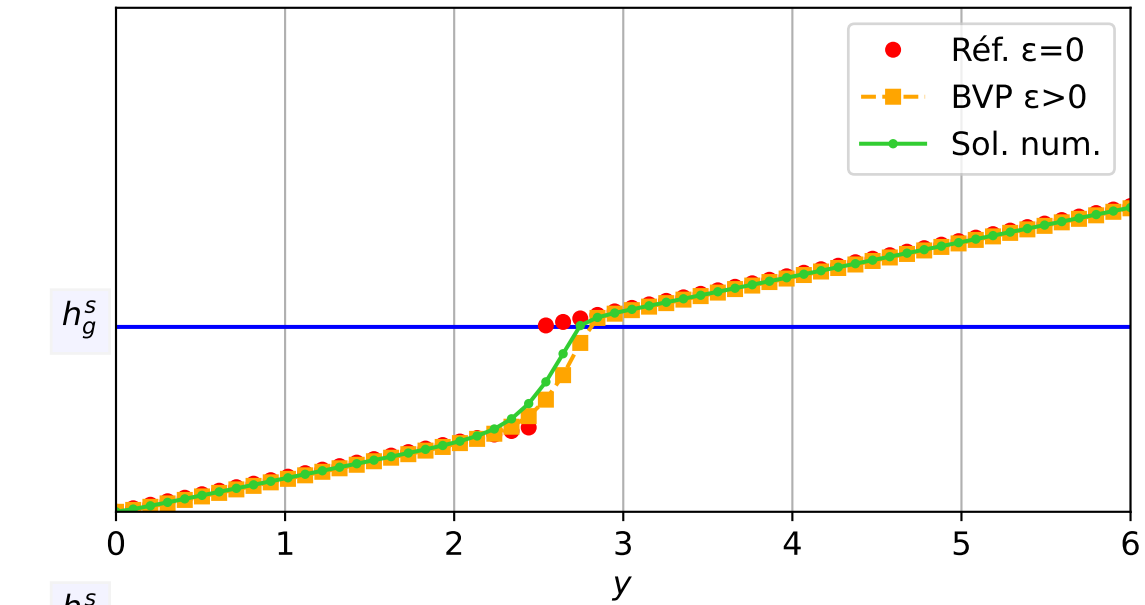
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 8.09e-04$  —  $t = 2.40019/12$  (7444 iter) — résidu eq. =  $8.05e-06$



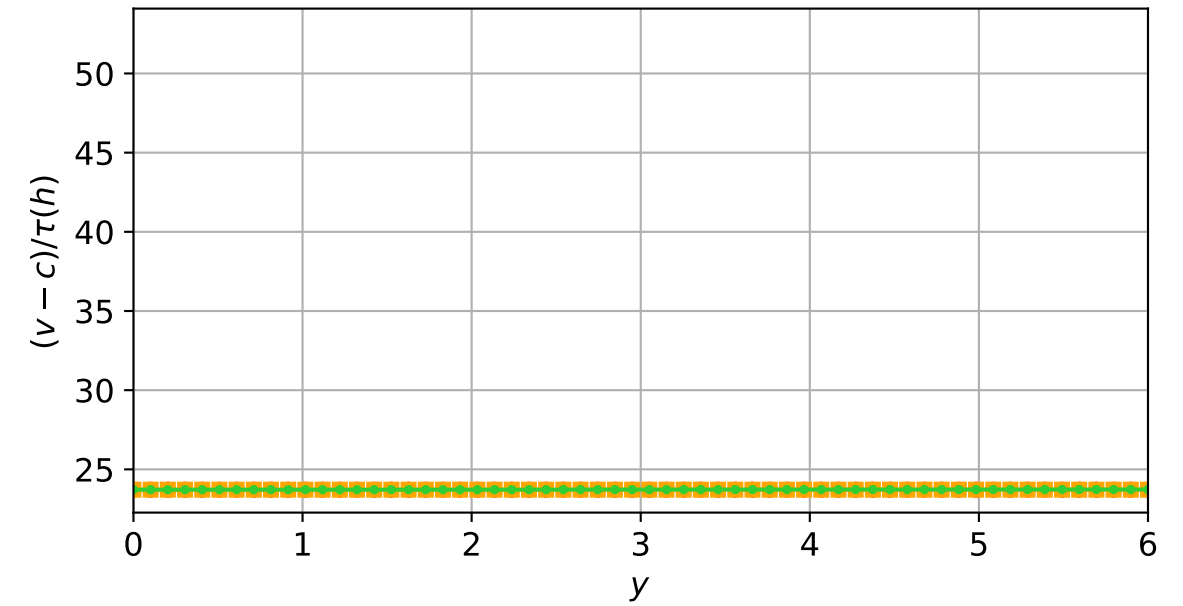
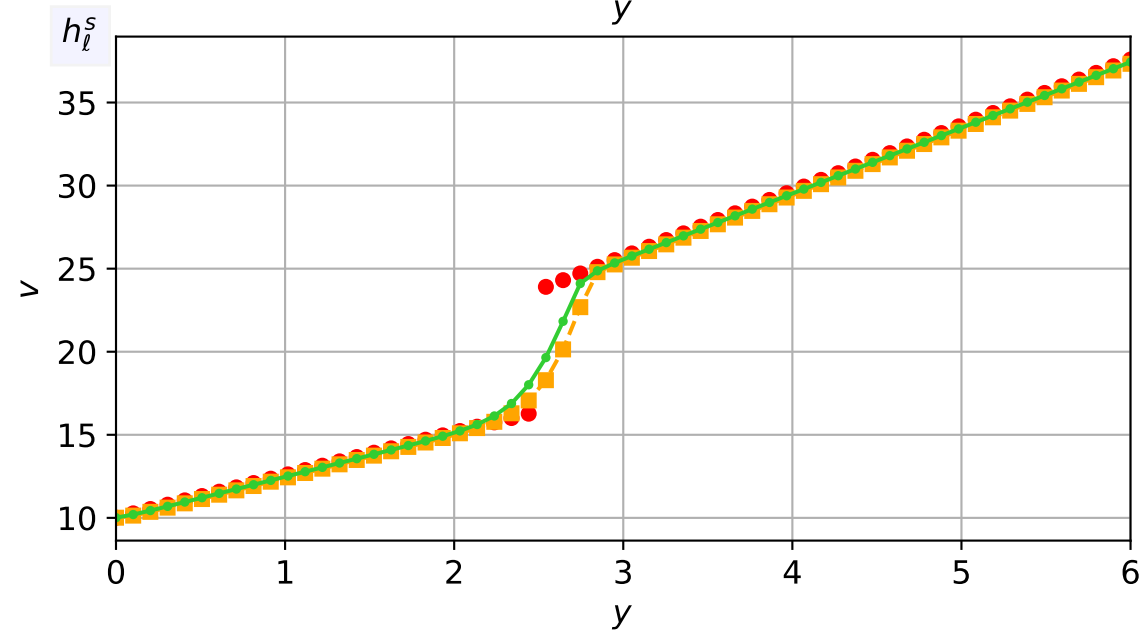
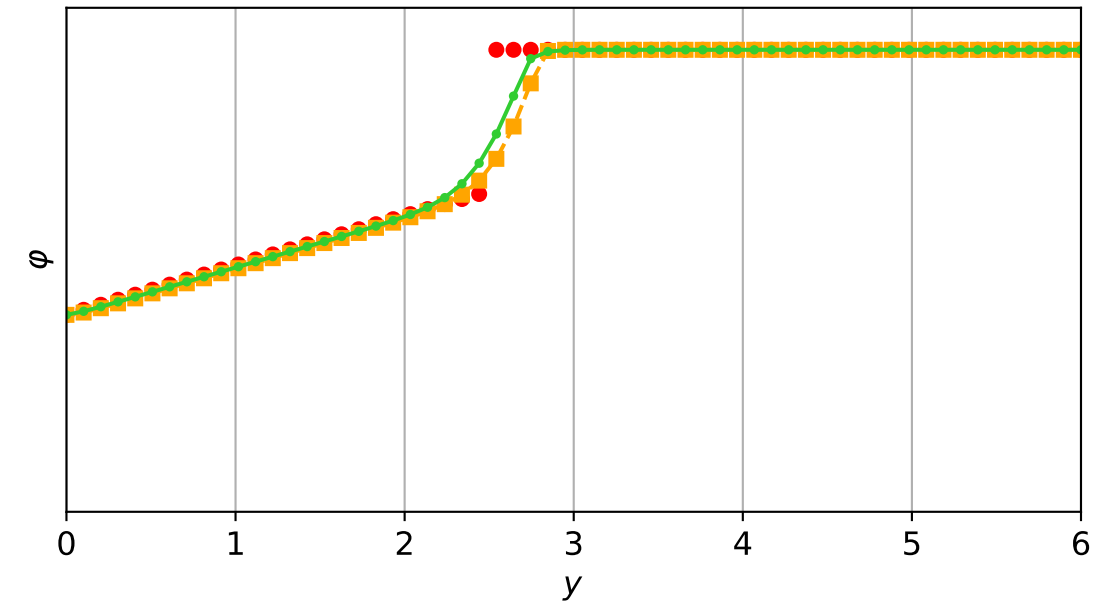
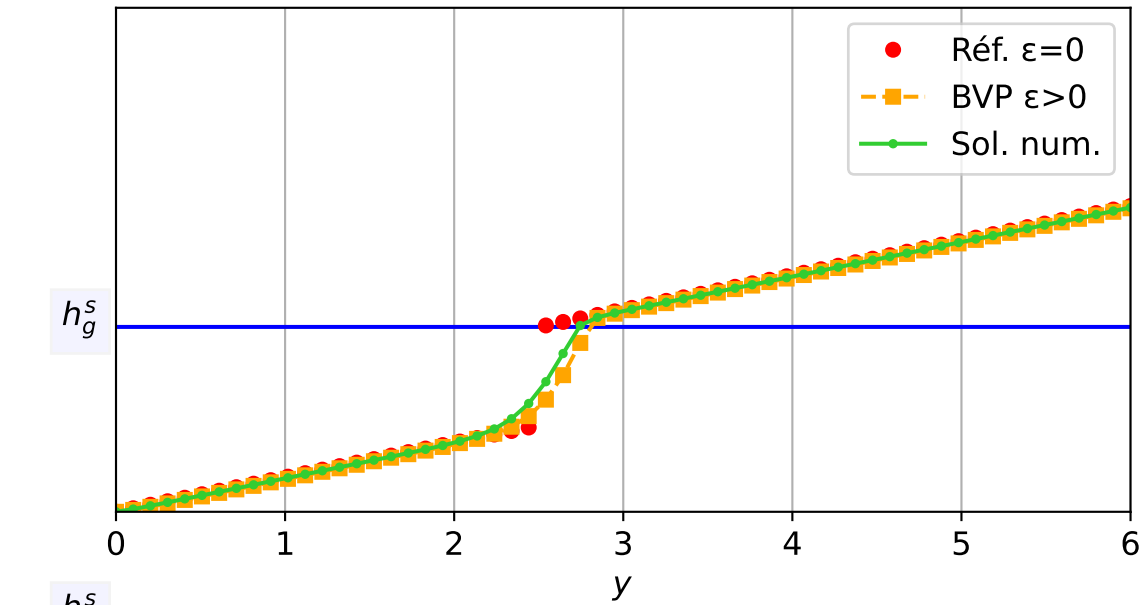
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 8.09e-04$  —  $t = 2.45037/12$  (7506 iter) — résidu eq. =  $7.10e-06$



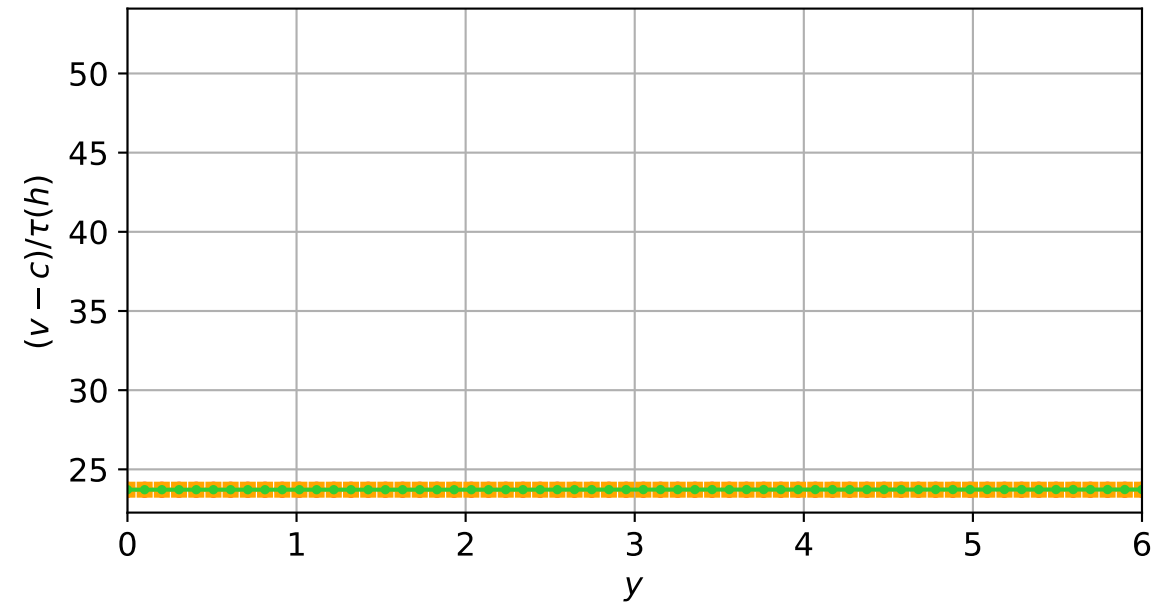
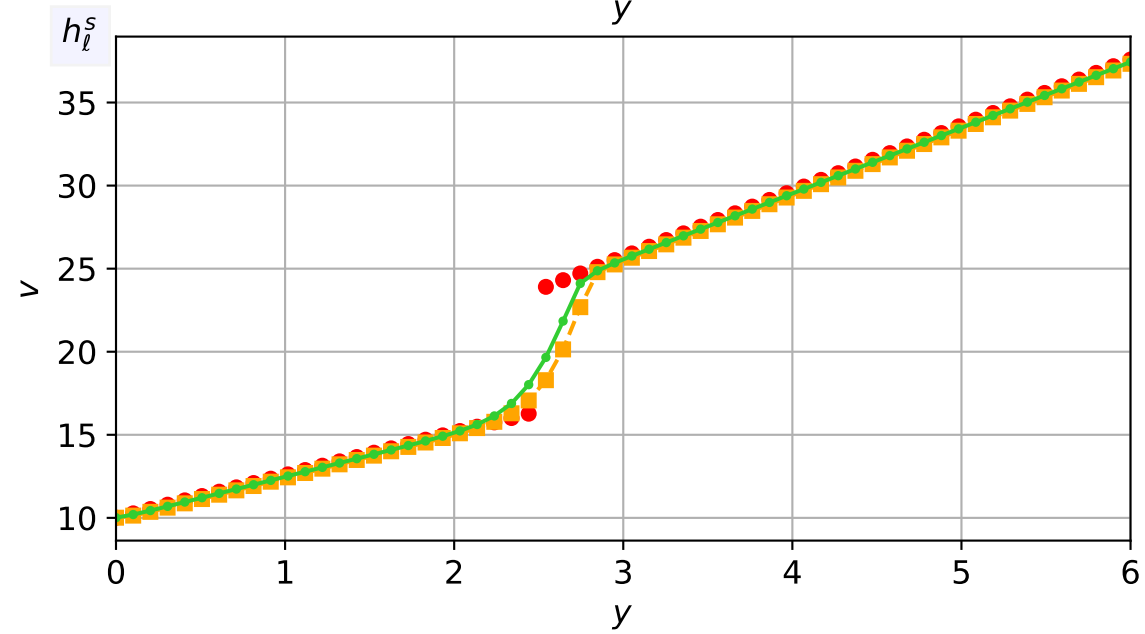
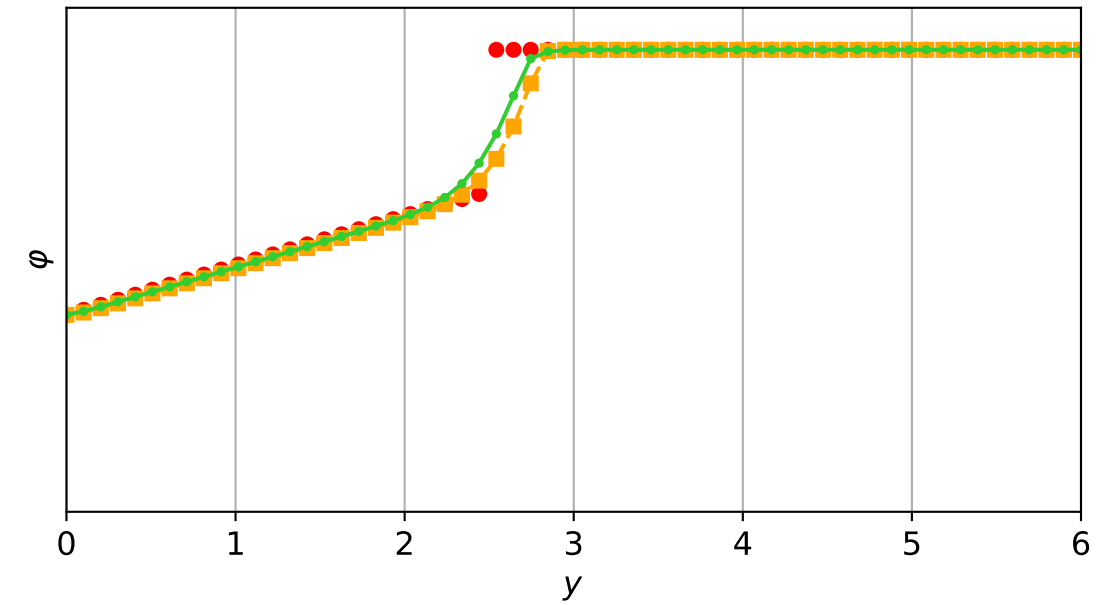
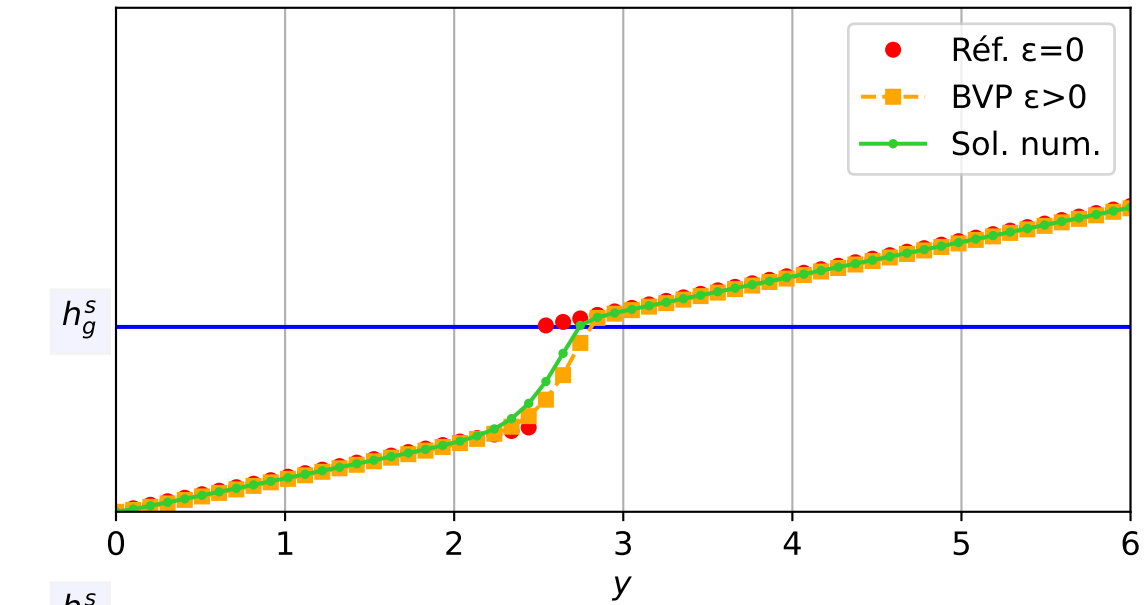
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 9.71e-04$  —  $t = 2.50024/12$  (7559 iter) — résidu eq. =  $8.52e-06$



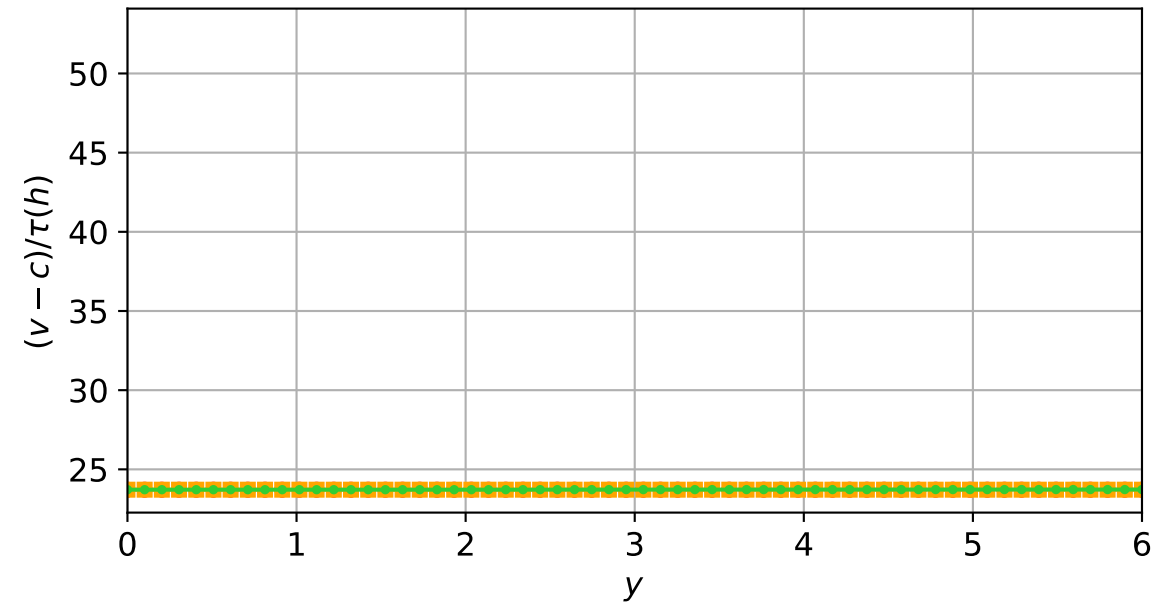
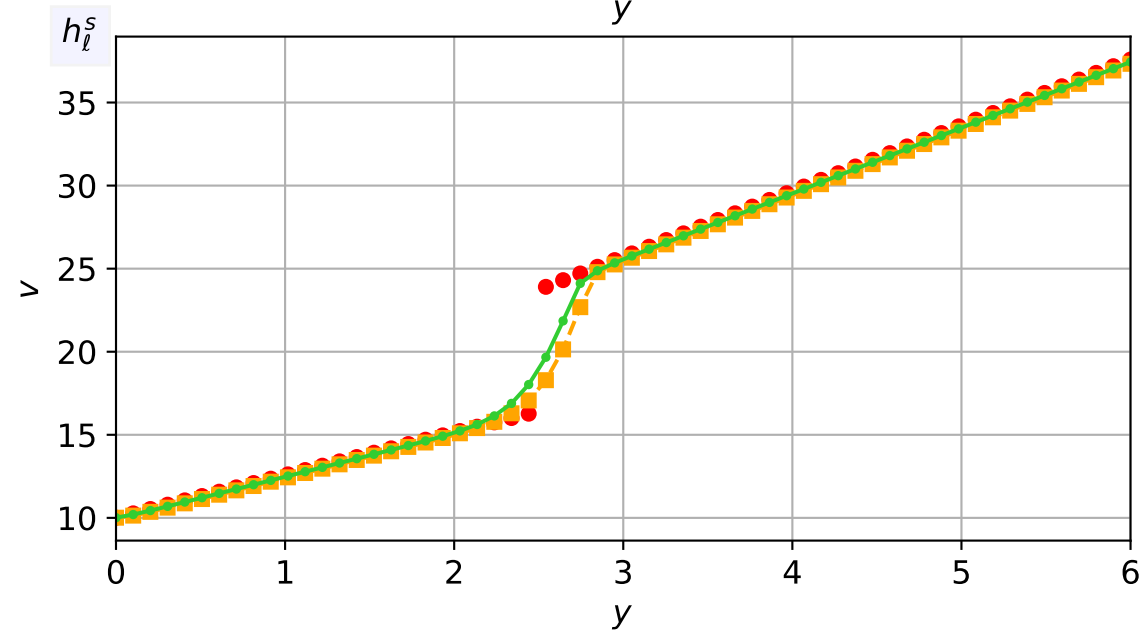
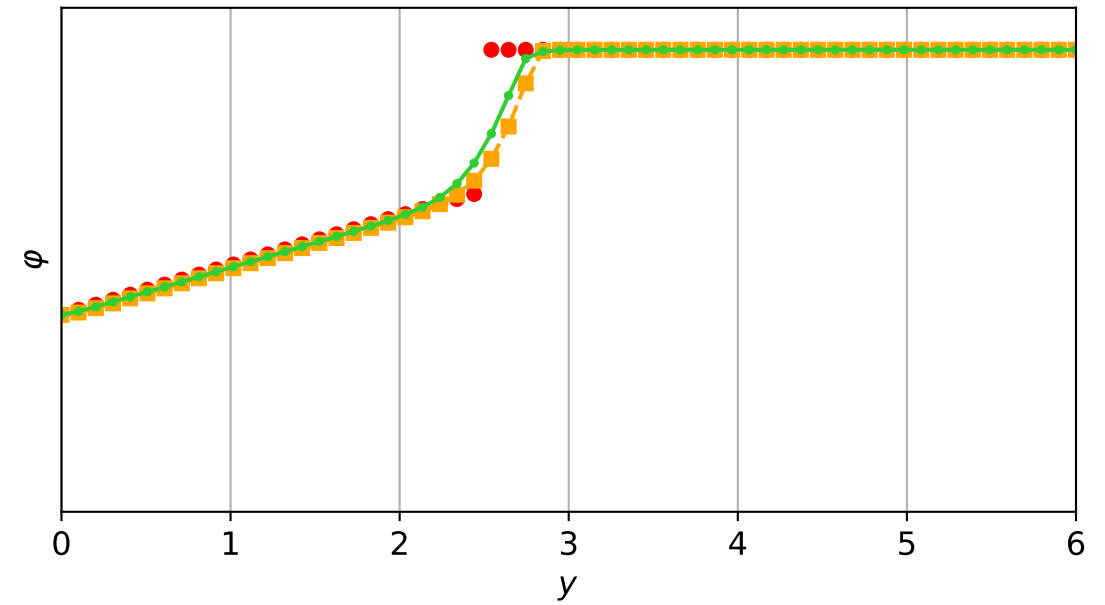
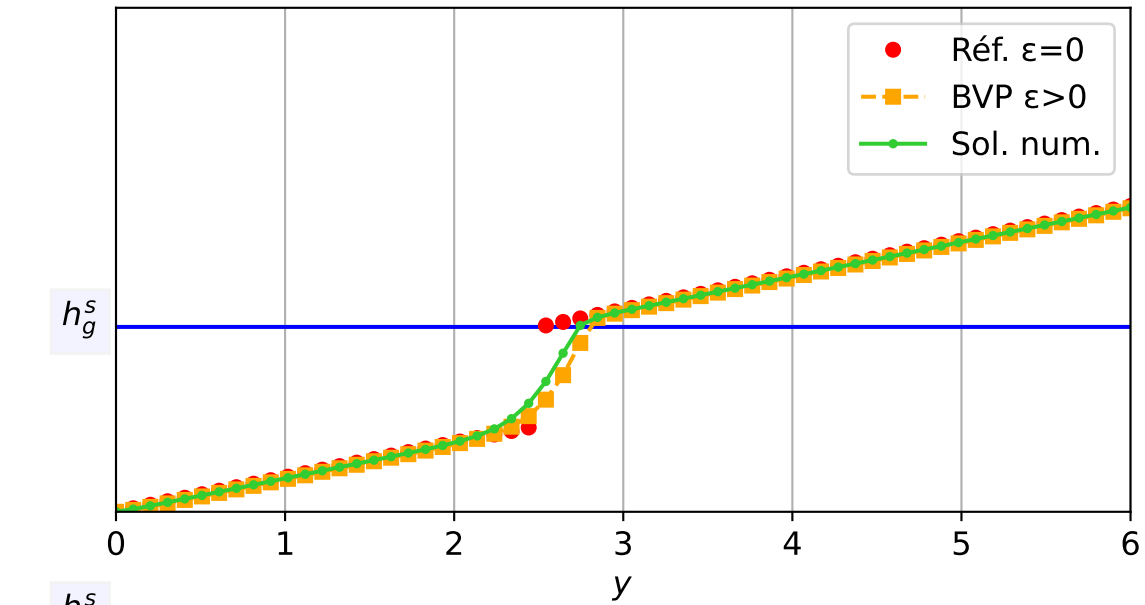
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 9.71e-04$  —  $t = 2.55074/12$  (7611 iter) — résidu eq. =  $7.52e-06$



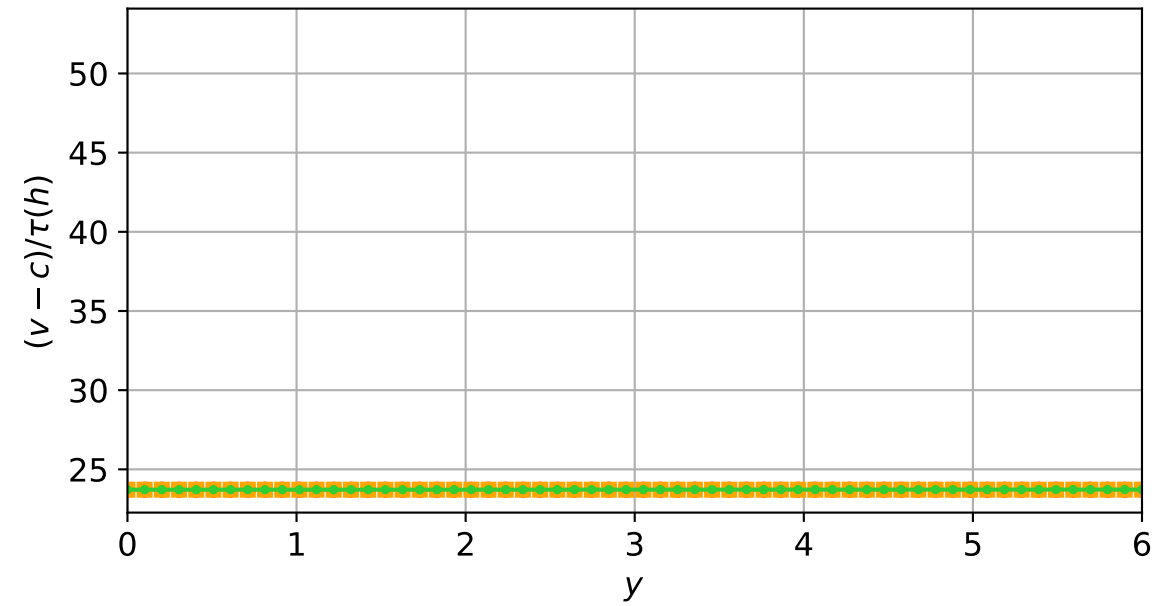
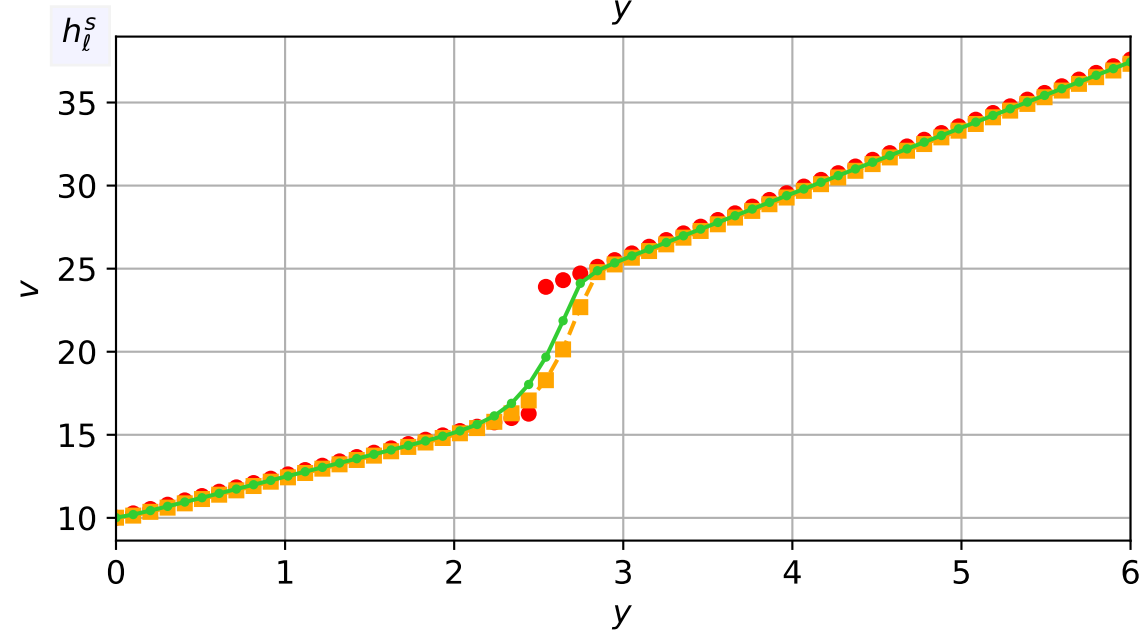
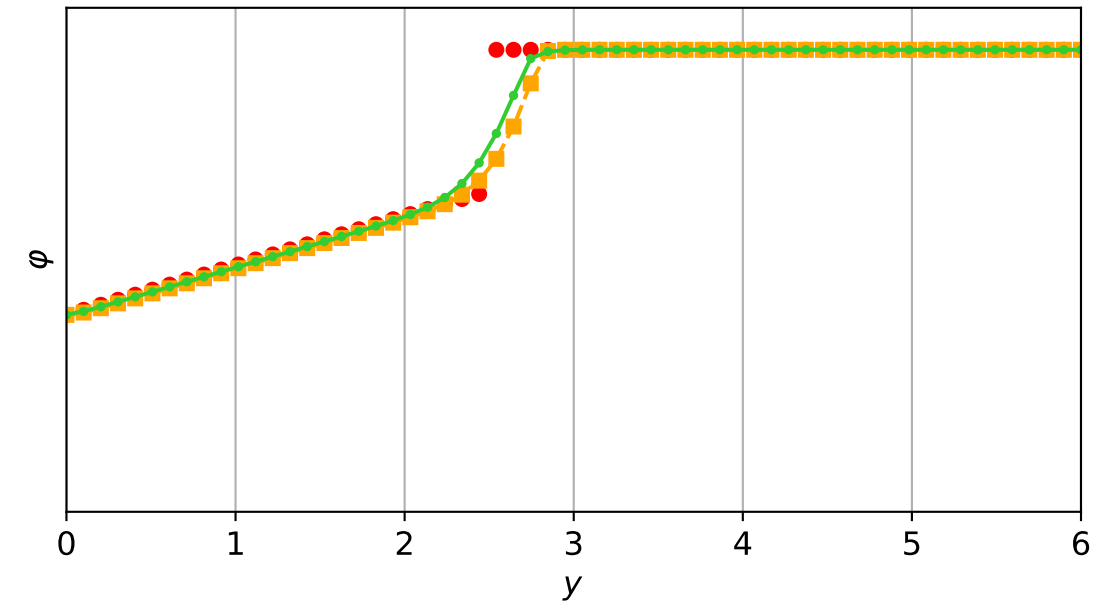
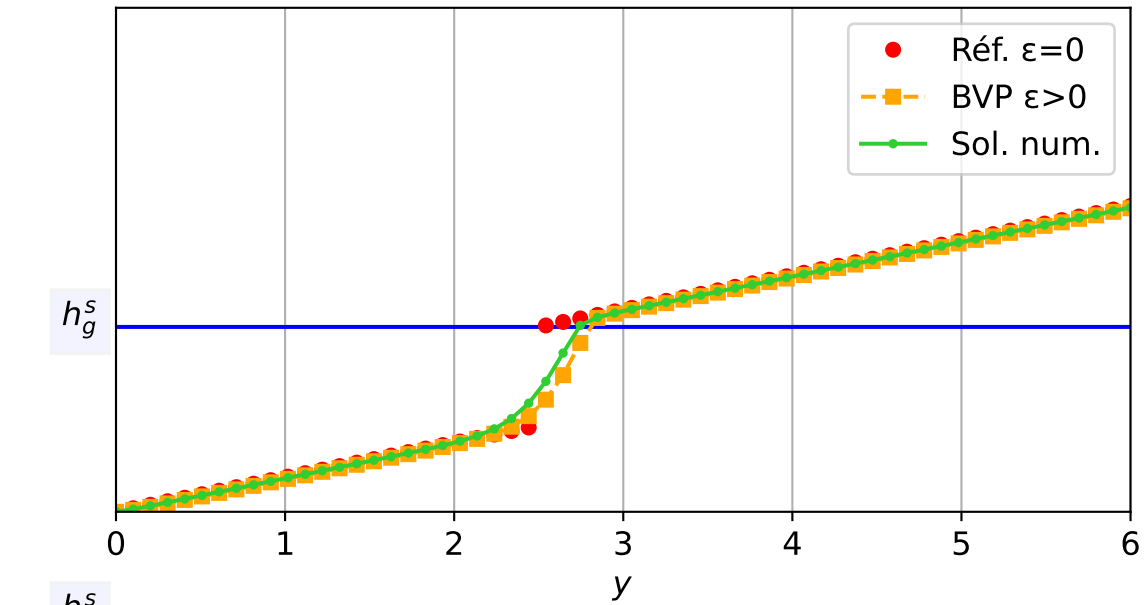
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.17e-03$  —  $t = 2.60048/12$  (7657 iter) — résidu eq. =  $8.90e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.17e-03$  —  $t = 2.6506/12$  (7700 iter) — résidu eq. =  $7.86e-06$

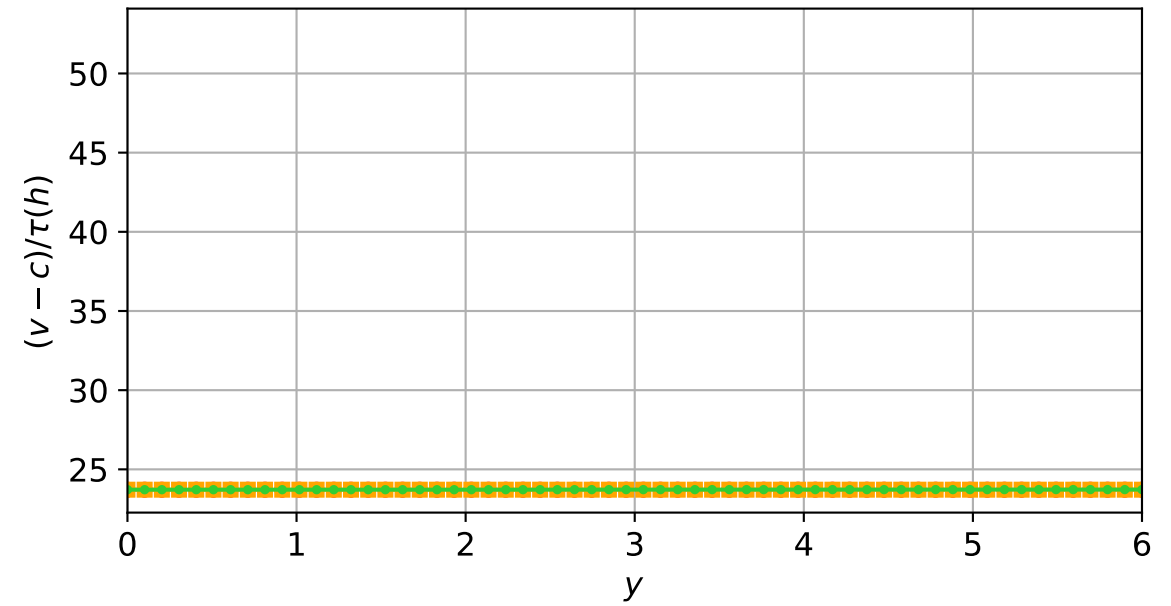
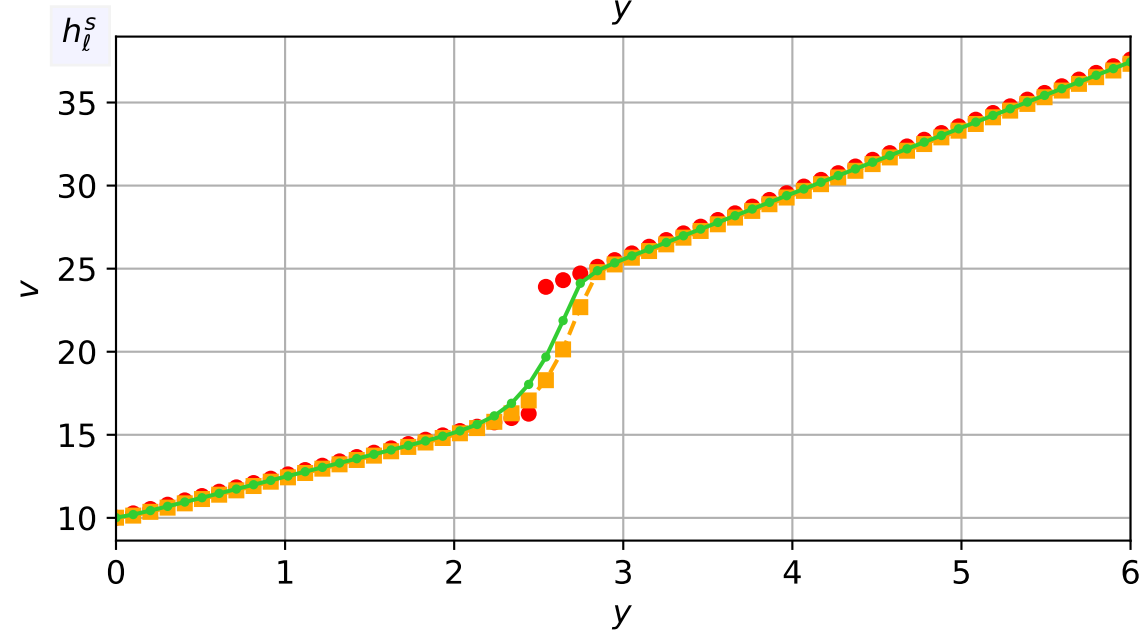
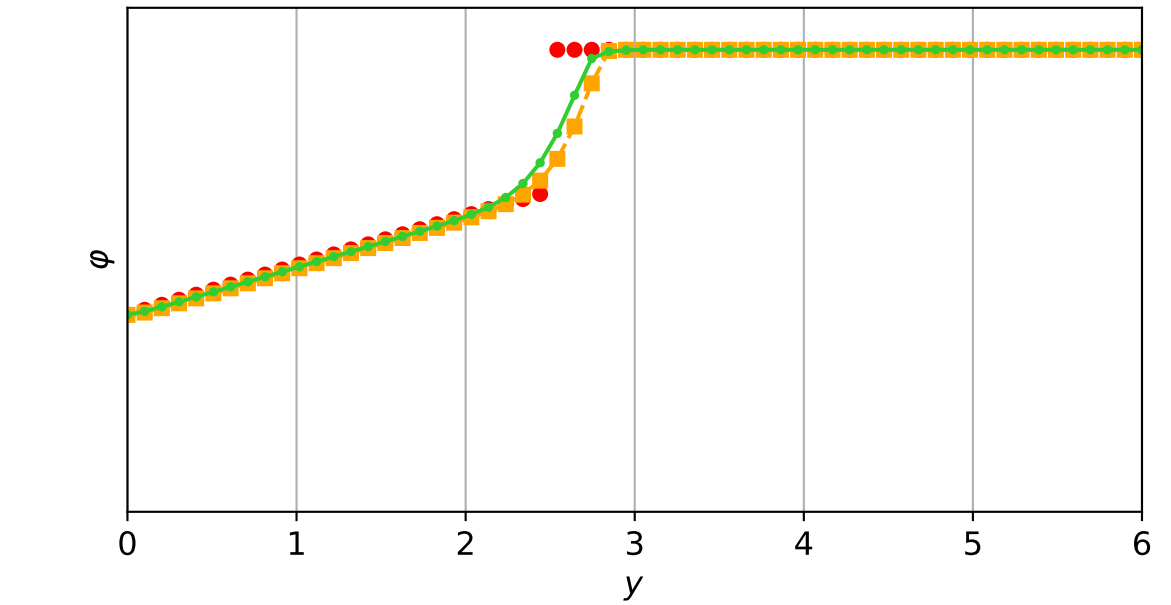
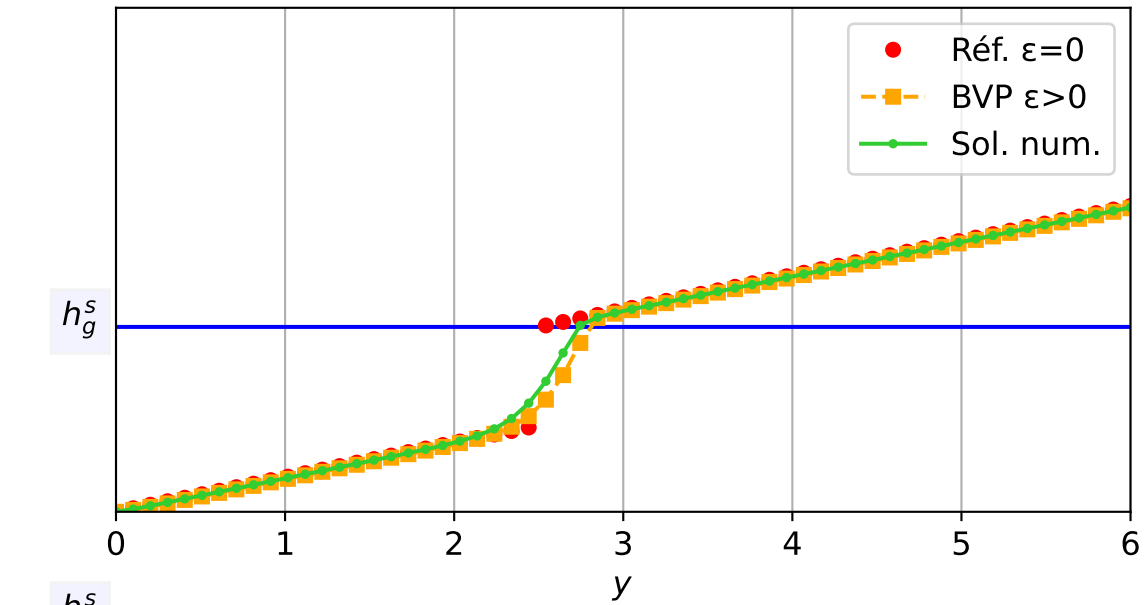


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
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 $dt = 1.40e-03$  —  $t = 2.70119/12$  (7740 iter) — résidu eq. =  $9.17e-06$

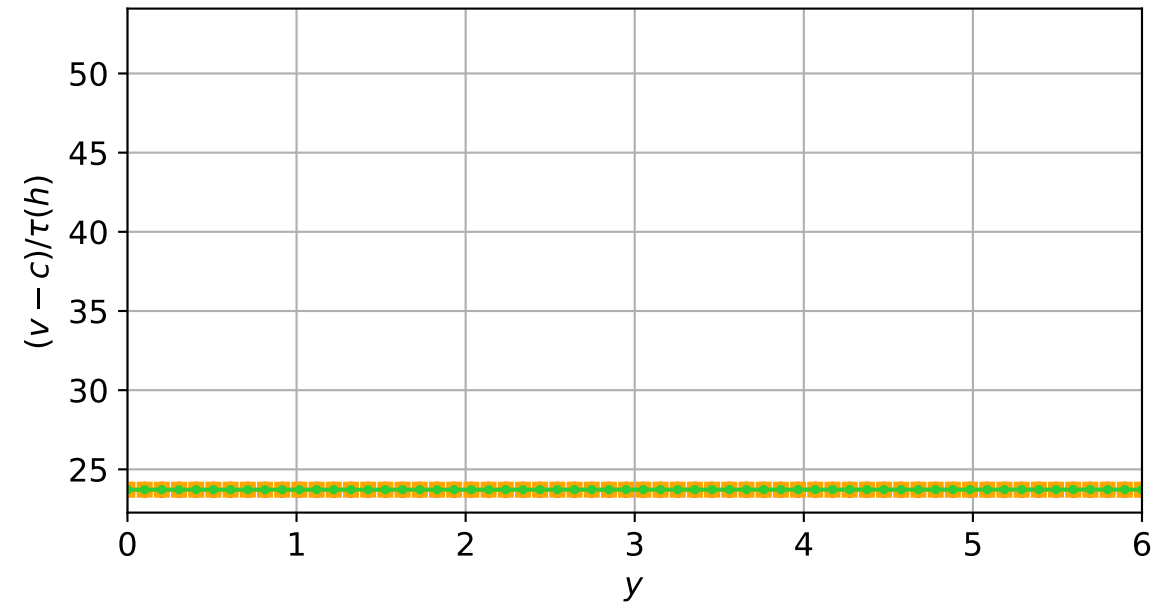
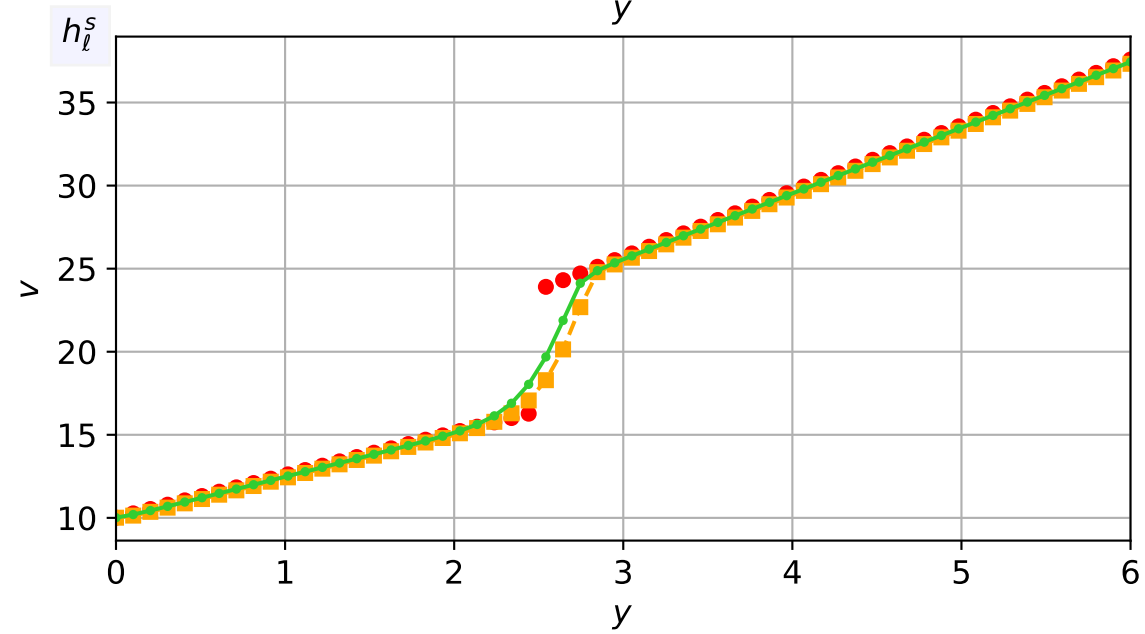
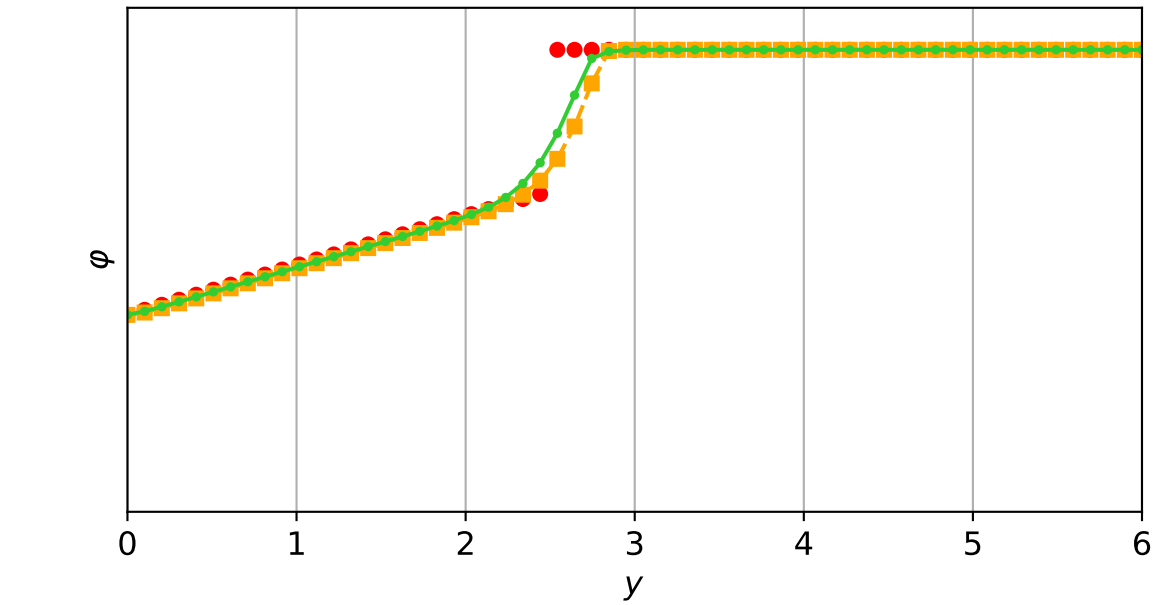
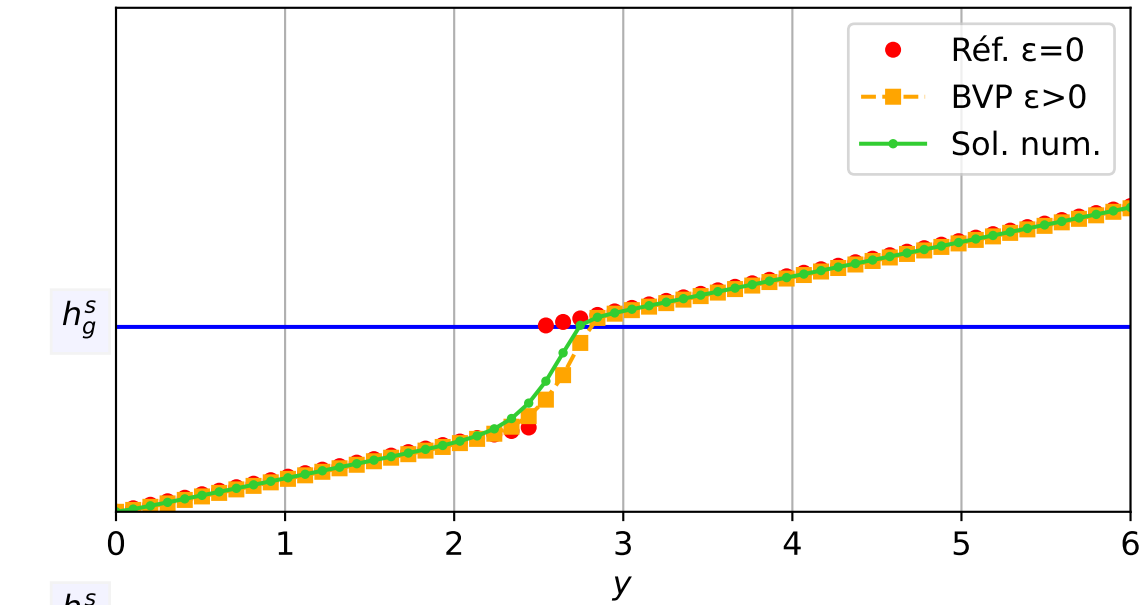




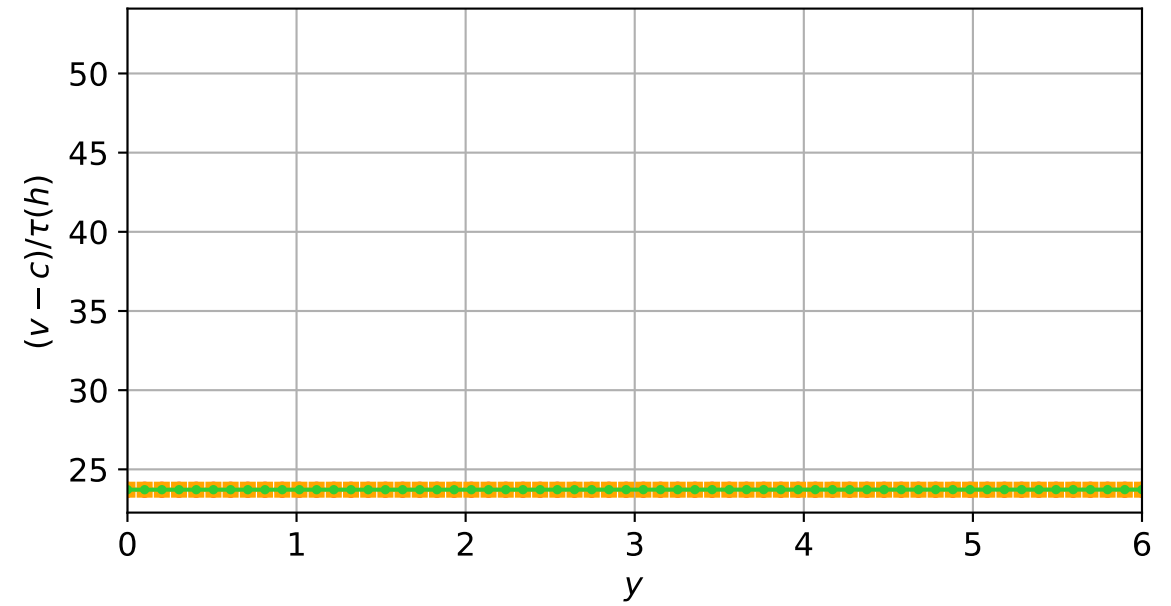
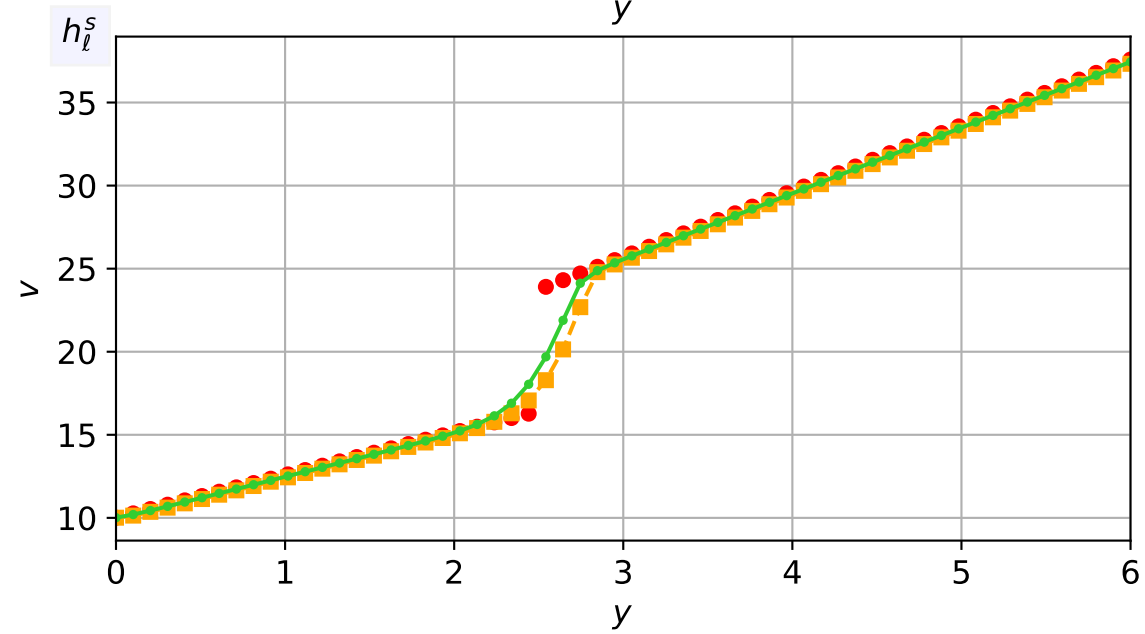
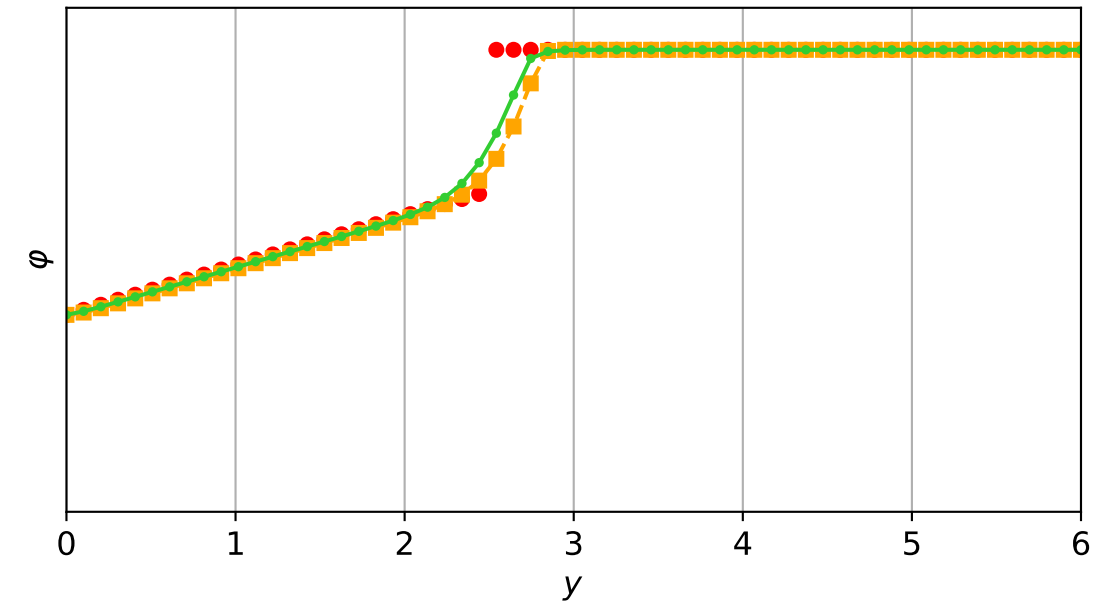
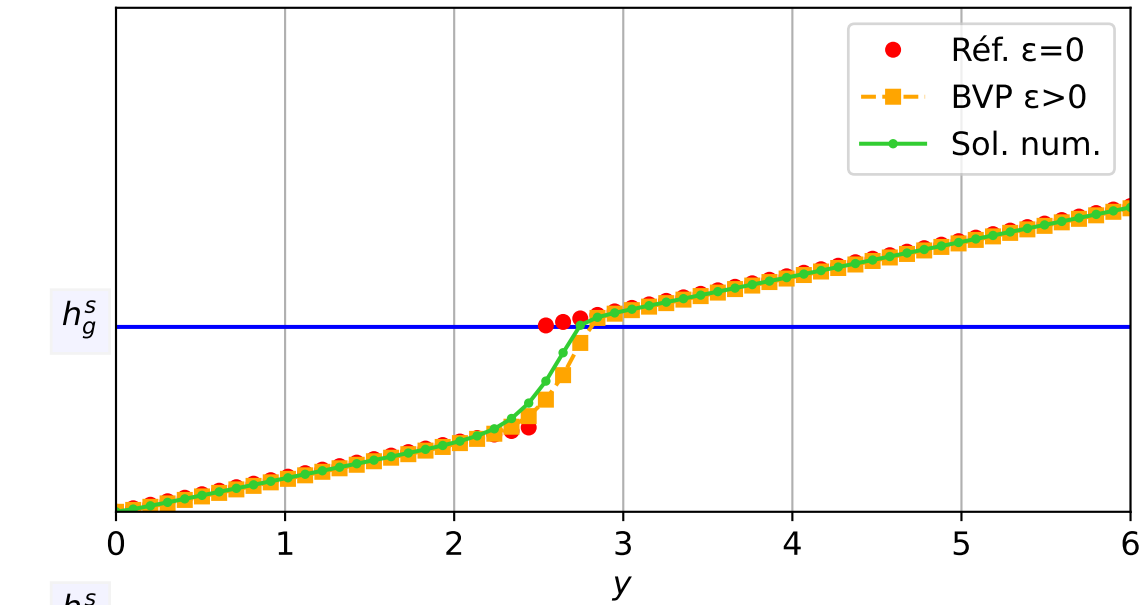
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.40e-03$  —  $t = 2.75014/12$  (7775 iter) — résidu eq. =  $8.14e-06$



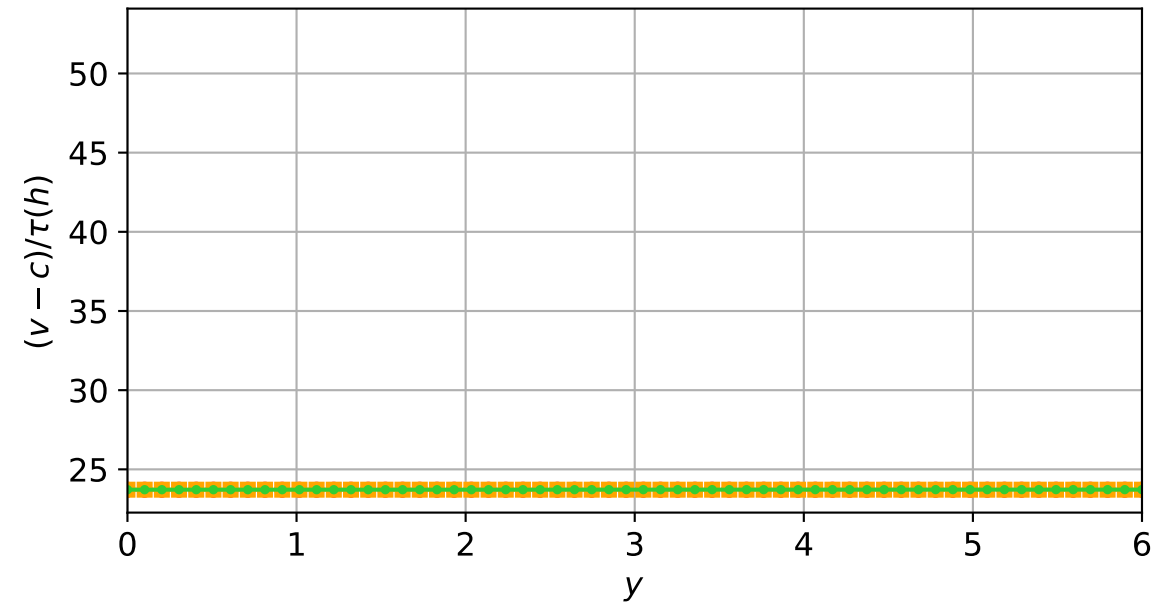
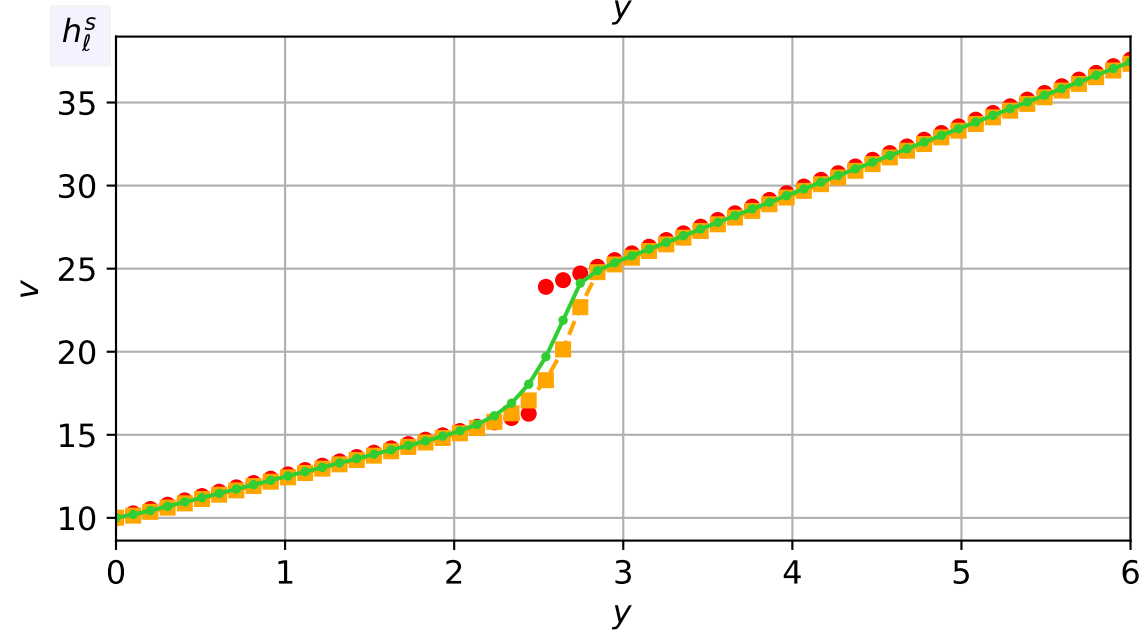
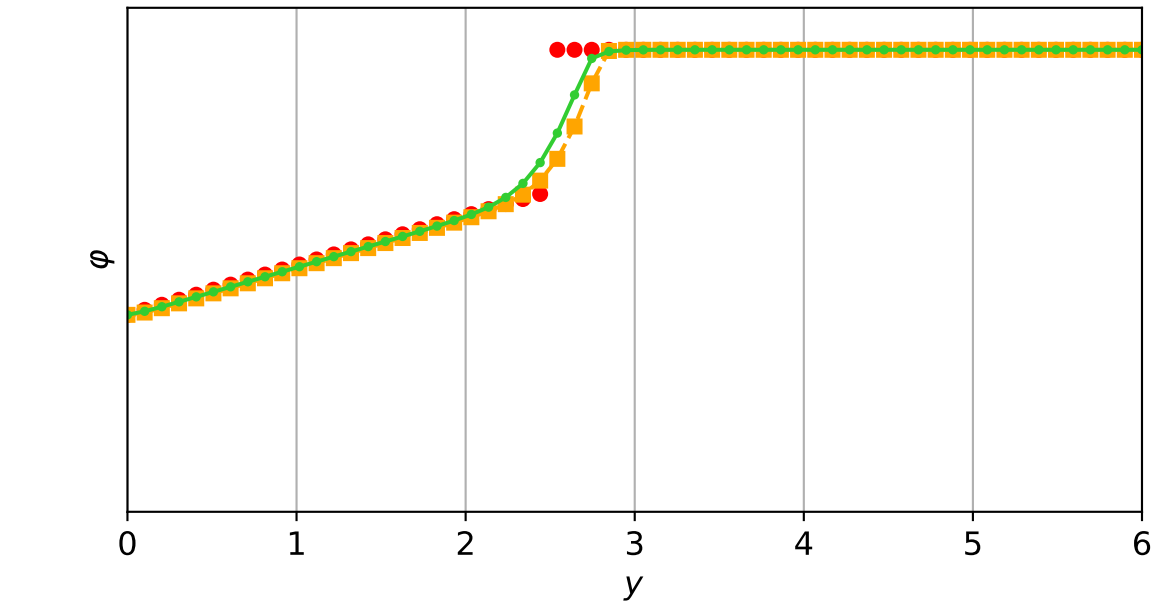
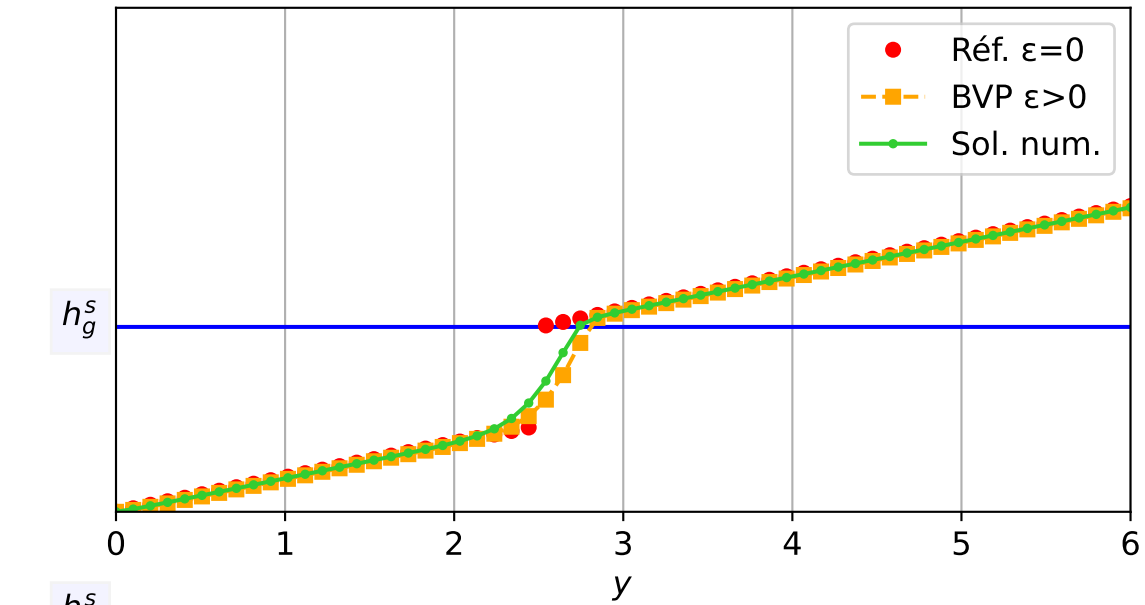
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.68e-03$  —  $t = 2.80106/12$  (7809 iter) — résidu eq. =  $9.38e-06$



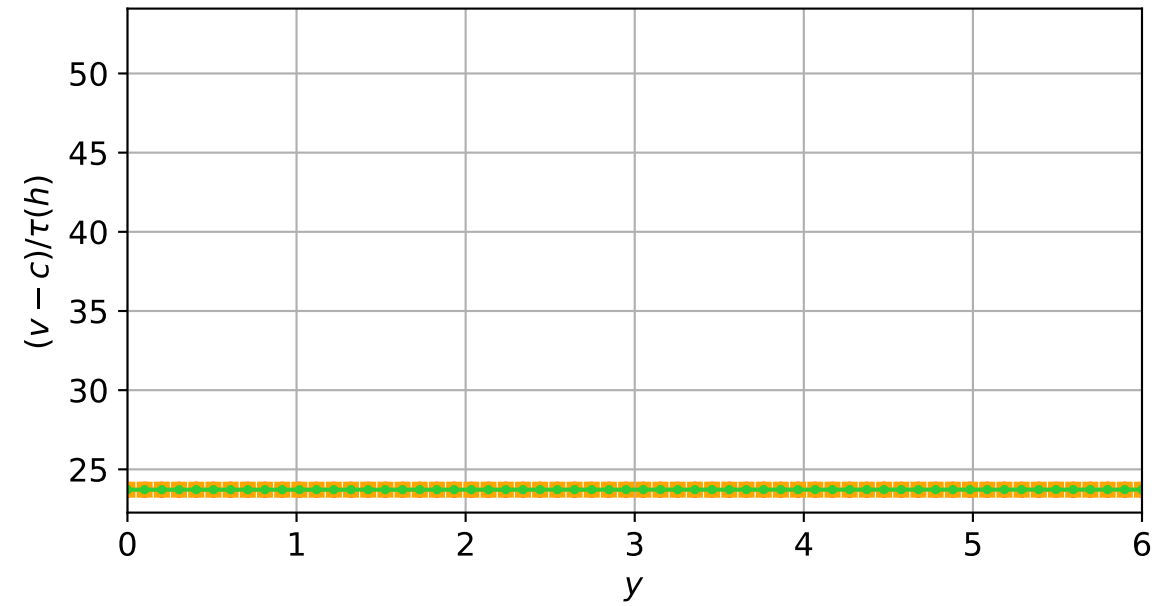
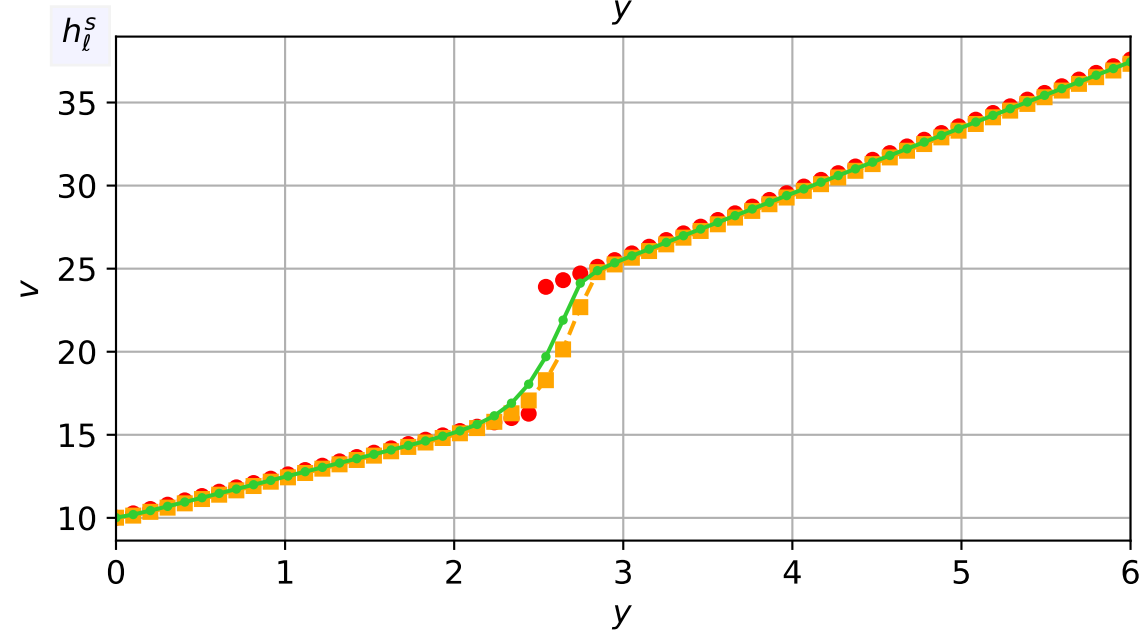
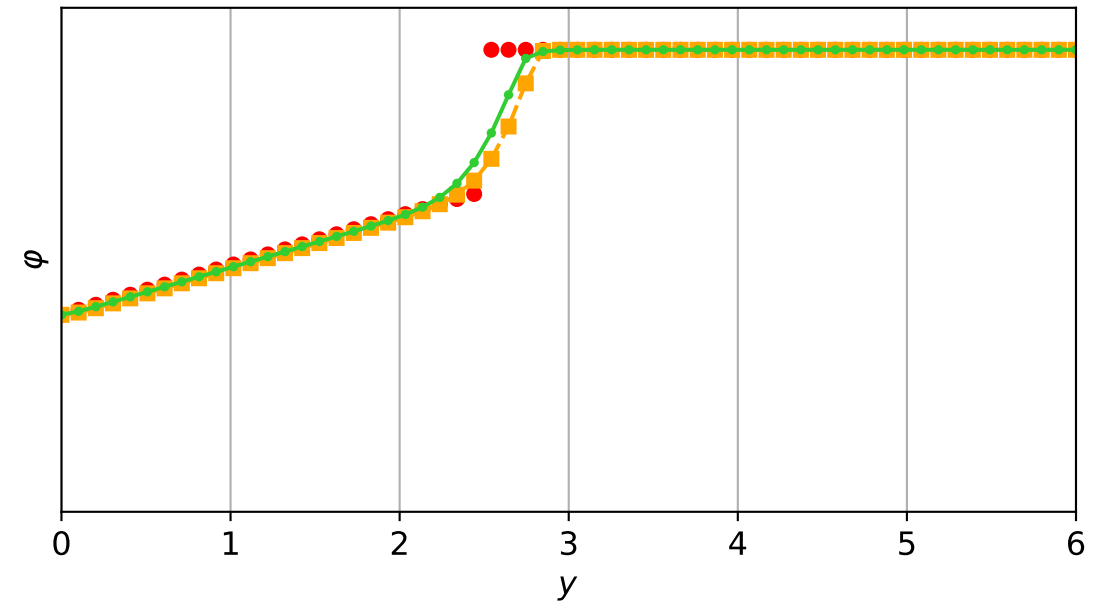
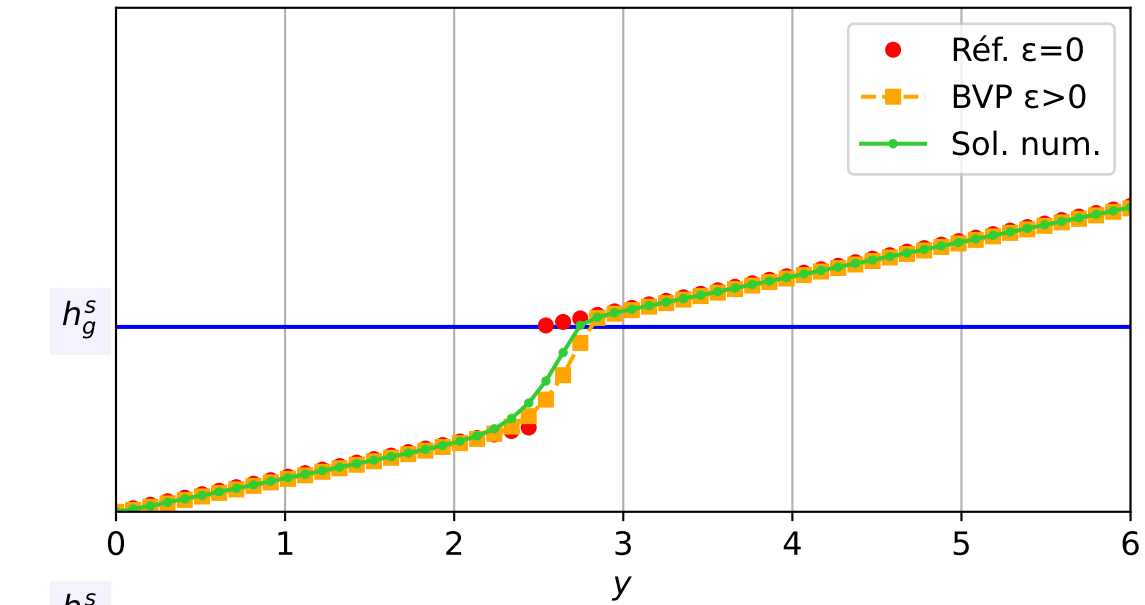
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.68e-03$  —  $t = 2.85141/12$  (7839 iter) — résidu eq. =  $8.31e-06$



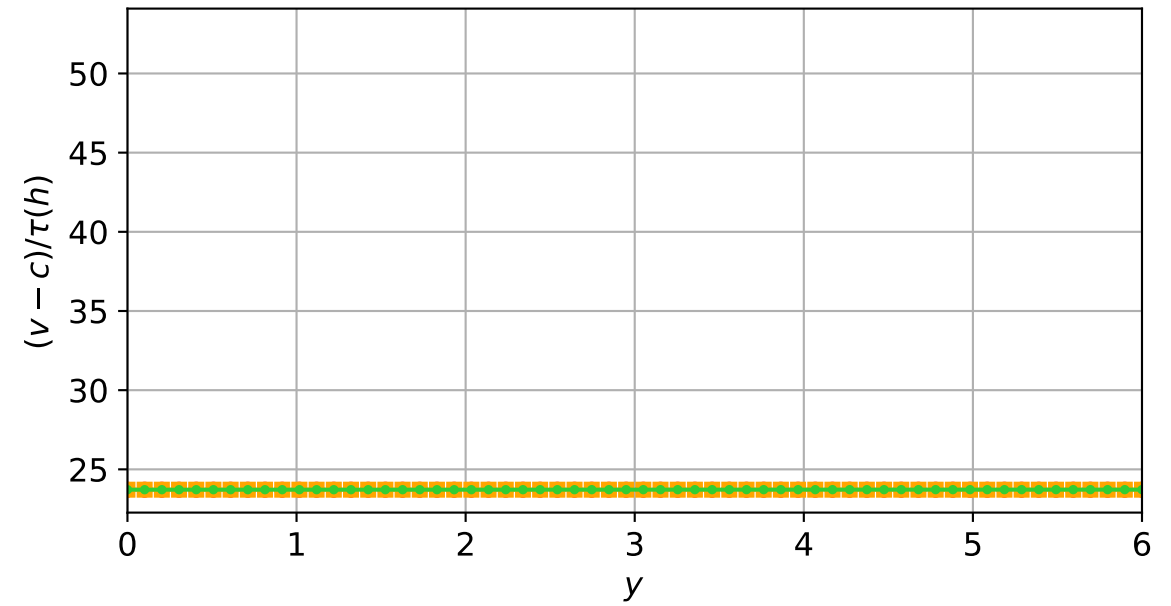
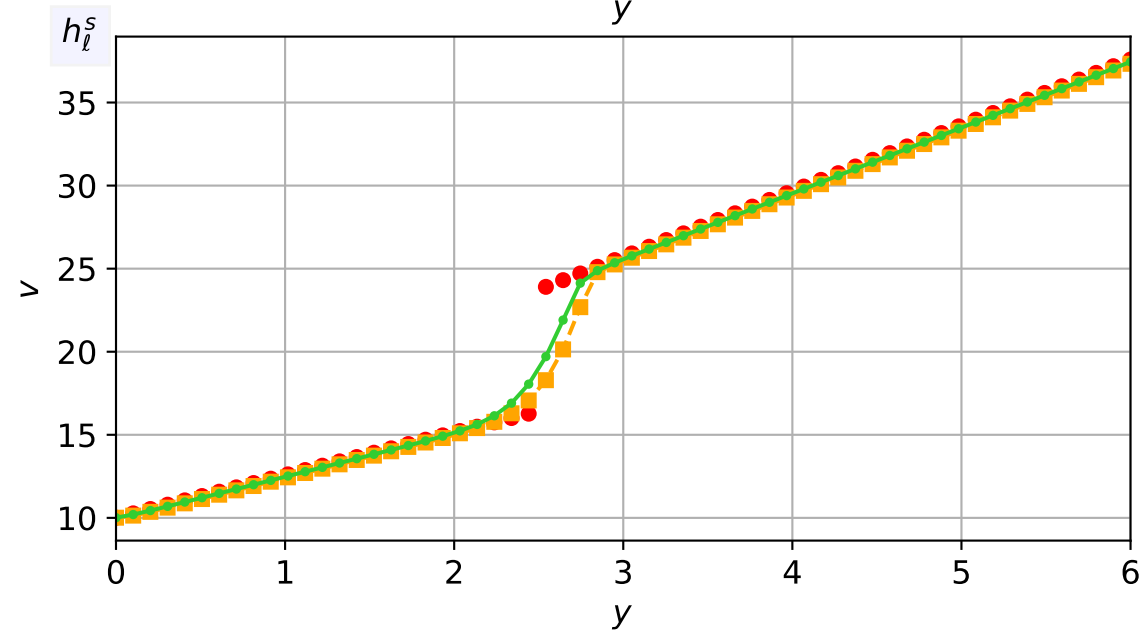
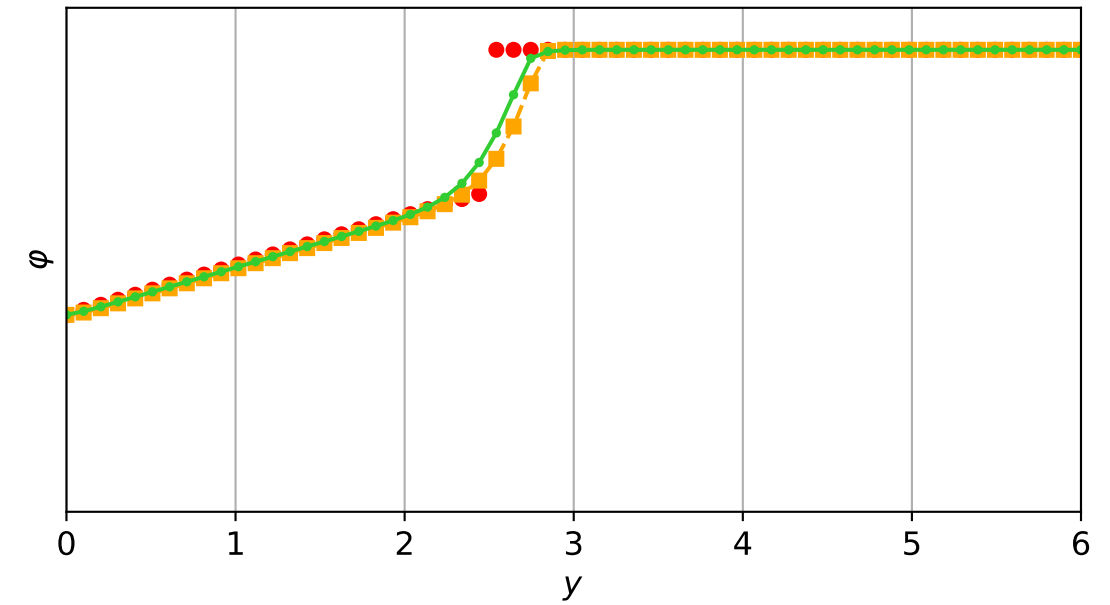
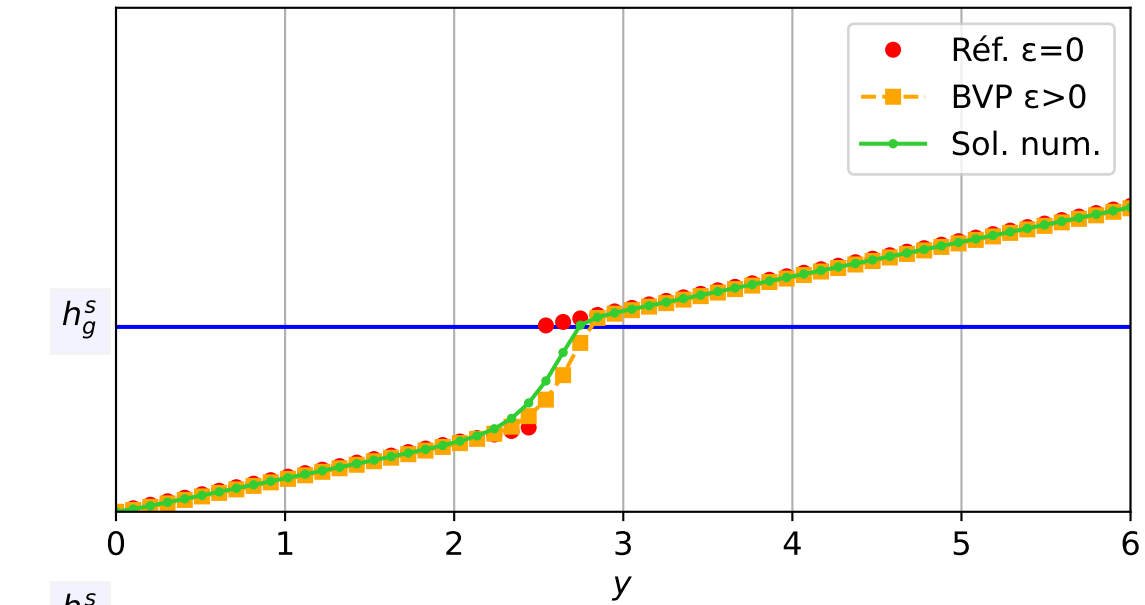
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
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 $dt = 2.01e-03$  —  $t = 2.90176/12$  (7867 iter) — résidu eq. =  $9.50e-06$



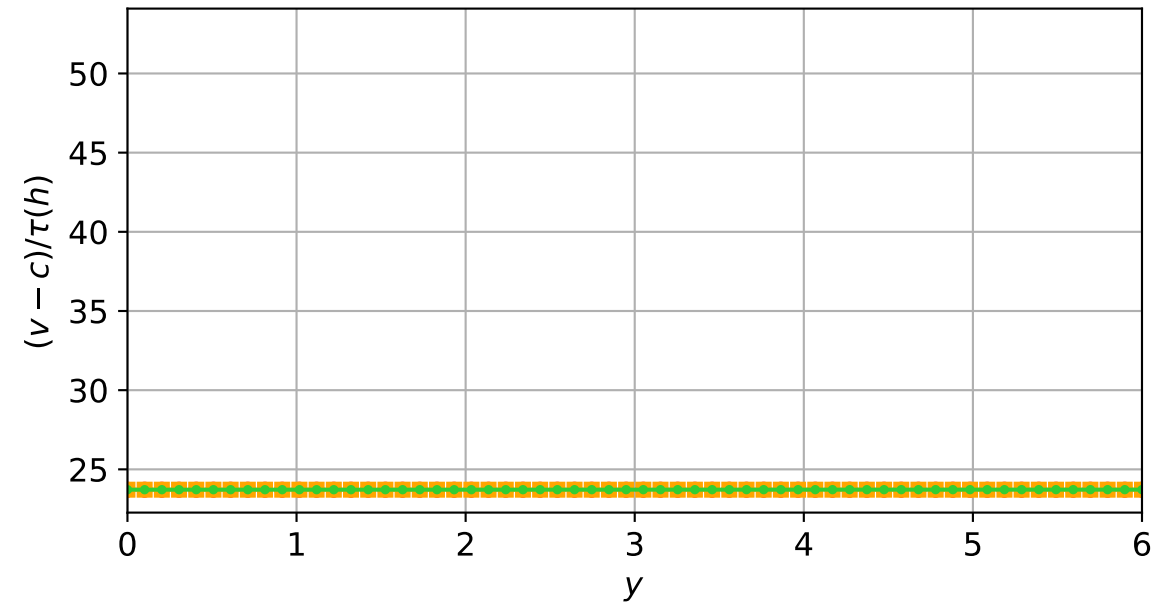
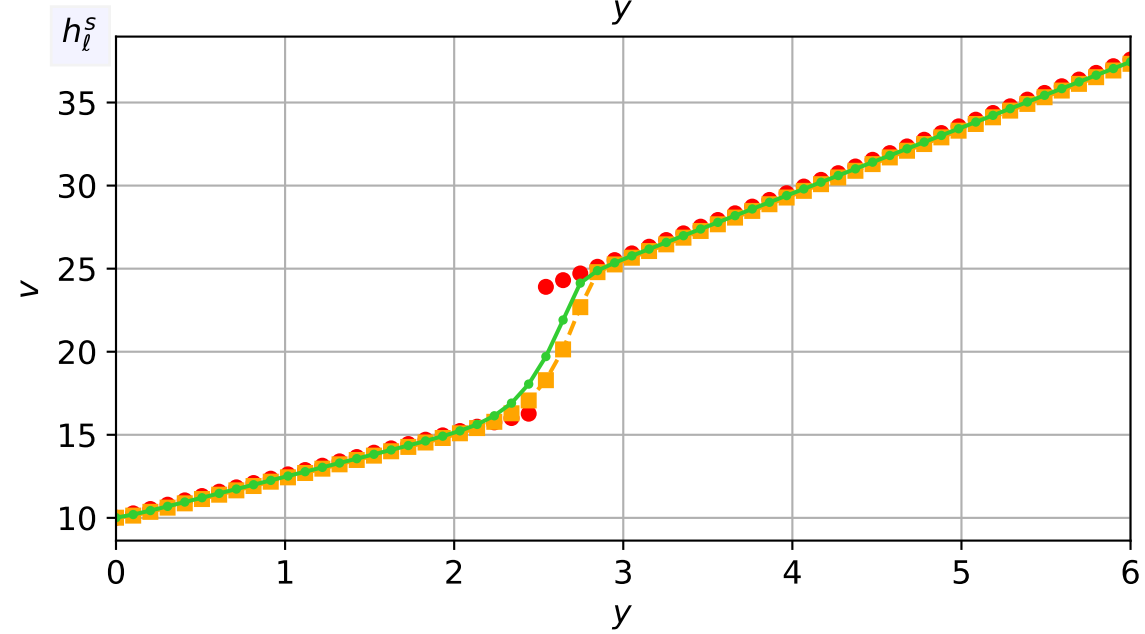
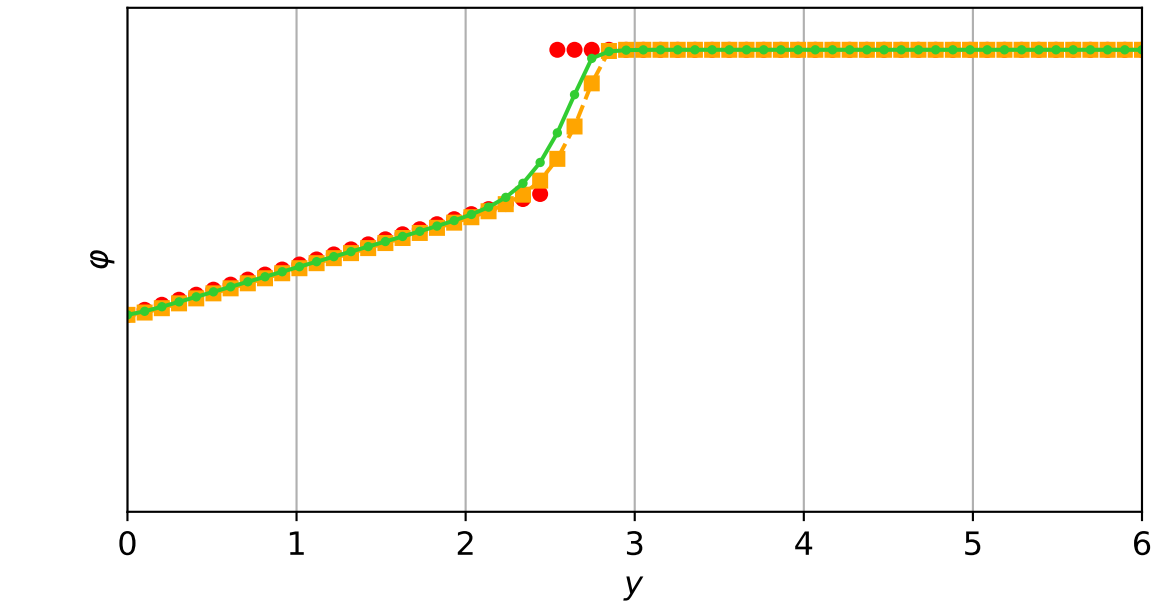
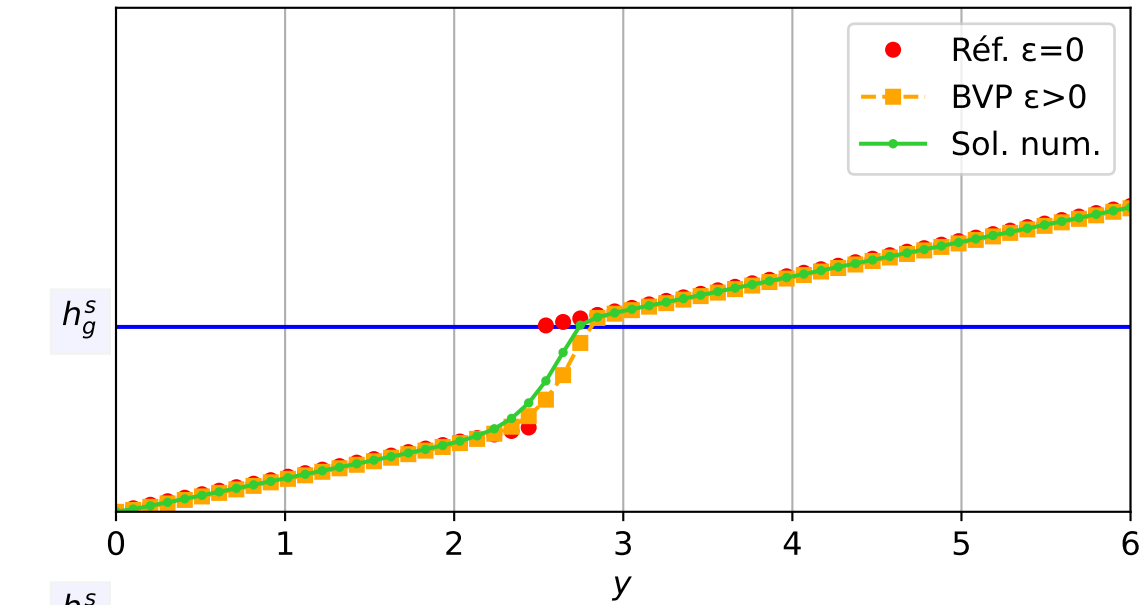
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.01e-03$  — t = 2.9501/12 (7891 iter) — résidu eq. =  $8.47e-06$



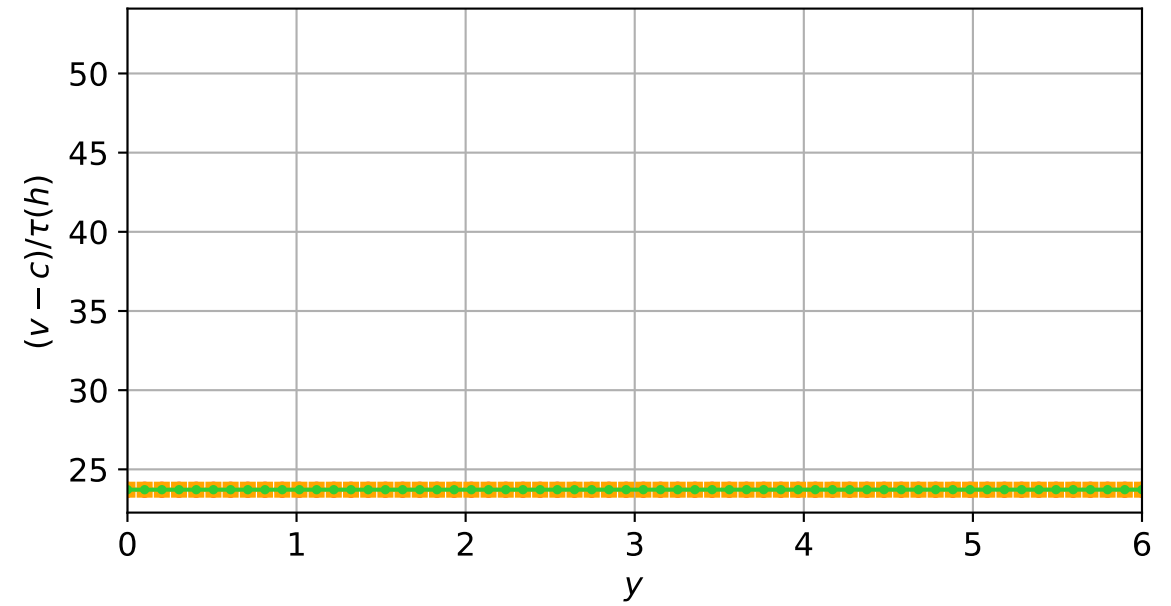
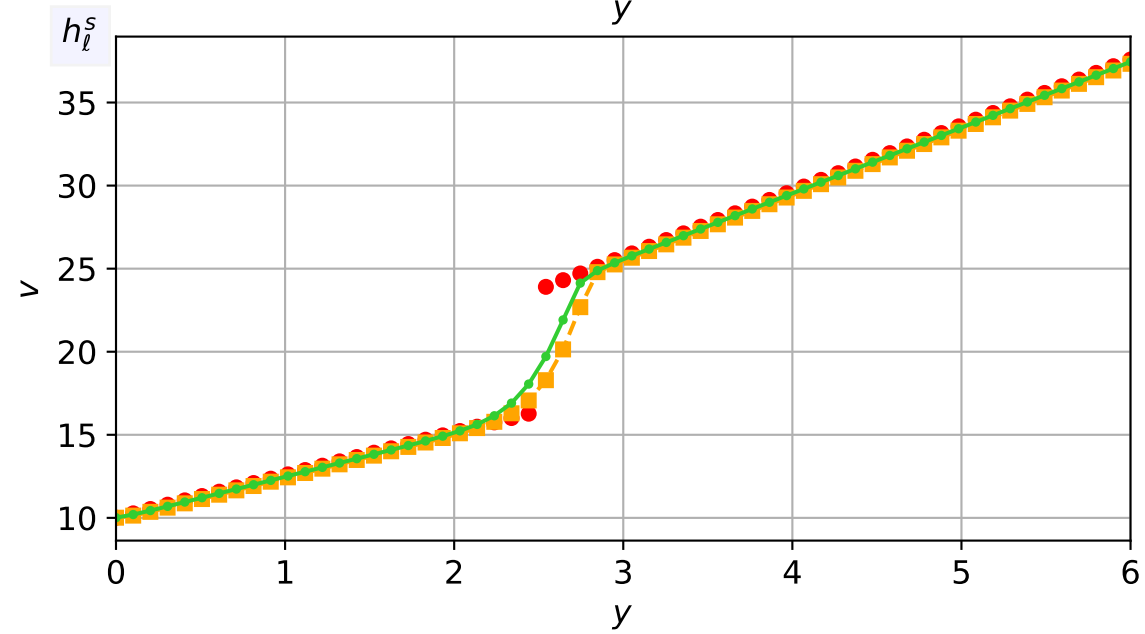
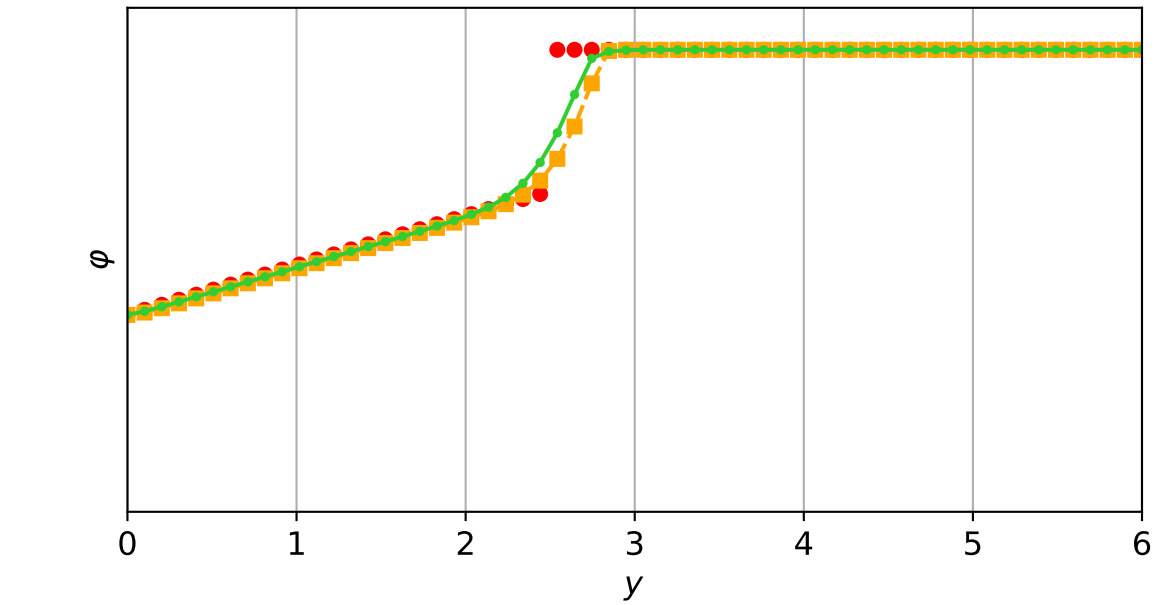
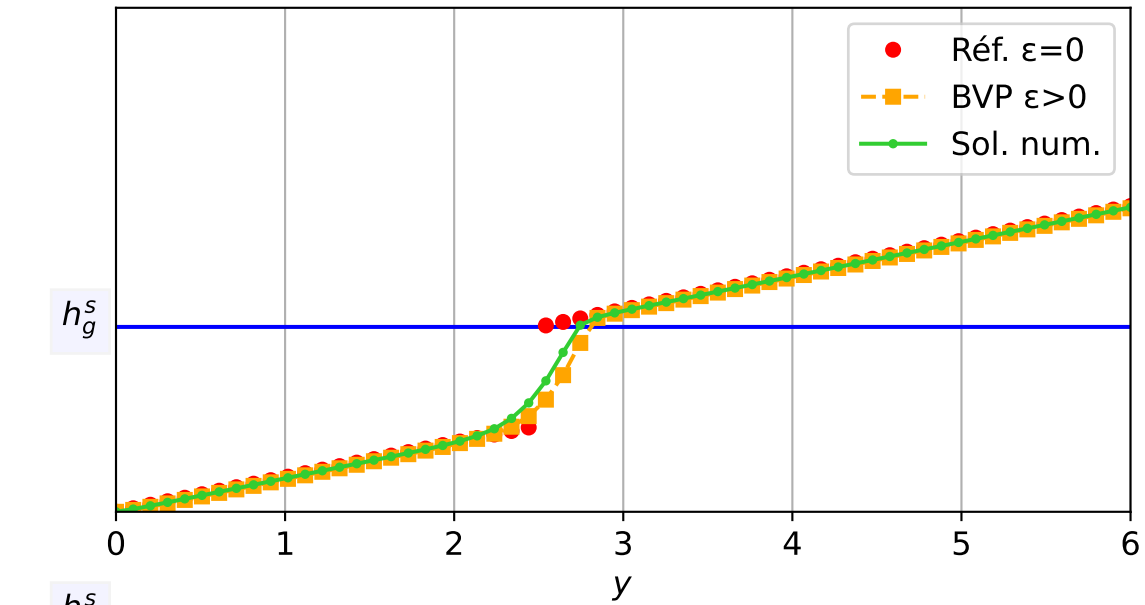
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.42e-03$  —  $t = 3.00207/12$  (7915 iter) — résidu eq. =  $9.57e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.42e-03$  —  $t = 3.05041/12$  (7935 iter) — résidu eq. =  $8.54e-06$

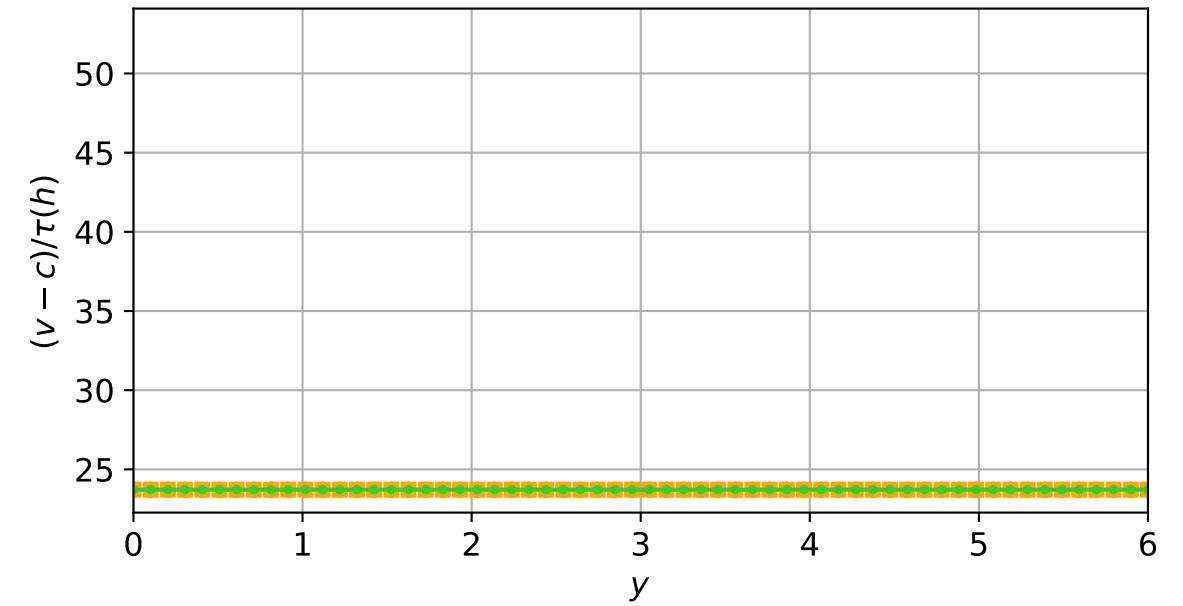
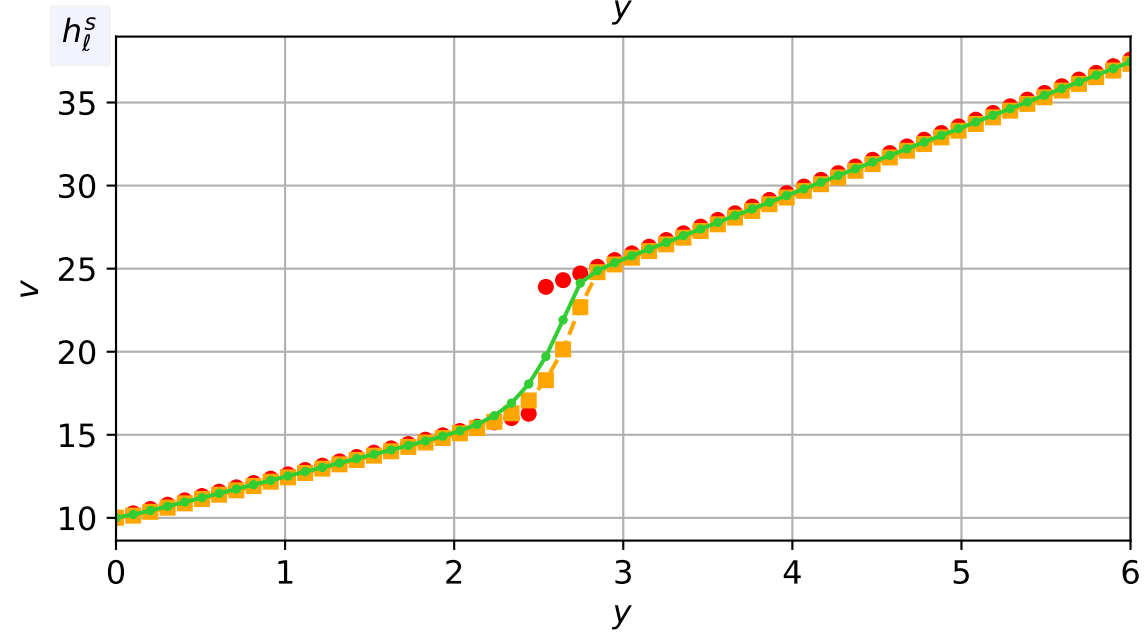
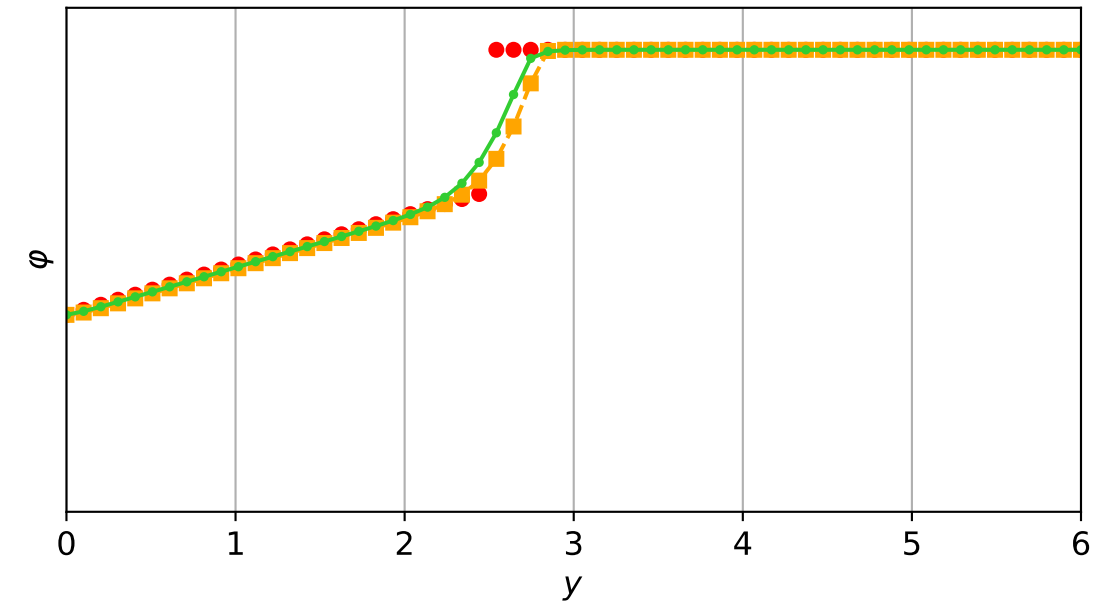
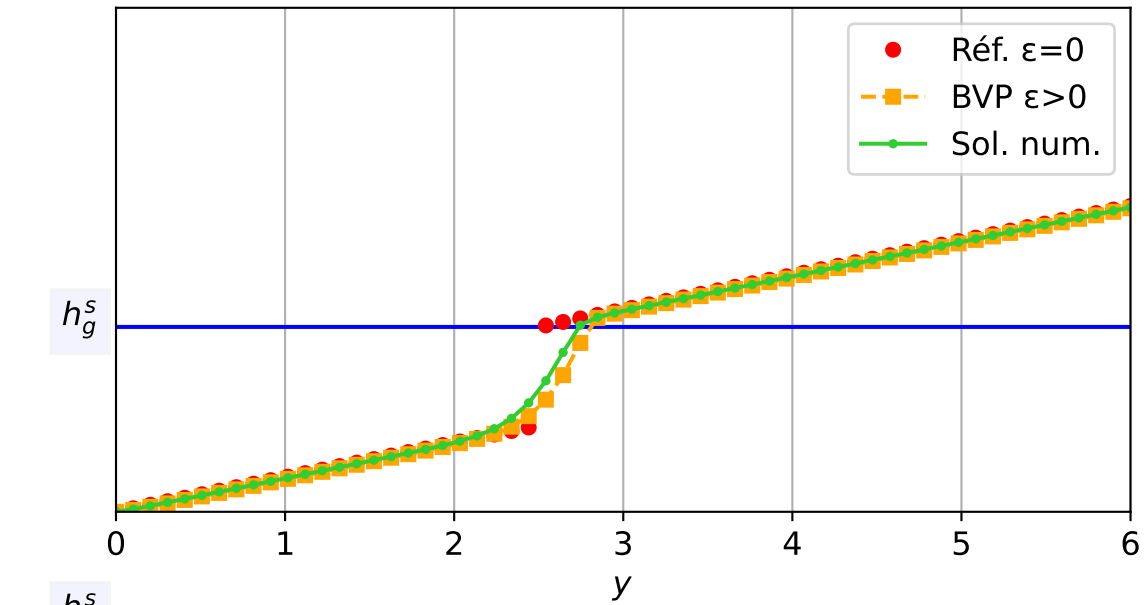


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.10144/12$  (7956 iter) — résidu eq. =  $9.73e-06$

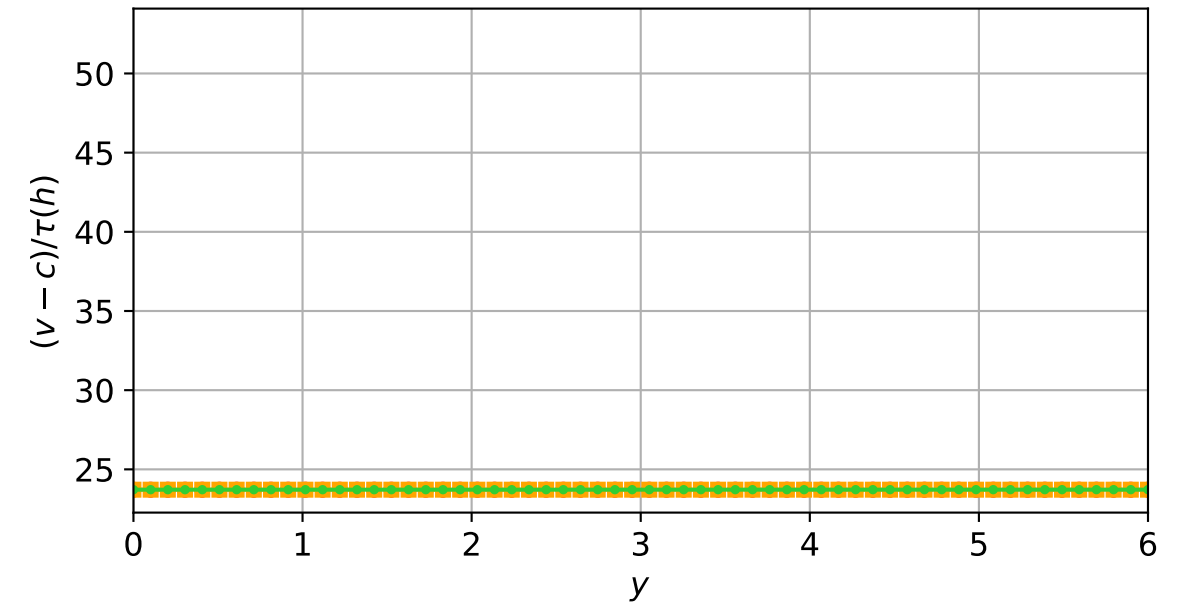
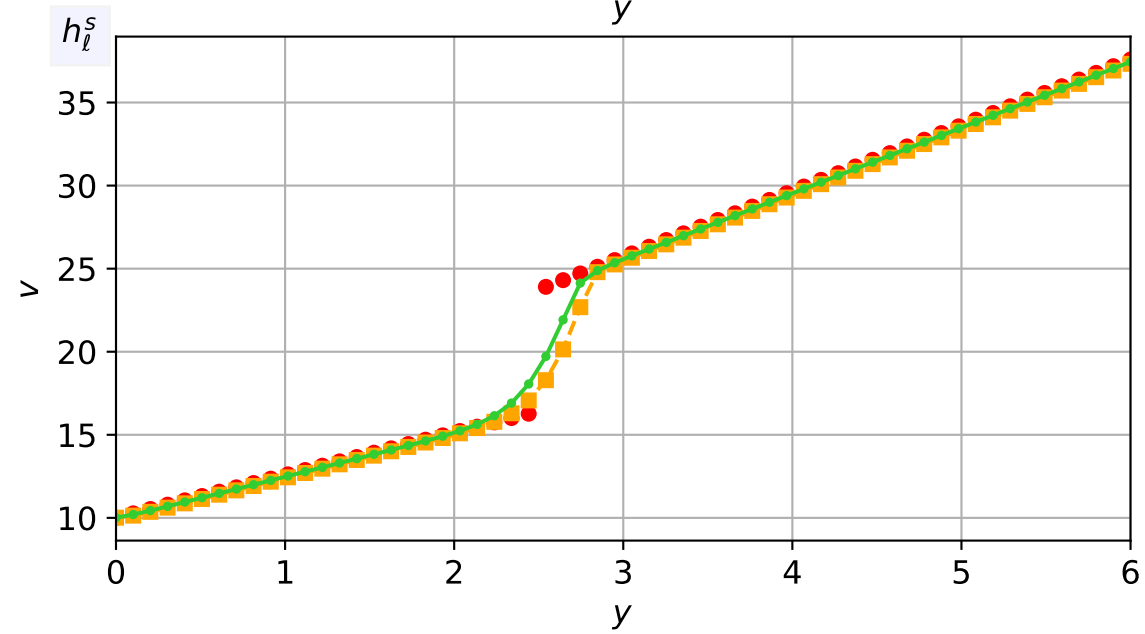
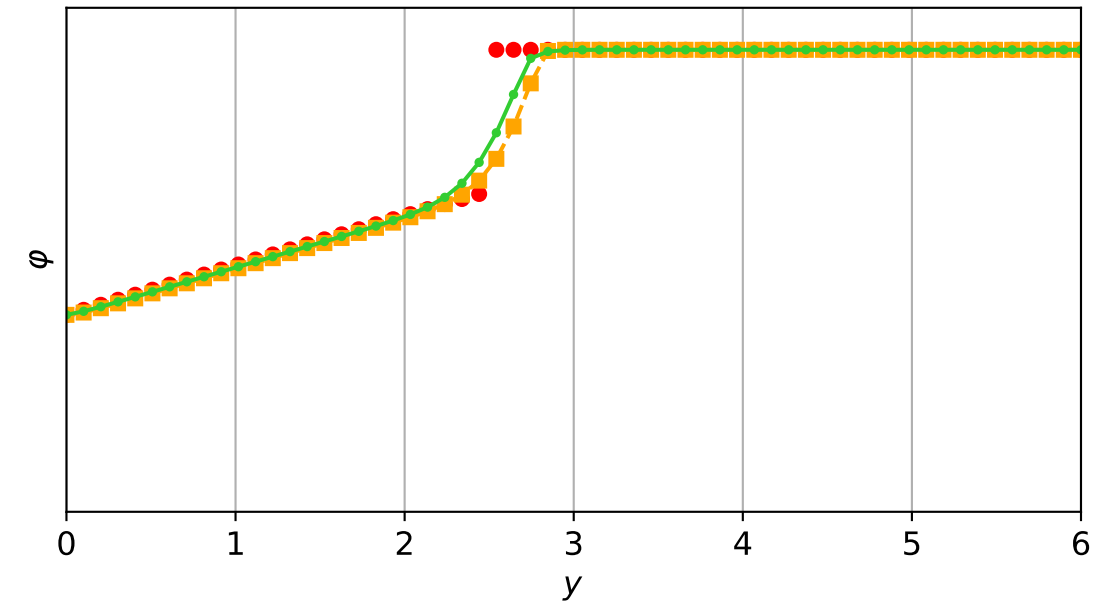
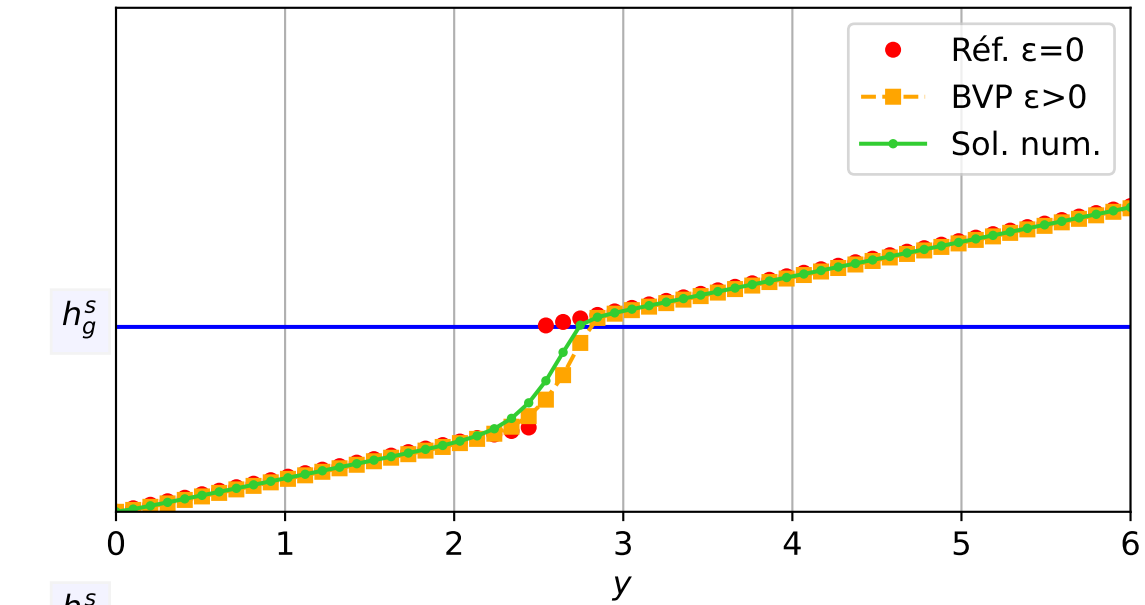




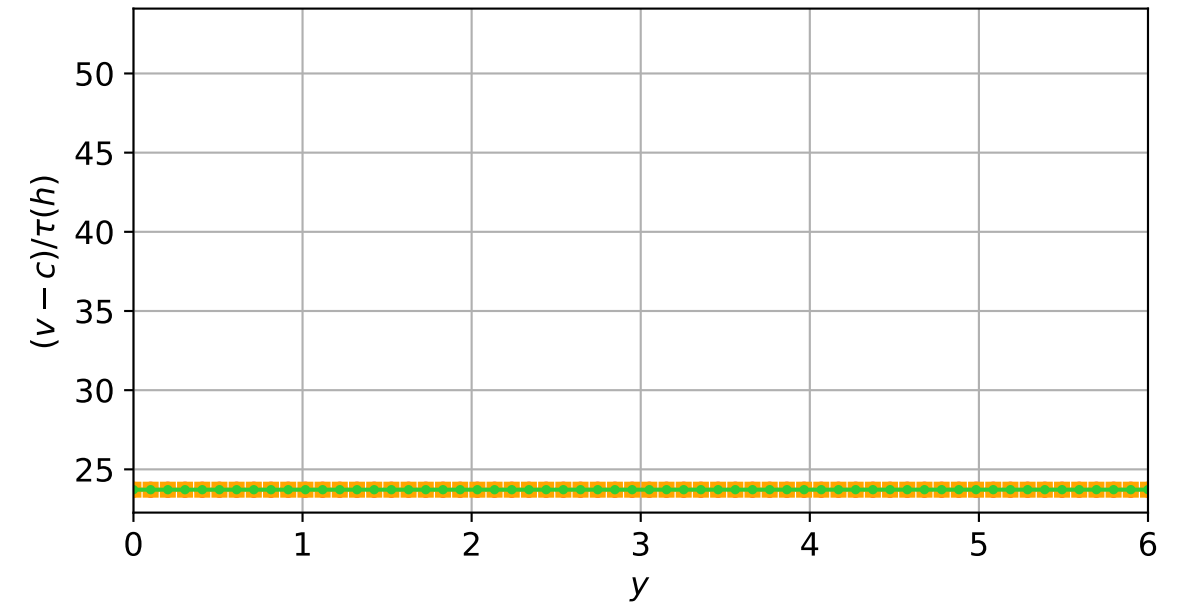
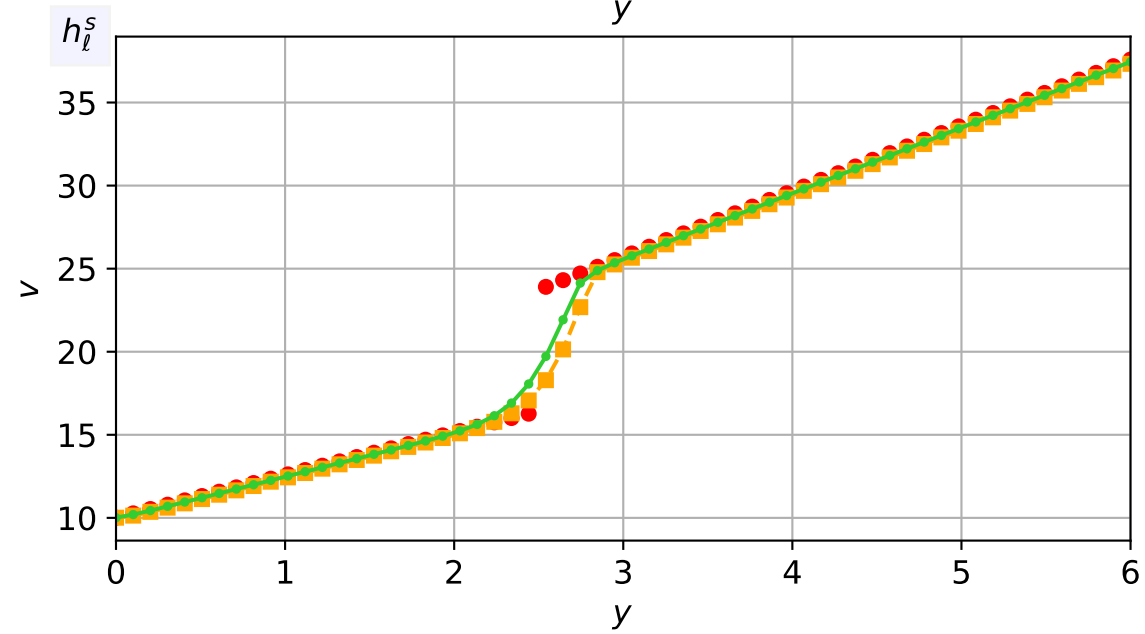
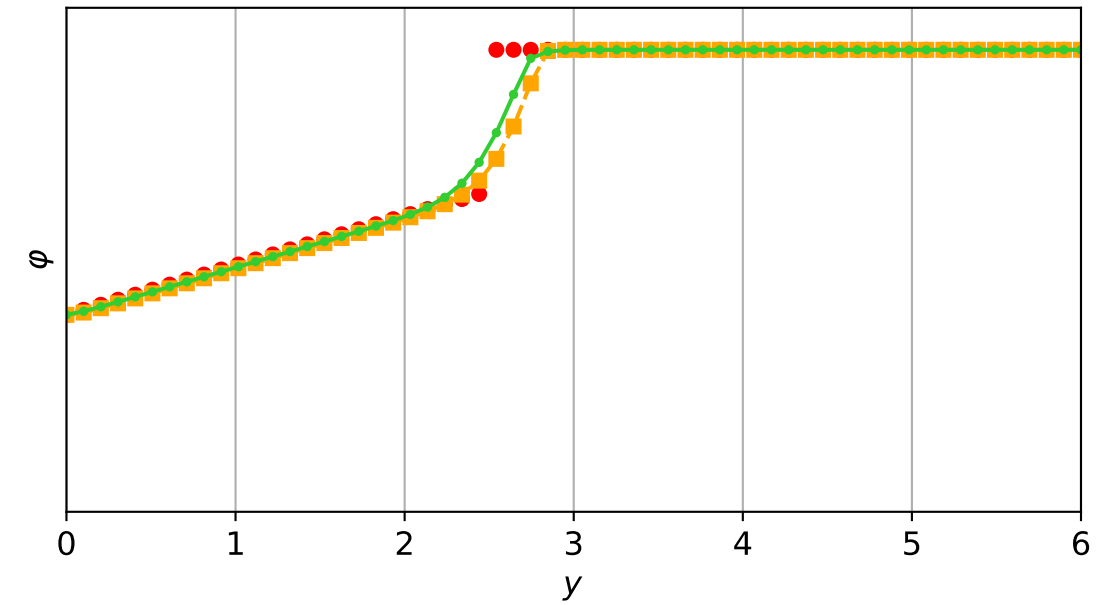
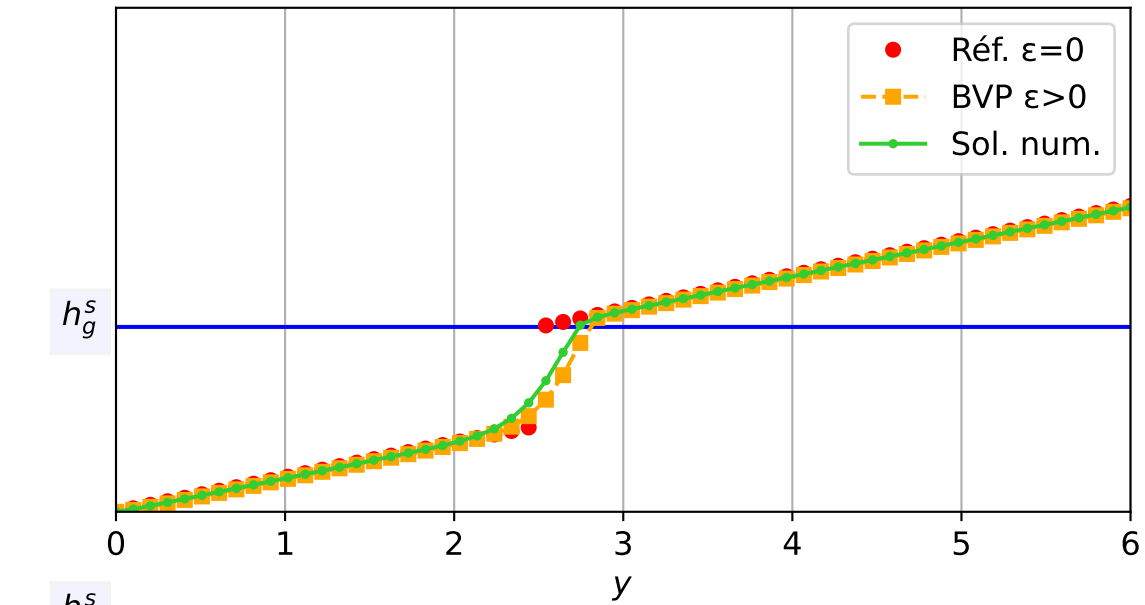
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.15032/12$  (7976 iter) — résidu eq. =  $9.01e-06$



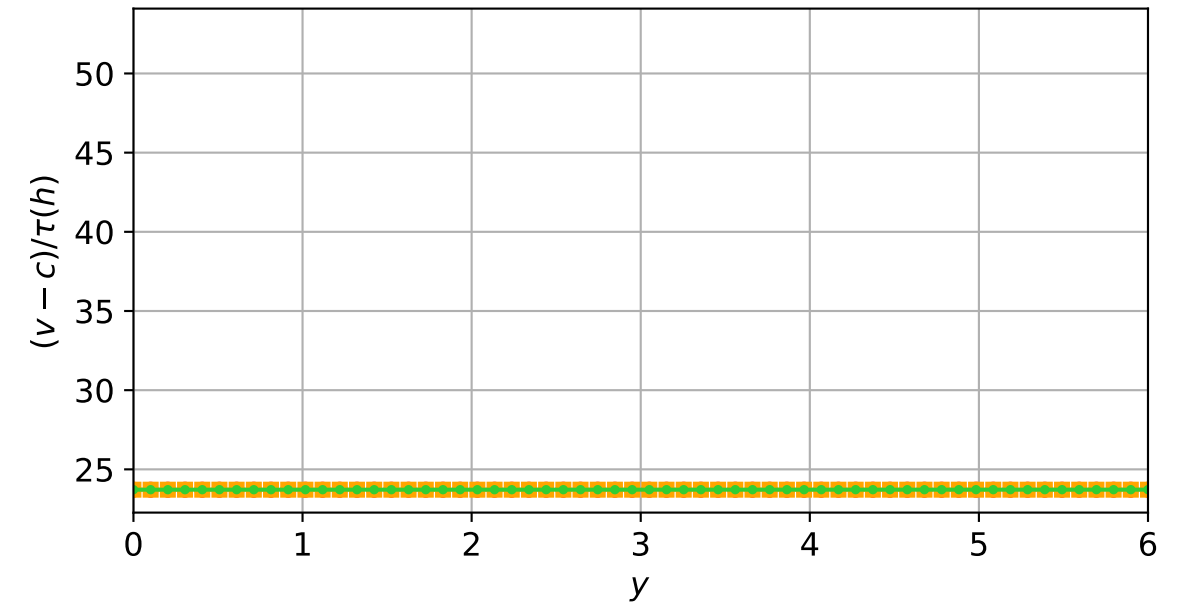
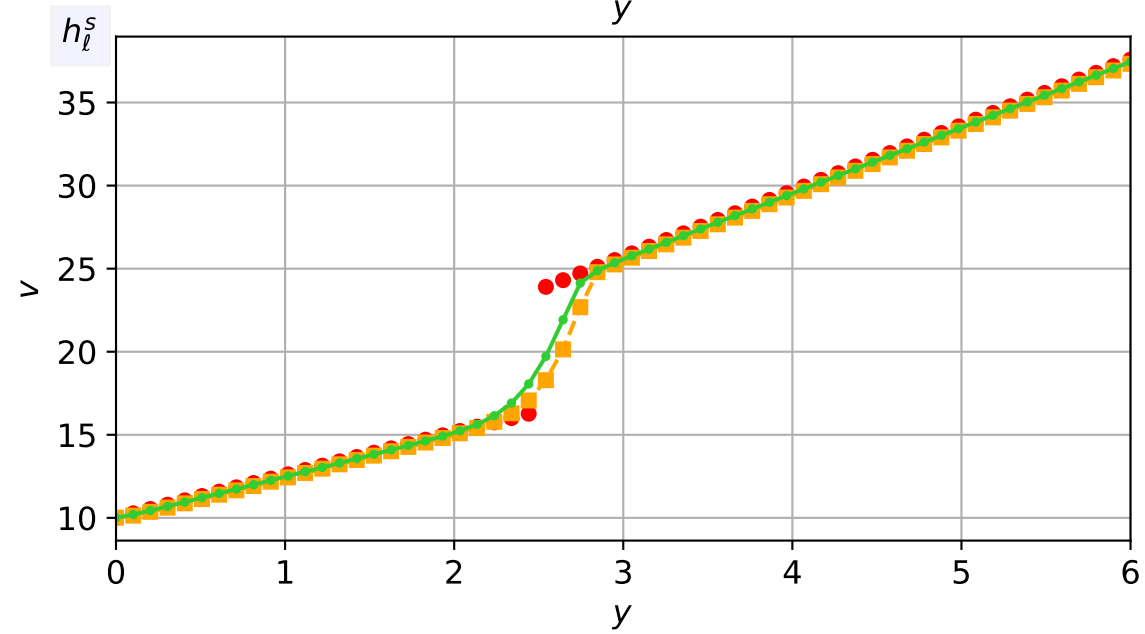
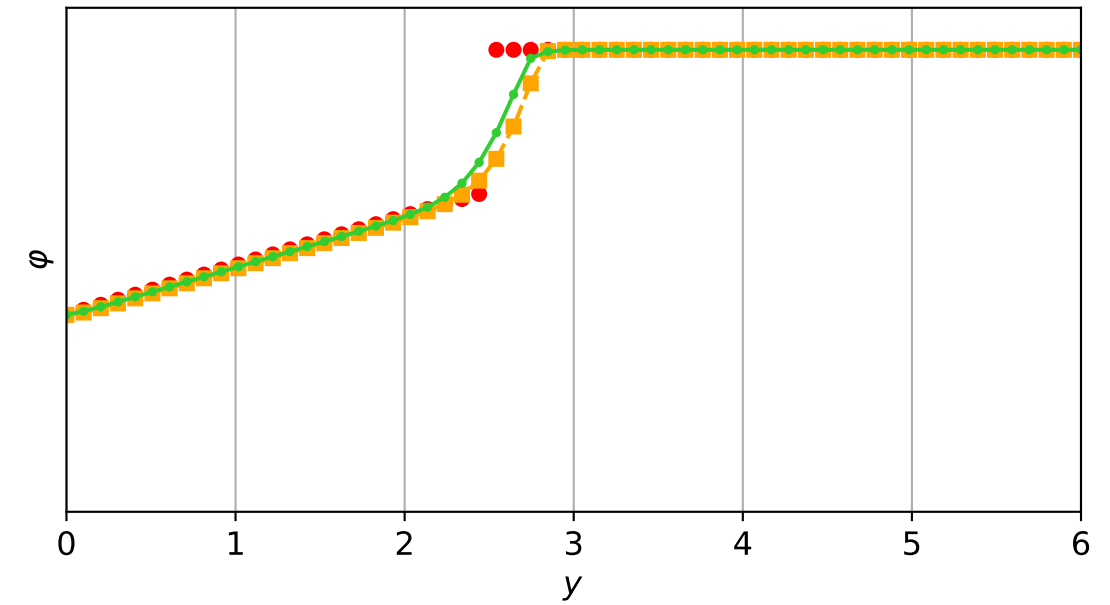
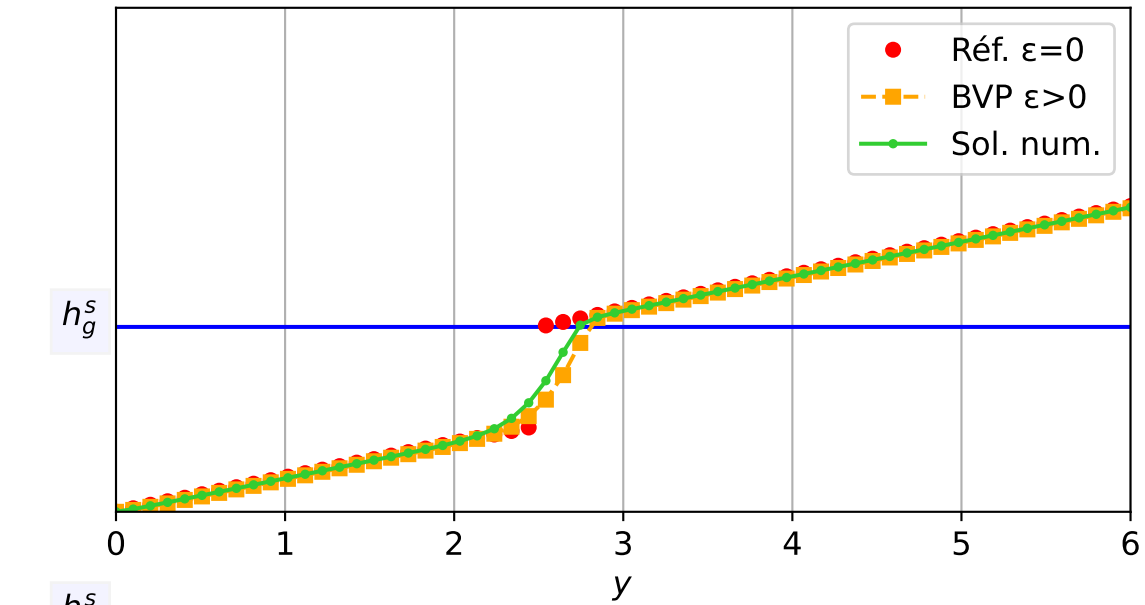
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.20164/12$  (7997 iter) — résidu eq. =  $8.22e-06$



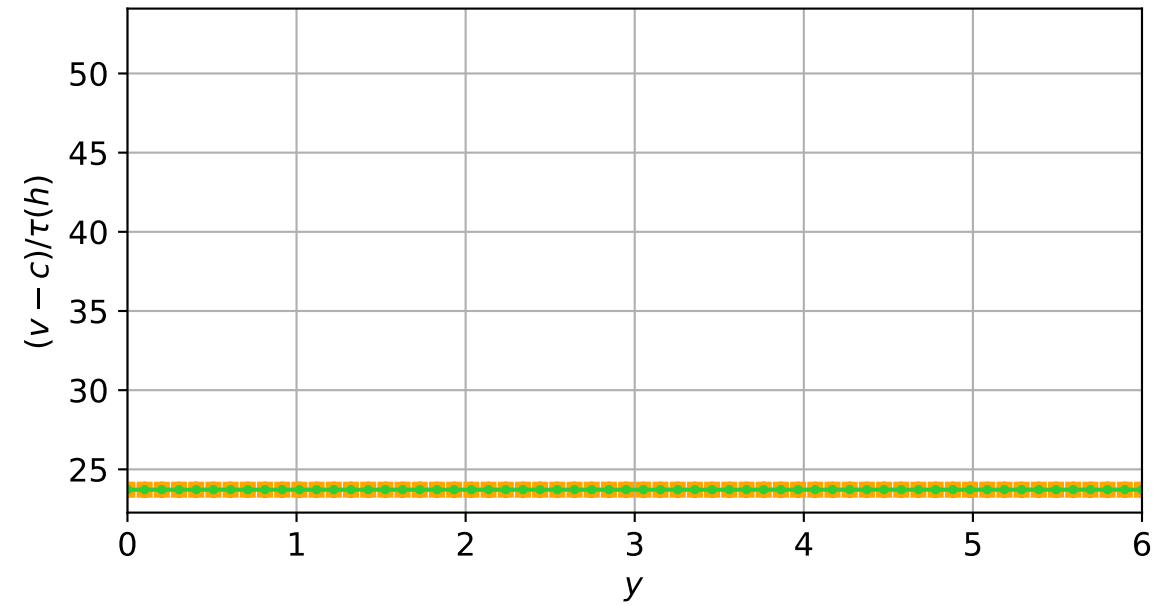
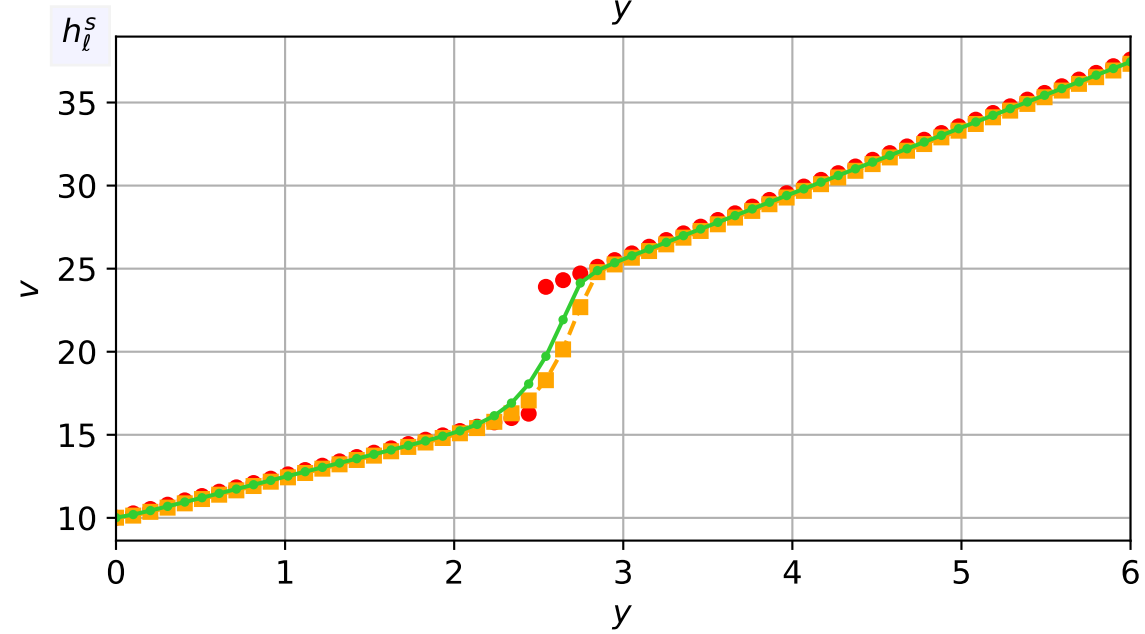
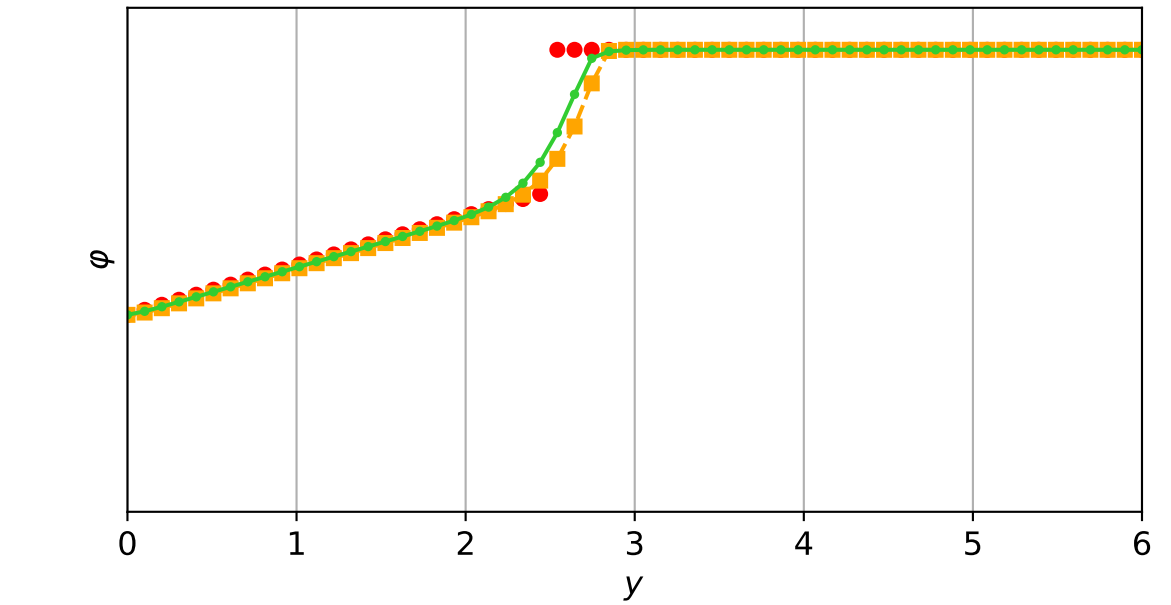
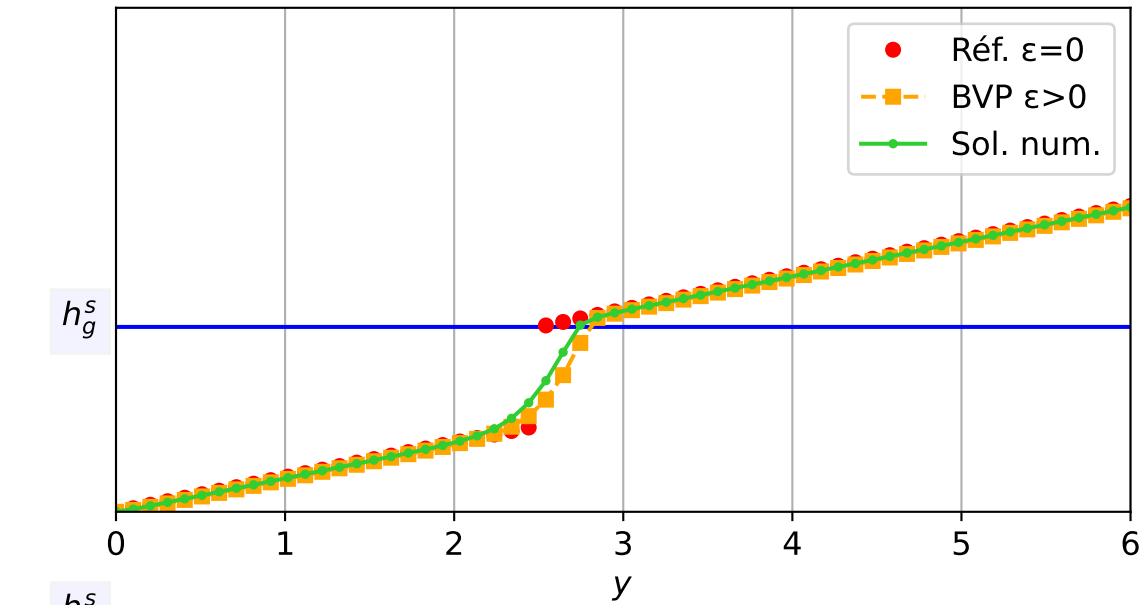
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.25052/12$  (8017 iter) — résidu eq. =  $9.55e-06$



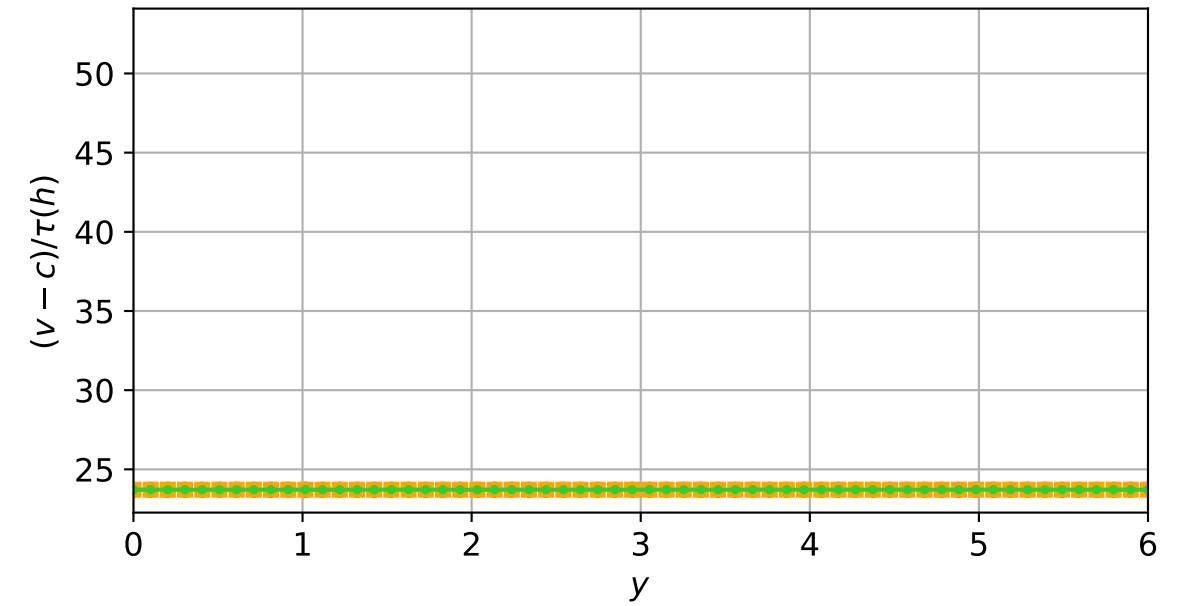
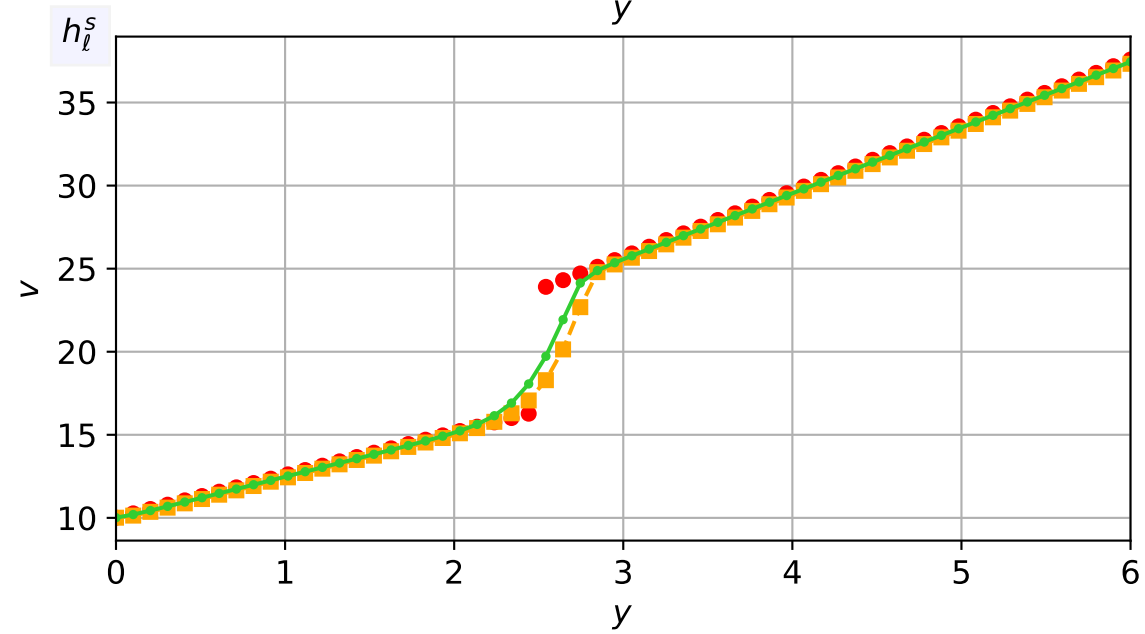
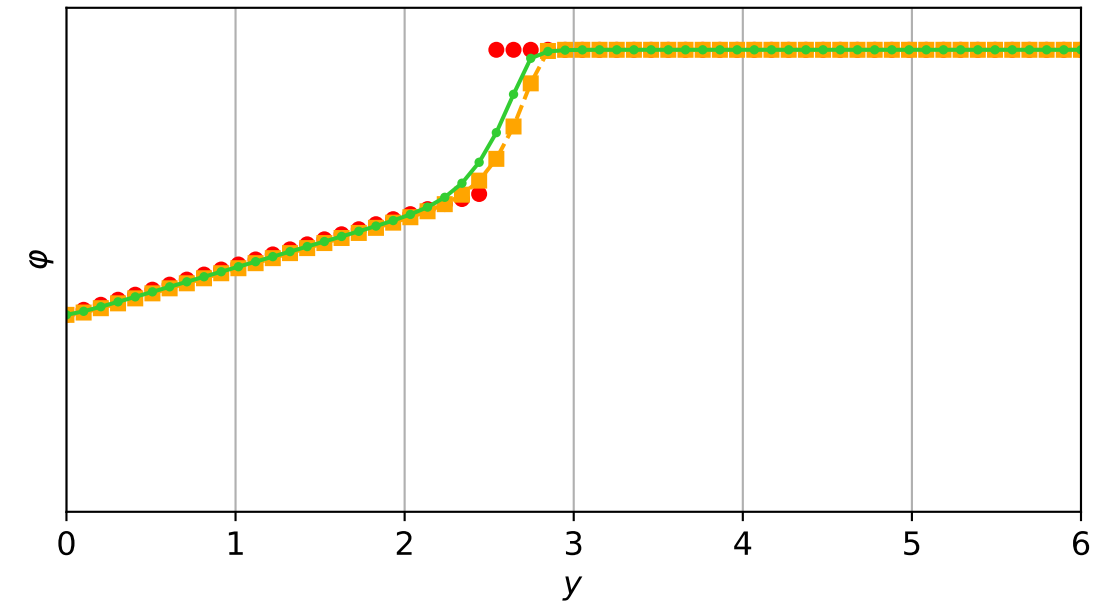
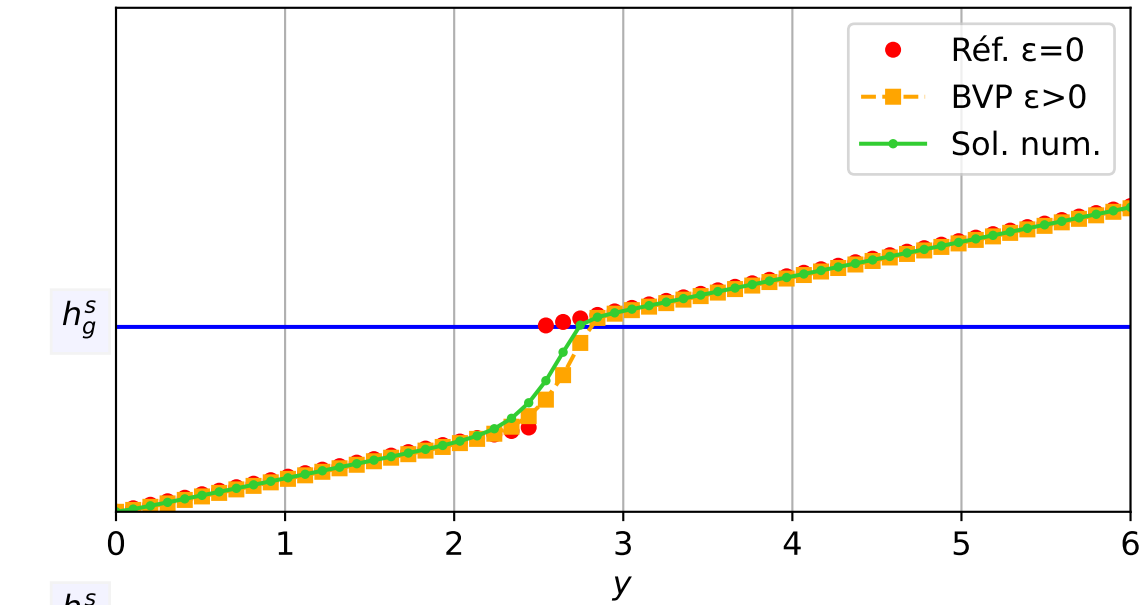
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.30185/12$  (8038 iter) — résidu eq. =  $8.48e-06$



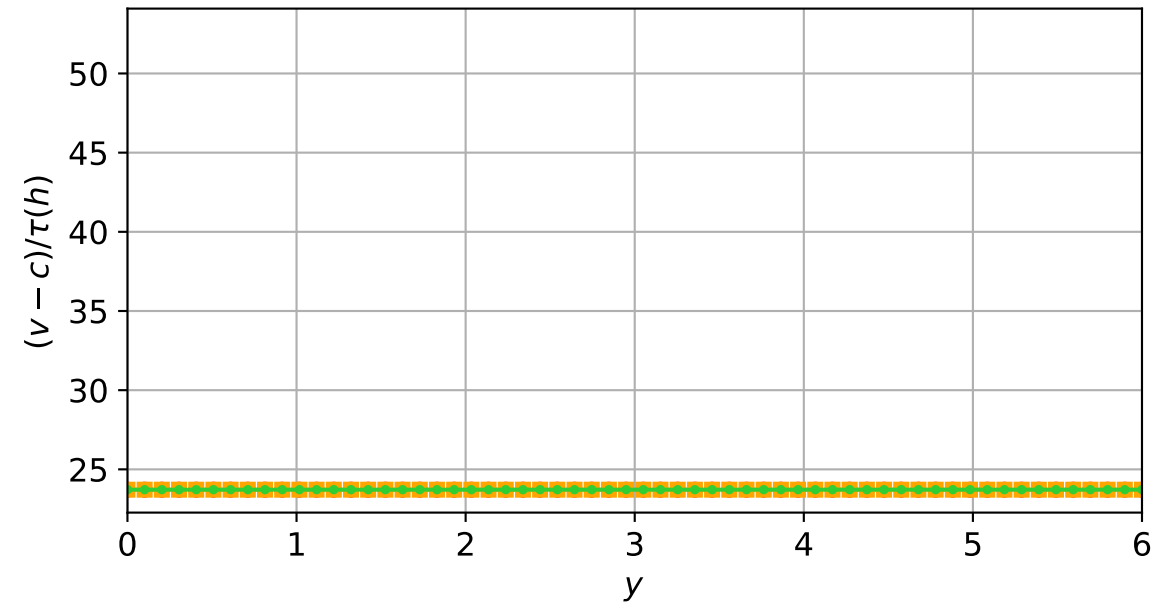
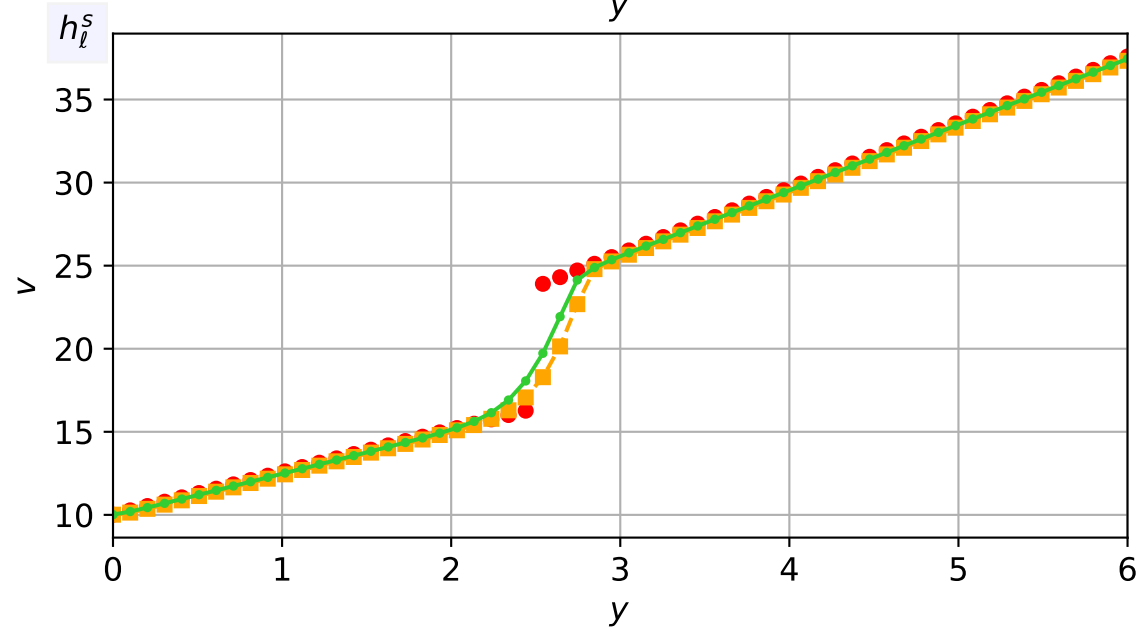
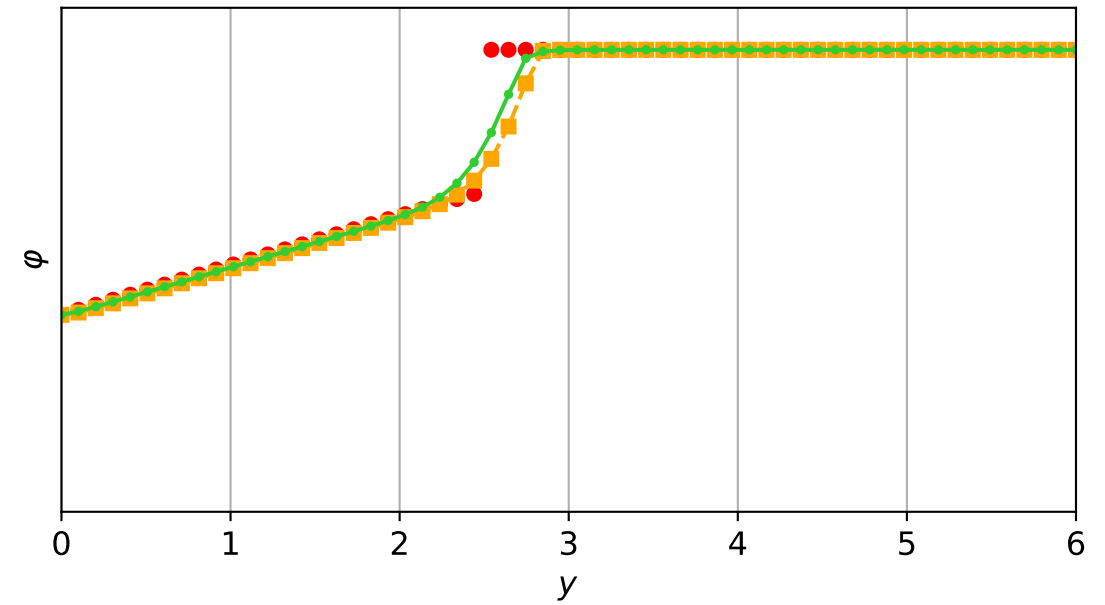
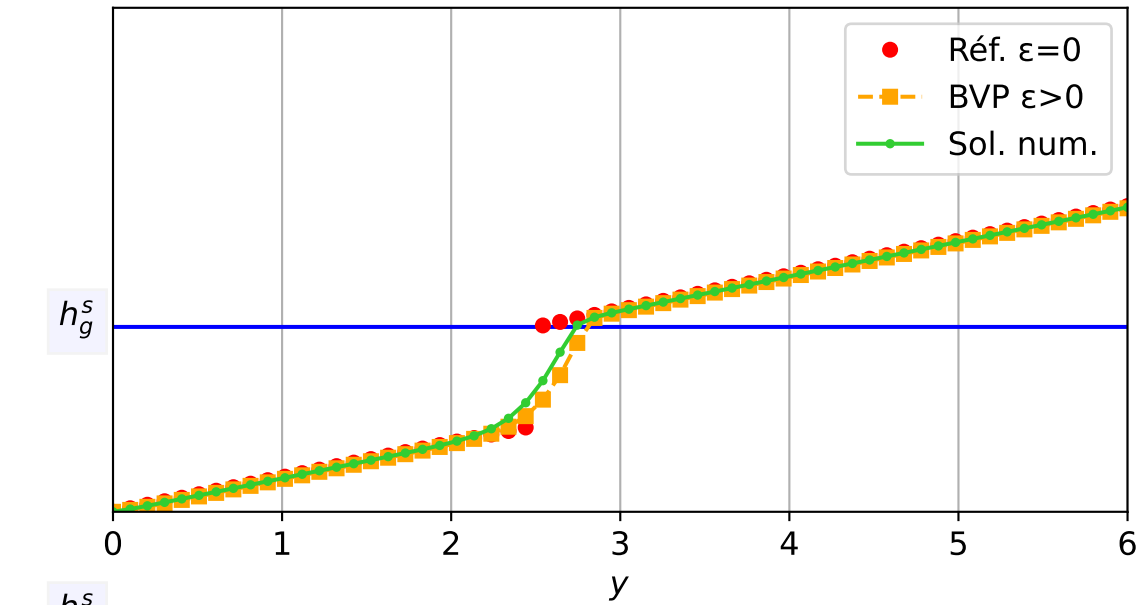
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.35073/12$  (8058 iter) — résidu eq. =  $9.72e-06$



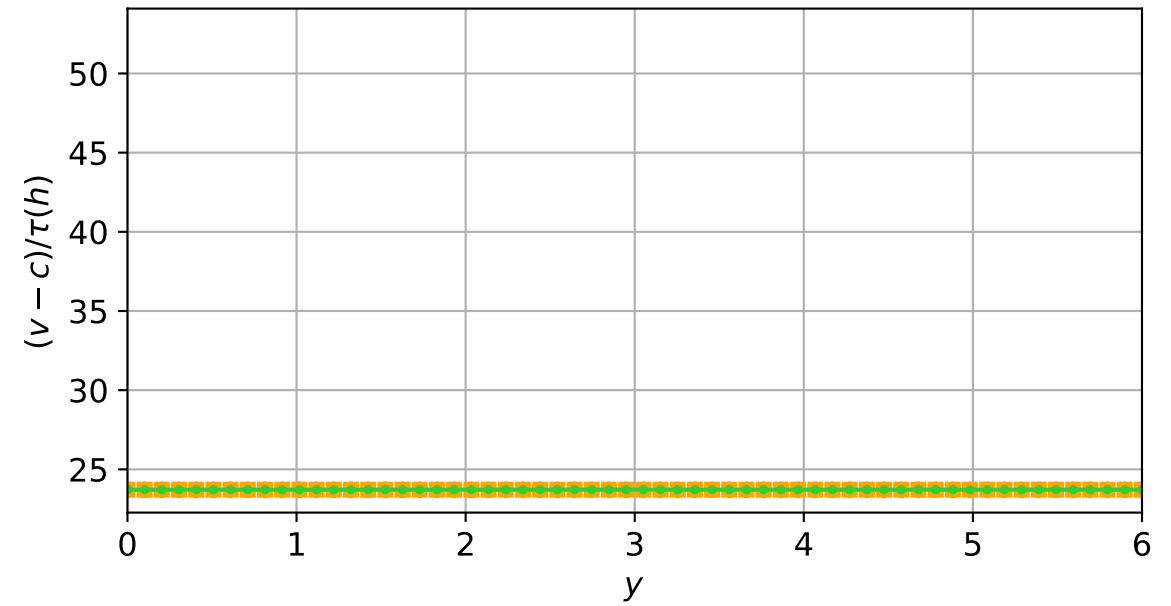
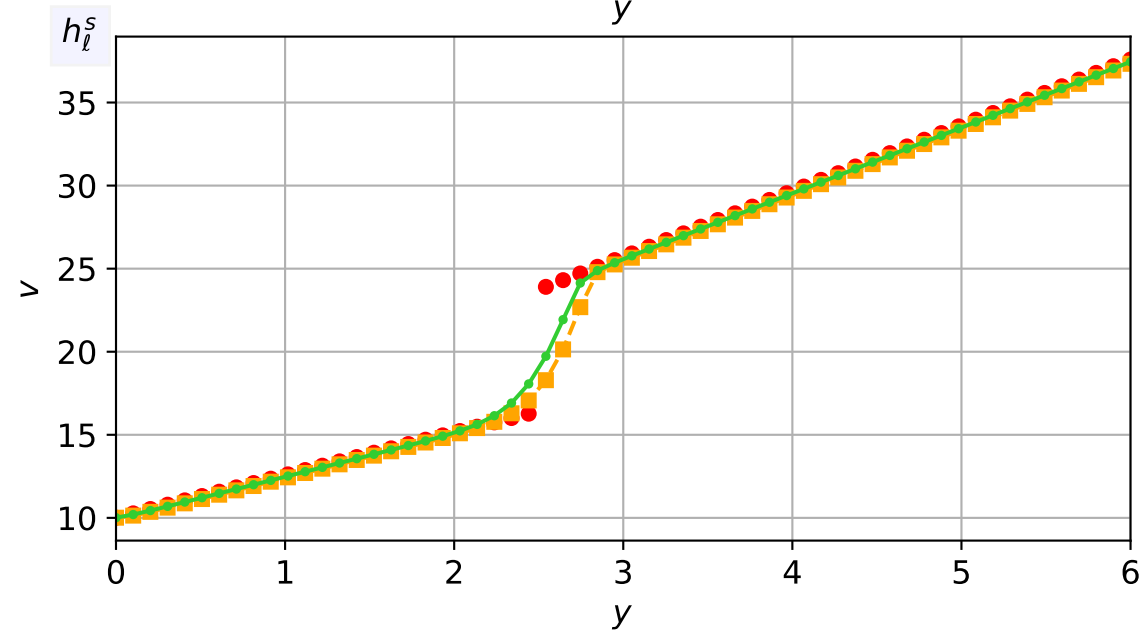
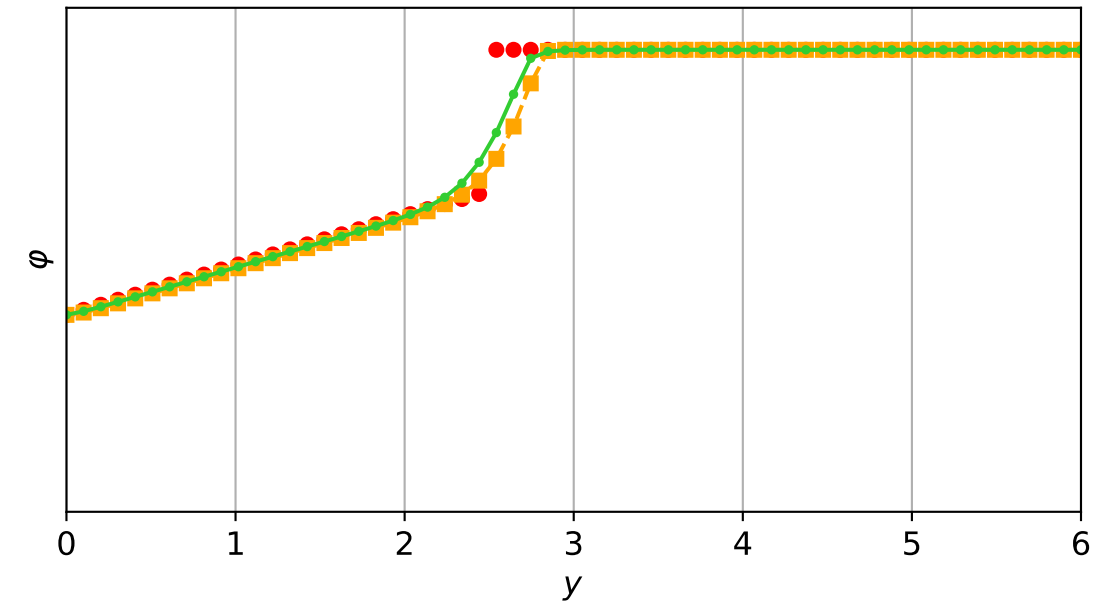
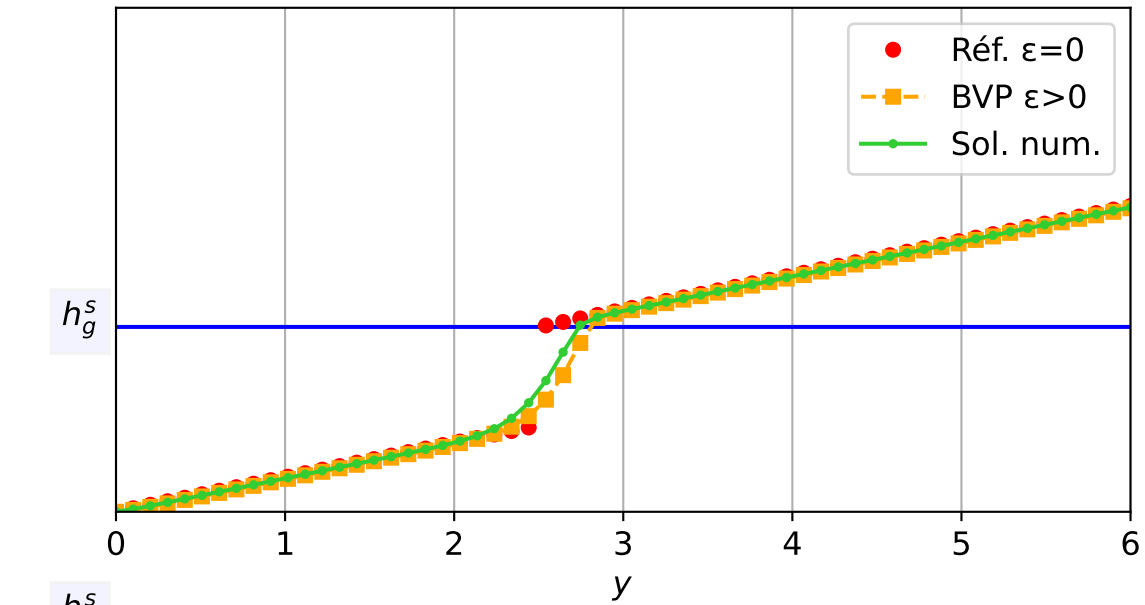
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.40205/12$  (8079 iter) — résidu eq. =  $9.32e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.45093/12$  (8099 iter) — résidu eq. =  $8.33e-06$

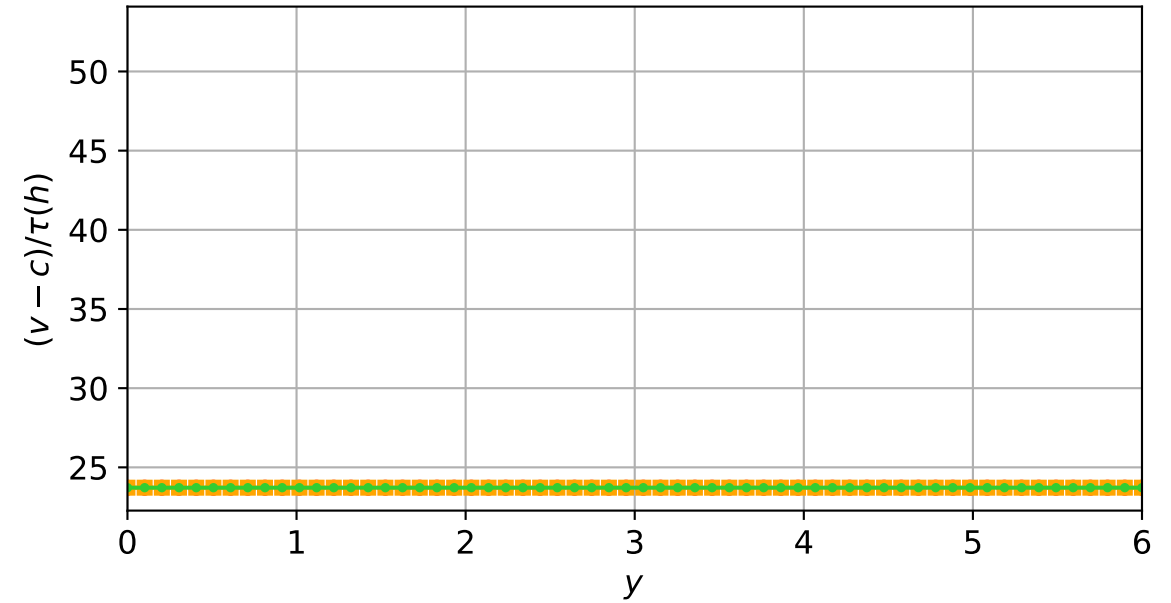
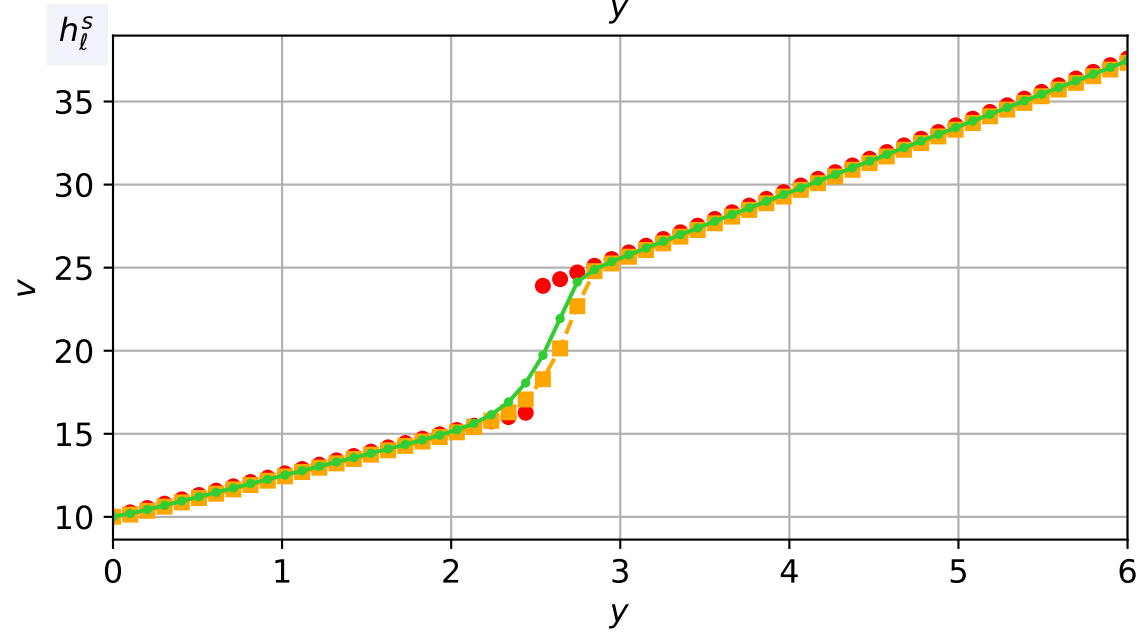
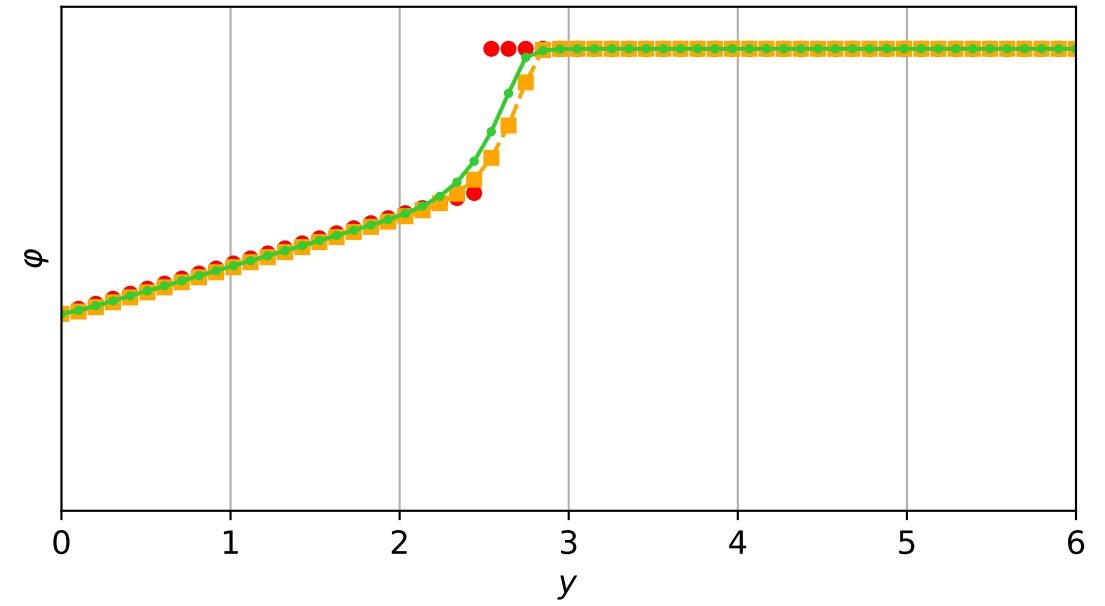
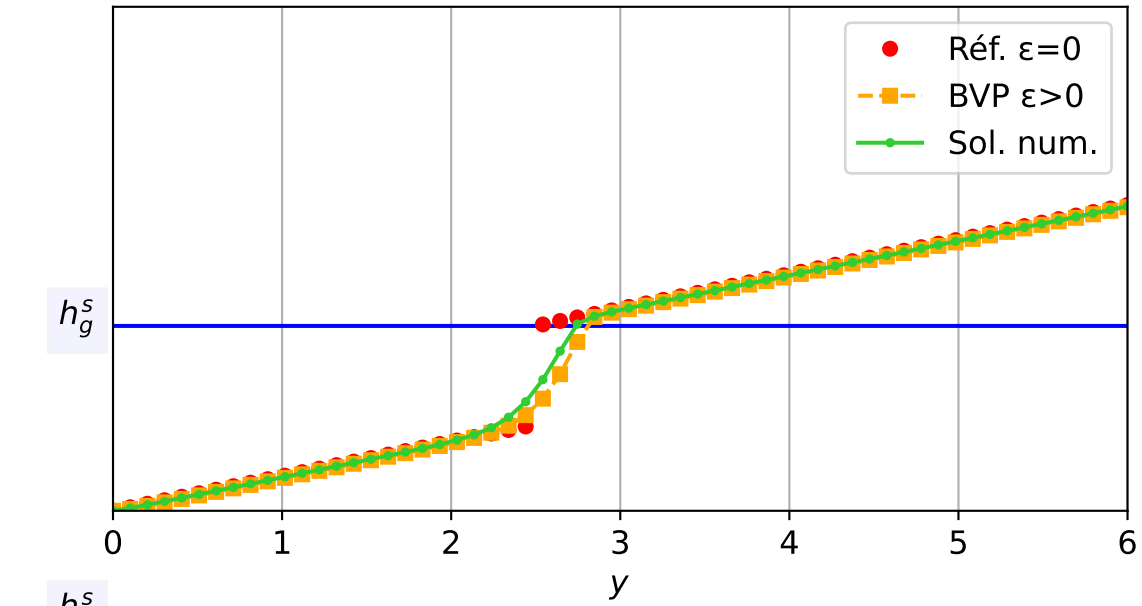


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.50225/12$  (8120 iter) — résidu eq. =  $7.37e-06$

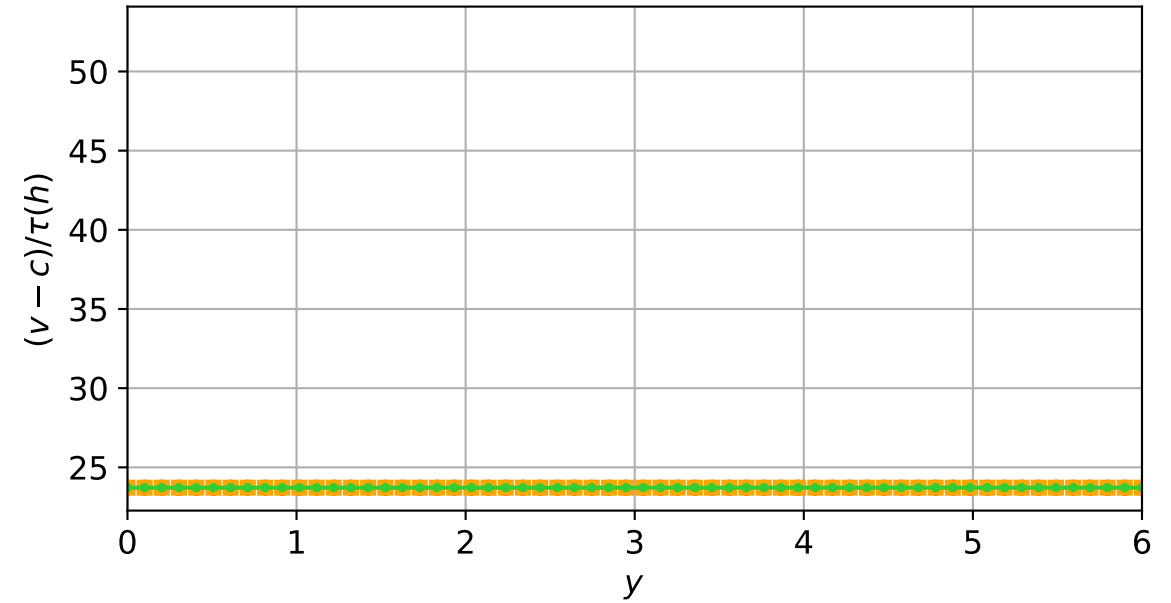
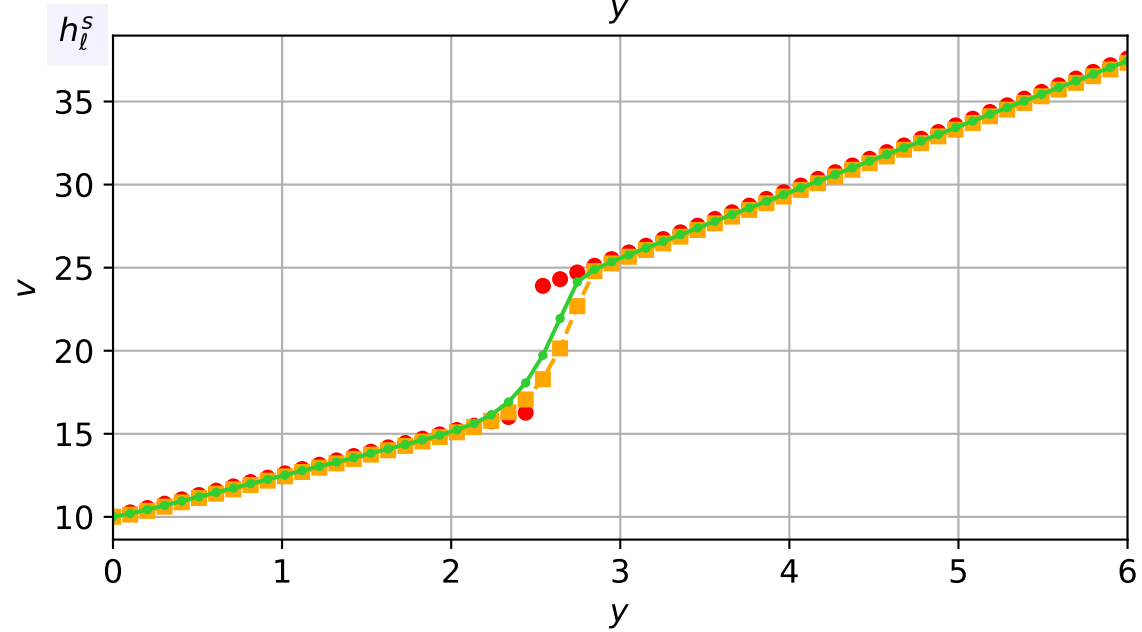
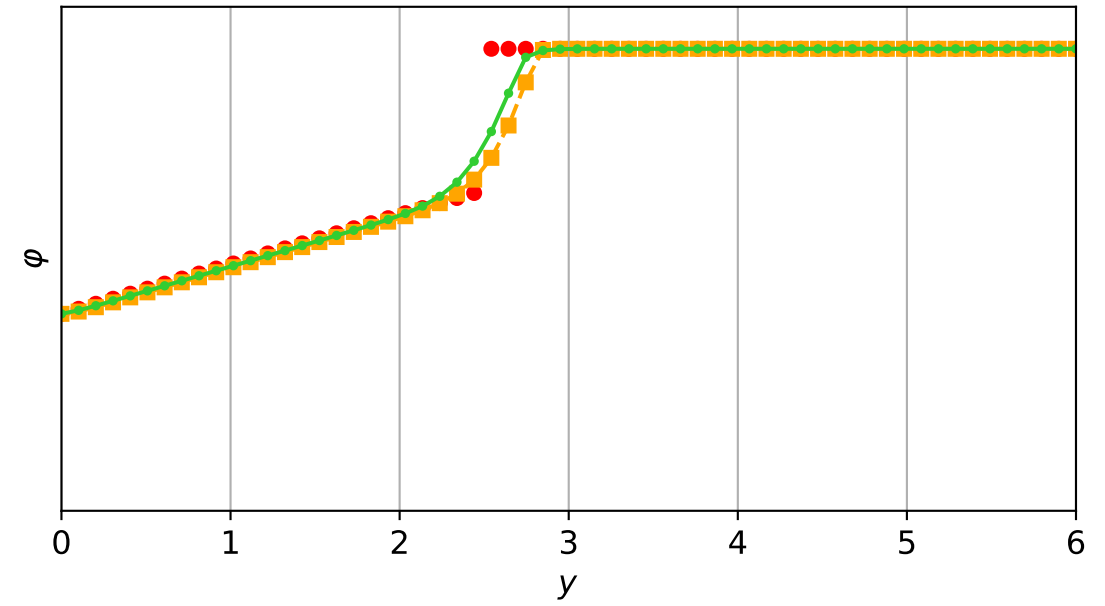
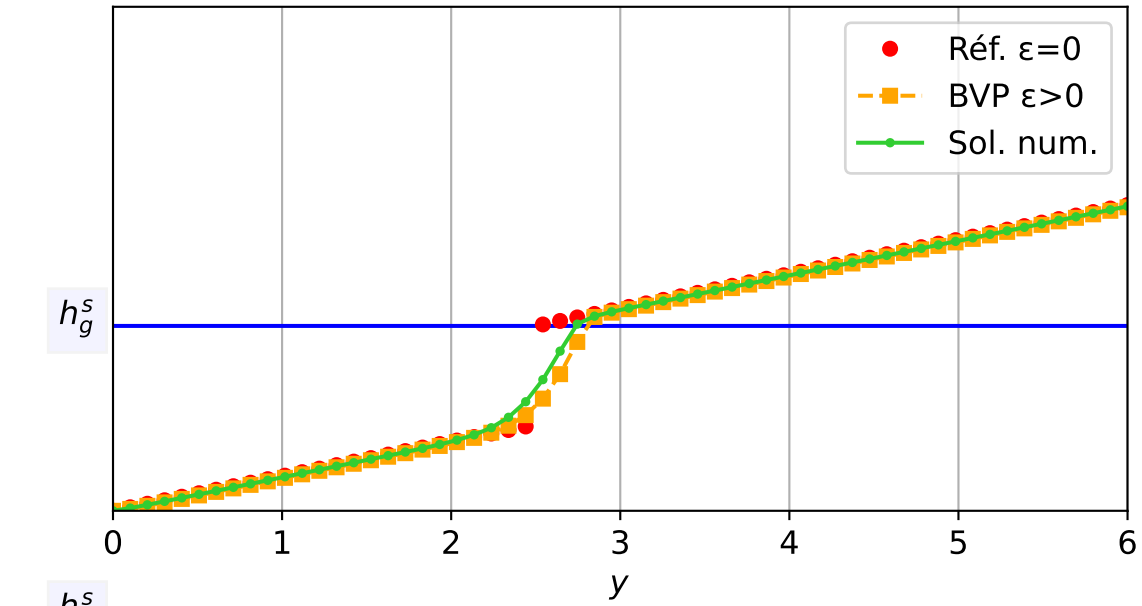




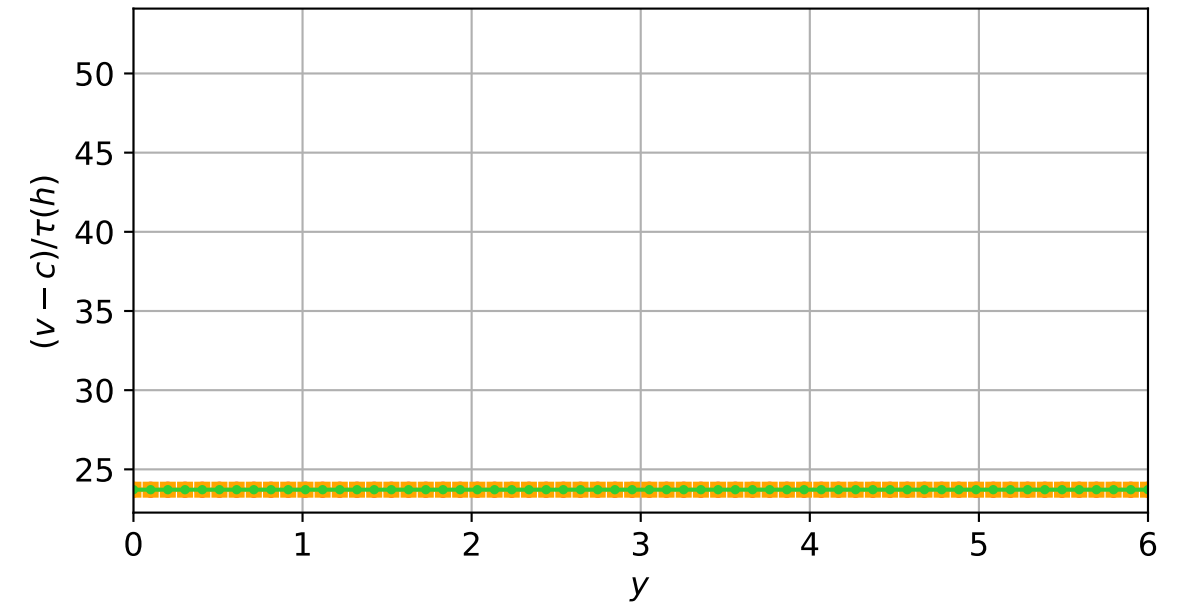
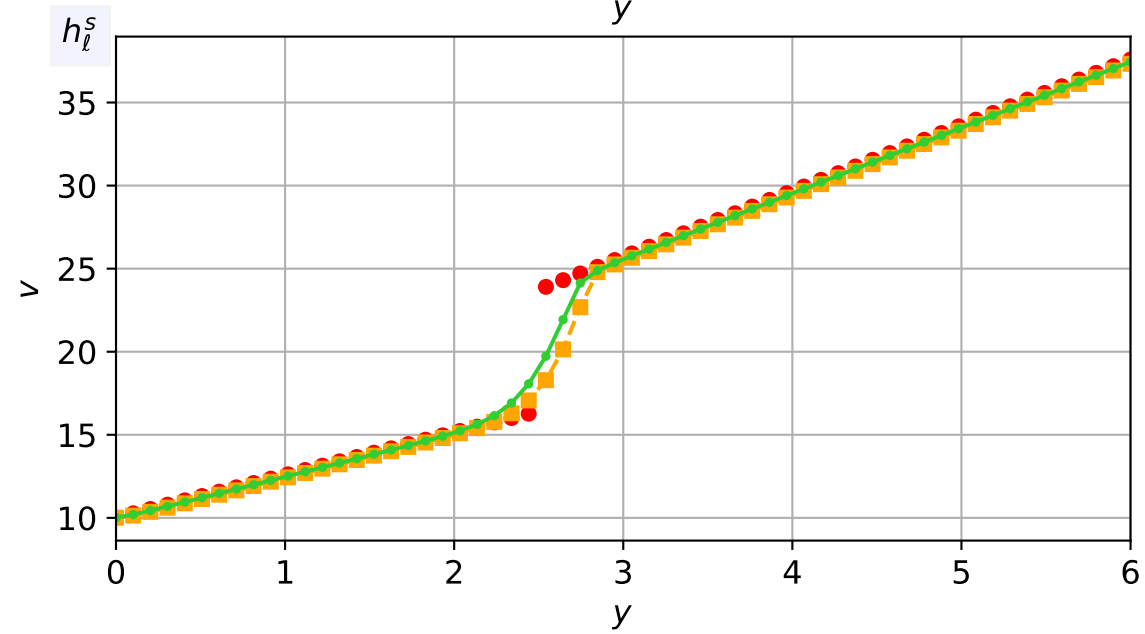
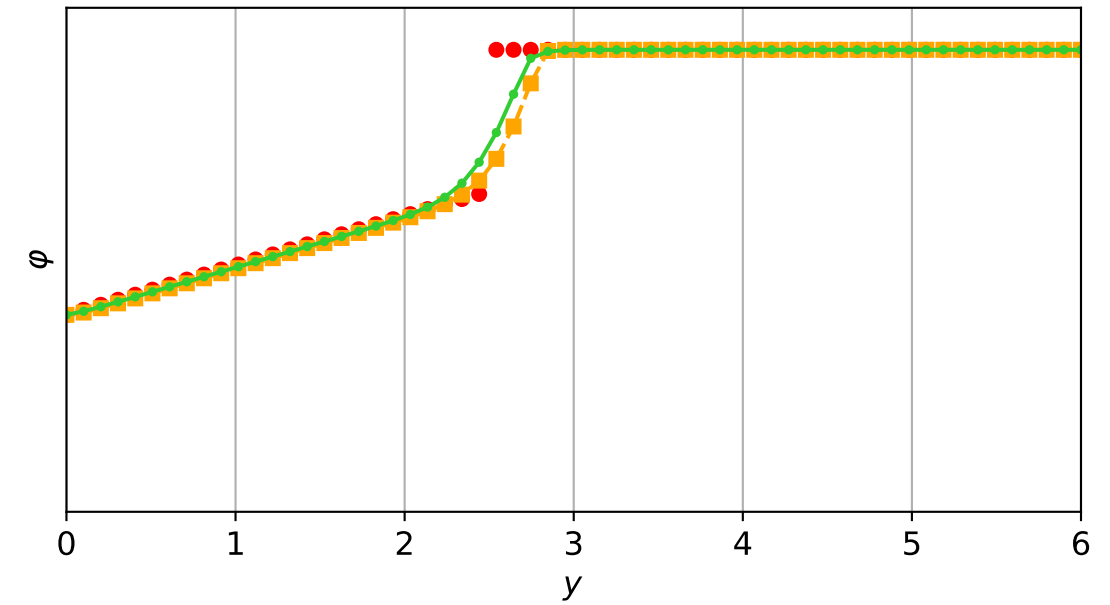
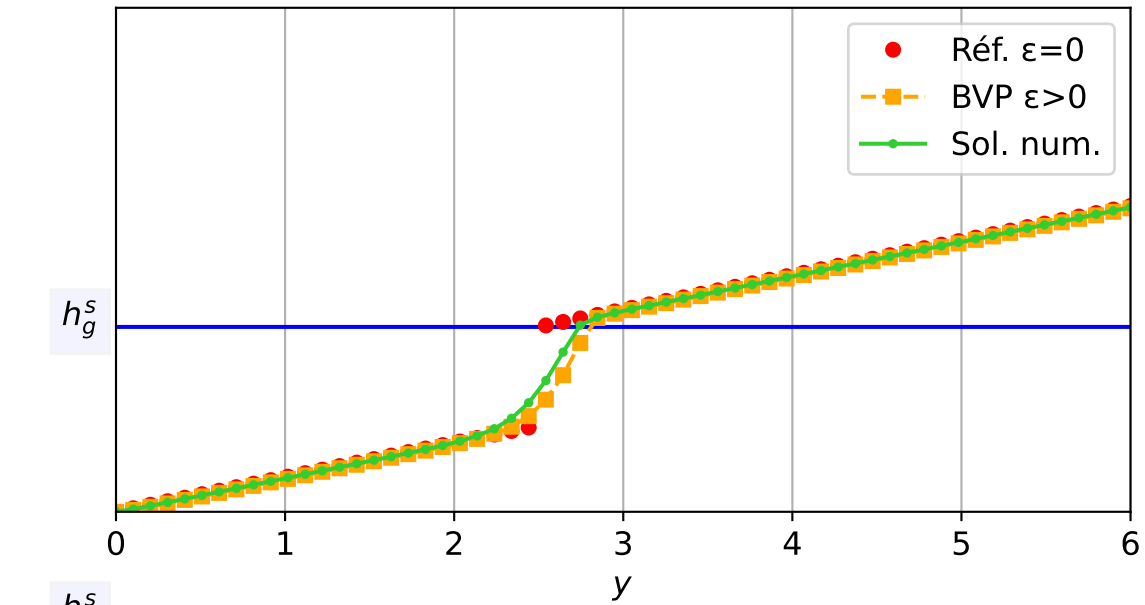
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.55113/12$  (8140 iter) — résidu eq. =  $6.54e-06$



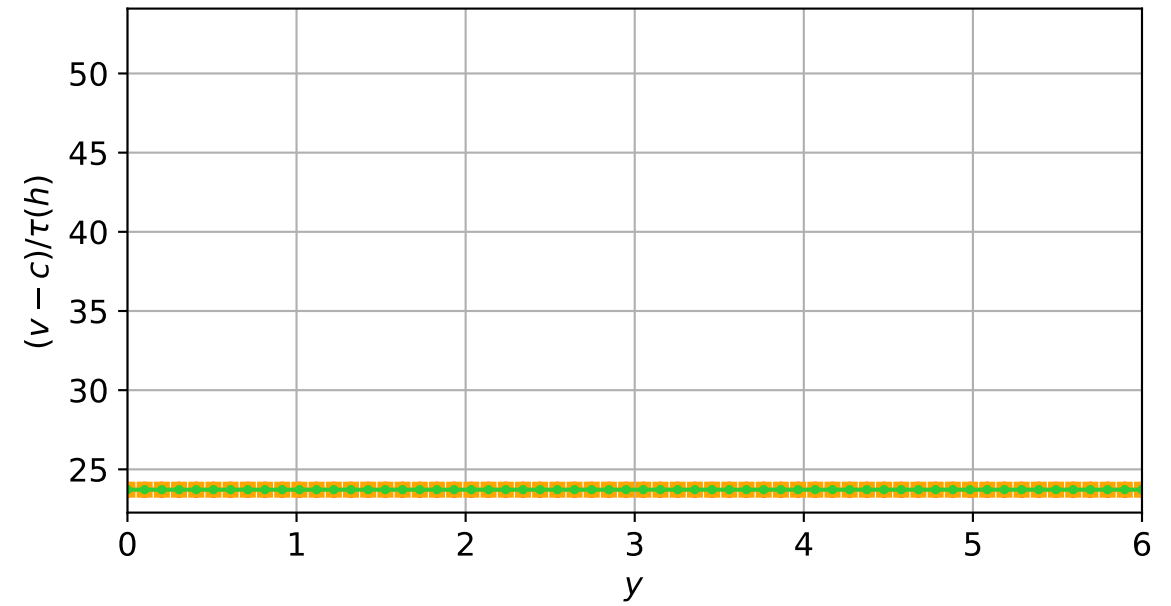
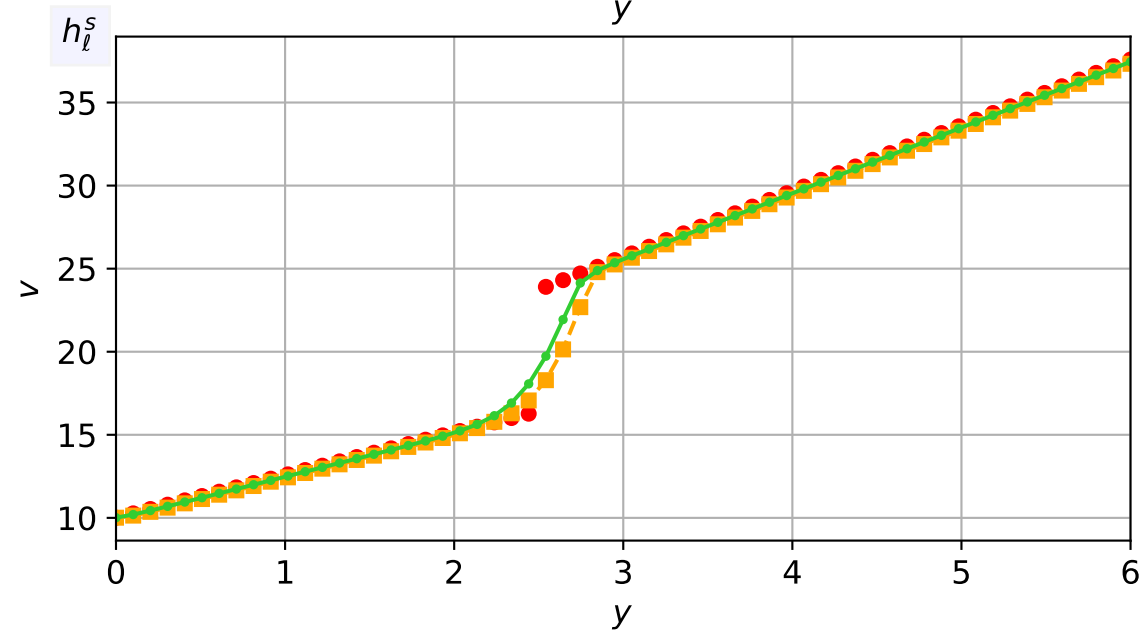
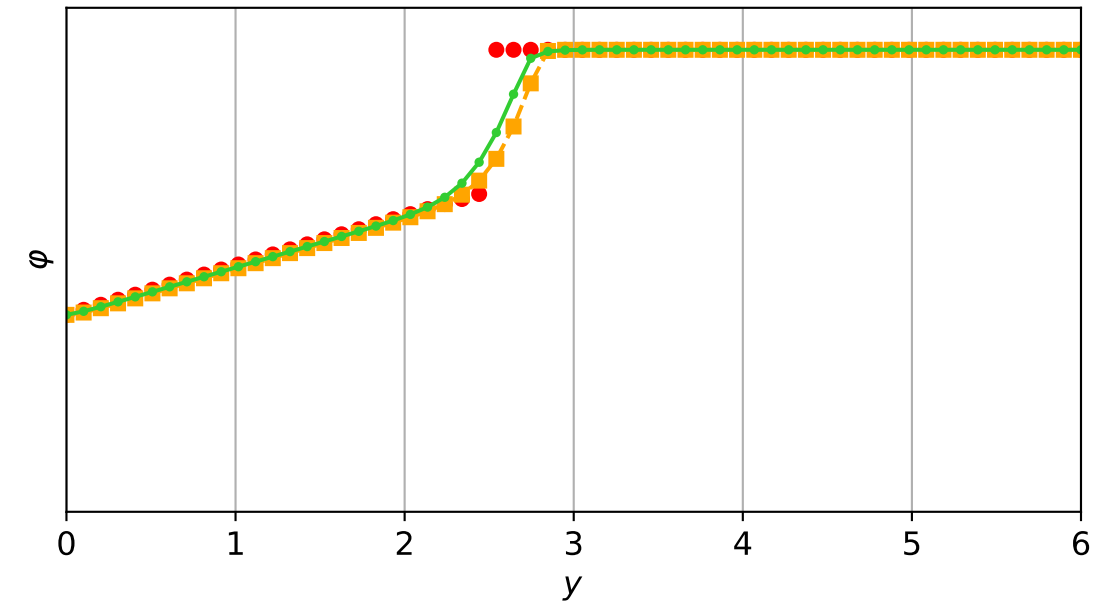
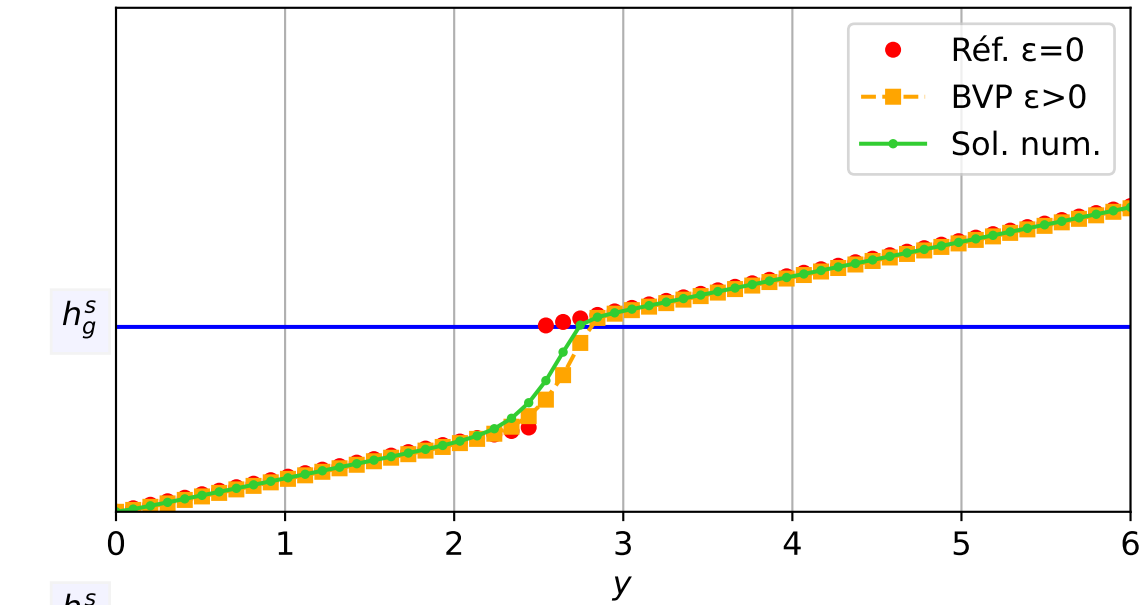
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.60001/12$  (8160 iter) — résidu eq. =  $5.80e-06$



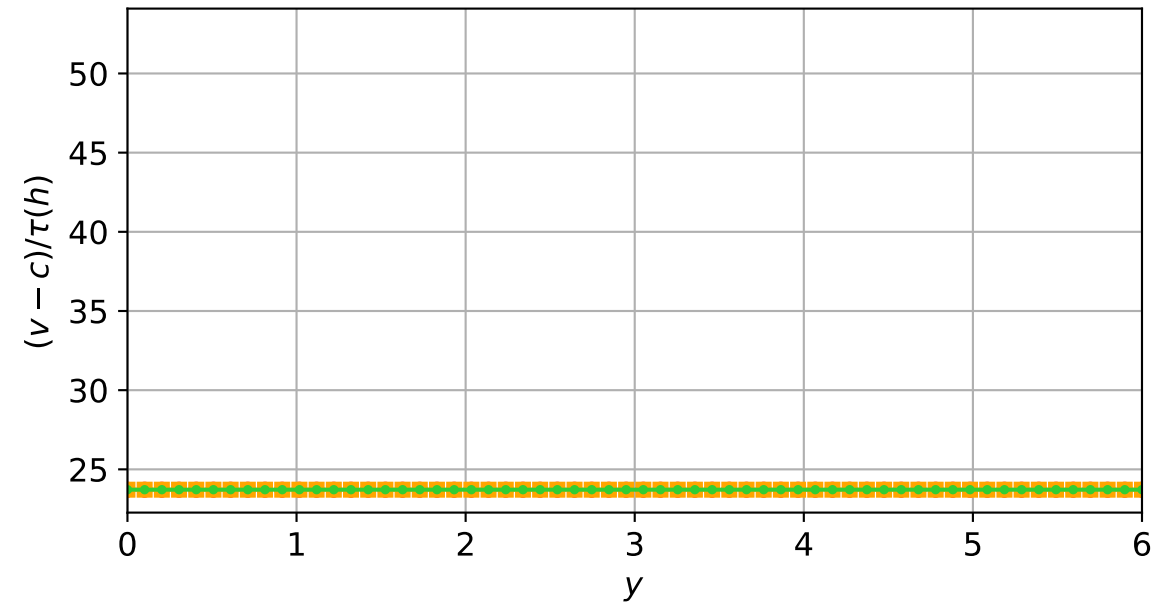
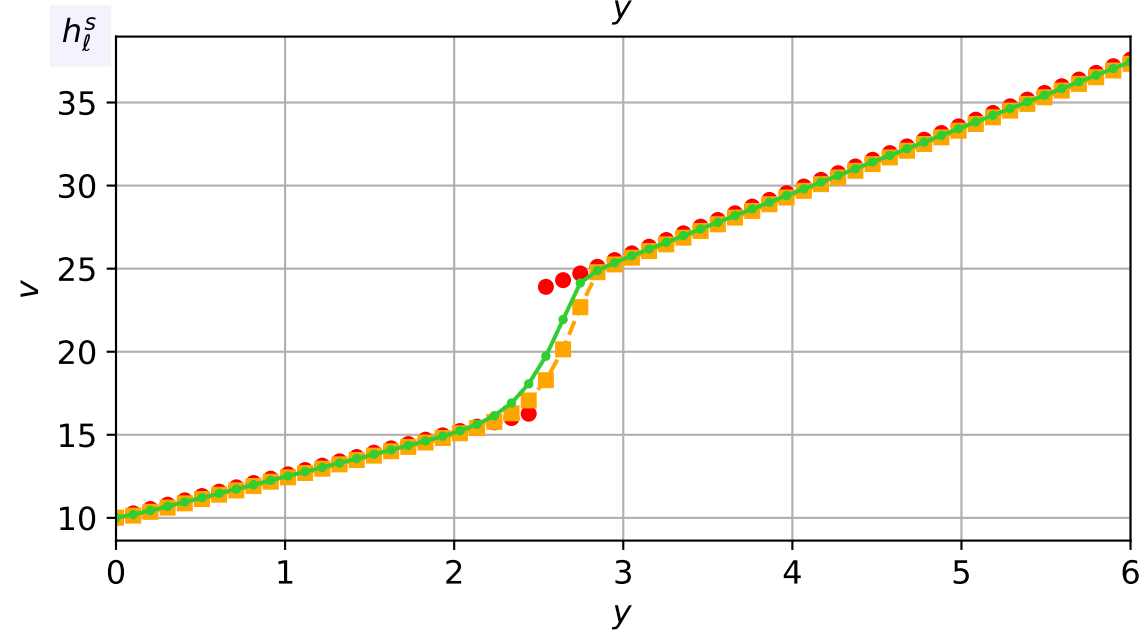
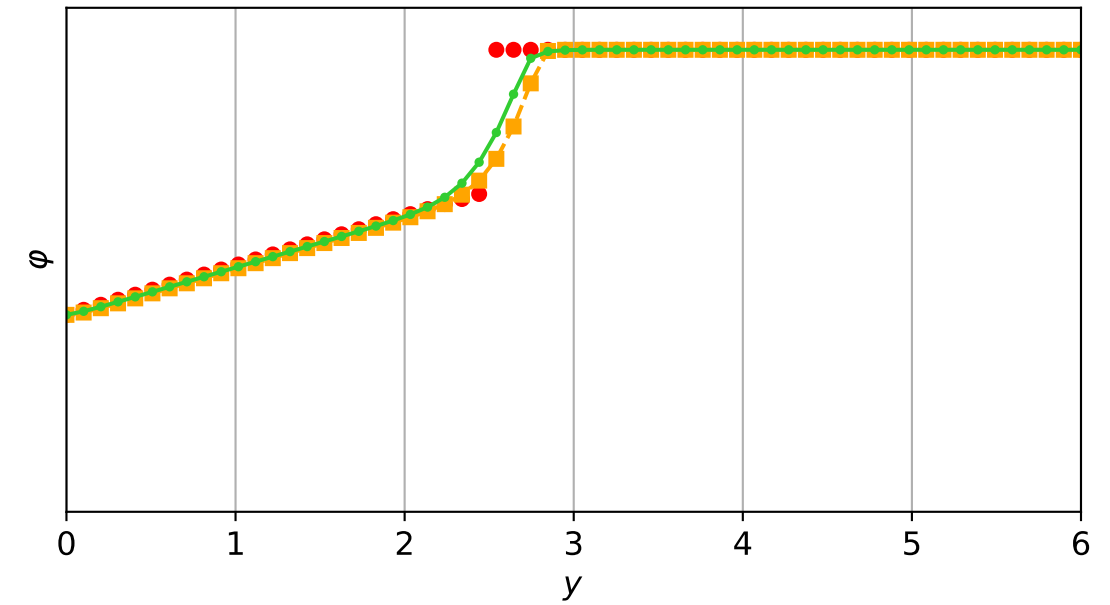
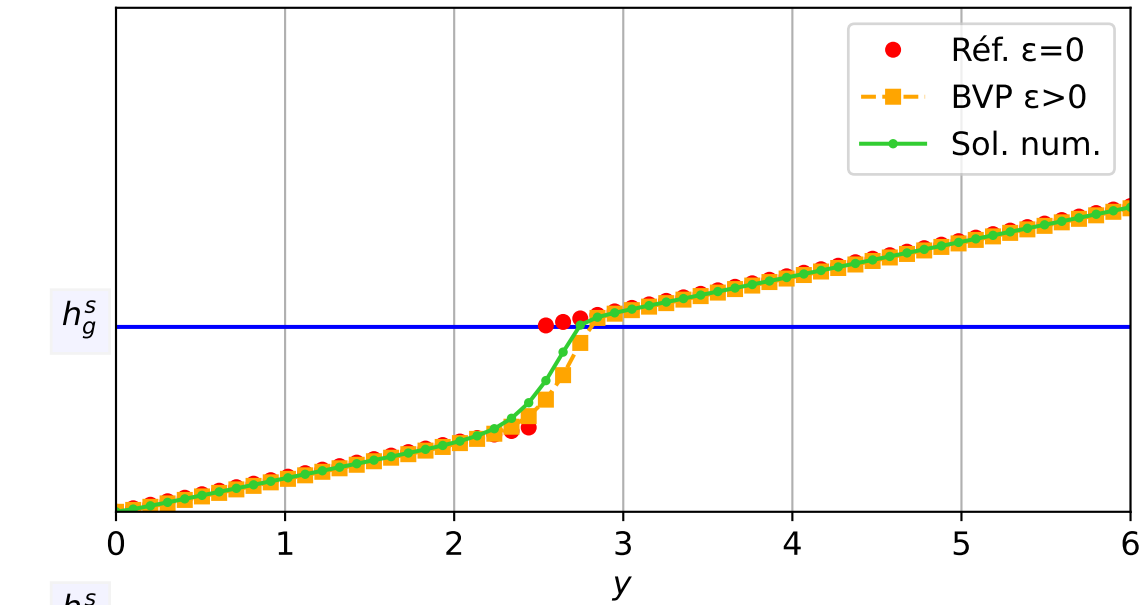
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.65133/12$  (8181 iter) — résidu eq. =  $5.11e-06$



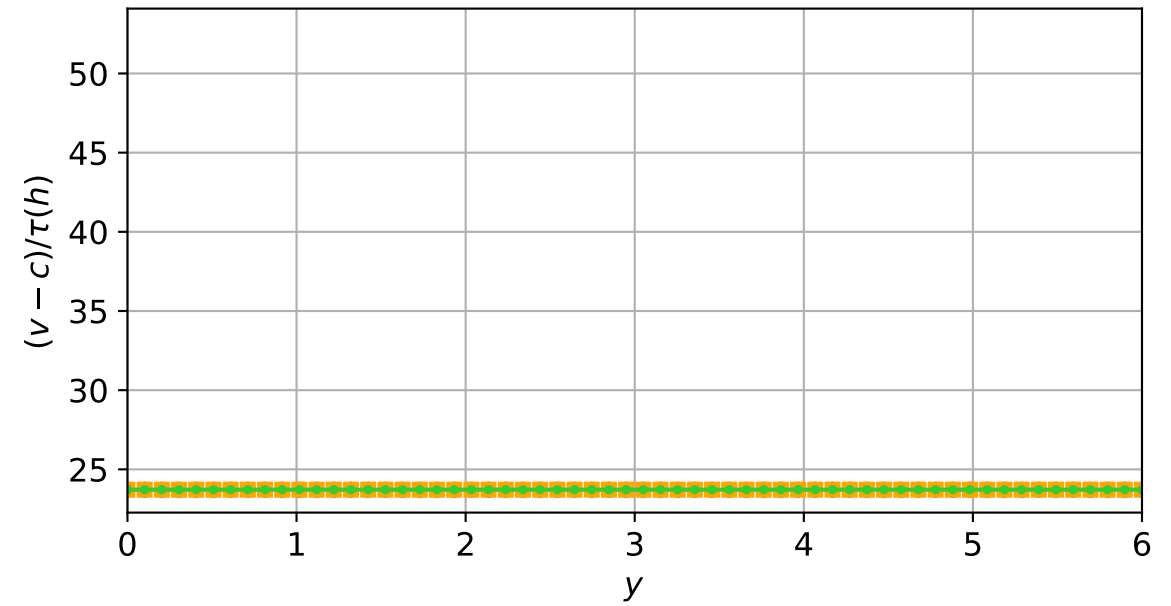
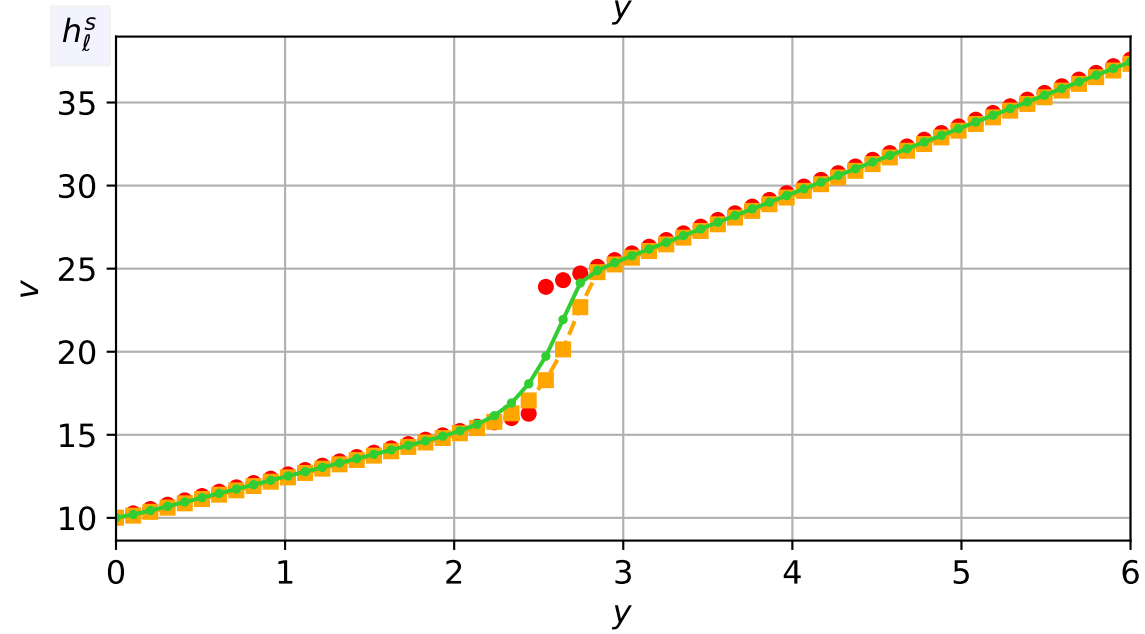
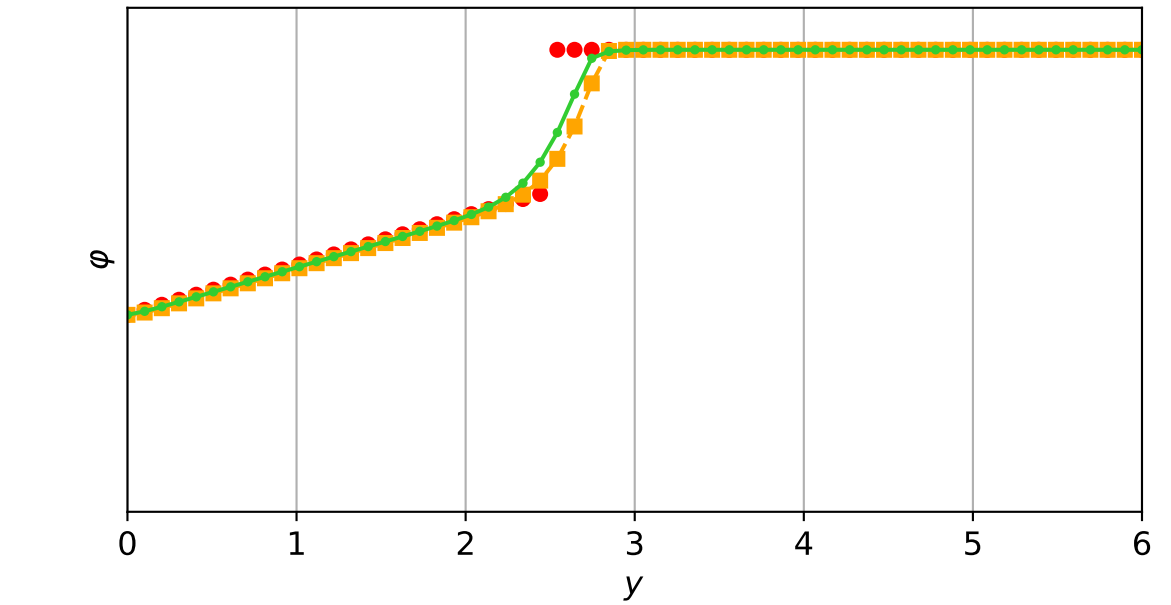
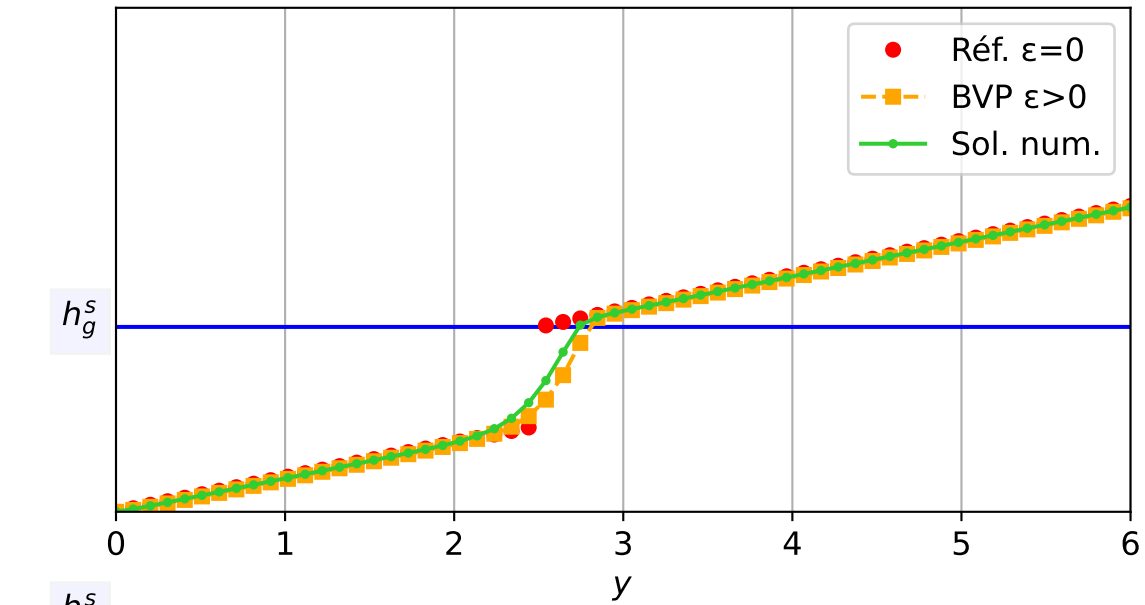
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.70021/12$  (8201 iter) — résidu eq. =  $4.53e-06$



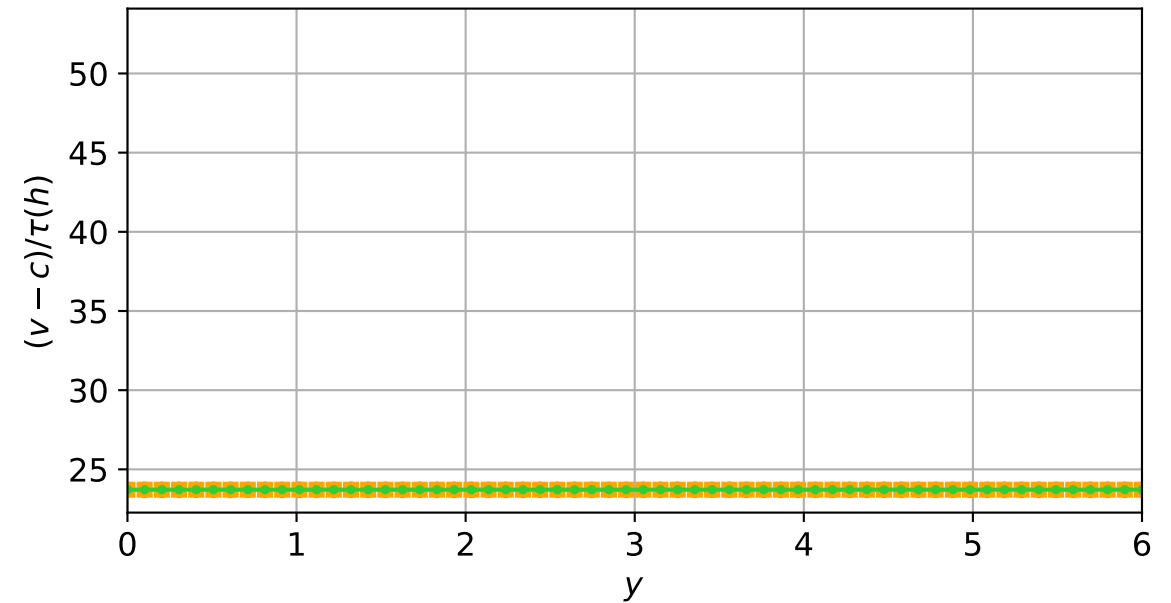
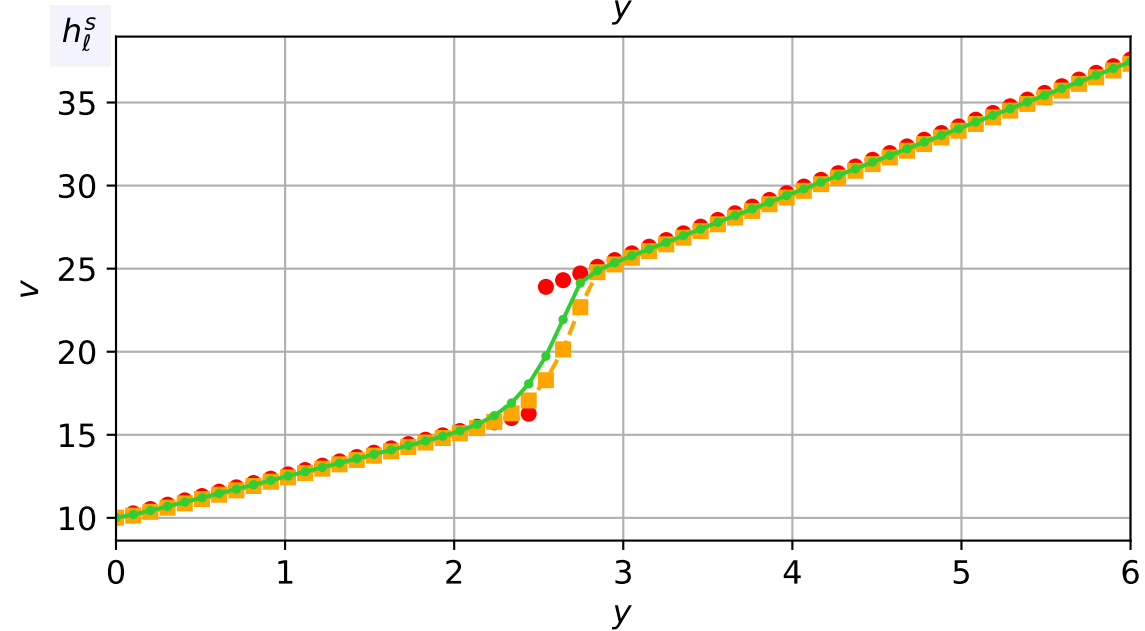
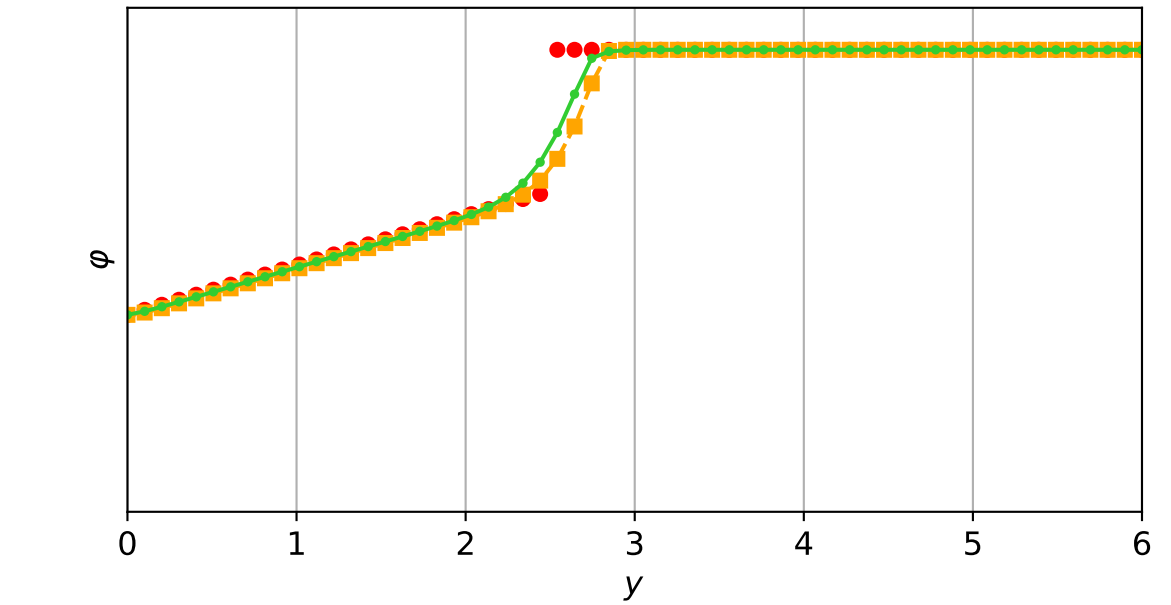
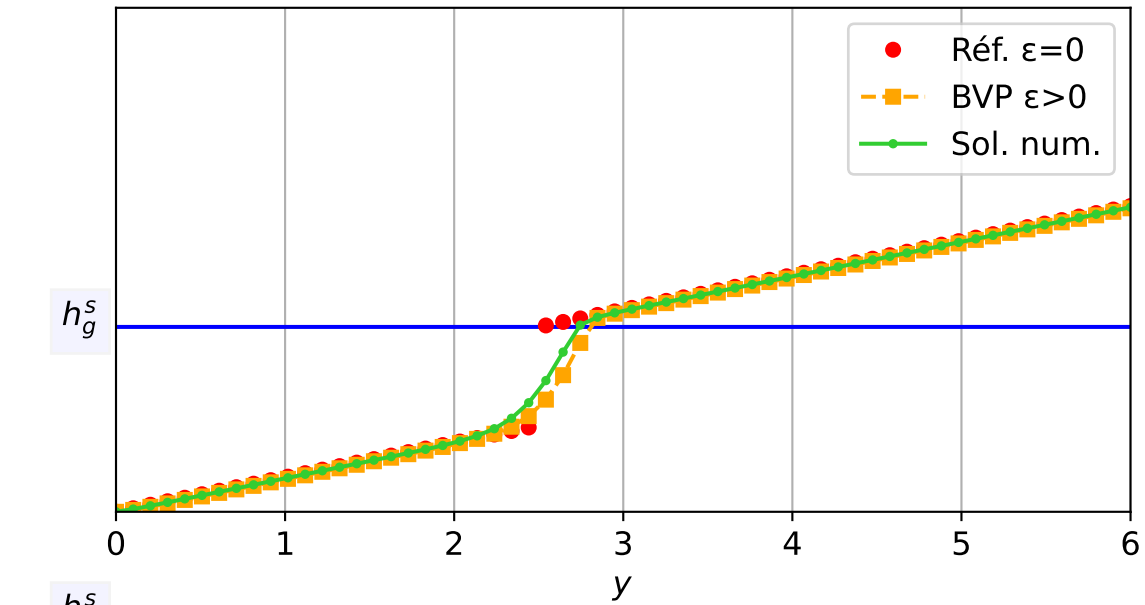
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.75153/12$  (8222 iter) — résidu eq. =  $3.99e-06$



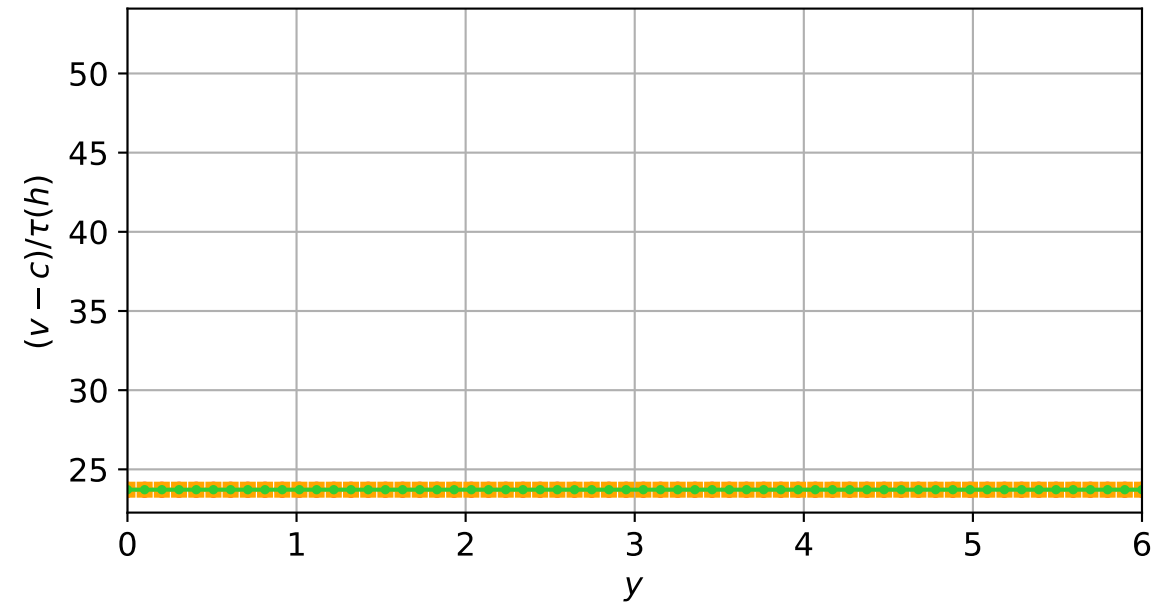
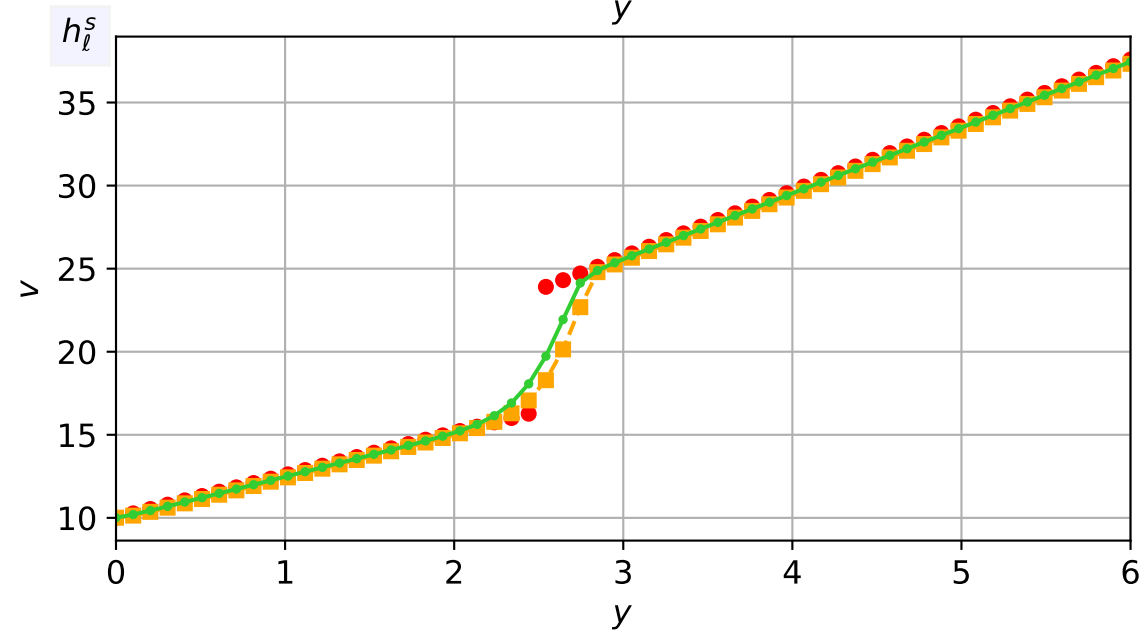
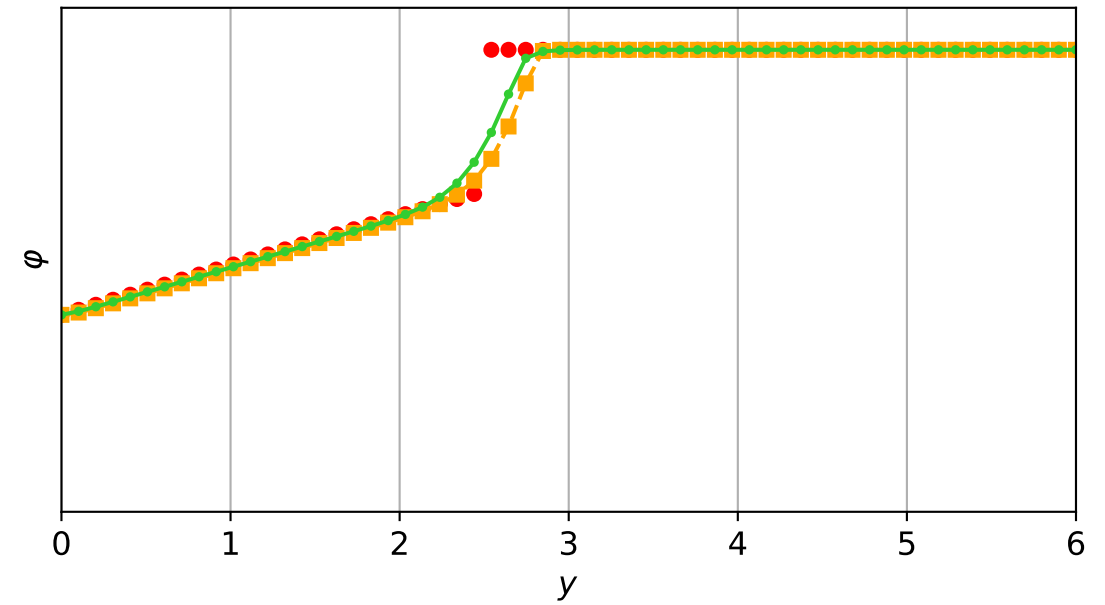
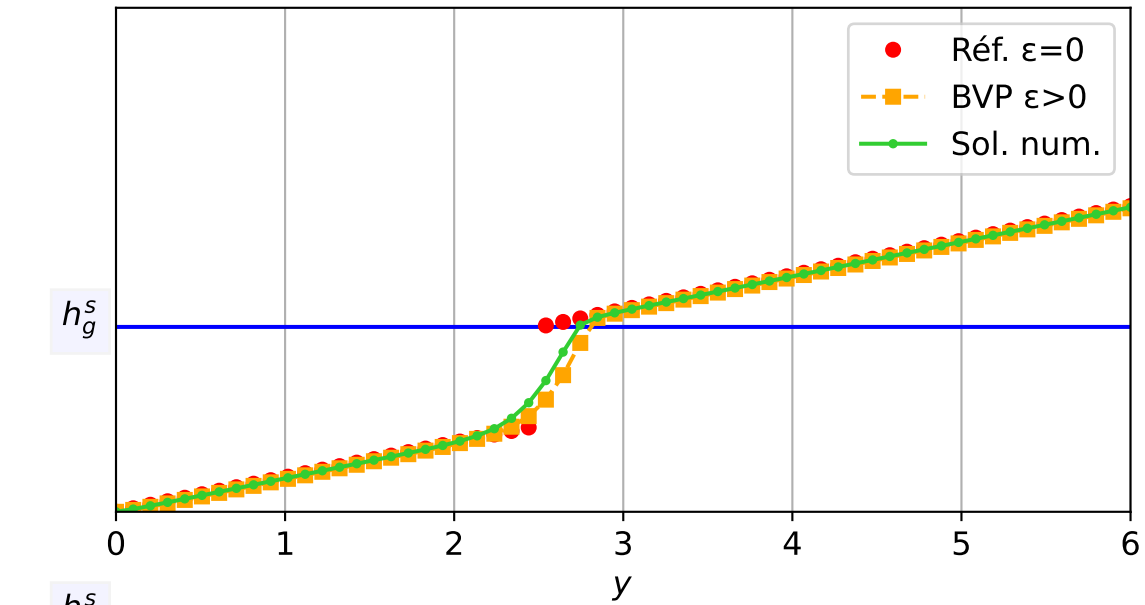
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.80041/12$  (8242 iter) — résidu eq. =  $3.53e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.85173/12$  (8263 iter) — résidu eq. =  $3.11e-06$

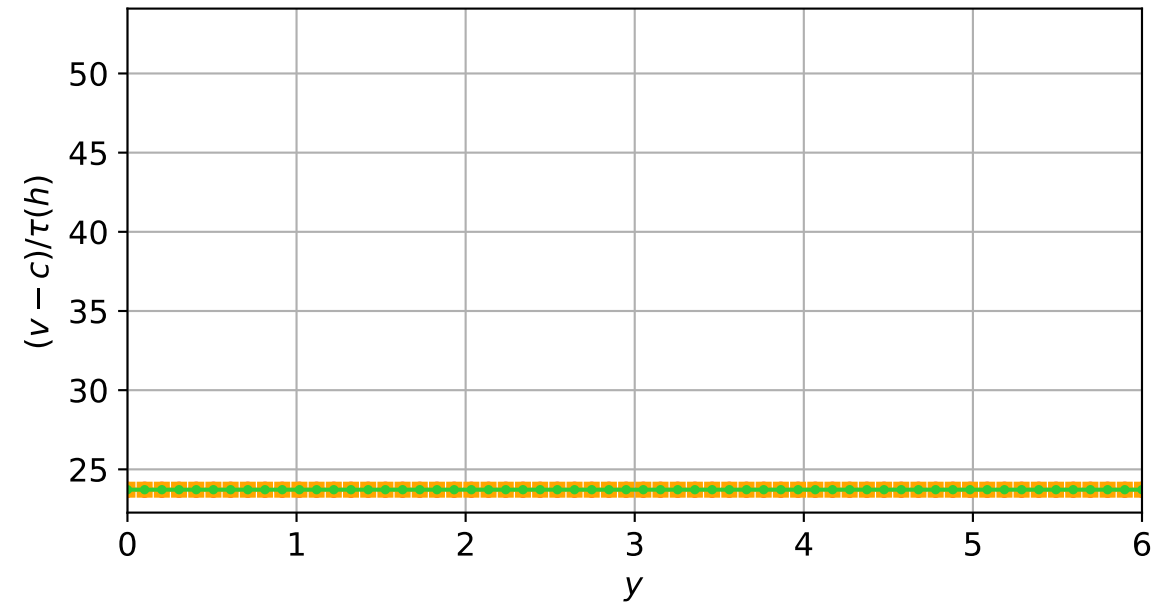
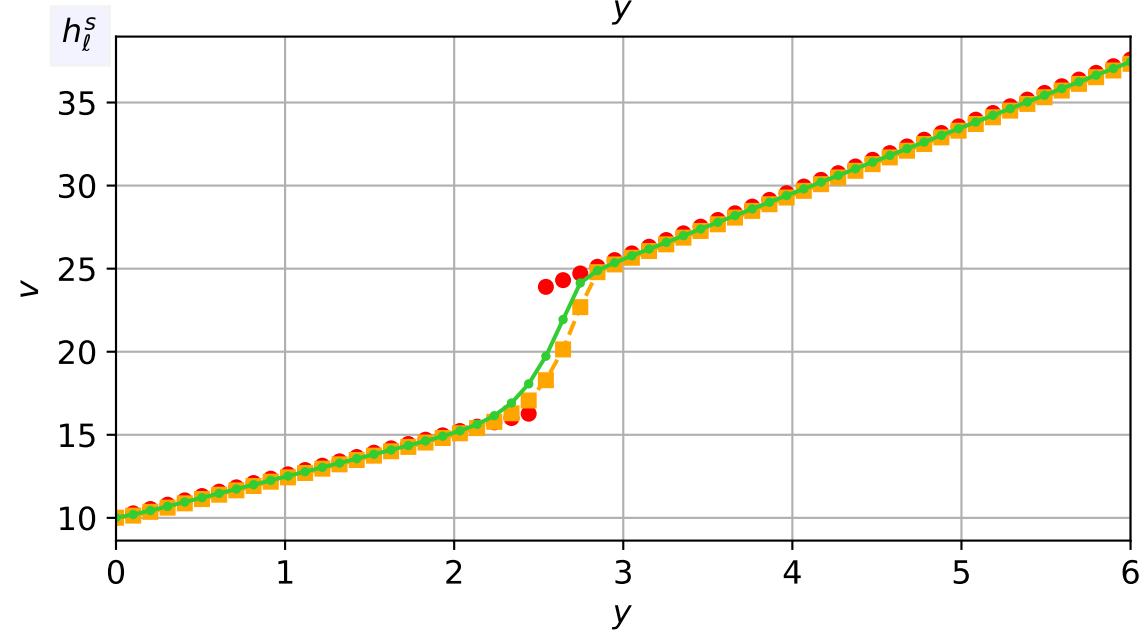
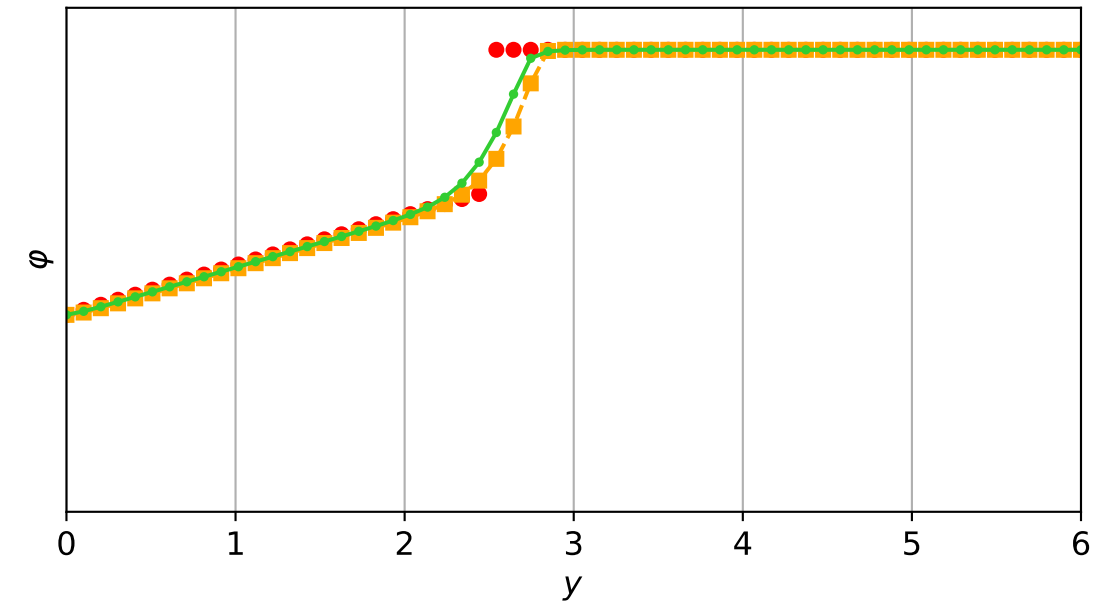
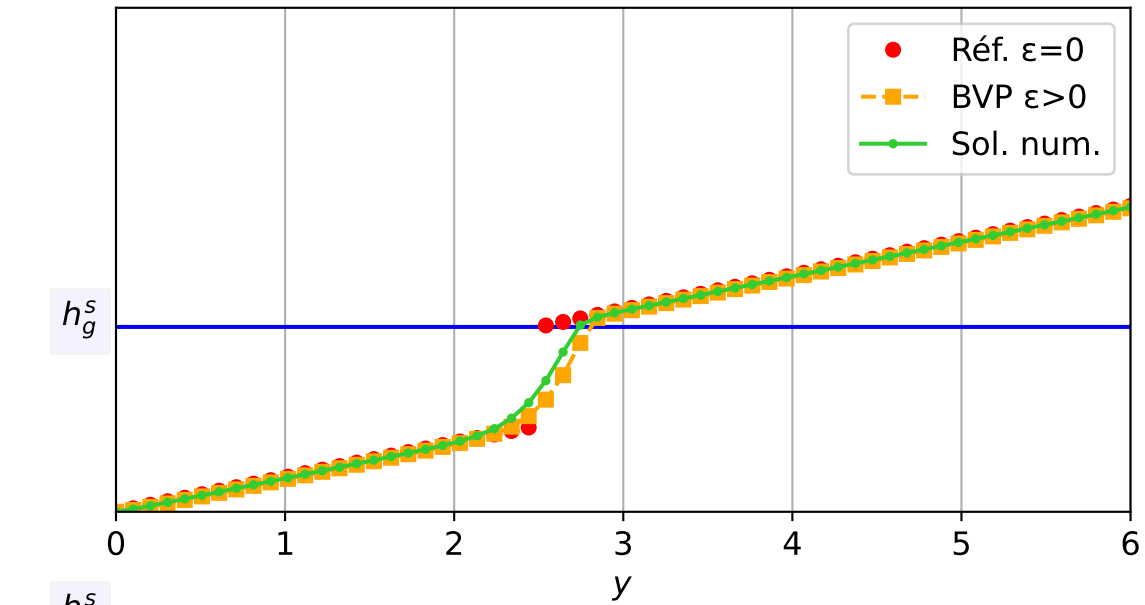


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.9006/12$  (8283 iter) — résidu eq. =  $2.76e-06$

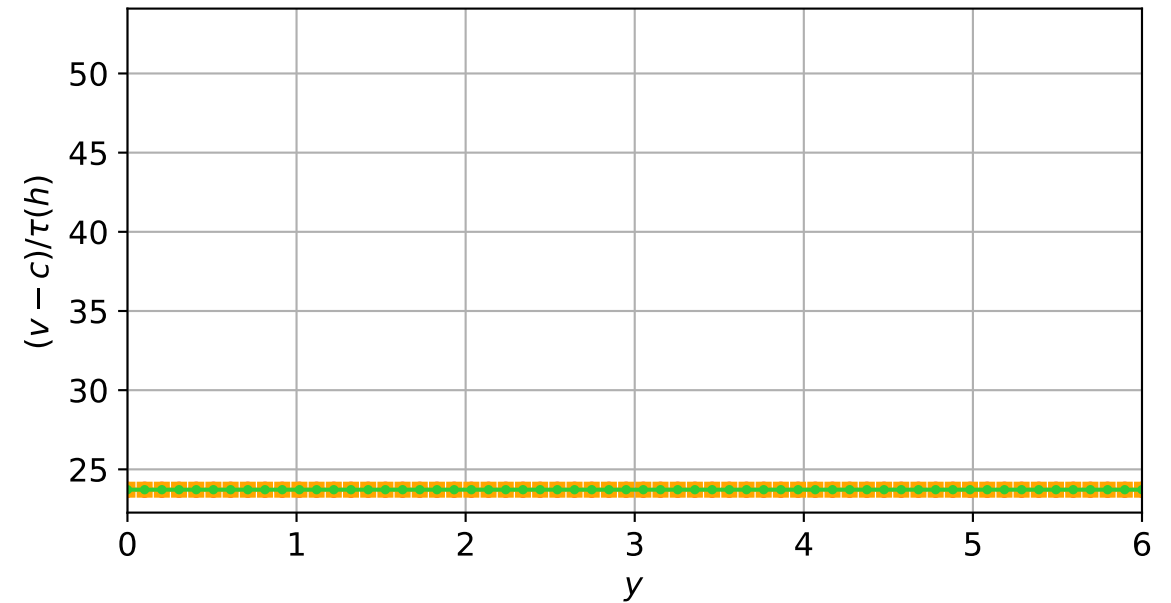
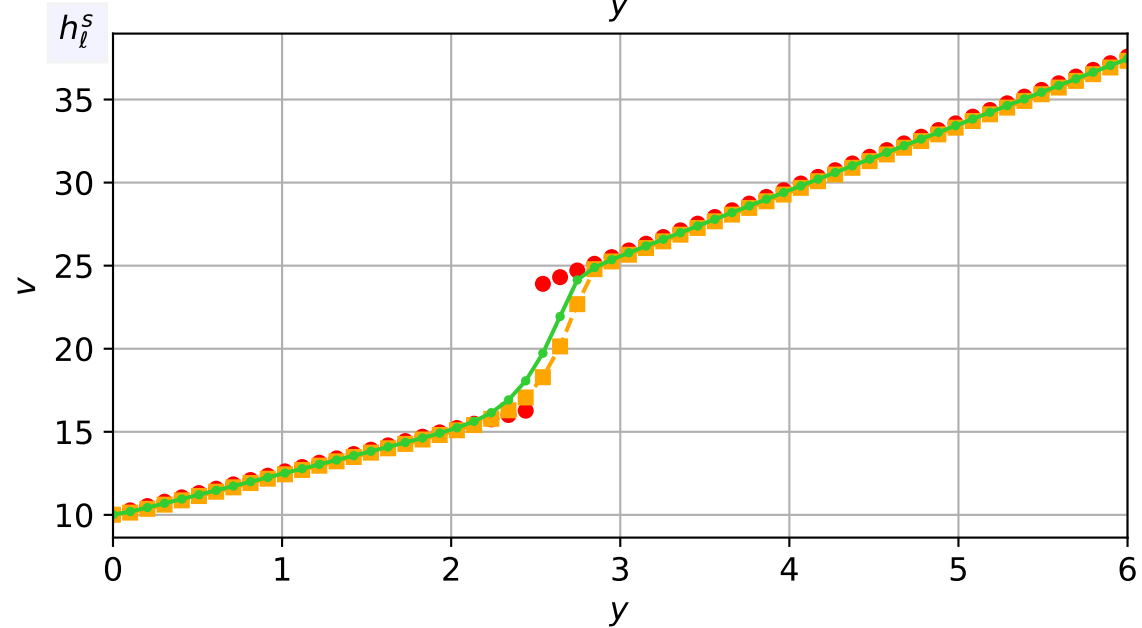
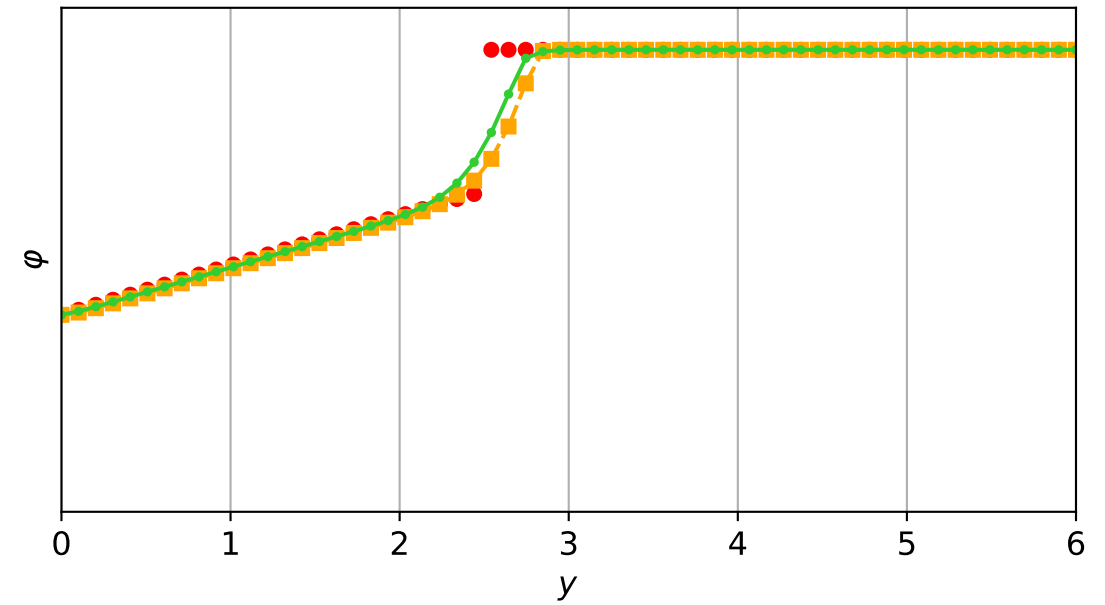
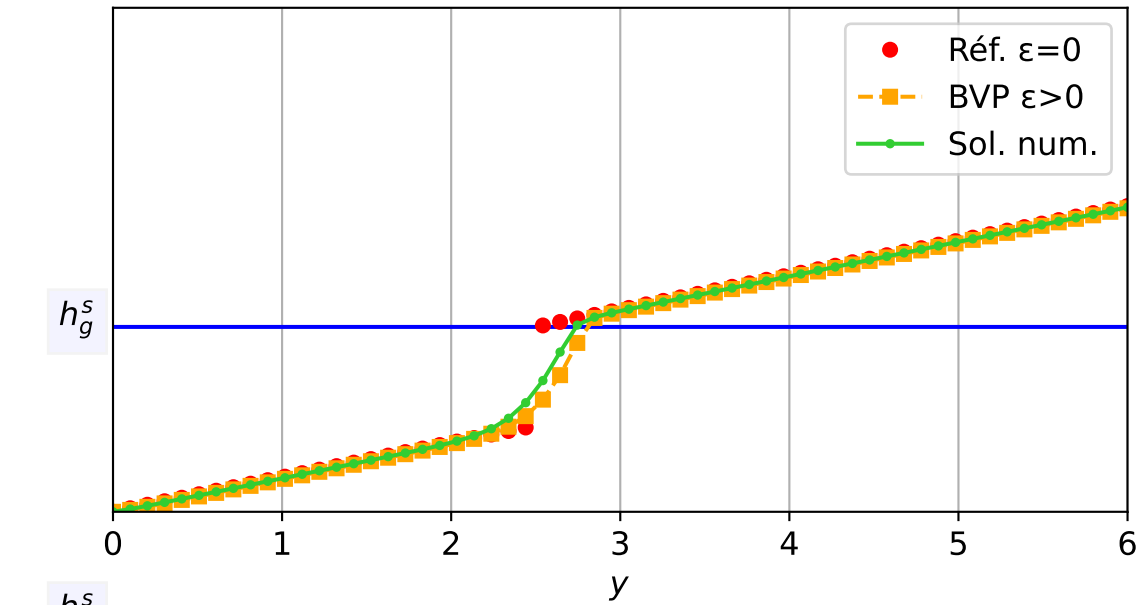




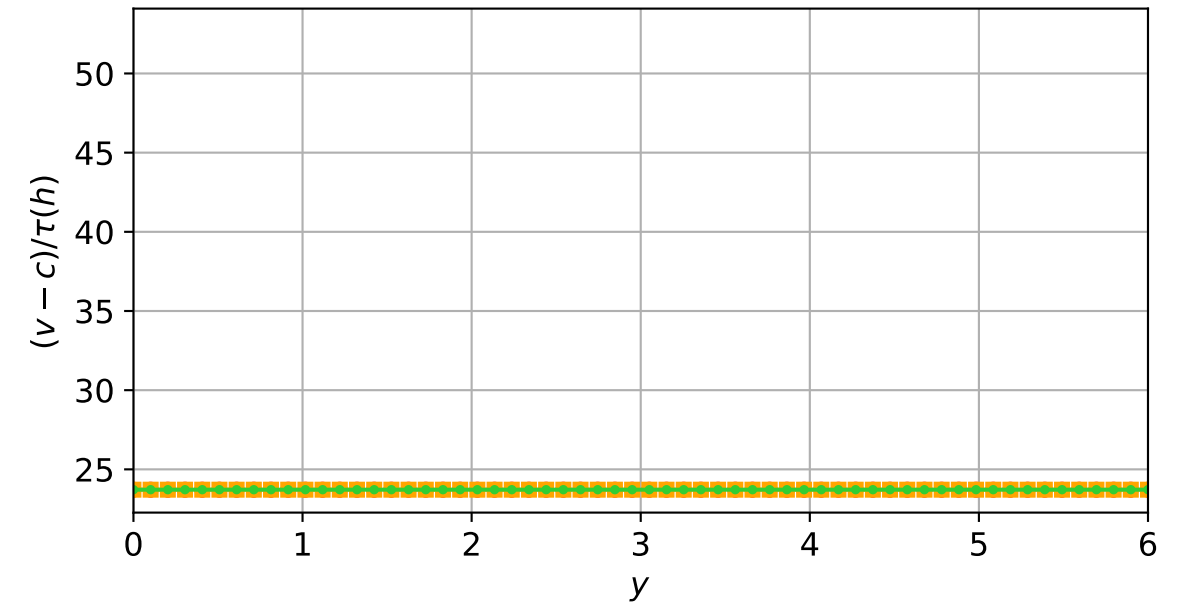
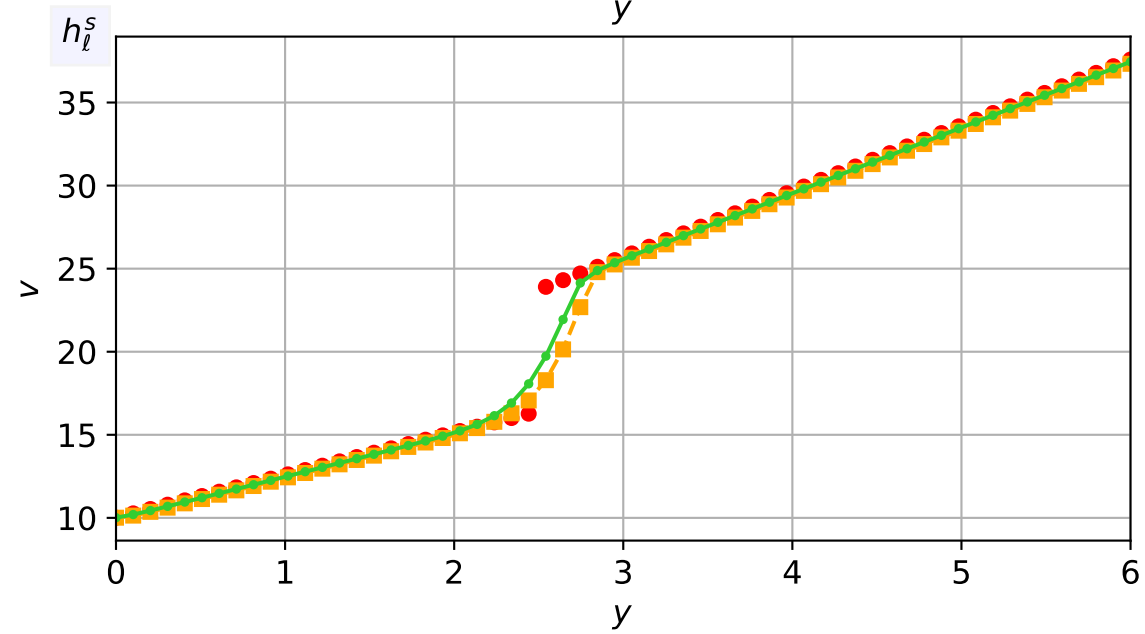
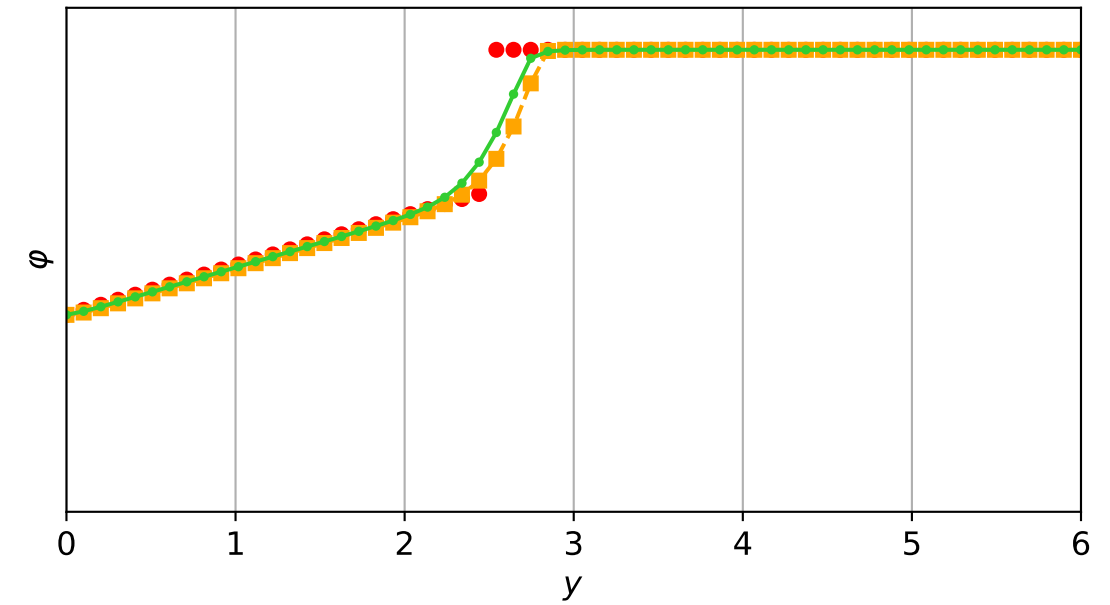
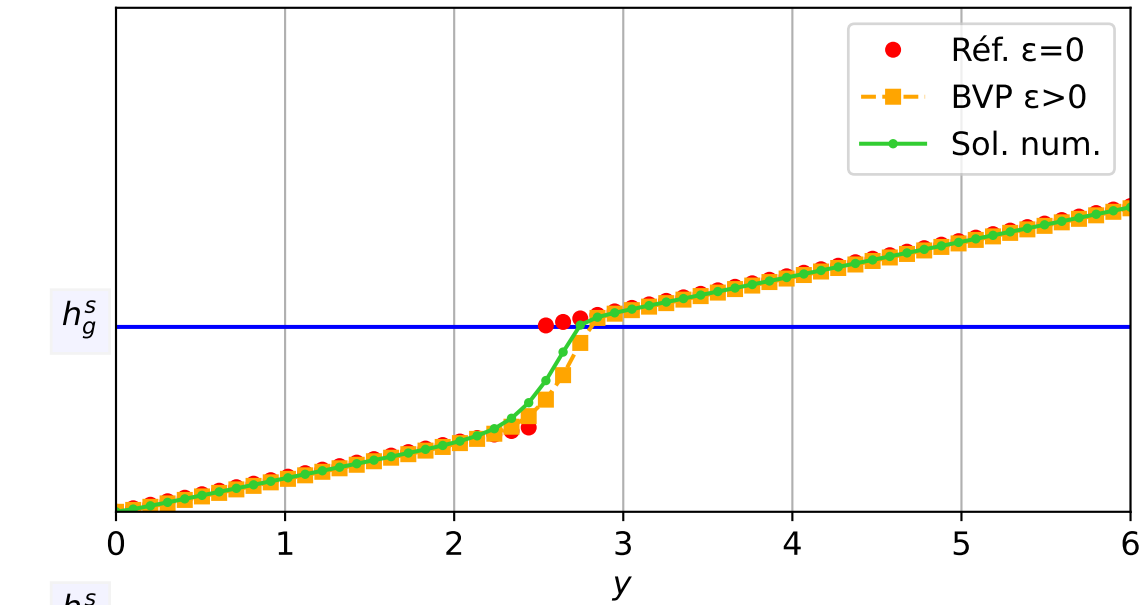
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.95193/12$  (8304 iter) — résidu eq. =  $9.69e-06$



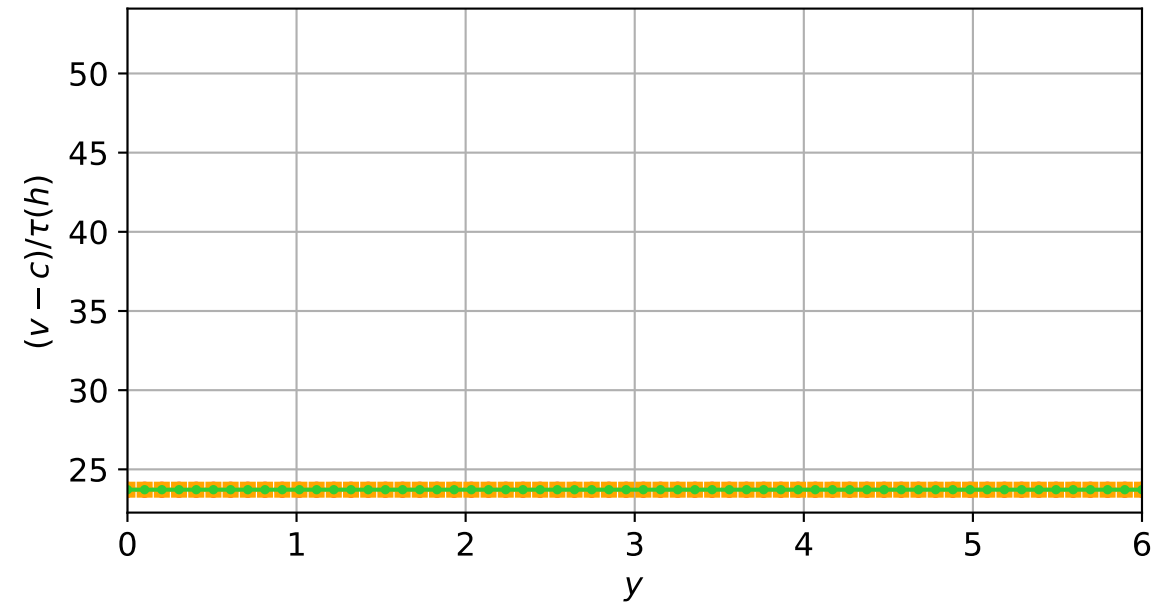
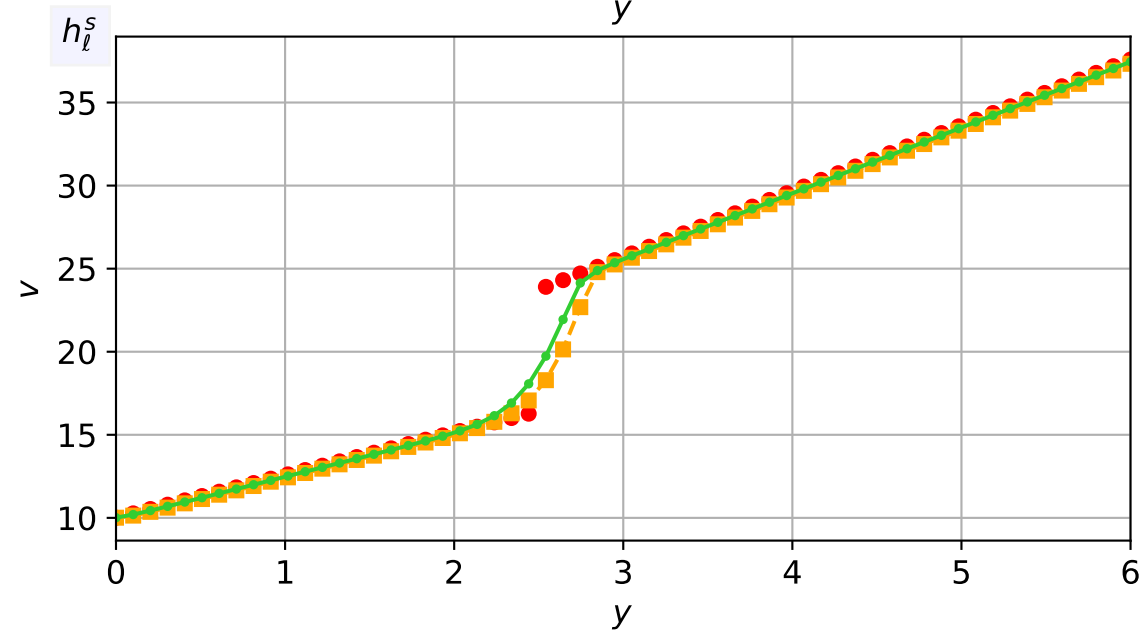
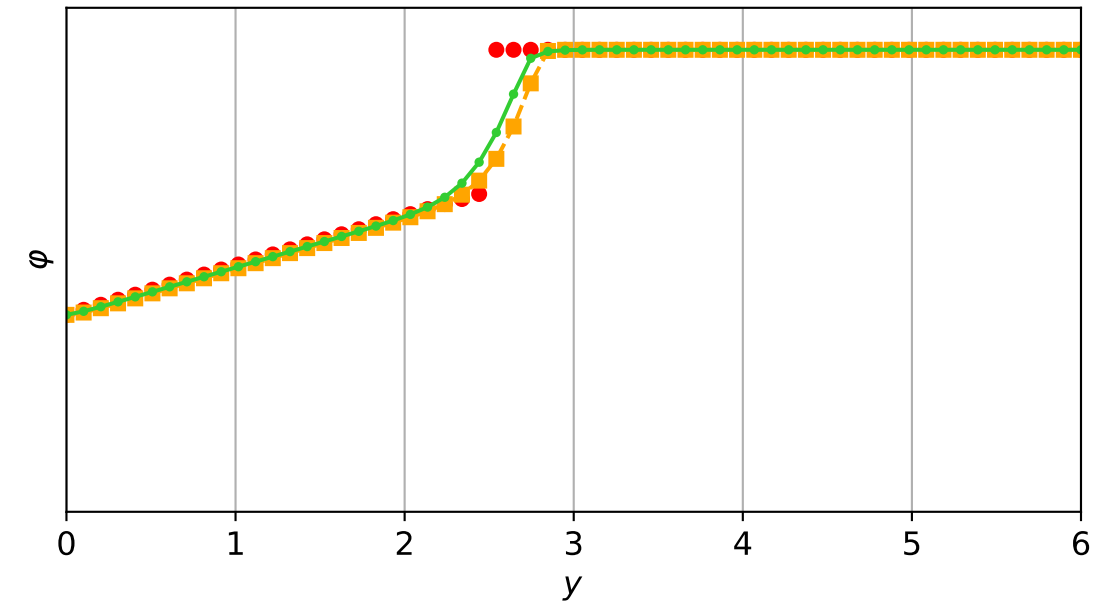
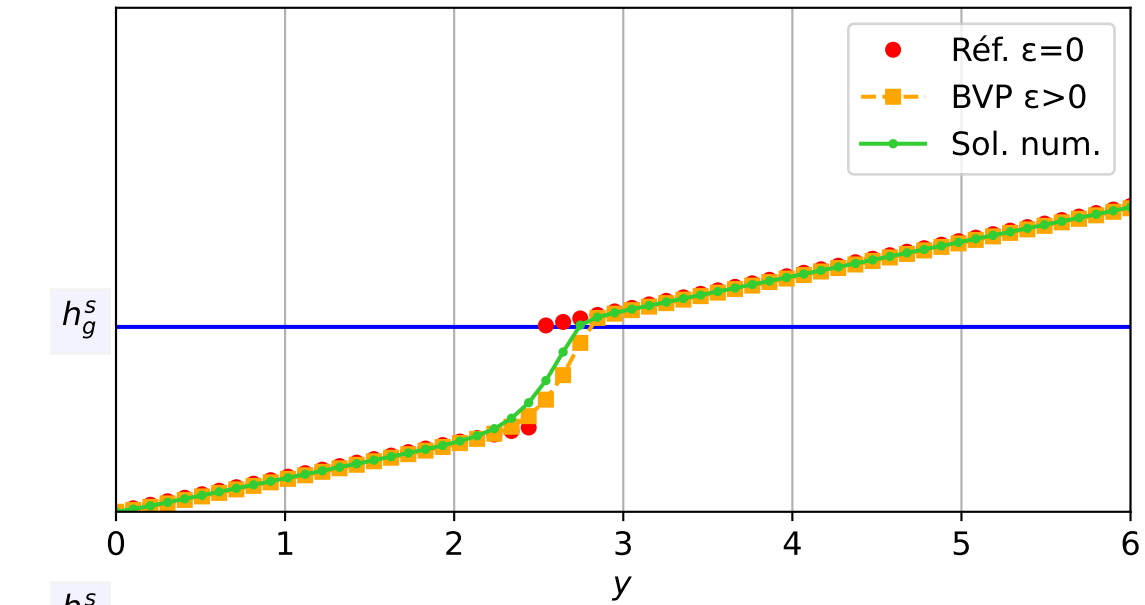
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.0008/12$  (8324 iter) — résidu eq. =  $8.57e-06$



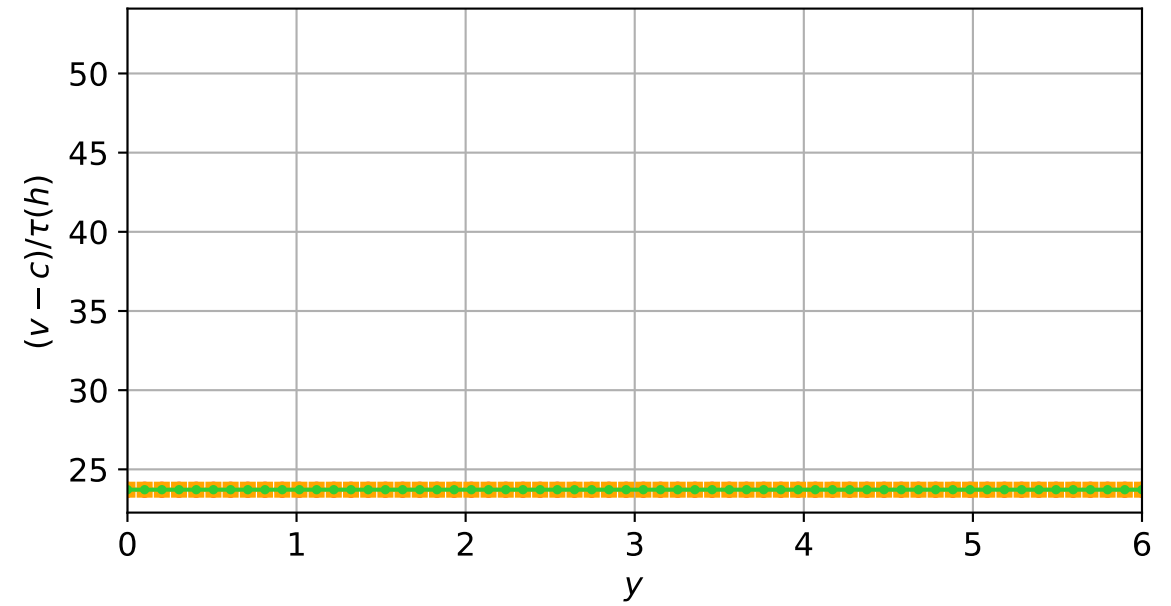
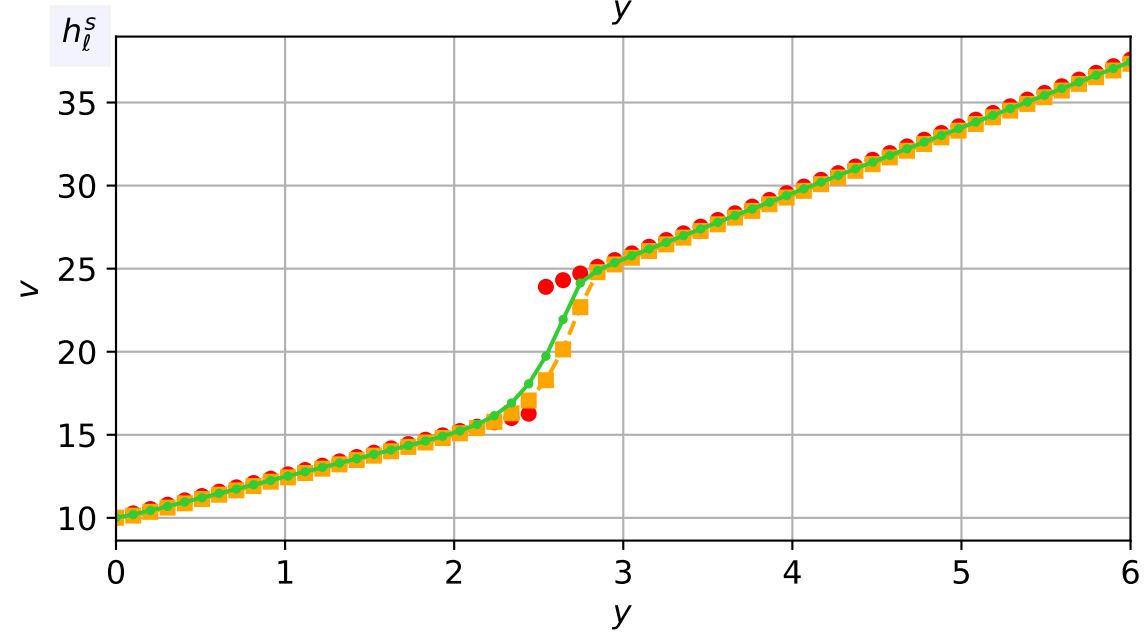
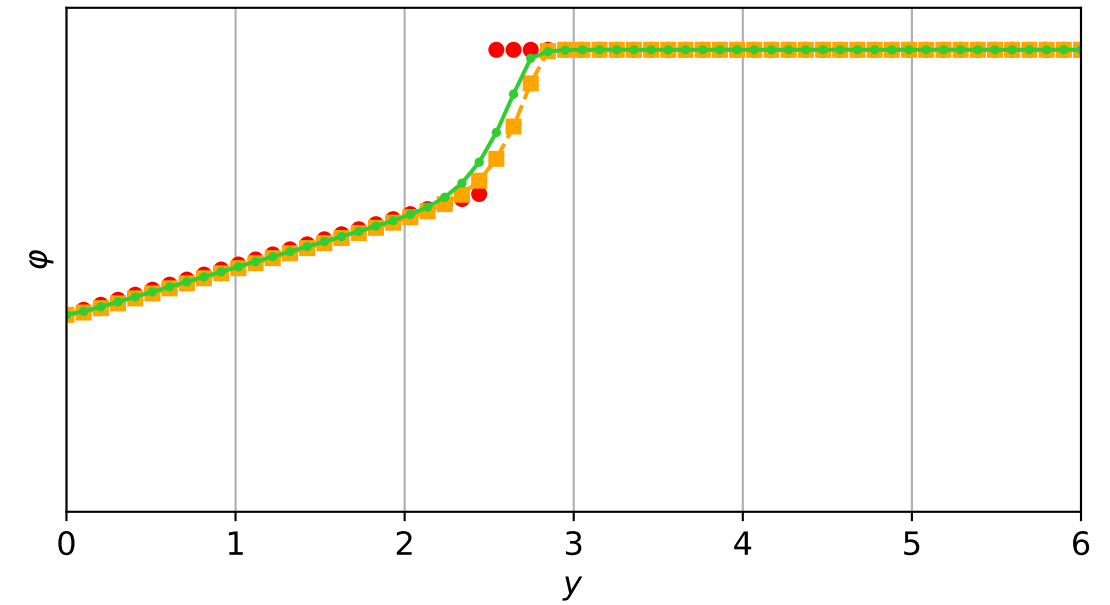
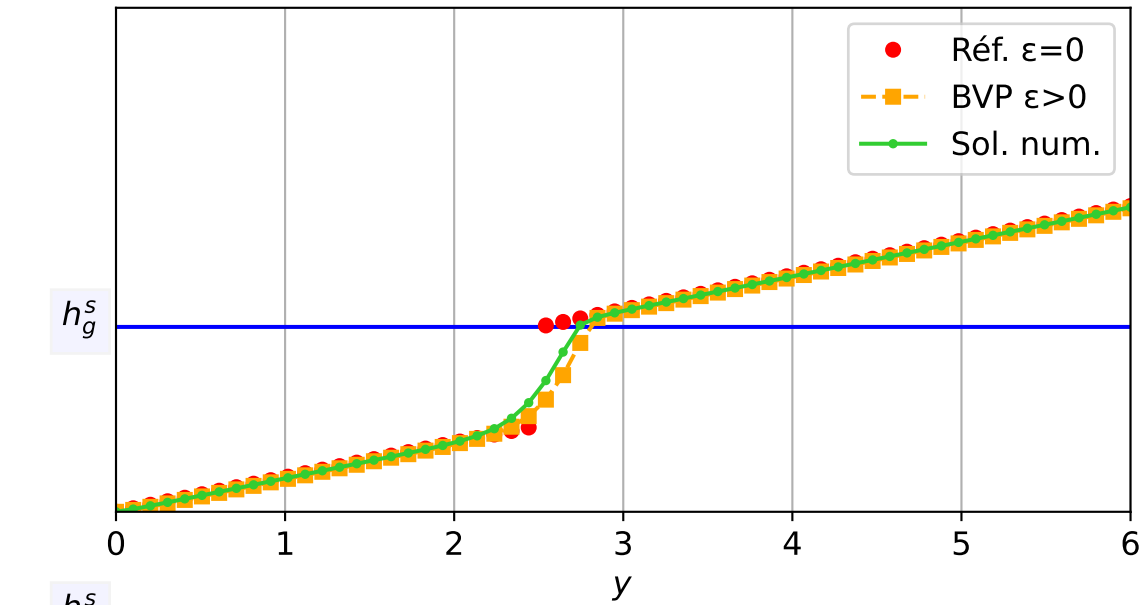
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.05212/12$  (8345 iter) — résidu eq. =  $7.54e-06$



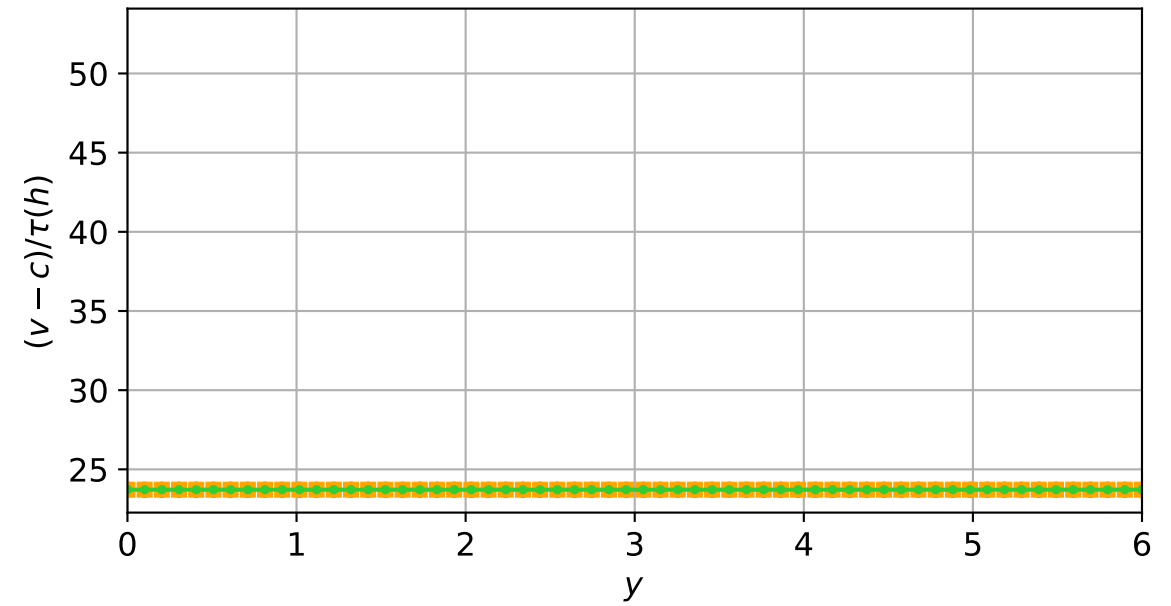
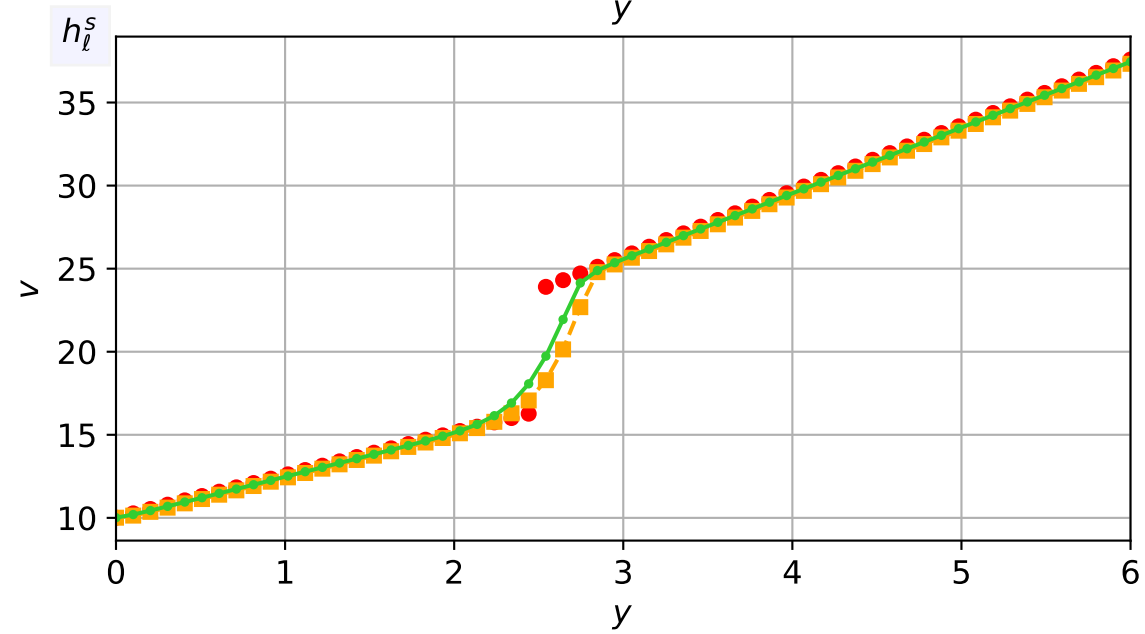
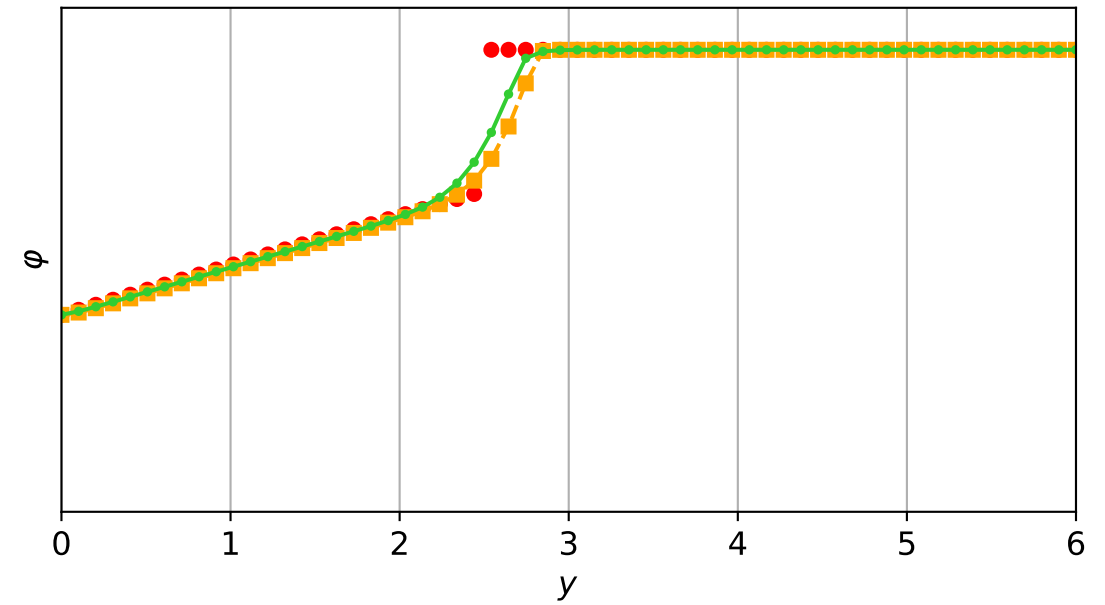
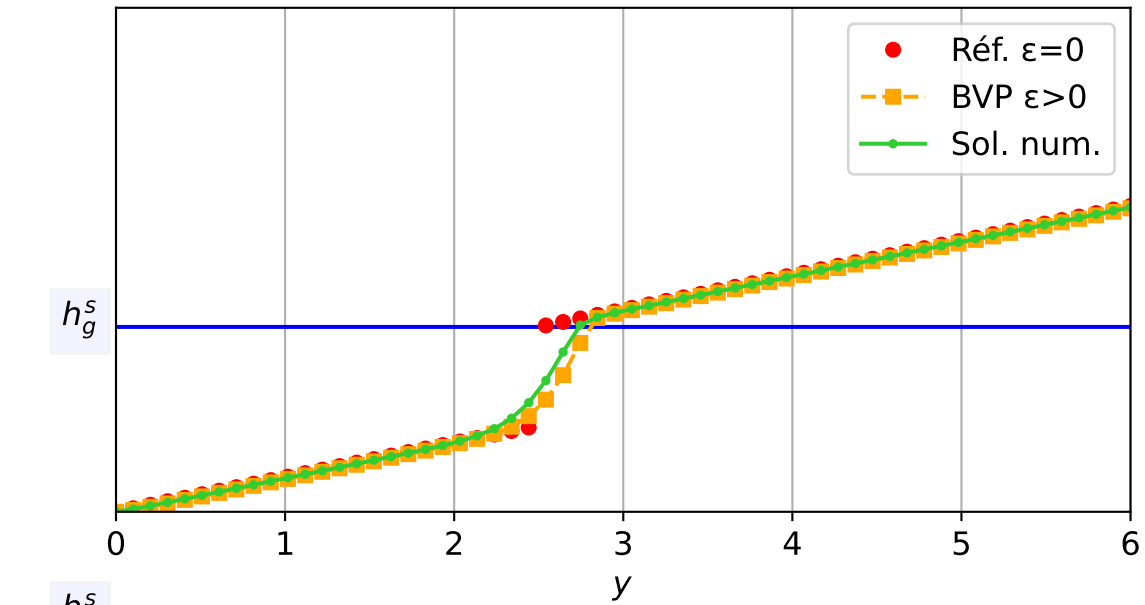
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.101/12$  (8365 iter) — résidu eq. =  $6.68e-06$



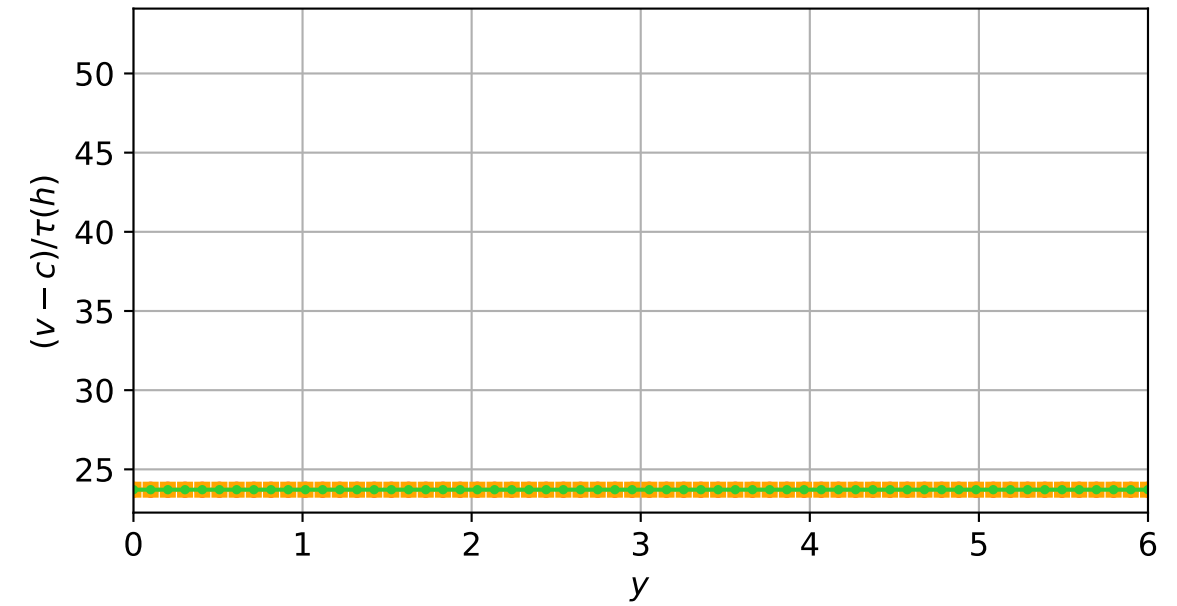
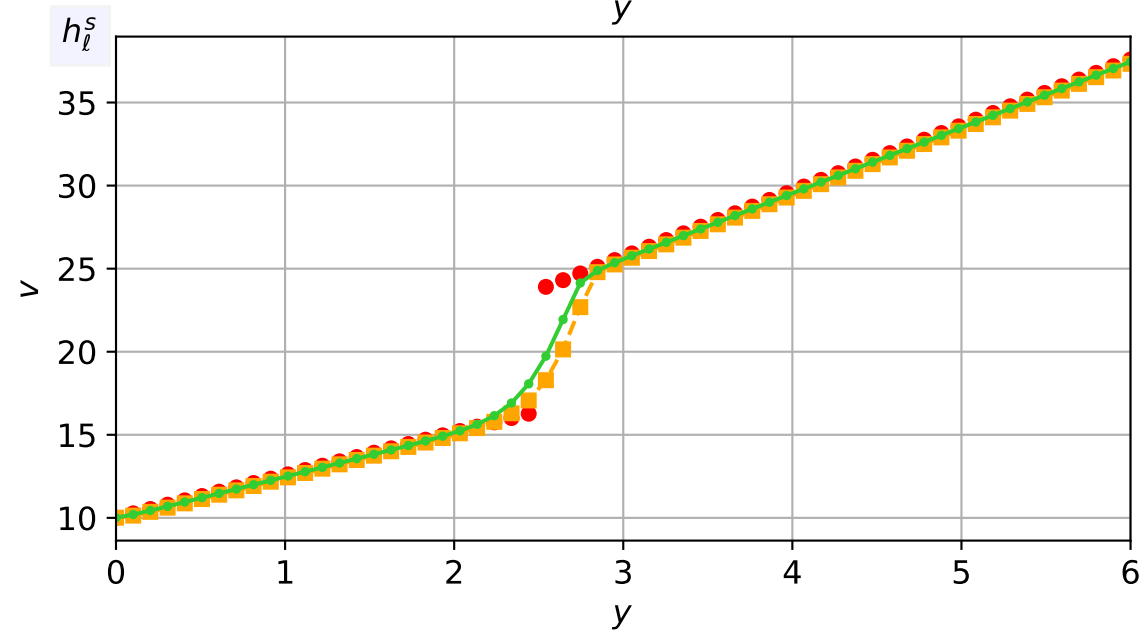
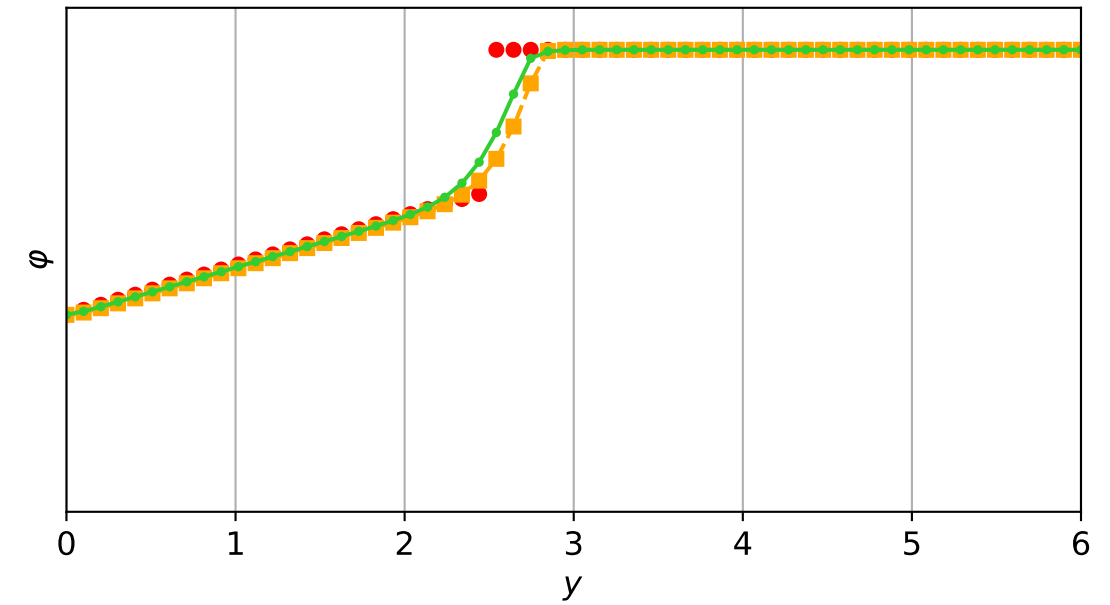
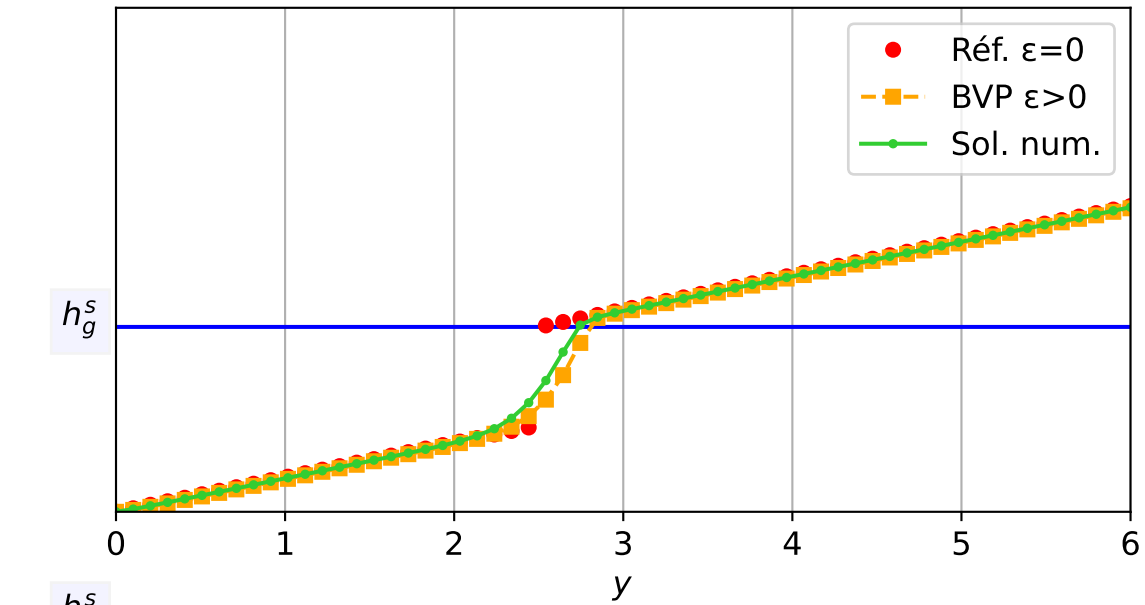
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.15232/12$  (8386 iter) — résidu eq. =  $5.88e-06$



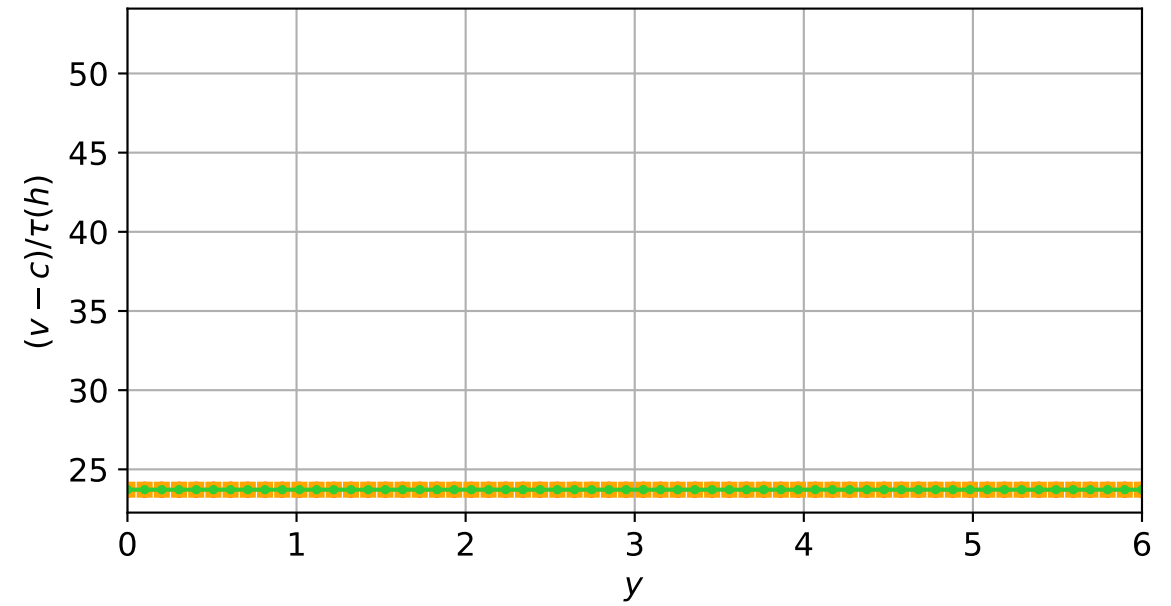
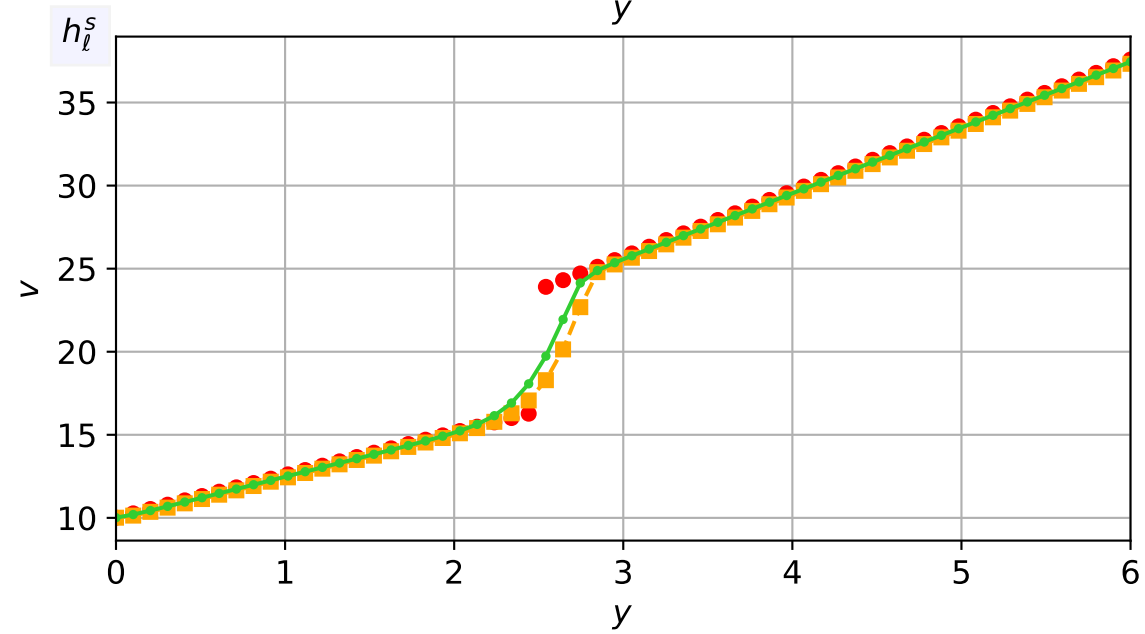
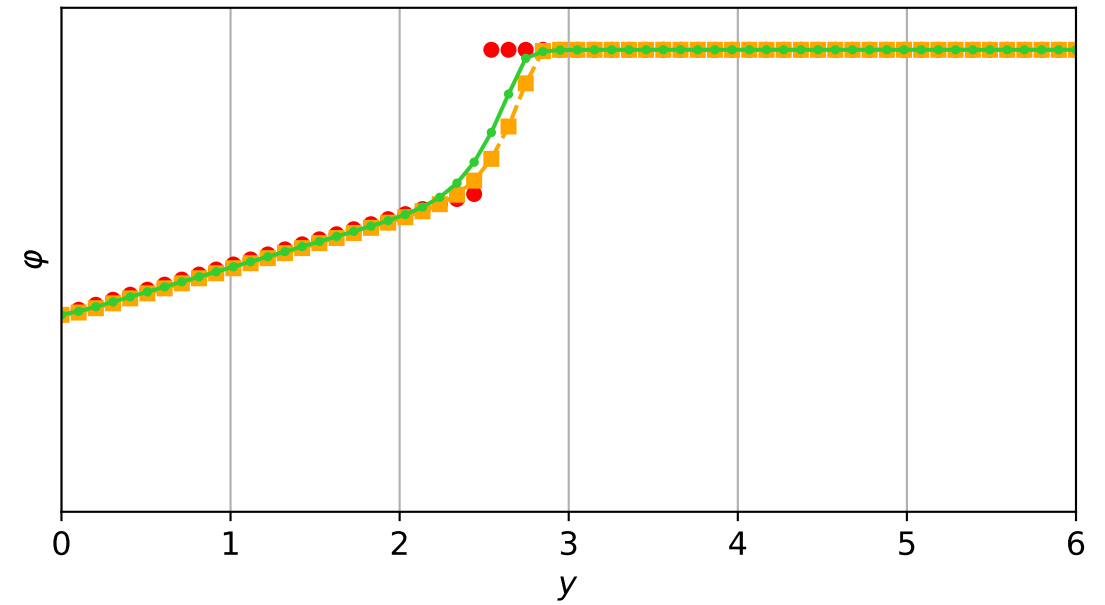
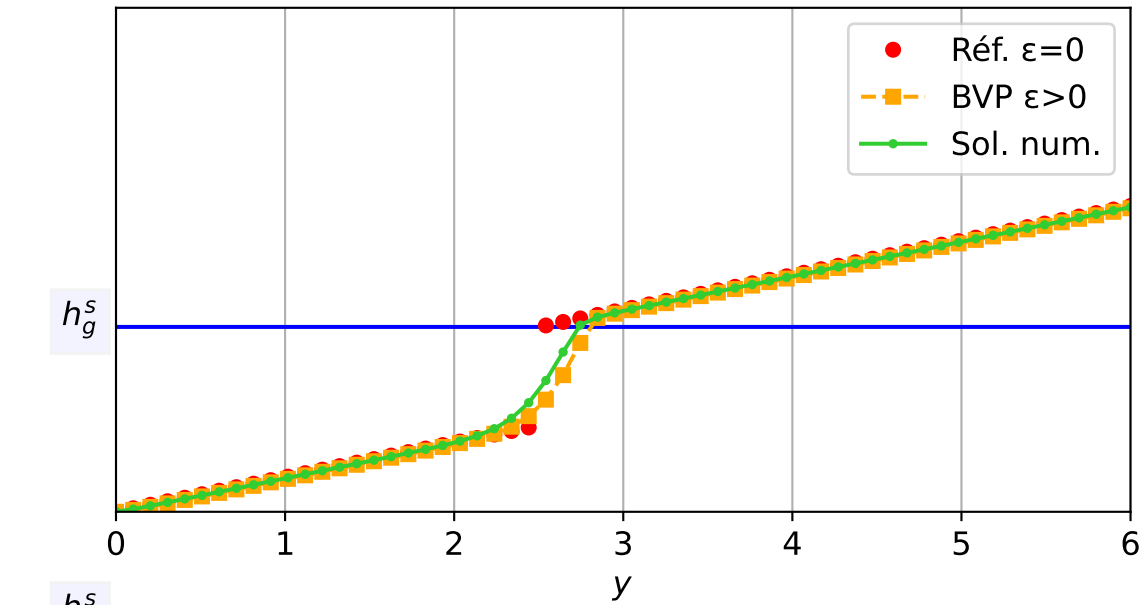
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.2012/12$  (8406 iter) — résidu eq. =  $5.21e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.25008/12$  (8426 iter) — résidu eq. =  $4.62e-06$

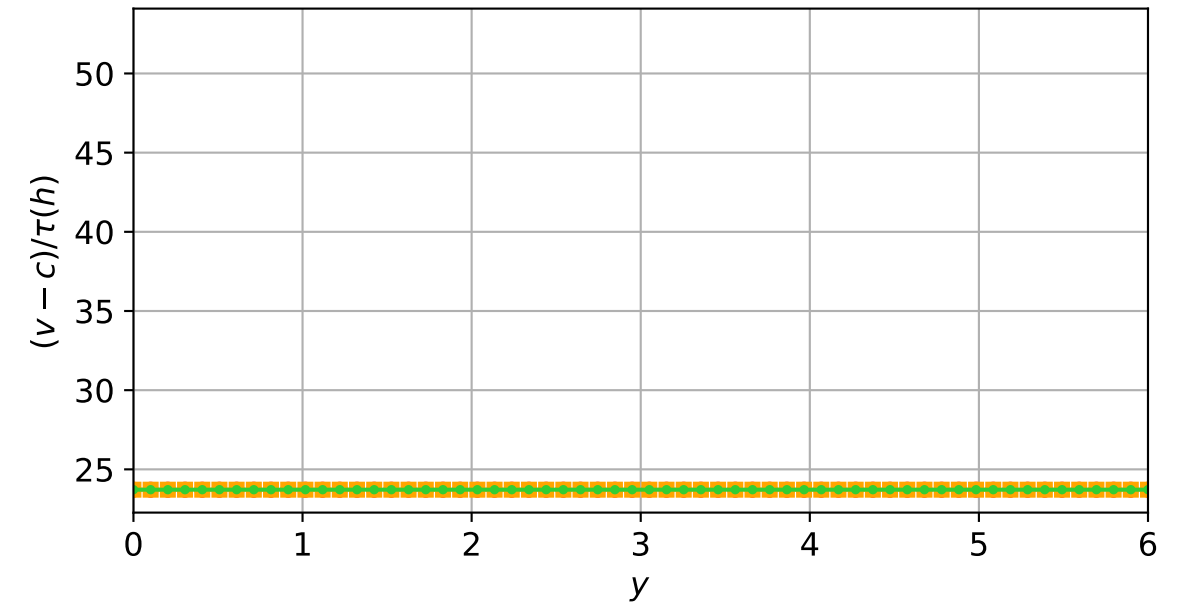
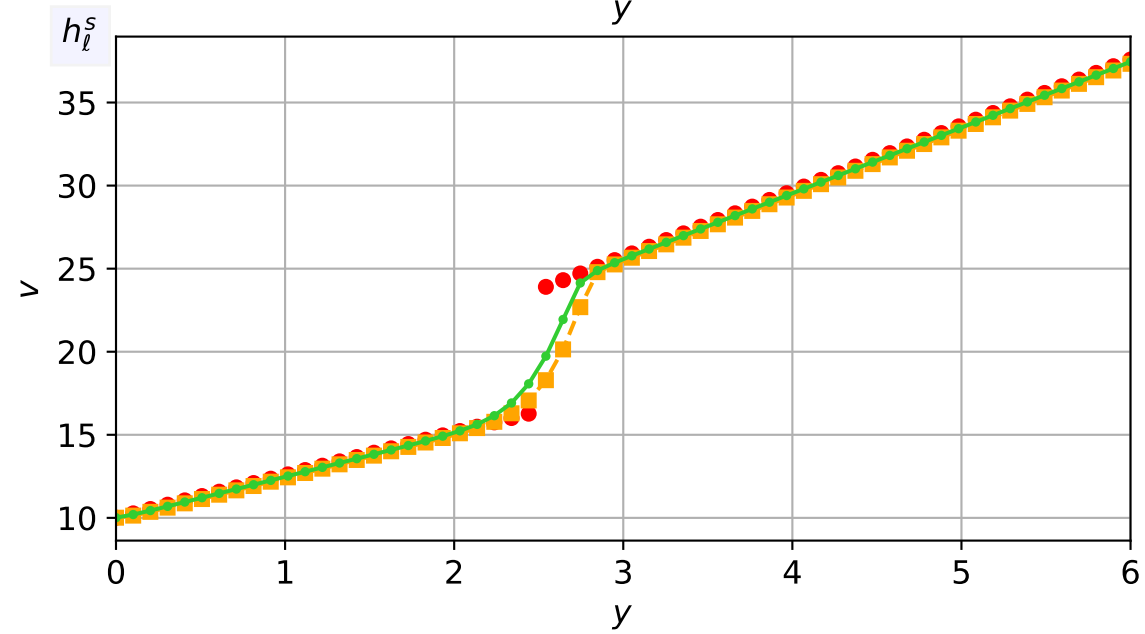
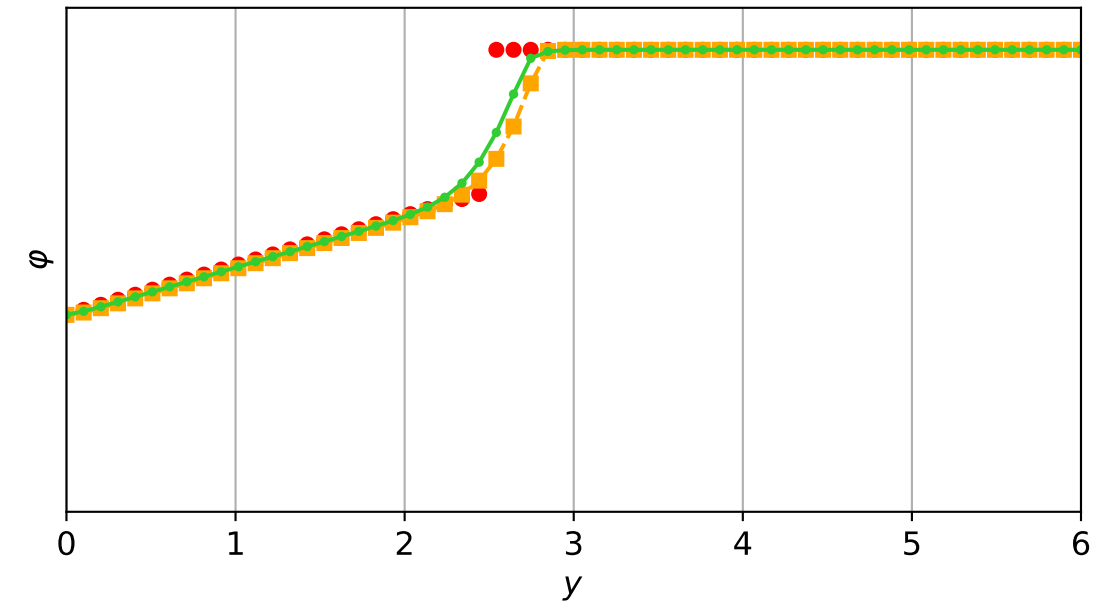
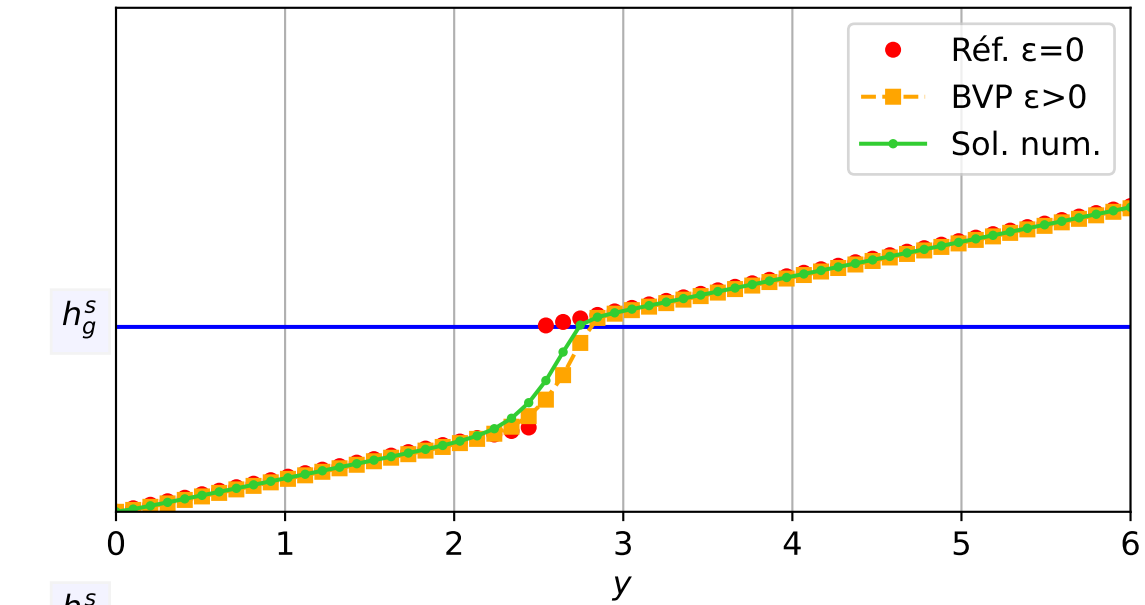


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.3014/12$  (8447 iter) — résidu eq. =  $4.07e-06$

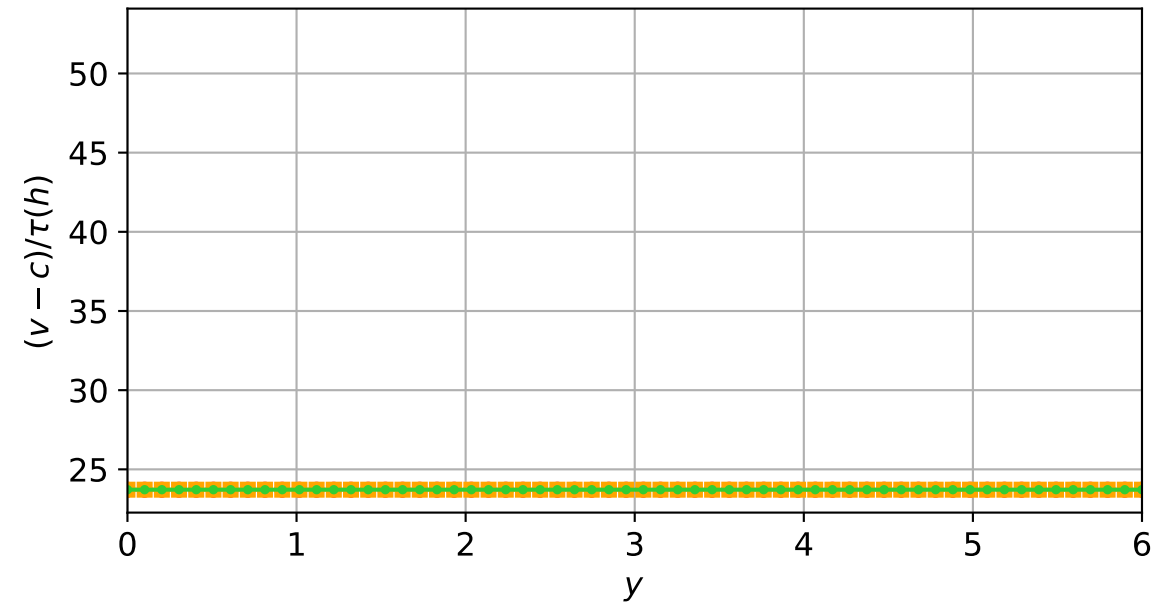
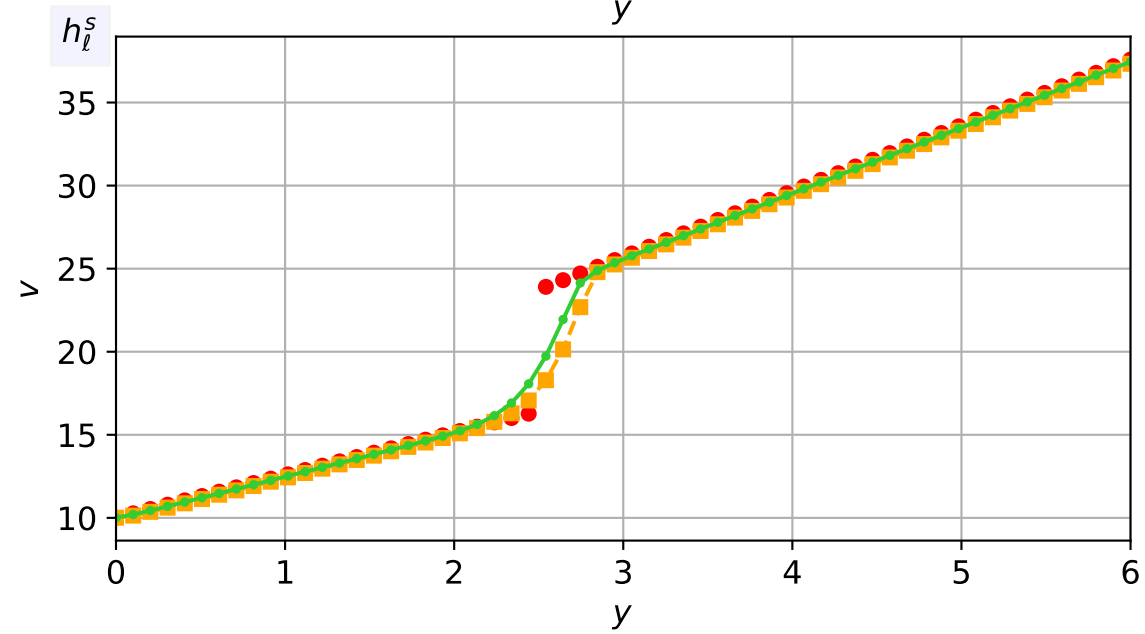
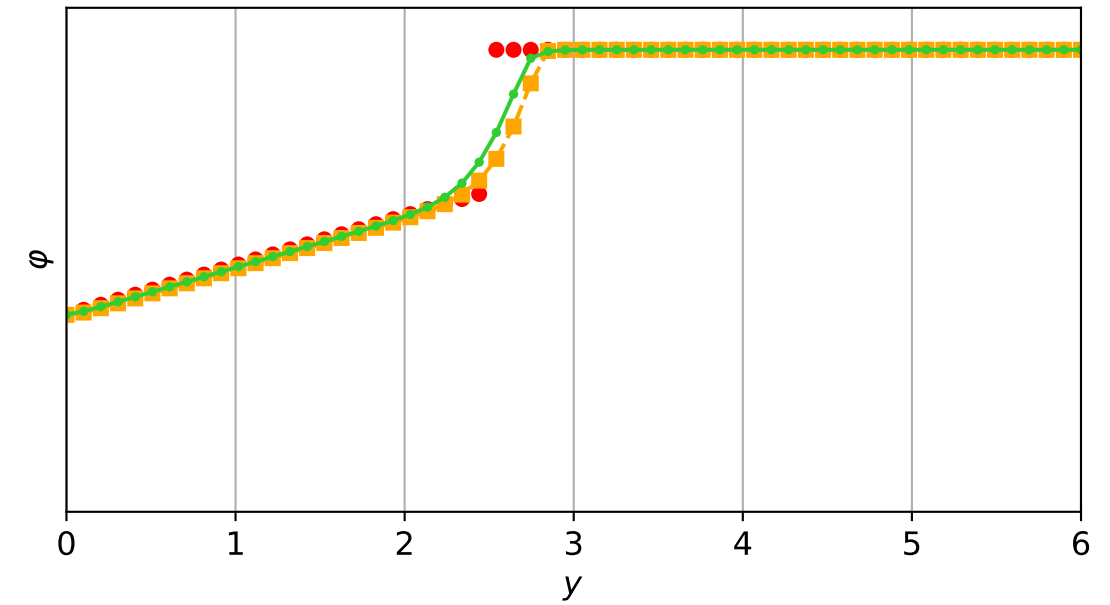
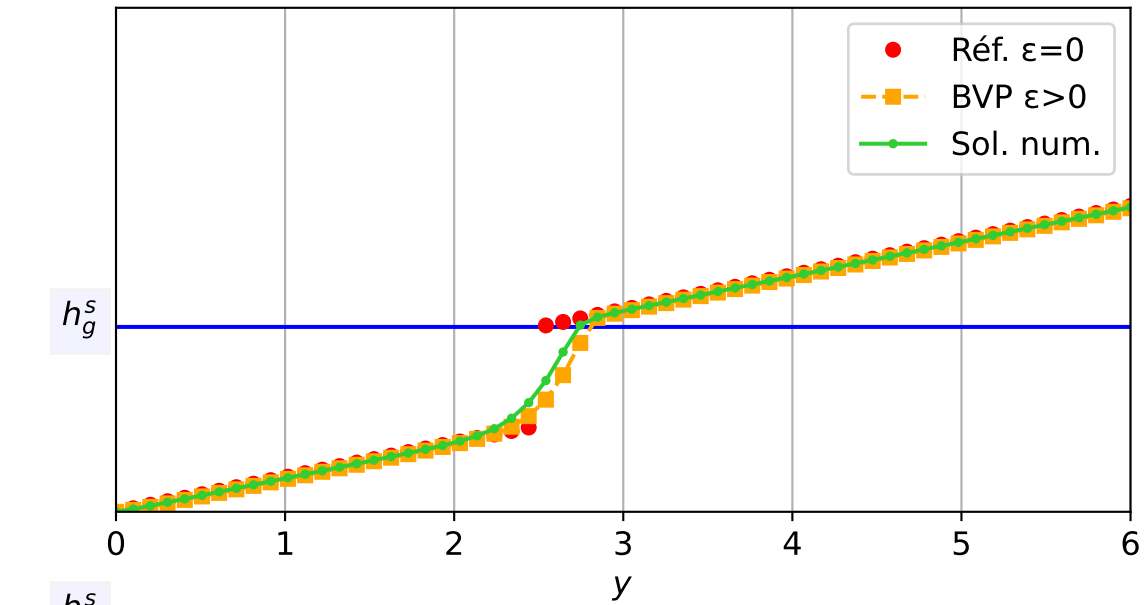




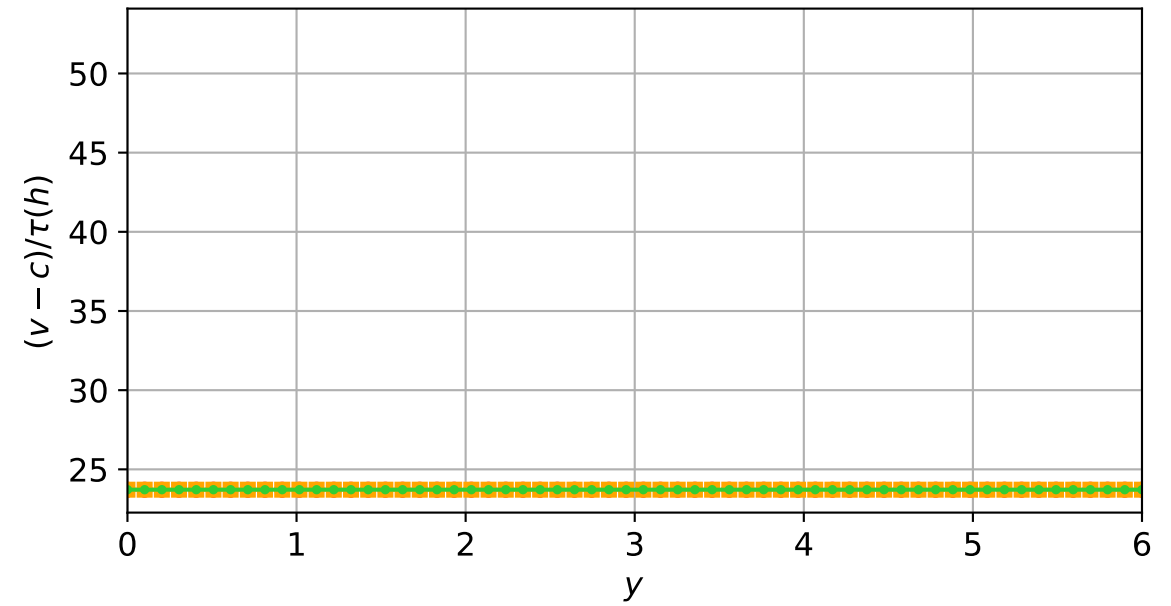
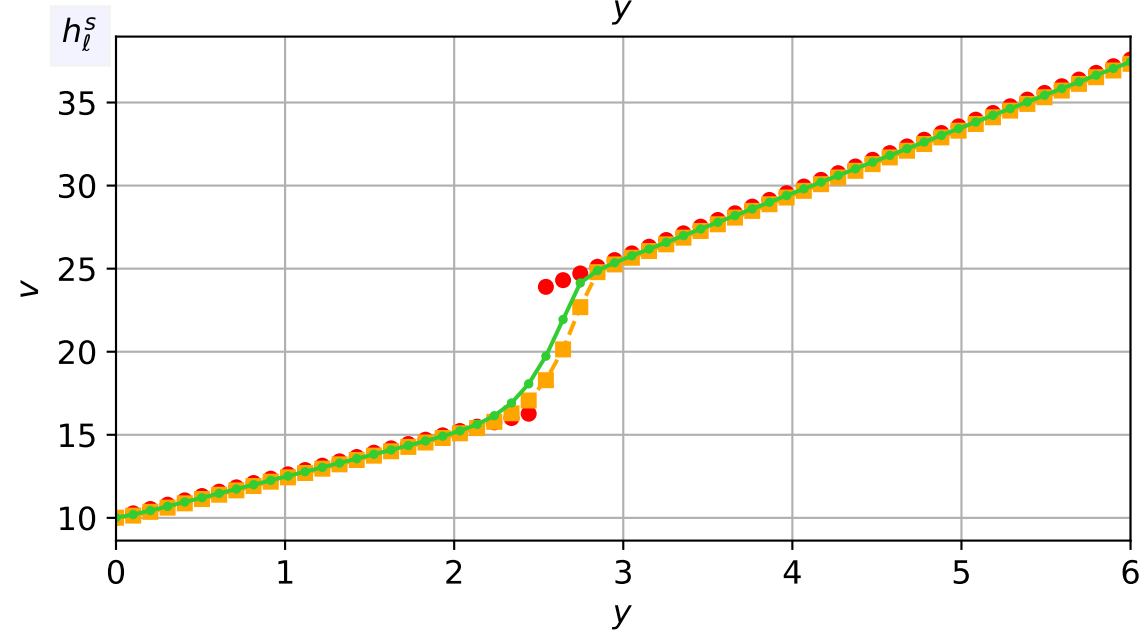
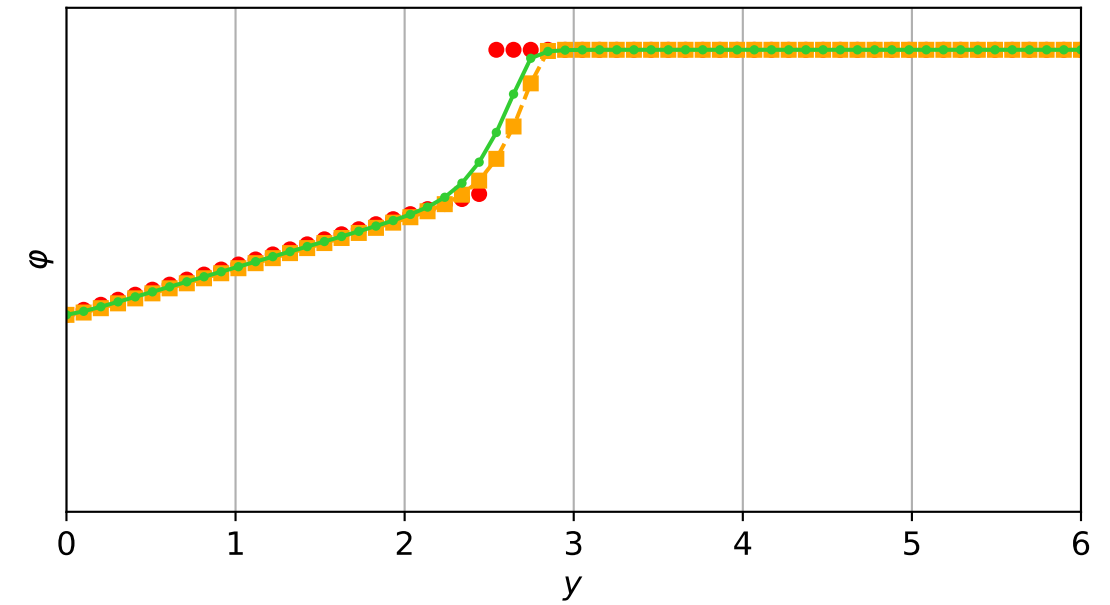
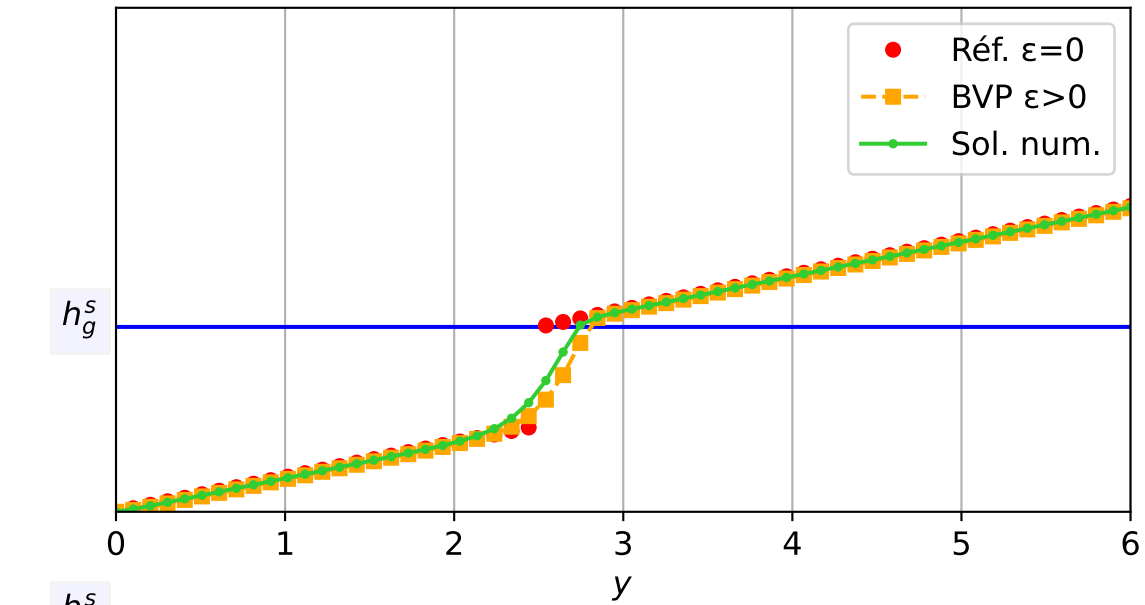
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.35027/12$  (8467 iter) — résidu eq. =  $3.60e-06$



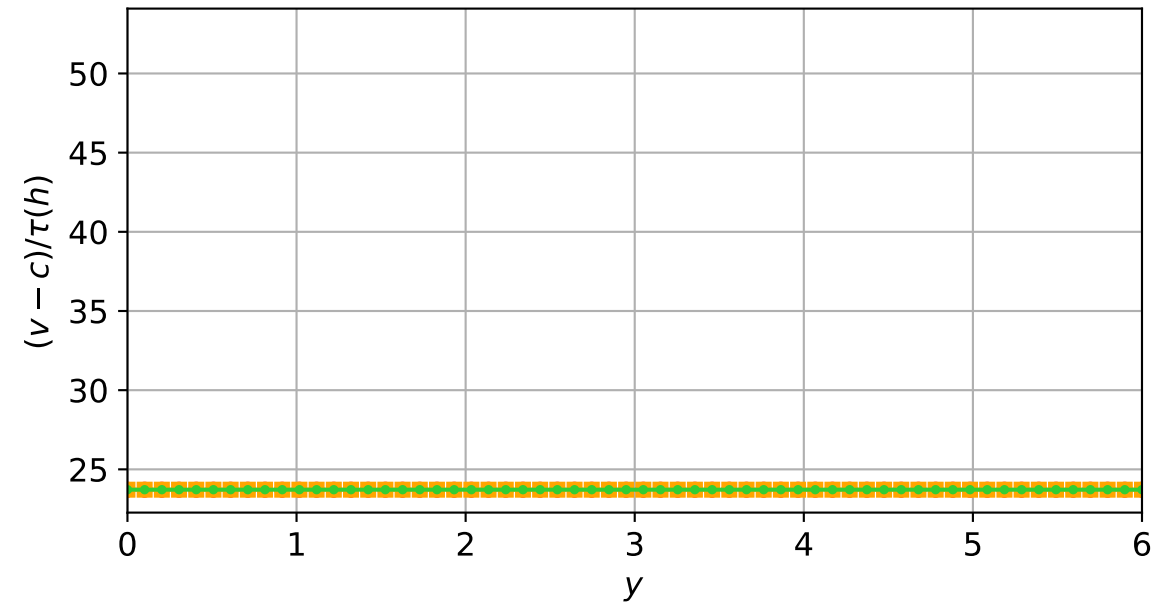
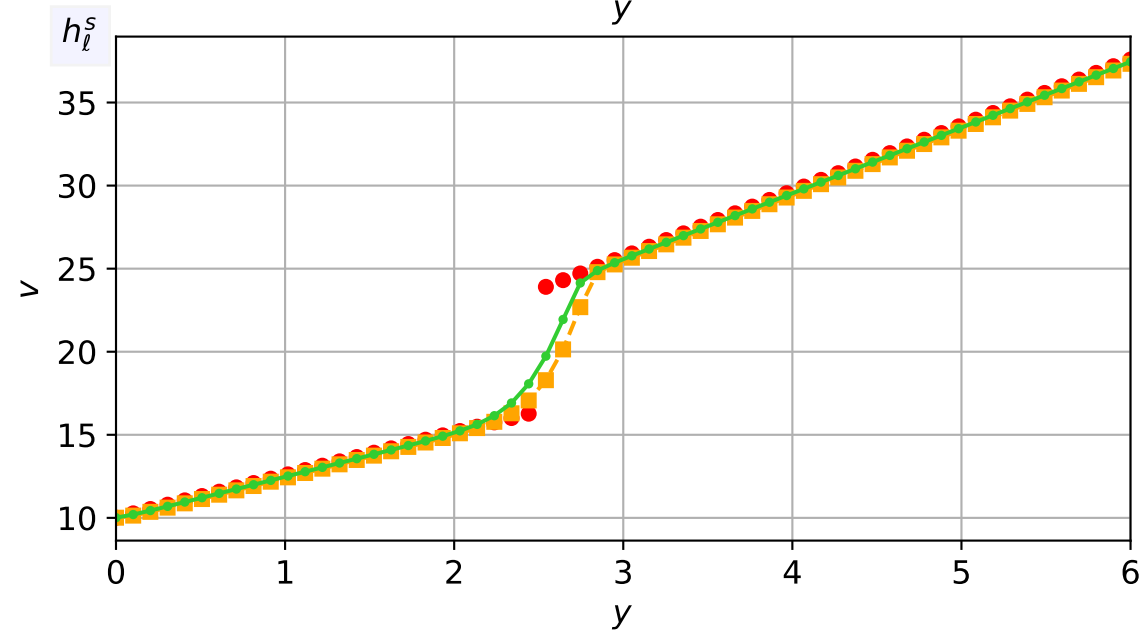
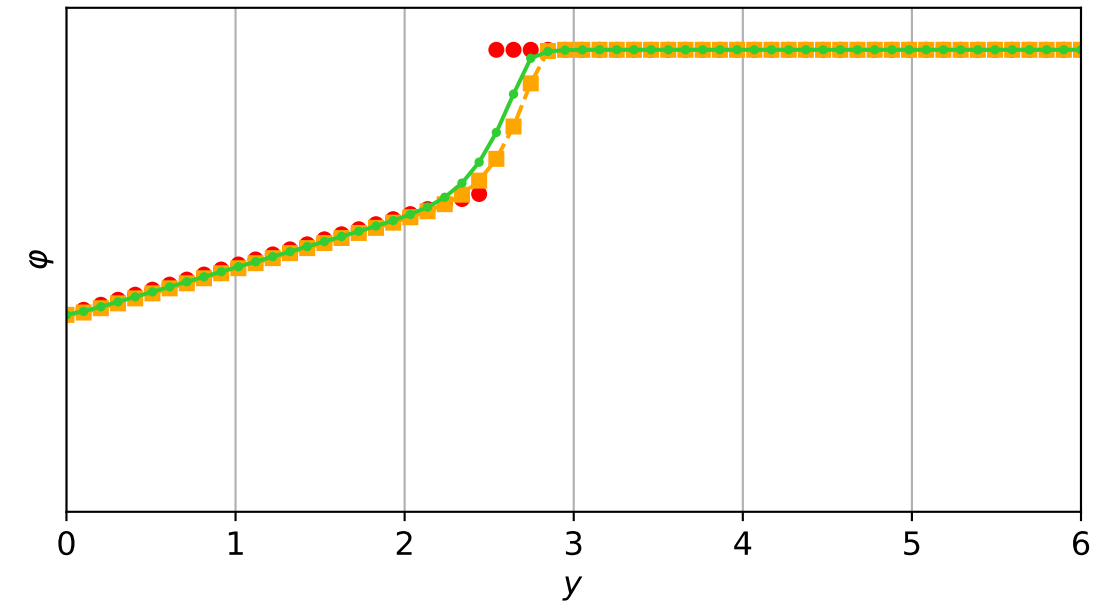
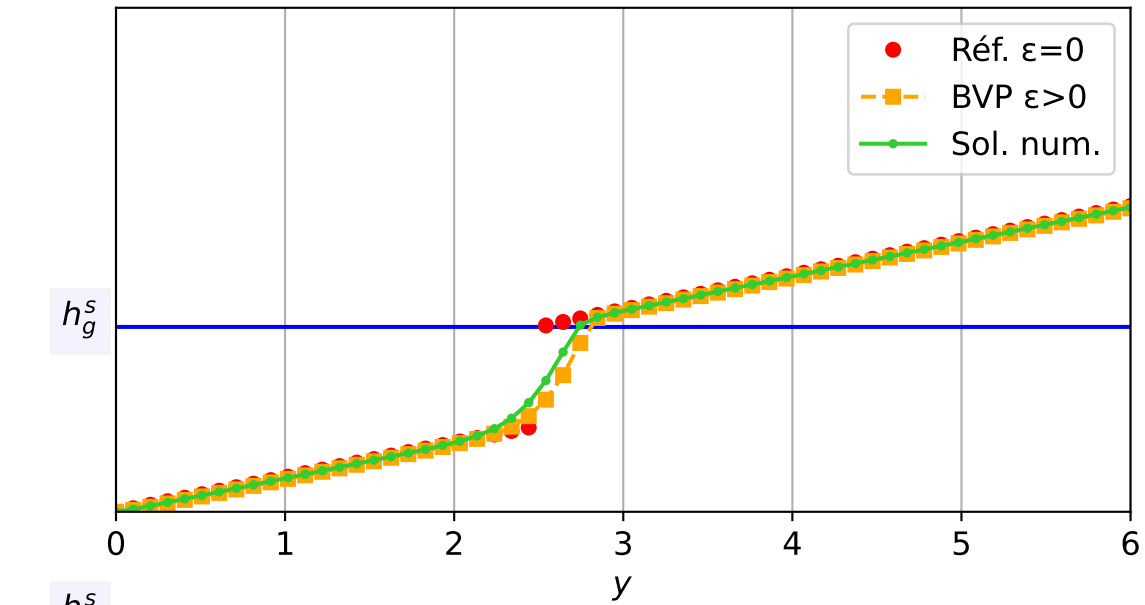
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.40159/12$  (8488 iter) — résidu eq. =  $3.18e-06$



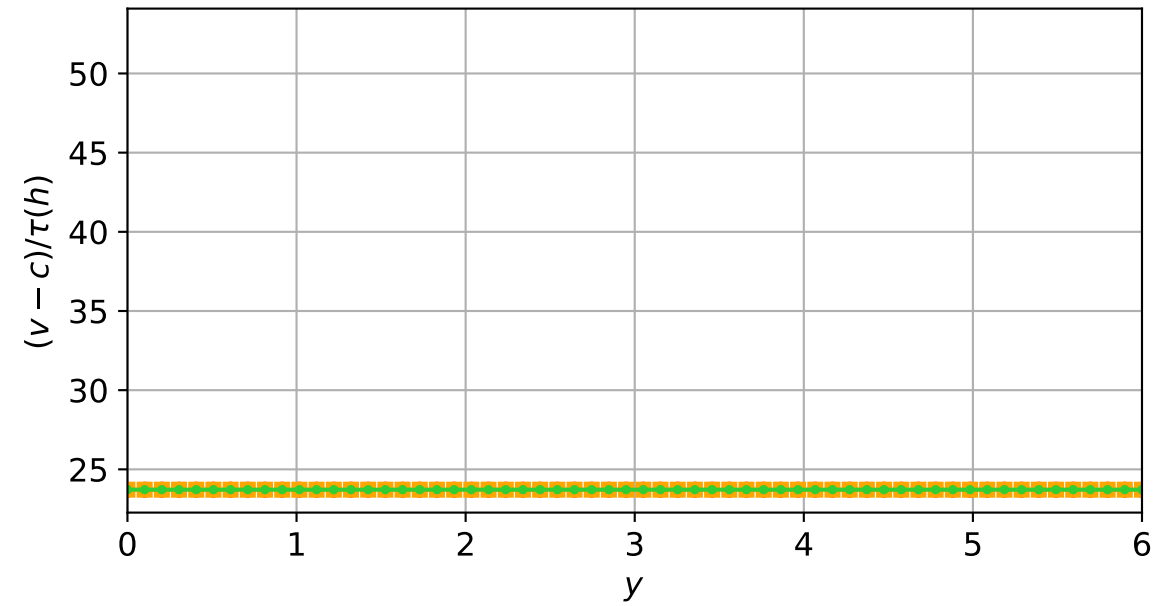
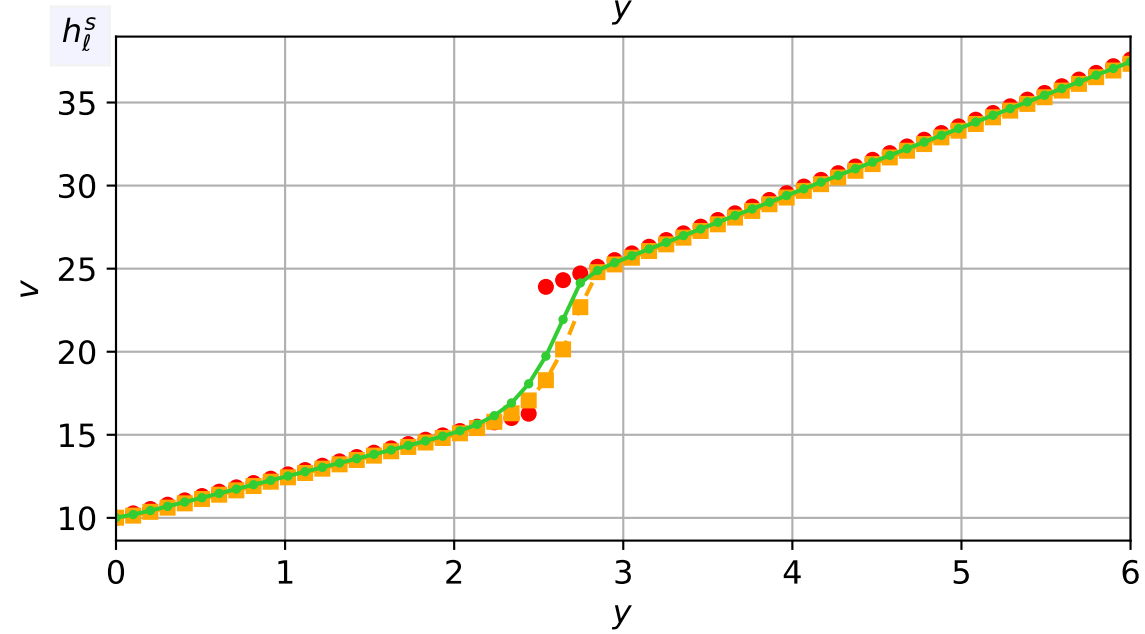
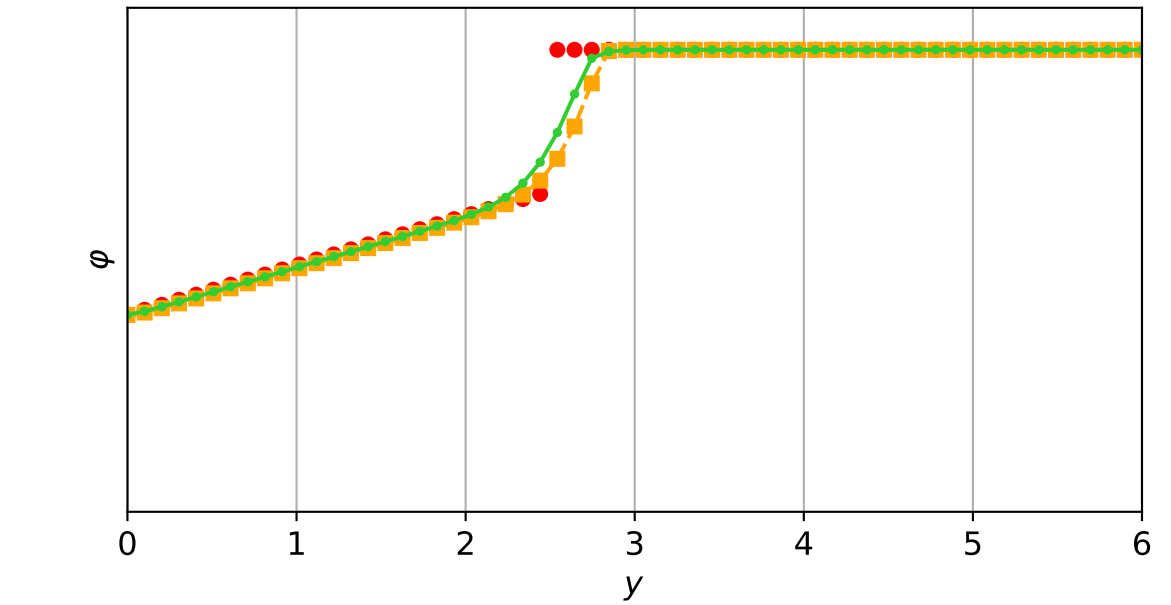
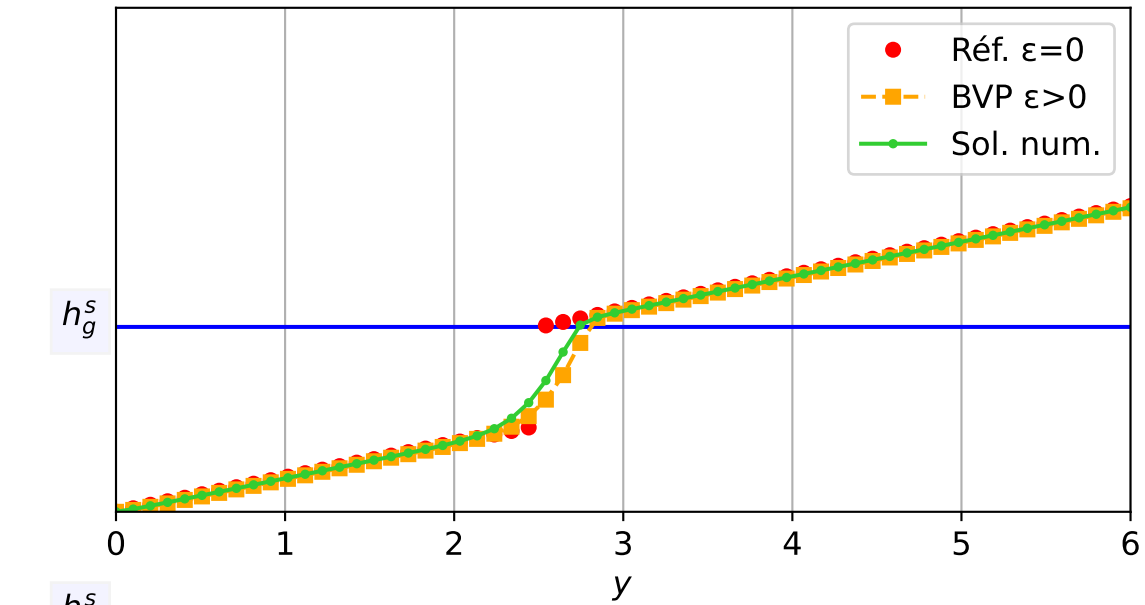
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.45047/12$  (8508 iter) — résidu eq. =  $2.81e-06$



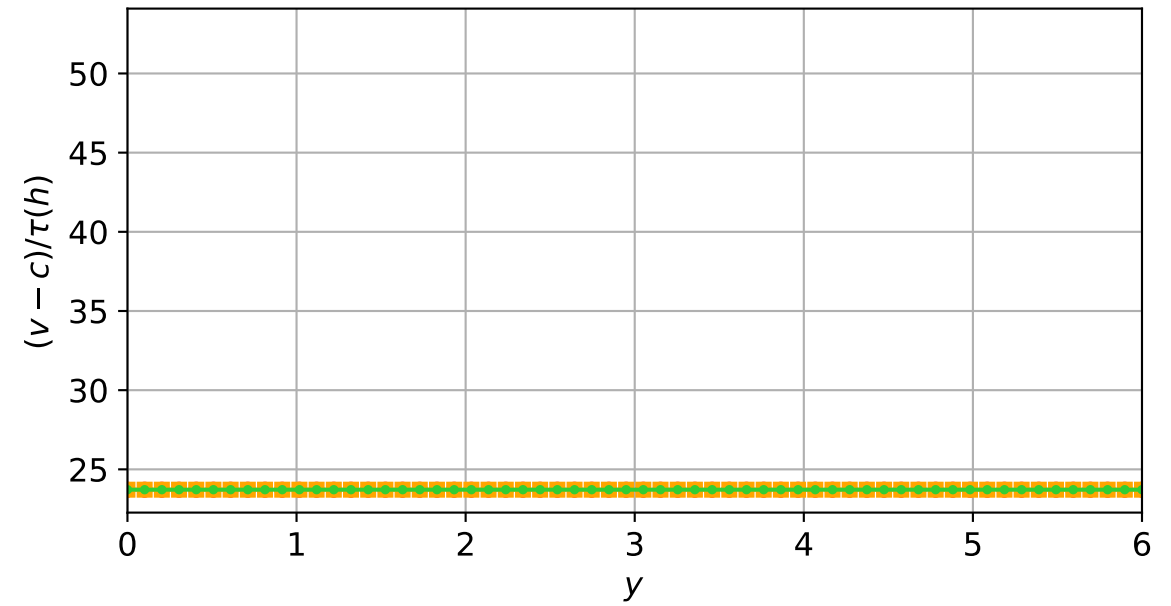
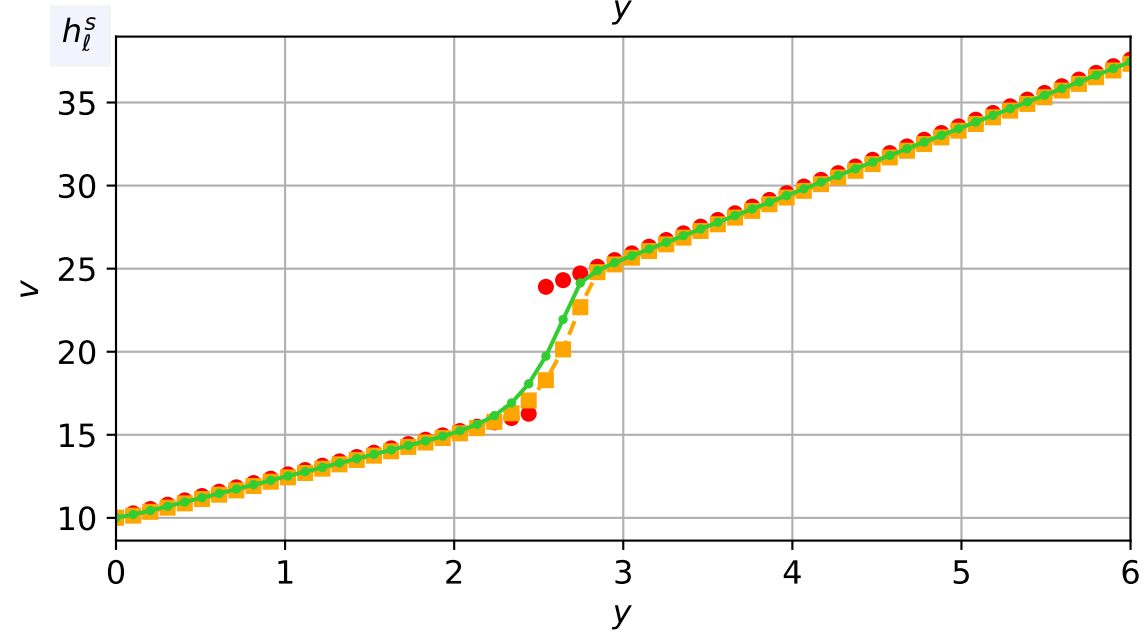
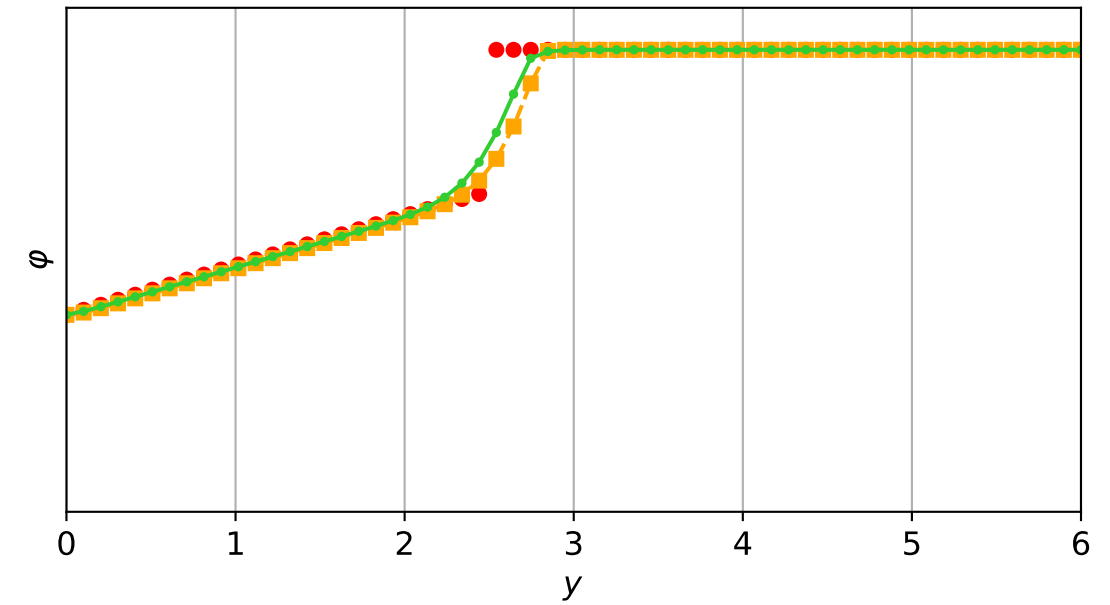
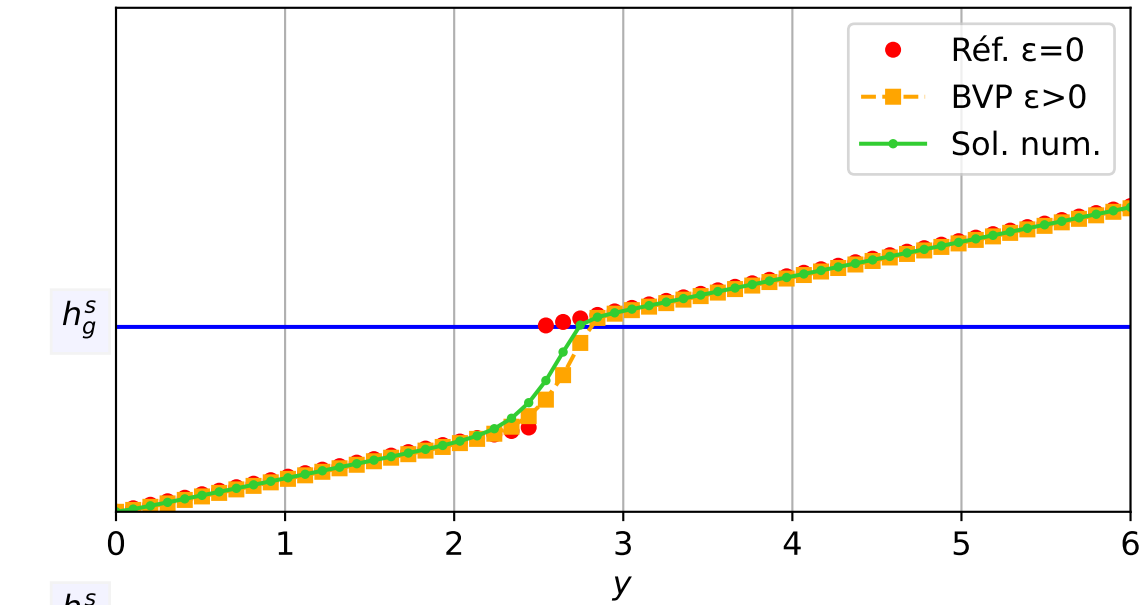
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.50179/12$  (8529 iter) — résidu eq. =  $2.48e-06$



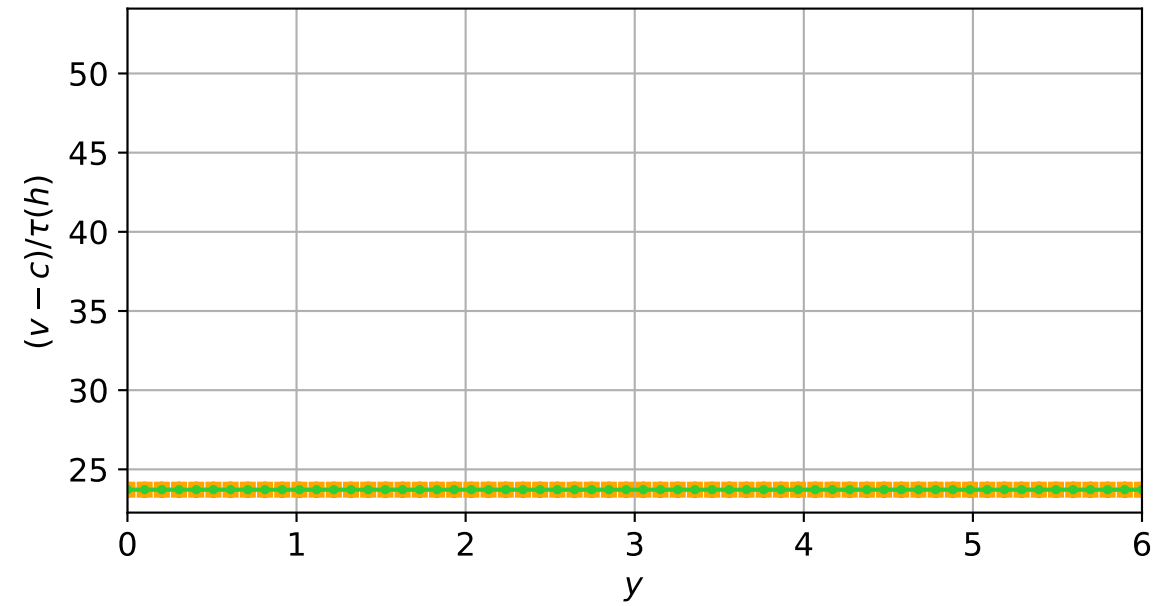
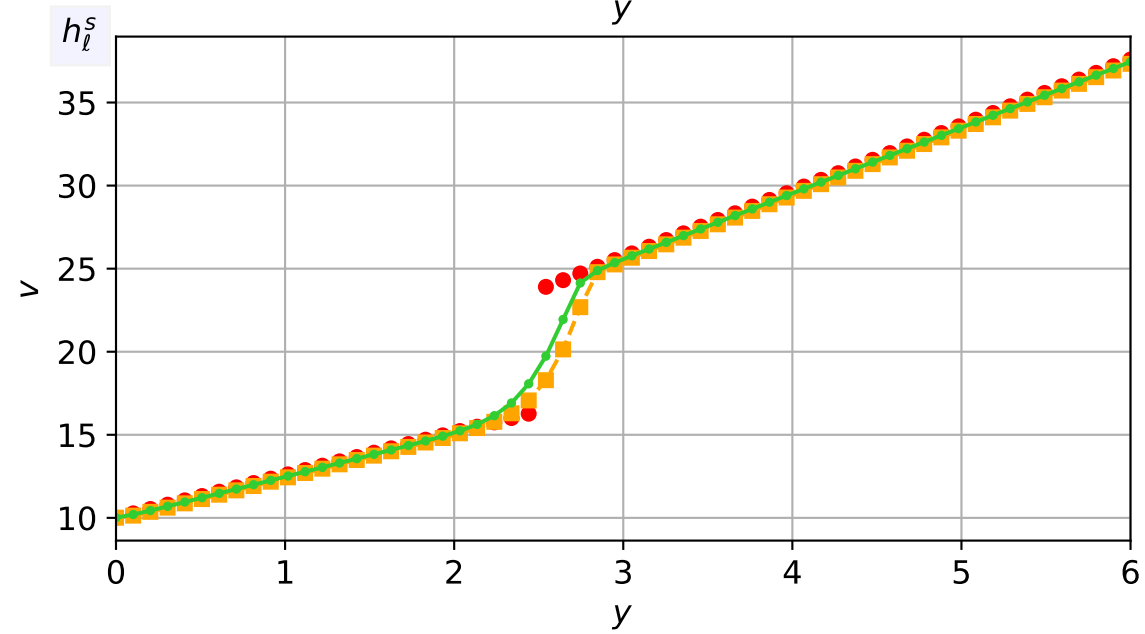
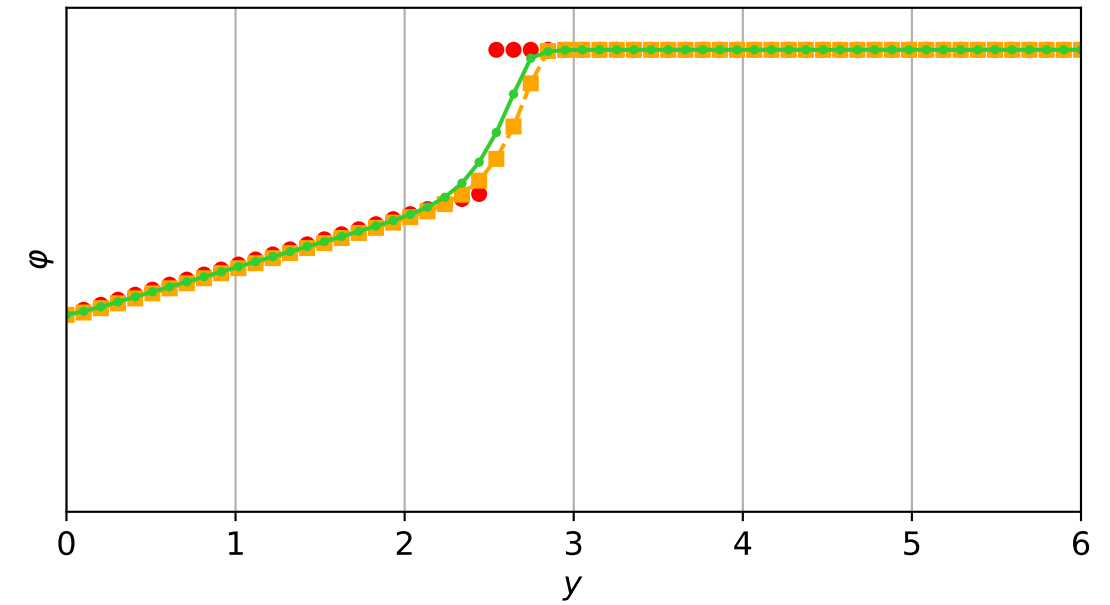
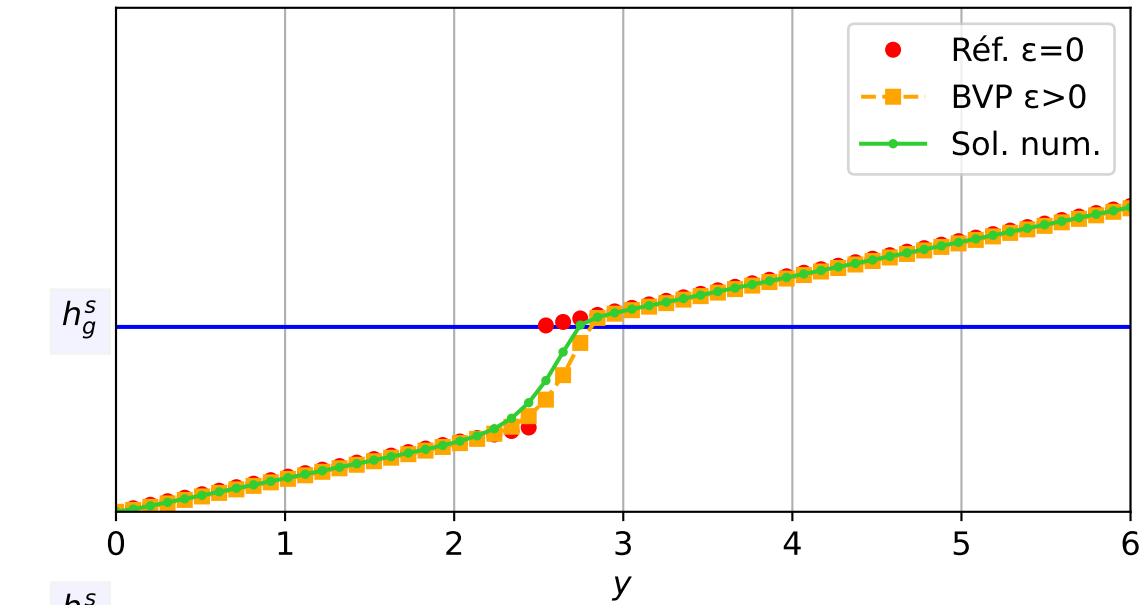
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.55067/12$  (8549 iter) — résidu eq. =  $2.20e-06$



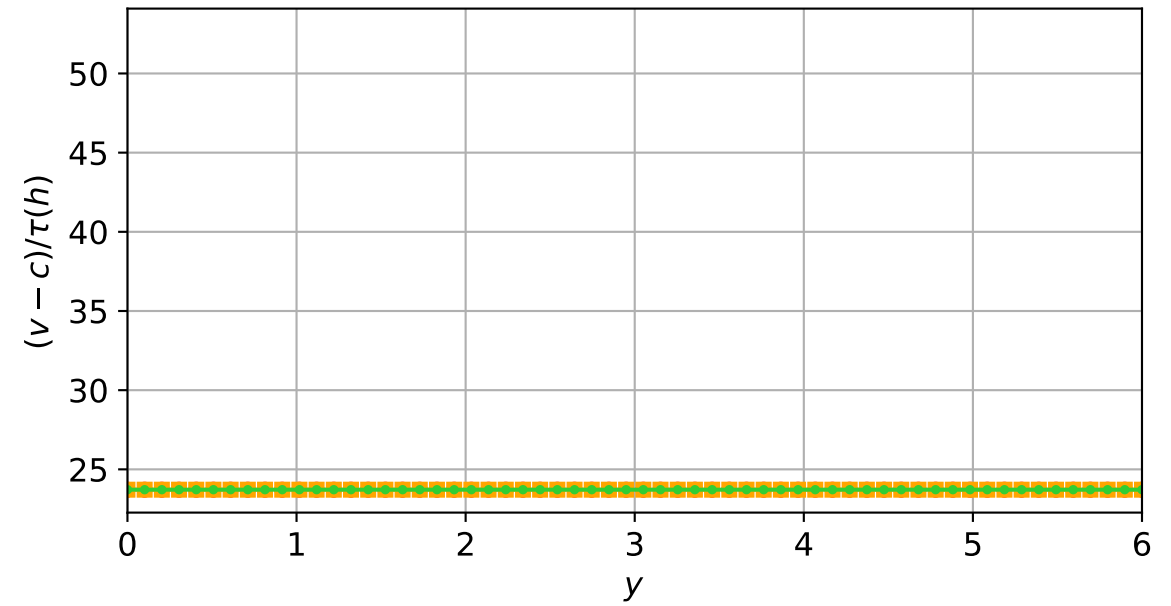
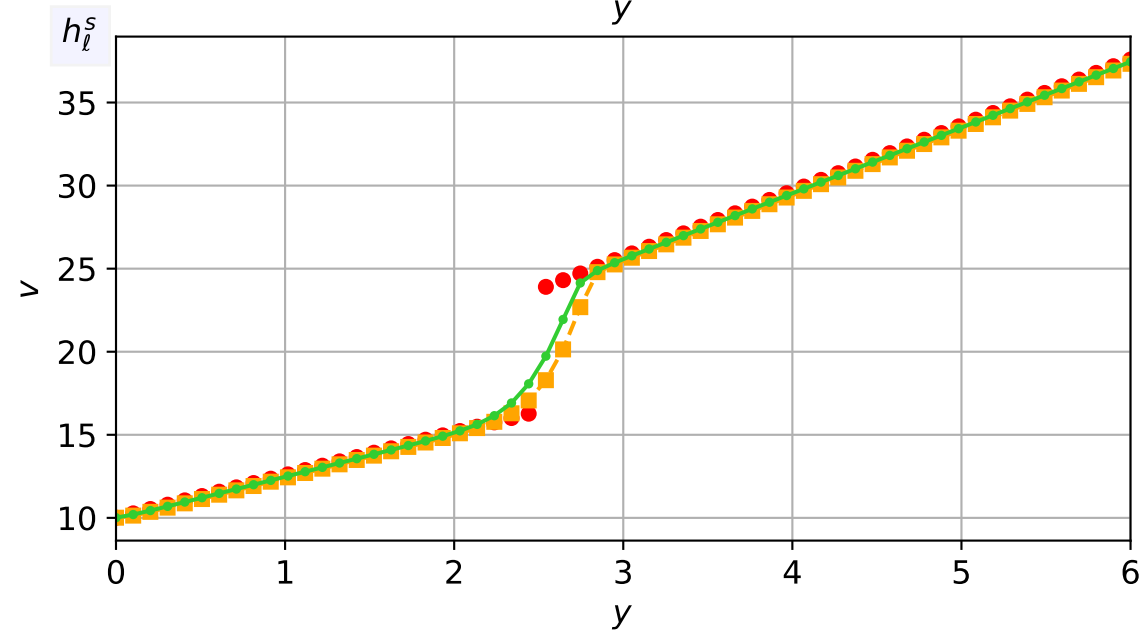
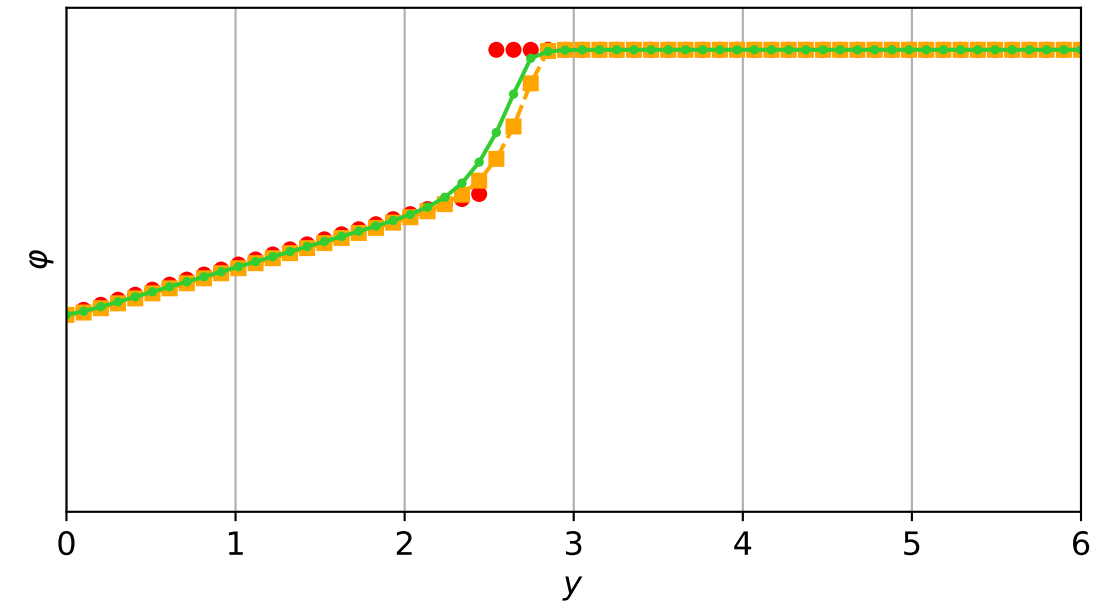
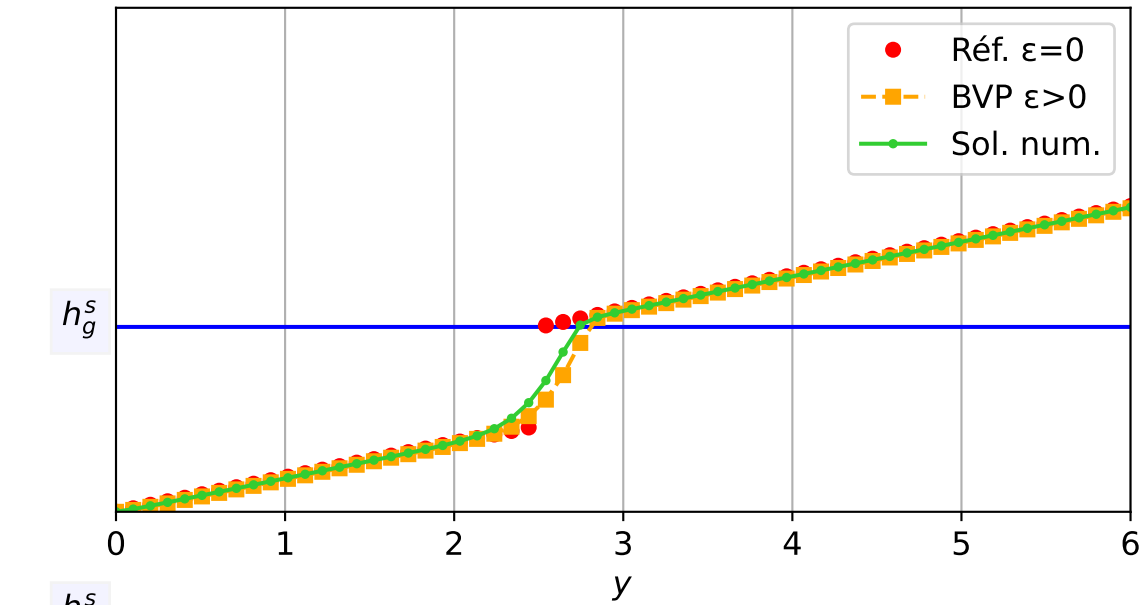
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.60199/12$  (8570 iter) — résidu eq. =  $1.94e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.65087/12$  (8590 iter) — résidu eq. =  $1.71e-06$

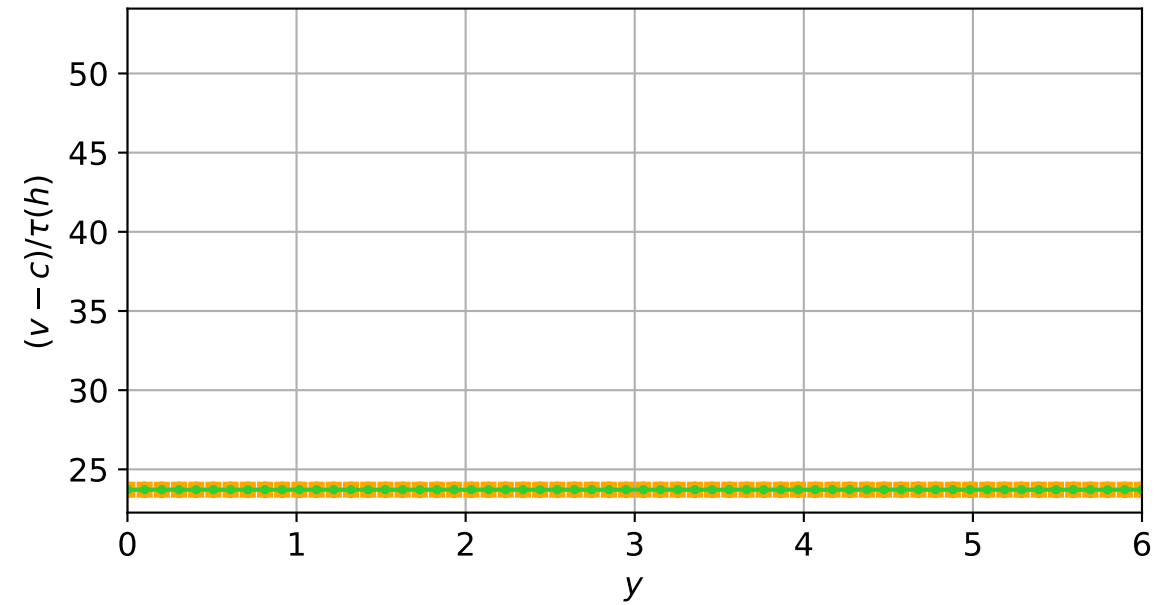
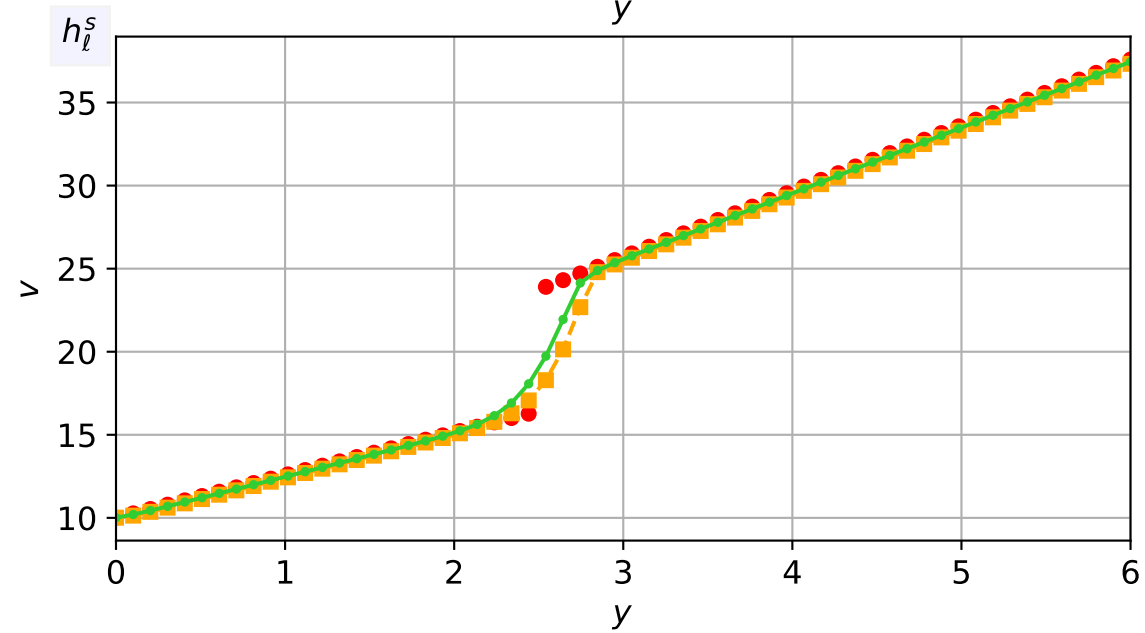
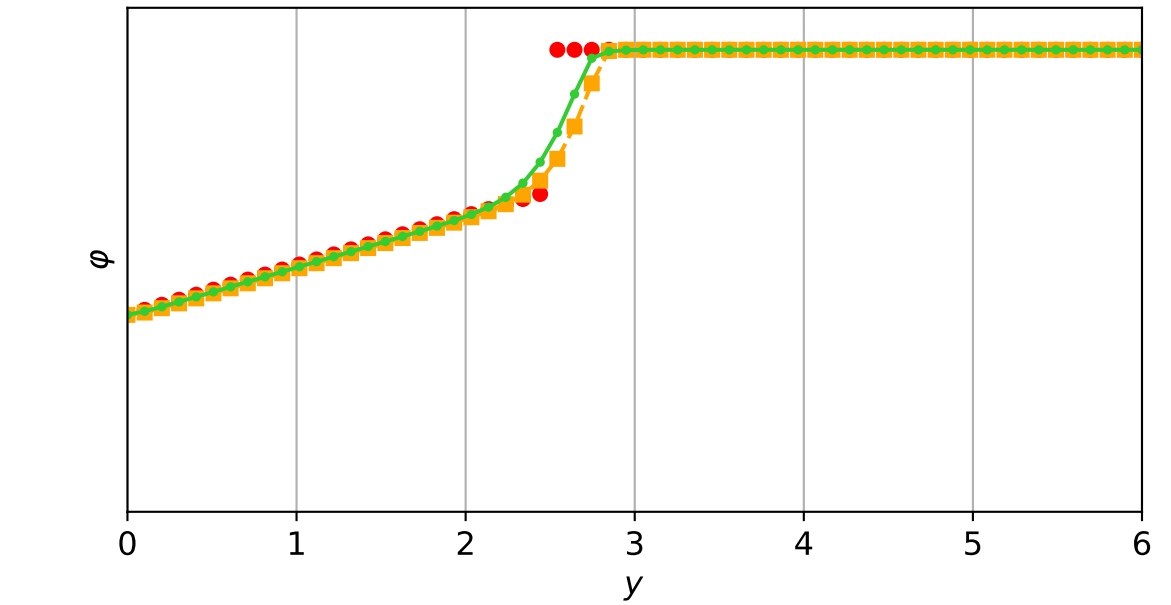
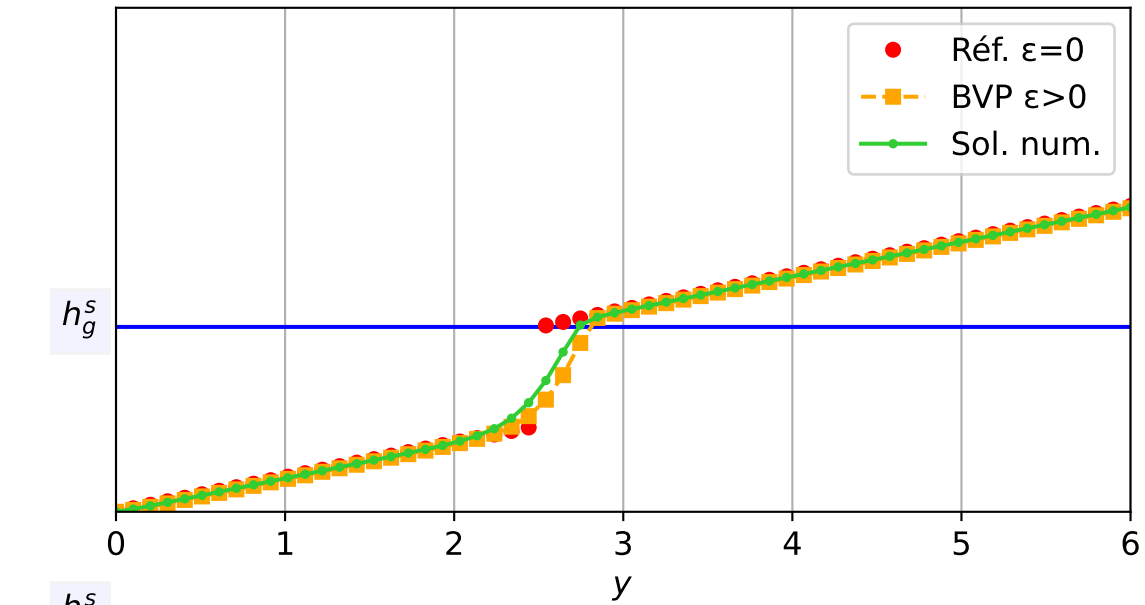


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.70219/12$  (8611 iter) — résidu eq. =  $1.51e-06$

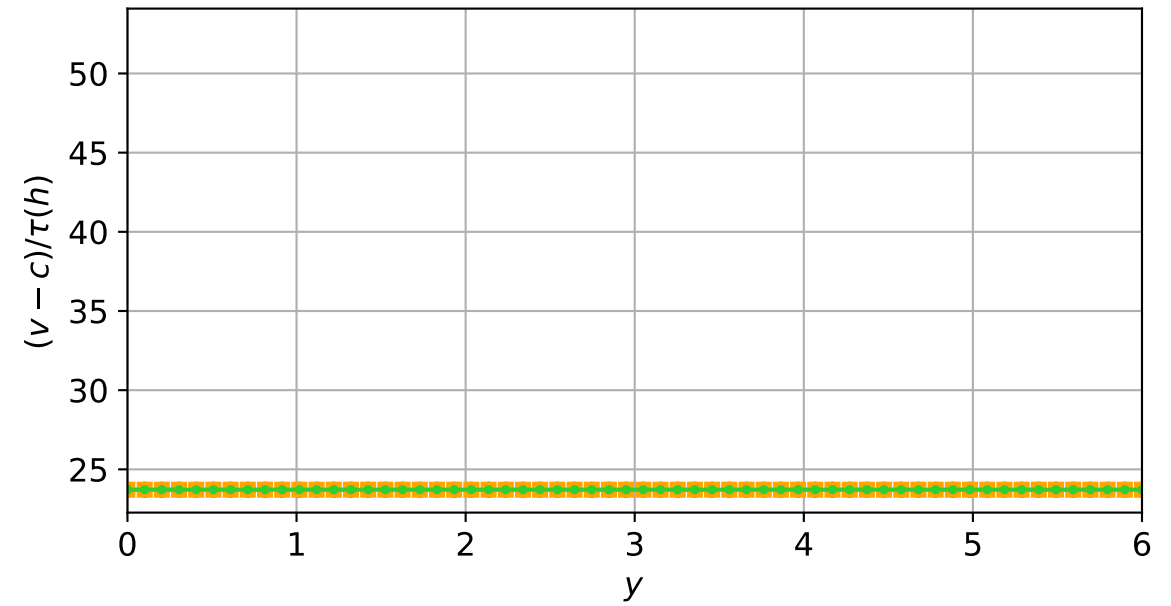
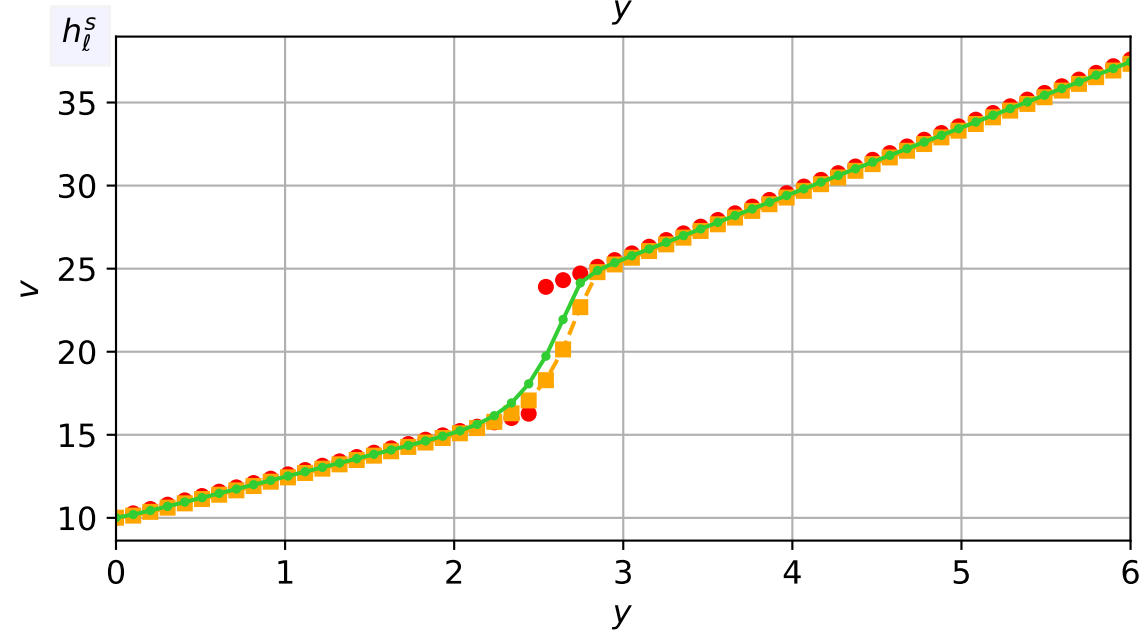
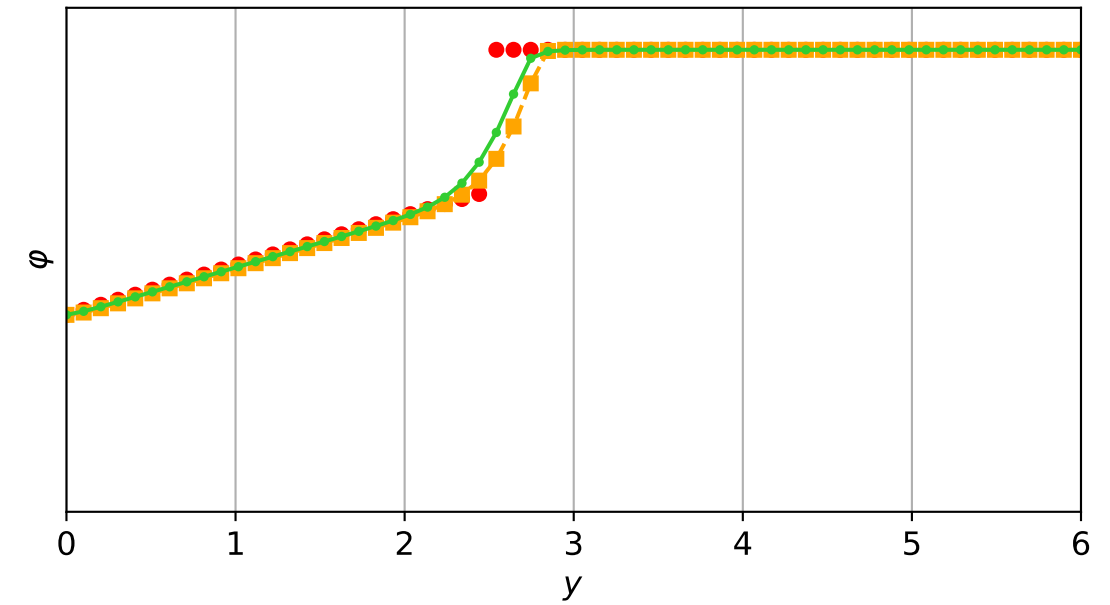
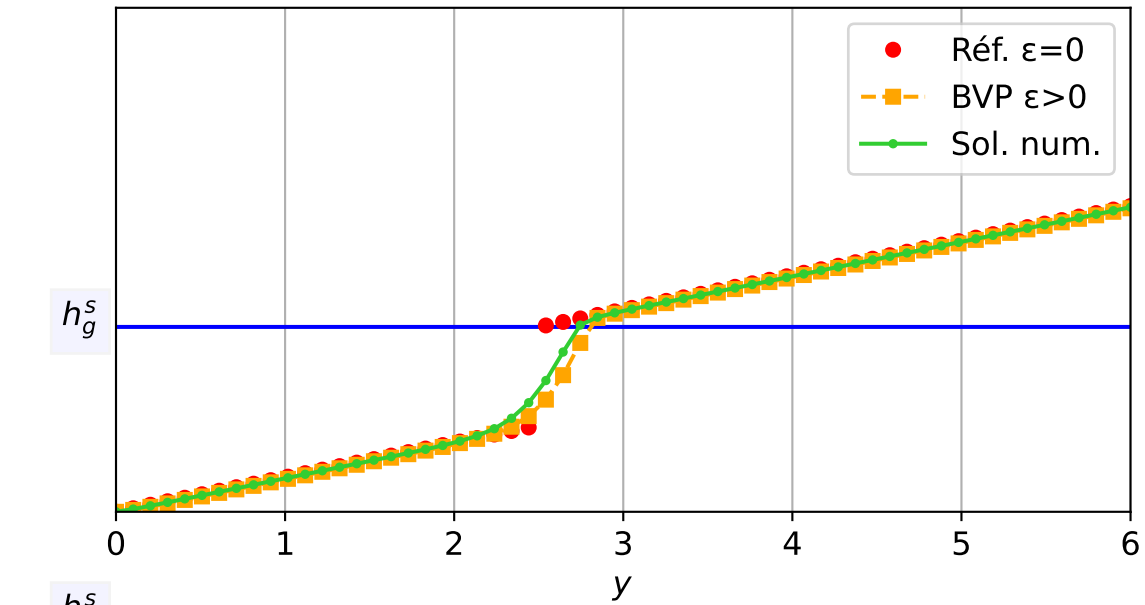




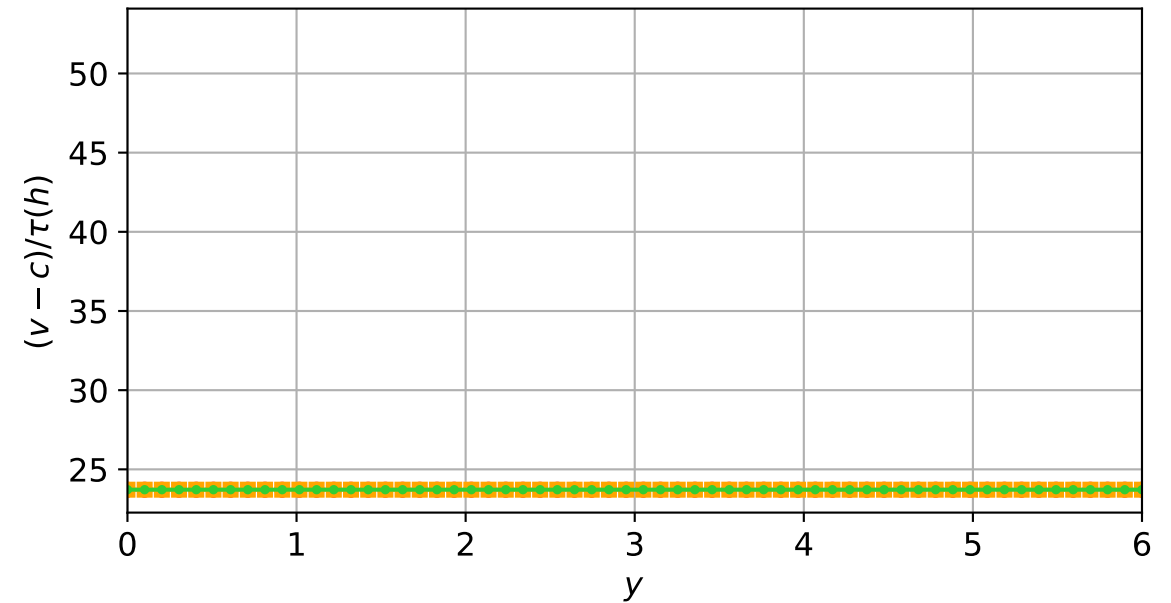
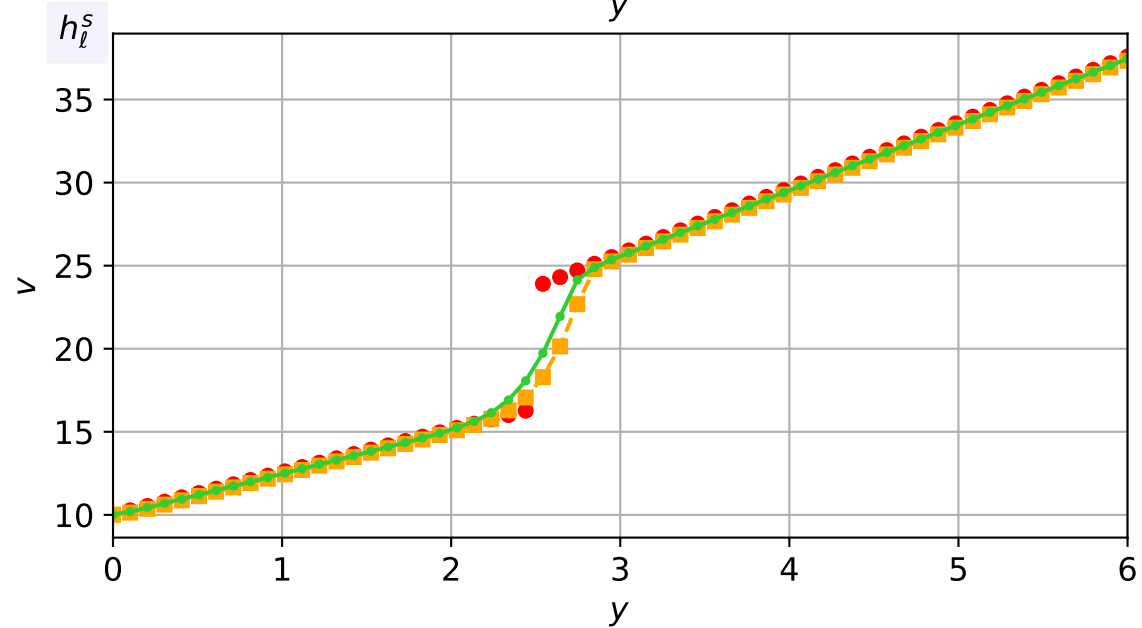
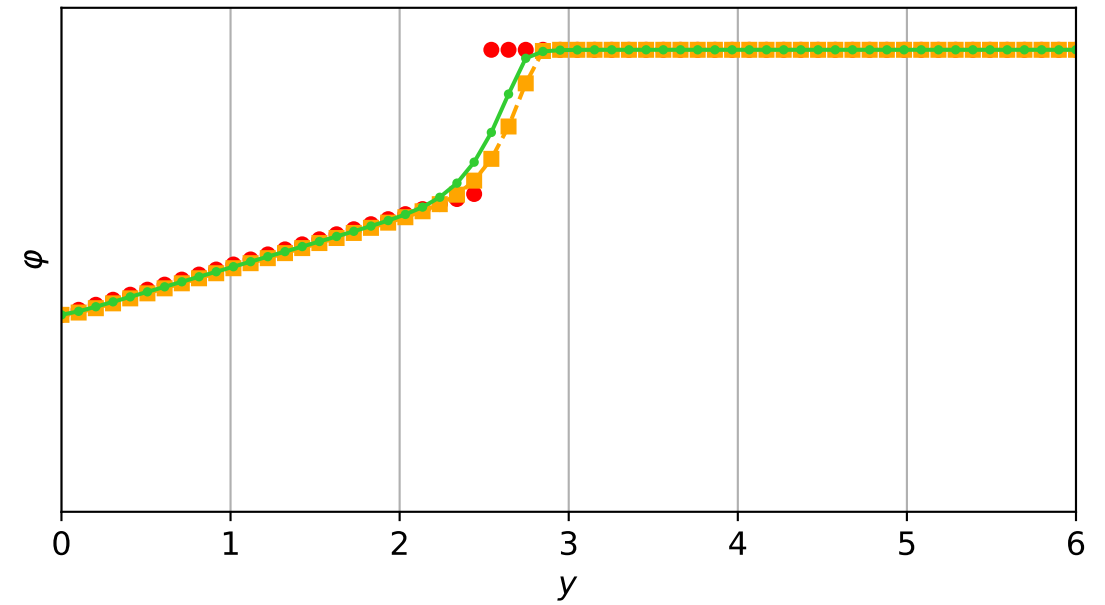
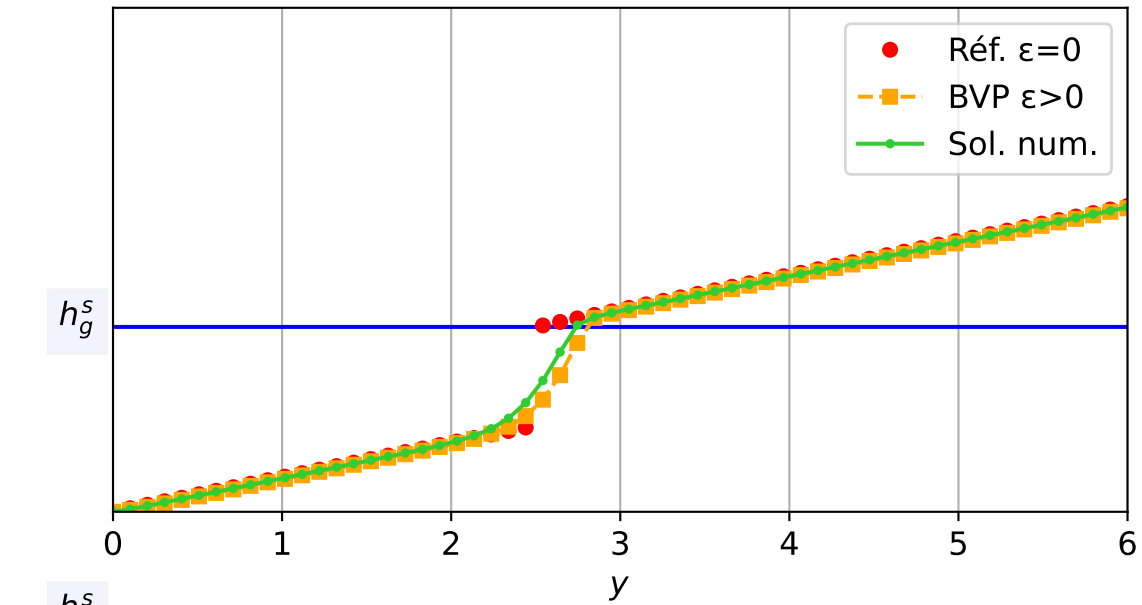
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.75106/12$  (8631 iter) — résidu eq. =  $1.34e-06$



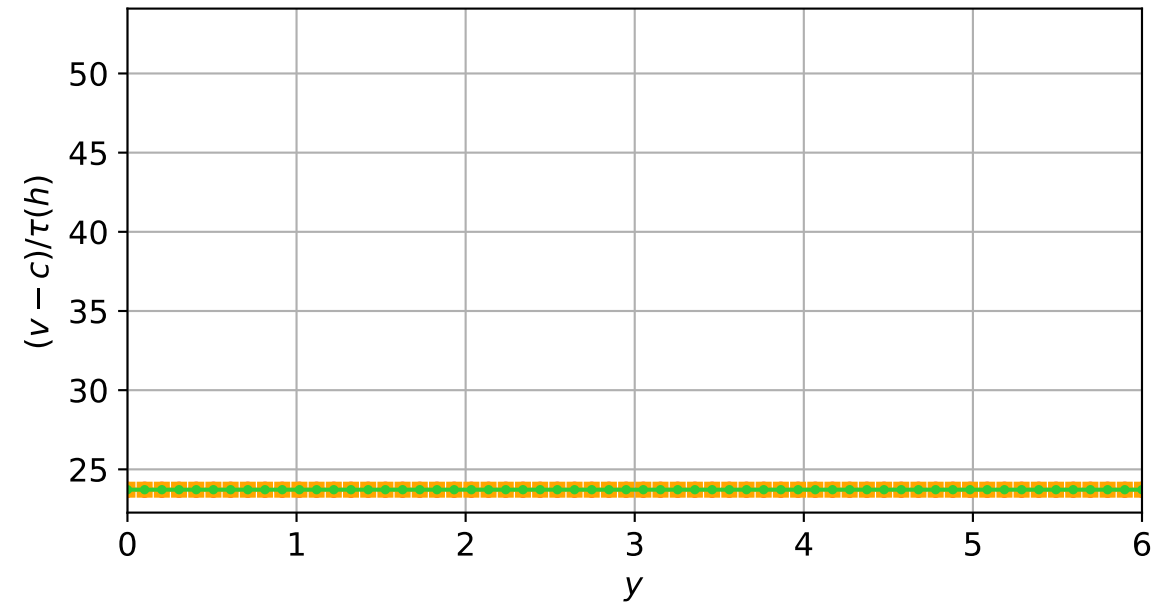
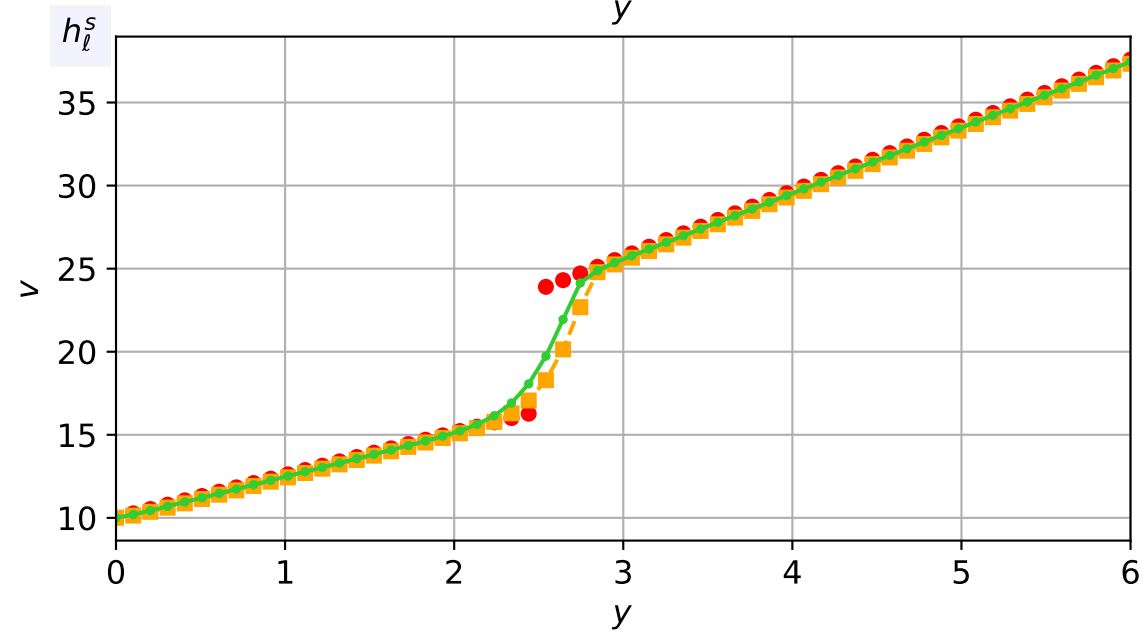
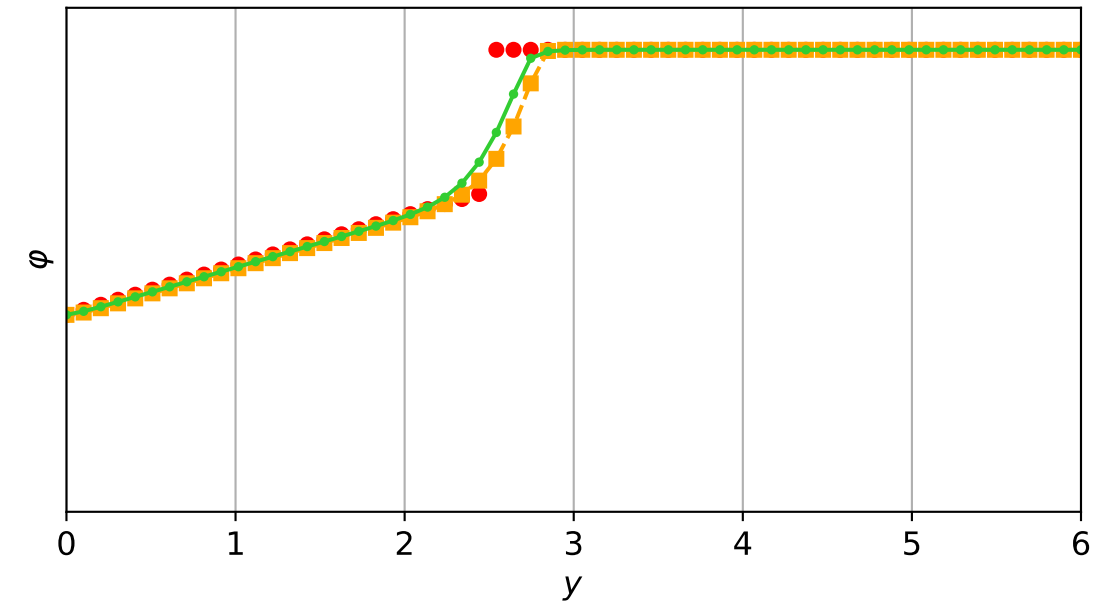
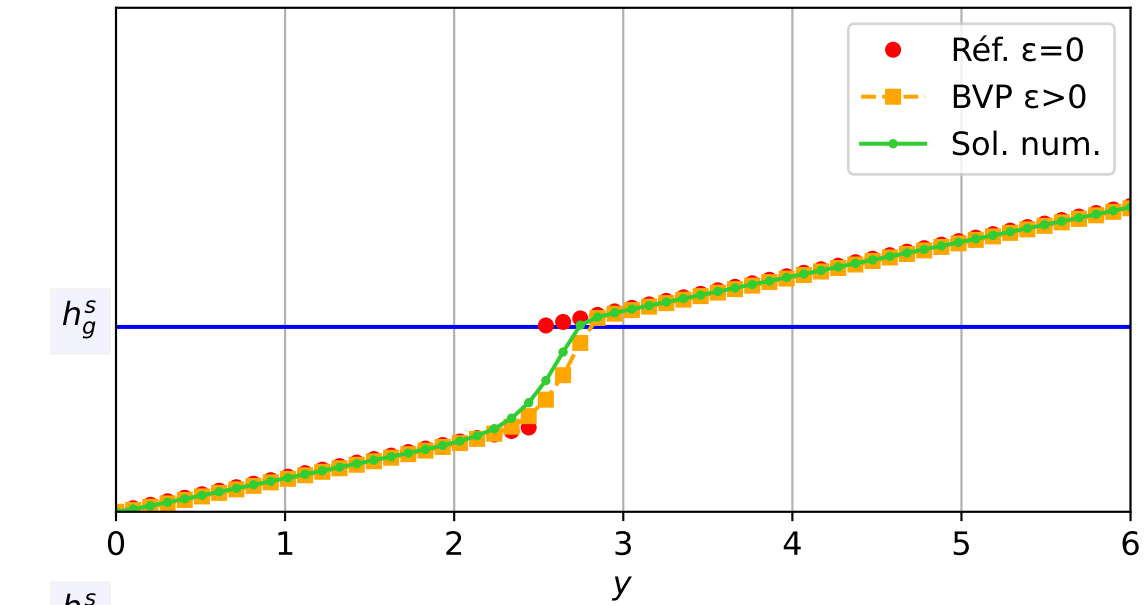
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.80238/12$  (8652 iter) — résidu eq. =  $8.91e-06$



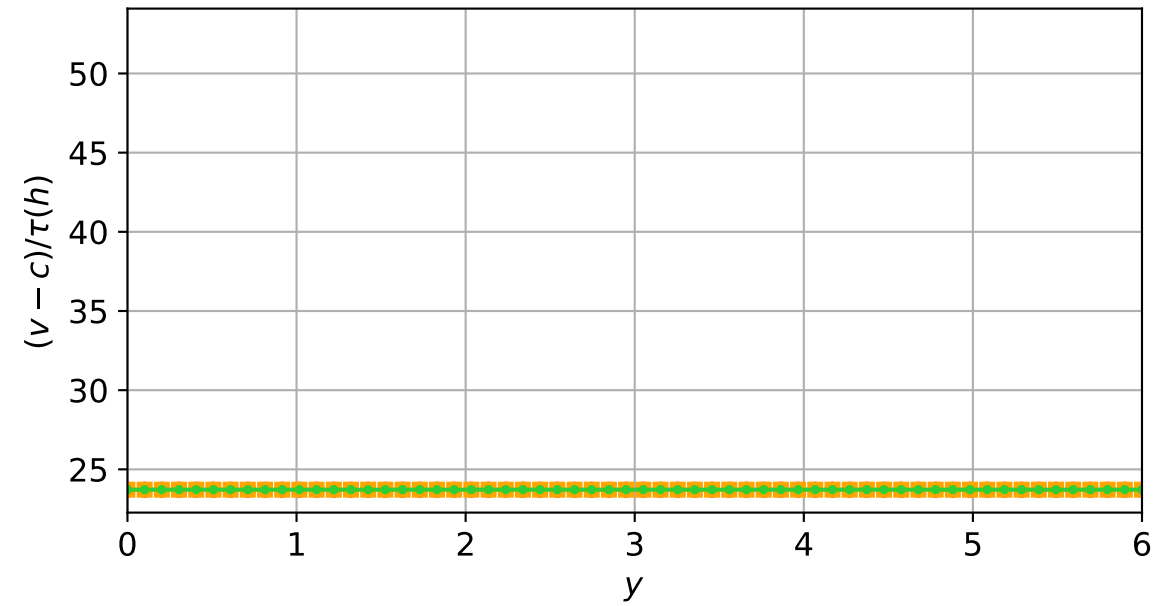
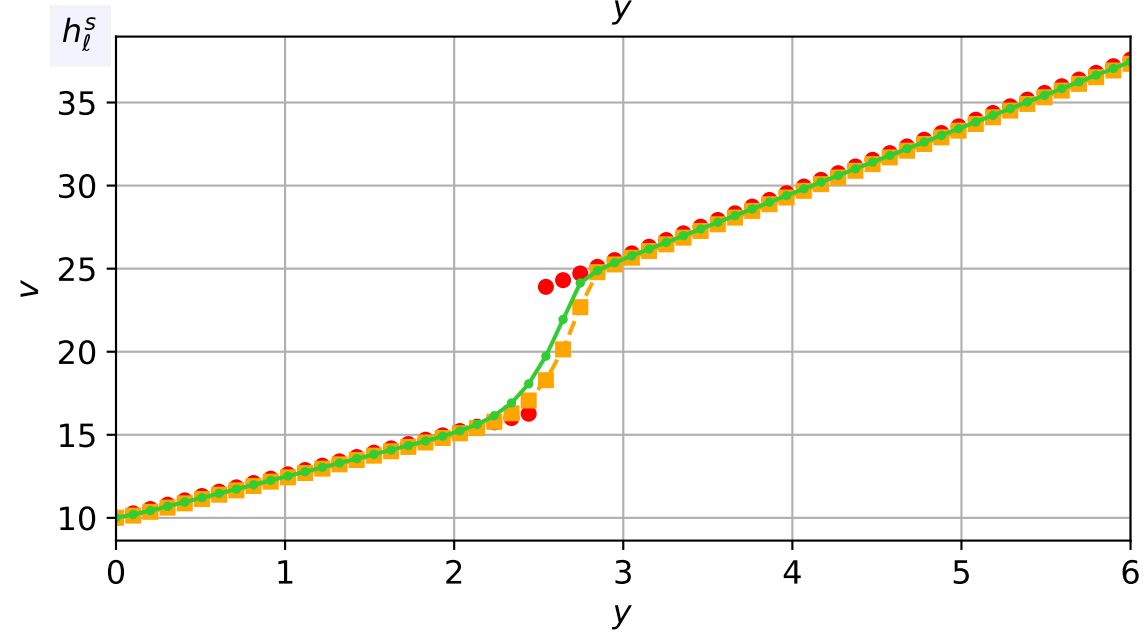
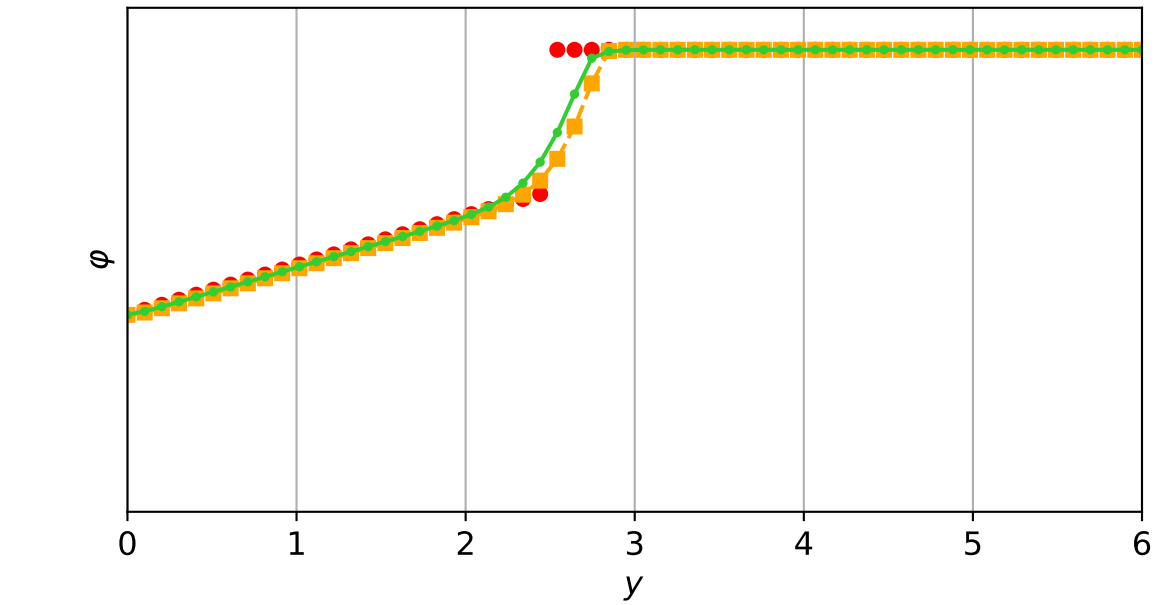
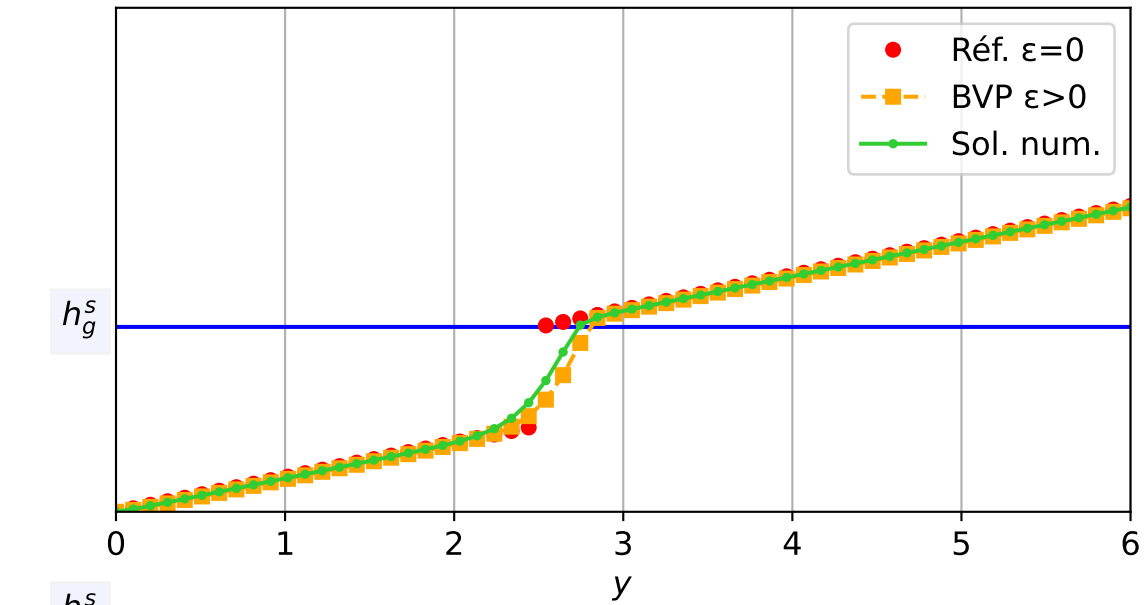
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.85126/12$  (8672 iter) — résidu eq. =  $7.88e-06$



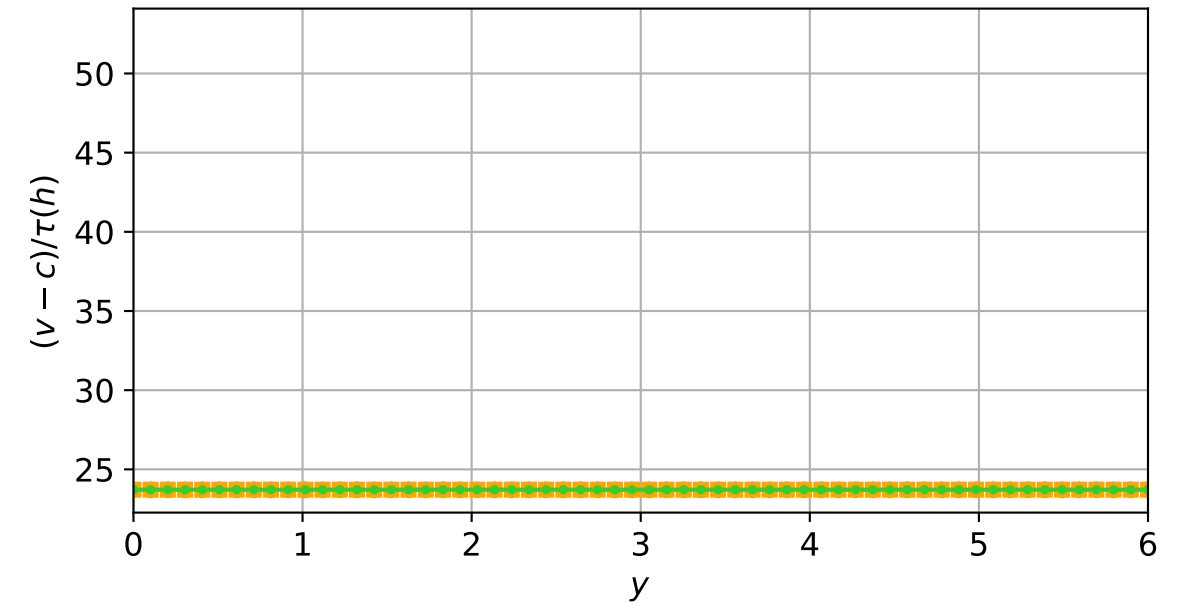
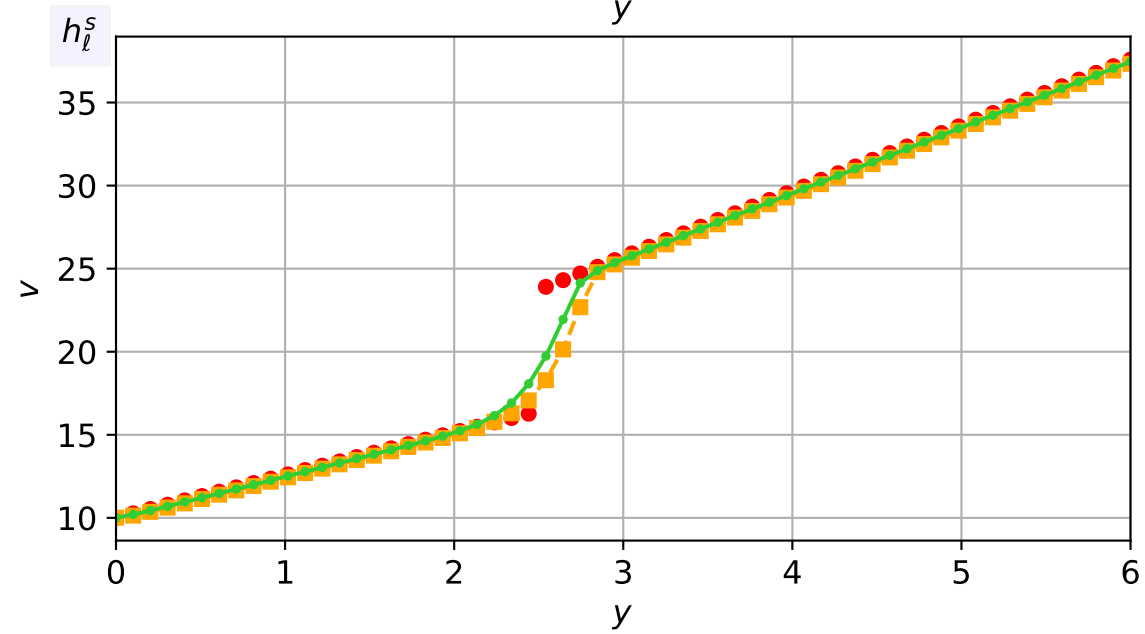
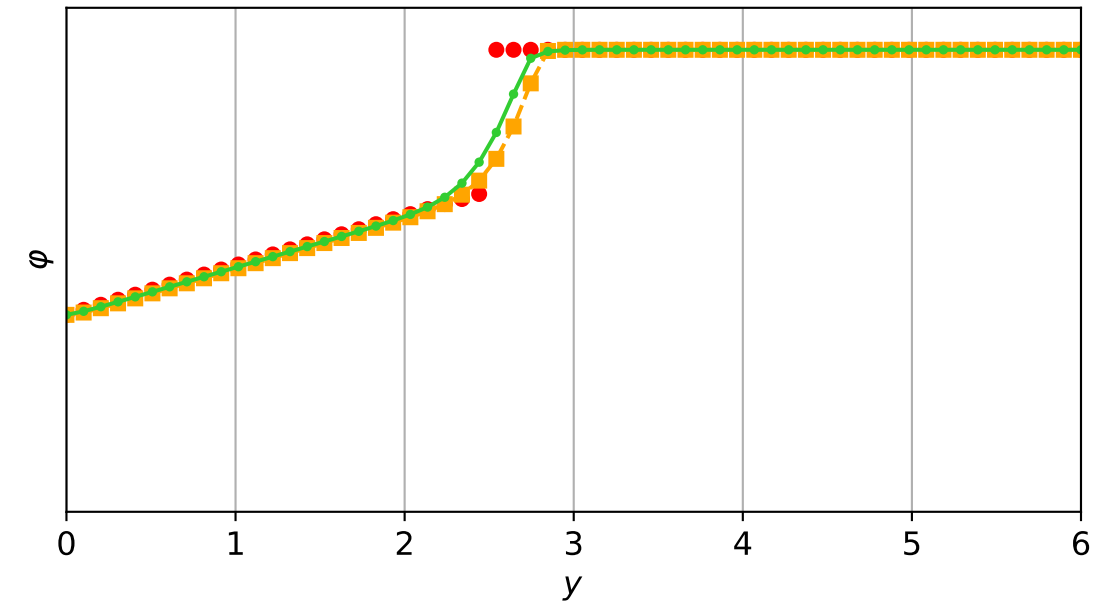
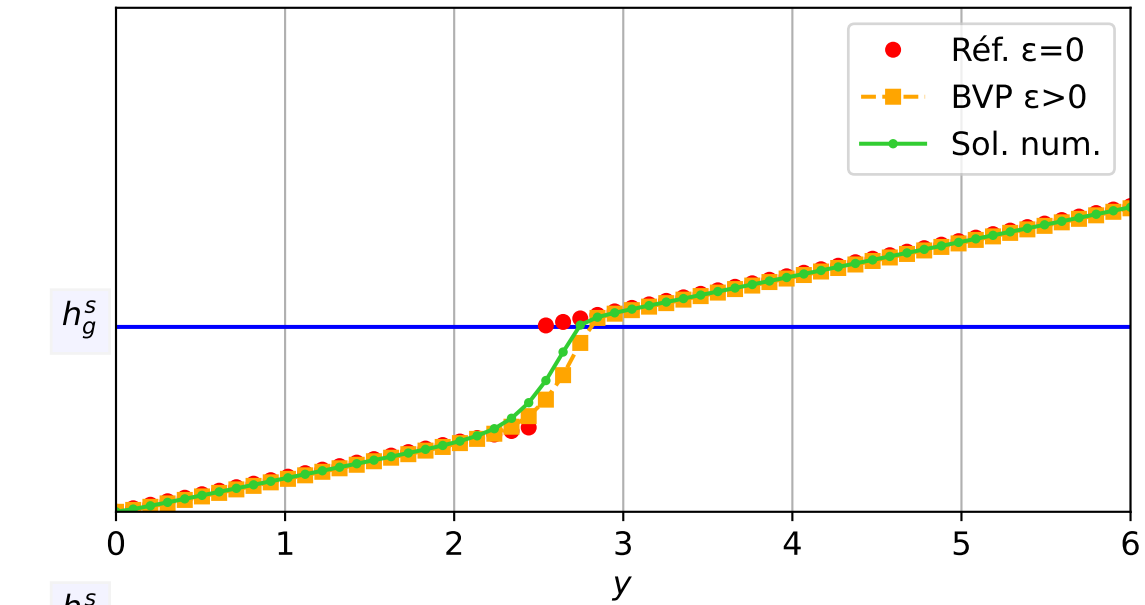
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.90014/12$  (8692 iter) — résidu eq. =  $6.97e-06$



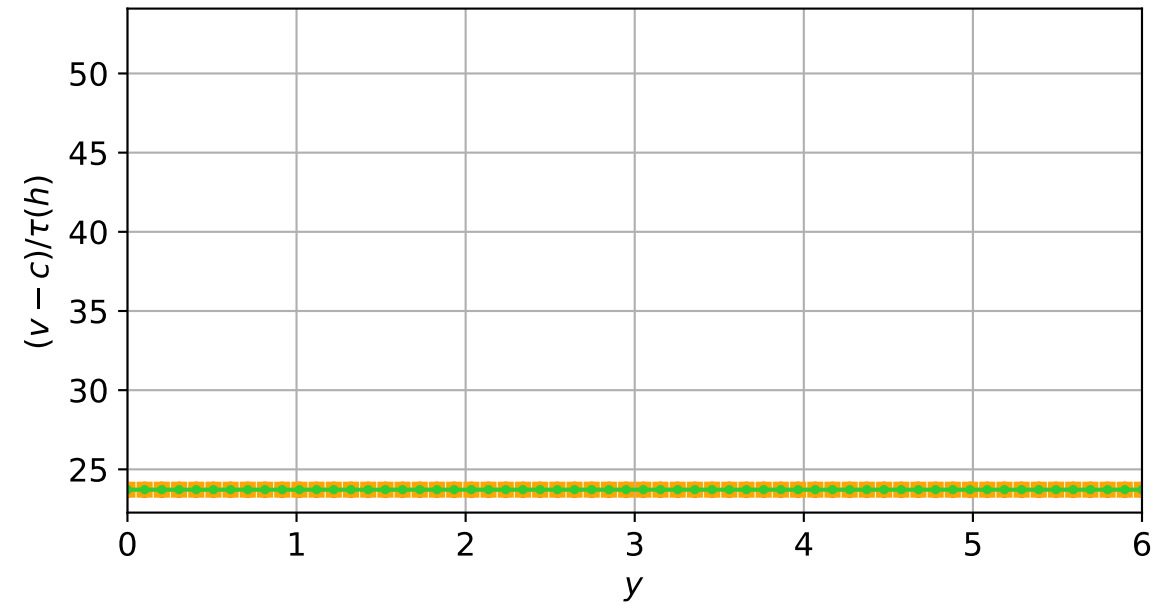
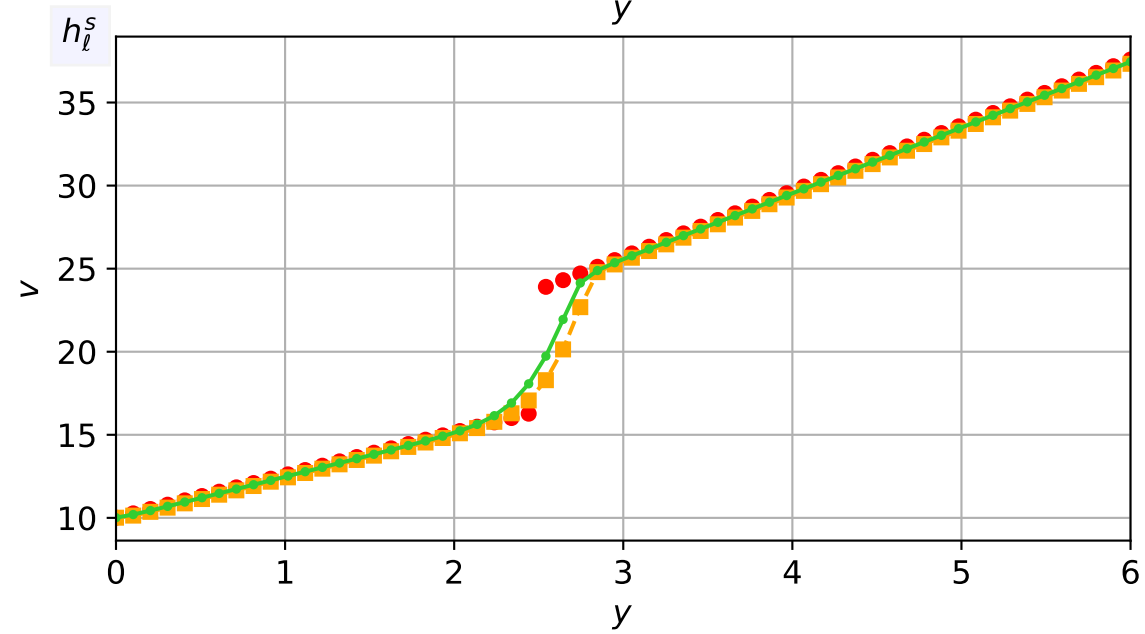
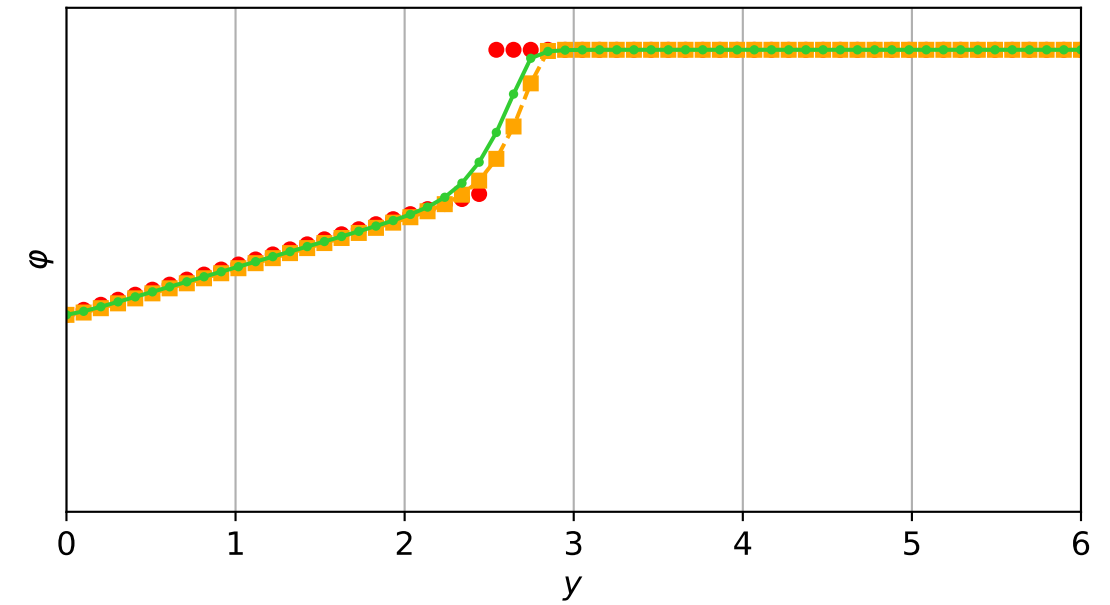
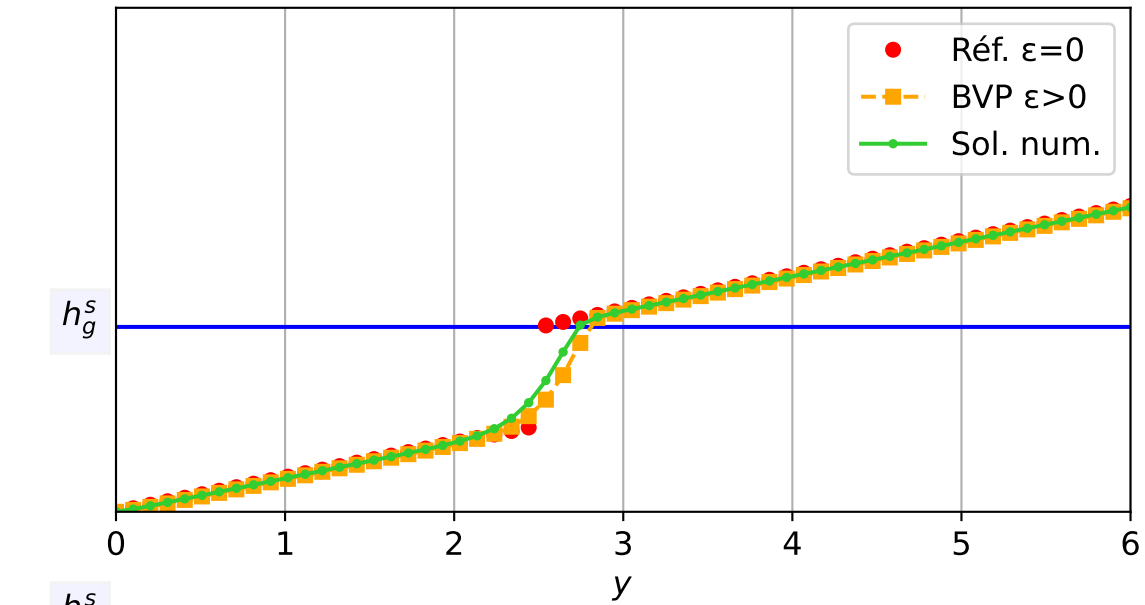
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.95146/12$  (8713 iter) — résidu eq. =  $6.14e-06$



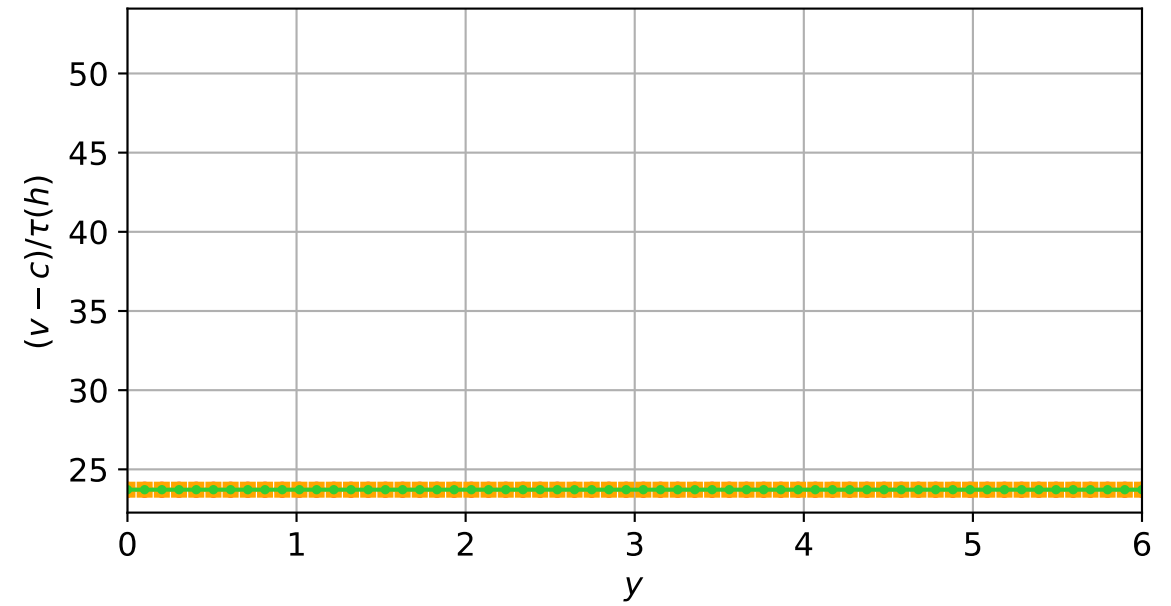
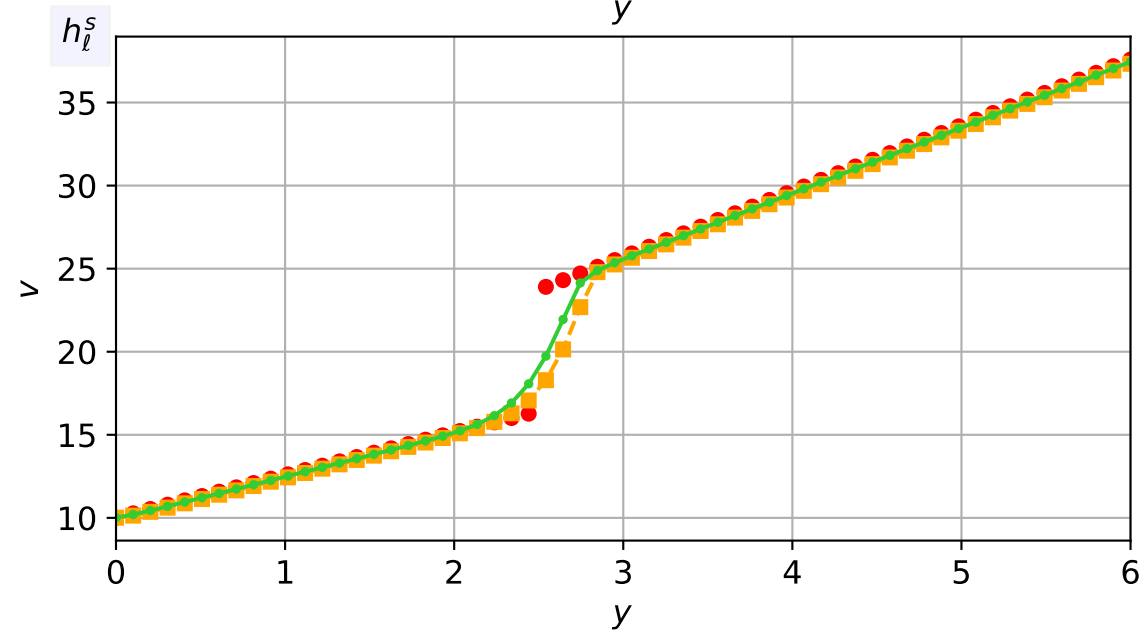
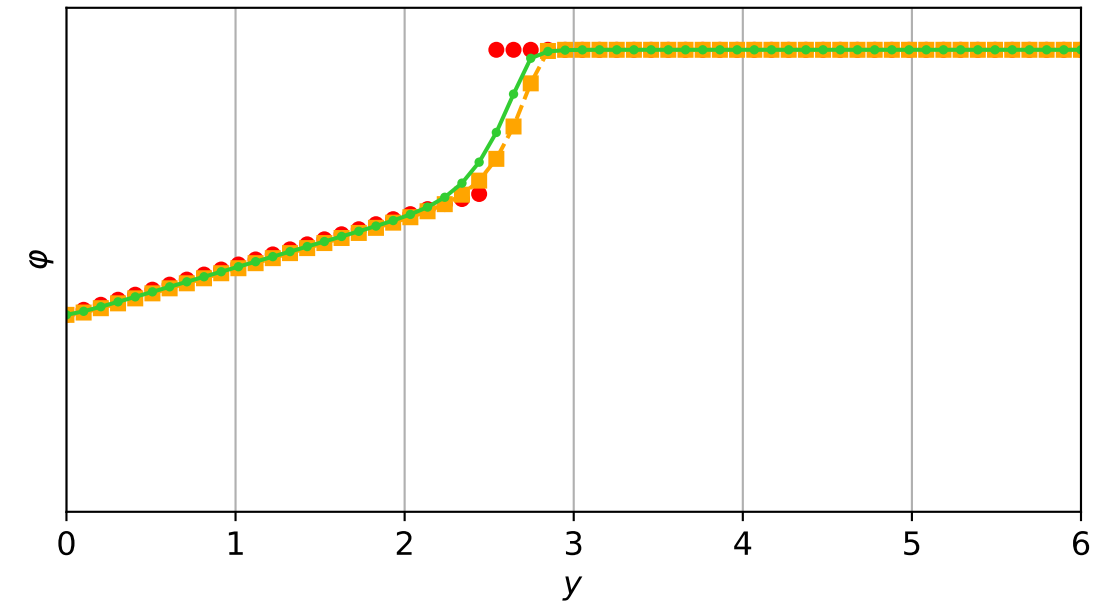
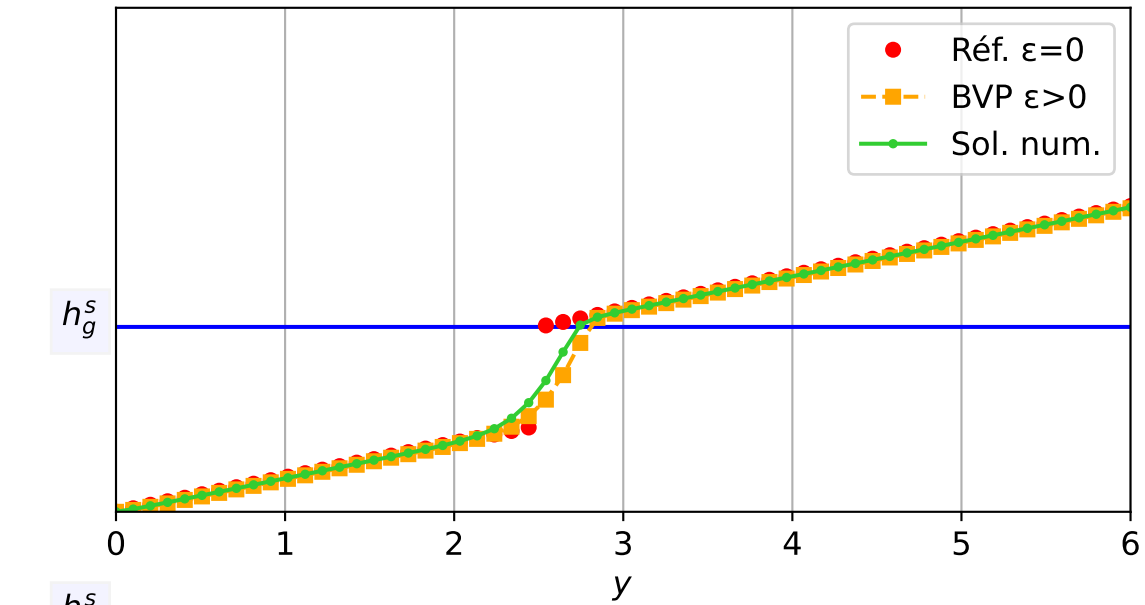
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.00033/12$  (8733 iter) — résidu eq. =  $5.44e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.05165/12$  (8754 iter) — résidu eq. =  $4.80e-06$

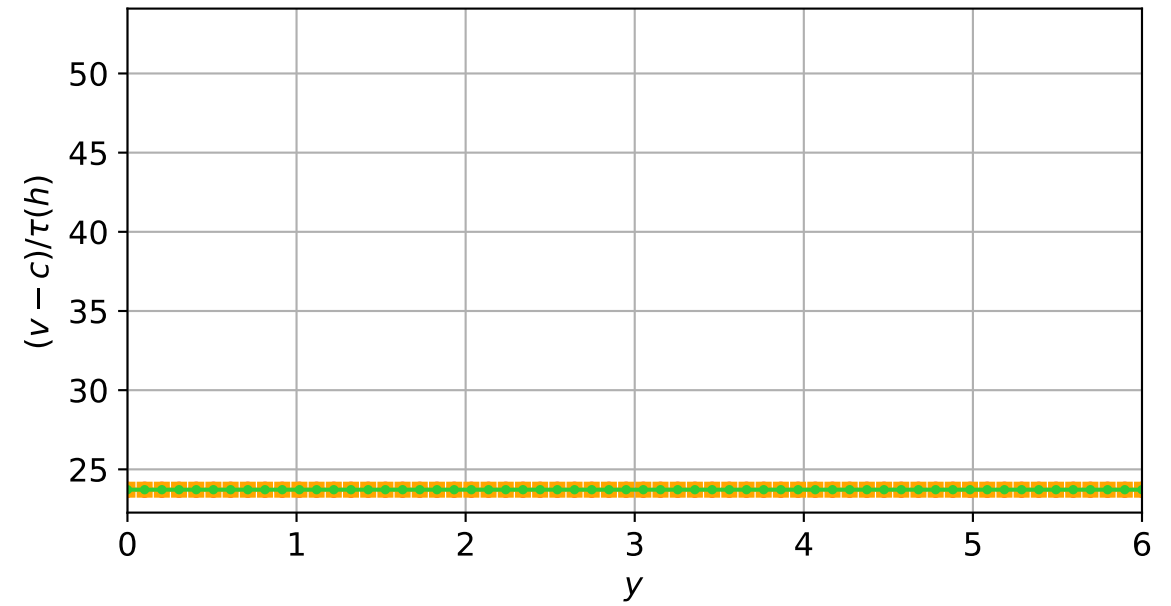
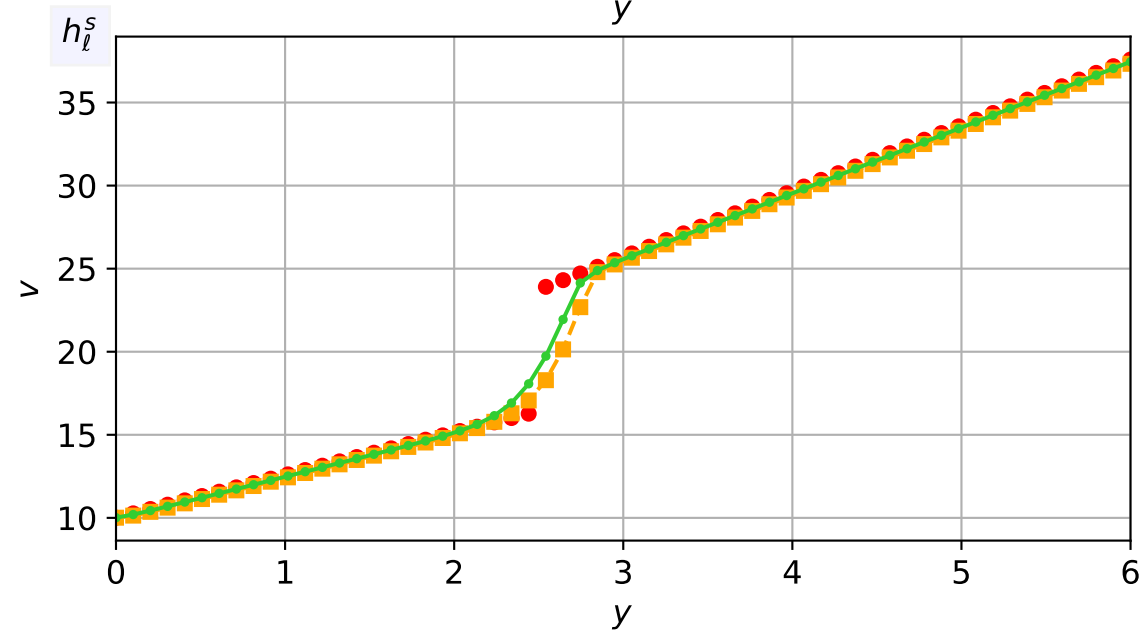
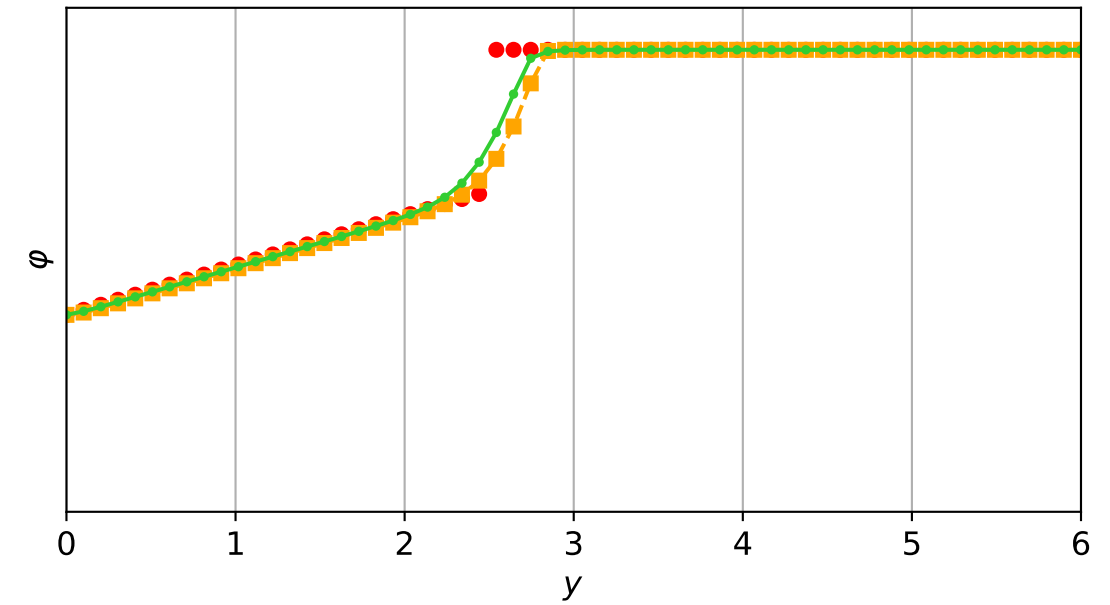
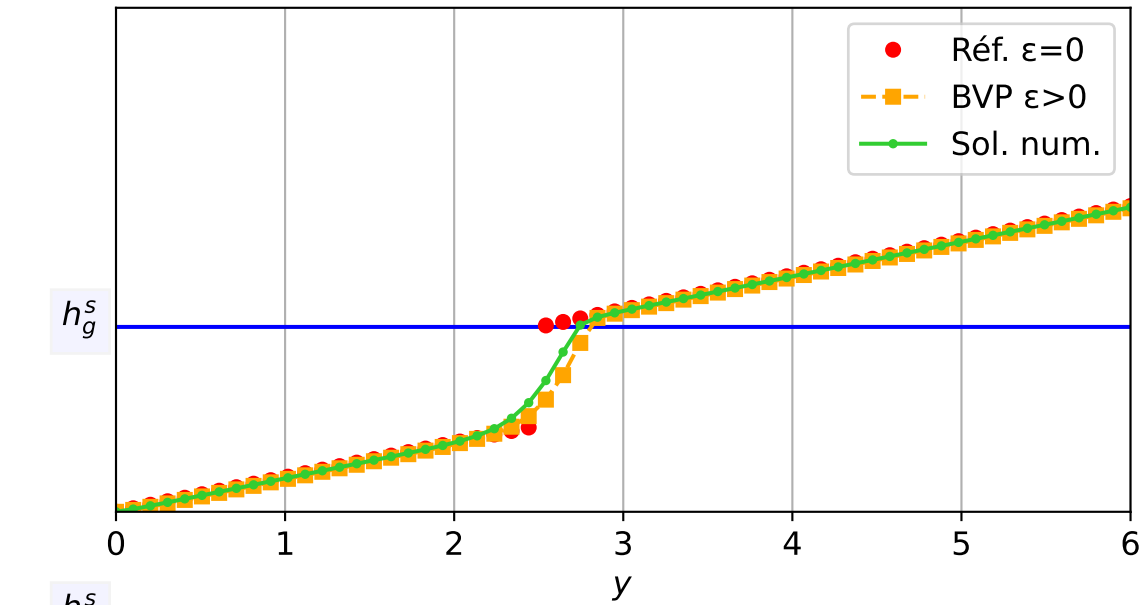


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.10053/12$  (8774 iter) — résidu eq. =  $4.26e-06$

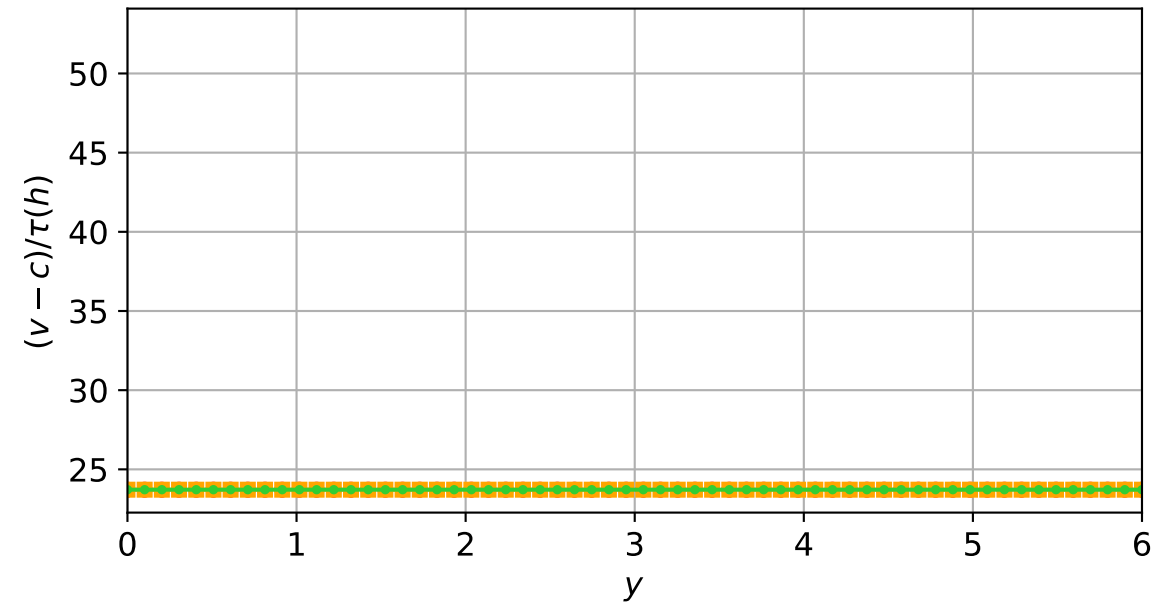
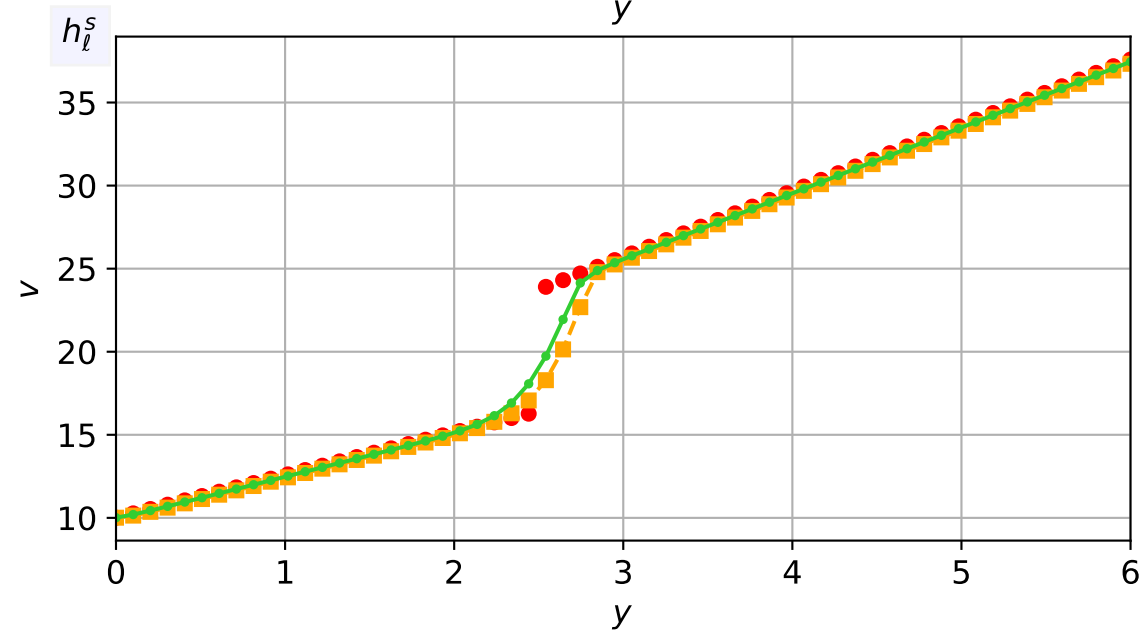
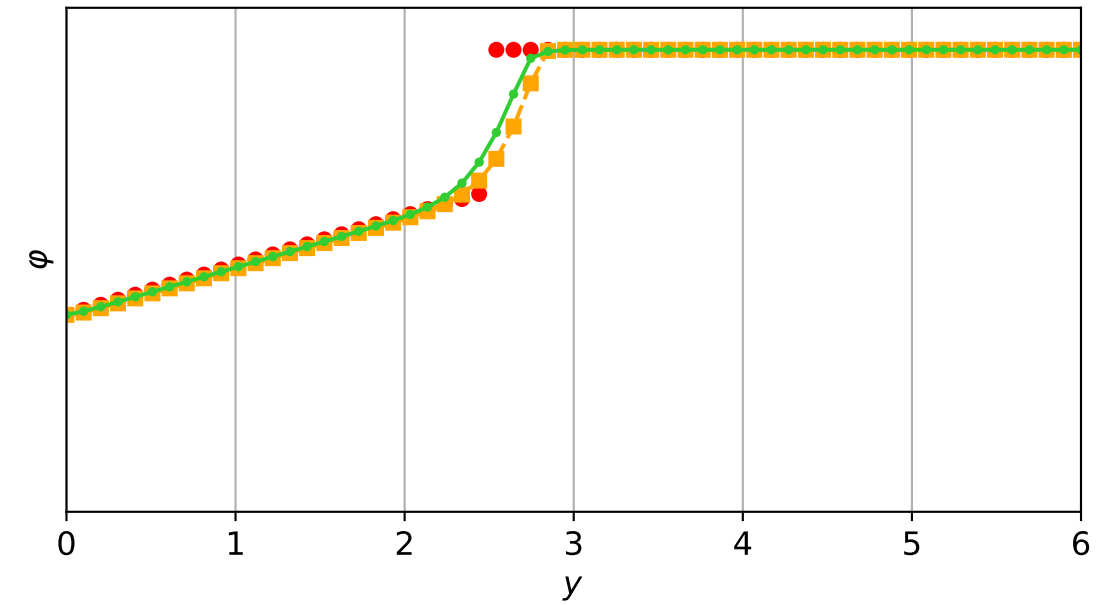
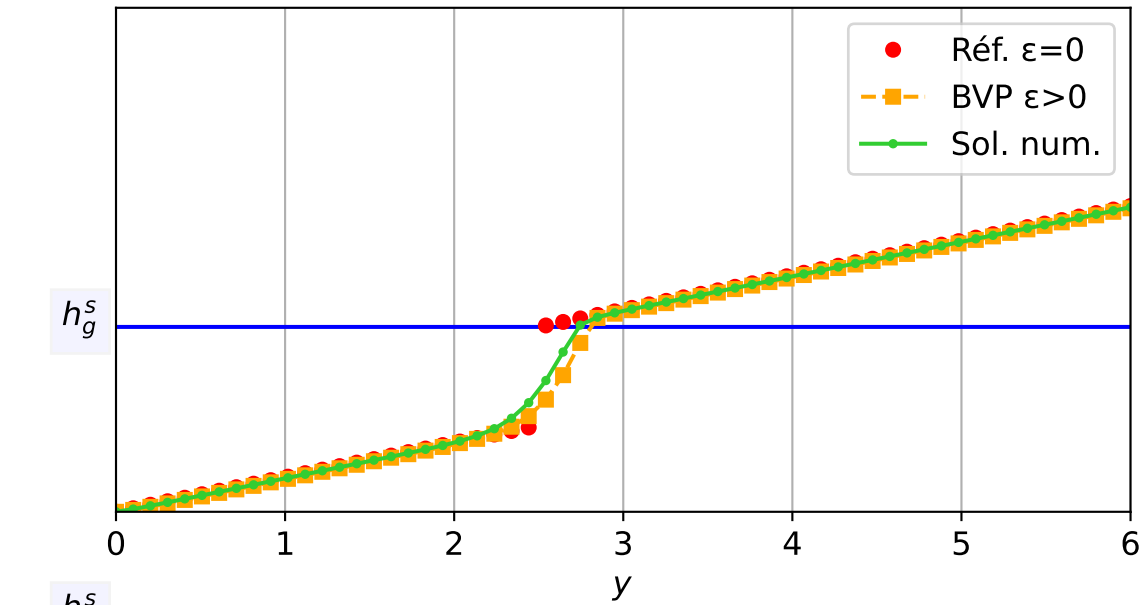




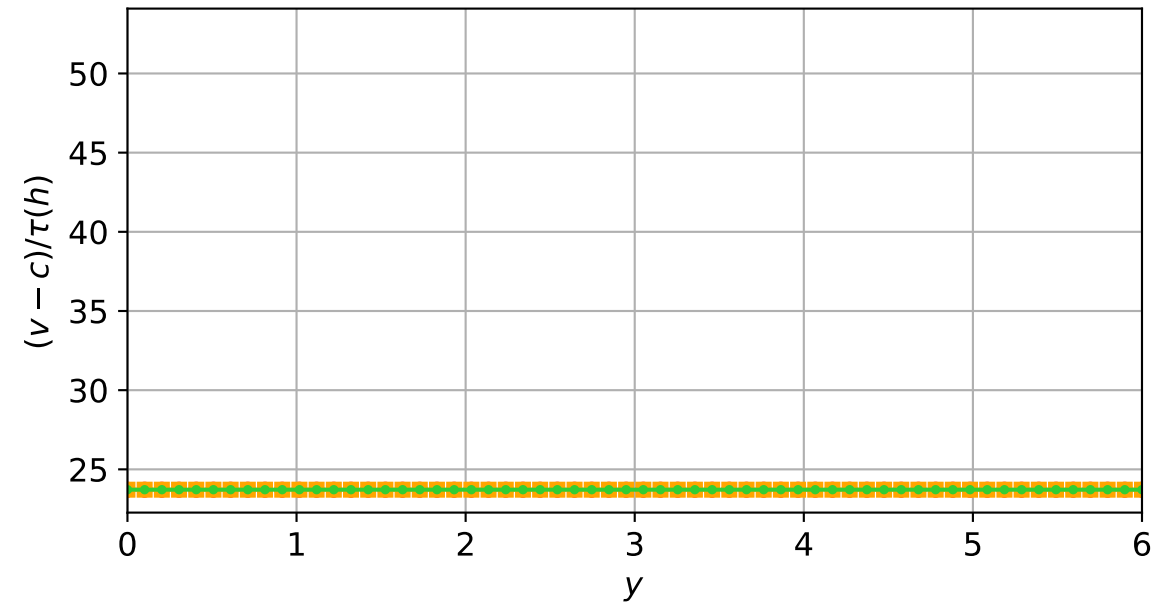
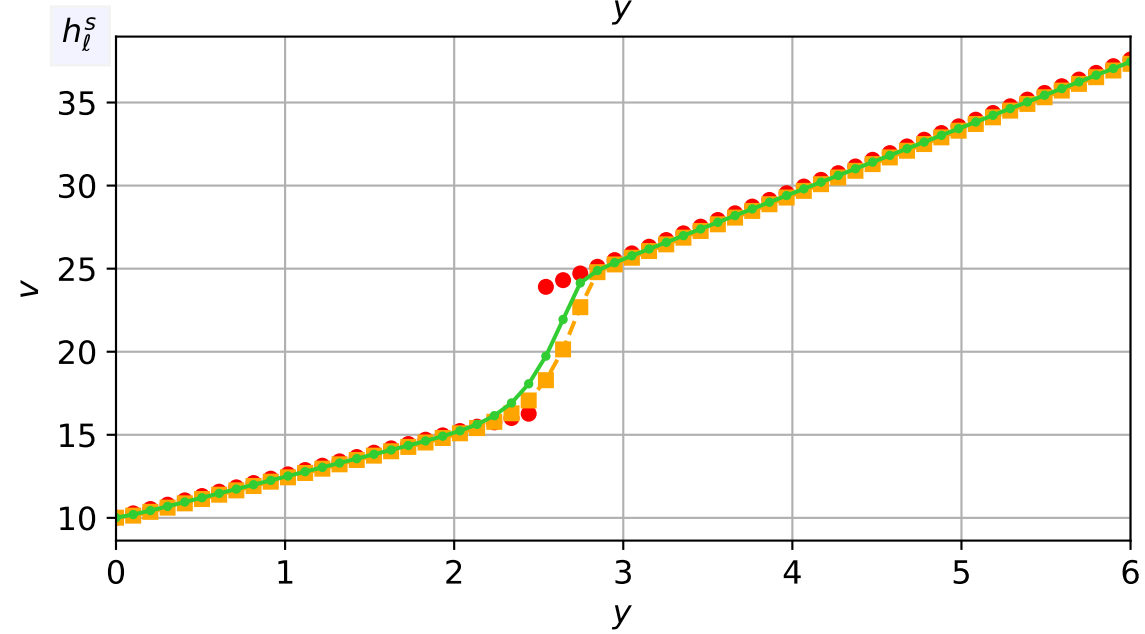
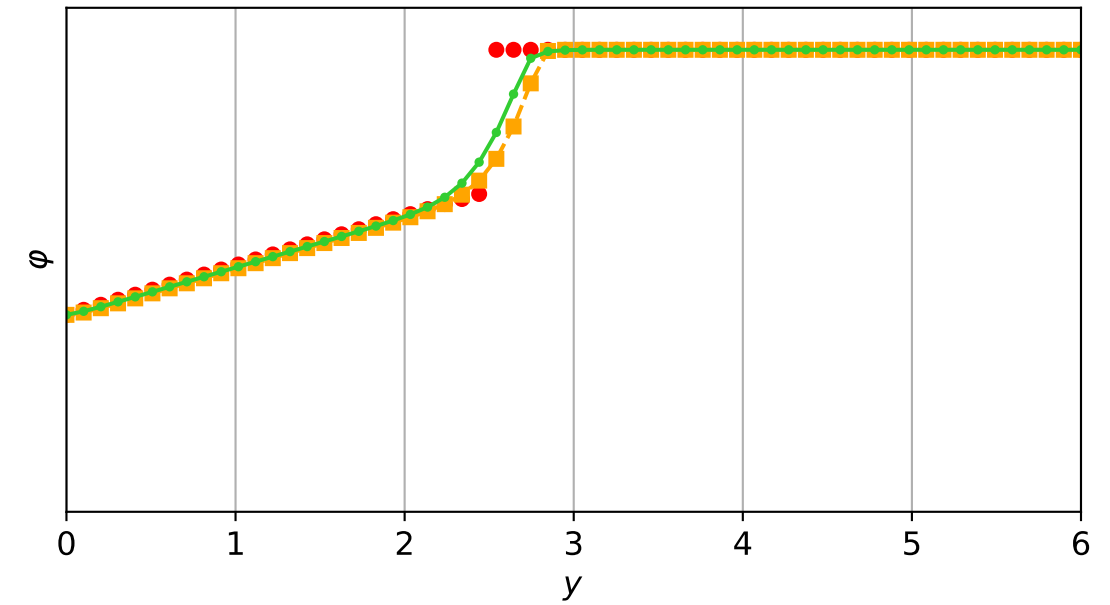
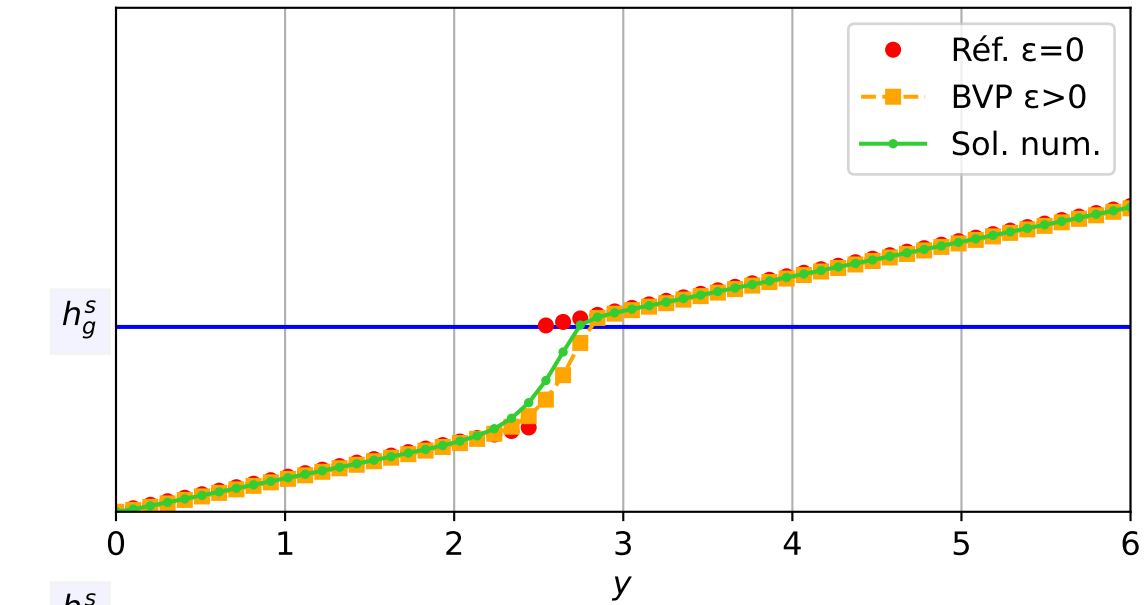
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.15185/12$  (8795 iter) — résidu eq. =  $3.75e-06$



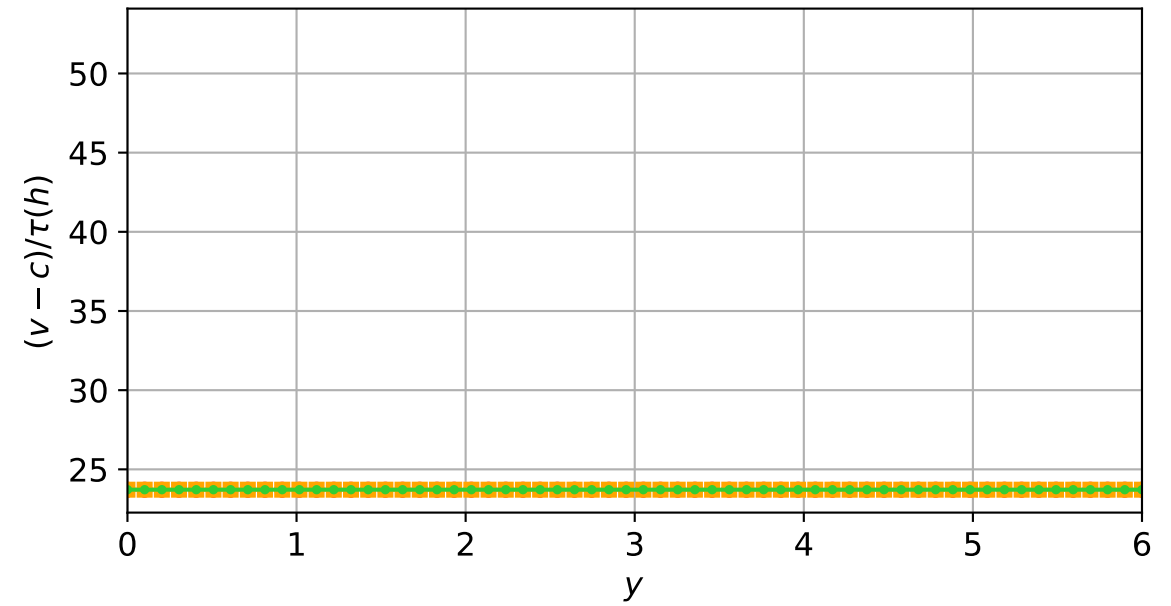
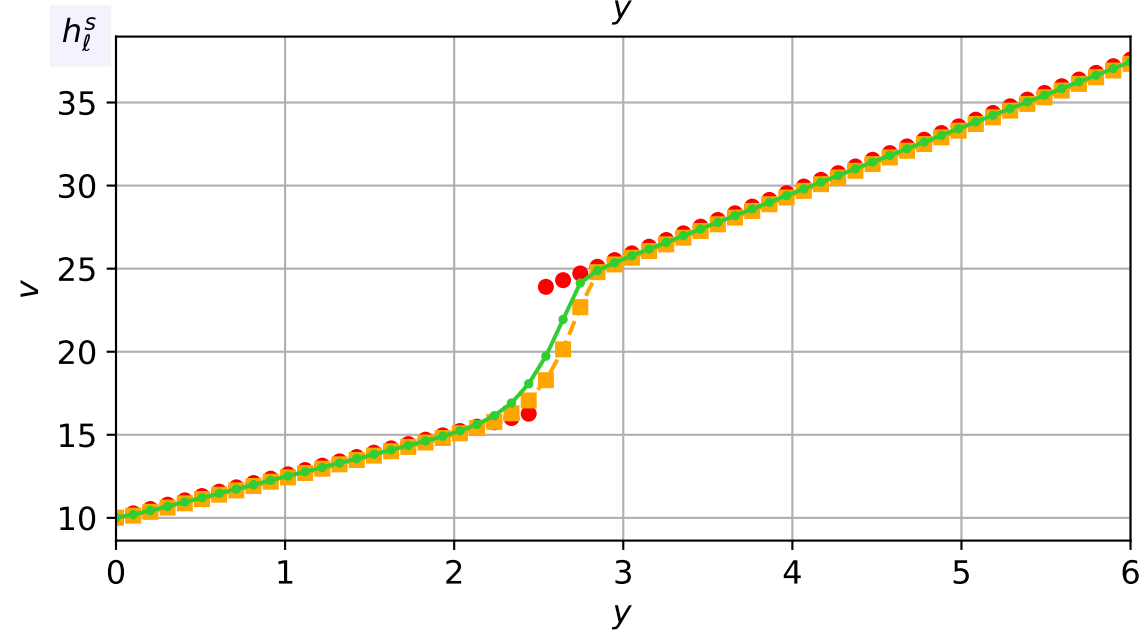
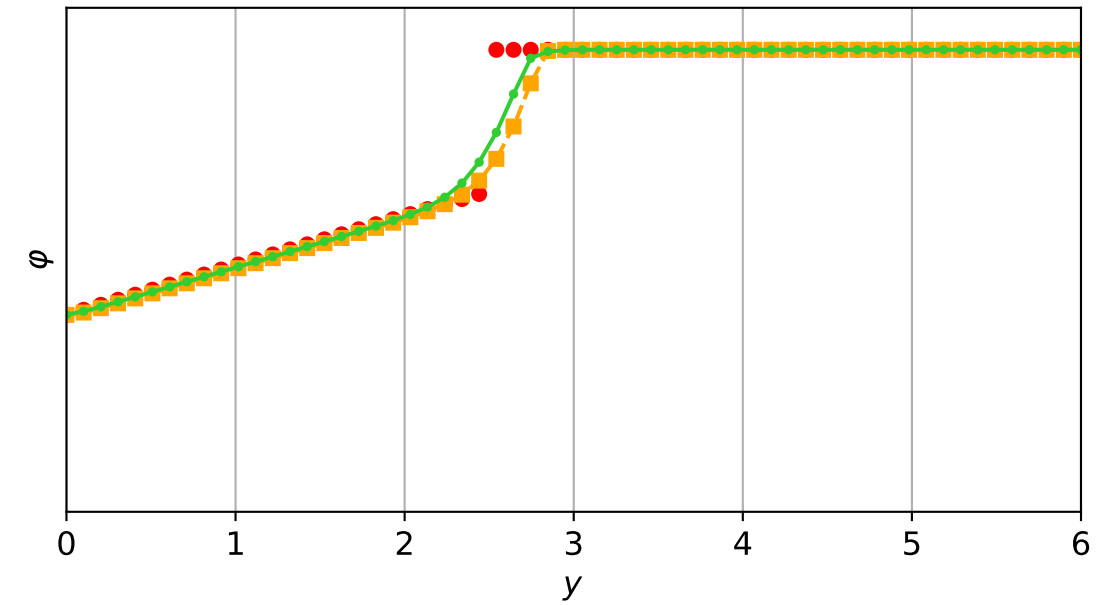
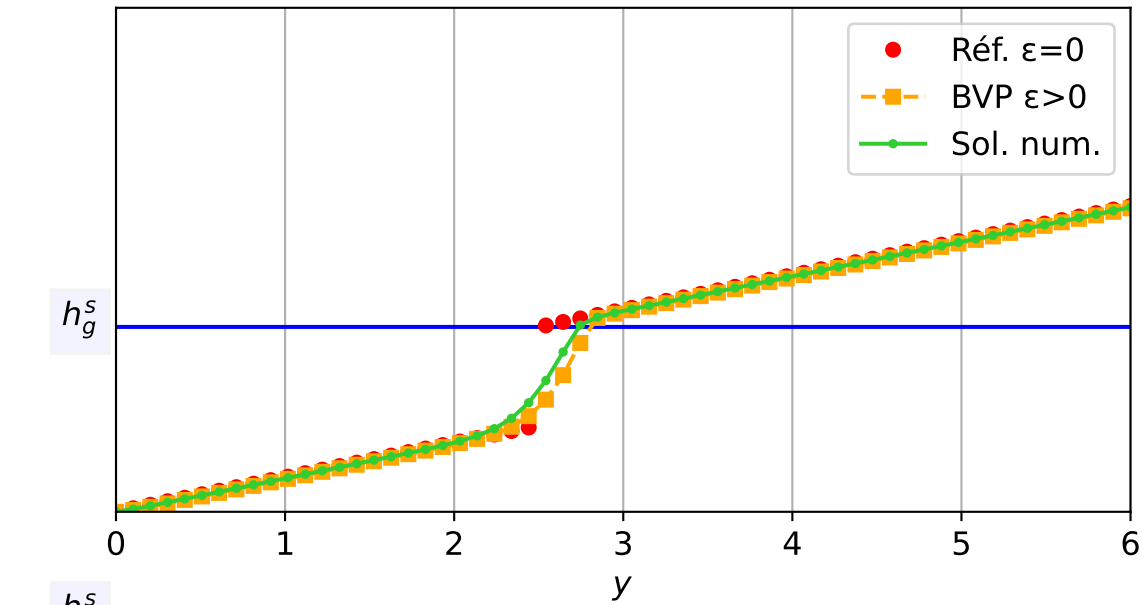
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.20073/12$  (8815 iter) — résidu eq. =  $3.33e-06$



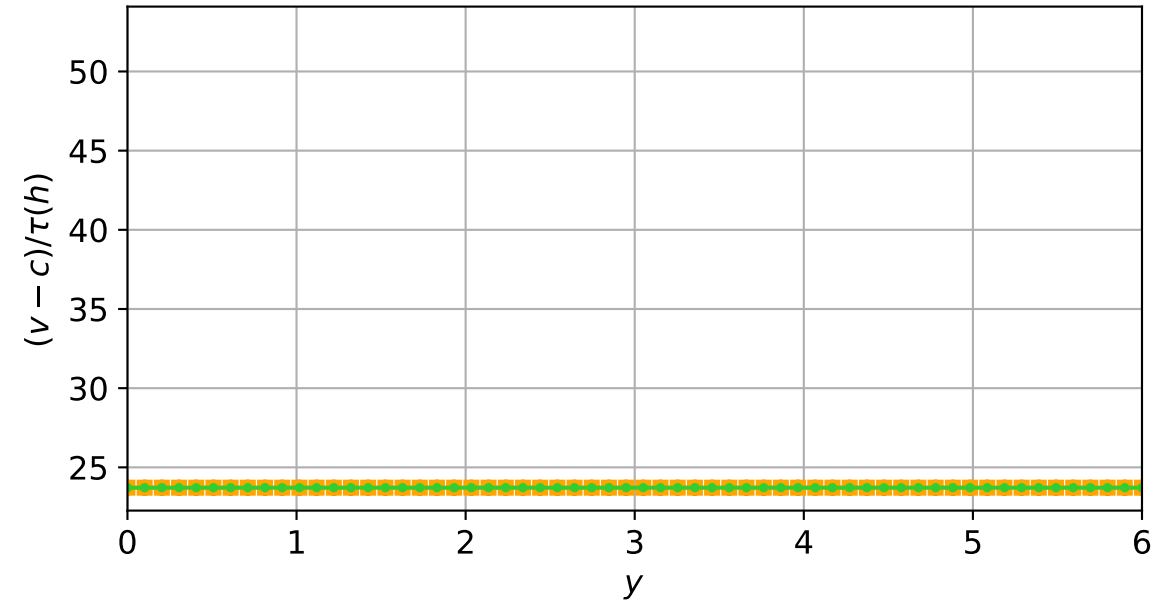
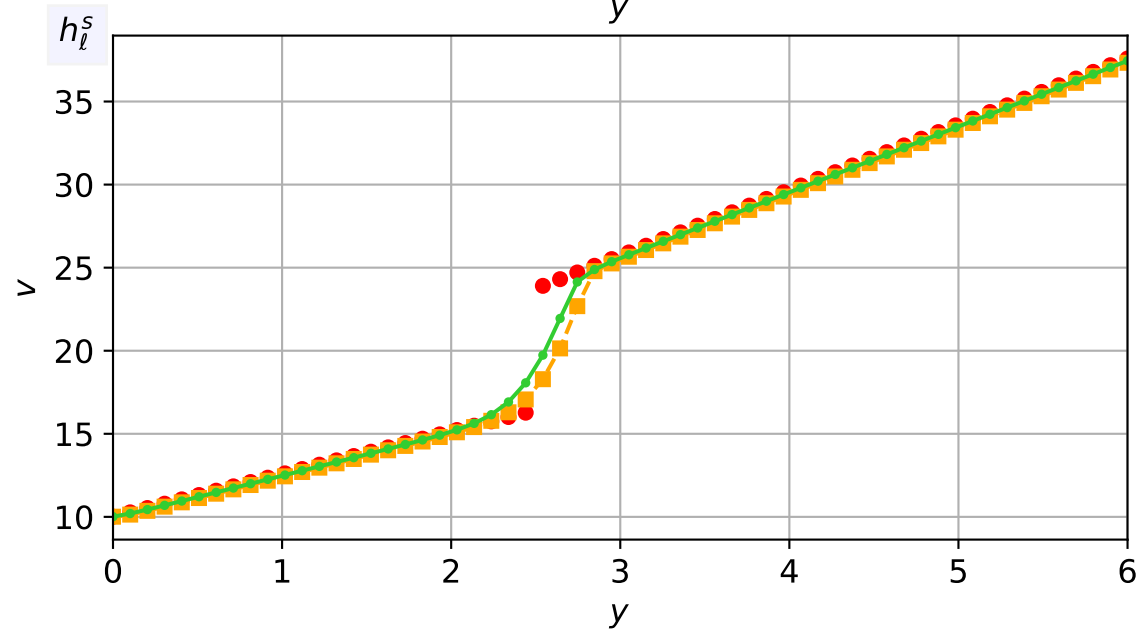
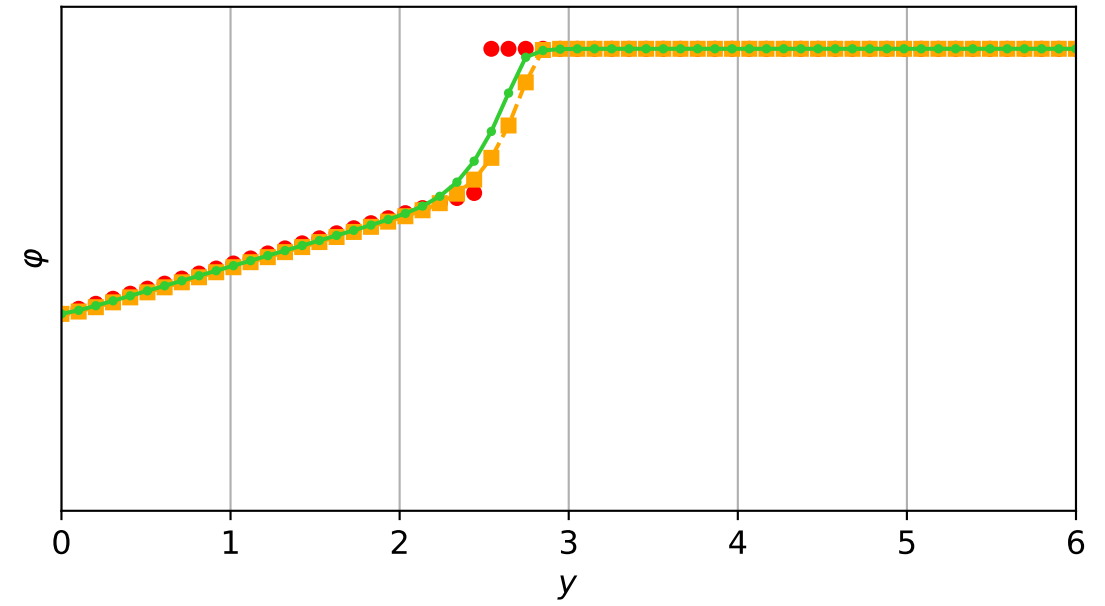
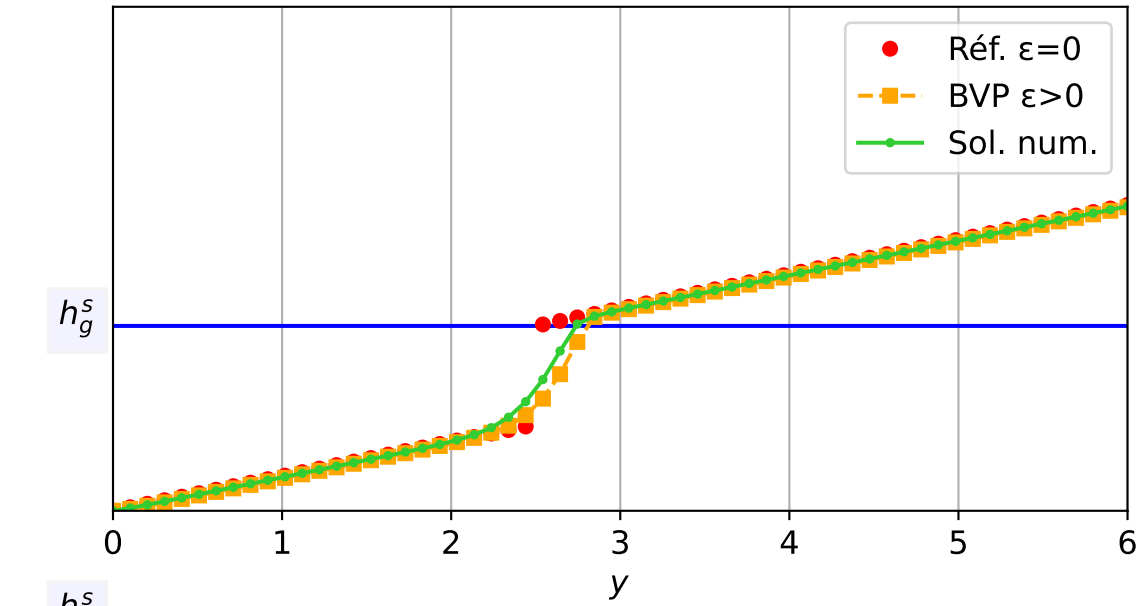
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.25205/12$  (8836 iter) — résidu eq. =  $2.93e-06$



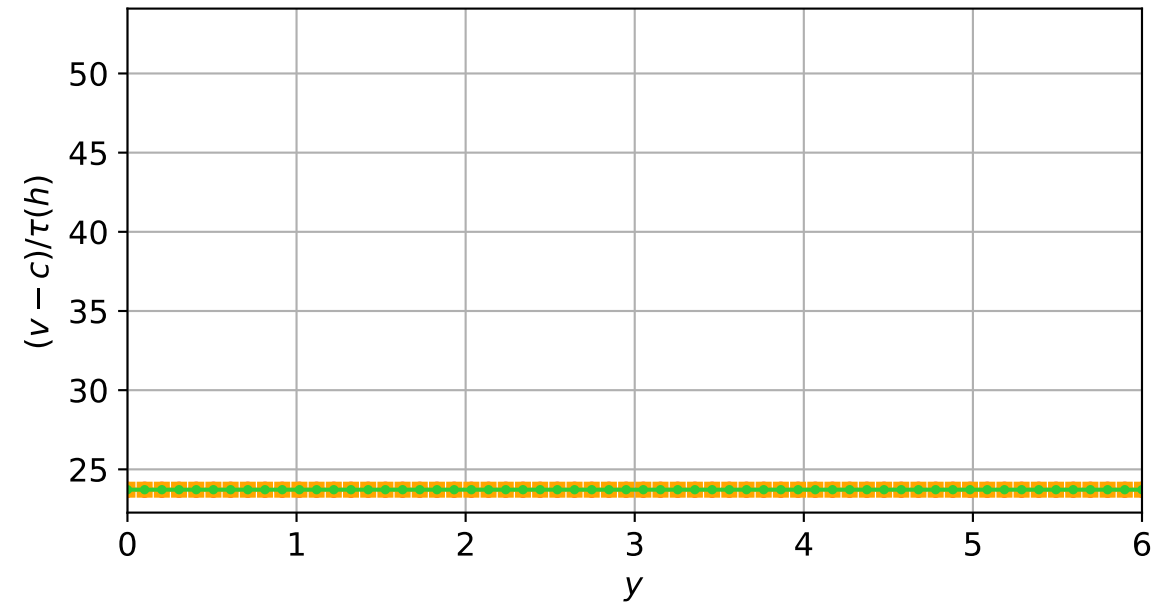
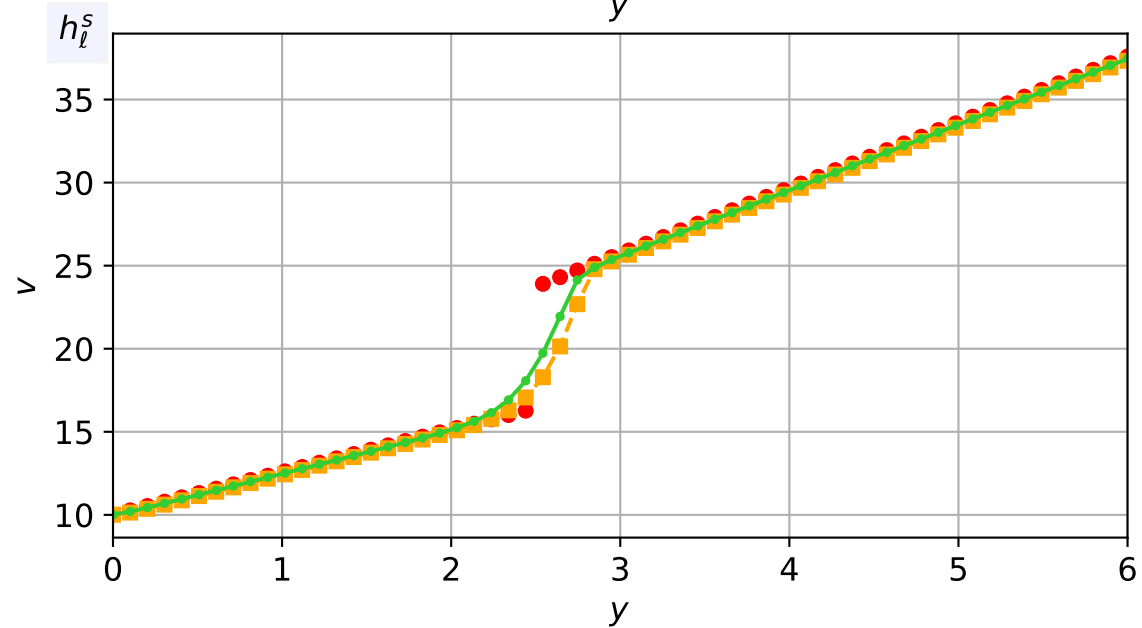
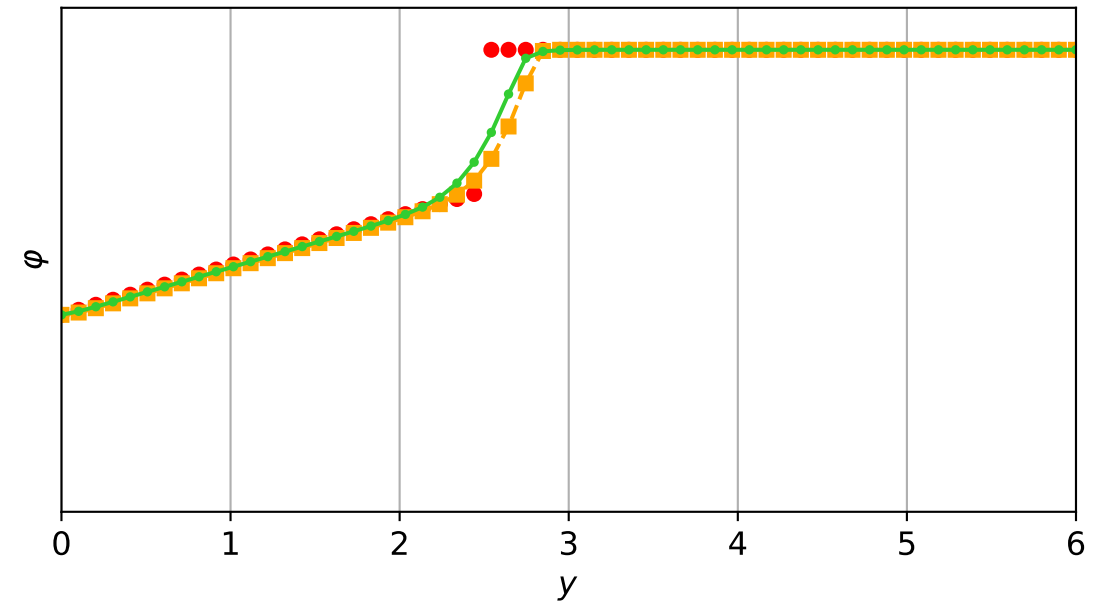
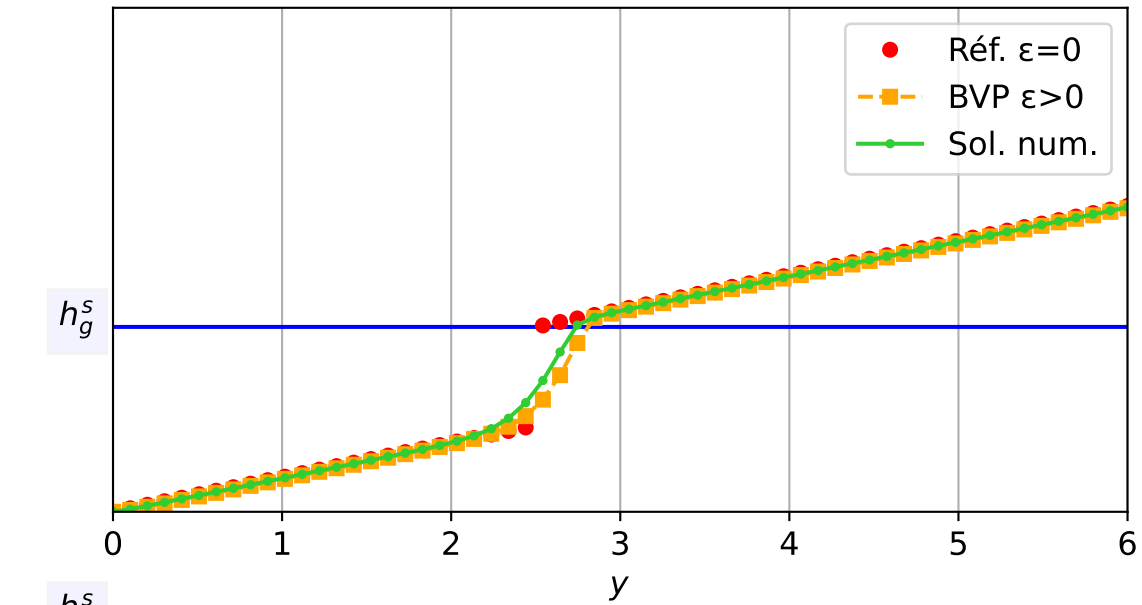
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.30093/12$  (8856 iter) — résidu eq. =  $2.60e-06$



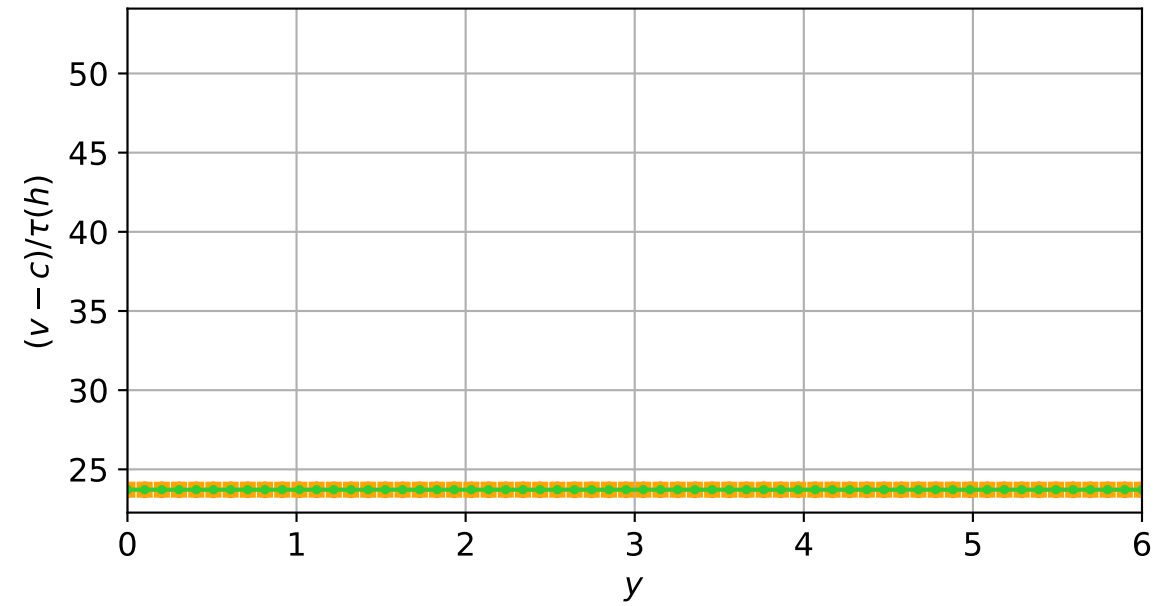
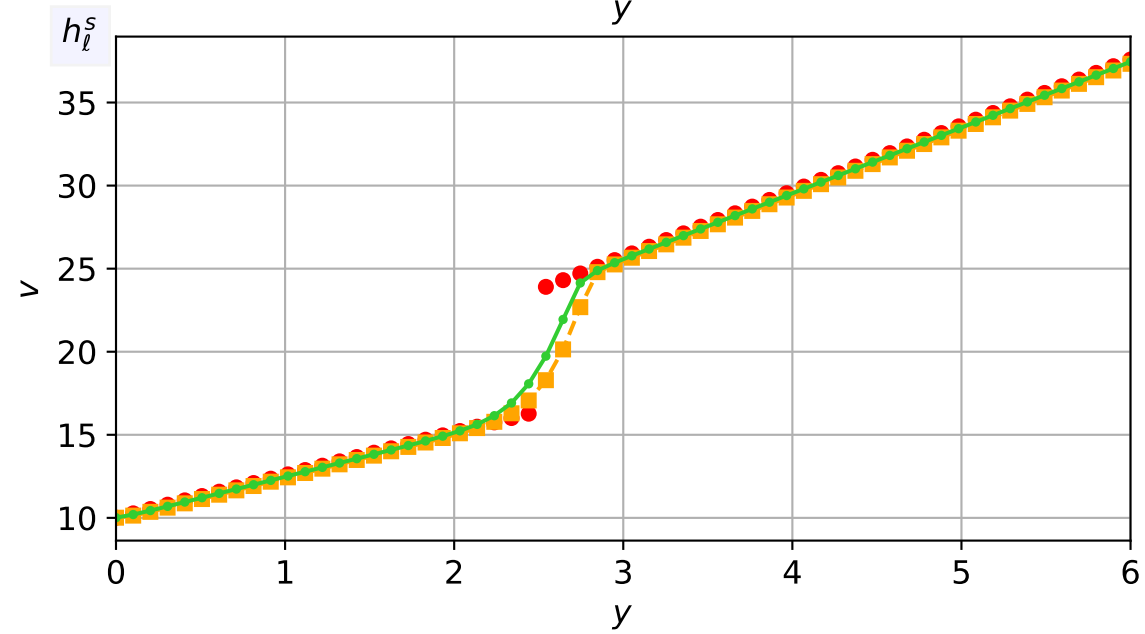
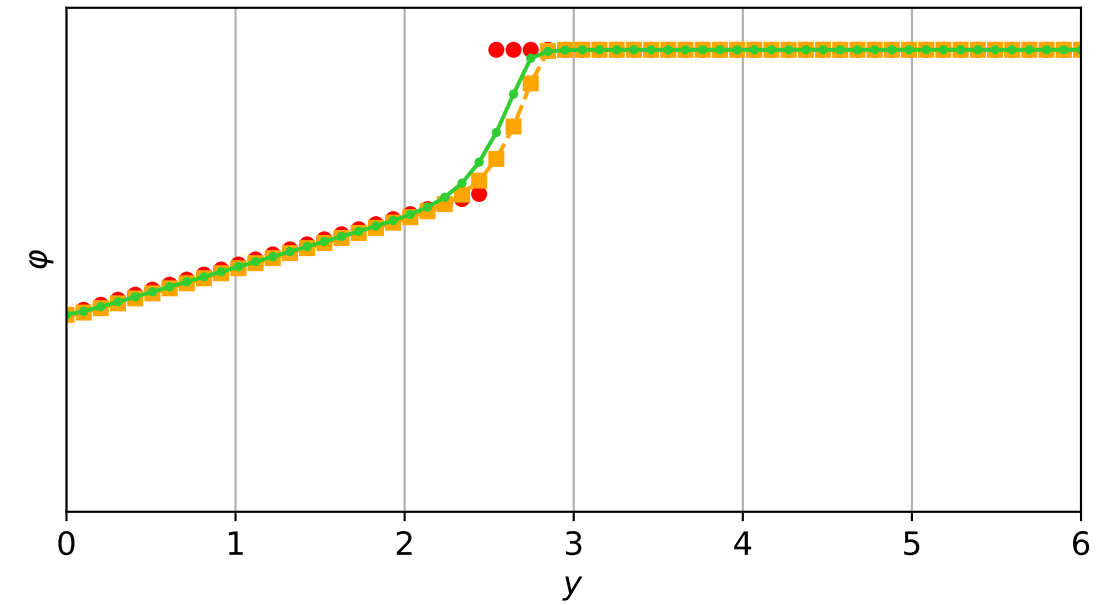
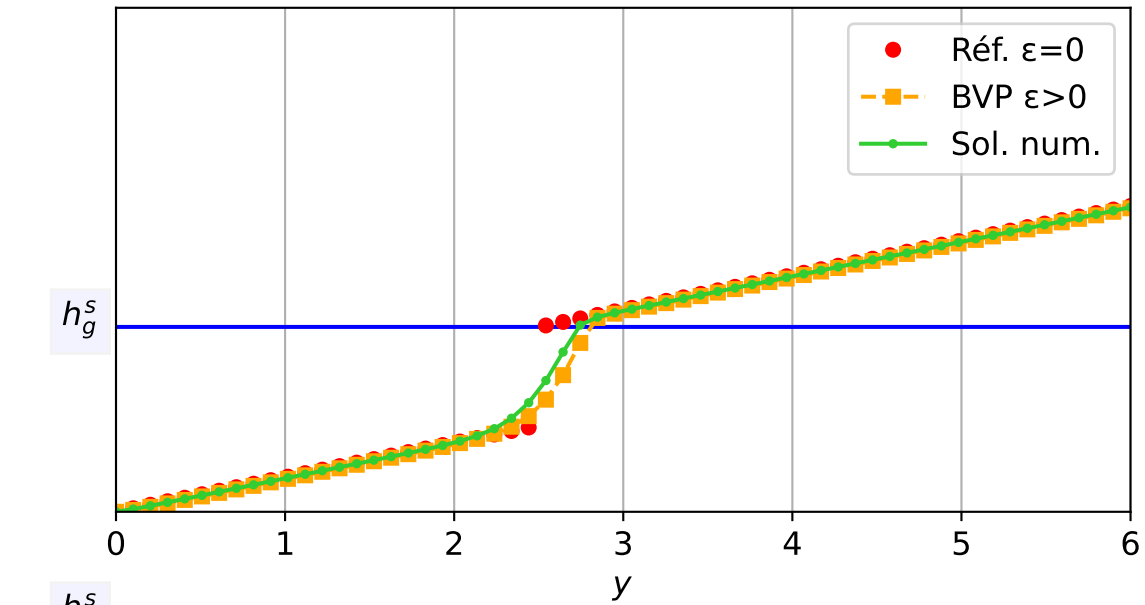
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.35225/12$  (8877 iter) — résidu eq. =  $2.30e-06$



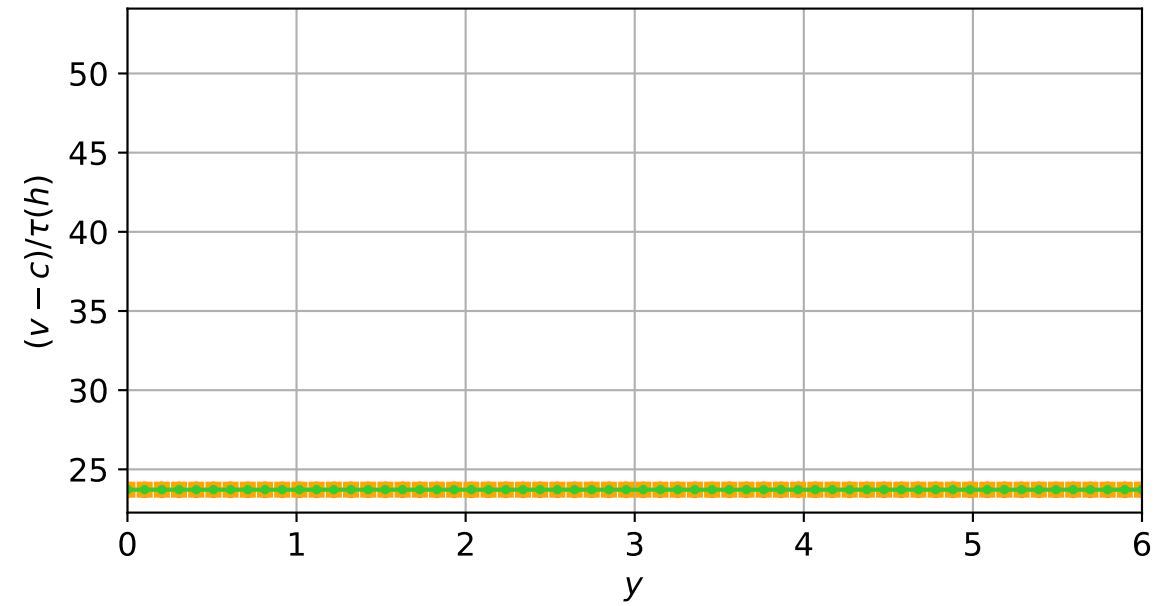
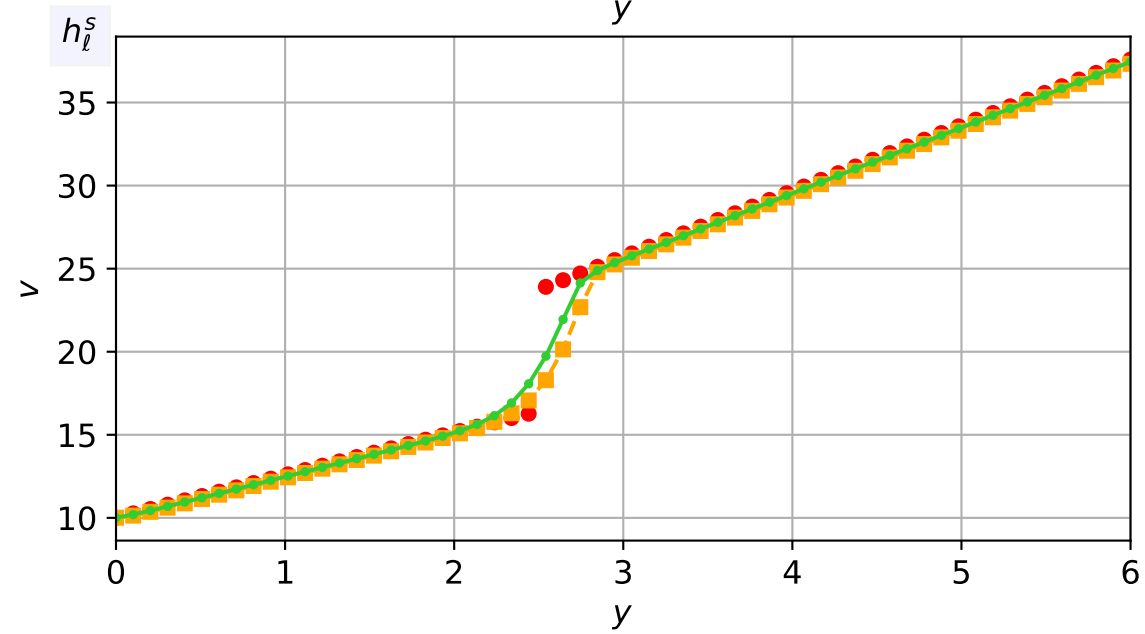
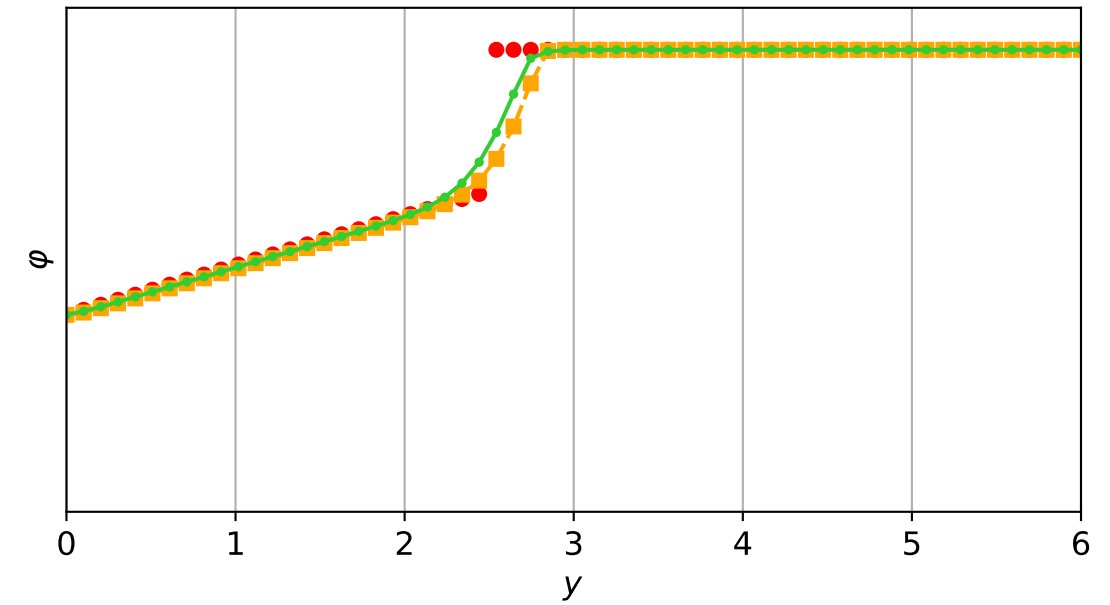
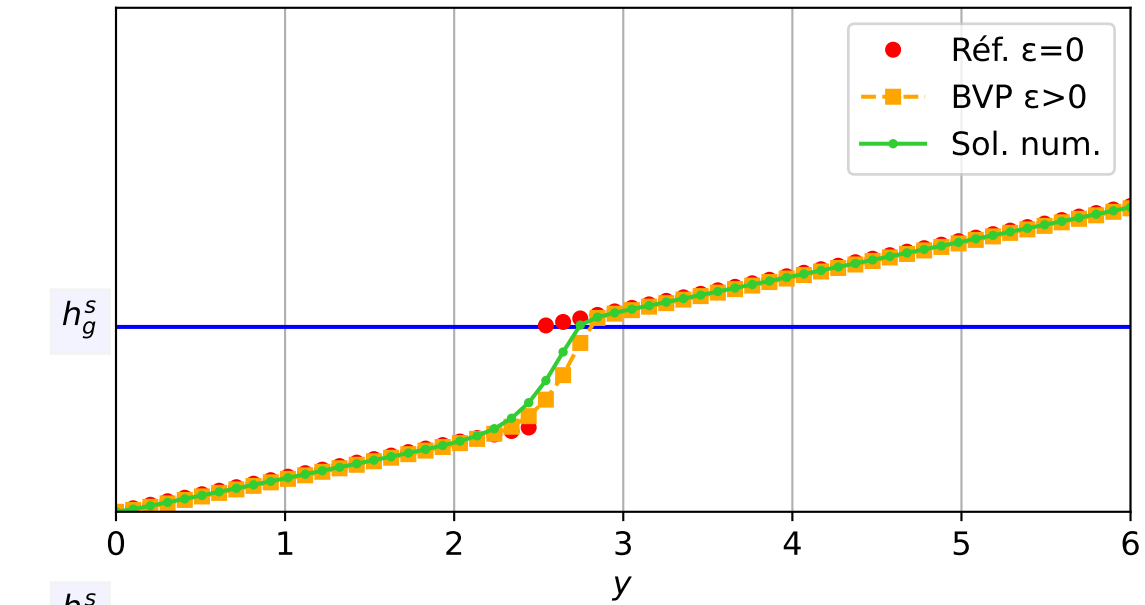
TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.40112/12$  (8897 iter) — résidu eq. =  $2.04e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.45244/12$  (8918 iter) — résidu eq. =  $1.80e-06$

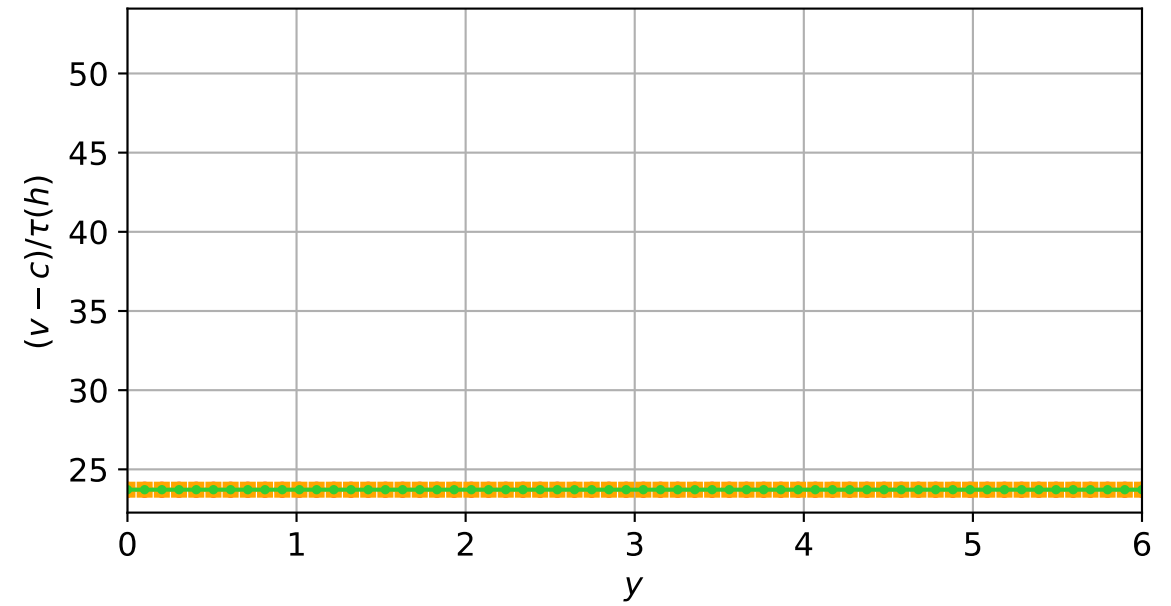
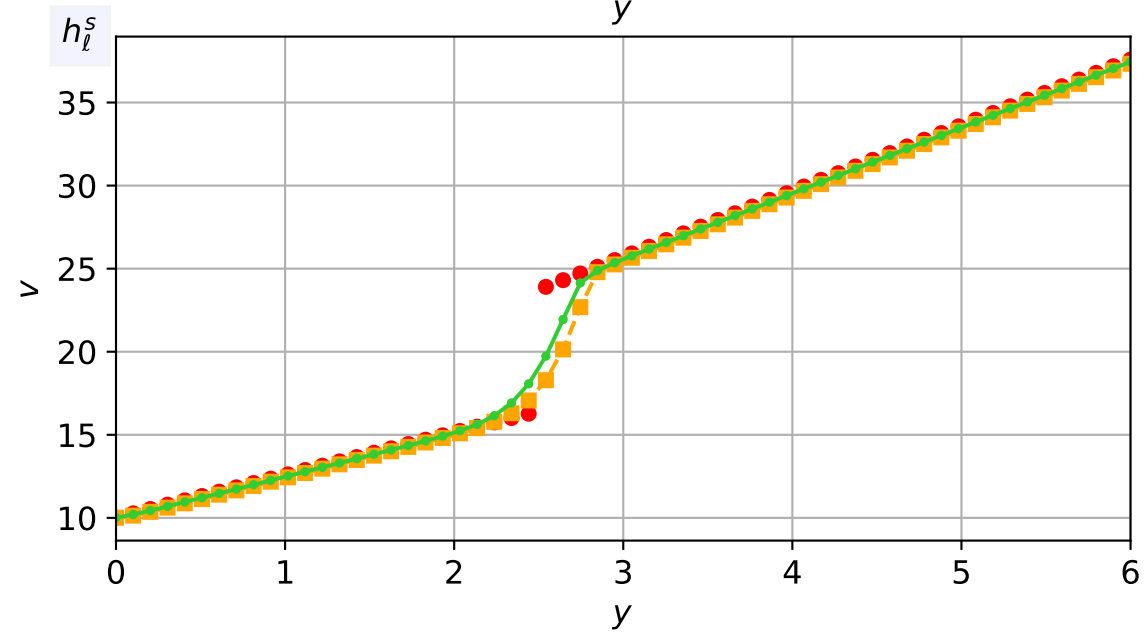
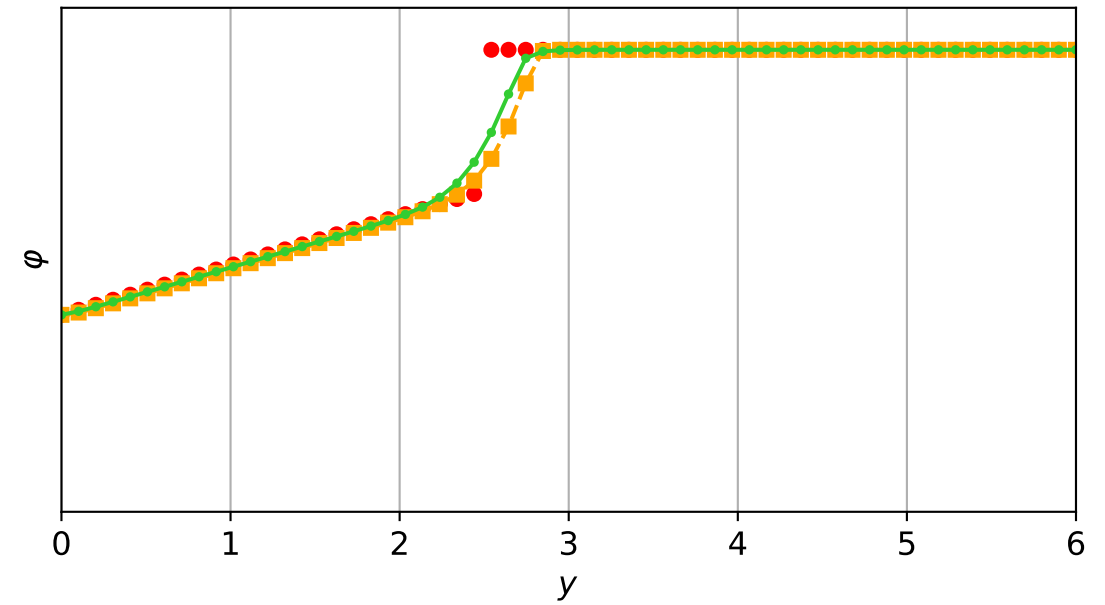
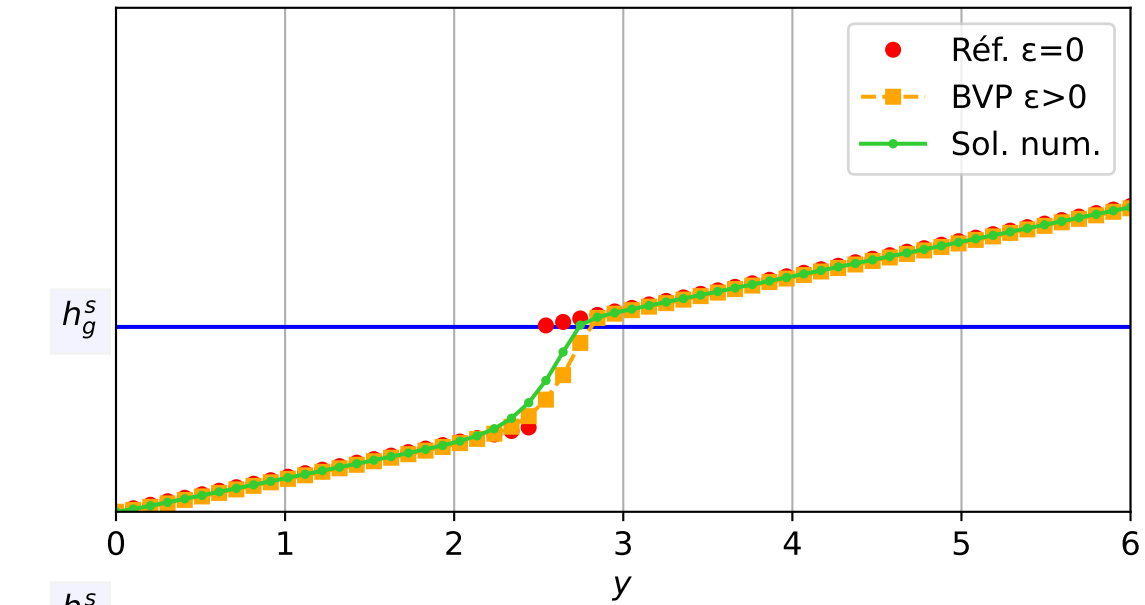


TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.50132/12$  (8938 iter) — résidu eq. =  $8.50e-06$





TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.5502/12$  (8958 iter) — résidu eq. =  $7.42e-06$



TEST-3\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-03$   
 $\bar{\epsilon} = 1e-03$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.55508/12$  (8960 iter) — résidu eq. =  $9.99e-06$

